Cambridge International A Level

Hardest Questions Mathematics 9709

Pure Mathematics 1 & Pure Mathematics 3

Total Questions	**541**
Pure Mathematics 1	**215**
Pure Mathematics 3	**326**
Years Covered	**2009–2025**
Source	**CIE Official Examiner Reports**

Each question is proven hard via the official Examiner Report commentary, and is accompanied by its full mark scheme and examiner remarks.

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Preface — Question Index by Year

YEAR	SESSION	PAPER	QUESTIONS
2009	Oct/Nov 2009	Paper 3 v32	Q3, Q5, Q6, Q8, Q9
2010	May/June 2010	Paper 3 v31	Q1, Q4, Q6, Q7, Q10
	Oct/Nov 2010	Paper 1 v11	Q1, Q2, Q3, Q4, Q5, Q7
		Paper 1 v12	Q1, Q4, Q5, Q8, Q9, Q11
		Paper 1 v13	Q8
		Paper 3 v31	Q2, Q3, Q5, Q7
		Paper 3 v32	Q2, Q3, Q5, Q10
		Paper 3 v33	Q1, Q2, Q4, Q7, Q8, Q9
2011	May/June 2011	Paper 3 v31	Q3, Q4, Q6, Q9
2012	May/June 2012	Paper 1 v12	Q2, Q7, Q8, Q9, Q10
		Paper 3 v31	Q1, Q3, Q5, Q6, Q8, Q9
	Oct/Nov 2012	Paper 3 v31	Q2, Q5, Q6, Q7, Q10
2013	May/June 2013	Paper 3 v31	Q1, Q4, Q6, Q7, Q8, Q9, Q10
	Oct/Nov 2013	Paper 3 v32	Q6, Q10
2014	May/June 2014	Paper 3 v31	Q1, Q4, Q6, Q8
	Oct/Nov 2014	Paper 1 v13	Q4, Q10
		Paper 3 v31	Q2, Q4, Q5, Q6
		Paper 3 v33	Q2, Q5, Q6, Q8, Q9
2015	May/June 2015	Paper 3 v31	Q2, Q3, Q9
	Oct/Nov 2015	Paper 1 v11	Q5, Q11
		Paper 1 v12	Q1, Q3, Q4, Q6, Q8, Q9, Q10
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		Paper 3 v31	Q4, Q6
		Paper 3 v32	Q4, Q6
		Paper 3 v33	Q2, Q6, Q7, Q9, Q10
2016	Feb/Mar 2016	Paper 1 v12	Q1, Q2, Q3, Q4, Q6, Q7, Q8, Q9
		Paper 3 v32	Q4, Q8, Q9
	May/June 2016	Paper 1 v12	Q5, Q6, Q7, Q10, Q11
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		Paper 3 v31	Q6, Q7, Q9, Q10
		Paper 3 v32	Q1, Q9, Q10
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	Oct/Nov 2016	Paper 1 v11	Q7
		Paper 1 v12	Q1, Q2, Q3, Q5, Q8
		Paper 1 v13	Q1, Q3, Q5, Q6, Q8, Q9, Q11
		Paper 3 v31	Q1, Q2, Q4, Q5, Q6, Q7, Q8, Q9, Q10
		Paper 3 v32	Q1, Q2, Q4, Q5, Q6, Q7, Q8, Q9, Q10
		Paper 3 v33	Q1, Q4, Q8, Q10
2017	Feb/Mar 2017	Paper 1 v12	Q1, Q3, Q5
		Paper 3 v32	Q1, Q4, Q5, Q7, Q8
	May/June 2017	Paper 1 v11	Q6, Q9
		Paper 1 v12	Q1, Q3, Q5, Q8, Q9, Q10
		Paper 1 v13	Q8, Q9, Q10
		Paper 3 v31	Q2, Q3, Q4, Q5, Q6, Q7
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		Paper 3 v33	Q3, Q5, Q6, Q7, Q8, Q9
	Oct/Nov 2017	Paper 1 v11	Q6
		Paper 1 v12	Q2, Q3, Q4, Q5, Q6, Q9
		Paper 1 v13	Q2, Q3, Q7, Q8, Q9
		Paper 3 v31	Q2, Q4, Q6
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		Paper 3 v33	Q2, Q4, Q6
2018	Feb/Mar 2018	Paper 3 v32	Q3, Q4, Q5, Q6, Q7, Q9
	May/June 2018	Paper 1 v11	Q8, Q9
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		Paper 3 v31	Q1, Q2, Q5, Q6, Q7, Q8, Q10
		Paper 3 v32	Q1, Q2, Q4, Q5, Q6, Q7
		Paper 3 v33	Q1, Q5, Q6, Q8, Q9, Q10
	Oct/Nov 2018	Paper 1 v12	Q3, Q4, Q6, Q7, Q10, Q11
		Paper 1 v13	Q1, Q3, Q4, Q6, Q8, Q10
		Paper 3 v31	Q1, Q2, Q3, Q5, Q6, Q7, Q8, Q10
		Paper 3 v32	Q1, Q4, Q6, Q7, Q9, Q10
		Paper 3 v33	Q1, Q2, Q3, Q5, Q6, Q7, Q8, Q10
2019	Feb/Mar 2019	Paper 1 v12	Q8
		Paper 3 v32	Q2, Q3, Q5
	May/June 2019	Paper 1 v11	Q2, Q7
		Paper 1 v12	Q4, Q5, Q8, Q9, Q10
		Paper 1 v13	Q3, Q4, Q8, Q9, Q10
		Paper 3 v31	Q1, Q2, Q3, Q4, Q6
		Paper 3 v32	Q1, Q2, Q5, Q6, Q9
		Paper 3 v33	Q1, Q4
	Oct/Nov 2019	Paper 1 v11	Q4, Q6, Q10

YEAR	SESSION	PAPER	QUESTIONS
		Paper 1 v12	Q3, Q4, Q6, Q10
		Paper 1 v13	Q2, Q9, Q10, Q11
		Paper 3 v31	Q1, Q4, Q7, Q8
		Paper 3 v32	Q2, Q3, Q4, Q6, Q7, Q8
		Paper 3 v33	Q2, Q4, Q5, Q8, Q9, Q10
2020	Oct/Nov 2020	Paper 1 v12	Q4, Q6, Q8, Q9, Q10, Q11
		Paper 1 v13	Q1, Q2, Q3, Q4, Q5, Q6, Q8, Q11
		Paper 3 v31	Q1, Q5, Q7, Q9, Q10
		Paper 3 v32	Q1, Q2, Q4, Q6, Q7, Q9, Q10
		Paper 3 v33	Q1, Q5, Q7, Q9, Q10
2021	Feb/Mar 2021	Paper 1 v12	Q1, Q8, Q10
		Paper 3 v32	Q1, Q4, Q5, Q6, Q7, Q9, Q10
	May/June 2021	Paper 1 v11	Q2, Q4, Q7, Q8, Q9, Q10, Q11
		Paper 1 v12	Q3, Q7, Q12
		Paper 1 v13	Q4, Q6, Q10, Q11
		Paper 3 v31	Q1, Q3, Q4, Q5
		Paper 3 v32	Q1, Q2, Q4
		Paper 3 v33	Q3, Q6, Q7, Q8, Q9
2022	Feb/Mar 2022	Paper 1 v12	Q11
		Paper 3 v32	Q3, Q4, Q6, Q7, Q9, Q11
	May/June 2022	Paper 1 v11	Q3, Q5, Q6, Q7, Q8, Q10
		Paper 1 v12	Q5, Q6, Q7, Q10, Q11
		Paper 1 v13	Q4, Q11
		Paper 3 v31	Q1, Q2, Q3, Q7, Q8, Q9
		Paper 3 v32	Q1, Q3, Q4, Q7, Q9
		Paper 3 v33	Q2, Q3, Q5, Q8, Q9
	Oct/Nov 2022	Paper 1 v11	Q2, Q3, Q4, Q10, Q11
		Paper 1 v12	Q1, Q3, Q5, Q6, Q9, Q10, Q11
		Paper 1 v13	Q4, Q5, Q10, Q11
		Paper 3 v31	Q1, Q3, Q5, Q6, Q7, Q8, Q10, Q11
		Paper 3 v32	Q1, Q2, Q4, Q5, Q6, Q7, Q8, Q9, Q10
		Paper 3 v33	Q3, Q5, Q9, Q10
2023	Feb/Mar 2023	Paper 1 v12	Q2, Q4, Q11
		Paper 3 v32	Q1, Q3, Q4, Q9, Q10, Q11
	May/June 2023	Paper 1 v11	Q5, Q8, Q9, Q10, Q11, Q12
		Paper 1 v12	Q3, Q4, Q5, Q6, Q8, Q10
		Paper 1 v13	Q1, Q10
		Paper 3 v31	Q1, Q8, Q9, Q10
		Paper 3 v32	Q6, Q7, Q8, Q9, Q10, Q11
		Paper 3 v33	Q3, Q6, Q7, Q8, Q9, Q10
2025	Feb/Mar 2025	Paper 1 v12	Q3, Q7, Q9, Q10
		Paper 3 v32	Q1, Q2, Q3, Q9

Question 3

Feb/Mar 2025 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> “Not dealing with the minus sign correctly or ignoring it completely was a common error.”

Source: 9709_m25_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

3 (a) Find the complete expansion of $\left(2x - \frac{3}{x}\right)^4$. [4]

(b) Hence determine the coefficient of x^2 in the expansion of $\left(x^2 + 5\right)\left(2x - \frac{3}{x}\right)^4$. [2]

3(a)

$(2x)^4$ and $\left(\frac{\pm 3}{x}\right)^4$ B1

$4(2x)^3\left(\frac{-3}{x}\right)+6(2x)^2\left(\frac{-3}{x}\right)^2+4(2x)\left(\frac{-3}{x}\right)^3$ B1

$\left[16x^4\right]+k_1x^2+k_2\left[x^0\right]+k_3x^{-2}\left[+81x^{-4}\right]$ M1 OE

$16x^4-96x^2+216-216x^{-2}+81x^{-4}$ A1 OE

Total: 4

3(b)

Use of (their 216) $+5\times$ (their -96) only, to arrive at the coefficient of x^2 M1

-264 A1 Accept $-264x^2$ as the final answer.

Total: 2

Question 7

Feb/Mar 2025 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "The other solution for sin2θ should have produced two values for sinθ, which many candidates did not realise and consequently did not find all three solutions."

Source: 9709_m25_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 12), Question 7

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7 **(a)** Show that $3\tan^2\theta + 5\sin^2\theta \equiv \dfrac{8\sin^2\theta - 5\sin^4\theta}{1-\sin^2\theta}$. [3]

(b) Hence solve the equation $3\tan^2\theta + 5\sin^2\theta = 9$ for $0° < \theta < 270°$. [4]

Coaching With Shuja

7(a)

Use of $\tan^2\theta = \dfrac{\sin^2\theta}{\cos^2\theta}$ **M1**

Relevant use of $\cos^2\theta = 1 - \sin^2\theta$ at least once **M1**

$\dfrac{8\sin^2\theta - 5\sin^4\theta}{1 - \sin^2\theta}$ **A1** AG All necessary detail needed.

Total: 3

7(b)

Attempt to solve their $5\sin^4\theta - 17\sin^2\theta + 9 = 0$ *M1

$\sin\theta = [\pm]0.81[0]$ or $\sqrt{\dfrac{17 - \sqrt{109}}{10}}$ **A1**

Their **54.1**, and **180−** their **54.1** or **180+** their **54.1** **DM1**

54.1, 125.9, 234.1 **A1** AWRT

Total: 4

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Question 9

Feb/Mar 2025 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> “dy A significant number of candidates incorrectly used = 9 after the first integration, whilst others dx ignored the requirement for a constant of integration at that stage altogether.”

Source: 9709_m25_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 12), Question 9

Coaching With Shuja

9 A curve is such that $\frac{d^2y}{dx^2} = \frac{6}{x^4} - \frac{5}{x^3}$. It is given that the curve has a stationary point at $\left(\frac{1}{2}, 9\right)$.

(a) Use the expression for $\frac{d^2y}{dx^2}$ to determine whether the stationary point is a maximum or a minimum point. [2]

(b) Find the equation of the curve. [7]

Coaching With Shuja

9(a)

$\frac{6}{\left(\frac{1}{2}\right)^4} - \frac{5}{\left(\frac{1}{2}\right)^3}$ **M1** Substitute $x = \frac{1}{2}$ and evaluate second derivative.

$96 - 40\,[= 56] > 0 \Rightarrow$ Minimum **A1** CWO. Evidence and conclusion.

Total: 2

9(b)

$\left\{\frac{6}{-3}x^{-3}\right\}\left\{-\frac{5}{-2}x^{-2}\right\}[+c_1]$ **B1 B1**

$0 = \frac{6}{-3}\left(\frac{1}{2}\right)^{-3} - \frac{5}{-2}\left(\frac{1}{2}\right)^{-2} + c_1$ **M1**

$c_1 = 6$ **A1**

$k_1x^{-2} + k_2x^{-1} + k_3x\,[+c_2]$ **M1**

$9 = \frac{1}{\left(\frac{1}{2}\right)^2} - \frac{5}{2\left(\frac{1}{2}\right)} + 6\left(\frac{1}{2}\right) + c_2$ **M1**

$y = x^{-2} - \frac{5}{2}x^{-1} + 6x + 7$ **A1** OE

Total: 7

Coaching With Shuja

Question 10

Feb/Mar 2025 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

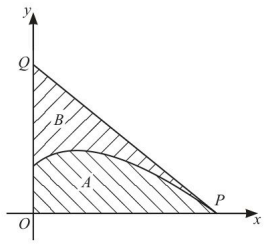
> "A more common error was that some candidates did not realise that finding was dx initially essential, and further progress was significantly hindered as a result."

Source: 9709_m25_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

Coaching With Shuja

10.

10



The diagram shows the curve with equation

$$y = 4(3x+4)^{\frac{1}{2}} - 2x - 6$$

for values of x such that $0 \leqslant x \leqslant 7$. The tangent to the curve at the point $P\,(7,0)$ meets the y-axis at the point Q. Region A is bounded by the curve and the two axes. Region B is bounded by the curve, the line segment PQ and the y-axis.

(a) Find the area of region A. [4]

The diagram shows the curve with equation

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for values of x such that $0 \leqslant x \leqslant 7$. The tangent to the curve at the point $P\,(7,0)$ meets the y-axis at the point Q. Region A is bounded by the curve and the two axes. Region B is bounded by the curve, the line segment PQ and the y-axis.

(a) Find the area of region A. [4]

(b) Find the area of region B. [5]

10(a)

$\left\{\frac{2}{3} \times \frac{4}{3}(3x+4)^{\frac{3}{2}}\right\}\left\{-\frac{2x^2}{2} - 6x\right\}$ **B1 B1**

$\left[A =\right] \left(\dfrac{8}{9}(21+4)^{\frac{3}{2}} - 7^2 - 6 \times 7\right) - \left(\dfrac{8}{9}(4)^{\frac{3}{2}}\right)$ **M1**

13 A1

Total: 4

10(b)

$\left(2(3x+4)^{-\frac{1}{2}} \times 3\right) - 2$ **B1**

$y - 0 = \left(\text{their } \frac{dy}{dx} \text{ with } x = 7\right)(x - 7)$ ***M1**

$[Q \text{ is}]\ \left(0,\ \frac{28}{5}\right)$ **DM1**

$\left[\dfrac{1}{2} \times 7 \times \text{their}\;\dfrac{28}{5}\right] - (\text{their}\;13\;\text{from (a)}) = \dfrac{33}{5}$ **A1 FT**

Total: 5

Question 1

Feb/Mar 2025 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A significant number of candidates struggled with this question, demonstrating a lack of knowledge of the rules of logarithms and how to apply them when solving equations."

Source: 9709_m25_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

Solve the equation

$$\ln(1 - e^{-2x}) + 3 = 0.$$

Give your final answer correct to 4 decimal places. [3]

Coaching With Shuja

State that $\mathbf{1 - e^{-2x} = e^{-3}}$ **B1** OE, with **ln** removed.

Use correct method to solve an equation of the form $\boldsymbol{e^{\pm 2x} = a}$, where $\boldsymbol{a > 0}$, and a reasonable attempt at the B1, for $\boldsymbol{\pm 2x \ln e}$ or $\boldsymbol{\pm x \ln e}$ **M1** E.g. $\boldsymbol{e^{-2x} = 1 - e^{-3}}$, $\boldsymbol{-2x = \ln(1 - e^{-3})}$ OE. Can be numerical $\boldsymbol{\ln(1 - 0.049787) = \ln 0.9502}$. Evidence of method must be seen.

Obtain answer **0.0255 A1** CAO. Must be 4 decimal places. No working seen scores 0.

Alternative Method for Question 1

State that $\mathbf{1 - e^{-2x} = e^{-3}}$ **B1** OE, without **ln**.

Rearrange to obtain an expression for $\boldsymbol{e^x}$ and solve an equation of the form $\boldsymbol{e^{\pm x} = a}$, where $\boldsymbol{a > 0}$, and a reasonable attempt at the B1, for $\boldsymbol{x}$ **M1** e.g. $\boldsymbol{e^x = \left(\frac{1}{1-e^{-3}}\right)^{1/2}}$, $\boldsymbol{x = \ln\left(\frac{1}{\sqrt{1-e^{-3}}}\right)}$. Can be numerical. Evidence of method must be seen.

Obtain answer **0.0255 A1** CAO. Must be 4 decimal places. No working seen scores 0.

Total: 3

Coaching with Shuja

Question 2

Feb/Mar 2025 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "However, many found it challenging to differentiate $\ln(x + 2y)$ correctly and therefore, unless they found the correct value for y at $x = 0$, gained no further marks."

> "A few candidates were able to gain marks following this rearrangement."

Source: 9709_m25_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Coaching With Shuja

The equation of a curve is

$$xy^2 + \ln(x + 2y) = 1.$$

Find the gradient of the curve at the point where $x = 0$. [5]

Coaching With Shuja

Mark Scheme

State or imply $\frac{1+2\frac{dy}{dx}}{x+2y}$ as the derivative of $\ln(x+2y)$ **B1**

State derivative of xy^2 is $x \cdot 2y\frac{dy}{dx} + y^2$ **B1**

Obtain $y^2 + 2xy\frac{dy}{dx} + \frac{1+2\frac{dy}{dx}}{x+2y} = 0$ **B1** OE. May be implied by correct final answer.

Obtain $y = \frac{1}{2}e$ when $x = 0$ **B1** OE. Allow $\frac{1}{2} \times 2.718$ or 1.36 or better e.g. 1.359... May be implied by correct final answer.

Obtain $\frac{dy}{dx} = -\frac{1}{8}(3e^3 + 4)$ **B1** OE. Accept AWRT -3.01. ISW.

Total: 5

Coaching With Shuja

Question 3

Feb/Mar 2025 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 3 (a) This proved to be a challenging question and relatively few candidates gained full marks."

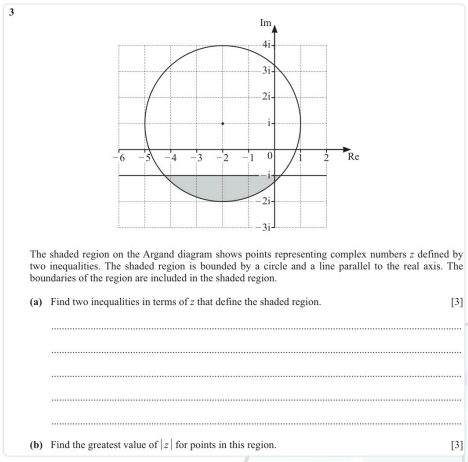
> "A common mistake was to use Im(z) ≤ i."

> "(b) This part also proved challenging for candidates, with many not attempting a solution at all and only very few gaining full marks."

Source: 9709_m25_er.pdf, Page 7, Section: Comments on specific questions – Paper 3 (variant 32), Question 3

Coaching With Shuja

3



The shaded region on the Argand diagram shows points representing complex numbers z defined by two inequalities. The shaded region is bounded by a circle and a line parallel to the real axis. The boundaries of the region are included in the shaded region.

(a) Find two inequalities in terms of z that define the shaded region. [3]

(b) Find the greatest value of $|z|$ for points in this region. [3]

(a)

Obtain $\mathbf{Im}(z) \leq -1$ **B1** Condone strict inequalities throughout (a).

Obtain answer of the form $|z-a| \leq b$ **M1** Accept equation or any inequality sign. $a = \pm 2 \pm i$ and $b = 3$, e.g. $|z+2-i| = 3$ or $|z+2-i| < 3$.

Obtain answer $|z+2-i| \leq 3$ **A1** Accept $|z-(-2+i)| \leq 3$ as final answer. Do not ISW.

Total: 3

(b)

Identify the coordinates of correct point $(-2-\sqrt{5}, -1)$ **M1*** From solving $(x \pm 2)^2 + (y \pm 1)^2 = 3^2$ with $y = -1$, or attempt to get $2+\sqrt{5}$ using a right-angled triangle.

Carry out a correct method for finding the greatest value of $|z|$ **DM1**

Obtain answer **4.35** or $\sqrt{10+4\sqrt{5}}$ **A1** AWRT 4.35, e.g. 4.3525...

Total: 3

Coaching With Shuja

Question 9

Feb/Mar 2025 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "(c) Very few candidates were fully successful in this part, perhaps because it relied on correct answers in both parts (a) and (b)."

Source: 9709_m25_er.pdf, Page 8, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Question Paper Feb/Mar 2025 • Paper 3 (variant 32) • Q9

The polynomial $6x^3 + ax^2 + bx + 9$ is denoted by $p(x)$, where a and b are constants. It is given that $(x - 3)$ is a factor of $p(x)$, and when the first derivative $p'(x)$ is divided by $(x - 3)$ the remainder is 72.

(a) Find the values of a and b. [5]

(b) When a and b have the values found in part (a), factorise $p(x)$ completely. [3]

(c) Hence solve the inequality $p(x) < 0$. [2]

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(a)

Substitute $x = 3$ or -3 into $p(x)$ and equate to 0 or into $p'(x)$ and equate to 72 **M1***

Obtain $162 + 9a + 3b + 9 = 0$ **A1** OE

Obtain $162 + 6a + b = 72$ **A1** OE

Solve simultaneous equations to obtain either a or b after using $p(\pm 3) = 0$ and $p'(\pm 3) = 72$ **DM1**

Obtain $a = -11$ and $b = -24$ **A1**

Total: 5

(b)

Equate $(x - 3)(6x^2 + Ax + B)$ to $6x^3 - 11x^2 - 24x + 9$ and obtain equations to solve for A and B, or divide $6x^3 - 11x^2 - 24x + 9$ by $x - 3$ and reach $6x^2 \pm 7x$ **M1** Using their a and their b.

$(x - 3)(6x^2 + 7x - 3)$ **A1** SOI

Obtain $(x - 3)(2x + 3)(3x - 1)$ **A1**

Special Case: If only $(x - 3)(2x + 3)(3x - 1)$ seen, SC B1 only, but can gain two marks in (c).

Total: 3

(c)

Obtain one correct region $x < -\frac{3}{2}$ or $\frac{1}{3} < x < 3$ **B1 FT** Must be final answer not in working. FT is on the last two brackets, not $(x - 3)$.

Obtain both regions $x < -\frac{3}{2}$, $\frac{1}{3} < x < 3$ **B1 FT** If incorrect factor or factors in (b) but correct regions here, allow SC B1 only.

Total: 2

Question 2

Feb/Mar 2023 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "The reflection proved to be most challenging of the three transformations, but other common errors included squaring 2x incorrectly and multiplying only some of the terms by 3."

> "A few candidates who carried out the three transformations correctly did not leave the result in the required form, so could not be awarded the final A1."

Source: 9709_m23_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 12), Question 2

Coaching With Shuja

Question Paper

Feb/Mar 2023 • Paper 1 (variant 12) • Q2

A function f is defined by $f(x) = x^2 - 2x + 5$ for $x \in \mathbb{R}$. A sequence of transformations is applied in the following order to the graph of $y = f(x)$ to give the graph of $y = g(x)$.

Stretch parallel to the x-axis with scale factor $\frac{1}{2}$

Reflection in the y-axis

Stretch parallel to the y-axis with scale factor 3

Find $g(x)$, giving your answer in the form $ax^2 + bx + c$, where a, b and c are constants. [4]

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Mark Scheme

Stretch: $(2x)^2 - 2(2x) + 5$ or $(2x-1)^2 + 4$ leading to $(2x-1)^2 + 4$ **M1** Replacing x by $2x$.

Reflection: $(-2x)^2 - 2(-2x) + 5$ or $(-2x-1)^2 + 4$ **M1** Replacing x by $-x$. FT on their stretch.

Stretch: $3\{(-2x)^2 - 2(-2x) + 5\}$ or $3\{(-2x-1)^2 + 4\}$ **M1** Multiplying the whole function by 3. FT on their stretch plus reflection.

$12x^2 + 12x + 15$ **A1**

Total: 4

Coaching With Shuja

Question 4

Feb/Mar 2023 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 4 (a) Most candidates did not take account of the fact that the first term should be 5.02 with n = 20 (or 5.00 with n = 21) and could only be awarded a mark for selection and use of the correct formula."

> "(b) Incorrect interpretation of the number of terms was a common error for most candidates."

Source: 9709_m23_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

Coaching With Shuja

Question Paper Feb/Mar 2023 • Paper 1 (variant 12) • Q4

The circumference round the trunk of a large tree is measured and found to be **5.00 m**. After one year the circumference is measured again and found to be **5.02 m**.

(a) Given that the circumferences at yearly intervals form an arithmetic progression, find the circumference 20 years after the first measurement. [2]

(b) Given instead that the circumferences at yearly intervals form a geometric progression, find the circumference 20 years after the first measurement. [3]

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(a)

$5.00 + 20 \times 0.02$ or $5.02 + 19 \times 0.02$ **M1** Allow for $a = 5$, $n = 20$ with $d = 0.02$ only. $a = 5$, $n = 21$ (OE) with $d = 0.2$ gets M1 only.

5.40 A1

Total: 2

(b)

$r = \frac{5.02}{5} = 1.004$ or $\frac{251}{250}$ **B1**

$5.00(1.004)^{20}$ or $5.02(1.004)^{19}$ **M1** Allow $a = 5$, $n = 20$.

5.42 A1 Any correct rounding of 5.41557108.

Total: 3

Coaching With Shuja

Question 11

Feb/Mar 2023 | Paper 1 (variant 12)

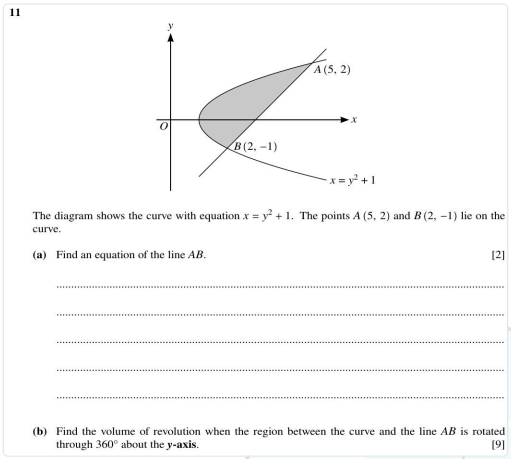
Examiner Report — Why students found this question difficult

> "The most common errors included inverting the gradient or swapping the x and y coordinates when using the straight-line equation."

Source: 9709_m23_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 12), Question 11

Coaching With Shuja

11



The diagram shows the curve with equation $x = y^2 + 1$. The points $A(5, 2)$ and $B(2, -1)$ lie on the curve.

(a) Find an equation of the line AB. [2]

(b) Find the volume of revolution when the region between the curve and the line AB is rotated through $360°$ about the y-axis. [9]

(a)

Gradient of $AB = \frac{2-(-1)}{5-2}$ **M1** Expect 1, must be from $\Delta y/\Delta x$.

Equation of AB is $y-2=1(x-5)$ or $y+1=1(x-2)$ **A1** OE. Expect $y=x-3$.

Total: 2

(b)

$\pi\int x^2\,dy = \pi\int(y^2+1)^2\,dy$ **M1** For curve: Attempt to square y^2+1 and attempt integration. Subtracting curve equation from line equation before squaring is M0. Integration before squaring M0.

$\pi\left[\frac{y^5}{5}+\frac{2y^3}{3}+y\right]$ **A2,1,0**

$\pi\int(y+3)^2\,dy = \pi\int(y^2+6y+9)\,dy$ **M1** For line: Attempt to square their $y+3$ and attempt integration.

$\pi\left[\frac{y^3}{3}+3y^2+9y\right]$ or $\pi\left[\frac{(y+3)^3}{3}\right]$ **A2,1,0** Not available for incorrect line equations.

$\pi\left\{\left(\frac{8}{3}+12+18\right)-\left(-\frac{1}{3}+3-9\right)\right\}$ or $\pi\left\{\left(\frac{32}{5}+\frac{16}{3}+2\right)-\left(-\frac{1}{5}-\frac{2}{3}-1\right)\right\}$ **DM1** Apply limits $-1\rightarrow 2$ to either integral providing they have been awarded M1. Expect $15\frac{3}{5}\pi$ and/or 39π. Some evidence of substitution of both -1 and 2 must be seen. Dependent on at least one of the first 2 M1 marks.

Volume $= \pi\left(39-15\frac{3}{5}\right)$ **DM1** Appropriate subtraction. Dependent on at least one of the first 2 M1 marks.

$= 23\frac{2}{5}\pi$ or $\frac{117\pi}{5}$ or awrt $73.5[1327]$ **A1**

Total: 9

Question 1

Feb/Mar 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 1 Most candidates scored full marks, however a few believed ln(a ± b) could be replaced by lna ± lnb and hence made no progress."

Source: 9709_m23_er.pdf, Page 5, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

Question Paper Feb/Mar 2023 • Paper 3 (variant 32) • Q1

It is given that $x = \ln(2y - 3) - \ln(y + 4)$.

Express y in terms of x. [3]

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Mark Scheme

Feb/Mar 2023 • Paper 3 (variant 32) • Q1

Use law of the logarithm of a quotient or express x as $\ln e^x$ **M1** $x = \ln\left(\frac{2y-3}{y+4}\right)$ or $\ln e^x = \ln(2y-3) - \ln(y+4)$.

Remove logarithms and obtain a correct equation e.g. $e^x = \frac{2y-3}{y+4}$ **A1**

Obtain answer $y = \frac{3+4e^x}{2-e^x}$ **A1** OE ISW

Total: 3

Coaching With Shuja

Question 3

Feb/Mar 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Candidates found it challenging to deal with the unknown constants in the polynomial when attempting long division, hence they rarely had a linear expression to equate to the given remainder."

> "This is something which candidates found difficult, as seen by their work in Question 4."

Source: 9709_m23_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 3

Coaching With Shuja

Question Paper Feb/Mar 2023 • Paper 3 (variant 32) • Q3

The polynomial $2x^4 + ax^3 + bx - 1$, where a and b are constants, is denoted by $p(x)$. When $p(x)$ is divided by $x^2 - x + 1$ the remainder is $3x + 2$.

Find the values of a and b. [5]

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Commence division and reach partial quotient $2x^2 + (a \pm 2)x$ **M1** Need $2x^2 + (a \pm 2)x$.

Obtain correct quotient $2x^2 + (a + 2)x + a$ **A1** Allow sign error e.g. in $b - 2$.

Set their linear remainder equal to part of $3x + 2$ and solve for a or for b **M1** Remainder $= 3x + 2 = (b - 2)x - 1 - a$. Allow for just equating x term or constant term.

Obtain answer $a = -3$ **A1**

Obtain answer $b = 5$ **A1**

Alternative method for Question 3

State $2x^4 + ax^3 + 0x^2 + bx - 1 = (x^2 - x + 1)(2x^2 + Ax + B) + 3x + 2$ and form and solve equation(s) to obtain A or B **M1** e.g. $0 = B - A + 2$ and $-1 = B + 2$.

Obtain $A = -1$, $B = -3$ **A1**

Form and solve equations for a or for b **M1** e.g. $a = A - 2$ or $b = -B + A + 3$.

Obtain answer $a = -3$ **A1**

Obtain answer $b = 5$ **A1**

Alternative method for Question 3

Use remainder theorem with $x = \frac{1 \pm \sqrt{3}i}{2}$ **M1** Allow correct use of a reasonable attempt at either root in exact or decimal form in the remainder theorem.

Obtain an exact equation for either root **A1**

Solve simultaneous equations, or single equation, for a or for b **M1**

Obtain answer $a = -3$ from exact working **A1**

Obtain answer $b = 5$ from exact working **A1**

Total: 5

Question 4

Feb/Mar 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “Question 4 Unfortunately, this was the third question in a row which candidates found challenging.”

> “Few candidates reached a correct simplified equation.”

Source: 9709_m23_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

Solve the equation

$$\frac{5z}{1+2i} - zz^* + 30 + 10i = 0,$$

giving your answers in the form $x + iy$, where x and y are real. [5]

Mark Scheme

Substitute $z = x + iy$ and $z^* = x - iy$ to obtain a correct equation, horizontal or with $(1 - 2i)/(1 - 2i)$ seen, in x and y **B1**

$$5(x + iy) - (x + iy)(x - iy)(1 + 2i) + (30 + 10i)(1 + 2i) = 0$$

or

$$\frac{5(x+iy)(1-2i)}{(1+2i)(1-2i)} - (x + iy)(x - iy) + (30 + 10i) = 0$$

Use $i^2 = -1$ at least once and equate real and imaginary parts to zero *M1 OE for their horizontal equation.

Obtain two correct equations e.g. $x + 2y - x^2 - y^2 + 30 = 0$ and $-2x + y + 10 = 0$ **A1**

Solve quadratic equation for x or for y **DM1** $x^2 - 9x + 18 = (x - 3)(x - 6) = 0$ or $y^2 + 2y - 8 = (y + 4)(y - 2) = 0$. DM0 if x or y imaginary.

Obtain answers $3 - 4i$ and $6 + 2i$ **A1**

Total: 5

Coaching with Shuja

Question 9

Feb/Mar 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “Question 9 This question proved extremely difficult for most.”

Source: 9709_m23_er.pdf, Page 7, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = e^{3y}\sin^2 2x.$$

It is given that $y = 0$ when $x = 0$.

Solve the differential equation and find the value of y when $x = \frac{1}{2}$. [7]

Mark Scheme

Separate variables correctly and obtain e^{-3y} and $\sin^2 2x$ on the opposite sides **B1**

Obtain term $-\frac{1}{3}e^{-3y}$ **B1**

Use correct double angle formula for $\sin^2 2x = \frac{1}{2}(1 - \cos 4x)$ **M1**

Obtain terms $\frac{1}{2}x - \frac{1}{8}\sin 4x$ OE **A1**

Use $x = 0$, $y = 0$ to evaluate a constant or as limits in a solution containing terms of the form ax and $b\sin 4x$ and $ce^{\pm 3y}$ **M1**

Obtain correct answer in any form e.g. $-\frac{1}{3}e^{-3y} = \frac{1}{2}x - \frac{1}{8}\sin 4x - \frac{1}{3}$ **A1**

Substitute $x = \frac{1}{2}$ and obtain $y = 0.175$ or $-\frac{1}{3}\ln\left(1 - \frac{3}{4} + \frac{3}{8}\sin 2\right)$ **A1** OE ISW

Total: 7

Coaching With Shuja

Question 10

Feb/Mar 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “However, the most common error was the incorrect notation on the left-hand side of the equation.”

> “not left blank, expressed in words or just denoted as l.”

Source: 9709_m23_er.pdf, Page 8, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

Question Paper Feb/Mar 2023 • Paper 3 (variant 32) • Q10

With respect to the origin O, the points A, B, C and D have position vectors given by

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}, \quad \overrightarrow{OC} = \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix}, \quad \overrightarrow{OD} = \begin{pmatrix} 5 \\ -6 \\ 11 \end{pmatrix}.$$

(a) Find the obtuse angle between the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$. [3]

The line l passes through the points A and B.

(b) Find a vector equation for the line l. [2]

(c) Find the position vector of the point of intersection of the line l and the line passing through C and D. [4]

Coaching With Shuja

(a)

Carry out correct process for evaluating the scalar product of $\overrightarrow{OA}$ and $\overrightarrow{OB}$ **M1**

$\pm(3,-1,2)\cdot(1,2,-3)=\pm(3-2-6)=[-5]$.

Using the correct process for the moduli, divide the scalar product by the product of the moduli and obtain $\cos^{-1}\left\{\frac{\pm(3-2-6)}{\sqrt{3^2+(-1)^2+2^2}\sqrt{1^2+2^2+(-3)^2}}\right\}$ **A1**

Obtain answer $110.9°$ or 1.94^c **A1**

Total: 3

(b)

Use a correct method to form an equation for line through AB **M1**

Obtain $\mathbf{r}=3\mathbf{i}-\mathbf{j}+2\mathbf{k}+\mu_1(2\mathbf{i}-3\mathbf{j}+5\mathbf{k})$ **A1** OE e.g. $\mathbf{r}=\mathbf{i}+2\mathbf{j}-3\mathbf{k}+\mu_2(-2\mathbf{i}+3\mathbf{j}-5\mathbf{k})$. Need $\mathbf{r}$ or (x,y,z).

Total: 2

(c)

Obtain a correct equation for line through CD, e.g. $\mathbf{r}=\mathbf{i}-2\mathbf{j}+5\mathbf{k}+\lambda_1(-4\mathbf{i}+4\mathbf{j}-6\mathbf{k})$ **B1** OE e.g. $\mathbf{r}=5\mathbf{i}-6\mathbf{j}+11\mathbf{k}+\lambda_2(-4\mathbf{i}+4\mathbf{j}-6\mathbf{k})$. $\mathbf{r}$ can be omitted or another symbol used.

Equate two pairs of components of general points on their l and their CD and solve for λ or for μ **M1**

Obtain e.g. $\lambda_1=-2$ or $\mu_1=3$ or $\lambda_2=-1$ or $\mu_2=-4$ **A1**

Obtain position vector $9\mathbf{i}-10\mathbf{j}+17\mathbf{k}$ **A1** Condone $(9,-10,17)$ but not $(9\mathbf{i},-10\mathbf{j},17\mathbf{k})$.

Total: 4

Question 11

Feb/Mar 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “A Bx+C (b) Virtually all candidates established the correct integral for , however the integral of 1+x 4+x2 Bx C proved challenging for most.”

Source: 9709_m23_er.pdf, Page 8, Section: Comments on specific questions – Paper 3 (variant 32), Question 11

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Let $f(x) = \frac{5x^2+x+11}{(4+x^2)(1+x)}$.

(a) Express $f(x)$ in partial fractions. [5]

(b) Hence show that $\int_0^2 f(x)\,dx = \ln 54 - \frac{1}{8}\pi$. [5]

Coaching With Shuja

(a)

State or imply the form $\frac{Ax+B}{4+x^2} + \frac{C}{1+x}$ **B1**

Use a correct method for finding a coefficient **M1** $(Ax+B)(1+x)+C(4+x^2) = 5x^2+x+11$.

Obtain one of $A = 2$, $B = -1$ and $C = 3$ **A1** If error present in above still allow A1 for C.

Obtain a second value **A1**

Obtain the third value **A1** If $A = 0$ then max M1 A1 for C.

Total: 5

(b)

Integrate and obtain terms $\frac{A}{2}\ln(4+x^2)$, $\frac{B}{2}\tan^{-1}\left(\frac{x}{2}\right)$ and $C\ln(1+x)$ **B1FT + B1FT + B1FT** The FT is on A, B and C. Integral of $\frac{Ax+B}{4+x^2}$ is $\frac{B}{2}\tan^{-1}\left(\frac{x}{2}\right)$ or $\frac{A}{2}\ln(4+x^2)$ only. BoFT unless clearly from single term.

Substitute limits 0 and 2 correctly in an integral of the form $a\ln(4+x^2)+b\tan^{-1}\left(\frac{x}{2}\right)+c\ln(1+x)$, where $abc \neq 0$ **M1** Allow one sign or substitution error. Allow omission of $b\tan^{-1}0 + c\ln 1$.

Obtain answer $\ln 54 - \frac{\pi}{8}$ after full and correct working **A1** AG. Work to combine or simplify at least 2 of $\ln$ terms is required CWO. A0 if any non-exact value(s) seen.

Total: 5

Question 5

May/June 2023 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Question 5 Candidates appeared to find this question challenging, with many of them unable to proceed with a method or introducing multiple algebraic and sign errors."

Source: 9709_s23_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 5

Coaching With Shuja

Question Paper May/June 2023 • Paper 1 (variant 11) • Q5

The line with equation $y = kx - k$, where k is a positive constant, is a tangent to the curve with equation $y = -\frac{1}{2x}$.

Find, in either order, the value of k and the coordinates of the point where the tangent meets the curve. [5]

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Method 1: Equating equations

$kx - k = -\frac{1}{2x} \Rightarrow 2kx^2 - 2kx + 1 = 0$ **M1**

OE e.g. $kx^2 - kx + \frac{1}{2} = 0$, $x^2 - x + \frac{1}{2k} = 0$

Equate line and curve to form 3-term quadratic (all terms on one side).

OR quadratic in y: $x = \frac{y+k}{k} \Rightarrow y = -\frac{1}{2\left(\frac{y+k}{k}\right)} \Rightarrow 2y^2 + 2ky + k = 0$ **M1**

$b^2 - 4ac = 0 \Rightarrow ([-]2k)^2 - 4(2k)(1) = 0$ **DM1**

or $4k^2 - 8k = 0 \Rightarrow 4k(k-2) = 0$

OR using equation in y: $(2k)^2 - 4(2)(k) = 0$

Use discriminant correctly with their a, b, c not in quadratic formula. DM0 if x still present.

May see $k^2 - 4(k)\left(\frac{1}{2}\right) = 0$ or $1 - 4\left(\frac{1}{2k}\right) = 0$.

$k = 2$ only **A1**

If DM0 then $k = 2$, award A0 XP then B0 B0

Allow A1 even if divides by k to solve.

If $k = 0$ also present but uses $k = 2$, award A1.

$4x^2 - 4x + 1 = 0 \Rightarrow [2x-1]^2 = 0 \Rightarrow x = \frac{1}{2}$ **B1**

$y = 2 \times \frac{1}{2} - 2 = -1$ **B1**

Total: 5

Alternative method for Q5

$\frac{dy}{dx} = \frac{1}{2x^2}$ or $\frac{1}{2}x^{-2}$ **M1**

Differentiate $-\frac{1}{2x}$ M0 for $2x^{-2}$. No errors.

$[y =]\frac{1}{2x^2}x - \frac{1}{2x^2} = -\frac{1}{2x}$ or $- = \frac{1}{2x^2}$ $[\Rightarrow 2x^2 - x = 0]$ **DM1**

Sub their $\frac{dy}{dx}$ into equation of line or set gradient $= k$

to form equation in x.

$x = \frac{1}{2}$ only **A1**

If DM0 then $x = \frac{1}{2}$, award A0XP then B0 B0.

$y = \left[2x\frac{1}{2} - 2\right] = -1$ **B1**

$k = 2$ **B1**

Total: 5

Question 8

May/June 2023 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "A common error was omitting to multiply the 1 when multiplying through by 7 – a to clear the denominator."

> "Finding b proved to be more difficult for those who seemed unsure how a composite function is formed."

> "Those who found f(5) first were generally more successful, though a common error was to omit the '1 +' in the formula for f(x)."

Source: 9709_s23_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 8

Coaching With Shuja

8 The functions f and g are defined as follows, where a and b are constants.

$$f(x) = 1 + \frac{2a}{x - a} \text{ for } x > a$$

$$g(x) = bx - 2 \text{ for } x \in \mathbb{R}$$

(a) Given that $f(7) = \frac{5}{2}$ and $gf(5) = 4$, find the values of a and b. **[4]**

For the rest of this question, you should use the value of a which you found in (a).

(b) Find the domain of f^{-1}. **[1]**

(c) Find an expression for $f^{-1}(x)$. **[3]**

(8)(a) $**M1** OE Substitute x = 7 then solve for a via legitimate mathematical steps. Condone sign errors only.$

$$1+\frac{2a}{7-a}=\frac{5}{2}\Rightarrow\frac{2a}{7-a}=\frac{3}{2}\Rightarrow 7a=21\Rightarrow a=\ldots$$

$$\text{OR } 1+\frac{2a}{7-a}=\frac{5}{2}\Rightarrow(7-a)+2a=\frac{5}{2}(7-a)[\Rightarrow 7a=21]\Rightarrow a=\ldots$$

$a=3$ $**A1**$ If M0, SC B1 for $a=3$ with no working.

$f(5)=1+\frac{2(\text{their3})}{5-\text{their3}}=4[\Rightarrow 4b-2=4]\Rightarrow b=\ldots$ $**M1**$ Evaluate $f(5)$, either separately or within gf then solve for b via legitimate mathematical steps. Cond

$$\text{OR } gf(5)=b\left(1+\frac{2(\text{their3})}{5-\text{their3}}\right)-2[\Rightarrow 4b-2=4]\Rightarrow b=\ldots$$

$b=\frac{3}{2}$ $**A1**$ OE e.g., $\frac{6}{4}$, 1.5.

Total: 4

(8)(b)

$x>1$ $**B1**$ Accept $(1,\infty)$ or $\{*;*>1\}$ where $*$ is any variable. B0 for $f^{-1}(x)>1$ or $f(x)>1$ or $y>1$.

Total: 1

(8)(c)

$$\text{EITHER } x-1=\frac{6}{y-3}\Rightarrow(y-3)(x-1)=6$$

$$\text{OR } x=1+\frac{6}{y-3}\Rightarrow x(y-3)=(y-3)+6$$

$y-3=\frac{6}{x-1}$ or $y(x-1)=3x+3$ $**DM1**$ OE $x-3=\frac{6}{y-1}$ or $x(y-1)=3y+3$. Allow DM1 for use of their 3 from (a).

$[f^{-1}(x)]=3+\frac{6}{x-1}$ $**A1**$ OE Correct answer e.g., $\frac{3x+3}{x-1}$ ISW. Must be in terms of x.

\[*M1 DM1 \text{ possible for ' } a \text{ ' used, but A0 so max 2/3.}\]

Total: 3

Question 9

May/June 2023 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Of those who did, a common error was to substitute h = 10 into the volume formula."

> "Many candidates did not answer this question, but those who did attempt it often did not realise differentiation was needed."

> "The most common error was to substitute h = 0.075."

Source: 9709_s23_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 9

Coaching With Shuja

9 Water is poured into a tank at a constant rate of $500\ \text{cm}^3$ per second. The depth of water in the tank, t seconds after filling starts, is h cm. When the depth of water in the tank is h cm, the volume, V cm^3, of water in the tank is given by the formula

$$V = \frac{4}{3}(25 + h)^3 - \frac{62500}{3}.$$

(a) Find the rate at which h is increasing at the instant when $h = 10$ cm. **[3]**

(b) At another instant, the rate at which h is increasing is 0.075 cm per second.

Find the value of V at this instant. **[3]**

(9)(a)

$\frac{dV}{dh} = \frac{4}{3} \times 3(25+h)^2 \quad [= 4900 \text{ when } h = 10]$ $**B1**$ Correct expression for $\frac{dV}{dh}$.

$\frac{dV}{dt} \times \frac{dh}{dt} \times \frac{dV}{dh} \Rightarrow \text{their } 'll4(25+10)^2'll \times \frac{dh}{dt} = 500 \Rightarrow \frac{dh}{dt} = \left[\frac{500}{4900}\right]$ $**M1**$ Use chain rule correctly to find a numerical expression for $\frac{dh}{dt}$. Accept e.g., $\frac{2500}{}$

$\frac{dh}{dt} = 0.102 [\text{cms}^{-1}]$ $**A1**$ AWRT OE e.g., $\frac{5}{49}$ ISW.

Total: 3

(9)(b)

$\frac{dV}{dt} \times \frac{dh}{dt} \Rightarrow 500 = \text{their } 'll4(25+h)^2'll \times 0.075$ $**M1**$ SOI Use chain rule correctly to form equation in h.

$(25+h)^2 = \frac{5000}{3} \Rightarrow h = [15.8248\ldots]$ $**DM1**$ Solve quadratic to find h.

Exact value of h is $\sqrt{\frac{5000}{3}} - 25$ or $\frac{50\sqrt{6}}{3} - 25$

$h + 25 = 40.82\ldots$

$V = 69900 \text{ cm}^3$ $**A1**$ AWRT ISW Look for 698(88.5).

Total: 3

Question 10

May/June 2023 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

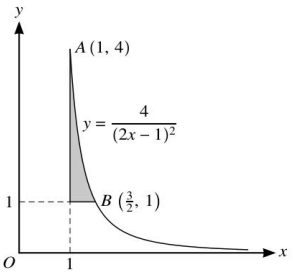
> "Some found the problem-solving aspects of this part more challenging, with many errors introduced, often through having an incorrect or no diagram."

Source: 9709_s23_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 10

Coaching With Shuja

10

10



The diagram shows part of the curve with equation $y = \frac{4}{(2x-1)^2}$ and parts of the lines $x = 1$ and $y = 1$.
The curve passes through the points $A\,(1, 4)$ and $B, \left(\frac{3}{2}, 1\right)$.

(a) Find the exact volume generated when the shaded region is rotated through 360° about the x-axis. [5]

The diagram shows part of the curve with equation

$$y = \frac{4}{(2x-1)^2}$$

and parts of the lines $x = 1$ and $y = 1$.

The curve passes through the points $A(1, 4)$ and $B\left(\frac{3}{2}, 1\right)$.

(a) Find the exact volume generated when the shaded region is rotated through 360° about the x-axis. **[5]**

(b) A triangle is formed from the tangent to the curve at B, the normal to the curve at B and the x-axis. Find the area of this triangle. **[6]**

(10)(a)

$[\pi]\int \frac{16}{(2x-1)^4}[\mathrm{dx}] = [\pi]\int 16(2x-1)^{-4}[\mathrm{dx}] = [\pi]\left(-\frac{16}{3\times 2\times (2x-1)^3}\right)$ ****M1** Integrate y^2 (power incr. by 1 or div by their new power). M0 if more than 1 er**

$[\pi]\left(-\frac{16}{3\times 2\times (2x-1)^3}\right)$ ****A1** OE e.g., $\left(-\frac{8}{3}(2x-1)^{-3}\right)$.**

$[\pi]\left(-\frac{16}{6\times 8}-\frac{16}{6\times 1}\right) = [\pi]\left[\frac{-112}{-7}\right] = [\pi]\frac{112}{-7}$ ****DM1** Sub correct limits into their integral: F$\left(\frac{3}{2}\right)$ − F(1).**

Must see at least $\left(-\frac{1}{3}+\frac{8}{3}\right)$. Allow 1 sign error.

Decimal: 2.33π or 7.33.

Volume of cylinder $= [\pi r^2 h]\times 2\times \frac{1}{2}] = \frac{1}{2}\pi$ **OR** $[\pi]\int_1^{1.5} 1[\mathrm{dx}] = \frac{1}{2}\pi$

Volume of revolution $= [\frac{7}{3}-\frac{1}{2}] = \frac{11}{6}\pi$ ****A1** A0 for 5.76 (not exact). If DM0 for insufficient substitution, or B0, SC B1 for $\frac{11}{6}\pi$.**

Total: 5

(10)(b) [Tangent and normal at point B calculation section omitted for brevity - involves finding derivative, gradient of tangent $= -\frac{3}{4}$, gradient of normal $= \frac{4}{3}$, equations of lines, intersection points, and triangle area calculation resulting in area = $\frac{25}{6}$]

Question 11

May/June 2023 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "A large number of arithmetical errors were seen in obtaining the coordinates, even from correct equations."

Source: 9709_s23_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 11), Question 11

Coaching With Shuja

11 The equation of a curve is such that

$$\frac{dy}{dx} = 6x^2 - 30x + 6a,$$

where a is a positive constant. The curve has a stationary point at $(a, -15)$.

(a) Find the value of a. **[2]**

(b) Determine the nature of this stationary point. **[2]**

(c) Find the equation of the curve. **[3]**

(d) Find the coordinates of any other stationary points on the curve. **[2]**

(11)(a)

$6a^2 - 30a + 6a = 0 \Rightarrow [6a(a-4) = 0]$ $**B1**$ Sub $x = a$ into $\frac{dy}{dx} = 0$. May see $a^2 - 5a + a = 0$.

$a = 4$ only $**B1**$

Total: 2

(11)(b)

$\frac{d^2y}{dx^2} = 12x - 30$ or correct values of $\frac{dy}{dx}$ either side of $x = 4$ $**M1**$ Differentiate $\frac{dy}{dx}$ (mult. by power or dec. power by 1)

M0 if no values of $\frac{dy}{dx}$, only signs.

At $x = 4$, $\frac{d^2y}{dx^2} > 0$; minimum or $\frac{d^2y}{dx^2} = 18$; minimum $**A1**$ WWW A0 XP if $a = 4$ obtained incorrectly in (a)

Must see 'minimum'.

If M0, SC B1 for 'minimum' from $\frac{dy}{dx}$ sign diagram.

Total: 2

(11)(c)

$[y =] \frac{6}{3}x^3 - \frac{30}{2}x^2 + 6(\text{their}a)[x] + [c]$ $**B1**$ FT Expect $2x^3 - 15x^2 + 24x + [c]$.

B1 poss. even if uses 'a' – no value in (a) – max 1/3.

$-15 = 2(\text{their } "4")^3 - 15(\text{their } "4")^2 + 6(\text{their } "4") + c$ $**M1**$ Sub $y =$ their "4", $y = -15$ into integral (must incl. $+c$)

Look for $-15 = 128 - 240 + 96 + c [\Rightarrow c = 1]$.

$y = 2x^3 - 15x^2 + 24x + 1$ $**A1**$ Coefficients must be correct and simplified.

Need to see '$y =$ ' or '$f(x) =$ ' in the working.

Total: 3

(11)(d)

$\frac{dy}{dx} = 6x^2 - 30x + 6(\text{their "4"})[= 0]$ $**M1**$ OE Forming a 3-term quadratic using the given $\frac{dy}{dx}$ and solving by factorisation, formula or completing the square. Check for working in (b).

If correct, $[6](x-1)(x-4) = [0]$ or $\frac{30 \pm \sqrt{(-30)^2 - 4(6)(24)}}{12}$

Coordinates $(1, 12)$ $**A1**$ Allow $x = 1, y = 12$ (ignore $x = 4$ if present).

If M0, award SC B1 for $(1, 12)$.

Total: 2

Question 12

May/June 2023 | Paper 1 (variant 11)

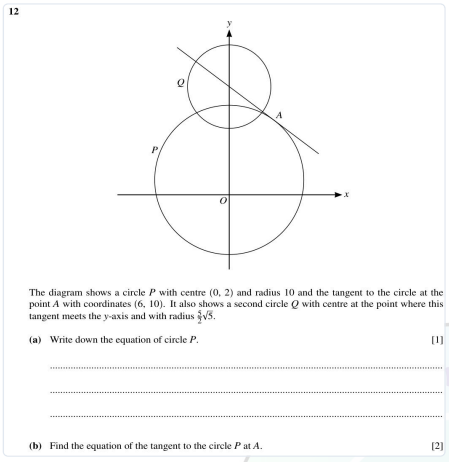
Examiner Report — Why students found this question difficult

> "Most candidates did not attempt to verify the y-coordinate of the points of intersection of the two circles."

Source: 9709_s23_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 11), Question 12

Coaching With Shuja

12



The diagram shows a circle P with centre $(0, 2)$ and radius 10 and the tangent to the circle at the point A with coordinates $(6, 10)$. It also shows a second circle Q with centre at the point where this tangent meets the y-axis and with radius $\frac{5}{2}\sqrt{5}$.

(a) Write down the equation of circle P. [1]

(b) Find the equation of the tangent to the circle P at A. [2]

(c) Find the equation of circle Q and hence verify that the y-coordinates of both of the points of intersection of the two circles are 11. [3]

(d) Find the coordinates of the points of intersection of the tangent and circle Q, giving the answers in surd form. [3]

(12)(a)

$x^2 + (y-2)^2 = 100$ **B1** OE e.g., $(x-0)^2 + (y-2)^2 = 10^2$ ISW.

Total: 1

(12)(b)

Gradient of radius $= \left[\frac{10-2}{6-0}\right] = \frac{4}{3}$ or gradient of tangent $= \frac{-3}{4}$ **M1** OE SOI Use coordinates to find gradient of radius or differentiate to find m_T

e.g., $2x + 2(y-2)\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-3}{4}$ at $(6, 10)$

Equation of tangent is $y - 10 = -\frac{3}{4}(x-6)$ $\left[\Rightarrow y = -\frac{3}{4}x + \frac{29}{2}\right]$ **A1** OE ISW Allow e.g., $\frac{58}{4}$.

Total: 2

(12)(c)

Coordinates of centre of circle Q are $\left(0, \text{their } \frac{29}{2}\right)$ **M1** SOI From a linear equation in (b).

Equation of circle Q is $x^2 + \left(y - \text{their } \frac{29}{2}\right)^2 = \left(\frac{5\sqrt{5}}{2}\right)^2 = \frac{125}{4}$ **A1FT** OE e.g., $(x-0)^2 + (y-14.5)^2 = 31.25$ ISW.

$x^2 + (11-2)^2 = 100 \Rightarrow x^2 + 19 = x^2 - 19$ [No] OR e.g., $x^2 + (11-11-2)^2 = \frac{125}{4} + (y-2)^2 = 100 \Rightarrow 25y = 275 \Rightarrow y = 11$

Correct argument to verify both y-coords are 11 **B1** OE e.g., $x^2 = [\pm]\sqrt{19}, x^2 - 19 = x^2 - 19$

Correct argument to verify both y-coords are 11 ISW.

Total: 3

(12)(d)

$x^2 + \left(-\frac{3}{4}x + \frac{29}{2} - \text{their } \frac{29}{2}\right)^2 = \frac{125}{4}$ $\left[\Rightarrow 25x^2 = 125 \Rightarrow x^2 = 20\right]$ **M1** Substitute equation of their tangent into equation of their circle. May see $y = \sqrt{31.25} - x^2 + 14.5$.

$x = \pm 2\sqrt{5}$ or $y = \frac{29 \pm 3\sqrt{5}}{2}$ **A1** OE e.g., $x = \pm\sqrt{20}$

For 2 x-values or 2 y-values or correct (x, y) pair.

$y\left[= \left(-\frac{3}{4}x \pm \sqrt{20}\right) + \frac{29}{2}\right] = \frac{29 \pm 3\sqrt{5}}{2}$ **A1** OE e.g., $\frac{58}{4} \pm \frac{3\sqrt{20}}{4}, \frac{58 \pm 3\sqrt{20}}{4}$ Correct (x, y) pairs.

Total: 3

Question 3

May/June 2023 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "These attempts were often successful initially, but simplifying the resulting inequality correctly proved challenging for many."

Source: 9709_s23_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

3 (a) Express $4x^2 - 24x + p$ in the form $a(x + b)^2 + c$, where a and b are integers and c is to be given in terms of the constant p. **[2]**

(b) Hence or otherwise find the set of values of p for which the equation $4x^2 - 24x + p = 0$ has no real roots. **[1]**

Coaching With Shuja

(3)(a)

$4(x-3)^2$ seen or $a=4$ and $b=-3$ $**B1**$ OE Award marks for the correct expression or their values

a, b and c. Condone $4(x-3)+p-36=0$ and $4\left(\frac{p}{4}-9\right)$.

$-36+p$ or $p-36$ seen or $c=p-36$ $**B1**$

Total: 2

(3)(b)

$p-36>0$ leading to $p>36$ or $24^2-4\times 4p(0 \Rightarrow p)36$ or $36<p<\infty$ $**B1**$ Allow $(36,\infty)$ or $36<p<\infty$. Consider final answer only.

Total: 1

Coaching With Shuja

Question 4

May/June 2023 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "The most common error however occurred 8 2 through using a calculator to solve the quadratic, rather than factorisation or other similar method."

Source: 9709_s23_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

Coaching With Shuja

4 Solve the equation $8x^6 + 215x^3 - 27 = 0$. **[3]**

(4)

$[8x^6 + 215x^3 - 27 = 0]$ leading to $(8x^3 - 1)(x^3 + 27) = [0]$ $**M1**$ OE

If a substitution is used then the correct coefficients must be retained. Condone substitution of $x = x^3$.

$$\frac{-215 \pm \sqrt{215^2 - 4.8.-27}}{2.8} \text{ or } \frac{-215 \pm \sqrt{47089}}{2.8}$$

$\frac{1}{8}, -27$ $**A1**$ Both correct values seen.

SC: if M0 scored SC B1 is available for sight of $\frac{1}{8}$ and -27 OE

$\frac{1}{2}$ or $0.5, -3$ $**A1**$ SC: if M0SCB1 scored then SCB1 is available for the correct answers and no others. Do not ISW if answers given as a range.

Total: 3

Question 5

May/June 2023 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

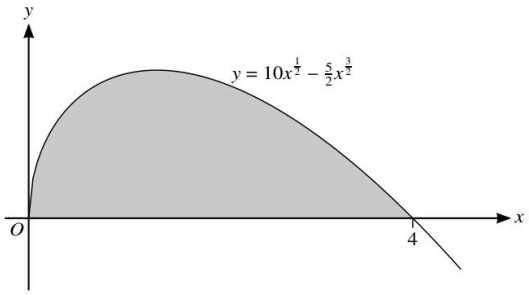
> "The 5 proved a challenge for some but most were able to integrate successfully."

Source: 9709_s23_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 5

Coaching With Shuja

5

5



The diagram shows the curve with equation $y = 10x^{\frac{1}{2}} - \frac{5}{2}x^{\frac{3}{2}}$ for $x > 0$. The curve meets the x-axis at the points (0, 0) and (4, 0).

Find the area of the shaded region. [4]

The diagram shows the curve with equation $\boldsymbol{y = 10x^{\frac{1}{2}} - \frac{5}{2}x^{\frac{3}{2}}}$ for $\boldsymbol{x > 0}$. The curve meets the $\boldsymbol{x}$-axis at the points $\boldsymbol{(0, 0)}$ and $\boldsymbol{(4, 0)}$.

Find the area of the shaded region. [4]

Shuja

(5)

$\left[\int\left(10x^{\frac{1}{2}}-\frac{5}{2}x^{\frac{3}{2}}\right)=\right]\left[\frac{10}{\frac{3}{2}}x^{\frac{3}{2}}-\frac{5}{2\times\frac{5}{2}}x^{\frac{5}{2}}\right]=\left[\frac{20}{3}x^{\frac{3}{2}}-x^{\frac{5}{2}}\right]$ **B1** B1 for contents of each {} then ISW.

$=\left[\text{their }\frac{20}{3}\times 8-32\right][-0]$ **M1** Using limit(s) correctly in an integrated expression (defined by one correct power). Minimum acceptable working is their $\left(\frac{160}{3}-32\right)$.

$[\text{Area of shaded region }]=\left[\frac{64}{3}, 21\frac{1}{3}\text{ or }21.3[333\ldots]\right]$ **A1** Condone the presence of π for the first 3 marks.

Condone using the limits the wrong way around for the M mark and if -21.3 is corrected to 21.3 allow the A mark.

SC: if M0 scored SCB1 is available for correct final answer

If $\int\left(10x^{\frac{1}{2}}-\frac{5}{2}x^{\frac{3}{2}}\right)=21.3$ and no integration seen B1 only.

Total: 4

Question 6

May/June 2023 | Paper 1 (variant 12)

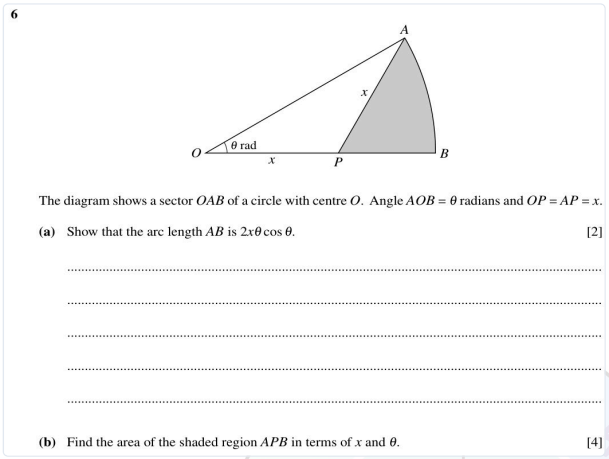
Examiner Report — Why students found this question difficult

> “Question 6 This question was the point where the paper started to be more challenging for many candidates.”

Source: 9709_s23_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 6

Coaching With Shuja

6



The diagram shows a sector OAB of a circle with centre O. Angle $AOB = \theta$ radians and $OP = AP = x$.

(a) Show that the arc length AB is $2x\theta\cos\theta$. [2]

(b) Find the area of the shaded region APB in terms of x and θ. [4]

Question 6 Mark Scheme

(a)

$$\frac{1}{2}OA = x\cos\theta \text{ or } \frac{OA}{\sin(\pi - 2\theta)} = \frac{\sin\theta}{x} \text{ or}$$

$$OA^2 = x^2 + x^2 - 2x\cos(\pi - 2\theta) \text{ or}$$

$$x^2 = r^2 + x^2 - 2rx\cos\theta \text{ or other valid method.}$$

B1 | Correct expression containing $\frac{1}{2}OA$, OA or OA^2 (allow p, a or r for OA) containing only terms with x and θ but not just $OA = 2x\cos\theta$. Do not condone $\sin\pi - 2\theta$ until missing brackets recovered or $\cos(180 - 2\theta)$ until it becomes $-\cos 2\theta$ etc.

$$OA = 2x\cos\theta \text{ leading to Arc length} = 2x\theta\cos\theta$$

DB1 | AG Complete correct method showing all necessary working. Condone $2x\cos\theta \times \theta$.

2 | If B0 but www then **SCB1** for $OA = 2x\cos\theta$ leading to Arc length $= 2x\theta\cos\theta$.

(b)

$$\text{Sector area} = \frac{1}{2}(2x\cos\theta)^2 \times \theta$$

M1 | OE Using sector formula with a correct OA. Condone $\cos\theta^2$ for $\cos^2\theta$ and missing brackets.

$$\text{Triangle area} = \frac{1}{2} \times 2x\cos\theta y\sin\theta \text{ OR } \frac{1}{2}x^2\sin(\pi - 2\theta)$$

M1 | Using a correct triangle formula for the correct triangle. Condone missing brackets and 180 for π.

$$[\text{Area } APB] = \text{Their sector area} - \text{their triangle area}$$

M1 | Both expressions must be areas involving terms with x^2 and θ only. Condone missing brackets and 180 for π for the triangle. Condone calling the sector a segment.

$$[\text{Area } APB] = \frac{1}{2}(2x\cos\theta)^2 \times \theta - \frac{1}{2}x^2\sin(\pi - 2\theta)$$

$$[= x^2(2\theta\cos^2\theta - \frac{1}{2}\sin 2\theta) \text{ or } x^2\cos\theta(2\theta\cos\theta - \sin\theta)]$$

A1 | OE A correct expression. Mark the first unsimplified result of subtraction and ISW any incorrect 'simplifications'.

Total: 4

Question 8

May/June 2023 | Paper 1 (variant 12)

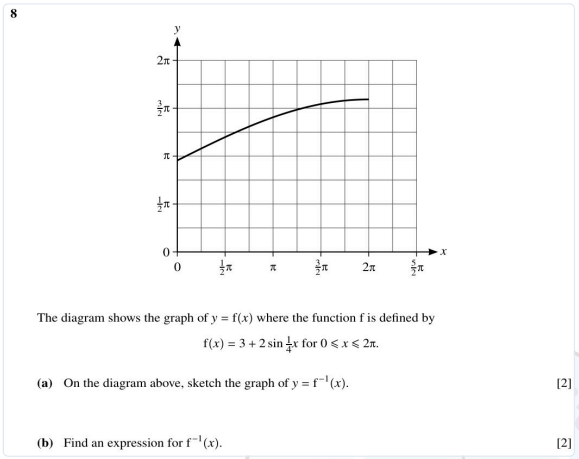
Examiner Report — Why students found this question difficult

> "Part (c) proved challenging with many responses 4 not showing a graph 'levelling out'."

Source: 9709_s23_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

Coaching With Shuja

8



The diagram shows the graph of $y = f(x)$ where the function f is defined by

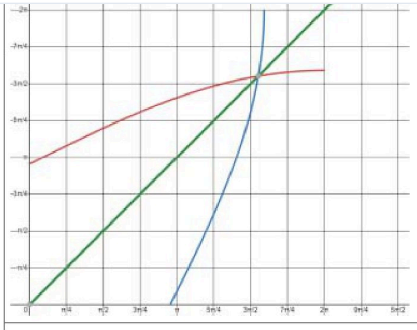
$$f(x) = 3 + 2\sin\frac{1}{4}x \text{ for } 0 \leq x \leq 2\pi.$$

(a) On the diagram above, sketch the graph of $y = f^{-1}(x)$. [2]

(b) Find an expression for $f^{-1}(x)$. [2]

8

(c)

8(a)		*B1	The line $y = x$ correctly drawn. Can be implied by reasonably correct graph of $f^{-1}(x)$.
		DB1	Fully correct (needs to reach $y = 2\pi$ and x-axis and cross the line $y = x$ in the correct squares).
		2	

The diagram above shows part of the graph of the function $g(x) = 3 + 2\sin\frac{1}{4}x$ for $-2\pi \leq x \leq 2\pi$.

Complete the sketch of the graph of $g(x)$ on the diagram above and hence explain whether the function g has an inverse. [2]

(d) Describe fully a sequence of three transformations which can be combined to transform the graph of $y = \sin x$ for $0 \leq x \leq \frac{1}{4}\pi$ to the graph of $y = f(x)$, making clear the order in which the transformations are applied. [6]

Question 10

May/June 2023 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Some candidates who found part (a) challenging omitted part (b) entirely but many more realised that the two parts were independent and were more successful in the second part."

Source: 9709_s23_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

Coaching With Shuja

Question 10

The equation of a circle is $(x-a)^2+(y-3)^2=20$. The line $y=\frac{1}{2}x+6$ is a tangent to the circle at the point P.

(a) Show that one possible value of a is 4 and find the other possible value. **[5]**

(b) For $a=4$, find the equation of the normal to the circle at P. **[4]**

(c) For $a=4$, find the equations of the two tangents to the circle which are parallel to the normal found in **(b)**. **[4]**

Question 10 Mark Scheme

(a)

$$(x-a)^2+\left(\frac{1}{2}x+6-3\right)^2=20 \text{ or using } x=2y-12$$

M1 | Obtaining an unsimplified equation in x or y only.

$$\frac{5}{4}x^2+(3-2a)x+a^2-11[=0]$$

A1 | OE e.g. $5x^2+4(3-2a)x+4a^2-44$. Rearranging to get a correct 3-term quadratic on one side. Condone terms not grouped together. $5y^2-y(54+4a)+133+a^2+24$.

$$(3-2a)^2-4\times\frac{5}{4}(a^2-11)[=0]$$

DM1 | OE Using b^2-4ac on **their** 3 term quadratic $[=0]$.

Using $a=4$: $(3-8)^2-5(5)=0$

A1 | Clearly substituting $a=4$.

$$a=-16$$

B1 | Condone no method shown for this value.

Total: 5

(b)

Centre $(4,3)$ identified or used or the point P is $(2,7)$

B1 |

$\because$ gradient of normal $=-2$

B1 | SOI

Forming normal equation using their gradient (not 0.5) and their centre or P

M1 | Condone use of $(\pm4,\pm3)$.

$$\frac{y-3}{(x-4)}=-2 \text{ or } y-7=-2(x-2)$$

A1 | OE Condone $f(x)=$.

Total: 4

(c)

Method 1 for Question 10(c)

Diameter: $y-3=\frac{1}{2}(x-4)$ [leading to $y=\frac{1}{2}x+1$]

Or

$$2(x-4)+2(y-3)\frac{dy}{dx}=0 \text{ [leading to } y=\frac{1}{2}x+1]$$

M1 | Using gradient $\frac{1}{2}$ with their centre.

$$(x-4)^2+\left(\frac{1}{2}x+1-3\right)^2=20 \text{ [leading to } \left[\frac{5}{4}x^2-10x=0\right]$$

DM1 | Obtaining an unsimplified equation in x or y only.

$$x=0 \text{ or } 8,\ y=1 \text{ or } 5 \text{ [(0, 1) and (8, 5)]}$$

A1 | Correct co-ordinates for both points. Condone no method shown for solution.

Equations are $y-1=-2x$ and $y-5=-2(x-8)$

A1 | $2x+y=1$ and $2x+y=21$.

Method 2 for Question 10(c)

Coordinates of points at which tangents meet curve are $(4+3-2)=(0,1)$ and $(4-3-2)=(0,1)$

M1 A1 | Vector approach using their centre and gradient $=0.5$. Condone answers only with no working.

Equations are $y - 5 = -2(x - 8)$ and $y - 1 = -2x$

DM1 A1 | Forming equations of tangents using **their** $(0, 1)$ and $(8, 5)$.

Method 3 for Question 10(c)

$$(x - 4)^2 + (-2x + c)^2 = 20$$

$$[5x^2 + (4 - 4c)x + (c - 3)^2 - 4] = 0$$

M1 | Obtaining an unsimplified equation in x only using equation of circle with $y = -2x + c$.

$$(4 - 4c)^2 - 20[[(c - 3)^2 - 4]][= 0]$$

DM1 | Using $b^2 - 4ac[= 0]$.

$$[-4c^2 + 32c + 120c + 16 - 100] \text{ [leading to } -4c^2 - 32c + 120c + 16 - 100]$$

$$c = 1 \text{ or } c = 21$$

A1 | Obtaining two solutions for c from their discriminant.

Equations: $y = -2x + 1$ and $y = -2x + 21$, i.e. $2x + y = 1$ and $2x + y = 21$.

A1 |

Total: 4

Question 1

May/June 2023 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

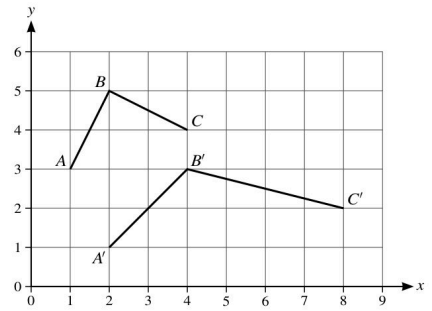
> "Those who chose not to use a vector to describe the translation often struggled to describe it using acceptable language and the direction of the stretch and its magnitude were regularly given incorrectly."

Source: 9709_s23_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 1

Coaching With Shuja

Question 1

1



The diagram shows the graph of $y = f(x)$, which consists of the two straight lines AB and BC. The lines $A'B'$ and $B'C'$ form the graph of $y = g(x)$, which is the result of applying a sequence of two transformations, in either order, to $y = f(x)$.

State fully the two transformations. [4]

The diagram shows the graph of $y = f(x)$, which consists of the two straight lines AB and BC. The lines $A'B'$ and $B'C'$ form the graph of $y = g(x)$, which is the result of applying a sequence of two transformations, in either order, to $y = f(x)$.

State fully the two transformations. **[4]**

Shuja

Question 1 Mark Scheme

$$\{\text{Translation}\}\left\{\begin{pmatrix}0\\-2\end{pmatrix}\right\}$$

B2, 1, 0 | B2 for fully correct, B1 with two elements correct. **{}** indicates different elements.

{Stretch}{[scale] factor 2}{parallel to x-axis}

B2, 1, 0 | B2 for fully correct, B1 with two elements correct.

4 | Transformations can be in either order.

Question 10

May/June 2023 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

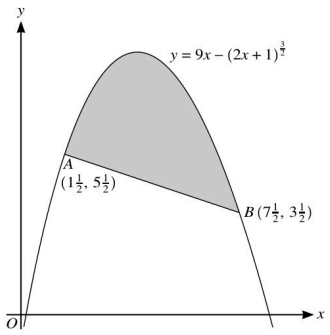
> “Common errors were sign errors and mistakes in squaring.”

Source: 9709_s23_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 10

Coaching With Shuja

Question 10

10



The diagram shows the points $A\,(1\frac{1}{2},\ 5\frac{1}{2})$ and $B\,(7\frac{1}{2},\ 3\frac{1}{2})$ lying on the curve with equation $y = 9x - (2x+1)^{\frac{3}{2}}$.

(a) Find the coordinates of the maximum point of the curve. [4]

The diagram shows the points $A\left(1\frac{1}{2}, 5\frac{1}{2}\right)$ and $B\left(7\frac{1}{2}, 3\frac{1}{2}\right)$ lying on the curve with equation

$$y = 9x - (2x+1)^{\frac{3}{2}}$$

(a) Find the coordinates of the maximum point of the curve. **[4]**

(b) Verify that the line AB is the normal to the curve at A. **[3]**

(c) Find the area of the shaded region. **[5]**

Question 10 Mark Scheme

(a)

$$\frac{dy}{dx} = \{9\} + \left\{\frac{3}{2}(2x+1)^{1/2} \times 2\right\}$$

B1, B1 | Including '+c' makes the second term Bo.

$$9 - 3(2x+1)^{1/2} = 0 \text{ leading to } 2x+1 = 9$$

M1 | Set differential to zero and solve by squaring SOI. Beware $9^2 - 3^2(2x+1) = 0$ MoAo.

Max point $= (4, 9)$

A1 | WWW $y = 9$ must come from original equation.

Total: 4

(b)

When $x = 1\frac{1}{2}$, shows substitution or $\frac{dy}{dx} = 3$

M1 | Substituting $x = 1\frac{1}{2}$ into their $\frac{dy}{dx}$.

$$\text{Gradient of } AB \text{ is } \frac{5\frac{1}{2} - 3\frac{1}{2}}{1\frac{1}{2} - 7\frac{1}{2}} = \left[\frac{-1}{3}\right]$$

M1 | Substituting into a correct expression for m_{AB}.

$$\frac{1}{3}x^3 = -1. \text{ [Hence } AB \text{ is the normal]}$$

A1 |

Total: 3

(c)

$$\left\{\frac{9x^2}{2}\right\} + \left\{\frac{-(2x+1)^{\frac{5}{2}}}{\frac{5}{2} \times 2}\right\}$$

B1 B1 | Integrating y with respect to x.

$$\left\{\frac{9}{2} \times 7.5^2 - \frac{1}{5}(2 \times 7.5 + 1)^{2.5}\right\} - \left\{\frac{9}{2} \times 1.5^2 - \frac{1}{5}(2 \times 1.5 + 1)^{2.5}\right\}$$

or

$$\left\{\frac{9}{2} \times 1.5^2 - \frac{1}{5}(2 \times 1.5 + 1)^{2.5}\right\} - \left\{\frac{9}{2} \times 1.5^2 - \frac{1}{5}(2 \times 1.5 + 1)^{2.5}\right\}$$

or $48.325 - 3.725$

M1 | OE Apply limits $1\frac{1}{2}$ to $7\frac{1}{2}$ to an integral. Working must be seen. Expect **44.6**.

$$\frac{1}{2}\left(5\frac{1}{2} + 3\frac{1}{2}\right) \times 6 \text{ or } \left(\frac{-1}{6}\left(\frac{15}{2}\right)^2 + 6 \times \frac{15}{2}\right) - \left(\frac{-1}{6}\left(\frac{3}{2}\right)^2 + 6 \times \frac{3}{2}\right)$$

or **27**

B1 | SOI Area of trapezium. May be seen combined with the area under the curve integral.

$$[\text{Shaded area} = 44.6 - 27] = 17.6$$

A1 | SC B1 if no substitution of the limits seen.

Total: 5

Question 1

May/June 2023 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "In solutions with a correct initial method, the most common error was to try to process a negative value for ex."

Source: 9709_s23_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

Question 1

Solve the equation

$$3e^{2x} - 4e^{-2x} = 5$$

Give the answer correct to 3 decimal places. **[3]**

Coaching With Shuja

Question 1 Mark Scheme

$$3(e^{2x})^2 - 5(e^{2x}) - 4 = 0$$

B1 | OE Form 3 term quadratic in e^{2x}.

$$e^{2x} = \frac{5 \pm \sqrt{73}}{6}, \quad x = \frac{1}{2}\ln\left(\frac{5+\sqrt{73}}{6}\right)$$

M1 | Use correct method to solve for x.

$$x = 0.407$$

A1 | Only

Total: 3

Question 8

May/June 2023 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "For candidates starting with a correct form, the most common errors were due to slips in the arithmetic."

> "Candidates with only two fractions often dealt with one correctly and made no progress with the second."

Source: 9709_s23_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 8

Coaching With Shuja

Question 8

Let $f(x) = \frac{5-5x^2}{(2x+1)(x+2)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence find the exact value of $\int_0^4 f(x)\,dx$, giving your answer in the form $a + b\ln c$, where a, b and c are integers. **[5]**

Question 8 Mark Scheme

(a)

State or imply the form $\frac{A}{2x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$

B1 | Accept $\frac{A}{2x+1} + \frac{Dx+E}{(x+2)^2}$.

Use a correct method for finding a constant

M1 |

Obtain one of $A = 1, B = -2, C = 3$

A1 | For alternative form: $A = 1, D = -2, E = -1$.

Obtain a second value

A1 |

Obtain the third value

A1 |

Total: 5

(b)

Integrate and obtain one of $\frac{1}{2}\ln(2x+1), -2\ln(x+2), \frac{-3}{x+2}$

B1 FT | The follow through is on **their** A, B, C.

Obtain a second term

B1 FT | If the alternative form is used, then either need to use integration by parts or split the fraction further.

Obtain the third term

B1 FT |

Substitute limits correctly in an integral with at least two terms of the form $\frac{1}{2}\ln(2x+1), -2\ln(x+2)$ and $\frac{-3}{x+2}$ and subtract in correct order

M1 | The terms used need to have been obtained correctly. Must be exact values, not decimals.

Obtain $\mathbf{1 - \ln 3}$

A1 |

Total: 5

Question 9

May/June 2023 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Using the limits to obtain the given equation proved to be more challenging, with several candidates not attempting to rearrange their equation."

> "For candidates using the iterative process correctly, the most common errors were due to not working to the required accuracy, or drawing the incorrect conclusion from correct work."

Source: 9709_s23_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 9

Coaching With Shuja

Question 9

The constant a is such that $\int_0^a xe^{-2x}\,dx = \frac{1}{8}$.

(a) Show that $a = \frac{1}{2}\ln(4a+2)$. **[5]**

(b) Verify by calculation that a lies between 0.5 and 1. **[2]**

(c) Use an iterative formula based on the equation in **(a)** to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

Question 9 Mark Scheme

(a)

Commence integration and reach $pxe^{-2x} + q\int e^{-2x}\,dx$

M1 | OE

Obtain $-\frac{1}{2}xe^{-2x} + \frac{1}{4}\int e^{-2x}\,dx$

A1 | OE

Complete integration and obtain $-\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}$

A1 |

Use limits correctly and equate to $\frac{1}{8}$, having integrated twice

DM1 | $-\frac{1}{2}ae^{-2a} - \frac{1}{4}e^{-2a} + \frac{1}{4} - \frac{1}{8}$.

Obtain $a = \frac{1}{2}\ln(4a+2)$ correctly

A1 | AG

Total: 5

(b)

Calculate the values of a relevant expression or pair of expressions at $a = 0.5$ and $a = 1$

M1 |

Justify the given statement with correct calculated values

A1 | e.g. $0.5 < 0.69\ldots, 1 > 0.89\ldots$ 0.193 > 0, -1.105 < 0. 0.066 < 0.125, 0.148 > 0.125 if put limits in the integral. Condone if they use calculator for the definite integral.

Total: 2

(c)

Use the iterative process $a_{n+1} = \frac{1}{2}\ln(4a_n + 2)$ correctly at least once.

M1 |

Obtain final answer 0.84

A1 |

Show sufficient iterations to at least 4 d.p. to justify 0.84 to 2 d.p. or show that there is a sign change in (0.835, 0.845)

A1 | e.g. 0.75, 0.8047, 0.8261, 0.8343, 0.8373, 0.8385 1, 0.8959, 0.8599, 0.8469, 0.8420, 0.8402.

Total: 3

Question 10

May/June 2023 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "There were several blank responses."

Source: 9709_s23_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 10

Coaching With Shuja

Question 10

The polynomial $x^3 + 5x^2 + 31x + 75$ is denoted by $p(x)$.

(a) Show that $(x + 3)$ is a factor of $p(x)$. **[2]**

(b) Show that $z = -1 + 2\sqrt{6}i$ is a root of $p(z) = 0$. **[3]**

(c) Hence find the complex numbers z which are roots of $p(z^2) = 0$. **[7]**

Total: 12

Coaching With Shuja

Question 10 - Mark Scheme

(a)

- **M1**: Substitute $x = -3$ to obtain value of $p(-3)$
- **A1**: Obtain $p(-3) = 0$ and hence given result

Alternative method for Question 10(a)

- **M1**: Divide $p(x)$ by $(x + 3)$ to obtain quotient $x^2 \pm 2x + \ldots$
- **A1**: Obtain quotient $x^2 + 2x + 25$, with zero remainder and hence given result

Total for (a): 2

(b)

- **M1**: Substitute $z = -1 + 2\sqrt{6}i$ and attempt expansions of z^2 and z^3
- **M1**: Use $i^2 = -1$
- **A1**: Obtain $p(z) = 0$ and hence given result (SC B1 if there is no evidence of working for the square or the cube. Total 1/3.)

Alternative Method 1

- **M1**: Use roots $z = -1 + 2\sqrt{6}i$ to form quadratic factor
- **M1**: Divide $p(z)$ by their quadratic factor
- **A1**: Obtain zero remainder and hence given result.

Alternative Method 2

- **M1**: Set their quadratic factor from (a) equal to zero
- **M1**: Solve for z
- **A1**: Obtain $z = -1 + 2\sqrt{6}i$ (and $z = -1 - 2\sqrt{6}i$)

Alternative Method 3

- **M1**: Substitute $z = -1 + 2\sqrt{6}i$ into their quadratic factor and attempt expansion of z^2
- **M1**: Use $i^2 = -1$
- **A1**: Obtain 0 and hence given result

Total for (b): 3

(c)

- **B1**: State $z_1 = \sqrt{3}i$ and $z_2 = -\sqrt{3}i$
- **M1**: Expand $(x + iy)^2 = -1 + 2\sqrt{6}i$ and compare real and imaginary parts
- **A1**: Obtain $x^2 - y^2 = -1$ and $xy = \sqrt{6}$
- **M1**: Solve to obtain x and y
- **A1**: Obtain $z_3 = \sqrt{2} + \sqrt{3}i$ and $z_4 = -\sqrt{2} - \sqrt{3}i$
- **M1**: Use $z^2 = -1 - 2\sqrt{6}i$ to obtain z_5 and z_6
- **A1**: Obtain $z_5 = \sqrt{2} - \sqrt{3}i$ and $z_6 = -\sqrt{2} + \sqrt{3}i$

Total for (c): 7

Question Total: 12

Question 6

May/June 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A common error was to round the final answer to 0.80 rather than 0.79."

Source: 9709_s23_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

Question 6

The equation $\cot \frac{x}{2} = 3x$ has one root in the interval $0 < x < \pi$, denoted by α.

(a) Show by calculation that α lies between 0.5 and 1. **[2]**

(b) Show that, if a sequence of positive values given by the iterative formula

$$x_{n+1} = \frac{1}{3}\left(x_n + 4\tan^{-1}\left(\frac{1}{3x_n}\right)\right)$$

converges, then it converges to α. **[2]**

(c) Use this iterative formula to calculate α correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

Total: 7

Coaching With Shuja

Question 6 - Mark Scheme

(a)

- **M1**: Calculate the values of a relevant expression or pair of expressions at $x = 0.5$ and $x = 1$. (Need to evaluate at both points, but **M1** still available if one value incorrect. Use of degrees is M0. Correct use of a smaller interval is **M1**.)
- **A1**: Complete the argument correctly with conclusion about change of sign or change of inequalities and with correct calculated values. (Can all be in symbols – an explanation in words is not required. Embedded values 0.5 and 1 are not sufficient. 3.92 and 1.83 alone are not sufficient.)

Total for (a): 2

(b)

- **M1**: State $x = \frac{1}{3}\left(x_n + 4\tan^{-1}\left(\frac{1}{3x_n}\right)\right)$
- **A1**: Rearrange to the given equation $\cot\left(\frac{x}{2}\right) = 3x$ as far as $2y = 4\tan^{-1}\left(\frac{1}{3x}\right)$ or continue rearrangement to $x = \frac{1}{3}\left(x_n + 4\tan^{-1}\left(\frac{1}{3x_n}\right)\right)$ and state iterative formula of $x_{n+1} = \frac{1}{3}\left(x_n + 4\tan^{-1}\frac{1}{3x_n}\right)$. (Need intermediate step between $\frac{x}{2} = \tan^{-1}\frac{1}{3x}$ and $\cot\left(\frac{x}{2}\right) = 3x$)

Total for (b): 2

(c)

- **M1**: Use the iterative process correctly at least once
- **A1**: Obtain final answer 0.79
- **A1**: Show sufficient iterations to at least 4 d.p. to justify 0.79 to 2 d.p. or show there is a sign change in the interval (0.785, 0.795). Example iterations: 1, 0.7623, 0.8037, 0.7921, 0.7951, 0.7943, 0.7945 or 0.5, 0.9506, 0.7665, 0.8024, 0.7924, 0.7950, 0.7944, 0.7945 or 0.75, 0.8076, 0.7911, 0.7954, 0.7943, 0.7946, 0.7945. Condone truncation. Allow recovery. Condone minor differences in the final d.p.

Total for (c): 3

Question Total: 7

Question 7

May/June 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common error was to substitute $y = -2x$ into the original equation."

Source: 9709_s23_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

Question 7

The equation of a curve is $3x^2 + 4xy + 3y^2 = 5$.

(a) Show that $\frac{dy}{dx} = -\frac{3x+2y}{2x+3y}$. **[4]**

(b) Hence find the exact coordinates of the two points on the curve at which the tangent is parallel to $y + 2x = 0$. **[5]**

Total: 9

Coaching With Shuja

Question 7 - Mark Scheme

(a)

- **B1**: State or imply $6y\frac{dy}{dx}$ as the derivative of $3y^2$
- **B1**: State or imply $4x\frac{dy}{dx} + 4y$ as the derivative of $4xy$
- **M1**: Equate derivative of LHS to zero and solve for $\frac{dy}{dx}$
- **A1**: Obtain $\frac{dy}{dx} = -\frac{3x+2y}{2x+3y}$

Total for (a): 4

(b)

- *__M1__: Equate $\frac{dy}{dx}$ to -2 and solve for x in terms of y or for y in terms of x
- **A1**: Obtain $x = -4y$ or $y = -\frac{x}{4}$
- **DM1**: Substitute their $x = -4y$ or their $y = -\frac{x}{4}$ in curve equation (Allow unsimplified.)
- **A1**: Obtain $y = \pm\frac{1}{\sqrt{7}}$ or $x = \pm\frac{4}{\sqrt{7}}$
- **A1**: Obtain both pairs of values

Guidance: Or exact equivalent. Or $x = \pm\frac{4}{\sqrt{7}}$ and $y = -\frac{1}{\sqrt{7}}$ or exact equivalent. Or $x = -\frac{4}{\sqrt{7}}$ and $y = -\frac{1}{\sqrt{7}}$ or exact equivalent. A1 A0 for incorrect final pairing.

Total for (b): 5

Question Total: 9

Coaching With Shuja

Question 8

May/June 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “Common errors included writing e2x+1 as e2x +1 or obtaining e2x+1 = 4+9y2 .”

Source: 9709_s23_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 32), Question 8

Coaching With Shuja

Question 8

(a) The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = \frac{4 + 9y^2}{e^{2x+1}}$$

It is given that $y = 0$ when $x = 1$.

Solve the differential equation, obtaining an expression for y in terms of x. **[7]**

(b) State what happens to the value of y as x tends to infinity. Give your answer in an exact form. **[1]**

Total: 8

Question 8 - Mark Scheme

(a)

- **B1**: Separate variables correctly as $\int \frac{1}{4+9y^2}dy = \int e^{-(2x+1)}dx$

(Condone missing integral signs or dx and dy missing.)

- **B1**: Obtain term $-\frac{1}{2}e^{-2x-1}$ (OE e.g. $-\frac{1}{2e}e^{-2x}$.)

- **M1**: Obtain term of the form $a\tan^{-1}\left(\frac{3y}{2}\right)$

- **A1**: Obtain term $\frac{1}{6}\tan^{-1}\left(\frac{3y}{2}\right)$ (OE e.g. $\frac{1}{9}\cdot\frac{3}{2}\tan^{-1}\frac{3y}{2}$.)

- **M1**: Use $x = 1, y = 0$ to evaluate a constant or as limits in a solution containing or derived from terms of the form $a\tan^{-1}(by)$ and $ce^{(2x+1)}$

- **A1**: Obtain correct answer in any form

- **A1**: Obtain final answer $y = \frac{2}{3}\tan\left(3e^{-3}\cdot e^{-2x+1}\right)$ (OE Allow with $3e^{-3} = 0.149...$)

Total for (a): 7

(b)

- **B1 FT**: State that y approaches $\frac{2}{3}\tan\left(3e^{-3}\right)$

(Or exact equivalent. The FT is on correct work on a solution containing e^{2x-1}. Condone $y = ...$. Accept correct answer stated with minimal wording. 0.10032... is not exact so **B0**.)

Total for (b): 1

Question Total: 8

Question 9

May/June 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "(b) This part proved to be more of a challenge and there were several errors with signs and coefficients."

> "Many recognised that the two fractions with linear denominators integrated to log terms, but they struggled with k dx."

> "As mentioned in part (b), candidates with a fraction of the form Dx+E often made no progress with (2−x)2 the integration, but a few were successful in using integration by parts."

Source: 9709_s23_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Question 9

Let $f(x) = \frac{2x^2+17x-17}{(1+2x)(2-x)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence show that $\int_0^1 f(x)dx = \frac{5}{2} - \ln 72$. **[5]**

Total: 10

Coaching With Shuja

Question 9 - Mark Scheme

(a)

- **B1**: State or imply the form $\frac{A}{1+2x} + \frac{B}{2-x} + \frac{C}{(2-x)^2}$
(Alternative form: $\frac{A}{1+2x} + \frac{Dx+E}{(2-x)^2}$)
- **M1**: Use a correct method for finding a coefficient (e.g.
$A(2-x)^2 + B(1+2x)(2-x) + C(1+2x) = 2x^2 + 17x - 17$ and compare coefficients or substitute for x.)
- **A1**: Obtain one of $A = -4, B = -3$ and $C = 5$
- **A1**: Obtain a second value
- **A1**: Obtain the third value (Extra term in partial fractions, then **B0** unless recover at end. Allow the marks for any constants found correctly. Missing terms in partial fractions, **B0** but M1A1 is available for a correct method that obtains at least one correct constant (e.g. cover-up rule) Max 2/5. Ignore any substitution back into their original expression.)
(If alternative form used: $A = -4, D = 3$ and $E = -1$.)

Total for (a): 5

(b)

- **B1FT**: Integrate and obtain terms $-2\ln(1+2x) + 3\ln(2-x) + \frac{5}{2-x}$
- **B1FT**: [Multiple conditional marks based on form used]
- **B1FT**: [If using A, D, E form then B1 for the A term, but no further marks until partial fractions are used to split the second term or they use integration by parts]
- **M1**: Substitute limits correctly in an integral with two terms (obtained correctly) of the form
$a\ln(1+2x) + b\ln(2-x) + \frac{c}{2-x}$, where $abc \neq 0$
- **A1**: Obtain answer $\frac{5}{2} - \ln 72$ after full and correct working (AG – evidence of some correct work to combine or simplify logs is required e.g. allow from $-\ln 9 + \ln\frac{1}{8}$ or $-\ln 2^3 - \ln 3^2$.)

Total for (b): 5

Question Total: 10

Question 10

May/June 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors occurred in using the chain rule to differentiate the square root."

> "Some struggled with the two terms involving the square root."

> "In the resulting linear equation, the common error was to obtain $-x+5$ instead of $-x-5$, resulting from an error in removing brackets."

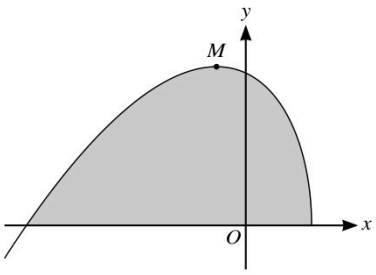
> "(b) This part provided a real challenge for candidates and there were very few fully correct solutions."

Source: 9709_s23_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

Question 10

10



The diagram shows the curve $y = (x + 5)\sqrt{3 - 2x}$ and its maximum point M.

(a) Find the exact coordinates of M. [5]

The diagram shows the curve $y = (x + 5)\sqrt{3 - 2x}$ and its maximum point M.

(a) Find the exact coordinates of M. **[5]**

(b) Using the substitution $u = 3 - 2x$, find by integration the area of the shaded region bounded by the curve and the x-axis. Give your answer in the form $a\sqrt{13}$, where a is a rational number. **[5]**

Total: 10

Question 10 - Mark Scheme

(a)

- ***M1**: Use the product rule correctly to obtain $p(x+5)(3-2x)^{''}+q(3-2x)^{\frac{1}{2}}$

(Allow with incorrect chain rule. BOD over sign errors unless an incorrect rule is quoted.)

- **A1**: Obtain correct derivative in any form (e.g. $-(x+5)(3-2x)^{-\frac{1}{2}}+(3-2x)^{\frac{1}{2}}$.)

- **DM1**: Equate derivative to zero and obtain a linear equation

(Allow with surd factor e.g. $(3-2x)^{-\frac{1}{2}}[-(x+5)+(3-2x)]=0$.)

- **A1**: Obtain a correct linear equation (e.g. $-(x+5)+3-2x=0$.)

- **A1**: Obtain answer $\left(-\frac{2}{3},\frac{13\sqrt{13}}{9}\right)$ or $\left(-\frac{2}{3},\frac{13\sqrt{13}}{9}\right)$ (Or exact equivalent e.g. $\left(-\frac{2}{3},\frac{2\sqrt{2197}}{27}\right)$.)

Alternative Method for Question 10(a)

- *M1*: Obtain y^2 and differentiate

- **A1**: Obtain correct derivative in any form (e.g. $-6x^2-34x-20$.)

- **DM1**: Equate derivative to zero and solve for x

- **A1**: Obtain $-\frac{2}{3}$

(Ignore -5 if seen.)

- **A1**: Obtain answer $\left(-\frac{2}{3},\frac{13\sqrt{13}}{9}\right)$ only

(ISW)

Total for (a): 5

(b)

- ***M1**: Use the given substitution and reach $a\int\left(\frac{13-u}{2}\right)u^{\frac{1}{2}}du$

(OE Need to see -2 or $-\frac{1}{2}$ used. Condone if du missing or the integral sign is missing.)

- **A1**: Obtain correct integral $-\frac{1}{2}\int\left(\frac{3-u}{2}\sqrt{u}du+5\int\sqrt{u}du\right)$.

(OE e.g. $-\frac{1}{2}\left[\int\frac{3-u}{2}\sqrt{u}du+5\int\sqrt{u}du\right]$.)

(Ignore limits at this stage. Condone if du missing.)

- **B1**: SOI e.g. by $u=13$ and 0. In any order and at any stage.

- **DM1**: Use correct limits the right way round in an integral of the form $a\left[\frac{26}{3}u^{\frac{3}{2}}-\frac{2}{5}u^{\frac{5}{2}}\right]$

- **A1**: Obtain answer $\frac{169}{15}\sqrt{13}$ or $a=\frac{169}{15}$

(or exact equivalents.)

Total for (b): 5

Question Total: 10

Question 11

May/June 2023 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Many candidates did not know how to get started – they often understood that a scalar product was required, but then did not use the correct vectors."

> "There were also a large number of blank responses."

Source: 9709_s23_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 11

Coaching With Shuja

Question 11

The points A and B have position vectors $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + \mathbf{k}$ respectively. The line l has equation

$$\mathbf{r} = \mathbf{i} - \mathbf{j} + 3\mathbf{k} + \mu(2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k})$$

(a) Show that l does not intersect the line passing through A and B. **[5]**

(b) Find the position vector of the foot of the perpendicular from A to l. **[4]**

Total: 9

Question 11 - Mark Scheme

(a)

- **M1**: Carry out correct method for finding a vector equation for $\overrightarrow{AB}$
- **A1**: Obtain $[\mathbf{r} =]\mathbf{i} + 2\mathbf{j} - 2\mathbf{k} + \lambda(\mathbf{i} - 3\mathbf{j} + 3\mathbf{k})$
(OE e.g. $\mathbf{r} = 2\mathbf{i} - \mathbf{j} + \mathbf{k} + \lambda(-\mathbf{i} + 3\mathbf{j} - 3\mathbf{k})$.)
- **M1**: Equate two pairs of components of general points on their AB and l and evaluate λ or μ
- **A1**: Obtain correct answer for λ or μ, e.g. $\lambda = -1, \mu = -2$
(Correct value from two correct component equations.)
- **A1**: Verify that all three equations are not satisfied and the lines fail to intersect ($\neq$ is sufficient justification e.g. $0 \neq -3$).
(Conclusion needs to follow correct values. Two possible cases are possible e.g. using **j** and **k** to get one parameter and then **i** to obtain the other. If two pairs of simultaneous equations and showing that the results are not the same.)

Alternatives

Case	λ	μ	Comparison	$\neq$
ij	2	1	$4 \neq 7$	-
ik	5	$\frac{5}{2}$	$-13 \neq -\frac{17}{2}$	-
jk	-1	-2	$0 \neq -3$	-

Total for (a): 5

(b)

- **B1**: Find $\overrightarrow{AP}$ for a general point P on l, e.g. $-3\mathbf{j} + 5\mathbf{k} + \mu(2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k})$
(Or equivalent e.g. $\overrightarrow{PA} = -2\mu\mathbf{i} + (3\mu + 3)\mathbf{j} - (4\mu + 5)\mathbf{k}$.)
- **M1**: Calculate scalar product of their $\overrightarrow{AP}$ and a direction vector for l and equate the result to zero
(e.g. $4\mu + (9 + 9\mu) + (20 + 16\mu) = 0$.)
(Mo if using $\overrightarrow{OP}$.)
(Mo if using parallel line through A.)
- **A1**: Obtain $\mu = -1$
- **A1**: Obtain answer $-\mathbf{i} + 2\mathbf{j} - \mathbf{k}$
(Accept coordinates in place of position vector.)

Alternative Method for Question 11(b)

- **B1**: Find $\overrightarrow{AP}$ for a general point P on l, e.g. $-3\mathbf{j} + 5\mathbf{k} + \mu(2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k})$
(Or equivalent e.g. $\overrightarrow{PA} = -2\mu\mathbf{i} + (3\mu + 3)\mathbf{j} - (4\mu + 5)\mathbf{k}$.)
- **M1**: Use Pythagoras and differentiate with respect to μ to obtain value of μ corresponding to minimum distance.
(No need to prove it is a minimum.)
(e.g. $\frac{d}{d\mu}\left[4\mu^2 + 9(\mu + 1)^2 + (4\mu + 5)^2\right] = 0$.)
- **A1**: Obtain $\mu = -1$
- **A1**: Obtain answer $-\mathbf{i} + 2\mathbf{j} - \mathbf{k}$
(Accept coordinates in place of position vector.)

Total for (b): 4

Question Total: 9

Question 3

May/June 2023 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The most common error seen was in omitting a scale from the diagram or showing the scale on only one axis."

Source: 9709_s23_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 33), Question 3

Coaching With Shuja

Question 3

On a sketch of an Argand diagram, shade the region whose points represent complex numbers z satisfying the inequalities $|z - 3 - i| \leq 3$ and $|z| \geq |z - 4i|$. **[4]**

Total: 4

Coaching With Shuja

Question 3 - Mark Scheme

Answer

- **B1**: Show a circle with centre $3 + i$

(Must be some evidence of scale on both axes or centre stated as $3 + i$ or (3, 1).)

- **B1**: Show a circle with radius 3 and centre not at the origin

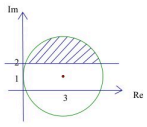
(Must be some evidence that radius = 3 or stated $r = 3$.)

- **B1**: Show the line $y = 2$

(Line $y = 2$ can be represented by 2 or correct dashes.)

- **B1**: Shade the correct region

(Line and circle must be correct.)

3	Show a circle with centre $3 + i$	**B1**	Must be some evidence of scale on both axes or centre stated as 3 + i or (3, 1).
	Show a circle with radius 3 and centre not at the origin	**B1**	Must be some evidence that radius = 3 or stated $r = 3$
	Show the line $y = 2$	**B1**	Line $y = 2$ can be represented by 2 or correct dashes.
	Shade the correct region	**B1**	Line and circle must be correct.
		**4**	Scales may be replaced by dashes on axes for all marks. Correct figure, with no scale on either axis then allow 1/3 and the B1 for correct shaded region Max 2/4. If B0 above for line but relatively correct position then B1 for correct shaded region Max 3/4. Re and Im axes interchanged but clearly labelled, allow **SCB1** for centre and radius of circle correct and **SCB1** for line and shading correct Max 2/4.

Total: 4

Guidance

Scales may be replaced by dashes on axes for all marks. Correct figure, with no scale on either axis then allow 1/3 and the **B1** for correct shaded region Max 2/4. If **B0** above for line but relatively correct position then **B1** for correct shaded region Max 3/4. Re and Im axes interchanged but clearly labelled, allow **SCB1** for centre and radius of circle correct and **SCB1** for line and shading correct Max 2/4.

Question 6

May/June 2023 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Using θ instead of 2θ or 2(θ − 23.41) instead of 2θ − 23.41 were common errors."

Source: 9709_s23_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 6

Coaching With Shuja

6(a) Express $3\cos x + 2\cos(x - 60^\circ)$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. State the exact value of R and give α correct to 2 decimal places. **[4]**

6(b) Hence solve the equation

$$3\cos 2\theta + 2\cos(2\theta - 60^\circ) = 2.5$$

for $0^\circ < \theta < 180^\circ$. **[4]**

6(a) B1 Need to see $A\cos x + B\sin x$ with A and B correct. A may be 4 or $3 + 2\cos 60^\circ$ and B may be $\sqrt{3}$ or $2\sin 60^\circ$.

B1 FT Follow through their 4 and $\sqrt{3}$. If coefficients are 3 and 2 then B0. $R = \sqrt{19}$ from $R = 4.36$ B0 but 4.36 seen after $\sqrt{19}$ ISW.

M1 Use correct trig formulae for their expansion to find α e.g. $\alpha = \tan^{-1}\frac{\sqrt{3}}{4}$ or $\cos^{-1}\frac{4}{\sqrt{19}}$ or $\sin^{-1}\frac{\sqrt{3}}{\sqrt{19}}$. If $\sin\alpha = \sqrt{3}\cos\alpha = 4$ seen then M0 A0. If $\tan\alpha = 23.41^\circ$ M0 A0 but can recover if $\alpha = 23.41^\circ$ seen later. $\alpha = \tan^{-1}\frac{2}{3}$ **M1** ($\alpha = 33.69^\circ$) but $\alpha = \tan^{-1}\frac{3}{2}$ M0

A1 Obtain $\alpha = 23.41^\circ$. Allow if x instead of α.

Total: 4

6(b) B1 FT $\cos^{-1}\left(\frac{2.5}{R}\right)$. SOI [55.0°]. Follow through their $\sqrt{19}$.

M1 Use a correct method to find a value of 2θ (not x) in the interval. Allow sign error in moving α to right side. $2\theta = \cos^{-1}\left(\frac{2.5}{R}\right) + 23.41^\circ$ or $2\theta = 360^\circ - \cos^{-1}\left(\frac{2.5}{R}\right) + 23.41^\circ$ with R substituted.

A1 Obtain one correct answer e.g. 39.2°. If working for **M1** not seen then **M1** implied by 39.2° or 164.2°. Must be at least 1 d.p.

A1 Obtain second correct answer e.g. 164.2° and no others in the interval. Must be at least 1 d.p. Ignore answers outside the given interval.

Total: 4

Question 7

May/June 2023 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The most common errors were integrating e2u incorrectly to obtain 2e2u or making one or more sign errors."

> "This meant they were faced with a far more difficult integral, and had to attempt to integrate by parts twice, often making little progress."

Source: 9709_s23_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 7

Coaching With Shuja

7(a) Use the substitution $u = \cos x$ to show that

$$\int_0^{\pi} \sin 2x e^{2\cos x} dx = \int_{-1}^{1} 2ue^{2u} du. \quad [4]$$

7(b) Hence find the exact value of $\int_0^{\pi} \sin 2x e^{2\cos x} dx$. **[4]**

Coaching With Shuja

7(a) B1 $\frac{du}{dx} = -\sin x$. SOI

M1 Use double angle formula and substitute for x and dx throughout the integral. All x's must be removed, can be coefficient errors provided 2 seen in working.

A1 Obtain $\pm \int 2ue^{2u}du$. Limits may be omitted, or left as 0 and π, during the change of variable stage.

A1 Justify new limits and obtain $\int_{-1}^{1} 2ue^{2u}du$ from correct working. **AG** Must see $x = 0, u = 1$ and $x = \pi, u = -1$. Inequalities alone e.g. $0 \leq x \leq \pi$ and $1 \leq u \leq -1$ or $-1 \leq u \leq 1$ for limits are insufficient A0. If sign in expression and order of limits incorrect then A0. If negative sign is present in the integrand then this can be removed and limits introduced in correct order in a single step.

Total: 4

7(b) M1* Condone dx.

A1 Complete integration and obtain $ue^{2u} - \frac{1}{2}e^{2u}$. **OE** Allow $\left(2u\frac{1}{2}e^{2u}\right) - \frac{1}{2}e^{2u}$.

DM1 Use correct limits correctly in $cue^{2u} + de^{2u}$ having integrated twice or in $c\cos xe^{2\cos x} + de^{2\cos x}$. 1 and -1 for u, 0 and π for x e.g. $ce^2 + de^2 - (-ce^{-2} + de^{-2})$. Not decimals. Allow one sign error at most in going from $cue^{2u} + de^{2u}$ or $c\cos xe^{2\cos x} + de^{2\cos x}$ to $ce^2 + de^2 - (-ce^{-2} + de^{-2})$. [e² - ½e² - (-e^{-2} - ½e^{-2})] Complete reversal of sign by converting back to $\cos x$ and not making $x = 0$ upper limit is DM0 A0.

A1 ISW. Obtain $\frac{1}{2}e^2 + \frac{3}{2}e^{-2}$. Or equivalent 2-term expression e.g. $\frac{e^4+3}{2e^2}$ or $\frac{1}{2}\left(e^2 + \frac{3}{e^2}\right)$.

Total: 4

Question 8

May/June 2023 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 8 There were many complete solutions seen, however a number of candidates found this question challenging 1 and could only separate the variables correctly and integrate ."

> "A small number chose to complete the y integrand by parts, but after undertaking this correctly once, for which M1 was available, little or no progress was made in the next integration step."

Source: 9709_s23_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 8

Coaching With Shuja

8 The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = \frac{x^2 + 4}{x(y + 4)}$$

for $x > 0$. It is given that $x = 4$ when $y = 2\sqrt{3}$.

Solve the differential equation to obtain the value of x when $y = 2$. **[8]**

Coaching With Shuja

8 B1 Separate the variables correctly. $\frac{y+4}{y^2+4}dy = \frac{1}{x}dx$.

B1 Obtain $\ln x$.

***M1** Split the fraction and integrate to obtain $p\ln(y^2+4)$ or $q\tan^{-1}\frac{y}{2}$ correctly. Only following subdivision into $\frac{y}{y^2+4} + \frac{4}{y^2+4}$. If no subdivision seen then both terms $p\ln(y^2+4)$ and $q\tan^{-1}\frac{y}{2}$ must be present.

A1 Obtain $\frac{1}{2}\ln(y^2+4)$.

A1 Obtain $2\tan^{-1}\frac{y}{2}$.

DM1 Use $(4, 2\sqrt{3})$ in an expression containing at least 2 of $a\ln x, b\ln(y^2+4)$ and $c\tan^{-1}\frac{y}{2}$ to obtain constant of integration. Allow one sign or arithmetic error e.g. $\frac{2\pi}{3}$. May use $(4, 2\sqrt{3})$ and $(x, 2)$ as limits to find x for the final 3 marks.

Correct solution (any form) e.g. $\frac{1}{2}\ln(y^2+4) + 2\tan^{-1}\frac{y}{2} = \ln x + \frac{2\pi}{3}$ or $\frac{1}{2}\ln(y^2+4) + 2\tan^{-1}\frac{y}{2} = \ln x + 2\tan^{-1}\sqrt{3} + \frac{1}{2}\ln 16 - \ln 4$.

A1 However solution not asked for so allow $\frac{1}{2}\ln 8 + 2\tan^{-1}1 = \ln x + 2\tan^{-1}\sqrt{3} + \frac{1}{2}\ln 16 - \ln 4$.

A1 ISW. Obtain $\sqrt{8}e^{-\frac{1}{6}}$ or 1.68 or more accurate or $2\sqrt{2}e^{-\frac{1}{6}}$ or $\frac{8}{\sqrt{e^{\frac{1}{3}}}}$ or $\frac{\sqrt{8}}{e^{\frac{1}{6}}}$.

Total: 8

Question 9

May/June 2023 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "However, the common error was for candidates to assume a point of intersection and solve by equating lines."

> "(b) Candidates found this part more difficult and many did not attempt it."

> "Common errors were using 2 instead of 3 (or 3 instead of 5), direction errors in 3 2 2 2 υυυρ υυυρ the relevant vectors, not squaring 3 when using magnitudes, or finding vector QR not OR."

Source: 9709_s23_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

Coaching With Shuja

9 The lines l and m have equations

$$l : \mathbf{r} = a\mathbf{i} + 3\mathbf{j} + b\mathbf{k} + \lambda(c\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}),$$

$$m : \mathbf{r} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k} + \mu(2\mathbf{i} - 3\mathbf{j} + \mathbf{k}).$$

Relative to the origin O, the position vector of the point P is $4\mathbf{i} + 7\mathbf{j} - 2\mathbf{k}$.

(a) Given that l is perpendicular to m and that P lies on l, find the values of the constants a, b and c. **[4]**

(b) The perpendicular from P meets line m at Q. The point R lies on PQ extended, with $PQ : QR = 2 : 3$. Find the position vector of R. **[6]**

Coaching With Shuja

9(a) M1 Perform scalar product of direction vectors and set result equal to zero. $2c + 6 + 4 = 0$.

M1 Use P to find the value of λ. $3 - 2\lambda = 7 \Rightarrow \lambda = -2$ [$a + \lambda c = 4, b + 4\lambda = -2$]. Equation for line l may contain $-\lambda$ instead of $+\lambda$ leading to $\lambda = 2$ all marks available.

A1 Obtain $c = -5$ or $b = 6$.

A1 $a = -6, b = 6$ and $c = -5$ all correct.

Total: 4 SC1: Use P to find the value of λ **M1**. Substitute $\lambda = -2$ into point P, so $a - 2c = 4, b + 4\lambda = -2$. Equation for line l may contain $-\lambda$ instead of $+\lambda$ leading to $\lambda = 2$ all marks available. All 3 values correct **A1**. Max 2/4.

9(b) B1 Find $\overline{PQ}$ (or $\overline{QP}$) for a general point Q on m: $= \pm \begin{bmatrix} -3 + 2\mu \\ -5 - 3\mu \\ 5 + \mu \end{bmatrix}$. Could be their a, b, c and λ values provided **M1 M1** gained in (a).

M1* Equate the scalar product of $\overline{PQ}$ (or $\overline{QP}$) and a direction vector for m to zero and obtain an equation in μ. Allow $\overline{PQ} = \overline{OQ} + \overline{QP}$ sign problem.

A1 $PQ^2 = (-3 + 2\mu)^2 + (-5 - 3\mu)^2 + (5 + \mu)^2 = [14(\mu + 1)^2 + 45]$. Min when $\mu = -1$ or by differentiation.

A1 Obtain $\overline{OQ} = -\mathbf{i} + 5\mathbf{j} + 2k$ or $\overline{PQ} = -5\mathbf{i} - \mathbf{j} + 2k$. Must be labelled correctly.

DM1 Carry out a method to find the position vector of R. e.g. Use $\overline{OR} = \overline{OP} + \frac{2}{3}\overline{PQ}$ or $\overline{OR} = \overline{OQ} + \frac{3}{2}\overline{PQ}$ or $\overline{OR} = \frac{5\overline{OO} - 2\overline{OQ}}{3} = \overline{OQ} + \frac{3}{2}\overline{PQ}$ where $\overline{OR} = (x, y, z)$. $\overline{OR} = \overline{OR} - \overline{OQ}$ Solve $|\overline{OR}|^2 = \frac{9}{4}|\overline{PQ}|^2$ or $|\overline{OR}| = \frac{3}{2}|\overline{PQ}| t = 2.5$.

Alternative method for DM1: $\overline{OR} = (4, 7, -2) + t(-5, -2, 4)$ $\overline{OR} = \overline{OR} - \overline{OQ}$ Solve $|\overline{OR}|^2 = \frac{9}{4}|\overline{PQ}|^2$ or $|\overline{OR}| = \frac{3}{2}|\overline{PQ}|$ $t = 2.5$.

A1 Accept coordinates. $\begin{pmatrix} -17/2 \\ 2 + 8k \end{pmatrix}$. Don't accept $-\frac{17}{2}\mathbf{i} + \frac{4}{2}\mathbf{j} + \frac{16}{2}k$.

Total: 6

SC2: Equate lines, attempt to find $\mu = -1$ or $\lambda = -1$ **M1***. Substitute $\lambda = -2$ into point P, so $\overline{OQ} = -\mathbf{i} + 5\mathbf{j} + 2k$ **A1**. Attempt to find $\overline{OQ}$ using the other parameter value **DM1**. $\overline{OQ} = -\mathbf{i} + 5\mathbf{j} + 2k$ therefore intersect **A1**. Then use main scheme for the final **DM1 A1**.

Question 10

May/June 2023 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The most common errors were extracting 3−1 instead of 3−2, extracting 3 and 9 instead of 3−1 and 3−2 or not evaluating the squared term(s) correctly."

Source: 9709_s23_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 10

Coaching With Shuja

10 Let $f(x) = \frac{21-8x-2x^2}{(1+2x)(3-x)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. **[5]**

10(a) B1 State or imply the form $\frac{A}{1+2x}+\frac{B}{3-x}+\frac{C}{(3-x)^2}$. Alternative form: $\frac{A}{1+2x}+\frac{Dx+E}{(3-x)^2}$.

M1 Use a correct method to find a constant.

A1 Obtain one of $A=2, B=2$ and $C=-3$.

A1 Obtain a second value.

A1 Obtain the third value. Do not need to substitute values back into original form.

Total: 5 If $\frac{A}{1+2x}+\frac{B}{3-x}+\frac{Cx+D}{(3-x)^2}$ Bo but **M1 A1** for A, A for B and **C** and **D**. If $C=0$ then recovers **B1** from above.

10(b) M1 Use a correct method to obtain the first two terms of one of the unsimplified expansions.
$(1+2x)^{-1}, \left(1-\frac{1}{3}x\right)^{-1}, \left(1-\frac{1}{3}x\right)^{-2}, (3-x)^{-1}, (3-x)^{-2}$.

$(1+2x)^{-1}=1+(-1)(2x)+\ldots$
$\left(1-\frac{1}{3}x\right)^{-1}=1+(-1)(-\frac{x}{3})+\ldots$
$\left(1-\frac{1}{3}x\right)^{-2}=1+(-2)(-\frac{x}{3})+((-2)(-3)/2)(-\frac{x}{3})^2+\ldots$
$(3-x)^{-1}=3^{-1}+(-1)3^{-2}(-x)\ldots$
$(3-x)^{-2}=3^{-2}+(-2)3^{-3}(-x)+\ldots$

A1 FT Follow through on their A, B, C.
$A(1+(-1)(2x)+((-1)(-2)/2)(2x)^2+\ldots)+\frac{B}{3}(1+\frac{x}{3}+(\frac{x}{3})^2+\ldots)$

A1 FT $\frac{C}{3^2}(1+(-2)(-\frac{x}{3})+((-2)(-3)/2)(-\frac{x}{3})^2+\ldots)$.

A1 FT Must be their coefficients from (a) but may be unsimplified expansions for **FT** marks. If correct, expect to see
$2(1-2x+(2x)^2)+\ldots$
$2-4x+8x^2$
or $2-x+\frac{x^2}{27}$
or $\frac{2}{3}+\frac{2x}{9}+\frac{2x^2}{27}$
or $-\frac{1}{3}\left(1+\frac{2x}{3}+(3)\left(\frac{x}{3}\right)^2+\ldots\right)$.

Obtain final answer $\frac{7}{3}-4x+\frac{215}{27}x^2$. Accept $2\frac{1}{3}-4x+7\frac{26}{27}x^2$. No **ISW**.

Question 11

Feb/Mar 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Setting this derivative to zero was often seen but algebraic errors resulted in few completely correct answers."

Source: 9709_m22_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 12), Question 11

Coaching With Shuja

Question Paper Feb/Mar 2022 • Paper 1 (variant 12) • Q11

It is given that a curve has equation $y = k(3x - k)^{-1} + 3x$, where k is a constant.

(a) Find, in terms of k, the values of x at which there is a stationary point. [4]

The function f has a stationary value at $x = a$ and is defined by

$$f(x) = 4(3x - 4)^{-1} + 3x$$

for $x \geq \frac{3}{2}$.

(b) Find the value of a and determine the nature of the stationary value. [3]

(c) The function g is defined by $g(x) = -(3x + 1)^{-1} + 3x$ for $x \geq 0$.
Determine, making your reasoning clear, whether g is an increasing function, a decreasing function or neither. [2]

Coaching With Shuja

(a)

$\frac{dy}{dx} = -k(3x-k)^{-2} \times 3 + 3$ **B2,1,0**

$-\frac{3k}{(3x-k)^2} + 3 = 0$ leading to $3(3x-k)^2 = 3k$ and hence $3x - k = \pm\sqrt{k}$ **M1** Set $\frac{dy}{dx} = 0$ and remove denominator

$x = \frac{k \pm \sqrt{k}}{3}$ **A1** OE

Total: 4

(b)

$a = \frac{4 \pm \sqrt{4}}{3}$ leading to $a = 2$ **B1** Substitute $x = a$ when $k = 4$. Allow $x = 2$.

$f''(x) = 72(3x-4)^{-3}$ **B1** Allow $18k(3x-k)^{-3}$

> 0 when $x = 2$, hence minimum **B1 FT**

Total: 3

(c)

Substitute $k = -1$ leading to $g'(x) = \frac{3}{(3x+1)^2} + 3$ **M1**

$g'(x) > 0$, or $g'(x)$ is always positive, hence g is an increasing function **A1**

Total: 2

Coaching With Shuja

Question 3

Feb/Mar 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

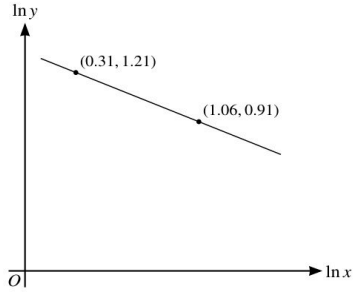
> "Question 3 Candidates generally found this question challenging but many were able to take logarithms of the given equation."

> "Common errors occurred when candidates either substituted lnx and lny into the given equation, for example used (0.31, 1.21), instead of the exponential values from the given coordinates."

Source: 9709_m22_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 3

Coaching With Shuja

3



The variables x and y satisfy the equation $x^n y^2 = C$, where n and C are constants. The graph of $\ln y$ against $\ln x$ is a straight line passing through the points (0.31, 1.21) and (1.06, 0.91), as shown in the diagram.

Find the value of n and find the value of C correct to 2 decimal places. [5]

The variables $\boldsymbol{x}$ and $\boldsymbol{y}$ satisfy the equation $\boldsymbol{x^n y^2 = C}$, where $\boldsymbol{n}$ and $\boldsymbol{C}$ are constants. The graph of $\boldsymbol{\ln y}$ against $\boldsymbol{\ln x}$ is a straight line passing through the points $\boldsymbol{(0.31, 1.21)}$ and $\boldsymbol{(1.06, 0.91)}$, as shown in the diagram.

Find the value of $\boldsymbol{n}$ and find the value of $\boldsymbol{C}$ correct to 2 decimal places. [5]

Mark Scheme Feb/Mar 2022 • Paper 3 (variant 32) • Q3

State or imply $n \ln x + 2 \ln y = \ln C$ **B1**

Substitute values of $\ln y$ and $\ln x$, or equate gradient of line to $\pm\frac{1}{2}n$, but not $\pm n$, and solve for n **M1** Using $\ln x$ and $\ln y$ values

Obtain $n = 0.8[0]$ or $0.8[00]$ or $\frac{4}{5}$ **A1**

Solve for C **M1** Using $\ln x$ and $\ln y$ values in equation of correct form, that is $\ln C$ not C. Allow $C = e^{2.668}$.

Obtain $C = 14.41$ **A1** Must be 2 d.p.

Alternative method for question 3

Obtain two correct equations in n and C by substituting x and y values in the given equation **B1**
$(2.886)^n \times (2.484)^2 = C$ and $(1.363)^n \times (3.353)^2 = C$

Solve for n **M1** Using x and y values

Obtain $n = 0.8[0]$ or $0.8[00]$ or $4/5$ **A1**

$\left(\frac{2.886}{1.363}\right)^n \times \left(\frac{2.484}{3.353}\right)^2 = 1$ leading to $n = 0.7995$

Solve for C **M1** Using x and y values

Obtain $C = 14.41$ **A1** Must be 2 d.p.

Total: 5

Coaching With Shuja

Question 4

Feb/Mar 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 4 Candidates generally found this question to be challenging, a common error was incorrect chain rule dy dy dθ application."

Source: 9709_m22_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Question Paper Feb/Mar 2022 • Paper 3 (variant 32) • Q4

The parametric equations of a curve are

$$x = 1 - \cos\theta, \qquad y = \cos\theta - \frac{1}{4}\cos 2\theta.$$

Show that $\frac{dy}{dx} = -2\sin^2\left(\frac{1}{2}\theta\right)$. [5]

Coaching With Shuja

Mark Scheme

Feb/Mar 2022 • Paper 3 (variant 32) • Q4

State $\frac{dx}{d\theta} = \sin\theta$ or $\frac{dy}{d\theta} = -\sin\theta + \frac{1}{2}\sin 2\theta$ **B1**

Use $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$ **M1**

Obtain correct answer in any form **A1** e.g. $\frac{-\sin\theta + \frac{1}{2}\sin 2\theta}{\sin\theta}$

Use double angle correctly to obtain $\frac{dy}{dx}$ in terms of θ **M1** $\sin 2\theta = 2\sin\theta\cos\theta$

Obtain the given answer with no errors seen $-2\sin^2\left(\frac{1}{2}\theta\right)$ **A1** AG. Requires correct cancellation of ALL $\sin\theta$ terms and $\cos\theta = 1 - 2\sin^2\left(\frac{1}{2}\theta\right)$ seen.

SC For incorrect signs, consistent throughout max. B0, M1, A0, M1, A1

Total: 5

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Question 6

Feb/Mar 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Candidates who tried to work with w and w* found it more difficult to do so."

Source: 9709_m22_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

Question Paper Feb/Mar 2022 • Paper 3 (variant 32) • Q6

Find the complex numbers w which satisfy the equation $w^2 + 2iw^* = 1$ and are such that $\mathrm{Re}\, w \leq 0$.
Give your answers in the form $x + iy$, where x and y are real. [6]

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Mark Scheme

Feb/Mar 2022 • Paper 3 (variant 32) • Q6

Substitute and obtain a correct equation in x and y **B1** $(x+iy)^2+2i(x-iy)=1$

Use $i^2=-1$ at least once and equate real and imaginary parts **M1**

Obtain two correct equations, e.g. $x^2-y^2+2y=1$ and $2xy+2x=0$ **A1**

Solve for x or for y **M1**

Using $y=-1$, obtain answer $w=-2-i$ only **A1** Ao if $w=2-i$ as well

Using $x=0$, obtain answer $w=i$ **A1**

Total: 6

Question 7

Feb/Mar 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “1 2 2 1 However, it should be noted that there are many other functions that can be used at these values, 1 for example x and 4−sec1 x 2 2 (c) Many candidates used degrees to answer this question, therefore were unable to get the correct result.”

Source: 9709_m22_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

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Question Paper Feb/Mar 2022 • Paper 3 (variant 32) • Q7

(a) By sketching a suitable pair of graphs, show that the equation $4 - x^2 = \sec \frac{1}{2}x$ has exactly one root in the interval $0 \leq x < \pi$. [2]

(b) Verify by calculation that this root lies between 1 and 2. [2]

(c) Use the iterative formula $x_{n+1} = \sqrt{4 - \sec \frac{1}{2}x_n}$ to determine the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

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(a)

Sketch a relevant graph, e.g. $y = 4 - x^2$ **B1** Needs $(0, 4)$ or marks on axis and $(2, 0)$ or $(\pi, 0)$

Sketch a second relevant graph, e.g. $y = \sec \frac{1}{2}x$, and justify the given statement **B1** Needs $(0, 1)$ or mark on axis and $(\pi, 0)$. Asymptote NOT required, but must NOT reach $x = \pi$. Sec graph must exist over at least interval $\left[0, \frac{3\pi}{4}\right]$ and quadratic graph over $[0, 2.5]$.

Total: 2

(b)

Calculate the value of a relevant expression or values of a pair of relevant expressions at $x = 1$ and $x = 2$ **M1** Need all 4 values or the 2 values correct for M1. Angles in degrees score M0.

Complete the argument with correct calculated values **A1**

Total: 2

(c)

Use the iterative process correctly at least twice **M1**

Obtain final answer **1.60 A1** Must be 2 d.p.

Show sufficient iterations to 4 d.p. to justify **1.60** to 2 d.p. or show there is a sign change in the interval $(1.595, 1.605)$ **A1**

Total: 3

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Question 9

Feb/Mar 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “1 Most of these candidates were successful with the integration of but struggled with the coefficient x+1 1 when integrating .”

Source: 9709_m22_er.pdf, Page 7, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Question Paper Feb/Mar 2022 • Paper 3 (variant 32) • Q9

The variables x and y satisfy the differential equation

$$(x+1)(3x+1)\frac{dy}{dx} = y,$$

and it is given that $y = 1$ when $x = 1$.

Solve the differential equation and find the exact value of y when $x = 3$, giving your answer in a simplified form. [9]

Mark Scheme Feb/Mar 2022 • Paper 3 (variant 32) • Q9

Correctly separate variables and integrate at least one side **M1** To obtain $a \ln y$ or $b \ln(x+1) + c \ln(3x+1)$

Obtain term $\ln y$ from integral of $1/y$ **B1**

State or imply the form $\frac{A}{x+1} + \frac{B}{3x+1}$ and use a correct method to find a constant **M1**

Obtain $A = -\frac{1}{2}$ and $B = \frac{3}{2}$ **A1**

Obtain terms $-\frac{1}{2}\ln(x+1) + \frac{1}{2}\ln(3x+1)$ or $-\frac{1}{2}\ln(2x+2) + \frac{1}{2}\ln(6x+2)$ or combination of these terms **A1 FT + A1 FT** The FT is on the values of A and B.

Use $x = 1$ and $y = 1$ to evaluate a constant, or expression for a constant, or as limits, in a solution containing terms $a \ln y$, $b \ln(x+1)$ and $c \ln(3x+1)$, where $abc \neq 0$ *M1 e.g. $\ln y = -\frac{1}{2}\ln(x+1) + \frac{1}{2}\ln(3x+1) - \frac{1}{2}\ln 2$

Obtain an expression for y or y^2 and substitute $x = 3$ **DM1** Do not accept decimal equivalent of $\ln$ terms

Obtain answer $y = \frac{1}{2}\sqrt{5}$ or $\sqrt{\frac{5}{4}}$ or $\sqrt{\frac{10}{8}}$ **A1** ISW. Must be simplified and exact, do not allow 1.118 or $e^{\frac{1}{2}\ln\frac{5}{4}}$.

Total: 9

Coaching with Shuja

Question 11

Feb/Mar 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "2sin2x was often replaced by 2sinxcosx making it very difficult to know if the double angle substitution was correct."

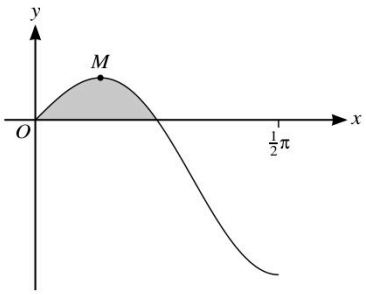
> "(b) Many candidates struggled to establish a correct integral in terms of u, however many good attempts were seen."

> "A common error was the inaccurate replacement of cos2x by 2cos2x – 1 accurately."

Source: 9709_m22_er.pdf, Page 8, Section: Comments on specific questions – Paper 3 (variant 32), Question 11

Coaching With Shuja

11



The diagram shows the curve $y = \sin x \cos 2x$ for $0 \leqslant x \leqslant \frac{1}{2}\pi$, and its maximum point M.

(a) Find the x-coordinate of M, giving your answer correct to 3 significant figures. [6]

The diagram shows the curve $y = \sin x \cos 2x$ for $0 \leq x \leq \frac{1}{2}\pi$, and its maximum point M.

(a) Find the x-coordinate of M, giving your answer correct to 3 significant figures. [6]

(b) Using the substitution $u = \cos x$, find the area of the shaded region enclosed by the curve and the x-axis in the first quadrant, giving your answer in a simplified exact form. [5]

(a)

Use correct product rule or chain rule **M1**

Obtain correct derivative in any form **A1** $\cos x \cos 2x - \sin x\, 2 \sin 2x$

Equate derivative to zero and use a correct double angle formula ***M1** If chain rule used then derivative set to 0 gains M1 since correct double angle formula has already been used.

Obtain an equation in one trigonometric variable **DM1** Allow following from coefficient errors in differentiation only

Obtain $6 \sin^2 x = 1$, $6 \cos^2 x = 5$ or $5 \tan^2 x = 1$ **A1** One of these 3 expressions

Obtain final answer $x = 0.421$ **A1** Must be 3 s.f.

Total: 6

(b)

State or imply $du = -\sin x\, dx$ **B1**

Using double angle formula, express integral in terms of u and du **M1** Use $\cos 2x = 2\cos^2 x - 1$

Integrate and obtain $\pm \left(u - \frac{2}{3}u^3\right)$ **A1**

Use limits $u = 1$, $u = \frac{1}{\sqrt{2}}$ in an integral of the form $au + bu^3$, where $ab \neq 0$ **M1** Require both limits substituted twice in $au + bu^3$ for M1. Do not condone decimals.

Obtain $\frac{1}{3}(\sqrt{2} - 1)$ or $\frac{1}{3}\sqrt{2} - \frac{1}{3}$ or $\frac{2}{3}\left(\frac{1}{\sqrt{2}}\right) - \frac{1}{3}$ or simplified equivalent **A1** ISW

Total: 5

Coaching With Shuja

Question 3

May/June 2022 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Question 3 (a) This question proved challenging for a number of candidates who had difficulty in identifying the correct term to find a, often providing a full expansion which was not necessary."

Source: 9709_s22_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 11), Question 3

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3 The coefficient of x^4 in the expansion of $\left(2x^2 + \frac{k}{x}\right)$ is a. The coefficient of x^2 in the expansion of $(2kx - 1)^4$ is b.

(a) Find a and b in terms of the constant k. **[3]**

(b) Given that $a + b = 216$, find the possible values of k. **[3]**

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3(a) M1 For selecting the term in x^4.
x^4 term is $[10\times](2x^2)^3\left(\frac{k^2}{x}\right) = 80k^4x^4 \Rightarrow a = 80k^4$.

A1 For correct value of a. Allow $80k^4x^4$.

B1 For correct value of b. Allow $24k^2x^2$.

Total: 3

3(b) M1 Forming a 3-term equation in k (all terms on one side) with their a and b and no x's.
$80k^4 + 24k^2 - 216[= 0] \Rightarrow 10k^4 + 3k^2 - 27 = 0$.

M1 Attempt to solve 3-term quartic (or quadratic in another variable) by factorisation, formula or completing the square - see guidance.

$(2k^2 - 3)(5k^2 + 9)[= 0][\Rightarrow k^2 = \frac{3}{2}$ or $-\frac{9}{5}]$.

A1 $[k] = \pm\sqrt{\frac{3}{2}}$ or $\pm\frac{\sqrt{6}}{2}$, $\pm\sqrt{1.5}$, **AWRT** ± 1.22.

Omission of $\pm$ A0. Additional answers A0. If **M1** M0, **SC B1** can be awarded for correct final answer, max 2/3.

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Question 5

May/June 2022 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Question 5 (a) Most candidates found a correct expression for the area of the sector, though the area of the triangle proved more difficult."

Source: 9709_s22_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 5

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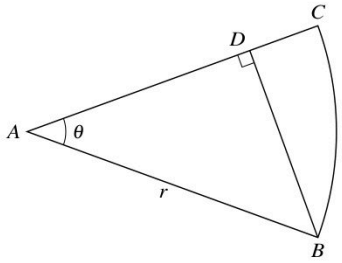
5

5



The diagram shows a sector ABC of a circle with centre A and radius r. The line BD is perpendicular to AC. Angle CAB is θ radians.

(a) Given that $\theta = \frac{1}{6}\pi$, find the exact area of BCD in terms of r. [3]

The diagram shows a sector $\boldsymbol{ABC}$ of a circle with centre $\boldsymbol{A}$ and radius $\boldsymbol{r}$. The line $\boldsymbol{BD}$ is perpendicular to $\boldsymbol{AC}$. Angle $\boldsymbol{CAB}$ is $\boldsymbol{\theta}$ radians.

(a) Given that $\boldsymbol{\theta = \frac{\pi}{6}}$, find the exact area of $\boldsymbol{BCD}$ in terms of $\boldsymbol{r}$. **[3]**

(b) Given instead that the length of $\boldsymbol{BD}$ is $\boldsymbol{\frac{\sqrt{3}}{2}r}$, find the exact perimeter of $\boldsymbol{BCD}$ in terms of $\boldsymbol{r}$. **[4]**

5(a) B1 Sector area = $\frac{1}{2}r^2\left(\frac{\pi}{6}\right) = \left[\frac{\pi}{12}r^2\right]$ using $\frac{1}{2}r^2\theta$ with θ in radians **SOI**. Bo if using a value for r.

B1 SOI Finding triangle area. Decimals Bo unless exact values seen in working.
$BD = \sin\frac{\pi}{6}r = \left[\frac{1}{2}r\right]$ and $AD = \cos\frac{\pi}{6}r = \left[\frac{\sqrt{3}}{2}r\right]$.

so triangle area = $\frac{1}{2}\left(\sin\frac{\pi}{6}r\right)\left(\cos\frac{\pi}{6}r\right) = \left[\frac{1}{2} \times \frac{1}{2}r \times \frac{\sqrt{3}}{2}r\right] = \left[\frac{1}{2}r \times \frac{\sqrt{3}}{2}r \times \frac{1}{2}\right]$

or $\frac{1}{2}r\left(\cos\frac{\pi}{6}r\right)\left[\sin\frac{\pi}{6}\right] = \left[\frac{1}{2}r \times \frac{\sqrt{3}}{2}r \times \frac{1}{2}\right]$.

B1 Area of $BCD = \frac{1}{12}\pi r^2 - \frac{\sqrt{3}}{8}r^2$. **OE** e.g. $\frac{r^2}{4}\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right)$ with $\cos\frac{\pi}{6}$ and $\sin\frac{\pi}{6}$ evaluated. Must be exact, in terms of r^2. **ISW**

Total: 3

5(b) B1 SOI by length of AD, CD or arc, or by perimeter.

Angle $BAC = \sin^{-1}\left(\frac{\sqrt{3}r}{2r}\right) = \left[\frac{\pi}{3}\right]$.

M1 SOI Finding length by Pythagoras, or by trigonometry with their angle BAC, provided $BAC \neq \frac{\pi}{6}$.

Length $AD = \cos\frac{\pi}{3}r = \left[\frac{1}{2}r\right]$ [so length $CD = \frac{1}{2}r$].

M1 SOI Using $r\theta$ with θ in radians. Condone $\theta = \frac{\pi}{6}$.

Length of arc $BC = r \times \frac{\pi}{3}$.

A1 Perimeter of $BCD = \frac{\sqrt{3}}{2}r + \frac{1}{2}r + \frac{\pi}{3}r$. **OE** e.g. $r\left(\frac{\sqrt{3}+1}{2} + \frac{\pi}{3}\right)$ with e.g. $\cos\frac{\pi}{3}$ evaluated.

Must be exact, in terms of r. **ISW**

Total: 4

Question 6

May/June 2022 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "A common error was giving the final result as ± x−1 instead of just the positive answer."

Source: 9709_s22_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 6

Coaching With Shuja

6 The function f is defined as follows:

$$f(x) = \frac{x^2 - 4}{x^2 + 4} \quad \text{for } x > 2.$$

(a) Find an expression for $f^{-1}(x)$. **[3]**

(b) Show that $1 - \frac{8}{x^2+4}$ can be expressed as $\frac{x^2-4}{x^2+4}$ and hence state the range of f. **[4]**

(c) Explain why the composite function ff cannot be formed. **[1]**

6(a) ***M1** For clearing denominator and expanding brackets. If swap variables first, look for $y^2x + 4x = y^2 - 4$.

DM1 For making x^2 the subject. If swap variables first, look for $y^2(1-x) = 4x + 4 \Rightarrow y^2 = \ldots$

A1 OE e.g. $\sqrt{\frac{4x+4}{1-x}}$ or $\sqrt{\frac{-4x-4}{x-1}}$ without $\pm$ in final answer.

Total: 3

6(b) $y = \frac{x^2-4}{x^2+4}$ leading to $(x^2+4)y = (x^2-4)$ leading to $x^2(y-1) = 4y + 4$ leading to $x^2 = \ldots$

DM1 For making x^2 the subject. If swap variables first, look for $y^2(1-x) = 4x + 4 \Rightarrow y^2 = \ldots$

A1 OE without $\pm$ in final answer.

Alternative method for Q6(a):
$x = \frac{y^2-4}{y^2+4}$ leading to $x = 1 - \frac{8}{y^2+4}$ leading to $x - 1 = \frac{-8}{y^2+4}$ leading to $[f^{-1}(x)] = \sqrt{\frac{-8}{x-1} - 4}$.

Total: 3

6(c) Explain why the composite function ff cannot be formed. **[1]**

Based on provided answer images: Range of f is all real values such that $f(x) > -1$ (since f is increasing for $x > 2$ and $\lim_{x \to 2^+} f(x) = 0$ while $\lim_{x \to \infty} f(x) = 1$). The range is not in the domain of f (since domain is $x > 2$), therefore ff cannot be formed.

Question 7

May/June 2022 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

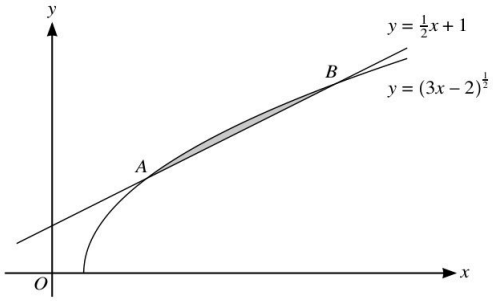
> "A very common error was to square the individual terms x and 1."

Source: 9709_s22_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 7

Coaching With Shuja

Question 7

7



The diagram shows the curve with equation $y = (3x - 2)^{\frac{1}{2}}$ and the line $y = \frac{1}{2}x + 1$. The curve and the line intersect at points A and B.

(a) Find the coordinates of A and B. [4]

The diagram shows the curve with equation $\boldsymbol{y = (3x - 2)^{\frac{1}{2}}}$ and the line $\boldsymbol{y = \frac{1}{2}x + 1}$. The curve and the line intersect at points $\boldsymbol{A}$ and $\boldsymbol{B}$.

(a) Find the coordinates of $\boldsymbol{A}$ and $\boldsymbol{B}$. **[4]**

(b) Hence find the area of the region enclosed between the curve and the line. **[5]**

Question 7(a):

M1 $(3x-2)^{\frac{1}{2}} = \frac{1}{2}x+1 \Rightarrow 3x-2 = \left(\frac{1}{2}x+1\right)^2 = \frac{1}{4}x^2+x+1$

M1 $\Rightarrow \frac{1}{4}x^2-2x+3=0 \Rightarrow x^2-8x+12=0 \Rightarrow (x-6)(x-2)=0$

A1 A1 $(2, 2)$ and $(6, 4)$ — A1 for each point, or A1 A0 for two correct x-values. If M0 for solving, **SC B2** possible: B1 for each point or B1 B0 for two correct x-values.

Total: 4

Question 7(b):

***M1** Area $= \pm\left[\int_{[2]}^{[6]}\left((3x-2)^{\frac{1}{2}} - \left(\frac{1}{2}x+1\right)\right)[dx]\right]$ — For intention to integrate and subtract (M0 if squared).

B1 B1 $\pm\left[\frac{2}{9}(3x-2)^{\frac{3}{2}} - \left(\frac{1}{4}x^2+x\right)\right]_2^6$ — B1 for each bracket integrated correctly (in any form).

DM1 $\pm\left[\left(\frac{2}{9}(16)^{\frac{3}{2}} - \left(\frac{1}{4}\times 36 + 6\right)\right) - \left(\frac{2}{9}(4)^{\frac{3}{2}} - \left(\frac{1}{4}\times 4 + 2\right)\right)\right]$—$\pm$ F(their 6) - F(their 2)$ with their integral. Allow 1 sign error.

A1 $\frac{4}{9}$ — AWRT 0.444. **SC1 B1** for $\frac{4}{9}$ if ***\M1 B1 B1 DM0. SC2 B1** for $\frac{4}{9}$ if* **\M1 B0 B0 DM0**, provided limits stated.

Alternative method for question 7(b)

Area $= \pm\int_{[2]}^{[6]}(3x-2)^{\frac{1}{2}}[dx]$ – area of trapezium (or triangle + rectangle)

B1 $\pm\left[\frac{2}{9}(3x-2)^{\frac{3}{2}}\right]_2^6 - 4\left(\frac{2+4}{2}\right)$ or $\pm\left[\frac{2}{9}(3x-2)^{\frac{3}{2}}\right]_2^6 - \left(\frac{2+4}{2}(2\times 4)\right)$ — B1 for bracket integrated correctly (in any form). B1 **FT** for using correct formula with their values.

DM1 $\pm\left(\frac{2}{9}(16)^{\frac{3}{2}} - \frac{2}{9}(4)^{\frac{3}{2}}\right) - 12$ – $\pm F(\text{their } 6) - F(\text{their } 2)$ using their integral. Allow 1 sign error.

Total: 5

Question 8

May/June 2022 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> " 0 0 (b) While many correct responses were seen, this question proved particularly challenging for some 2 candidates who could not rearrange to reach sin−1 ."

Source: 9709_s22_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 8

Coaching With Shuja

Question 8

(a) The curve $y = \sin x$ is transformed to the curve $y = 4\sin\left(\frac{1}{2}x - 30^\circ\right)$.

Describe fully a sequence of transformations that have been combined, making clear the order in which the transformations are applied. **[5]**

(b) Find the exact solutions of the equation $4\sin\left(\frac{1}{2}x - 30^\circ\right) = 2\sqrt{2}$ for $0^\circ \leq x \leq 360^\circ$. **[3]**

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Question 8(a):

B2,1,0 B2 for fully correct, **B1** with two elements correct. \{\} indicates different elements. Accept angle in radians.

EITHER (1) \{Translation\} $\begin{pmatrix} 30^\circ \\ 0 \end{pmatrix}$ **OR (2)** \{Translation\} $\begin{pmatrix} 60^\circ \\ 0 \end{pmatrix}$

B2,1,0 B2 for fully correct, **B1** with two elements correct. \{\} indicates different elements.

(3) \{Stretch\} \{factor 2\} \{in x-direction\}

B1 Stretch, y-direction and factor **and** correct order. Correct order is **either (1)** then **(3) or (3)** then **(2)**. **(4)** can be anywhere in the sequence.

(4) Stretch factor 4 in y-direction **and** correct order

Total: 5

Question 8(b):

M1 $4\sin\left(\frac{1}{2}x - 30^\circ\right) = 2\sqrt{2} \Rightarrow \sin^{-1}\left(\frac{\sqrt{2}}{2}\right) = [45] - \textbf{SOI}$

M1 $\frac{1}{2}x - 30 = 45$ or $135 \Rightarrow x = 2(45 + 30)$ or $x = 2(135 + 30) - \textbf{SOI}$. The M marks are independent.

A1 $x = 150^\circ, x = 330^\circ$ — Both exact values, condone $\frac{5\pi}{6}, \frac{11\pi}{6}$. **A0** if extra solutions in the interval. Ignore other solutions outside $[0^\circ, 360^\circ]$.

Total: 3

Question 10

May/June 2022 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "(b) Most candidates found this question to be particularly challenging, so only a small number of fully correct solutions were seen."

> "Common errors included omitting the constant of integration when dy dy 9 finding and/or y, or substituting = (instead of 0)."

> "dy (c) Many candidates set their expression for to 0 but were either unable to solve their equation or dx did not justify the statement given in the question."

Source: 9709_s22_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 10

Coaching With Shuja

Question 10

The equation of a curve is such that $\frac{d^2y}{dx^2} = 6x^2 - \frac{4}{x^3}$. The curve has a stationary point at $\left(-1, \frac{9}{2}\right)$.

(a) Determine the nature of the stationary point at $\left(-1, \frac{9}{2}\right)$. **[1]**

(b) Find the equation of the curve. **[5]**

(c) Show that the curve has no other stationary points. **[3]**

(d) A point A is moving along the curve and the y-coordinate of A is increasing at a rate of 5 units per second.

Find the rate of increase of the x-coordinate of A at the point where $x = 1$. **[3]**

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Question 10(a):

B1 Sub $x = -1$ into $\frac{d^2y}{dx^2}$, correct conclusion. **WWW**

$\frac{d^2y}{dx^2} = 6(-1)^2 - \frac{4}{(-1)^3} > 0$ \; minimum or $\frac{d^2y}{dx^2} = 10$ \; minimum

Total: 1

Question 10(b):

***M1** Integrating $\frac{d^2y}{dx^2}$ (at least one term correct).

$\frac{dy}{dx} = 2x^3 + \frac{2}{x^2}[+c]$

DM1 Substituting $x = -1$, $\frac{dy}{dx} = 0$ (need to see) to evaluate c. **DM0** if simply state $c = 0$ or omit $+c$.

A1 FT Integrated. **FT** their non-zero value of c if **DM1** awarded.

$y = \frac{1}{2}x^4 - \frac{2}{x} + (\text{their } c)x + k$

DM1 Substituting $x = -1$, $y = \frac{9}{2}$ to evaluate k (dep on ***M1**).

$\frac{9}{2} = \frac{1}{2} - 2 + k$ leading to $k = [2]$

A1 $y = \frac{1}{2}x^4 - \frac{2}{x} + 2$

OE e.g. $2x^{-1}$ or $\frac{4}{2x}$.

A0 (wrong process) if c not evaluated but correct answer obtained.

Total: 5

Question 10(c):

M1 $\frac{dy}{dx} = 2x^3 + \frac{2}{x^2} = 0$

M1 Leading to $x^5 = -1$

A1 $x = -1$ and stating e.g. 'only' or 'no other solutions'.

Total: 3

Question 10(d):

***M1** Substituting $x = 1$ into **their** $\frac{dy}{dx}$.

$\frac{dy}{dx} = [4]$

DM1 OE Using chain rule correctly **SOI**.

$\frac{dx}{dt} = \frac{dx}{dy} \times \frac{dy}{dt} = \frac{1}{4} \times 5$

A1 OE e.g. 1.25.

$\frac{5}{4}$

Total: 3

Question 5

May/June 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 5 Both parts of this question proved to be challenging for many candidates."

Source: 9709_s22_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 5

Coaching With Shuja

Question 5

The equation of a curve is $y = 4x^2 - kx + \frac{1}{2}k^2$ and the equation of a line is $y = x - a$, where k and a are constants.

(a) Given that the curve and the line intersect at the points with x-coordinates 0 and $\frac{3}{4}$, find the values of k and a. **[4]**

(b) Given instead that $a = -\frac{7}{2}$, find the values of k for which the line is a tangent to the curve. **[5]**

Question 5(a):

M1 $4 \times 0^2 - 0 + \frac{1}{2}k^2 = 0 - a$ — Equating the equations of curve and line and substituting $x = 0$. Condone slight errors e.g. ± sign errors.

M1 $4 \times \left(\frac{3}{4}\right)^2 - \frac{3}{4}k + \frac{1}{2}k^2 = \frac{3}{4} - a$ — Equating the equations of curve and line and substituting $x = \frac{3}{4}$. Condone slight errors e.g. ± sign errors.

A1 A1 $k = 2, a = -2$ — **WWW**

Alternative method for question 5(a)

***M1** $(x - 0)\left(x - \frac{3}{4}\right) = 0$ or $x(4x - 3) = 0$ — [$\Rightarrow 4x^2 - 3x = 0$] Use 0, $\frac{3}{4}$ to form a quadratic equation. **Do not allow** $(x + 0)\left(x + \frac{3}{4}\right) = 0$.

DM1 $4x^2 - kx + \frac{1}{2}k^2 = x - a$ leading to $4x^2 - (k + 1)x + \frac{1}{2}k^2 + a[= 0]$ — Equating the equations of curve and line and rearranging so that terms are all on same side. Condone slight errors e.g. ± sign errors.

A1 A1 $k = 2, a = -2$ — **WWW**

Alternative method for question 5(a)

***M1** Using sum and product of roots. Condone ± sign errors.

$\frac{b}{a} = \frac{3}{4} + 0$ and $\frac{c}{a} = 0 \times \frac{3}{4}$

DM1 Equating the equations of curve and line and equating to $\frac{3}{4}$ and 0. — Equating the equations of curve and line and equating to $\frac{3}{4}$ and 0.

$\frac{k+1}{4} = \frac{3}{4}$ and $\frac{1}{2}k^2 + a = 0$

A1 A1 $k = 2, a = -2$ — **WWW**

Total: 4

Question 5(b):

***M1** $4x^2 - kx + \frac{1}{2}k^2 = x + \frac{7}{2}$ — **OE** Substitute $a = -\frac{7}{2}$ and rearrange so that terms are all on same side, condone ± sign errors. Watch for multiples.

***DM1** Use of $b^2 - 4ac$ with the coefficients from **their** 3-term quadratic. Both coefficients 'b' and 'c' must consist of two components.

$(k + 1)^2 - 4 \times 4 \times \left(\frac{1}{2}k^2 - \frac{7}{2}\right)$

$\Rightarrow 7k^2 - 2k - 57$

A1 OE Correct 3 term quadratic in k or k only.

$(k - 3)(7k + 19)$ or other valid method

DM1 Factorising or use of the formula or completing the square. Must be evidence of an attempt to solve for this mark. Dependent upon both previous method marks.

A1 OE e.g. AWRT -2.71. No **ISW** if inequalities used.

$k = 3, k = -\frac{19}{7}$

SC: If second **DM1** not scored, **SC B1** available for correct final answers.

Total: 5

Question 6

May/June 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

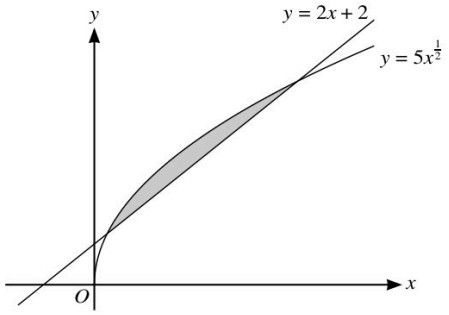
> "Most realised the need to equate the line and the curve but many then struggled to solve the resulting quadratic equation in x or as in Question 5(b), did not show the necessary working and used a calculator function instead."

Source: 9709_s22_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 6

Coaching With Shuja

Question 6

6



The diagram shows the curve with equation $y = 5x^{\frac{1}{2}}$ and the line with equation $y = 2x + 2$.

Find the exact area of the shaded region which is bounded by the line and the curve. [5]

The diagram shows the curve with equation $y = 5x^{\frac{1}{2}}$ and the line with equation $y = 2x + 2$.

Find the exact area of the shaded region which is bounded by the line and the curve. **[5]**

Shuja

Question 6:

M1 Line meets curve when: $2x + 2 = 5x^{\frac{1}{2}}$ leading to $2x - 5x^{\frac{1}{2}} + 2 = [0]$ — Equating line and curve and rearranging so that terms are all on same side, condone sign errors, and making a valid attempt to solve by factorising, using the formula or completing the square.

or $4x^2 + 8x + 4 = 25x$ leading to $4x^2 - 17x + 4 = [0]$ — Factors are: $(2x^2 - 1)(x^2 - 2)$, $(4x - 1)(x - 4)$ and $(2y - 5)(y - 10)$.

or $x = \frac{y^2}{25}$ leading to $2y^2 - 25y + 50 = [0]$

A1 SC: If **M1** not scored, **SC B1** available for correct answers, could just be seen as limits.

$x = \frac{1}{4}, x = 4$

*M1 Area $= \int 5x^{\frac{1}{2}} - (2x + 2)dx = \int 5x^{\frac{1}{2}} - 2x - 2\,dx$ — Intention to integrate and subtract areas. Condone missing brackets and/or subtraction wrong way around.

DM1 Integrating ($kx^{\frac{3}{2}}$ seen) and substituting **'their'** points of intersection (but limits need to be found, not assumed to be 0 and something else).

$$= \left[\frac{10}{3}x^{\frac{3}{2}} - x^2 - 2x\right]_{\frac{1}{4}}^{4} = \left(\left(\frac{10}{3} \times 8\right) - 16 - 8\right) - \left(\frac{10}{3} \times \frac{1}{8} - \frac{1}{16} - \frac{1}{2}\right)$$

or a trapezium using the correct formula (**'their'** $\frac{15}{4}$, must be **'their 4'** -- **'their'** $\frac{1}{4}$ but not 0).

A1 OE exact answer.

Condone $-\frac{45}{16}$ if corrected to $\frac{45}{16}$. **A0** for inclusion of π.

$\frac{45}{16}$ or $2\frac{13}{16}$ or 2.8125

SC: If *M1 **DM0** scored, **SC B1** available for correct answer.

Total: 5

Question 7

May/June 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

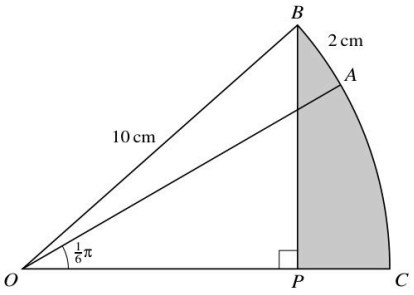
> "In part (b) many candidates used the formula for the area 6 5 30 of a sector correctly but found the area of the triangle OPB more challenging."

Source: 9709_s22_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 7

Coaching With Shuja

Question 7

7



The diagram shows a sector $OBAC$ of a circle with centre O and radius 10 cm. The point P lies on OC and BP is perpendicular to OC. Angle $AOC = \frac{1}{6}\pi$ and the length of the arc AB is 2 cm.

(a) Find the angle BOC. [2]

The diagram shows a sector $OBAC$ of a circle with centre O and radius 10 cm. The point P lies on OC and BP is perpendicular to OC. Angle $AOC = \frac{1}{6}\pi$ and the length of the arc AB is 2 cm.

(a) Find the angle BOC. **[2]**

(b) Hence find the area of the shaded region BPC giving your answer correct to 3 significant figures. **[4]**

Shuja

Question 7(a):

B1 OE

Sight of 0.2 from $s = r\theta$ but $10\theta = 2$ is not enough.

ISW if $\frac{2}{10} = \frac{\pi}{5}$.

$[\angle AOB] = \frac{2}{10}$

B1 OE e.g. 0.724° **AWRT** or 41.5 degrees **AWRT**.

But not $\frac{2+\frac{5\pi}{3}}{10}$ — fraction within a fraction.

ISW incorrect simplifications.

Alternative method for question 7(a)

OR [Arc AC] = $10 \times \frac{10\pi}{6}$ or [Arc BC] = $10 \times \frac{10\pi}{6} + 2$ or 7.2

B1 AWRT. Sight of $\frac{10\pi}{6}$ or 5.2 or 7.2.

$[\angle BOC] = \frac{5\pi+6}{30}$ or $\frac{1}{6}\pi + 0.2$ or $\frac{5\pi+6}{30}$ **AWRT**

B1 OE e.g. 0.724° **AWRT** or 41.5 degrees **AWRT**.

But not $\frac{2+\frac{5\pi}{3}}{10}$ — fraction within a fraction.

ISW incorrect simplifications.

Total: 2

Question 7(b):

M1 OE

Any correct method for **both** lengths, for their angle **BOC** (which may have been incorrectly 'simplified' but not 0.2) or length **BP** and $\widehat{OBP}$. May be seen as part of $\frac{1}{2}ab\sin C$.

Sight of correct method enough. Can be implied by the next **A1**.

$[BP] = 10\sin\left(\frac{5\pi+6}{30}\right)$ and $[OP] = 10\cos\left(\frac{5\pi+6}{30}\right)$ or $= [6.6208\ldots]$ and $= [7.494\ldots]$

OR

$[BP] = 10\sin\left(\frac{5\pi+6}{30}\right)$ and $[\widehat{OBP}] = \left(\frac{5\pi+6}{30}\right)$ or

A1 OE Can be implied by any answer in range (24.7, 24.9) or a final answer in the range (11.3, 11.5) **WWW**.

Area of $\triangle OBP = \frac{1}{2} \times 10 \sin\left(\frac{5\pi + 6}{30}\right) \times 10 \cos\left(\frac{5\pi + 6}{30}\right)$ *or* $\frac{1}{2} \times 10 \times 10 \sin\left(\frac{5\pi + 6}{30}\right) \times \sin\left(\frac{5\pi + 6}{30}\right)$ = [24.809]

M1 Use of $\frac{1}{2}r^2\theta$ with **their** angle **BOC** (may have been incorrectly 'simplified' but not 0.2).

[Sector BOC] $= \frac{1}{2} \times 10^2 \times$ **their** $\left(\frac{5\pi+6}{30}\right)$ = [36.1799...]

A1 CAO

Area of region $BPC = 11.4$

Total: 4

Question 10

May/June 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 10 Part (a) proved to be one of the most challenging questions on the paper with many candidates omitting it."

> "Many candidates were more confident with the standard inverse function requirements of part (b), although some, having struggled with part (a), missed out the whole of the Question 10."

Source: 9709_s22_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

Coaching With Shuja

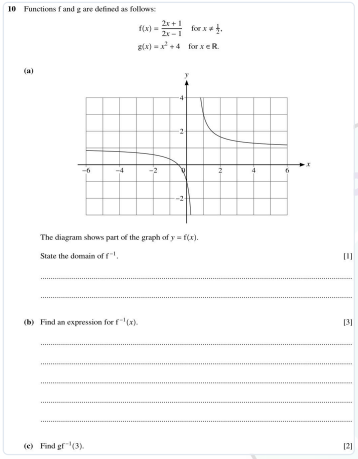
Question 10

Functions f and g are defined as follows:

$$f(x) = \frac{2x+1}{2x-1} \quad \text{for } x \neq \frac{1}{2}$$

$$g(x) = x^2 + 4 \quad \text{for } x \in \mathbb{R}$$

(a)


10 Functions f and g are defined as follows:

$f(x) = \frac{2x+1}{2x-1}$ for $x \neq \frac{1}{2}$,

$g(x) = x^2 + 4$ for $x \in \mathbb{R}$.

(a)

The diagram shows part of the graph of $y = f(x)$.

State the domain of f^{-1}. [1]

(b) Find an expression for $f^{-1}(x)$. [3]

(c) Find $gf^{-1}(3)$. [2]

The diagram shows part of the graph of $y = f(x)$.

State the domain of f^{-1}. **[1]**

(b) Find an expression for $f^{-1}(x)$. **[3]**

(c) Find $gf^{-1}(3)$. **[2]**

(d) Explain why $g^{-1}(x)$ cannot be found. **[1]**

(e) Show that $1 + \frac{2}{2x-1}$ can be expressed as $\frac{2x+1}{2x-1}$. Hence find the area of the triangle enclosed by the tangent to the curve $y = f(x)$ at the point where $x = 1$ and the x- and y-axes. **[6]**

Question 10(a):

B1 Must be $x \neq 1$ or $x < 1, x > 1$ or $(-\infty, 1), (1, \infty)$ [$x \in \mathbb{R}$]

Total: 1

Question 10(b):

***M1** Setting $y =$, removing fraction and expanding brackets.

$y = \frac{2x+1}{2x-1}$ leading to $(2x-1)y = 2x+1$ leading to $2xy - y = 2x+1$

DM1 Reorganising to get $x =$. Condone $\pm$ sign errors only.

$2xy - 2x = y + 1$ leading to $2x(y-1) = y+1$

leading to $x = \frac{y+1}{2(y-1)}$

A1 OE. Must be in terms of x. Do not allow $\frac{x+1}{x-1} \div 2$.

$[f^{-1}(x)] = \frac{x+1}{2(x-1)}$ or $\frac{x+1}{x-1} - \frac{1}{2}$ or $\frac{1}{2} + \frac{1}{x-1}$

Total: 3

Question 10(c):

M1 Correct order of operations and substitution of $x = 3$ needed.

$(\textbf{their } f^{-1}(3))$ leading to $(\textbf{their } f^{-1}(3))^2 + 4$ [$f^{-1}(3) = 1, 1 + 4 =$]

A1

5

Total: 2

Question 10(d):

B1 Any reason mentioning 2 values, or $+$ and $-$, such as: square root gives 2 values or horizontal line test crosses curve twice or 2 values because of turning point or 2 values because it is a quadratic.

Total: 1

Question 10(e):

B1 AG

Do not condone equating expressions and verification.

$f(x) = 1 + \frac{2}{2x-1} = \frac{2x-1}{2x-1} + \frac{2}{2x-1} = \frac{2x+1}{2x-1}$

***M1** For $k(2x-1)^{-2}$ and no other terms or correct use of the product or quotient rule then **ISW**.

$f'(x) = -4(2x-1)^{-2}$ or $2(2x-1)^{-1} + \{-(2x+1)2(2x-1)^{-2}\}$ or $\frac{(2x-1)2-2(2x+1)}{(2x-1)^2}$

A1 Differentiation must have clearly taken place.

Gradient $m = -4$

DM1 Using (1, 3) in the equation of a line with **their** gradient.

Equation of tangent is $y - 3 = -4(x-1)$ [$\Rightarrow y = -4x + 7$]

A1 FT SOI from **their** straight line or by integration from 0 to **'their' 7/4**.

Crosses axes at $\left(\frac{7}{4}, 0\right)$ and $(0, 7)$

[**Area**] $= \frac{49}{8}$

A1 OE e.g. 6.13 **AWRT**.

If M0 A0 DM0, **SC B2** available for correct answer.

Total: 6

Question 11

May/June 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (b) proved to be the most challenging question on the paper with many candidates missing it out and very few good attempts seen."

Source: 9709_s22_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 11

Coaching With Shuja

Question 11

The function f is given by $f(x) = 4\cos^4 x + \cos^2 x - k$ for $0 \leq x \leq 2\pi$, where k is a constant.

(a) Given that $k = 3$, find the exact solutions of the equation $f(x) = 0$. **[5]**

(b) Use the quadratic formula to show that, when $k > 5$, the equation $f(x) = 0$ has no solutions. **[5]**

Question 11(a):

M1 Attempt to solve 3 term quartic (or quadratic in another variable).

$4\cos^4 x + \cos^2 x - 3 = 0 \Rightarrow (4\cos^2 x - 3)(\cos^2 x + 1) = 0$

A1 If M0 scored then **SC B1** is available for sight of $\frac{3}{4}$ [and -1].

$\Rightarrow [\cos^2 x =] \frac{3}{4}$ [$\cos^2 x = -1$]

M1 Square rooting **'their'** $\cos^2 x$. Allow without $\pm$. May be implied by correct final answer(s). Ignore $\sqrt{-1}$.

$\Rightarrow \cos x = [\pm]\sqrt{\text{their } \frac{3}{4}}$ **OE** $= [\pm]\frac{\sqrt{3}}{2}$

A1 FT Dependent on preceding **M1** only. Exact answers needed.

A1 for any 2 correct answers A1 A1 for 4 correct answers and no others inside the range $0 \leq x < 2\pi$
A0 A1 FT can be awarded for two exact answers that are 2π -- **'their'** $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ within the range $0 \leq x < 2\pi$.

$[x =] \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

SC: If all 4 answers given in degrees (30, 150, 210, 330) or non-exact (**AWRT** 0.524, 2.62, 3.67, 5.76 or 0.167π, 0.833π, 1.17π, 1.83) and no others then **SC B1**.

Total: 5

Question 11(b):

B1 State that this root is less than 0, needs to be linked to $\cos^2 x$. Can be achieved by substituting a value for $k > 0$.

$\cos^2 x = \frac{-1-\sqrt{1+16k}}{8} < 0$ [$\therefore$ no solutions].

***M1** Must use quadratic formula. Allow any value of k **but not** ± 3. Condone $+$ rather than $\pm$.

$[\cos^2 x] = \frac{-1\pm\sqrt{1+16k}}{8}$

DM1 $*$ represents any inequality or $=$. Or argue logically if $k > 5 \Rightarrow 1 + 16k > 81 \Rightarrow > 1$.

$\frac{-1+\sqrt{1+16k}}{8} * 1 \Rightarrow -1 + \sqrt{1+16k} * 8 \Rightarrow 1 + 16k * 81$

A1 Needs to be linked to $\cos^2 x$.

$\cos^2 x = 1$ or $\cos^2 x >$ or ≥ 1

A1 Concluding statement having considered both $\pm$ cases. $\therefore$ no solutions

Total: 5

Question 4

May/June 2022 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "(b) In contrast to part (a), fewer completely correct descriptions were seen in this part."

Source: 9709_s22_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 4

Coaching With Shuja

(a) The curve with equation $y = x^2 + 2x - 5$ is translated by $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$.

Find the equation of the translated curve, giving your answer in the form $y = ax^2 + bx + c$. **[3]**

(b) The curve with equation $y = x^2 + 2x - 5$ is transformed to a curve with equation $y = 4x^2 + 4x - 5$.

Describe fully the single transformation that has been applied. **[2]**

Coaching With Shuja

(a) $\{(x+1)^2+2(x+1)-5\}+\{3\}$, or $\{(x+1+1)^2\}+\{-6+3\}$ **M1 M1**

M1 for dealing with $\begin{pmatrix}-1\\0\end{pmatrix}$ and M1 for dealing with $\begin{pmatrix}0\\3\end{pmatrix}$.

$[y=]x^2+4x+1$ **A1**

Answer only given full marks.

Total: 3

(b) [Implicit from part (a)]

Coaching With Shuja

Question 11

May/June 2022 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "(b) This question proved challenging to candidates, many of whom had not reached a correct answer or any answer in part (a)."

> "Only a few candidates were able to factorise a quadratic expression with algebraic coefficients to find the coordinates of Q."

Source: 9709_s22_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 11

Question Paper May/June 2022 • Paper 1 (variant 13) • Q11

The point P lies on the line with equation $y = mx + c$, where m and c are positive constants. A curve has equation $y = -\frac{m}{x}$. There is a single point P on the curve such that the straight line is a tangent to the curve at P.

(a) Find the coordinates of P, giving the y-coordinate in terms of m. **[6]**

The normal to the curve at P intersects the curve again at the point Q.

(b) Find the coordinates of Q in terms of m. **[4]**

Coaching With Shuja

(a) $mx + c = -\frac{m}{x} \Rightarrow mx^2 + cx + m = 0$ **M1**

All x terms in the numerator. OE e.g. $mx^2 + cx = -m$.

$b^2 - 4ac = 0 \Rightarrow c^2 - 4m^2 = 0$ **M1**

OE $b^2 - 4ac = 0$ is implied by $c^2 - 4m^2 = 0$.

$c = [\pm]2m$ **A1**

SOL. Allow $\pm$ at this stage.

$mx^2[\pm]2mx + m = 0 \Rightarrow x^2[\pm]2x + 1 = 0$ **M1**

Sub $c = \pm 2m$. Ignore substitution of $-2m$.

$(x + 1)^2 = 0 \Rightarrow x = -1$ only **A1**

$y = m$ only or $(-1, m)$ only **A1**

Total: 6

(b) Equation of normal is $y - m = -\frac{1}{m}(x + 1)$ *M1

Through *their* P with gradient $-\frac{1}{m}$, OE

e.g. $y = -\frac{1}{m}x + \frac{m^2 - 1}{m}$.

Allow use of the gradient of the curve as $-\frac{1}{(\text{their } x)^2}$ with their P.

Coordinates of P must be in terms of m only.

$-\frac{x}{m} - \frac{1}{m} + m = -\frac{m}{x} \Rightarrow x^2 + x(1 - m^2) - m^2[= 0]$ **DM1**

OE Equating *their* normal equation to the equation of the curve and removing x from the denominator.

$(x + 1)(x - m^2)[= 0] \Rightarrow x = m^2$ **A1**

or

$$x = \frac{m^2 - 1 \pm \sqrt{1 - 2m^2 + m^4 + 4m^2}}{2} = \frac{m^2 - 1 \pm (m^2 + 1)}{2} = m^2$$

$y = -\frac{m}{m^2} = -\frac{1}{m}$ **A1**

or $(m^2, -\frac{1}{m})$, ignore the coordinates of P.

Total: 4

Question 1

May/June 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "A few candidates were successful in 32x reducing the given equation to =6, which is relatively straightforward to solve."

Source: 9709_s22_er.pdf, Page 19, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

Question Paper May/June 2022 • Paper 3 (variant 31) • Q1

Solve the equation $2(3^{2x-1}) = 4^{x+1}$, giving your answer correct to 2 decimal places. **[4]**

Coaching With Shuja

Use law of the logarithm of a product or a quotient or a power *M1

Obtain a correct linear equation in any form A1

e.g. $\ln 2 + (2x-1)\ln 3 = (x+1)\ln 4$ or $\log_3 2 + (2x-1)\log_3 3 = (2x+2)\log_2 2$

Solve for x DM1

Allow for unsimplified expression $x = \ldots$. Allow M1 M1 for $x = 1.45$ from $6^{2x-1} = 4^{x+1}$.

Obtain answer $x = 2.21$ A1

The question asks for 2 dp.

Alternative method for question 1

Correct use of indices to obtain $2.25^x = 6$ or $1.5^{2x} = 6$ M1 A1

Correct use of logarithms to solve for x M1

Allow solution of $2.25^x = 6$ by trial and improvement as far as 2.2...

Obtain answer $x = 2.21$ A1

Need to see an intermediate step / sequence of iterations.

Total: 4

Question 2

May/June 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common errors were in taking out the factor of 2 incorrectly, either not obtaining 2−2 or going on to expand(1−x2)−2 ."

> "(b) Many candidates did not attempt this question."

Source: 9709_s22_er.pdf, Page 19, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

Coaching With Shuja

(a) Expand $(2 - x^2)^{-2}$ in ascending powers of x, up to and including the term in x^4, simplifying the coefficients. **[4]**

(b) State the set of values of x for which the expansion is valid. **[1]**

Coaching With Shuja

(a) State a correct unsimplified version of the x^2 or the x^4 term of the expansion of $(2-x^2)^{-2}$ or $\left(1-\frac{1}{2}x^2\right)^{-2}$ **M1**

$$\frac{1}{4}\left(1+2\frac{x^2}{2}+\frac{-2\cdot-3}{2}\left(\frac{x^2}{2}\right)^2\ldots\right)$$

Symbolic binomial coefficients are not sufficient for the M1.

State correct first term **B1**

Accept 2^{-2}.

Obtain the next two terms **A1 A1**

$$\frac{1}{4}x^2+\frac{3}{16}x^4$$

A1 for each one correct ISW. Full marks for $\frac{1}{4}\left(1+x^2+\frac{3}{4}x^4\right)$ ISW.

SC allow M1 A1 A1 for $\frac{2}{4}$ and $1+x^2+\frac{3}{4}x^4$ SOI.

SC allow M1 A1 for $1+x^2+\frac{3}{4}x^4$

Total: 4

(b) State answer $|x|<\sqrt{2}$ **B1**

Or $-\sqrt{2}<x<\sqrt{2}$.

Total: 1

Coaching with Shuja

Question 3

May/June 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common errors seen were in the algebra when simplifying to obtain the quadratic 1 1 1 equation."

Source: 9709_s22_er.pdf, Page 19, Section: Comments on specific questions – Paper 3 (variant 31), Question 3

Coaching With Shuja

Solve the equation $2\cot 2x + 3\cot x = 5$, for $0^\circ < x < 180^\circ$. **[6]**

Coaching With Shuja

Use correct trigonometric formula to form an equation in $\tan x$ *M1

e.g. $\frac{1-\tan^2 x}{\tan x} + \frac{3}{\tan x} = 5$

Obtain a correct linear equation in any form A1

$1 - \tan^2 x + 3 = 5\tan x$

Reduce equation to a 3-term quadratic A1

$\tan^2 x + 5\tan x - 4 = 0$, or 3-term equivalent

Solve a 3-term quadratic in $\tan x$ and obtain a value of x DM1

Obtain answer, e.g. $x = 35.1^\circ$ A1

Obtain second answer, e.g. $x = 99.9^\circ$, and no other in $(0^\circ, 180^\circ)$ A1

Ignore answers outside $(0^\circ, 180^\circ)$. Treat answers in radians $(0.612, 1.74)$ as a misread.

Alternative method for question 3

Use correct formulae for $\sin 2x$ and $\cos 2x$ to form an equation in $\sin x$ and $\cos x$ *M1

Obtain $4\frac{\cos x}{\sin x} - \frac{\sin x}{\cos x} = 5$ A1

Reduce equation to a 3-term quadratic A1

$\tan^2 x + 5\tan x - 4 = 0$, or 3-term equivalent

Solve a 3-term quadratic in $\tan x$ and obtain a value of x DM1

Obtain answer, e.g. $x = 35.1^\circ$ A1

Obtain second answer, e.g. $x = 99.9^\circ$, and no other in $(0^\circ, 180^\circ)$ A1

Ignore answers outside $(0^\circ, 180^\circ)$. Treat answers in radians $(0.612, 1.74)$ as a misread.

Total: 6

Question 7

May/June 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(c) A few candidates followed through their answer to part (b) correctly."

Source: 9709_s22_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

Question Paper May/June 2022 • Paper 3 (variant 31) • Q7

The complex number u is defined by $u = \frac{\sqrt{2}-av2i}{1+2i}$, where a is a positive integer.

(a) Express u in terms of a, in the form $x + iy$, where x and y are real and exact. **[3]**

It is now given that $a = 3$.

(b) Express u in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$, giving the exact values of r and θ. **[2]**

(c) Using your answer to part **(b)**, find the two square roots of u. Give your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$, giving the exact values of r and θ. **[3]**

(a) Multiply numerator and denominator by $1-2i$, or equivalent **M1**

At least one multiplication completed.

Obtain correct numerator $(1-2a)\sqrt{2}-(2+a)\sqrt{2}i$ **A1**

OE

Obtain final answer $\frac{1-2a}{5}\sqrt{2}-\frac{2+a}{5}\sqrt{2}i$ **A1**

OE

Alternative method for question 7(a)

Multiply $x+iy$ by $1+2i$ and compare real and imaginary parts **M1**

Obtain $x-2y=\sqrt{2}$ and $2x+y=a\sqrt{2}$ **A1**

Obtain final answer $\frac{1-2a}{5}\sqrt{2}-\frac{2+a}{5}\sqrt{2}i$ **A1**

OE

Total: 3

(b) Obtain $r=2$ **B1 FT**

Obtain $\theta=-\frac{3}{4}\pi$ **B1**

Total: 2

(c) Use correct method to find r or θ **M1**

State answer $\sqrt{2}e^{\frac{3\pi}{8}i}$ **A1 FT**

State answer $\sqrt{2}e^{\frac{5\pi}{8}i}$ **A1 FT**

Total: 3

Coaching with Shuja

Question 8

May/June 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(b) This was an unusual question, and many candidates did not know where to start."

Source: 9709_s22_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 8

Coaching With Shuja

The equation of a curve is $x^3 + y^3 + 2xy + 8 = 0$.

(a) Express $\frac{dy}{dx}$ in terms of x and y. **[4]**

The tangent to the curve at the point where $x = 0$ and the tangent at the point where $y = 0$ intersect at the acute angle α.

(b) Find the exact value of $\tan\alpha$. **[5]**

Coaching With Shuja

(a) State or imply $3y^2\frac{dy}{dx}$ as derivative of y^3 **B1**

State or imply $2y + 2x\frac{dy}{dx}$ as derivative of $2xy$ **B1**

Complete differentiation and equate attempted derivative to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain answer $-\frac{3x^2+2y}{3y^2+2x}$ **A1**

Total: 4

(b) Find gradient at either $(0, -2)$ or $(-2, 0)$ **M1**

Obtain answers $-\frac{1}{3}$ and **3** **A1** **A1**

Use $\tan(A \pm B)$ formula to find $\tan\alpha$ **M1**

Obtain answer $\tan\alpha = \frac{4}{3}$ **A1**

Total: 5

Coaching With Shuja

Question 9

May/June 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

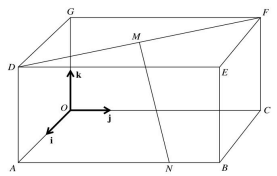
> "The most common error was in writing the equation without r =...."

> "(c) This part of the question was more challenging."

Source: 9709_s22_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 9

Coaching With Shuja

9



In the diagram, $OABCDEFG$ is a cuboid in which $OA = 2$ units, $OC = 4$ units and $OG = 2$ units. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OC and OG respectively. The point M is the midpoint of DF. The point N on AB is such that $AN = 3NB$.

(a) Express the vectors $\overrightarrow{OM}$ and $\overrightarrow{MN}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. [3]

(b) Find a vector equation for the line through M and N. [2]

In the diagram, $OABCDEFG$ is a cuboid in which $OA = 2$ units, $OC = 4$ units and $OG = 2$ units. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OC and OG respectively. The point M is the midpoint of DF. The point N on AB is such that $AN = 3NB$.

(a) Express the vectors $\overrightarrow{OM}$ and $\overrightarrow{MN}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. **[3]**

(b) Find a vector equation for the line through M and N. **[2]**

(c) Show that the length of the perpendicular from O to the line through M and N is $\sqrt{\frac{53}{6}}$. **[4]**

(a) Obtain $\overrightarrow{OM} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ **B1**

Use a correct method to find $\overrightarrow{MN}$ **M1**

e.g. $\overrightarrow{MO} + \overrightarrow{OA} + \overrightarrow{AN}$ or $\overrightarrow{MO} + \overrightarrow{ON}$

Obtain $\overrightarrow{MN} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$ **A1**

Accept any notation.

Total: 3

(b) Use a correct method to form an equation for $\overrightarrow{MN}$ **M1**

Allow without $r = \ldots$

Obtain $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} + \lambda(\mathbf{i} + \mathbf{j} - 2\mathbf{k})$ **A1 FT**

OE e.g. $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k} + \mu(\mathbf{i} + \mathbf{j} - 2\mathbf{k})$

Must have $\mathbf{r} = \ldots$ Follow *their* answers to part 9(a).

Total: 2

(c) State $\overrightarrow{OP}$ for a general point P on MN in component form, e.g. $(2 + \lambda, 3 + \lambda, -2\lambda)$ **B1**

Equate scalar product of $\overrightarrow{OP}$ and a direction vector for MN to zero and solve for λ **M1**

Obtain $\lambda = -\frac{5}{6}$ **A1**

OE e.g. $\mu = \frac{1}{6}$

Obtain $\sqrt{\frac{53}{6}}$ correctly **A1**

AG e.g. from $\sqrt{\left(\frac{7}{6}\right)^2 + \left(\frac{13}{6}\right)^2 + \left(\frac{5}{3}\right)^2}$

Total: 4

Coaching With Shuja

Question 1

May/June 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "( ) The most common error was an incorrect initial statement such asln e2x +3 =lne2x +ln3."

Source: 9709_s22_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

1 Solve the equation $\ln(e^{2x} + 3) = 2x + \ln 3$, giving your answer correct to 3 decimal places. **[4]**

Coaching With Shuja

Step	Description	Mark	Notes
	Use law of the logarithm of a product, power or quotient or a law of indices (on an expression that is relevant to the question)	**M1**	e.g. $\ln(e^{2x}+3)-\ln 3=\ln\left(\frac{e^{2x}+3}{3}\right)$ or $e^{(2x+\ln 3)}=e^{2x}e^{\ln 3}$
	State a correct equation without logs (in any form)	**A1**	e.g. $3+e^{2x}=3e^{2x}$
	Carry out correct method to solve an equation of the form $e^{2x}=a$, where $a>0$, or for solving $e^{x}=b$ ($b>0$) if they have already taken the square root	**M1**	Allow for $x=\frac{1}{2}\ln\frac{1}{2}$. **M1** can be implied by correct answer.
	Obtain answer $x=0.203$	**A1**	CAO. The question requires 3 d.p. Answer only with no working shown is 0/4.
		4	

Total: 4

Coaching With Shuja

Question 3

May/June 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “A common mistake was to substitute x =3 into the polynomial instead of x =−2 (from 5−2=3).”

Source: 9709_s22_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 32), Question 3

Coaching With Shuja

3 The polynomial $ax^3 + x^2 + bx + 3$ is denoted by p(x). It is given that p(x) is divisible by $(2x - 1)$ and that when p(x) is divided by $(x + 2)$ the remainder is 5.

Find the values of a and b. **[5]**

Coaching With Shuja

Step	Description	Mark	Notes
	Substitute $x = \frac{1}{2}$, equate result to zero	**M1**	Or divide by $2x - 1$ and equate constant remainder to zero.
	Obtain a correct simplified equation	**A1**	e.g. $\frac{1}{8}a + \frac{1}{4} + \frac{1}{2}b + 3 = 0$ or $a + 4b = -26$
	Substitute $x = -2$, equate result to 5	**M1**	Or divide by $x + 2$ and equate constant remainder to 5.
	Obtain a correct simplified equation	**A1**	e.g. $-8a + 4 - 2b + 3 = 5$ or $8a + 2b = 2$
	Obtain $a = 2$ and $b = -7$	**A1**	WWW
		5	

Total: 5

Coaching With Shuja

Question 4

May/June 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 4 The differentiation in this question proved to be quite challenging."

Source: 9709_s22_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

4 The equation of a curve is $y = \cos^3 x\sqrt{\sin x}$. It is given that the curve has one stationary point in the interval $0 < x < \frac{1}{2}\pi$.

Find the x-coordinate of this stationary point, giving your answer correct to 3 significant figures. **[6]**

Coaching With Shuja

Step	Description	Mark	Notes
	Use the correct product rule and then the chain rule to differentiate either $\cos^3 x$ or $\sqrt{\sin x}$	**M1**	e.g. two terms with one part of $\frac{dy}{dx} = p\cos^2 x \sin x\sqrt{\sin x} + q\frac{\cos^3 x \cos x}{\sqrt{\sin x}}$.
	Obtain correct derivative in any form e.g. $\frac{dy}{dx} = -3\cos^2 x \sin x\sqrt{\sin x} + \frac{\cos^3 x \cos x}{2\sqrt{\sin x}}$	**A1 A1**	**A1** for each correct term substituted in the complete derivative.
	Equate their derivative to zero and obtain a horizontal equation with positive integer powers of $\sin x$ and/or $\cos x$ from an equation including $\sqrt{\sin x}$ or $\frac{1}{\sqrt{\sin x}}$ using sensible algebra.	**M1**	e.g. $-3\cos^2 x \sin^2 x + \frac{1}{2}\cos^4 x = 0$
	Use correct formula(s) to express their equation/derivative in terms of one trigonometric function	**M1**	Can be awarded before the previous **M1**. May involve more than one trigonometric term.
	Obtain $7\cos^2 x = 6$, $7\sin^2 x = 1$, or $6\tan^2 x = 1$, or equivalent, and obtain answer $x = 0.388$	**A1**	CAO. The question asks for 3 sf. Ignore additional answers outside $\left(0, \frac{\pi}{2}\right)$. $22.2°$ is A0.
		6	

Total: 6

Coaching with Shuja

Question 7

May/June 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A significant minority of candidates was not familiar with implicit differentiation and so could make no progress."

> "The most common error to overlook the factor of x in the numerator, or to reject it as not relevant."

Source: 9709_s22_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

7 The equation of a curve is $x^3 + 3x^2y - y^3 = 3$.

(a) Show that $\frac{dy}{dx} = \frac{x^2+2xy}{y^2-x^2}$. **[4]**

(b) Find the coordinates of the points on the curve where the tangent is parallel to the x-axis. **[5]**

Part	Step	Description	Mark	Notes
7(a)		State or imply $6xy + 3x^2\frac{dy}{dx}$ as derivative of $3x^2y$	**B1**	Allow **B1 B1** for $(3x^2dx + 6xydy + 3x^2dy - 3y^2dy = [0])$
		State or imply $3y^2\frac{dy}{dx}$ as derivative of y^3	**B1**	
		Equate attempted derivative of left-hand side to zero and solve to obtain an equation with $\frac{dy}{dx}$ as subject	**M1**	Allow if zero implied by subsequent working. Allow if recover from an extra $\frac{dy}{dx} = \ldots$ at the beginning of the left-hand side.
		Obtain $\frac{dy}{dx} = \frac{x^2+2xy}{y^2-x^2}$ correctly	**A1**	AG Accept y' for $\frac{dy}{dx}$.
			4	
7(b)		Equate numerator to zero	*M1**	Must be using the given derivative.
		Obtain $x = -2y$, or equivalent	**A1**	An equation with x or y as the subject SOI.
		Use $x^3 + 3x^2y - y^3 = 3$ to obtain an equation in x or y	**DM1**	$-8y^3 + 12y^3 - y^3 = 3$ or $x^3 - \frac{3}{8}x^3 + \frac{1}{8}x^3 = 3$ or any equivalent form (do not need to evaluate powers).
		Obtain the point $(-2, 1)$ and no others from solving their cubic equation	**A1**	Allow if each component stated separately. ISW.
		State the point $(0, -\sqrt{3})$, or equivalent from correct work	**B1**	Accept $(0, -\sqrt{3})$, or $(0, -1.44)$ (-1.44225). Allow if each component stated separately. ISW.
			5	

Total: 9

Question 9

May/June 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 9 (a) Although quite simple, this task was an unfamiliar one and proved challenging to many candidates."

> "The most common errors were in the arithmetic and algebra."

Source: 9709_s22_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

9 The lines l and m have vector equations

$$\mathbf{r} = -\mathbf{i} + 3\mathbf{j} + 4\mathbf{k} + \lambda(2\mathbf{i} - \mathbf{j} - \mathbf{k}) \quad \text{and} \quad \mathbf{r} = 5\mathbf{i} + 4\mathbf{j} + 3\mathbf{k} + \mu(a\mathbf{i} + b\mathbf{j} + \mathbf{k})$$

respectively, where a and b are constants.

(a) Given that l and m intersect, show that $2b - a = 4$. **[4]**

(b) Given also that l and m are perpendicular, find the values of a and b. **[4]**

(c) When a and b have these values, find the position vector of the point of intersection of l and m. **[2]**

Coaching With Shuja

Part	Step	Description	Mark	Notes
9(a)		Express general point of l or m in component form, i.e. $(-1+2\lambda, 3-\lambda, 4-\lambda)$ or $(5+a\mu, 4+b\mu, 3+\mu)$	B1	
		Equate components and eliminate either λ or μ	M1	e.g. $\mu = \frac{-\lambda}{1-b}$, $\lambda = \frac{-1-b}{1-b}$, $\mu = \frac{a+6}{a+2}$
		Eliminate the other parameter or obtain a second expression in the first parameters.	M1	λ and μ are not required to be the subject of the equations.
		Show intermediate steps to obtain $2b-a=4$	A1	AG
			4	
	Alternative method for question 9(a)			
		Express general point of l or m in component form, i.e. $(-1+2\lambda, 3-\lambda, 4-\lambda)$ or $(5+a\mu, 4+b\mu, 3+\mu)$	B1	
		Express a or b in terms of λ and μ	M1	$a = \frac{2\lambda-6}{\mu}$, $b = \frac{-1-\lambda}{\mu}$, $\mu = \frac{a+6}{a+2}$
		Use $\lambda = 1-\mu$	M1	
		Obtain $2b-a=4$	A1	AG
			4	
9(b)		Using the correct process equate the scalar product of the direction vectors to zero	*M1	$(2\mathbf{i}-\mathbf{j}-\mathbf{k})\cdot(a\mathbf{i}+b\mathbf{j}+\mathbf{k})=0$ SOI.
		Obtain $2a-b-1=0$	A1	OE e.g. $2(2b-4)-b-1=0$
		Solve simultaneous equations for a or for b	DM1	
		Obtain $a=2, b=3$	A1	
			4	
9(c)				
			2	

Total: 10

Question 2

May/June 2022 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "However, a common error was to show tan θ expressed as 6– 3 instead of its reciprocal, cosθ either through a simple algebraic error or mistakenly thinking that tanθ was ."

Source: 9709_s22_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 33), Question 2

Coaching With Shuja

2 Solve the equation $\cos(\theta - 60^\circ) = 3\sin\theta$, for $0^\circ \leq \theta \leq 360^\circ$. **[5]**

Coaching With Shuja

Step	Description	Mark	Notes
	Use correct $\cos(A-B)$ formula to obtain an equation in $\cos\theta$ and $\sin\theta$	**B1**	$\cos\theta\cos 60^\circ + \sin\theta\sin 60^\circ = 3\sin\theta$
	Use trigonometric formula and substitute values for $\cos 60$ and $\sin 60$ to obtain an equation in $\tan\theta$ (or $\cos\theta$ or $\sin\theta$)	**M1**	Allow $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$ interchanged. $\frac{1}{2} + \frac{\sqrt{3}}{2}\tan\theta = 3\tan\theta$ or $\frac{1}{4}\cos^2\theta = \left[3-\frac{\sqrt{3}}{2}\right]\left[3-\frac{\sqrt{3}}{2}\right](1-\cos^2\theta)$ or $\frac{1}{4}(1-\sin^2\theta) = \left[3-\frac{\sqrt{3}}{2}\right]\left[3-\frac{\sqrt{3}}{2}\right]\sin^2\theta$
	Obtain $\tan\theta = \frac{1}{6-\sqrt{3}}$ or $\tan\theta = \frac{6+\sqrt{3}}{33}$ or 0.2343, $\cos\theta = \frac{3\sqrt{3}}{2\sqrt{10-3\sqrt{3}}}$ or 0.9736 or $\sin\theta = \frac{1}{2\sqrt{10-3\sqrt{3}}}$ or 0.2281	**A1**	OE
	Obtain answer, e.g. $\theta = 13.2^\circ$	**A1**	May be more accurate, allow value rounding to 13.2°. $\theta = 13.1867^\circ$.
	Obtain second answer, e.g. $\theta = 193.2^\circ$ and no others in the given interval	**A1 FT**	May be more accurate. Allow value rounding to 193.2°. **FT** is on previous value of θ, must have scored **M1**. Note if θ is negative (e.g. -13.2): $-13.2 + 180 = 166.8$ A0 but $-13.2 + 360 = 346.8$ A1 FT. Ignore answers outside the given interval. Treat answers in radians as a misread. $0.23015, 3.3717$.
		5	

Total: 5

Question 3

May/June 2022 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "© 2022 Cambridge International Advanced Level 9709 Mathematics June 2022 Principal Examiner Report for Teachers (b) A considerable number of candidates found this question difficult."

Source: 9709_s22_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 33), Question 3

Question Paper May/June 2022 • Paper 3 (variant 33) • Q3

(a) Show that the equation $\log_3(2x+1) = 1 + 2\log_3(x-1)$ can be written as a quadratic equation in x. **[3]**

(b) Hence solve the equation $\log_3(4y+1) = 1 + 2\log_3(2y-1)$, giving your answer correct to 2 decimal places. **[2]**

Coaching With Shuja

Part	Step	Description	Mark	Notes
3(a)		Use law of logarithm of a power	**M1**	$\log_3(2x+1) = 1 + \log_3(x-1)^2$
		Use $\log_3 3 = 1$	**B1**	$\log_3(2x+1) = \log_3 3 + \log_3(x-1)$
				SC For candidates scoring M0 B0 due to combining logs before dealing with coefficient 2, and confusing coefficients, allow $\log_3(\ldots) = c$ leading to $(\ldots) = 3^c$ **B1**.
		Obtain $3x^2 - 8x + 2 = 0$ or $1.5x^2 - 4x + 1 = 0$	**A1**	OE 3 terms only and $= 0$ required.
			3	
3(b)		Solve 3-term quadratic equation from part **(a)** or restart to find y	**M1**	$\frac{4 \pm \sqrt{10}}{6}$ or $y = 1.1937\ldots$ or $y = 0.1396\ldots$ $(x = 2.3874$ or $x = 0.2792)$ May solve for x but must find $y = \frac{x}{2}$ to gain **M1**.
		Obtain answer 1.19	**A1**	CAO. 2 dp required.
			2	

Total: 5

Coaching With Shuja

Question 5

May/June 2022 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 5 (a) Most candidates correctly evaluated u* and u* – u, but some did not show any scale on their Argand diagram when plotting the positions of points A, B and C."

Source: 9709_s22_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 33), Question 5

Coaching With Shuja

5 The complex number $3 - i$ is denoted by u.

(a) Show, on an Argand diagram with origin O, the points A, B and C representing the complex numbers u, u^* and $u^* - u$ respectively.

State the type of quadrilateral formed by the points O, A, B and C. **[3]**

(b) Express $\frac{u^*}{u}$ in the form $x + iy$, where x and y are real. **[3]**

(c) By considering the argument of $\frac{u^*}{u}$, or otherwise, prove that $\tan^{-1}\left(\frac{3}{4}\right) = 2\tan^{-1}\left(\frac{1}{3}\right)$. **[2]**

Coaching With Shuja

Part	Step	Description	Mark	Notes
5(a)		Show u and u^* in relatively correct positions. Must have sense of scale on axes	**B1**	$u = 3 - i$, $u^* = 3 + i$ Ignore labels.
		Show $u^* - u$ in a relatively correct position. Must have sense of scale on axes	**B1**	$2i$. Scale only on Imaginary axis is sufficient for this mark.
		State that $OABC$ is a parallelogram [independent of previous marks]	**B1**	Ignore 'quadrilateral'. Allow 'trapezium' from correct work.
			3	
5(b)		Multiply their numerator and the given denominator by $3 + i$ and attempt to evaluate either	**M1**	Can have missing term and arithmetic errors but need $i^2 = -1$ once, seen or implied.
		Obtain numerator $8 + 6i$ or denominator 10	**A1**	
		State final answer $\frac{4}{5} + \frac{3}{5}i$ or $0.8 + 0.6i$	**A1**	Correct answer with no working scores 0/3.
			3	
5(c)		State or imply $\arg\frac{u^*}{u} = \arg u^* - \arg u$ or $2\arg u$ or $2\arg u^*$	**M1**	
		Justify the given statement correctly	**A1**	AG $\arg\frac{u^*}{u} = \tan^{-1}\frac{3}{4}$, $\arg u^* = \tan^{-1}\frac{1}{3}$ and $\arg u = \tan^{-1}\left(-\frac{1}{3}\right)$ (or $\arg u = -\tan^{-1}\frac{1}{3}$), needed if use first expression in **M1**; or $\arg\frac{u^*}{u} = 2\tan^{-1}\frac{3}{4}$ $\arg u^* = \tan^{-1}\frac{1}{3}$, needed if use second expression in **M1**.
			2	

Total: 8

Question 8

May/June 2022 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "A common error led to 0.02 in the numerator rather than the denominator."

> "(c) Again, candidates found the algebra difficult and struggled to convert to an expression for N in terms of t with their value of k substituted."

Source: 9709_s22_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 8

Question Paper May/June 2022 • Paper 3 (variant 33) • Q8

At time t days after the start of observations, the number of insects in a population is N. The variation in the number of insects is modelled by a differential equation of the form $\frac{dN}{dt} = kN^{\frac{3}{2}} \cos 0.02t$, where k is a constant and N is a continuous variable. It is given that when $t = 0$, $N = 100$.

(a) Solve the differential equation, obtaining a relation between N, k and t. **[5]**

(b) Given also that $N = 625$ when $t = 50$, find the value of k. **[2]**

(c) Obtain an expression for N in terms of t, and find the greatest value of N predicted by this model. **[2]**

Coaching With Shuja

8(a) Separate variables correctly. **B1**

$$\frac{dN}{N^{3/2}} = (k\cos 0.02t)dt$$

Allow without integral signs. **OE**

Obtain term $-\frac{2}{\sqrt{N}}$. **B1**

Obtain term $50\sin 0.02t$. **B1**

Use $t = 0, N = 100$ to evaluate a constant, or as limits, in a solution containing terms $\frac{a}{\sqrt{N}}$ and $b\sin 0.02t$, where $ab \neq 0$. **M1**

Obtain correct solution in any form, e.g. $-\frac{2}{\sqrt{N}} = 50k\sin 0.02t - 0.2$. **A1** ISW

e.g. $N = \frac{1}{(25k\sin 0.02t - 0.1)^2}$, $-2N^{-1/2} = \frac{k}{0.02}\sin 0.02t - \frac{1}{5}$, $\frac{2}{\sqrt{N}} = \frac{k}{50\sin 0.02t + \frac{1}{10}}$,
$50k\sin(\frac{1}{50}t) = -\frac{2}{\sqrt{N}} + \frac{1}{5}$.

Total: 5

8(b) Use the substitution $N = 625$ and $t = 50$ in expression of appropriate form to evaluate k. **M1**

Expression must contain $a + bk\sin 0.02t$, $(\sqrt{N})^{\pm n}$, where $n = -1, 1, 3$ or 5 and a and b are constants $ab \neq 0$ or $(a + bk\sin 0.02t)^{\pm n}$ and $(N)^{\pm n}$. Allow with k replaced by $\frac{1}{k}$. Error due to $k(N^{-3/2})$ when separating variables in **8(a)**. If invert term by term when 3 terms shown then M0.

Obtain $k = 0.00285[2148]$. **A1**

Total: 2

8(c) Rearrange and obtain $N = 4(0.2 - 0.142(607)\sin 0.02t)^{-2}$. **M1**
Substitution for k required.
Anything of the form $N = c(d - ek\sin 0.02t)^{-2}$, where c, d and ek are constants $cde \neq 0$ and value of k substituted. Allow with k replaced by $1/k$. Error due to $k(N^{-3/2})$ when separating variables in **8(a).**

OE ISW

e.g.

$$N = \left(-\frac{10}{0.7125\sin 0.02t - 1}\right)^2, \quad N = \frac{1}{(-0.0713\sin 0.02t + 0.1)^2}$$

$$N = \frac{100}{\left(\left(\frac{0.6}{\sin 1}\right)\sin 0.02t - 1\right)^2}, \quad N = \frac{1}{\left(-\frac{3}{-50\sin 1} \times \sin 0.02t + \frac{1}{10}\right)^2}$$

$$N = \left(-\frac{0.06}{\sin 1}\sin 0.02t + 0.1\right)^{-2}, \quad N = \left(\frac{800}{80 - 57\sin 0.02t}\right)^2$$

Do not need to substitute for $\sin(0.02t) = 1$, but must substitute for k.

Accept answers between 1209 and 1215. **A1** ISW

Substitute $\sin 0.02t = 1$ or $t = 50\sin^{-1} 1$ or 78.5 or 25π. Answer with no working (rubric) 0/2.

SC $N = \ldots$ not seen but correct numerical answer **B1** 1/2.

Total: 2

Question 9

May/June 2022 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "This would have avoided the very common error of setting to zero the scalar product of OP and the direction vector of l."

Source: 9709_s22_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

Coaching With Shuja

Question Paper May/June 2022 • Paper 3 (variant 33) • Q9

With respect to the origin O, the point A has position vector given by $\overrightarrow{OA} = \mathbf{i} + 5\mathbf{j} + 6\mathbf{k}$. The line l has vector equation $\mathbf{r} = 4\mathbf{i} + \mathbf{k} + \lambda(-\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$.

(a) Find in degrees the acute angle between the directions of $\overrightarrow{OA}$ and l. **[3]**

(b) Find the position vector of the foot of the perpendicular from A to l. **[4]**

(c) Hence find the position vector of the reflection of A in l. **[2]**

Coaching With Shuja

Mark Scheme May/June 2022 • Paper 3 (variant 33) • Q9

9(a) Using the correct process find the scalar product of direction vectors of l and $\overrightarrow{OA}$. **M1**

$(1, 5, 6) \cdot (-1, 2, 3) = -1.1 + 5.2 + 6.3 = -1 + 10 + 18$

Using the correct process for the moduli, divide the scalar product by the product of the moduli and find the inverse cosine of the result. **M1**

Their scalar product $\div \sqrt{(1^2 + 5^2 + 6^2)}\sqrt{(-1)^2 + 2^2 + 3^2}$.

Angle $= \cos^{-1} \frac{27}{\sqrt{62}\sqrt{14}}$

Obtain answer $23.6°$. **A1** AWRT $23.6°$.

$23.5889°$. Radians 0.412 scores Ao ($0.4117...$).

Total: 3

9(b) Taking a general point P on l, state $\overrightarrow{AP}$ (or $\overrightarrow{PA}$) in component form, e.g. $(3 - \lambda, -5 + 2\lambda, -5 + 3\lambda)$. **B1**

Note: $(4, 1, 0)$ or $(4, 1, 1)$, for $4\mathbf{i} + \mathbf{k}$ is not MR, but **M1** possible.

Either equate scalar product of $\overrightarrow{AP}$ and direction vector of l to zero and solve for λ or use Pythagoras in a relevant triangle and solve for λ. **M1**

$(3 - \lambda, -5 + 2\lambda, -5 + 3\lambda) \cdot (-1, 2, 3) = 0$

$-3 - 10 - 15 + \lambda + 4\lambda + 9\lambda = 0$

or let $\overrightarrow{OQ} = (4, 0, 1)$ so $\overrightarrow{AQ} = (3, -5, -5)$,
$\overrightarrow{QP} = (-\lambda, 2\lambda, 3\lambda)$, $\overrightarrow{AP} = (3 - \lambda, -5 + 2\lambda, -5 + 3\lambda)$ hence
$3^2 + (-5)^2 + (-5)^2 = (3 - \lambda)^2 + (-5 + 2\lambda)^2 + (-5 + 3\lambda)^2 + (-\lambda)^2 + (2\lambda)^2 + (3\lambda)^2$

Obtain $\lambda = 2$. **A1**

State that the position vector $\overrightarrow{OP}^*$ of the foot is $2\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}$. **A1** OE

Condone coordinates.

Total: 4

9(c) Set up a correct method for finding the position vector of the reflection of A in l. **M1**

For all methods, allow a sign error in one component only:
$\overrightarrow{OA}' = \overrightarrow{OP}^* + (\overrightarrow{OP}^* - \overrightarrow{OA})$

their $(2, 4, 7) + ({}_t heir\ 2, 4, 7 - 1, 5, 6)$

or $\overrightarrow{OA}' = \overrightarrow{OP}^* - (\overrightarrow{OA} - \overrightarrow{OP}^*)$

their $(2, 4, 7) - (1, 5, 6 -_t heir\ 2, 4, 7)$

or $\overrightarrow{OA}' = \overrightarrow{OA} + 2(\overrightarrow{OP}^* - \overrightarrow{OA})$

$(1 + 2({}_t heir 2 - 1))$

$5 + 2({}_t heir 4 - 5)$

$(6 + 2({}_t heir 7 - 6))$

or midpoint $\overrightarrow{OP}^* = (\overrightarrow{OA} + \overrightarrow{OA}')/2$ with _their_ λ value substituted.

$1 + x$

$5 + y$

$6 + z = 2 \times_t heir\ 2, 4, 7$

Obtain answer $3\mathbf{i} + 3\mathbf{j} + 8\mathbf{k}$ or $3\left(\mathbf{i} + \mathbf{j} + \frac{8}{3}\right)$. **A1** OE

Condone coordinates.

$x = 3, y = 3, z = 8$ A1.

No method shown and correct answer 2/2.

Total: 2

Question 2

Oct/Nov 2022 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "The most common errors related to integration, with incorrect powers often seen and many candidates forgetting to use the reverse chain rule to divide by 1â„2, the coefficient of x, as well as by the new power, –3."

Source: 9709_w22_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 11), Question 2

Coaching With Shuja

The equation of a curve is such that $\frac{dy}{dx} = 12\left(\frac{1}{2}x - 1\right)^{-4}$. It is given that the curve passes through the point $P(6, 4)$.

(a) Find the equation of the tangent to the curve at P. **[2]**

(b) Find the equation of the curve. **[4]**

2(a) Substitute $x = 6$ into $\frac{dy}{dx}$. **M1**

$$12\left(\frac{1}{2} \times 6 - 1\right)^{-4} = 12(2)^{-4} = \frac{3}{4}$$

SOI by gradient $\frac{3}{4}$ used.

OE e.g. $y = \frac{3}{4}x - \frac{1}{2}$ or evaluates c in $y = \frac{3}{4}x + c$ using $(6, 4)$ and gradient $\frac{3}{4}$. **A1** ISW

Total: 2

2(b) Integrate: **B1** (Marks shown are for general marking - refer to the specific question.)

Let $u = \frac{1}{2}x - 1$ then $\frac{du}{dx} = \frac{1}{2}$, so $\int 12u^{-4}du = 12 \times \frac{u^{-3}}{-3} = -4u^{-3} = -4\left(\frac{1}{2}x - 1\right)^{-3}$.

Or: $\int 12\left(\frac{1}{2}x - 1\right)^{-4} dx = \frac{12}{1/2} \times \frac{\left(\frac{1}{2}x - 1\right)^{-3}}{-3} = -8\left(\frac{1}{2}x - 1\right)^{-3}$

Use $(6, 4)$ to find constant: $4 = -4(2)^{-3} + c \Rightarrow c = \frac{9}{2}$.

Obtain $y = -4\left(\frac{1}{2}x - 1\right)^{-3} + \frac{9}{2}$ or equivalent. **A1** ISW

Coaching With Shuja

Question 3

Oct/Nov 2022 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "A number of candidates made no progress with solving the equation while some attempted to substitute their a-value into dy instead of into y."

Source: 9709_w22_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 3

Coaching With Shuja

Question Paper Oct/Nov 2022 • Paper 1 (variant 11) • Q3

A curve has equation $y = ax^{1/2} - 2x$, where $x > 0$ and a is a constant. The curve has a stationary point at the point P, which has x-coordinate 9.

Find the y-coordinate of P. **[5]**

Coaching With Shuja

$$\frac{dy}{dx} = \frac{1}{2}ax^{-1/2} - 2$$

B2, 1, 0

Substitute $x = 9$ and $\frac{dy}{dx} = 0$ into their derivative and solve a linear equation for a. **M1**

$$0 = \frac{1}{2}a(9)^{1/2} - 2 \Rightarrow \frac{a}{6} - 2 = 0 \Rightarrow a = [12]$$

A1

$$[y = \text{their } a \times (9)^{1/2} - 18 = [18]]$$

A1 FT on their a.

Total: 5

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Question 4

Oct/Nov 2022 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> “The most common error at this stage was to omit to square .”

Source: 9709_w22_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 4

Coaching With Shuja

The coefficient of x^2 in the expansion of $\left(1+\frac{2}{p}x\right)^5+(1+px)^6$ is 70.

Find the possible values of the constant p. **[6]**

Coaching With Shuja

Coefficient of x^2 in $\left(1+\frac{2}{p}x\right)^5$ is $10\left(\frac{2}{p}\right)^2 = \frac{10\times 2^2}{p^2} = \left[\frac{40}{p^2}\right]$. **B1**

Accept with x^2 present. Must evaluate ${}^5C_2$.

Coefficient of x^2 in $(1+px)^6$ is $15(p)^2 = [15p^2]$. **B1**

Accept with x^2 present. Must evaluate ${}^6C_2$.

$$\frac{40}{p^2} + 15p^2 = 70$$

***M1** Forming an equation in p with their coefficients, the given 70, no x terms and no extra terms.

$$15p^4 - 70p^2 + 40 = [0] \quad \text{or} \quad 3p^4 - 14p^2 + 8 = [0]$$

DM1 Forming a 3-term equation in p (or another variable) with all terms on one side and their coefficients.

$$[5](p^2-4)(3p^2-2) = [0] \quad \text{or} \quad \frac{70 \pm \sqrt{70^2 - 4(15)(40)}}{30}$$

or

$$\frac{14 \pm \sqrt{14^2 - 4(3)(8)}}{6}$$

DM1 Attempt to solve 3-term quartic (or quadratic in another variable) by factorisation, formula or completing the square.

$$p = \pm 2, \pm\sqrt{\frac{2}{3}}$$

A1 OE e.g. $\pm\frac{\sqrt{6}}{3}$ or AWRT ± 0.816

If ***M1 DM1 DM0**, allow SC **B1** for 4 correct values.

Total: 6

Question 10

Oct/Nov 2022 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

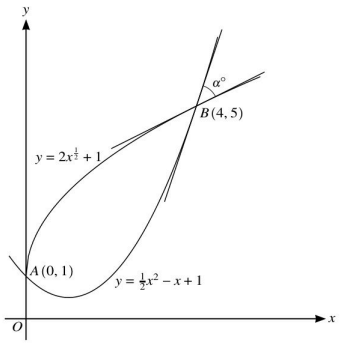
> “Common errors involved powers or fractions (e.g.”

> “2 (b) This problem-solving question proved to be very challenging for all but the most able candidates.”

Source: 9709_w22_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 11), Question 10

Coaching With Shuja

10



Curves with equations $y = 2x^{\frac{1}{2}} + 1$ and $y = \frac{1}{2}x^2 - x + 1$ intersect at $A\,(0,\,1)$ and $B\,(4,\,5)$, as shown in the diagram.

(a) Find the area of the region between the two curves. [5]

Curves with equations $y = 2x^{1/2} + 1$ and $y = \frac{1}{2}x^2 - x + 1$ intersect at $A(0, 1)$ and $B(4, 5)$, as shown in the diagram.

(a) Find the area of the region between the two curves. **[5]**

(b) Find α. **[5]**

10(a)

$$\pm\int[(2x^{1/2}+1)-(\frac{1}{2}x^2-x+1)]dx=\pm\int[2x^{1/2}-\frac{1}{2}x^2+xdx]$$

***M1**

$$\pm\left(\frac{4x^{3/2}}{3}+x-\frac{x^3}{6}-\frac{x^2}{2}\right)\quad\text{or}\quad\pm\left(\frac{4x^{3/2}}{3}-\frac{x^3}{6}+\frac{x^2}{2}\right)$$

B2, 1, 0 OE Coefficients may be unsimplified.

$$\pm\left(\frac{32}{3}-\frac{32}{3}+8\right)\quad\text{or}\quad\pm\left(\frac{44}{3}-0-\frac{20}{3}+0\right)$$

DM1 $\pm(F(4)-F(0))$ using their integral(s).

$=8$ **A1** Depends on all previous marks.
If ***M1 B2 DM0** and limits stated, **SC B1** for $+8$.

Total: 5

10(b) Upper curve: $\frac{dy}{dx}=x^{-1/2}$. Lower curve: $\frac{dy}{dx}=x-1$. **M1 A1** Attempt at differentiating one function.
A1 if both correct.

At $x=4$: gradient of upper curve $=\frac{1}{2}$, gradient of lower curve $=3$. **M1** Evaluate two gradients using $x=4$.

$$\alpha=\tan^{-1}3-\tan^{-1}\frac{1}{2}=[71.57-26.57]$$

M1 Use inverse tan to find angles then subtract.
OR find equations of both tangents then Pythagoras using a point on each e.g. on axes.
OR cosine rule using intercepts or proportion.

$$[\alpha=]45°$$

A1 AWRT

Total: 5

Question 11

Oct/Nov 2022 | Paper 1 (variant 11)

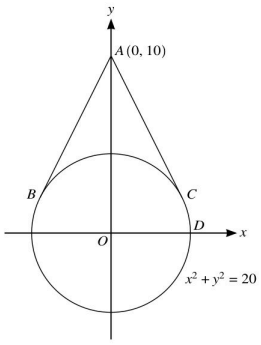
Examiner Report — Why students found this question difficult

> "(c) Almost all candidates found this question to be very challenging."

Source: 9709_w22_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 11), Question 11

Coaching With Shuja

11



The diagram shows the circle with equation $x^2 + y^2 = 20$. Tangents touching the circle at points B and C pass through the point A (0, 10).

(a) By letting the equation of a tangent be $y = mx + 10$, find the two possible values of m. [4]

The diagram shows the circle with equation $\boldsymbol{x^2 + y^2 = 20}$. Tangents touching the circle at points $\boldsymbol{B}$ and $\boldsymbol{C}$ pass through the point $\boldsymbol{A(0, 10)}$.

(a) By letting the equation of a tangent be $\boldsymbol{y = mx + 10}$, find the two possible values of $\boldsymbol{m}$. **[4]**

(b) Find the coordinates of $\boldsymbol{B}$ and $\boldsymbol{C}$. **[3]**

(c) Find angle $\boldsymbol{BDC}$ in degrees. **[3]**

Mark Scheme Oct/Nov 2022 • Paper 1 (variant 11) • Q11

11(a)

$$x^2 + (mx + 10)^2 = 20 \quad \text{or} \quad y^2 + \left(\frac{y - 10}{m}\right)^2 = 20 \quad \text{or} \quad mx + 10 = \sqrt{20 - x^2}$$

*M1 Substitute equation of line into equation of circle.

$$x^2(1 + m^2) + 20mx + 80 = [0] \quad \text{or}$$

$$y^2(m^2 + 1) - 20y + (100 - 20m^2) = [0]$$

A1 Collect terms into a 3 term quadratic.

$$(20m)^2 - 4(1 + m^2) \times 80 = 0 \Rightarrow 80m^2 - 320 = 0 \Rightarrow [80][m^2 - 4] = [0]$$

DM1 Use $b^2 - 4ac = [0]$.

$$m = \pm 2$$

A1 Two values for m.

Total: 4

11(b) Method 1: Use of quadratic

$$(1 + 2^2)x^2 + 20(2)x + 80 = 0 \Rightarrow 5x^2 \pm 40x + 80 = 0$$

$$[5](x \pm 4)^2 = 0 \Rightarrow x = \pm 4 \quad \text{or} \quad y = 2$$

A1 Correct solutions or one correct pair (x, y).

$$(-4, 2), (4, 2)$$

A1 Two correct points with x and y paired correctly.

Method 2: Using equation of normal

$$2x + 10 = -\frac{1}{2}x \quad \text{or} \quad -2x + 10 = \frac{1}{2}x$$

M1 Equate tangent and normal and solve for x.

$$x = \pm 4$$

A1 Two correct x-values or one correct pair (x, y).

$$(-4, 2), (4, 2)$$

A1 Two correct points with x and y paired correctly.

Total: 3

11(c) Method 1: Using angle at circumference

$$\cos BOA = \frac{\sqrt{20}}{10} \quad \text{or} \quad \sin BOA = \frac{\sqrt{80}}{10} \quad \text{or} \quad \tan BOA = \frac{\sqrt{80}}{\sqrt{20}} [= 2]$$

$$BOA = 63.4° \Rightarrow BOC = 126.8° \text{or} 126.9°$$

*M1 Use a trig function in triangle AOB.

$$BDC = \frac{BOC}{2} \quad \text{or} \quad BDC = \frac{126.8°}{2}$$

DM1 Strategy involving doubling.

$$[BDC =] 63.4°$$

A1 AWRT

Method 2: Using cosine rule

$$BC = 8, BD = \sqrt{(\sqrt{20} + 4)^2 + 2^2}, CD = \sqrt{(\sqrt{20} - 4)^2 + 2^2}$$

*M1 Calculate two lengths in triangle BCD.

$$64 = 80 - 16\sqrt{5} \cos BDC$$

DM1 Use cosine rule with their lengths.

$$\cos BDC = \frac{\sqrt{5}}{5} \Rightarrow [BDC =]63.4°$$

A1 AWRT

Method 3: Subtract angles from **90°**

Calculate one angle at $D = [13.28]$

***M1** ODB or angle between CD and the vertical from D.

Calculate a second angle at $D = [13.28]$ and subtract both from 90°

DM1

$$[BDC =]63.4°$$

A1 AWRT

Total: 3

Coaching With Shuja

Question 1

Oct/Nov 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Common mistakes were using A or B rather than the midpoint or failing to calculate the negative reciprocal."

Source: 9709_w22_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 1

Coaching With Shuja

Points A and B have coordinates $(5, 2)$ and $(10, -1)$ respectively.

(a) Find the equation of the perpendicular bisector of AB. **[3]**

(b) Find the equation of the circle with centre A which passes through B. **[3]**

Coaching With Shuja

1(a) Mid-point AB is $\left(\frac{10+5}{2}, \frac{2-1}{2}\right) = \left[\left(\frac{15}{2}, \frac{1}{2}\right)\right]$. **B1** Accept unsimplified.

Gradient of $AB = \frac{-1-2}{10-5} = \frac{-3}{5}$. Gradient perpendicular $= \frac{5}{3}$. **M1**

For use of $\frac{\text{Change in } y}{\text{Change in } x}$, condone inconsistent order of x and y, and $m_1 m_2 = -1$.

$$y - \frac{1}{2} = \frac{5}{3}\left(x - \frac{15}{2}\right)$$

A1 OE ISW

Any correct version e.g. $y = \frac{3}{5}x - 12$ or $5x - 3y = 36$.

Total: 3

1(b) [Radius] $= \sqrt{34}$ or 5.8 AWRT or $[\text{radius}]^2 = 34$. **B1**

Sight of $\sqrt{34}$ or 34. Condone confusion of r and r^2.

Sight of $(x-5)^2 + (y-2)^2$. **B1**

$$(x-5)^2 + (y-2)^2 = 34$$

B1 CAO ISW

Alternative method for Question 1(b)

$$x^2 + y^2 - 10x - 4y$$

B1

$[c] = 5$ or $[c] = -5$. **B1**

Substitution of $(10, -1)$ into $x^2 + y^2 - 10x - 4y + c = 0$.

$$x^2 + y^2 - 10x - 4y - 5 = 0$$

B1

Total: 3

Question 3

Oct/Nov 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Common mistakes were only considering positive values of k or deciding that the outside regions were the correct answer or that k = 8."

Source: 9709_w22_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

(a) Find the set of values of k for which the equation $8x^2 + kx + 2 = 0$ has no real roots. **[2]**

(b) Solve the equation $8\cos^2\theta - 10\cos\theta + 2 = 0$ for $0^\circ \leq \theta \leq 180^\circ$. **[3]**

Coaching With Shuja

Mark Scheme Oct/Nov 2022 • Paper 1 (variant 12) • Q3

(a) $k^2 - 4 \times 8 \times 2 < 0$ **M1** Use of $b^2 - 4ac$ but not just in the quadratic formula.

$-8 < k < 8$ or $-8 < k, k < 8$ or $|k| < 8$ or $(-8, 8)$ **A1** Condone '$-8 < k$ or $k < 8$', '$-8 < k$ and $k < 8$' but not $\sqrt{64}$.

Total: 2

(b) [Mark scheme omitted - diagram/graph working shown]

Coaching With Shuja

Question 5

Oct/Nov 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 5 This question was found to be challenging for many candidates."

Source: 9709_w22_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 5

Coaching With Shuja

Question Paper Oct/Nov 2022 • Paper 1 (variant 12) • Q5

The graph with equation $y = f(x)$ is transformed to the graph with equation $y = g(x)$ by a stretch in the x-direction with factor 0.5, followed by a translation of $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

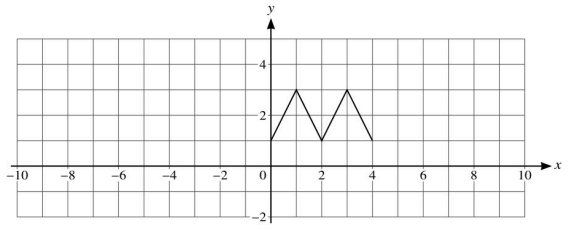
(a) The diagram below shows the graph of $y = f(x)$.

On the diagram sketch the graph of $y = g(x)$. **[3]**

5 The graph with equation $y = f(x)$ is transformed to the graph with equation $y = g(x)$ by a stretch in the x-direction with factor 0.5, followed by a translation of $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

(a) The diagram below shows the graph of $y = f(x)$.

On the diagram sketch the graph of $y = g(x)$. [3]



(b) Find an expression for $g(x)$ in terms of $f(x)$. [2]

(b) Find an expression for $g(x)$ in terms of $f(x)$. **[2]**

Shuja

(a) Three points at the bottom of their transformed graph plotted at $y = 2$ **B1** All 5 points of the graph must be connected.

Bottom three points of Λ at $x = 0, x = 1 \& x = 2$ **B1** Must be this shape.

All correct **B1** Condone extra cycles outside $0 \leq x \leq 2$.

Total: 3 SC: If Bo Bo scored, **B1** available for Λ in one of correct positions or all 5 points correctly plotted and not connected or correctly sized shape in the wrong position.

(b) $[g(x)] = f(2x) + 1$ **B1 B1** Award marks for their final answer as follows: $f(2x)$ **B1**, $+1$ **B1**. Condone $y =$ or $f'(x) =$.

Total: 2

Coaching With Shuja

Question 6

Oct/Nov 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "In part (c) many candidates did not use the information provided by the completed square form from part (a) and sometimes differentiated to find the stationary point."

Source: 9709_w22_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 6

Coaching With Shuja

Question Paper Oct/Nov 2022 • Paper 1 (variant 12) • Q6

The equation of a curve is $y = 4x^2 + 20x + 6$.

(a) Express the equation in the form $y = a(x + b)^2 + c$, where a, b and c are constants. **[3]**

(b) Hence solve the equation $4x^2 + 20x + 6 = 45$. **[3]**

(c) Sketch the graph of $y = 4x^2 + 20x + 6$ showing the coordinates of the stationary point. You are not required to indicate where the curve crosses the x- and y-axes. **[3]**

Coaching With Shuja

Mark Scheme Oct/Nov 2022 • Paper 1 (variant 12) • Q6

(a) $y = 4\left(x + \frac{5}{2}\right)^2 - 19$ **B1** $a = 4$ **B1** $b = \frac{5}{2}$ **OE B1** $c = -19$

There is no requirement for the candidate to list a, b and c. Look at values in their final expression, condone omission of ^{2}, and award marks as follows:

Total: 3

(b) Their$4\left(x + \frac{5}{2}\right)^2 - 19 = 45 \Rightarrow \left(x + \frac{5}{2}\right)^2 = 16$ **M1** Equate their quadratic completed square form from **(a)** to 45 or re-start and use completing the square.

Solve as far as $x =$ **DM1** Any valid method leading to two answers.

$\left[x = -\frac{3}{2}, \frac{13}{2}\right]$ **A1** SC: If M0 or M1 DM0 awarded, **B1** available for correct final answers.

Total: 3

(c) Quadratic curve that is the right way up (must be seen either side of stationary point) **B1** No axes required, ignore any axes even if incorrect.

Stationary point stated using any valid method or correctly labelled on their diagram. **B1 FT** FT their values from **(a)** as long as their expression is of the form $p(qx + r)^2 + s$. Expect $\left(-\frac{5}{2}, -19\right)$.

B1 FT form $p(qx + r)^2 + s$. Condone if stated correctly but plotted incorrectly.

Total: 3

Question 9

Oct/Nov 2022 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 9 Parts (a) and (b) were very well done on this question but candidates struggled with the explanations required in part (c) and part (e) especially."

> "In part (d) many candidates could find the required function although simplifying it proved challenging for many with the 5 sometimes appearing in the numerator rather than the denominator."

> "Part (e) proved to be the most challenging on the whole paper with only the very best candidates able to give a satisfactory explanation."

Source: 9709_w22_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 9

Coaching With Shuja

Functions f and g are defined by

$$f(x) = x + \frac{1}{x} \quad \text{for } x > 0,$$

$$g(x) = ax + 1 \quad \text{for } x \in \mathbb{R},$$

where a is a constant.

(a) Find an expression for $gf(x)$. **[1]**

(b) Given that $gf(2) = 11$, find the value of a. **[2]**

(c) Given that the graph of $y = f(x)$ has a minimum point when $x = 1$, explain whether or not f has an inverse. **[1]**

It is given instead that $a = 5$.

(d) Find and simplify an expression for $g^{-1}f(x)$. **[3]**

(e) Explain why the composite function fg cannot be formed. **[1]**

Coaching with Shuja

(a) $a\left(x+\frac{1}{x}\right)+1$ **B1** ISW

Total: 1

(b) $a\left(2+\frac{1}{2}\right)+1=11$ **M1** Substitute $x=2$ into their expression from **(a)** and equate to 11. This may be done in 2 stages: $f(2)=2.5, g(2.5)=11$.

$[a]=4$ **A1**

Total: 2

(c) No, [because it is] not one-one **B1** Or other suitable explanation that may include one to many or many to one.

Total: 1

(d) $[g^{-1}(x)]=\frac{x-1}{5}$ **B1** Condone use of a instead of 5.

$[g^{-1}f(x)]=\frac{x+\frac{1}{x}-1}{5}$ **M1** Correct combination of their $g^{-1}(x)$ with given $f(x)$ Condone use of a instead of 5.

$$\frac{x^3-x+1}{5x} \text{ or } \frac{1}{5}\left(x+x^{-1}-1\right) \text{ or } \frac{1}{5}\left(x+\frac{1}{x}-1\right) \text{ or } \frac{x+x^{-1}-1}{5} \text{ OE ISW}$$

A1 Must not contain unresolved fractions e.g. $\frac{x+x^{-1}-1}{5}$.

Total: 3

(e) The domain of f does not include the whole of the range of g. Or The range of g does not lie in the domain of f. **B1** Accept an answer that includes an example outside the domain of f, e.g. $g(-1)=-4$ but for f, $x>0$.

Total: 1

Question 10

Oct/Nov 2022 | Paper 1 (variant 12)

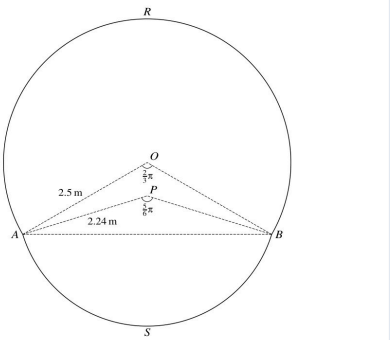
Examiner Report — Why students found this question difficult

> "Part (c) proved challenging with many candidates not realising that the answer from part (b) should be added to the two sector areas."

Source: 9709_w22_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

Coaching With Shuja

10



The diagram shows a cross-section $RASB$ of the body of an aircraft. The cross-section consists of a sector $OARB$ of a circle of radius 2.5 m, with centre O, a sector $PASB$ of another circle of radius 2.24 m with centre P and a quadrilateral $OAPB$. Angle $AOB = \frac{2}{3}\pi$ and angle $APB = \frac{5}{6}\pi$.

(a) Find the perimeter of the cross-section $RASB$, giving your answer correct to 2 decimal places. **[3]**

The diagram shows a cross-section $\boldsymbol{RASB}$ of the body of an aircraft. The cross-section consists of a sector $\boldsymbol{OARB}$ of a circle of radius 2.5 m, with centre $\boldsymbol{O}$, a sector $\boldsymbol{PASB}$ of another circle of radius 2.24 m with centre $\boldsymbol{P}$ and a quadrilateral $\boldsymbol{OAPB}$. Angle $\boldsymbol{AOB = \frac{2\pi}{3}}$ and angle $\boldsymbol{APB = \frac{5\pi}{6}}$.

(a) Find the perimeter of the cross-section $\boldsymbol{RASB}$, giving your answer correct to 2 decimal places. **[3]**

(b) Find the difference in area of the two triangles $\boldsymbol{AOB}$ and $\boldsymbol{APB}$, giving your answer correct to 2 decimal places. **[2]**

(c) Find the area of the cross-section $\boldsymbol{RASB}$, giving your answer correct to 1 decimal place. **[3]**

(a) $2.5 \times \frac{4\pi}{3} + 2.24 \times \frac{5\pi}{6} = [10.47][2] + 5.86[4]$ or $\frac{10\pi}{3} + \frac{28\pi}{15}$ **B1** For either arc correct. Arc ARB could be $AR + RB$.

16.34 or $\frac{26\pi}{5}$ **A1** AWRT Condone 16.33 only.

Total: 3 **M1** For adding two (or three) arc lengths using different radii and angles and nothing else. SOI

(b) Area $AOB = \frac{1}{2} \times 2.5^2 \sin\frac{2\pi}{3} = [2.706]$ **M1** For either $\triangle AOB$ or $\triangle APB$ ($AB = 4.33$, $h = 1.25, (0.58)$) or any other valid method.

Area $APB = \frac{1}{2} \times 2.24^2 \sin\frac{5\pi}{6} = [1.254]$ **A1** AWRT Condone 1.46 only.

[Difference =] 1.45

Total: 2

(c) Area $AOB = \frac{1}{2} \times 2.5^2 \times \frac{4\pi}{3} = [13.09]$ **B1** For either sector area correct

Area $APB = \frac{1}{2} \times 2.24^2 \times \frac{5\pi}{6} = [6.57]$ **M1** Adding two sector areas from different sectors and 'their10(b)' and nothing else. SOI

[Area of cross section =]

$$\frac{1}{2} \times 2.5^2 \times \frac{4\pi}{3} + \frac{1}{2} \times 2.24^2 \times \frac{5\pi}{6} + \text{'their10(b)'} = [13.09 + 6.57 + \text{'their10(b)'}]$$

21.1 **A1** CAO Condone slight inaccuracies in intermediate working if the correct answer is arrived at.

Total: 3

Question 11

Oct/Nov 2022 | Paper 1 (variant 12)

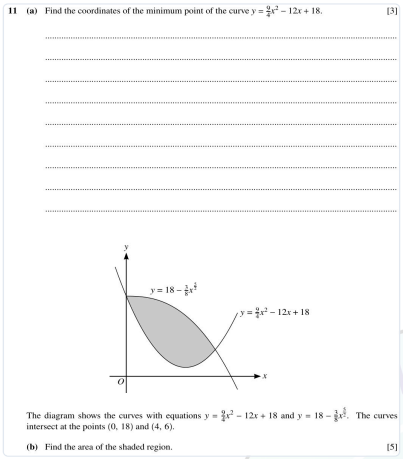
Examiner Report — Why students found this question difficult

> "A few candidates wrote down the correct answer with no working shown and were therefore penalised – see the key messages about calculator usage."

Source: 9709_w22_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 11

Coaching With Shuja

(a) Find the coordinates of the minimum point of the curve $y = \frac{9}{4}x^2 - 12x + 18$. **[3]**


11 (a) Find the coordinates of the minimum point of the curve $y = \frac{9}{4}x^2 - 12x + 18$. [3]

The diagram shows the curves with equations $y = \frac{9}{4}x^2 - 12x + 18$ and $y = 18 - \frac{3}{8}x^{\frac{5}{2}}$. The curves intersect at the points (0, 18) and (4, 6).

(b) Find the area of the shaded region. [5]

The diagram shows the curves with equations $y = \frac{9}{4}x^2 - 12x + 18$ and $y = 18 - \frac{3}{8}x^{5/2}$. The curves intersect at the points $(0, 18)$ and $(4, 6)$.

(b) Find the area of the shaded region. **[5]**

(c) A point P is moving along the curve $y = 18 - \frac{3}{8}x^{5/2}$ in such a way that the x-coordinate of P is increasing at a constant rate of 2 units per second.

Find the rate at which the y-coordinate of P is changing when $x = 4$. **[3]**

(a) $\left[\frac{dy}{dx}\right] = \frac{9}{2}x - 12[= 0]$ or $|y| = \frac{9}{4}\left(\left(x - \frac{8}{3}\right)^2 + \frac{9}{9}\right)$ **B1** OE Either $\frac{dy}{dx}$ or a correct expression in completed square form. Allow unsimplified.

$x = \frac{24}{9}$ **B1** OE Condone 2.67 AWRT.

$y = 2$ **B1** CAO

Note: $x = \frac{-b}{2a} = \frac{8}{3}$ B1; substitute $\frac{8}{3}$ for x in $y =$ B1; $y = 2$ B1.

Total: 3

(b) [Area =] $\int \left(18 - \frac{3}{8}x^2 - \left(\frac{9}{4}x^2 - 12x + 18\right)\right)dx$ **M1** Intention to integrate and subtract areas (either way around). Can be two separate functions or combined.

Using y^2 scores 0/5 but condone inclusion of π except for the final mark.

Note: Subtraction not required for these marks.

Either separately

$$\left[18x - \frac{3x^7}{28}\right] \cdot \left(\frac{9x^3}{4 \times 3} - \frac{12x^2}{2}\right) + [18x]$$

or

$$\left[18x - \frac{3x^2}{28} - \frac{9x^3}{4 \times 3} + \frac{12x^2}{2}\right] - [18x]$$

B1, B1 One mark for correct integration of each curve, allow unsimplified.

$$= \left(-\frac{3}{28} \times 4^2 - \frac{9}{4} \times 4^3 + 6 \times 4^2 - [18x]\right)$$

M1 Clear substitution of 4 into at least one integrated expression (defined by at least one correct power) which can be unsimplified.

$$= \frac{240}{7} \text{ or } 34.3 \text{ AWRT}$$

A1 SC: If all marks awarded except the final M1, SCB1 is available for the correct final answer.

Total: 5

(c) $\left[\frac{dy}{dx}\right] = \frac{-5 \times 3}{2 \times 8}x^{3/2} = \left[-\frac{15}{16}x^{3/2}\right]$ **B1** Allow unsimplified.

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} \Rightarrow \frac{dy}{dt} = -\frac{15}{16} \times 8 \times 2$$

M1 Substitute $x = 4$ into their $\frac{dy}{dx}$ and multiply by 2.

$$-15$$

A1 Accept decreasing [at/by] 15

Total: 3 Note: If incorrect curve used, this is not a MR and only M1 mark is available. Expect $\left(\frac{9(4)}{2} - 12\right) \times 2 = [= 12]$

Question 4

Oct/Nov 2022 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "The most common error at this stage was the omission of the multiplication by the dV derivative of 9−x."

Source: 9709_w22_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 4

Coaching With Shuja

Question Paper Oct/Nov 2022 • Paper 1 (variant 13) • Q4

A large industrial water tank is such that, when the depth of the water in the tank is x metres, the volume V m³ of water in the tank is given by $V = 243 - \frac{1}{3}(9 - x)^3$. Water is being pumped into the tank at a constant rate of 3.6 m³ per hour.

Find the rate of increase of the depth of the water when the depth is 4 m, giving your answer in cm per minute. **[5]**

Coaching With Shuja

$$\left[\frac{dv}{dx}\right] = (9-x)^2$$

B1 Allow unsimplified forms. Allow any or no notation

Substitute $x = 4$ into their differentiated V, **M1** Expect 25.

$$\frac{dx}{dt} = \frac{1}{\text{their derivative}} \times 3.6$$

(accept $\frac{dx}{dt} = \frac{\text{their derivative}}{3.6}$) **M1** Correct use of the chain rule, ignore incorrect conversions at this point. Expect 0.144

$$= \frac{1}{25} \times 3.6 \times \frac{100}{60} = 0.24$$

DM1 Correct use of the conversion factors.

A1

Coaching With Shuja

Question 5

Oct/Nov 2022 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

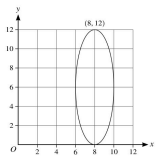
> "(a) This question proved challenging for many candidates."

> "3 (b) Candidates found this part less challenging."

Source: 9709_w22_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 5

Coaching With Shuja

5



The diagram shows a curve which has a maximum point at (8, 12) and a minimum point at (8, 0). The curve is the result of applying a combination of two transformations to a circle. The first transformation applied is a translation of $\begin{pmatrix} 7 \\ -3 \end{pmatrix}$. The second transformation applied is a stretch in the y-direction.

(a) State the scale factor of the stretch. [1]

.....

.....

(b) State the radius of the original circle. [1]

.....

.....

(c) State the coordinates of the centre of the circle after the translation has been completed but before the stretch is applied. [2]

.....

.....

.....

(d) State the coordinates of the centre of the original circle. [2]

The diagram shows a curve which has a maximum point at $(8, 12)$ and a minimum point at $(8, 0)$. The curve is the result of applying a combination of two transformations to a circle. The first transformation applied is a translation of $\begin{pmatrix} 7 \\ -3 \end{pmatrix}$. The second transformation applied is a stretch in the y-direction.

(a) State the scale factor of the stretch. **[1]**

(b) State the radius of the original circle. **[1]**

(c) State the coordinates of the centre of the circle after the translation has been completed but before the stretch is applied. **[2]**

(d) State the coordinates of the centre of the original circle. **[2]**

(a) 3 B1 Ignore any description.

Total: 1

(b) 2 B1 Ignore any description.

Total: 1

(c) (8, 2) B1 B1 Ignore any description. Allow vector notation and absence of brackets.

Total: 2

(d) [Not shown in provided image - coordinate centre before translation]

Total: [2]

Coaching With Shuja

Question 10

Oct/Nov 2022 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "(b) Candidates found this more challenging than part (a)."

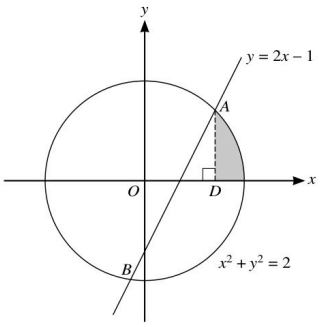
> "(c) This proved to be very challenging for many candidates."

Source: 9709_w22_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 10

Coaching With Shuja

10

10



The diagram shows the circle $x^2 + y^2 = 2$ and the straight line $y = 2x - 1$ intersecting at the points A and B. The point D on the x-axis is such that AD is perpendicular to the x-axis.

(a) Find the coordinates of A. [4]

The diagram shows the circle $\boldsymbol{x^2 + y^2 = 2}$ and the straight line $\boldsymbol{y = 2x - 1}$ intersecting at the points $\boldsymbol{A}$ and $\boldsymbol{B}$. The point $\boldsymbol{D}$ on the $\boldsymbol{x}$-axis is such that $\boldsymbol{AD}$ is perpendicular to the $\boldsymbol{x}$-axis.

(a) Find the coordinates of $\boldsymbol{A}$.

[4]

(b) Find the volume of revolution when the shaded region is rotated through $\boldsymbol{360^\circ}$ about the $\boldsymbol{x}$-axis.

Give your answer in the form $\frac{\pi}{a}(b\sqrt{c} - d)$, where a, b, c and d are integers.

[4]

(c) Find an exact expression for the perimeter of the shaded region.

[2]

10(a)

$x^2 + (2x - 1)^2 - 2[= 0] \rightarrow 5x^2 - 4x - 1[= 0]$
M1 A1 Or $5y^2 + 2y - 7[= 0]$.

$(5x + 1)(x - 1)[= 0]$ or $(5y + 7)(y - 1)[= 0]$
DM1 May see factors or formula or completing square.

$x = 1, y = 1$ or $(1, 1)$ only
A1 May be implied on the diagram.

Total: 4

Coaching With Shuja

Question 11

Oct/Nov 2022 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Most candidates formed correct expressions for the gradients of AC and BC, but some then struggled to set them up as a product equal to –1."

> "Common errors were to merely let the expressions for the gradients of AC and BC equal each other or to multiply one of the gradient expressions by –1 and then equate the gradient expressions."

Source: 9709_w22_er.pdf, Page 11, Section: Comments on specific questions – Paper 1 (variant 13), Question 11

11 The coordinates of points A, B and C are $A(5, -2), B(10, 3)$ and $C(2p, p)$, where p is a constant.

(a) Given that AC and BC are equal in length, find the value of the fraction p.
[3]

(b) It is now given instead that AC is perpendicular to BC and that p is an integer.

(i) Find the value of p.
[4]

(ii) Find the equation of the circle which passes through A, B and C, giving your answer in the form $x^2 + y^2 + ax + by + c = 0$, where a, b and c are constants.
[4]

Coaching With Shuja

11(a)

$(5-2p)^2+(-2-p)^2=(10-2p)^2+(3-p)^2$

M1 A1 Allow one sign error for M mark only.

$25-20p+4p^2+p^2+4p+4=100-40p+4p^2+9-6p+p^2$

A1 Allow 2.67 AWRT.

$30p=80 \rightarrow p=\frac{8}{3}$ oe

Total: 3

11(b)(i)

$m_{AC}=\frac{p+2}{2p-5}$, $m_{BC}=\frac{p-3}{2p-10}$

M1 Allow a sign error.

$\frac{p+2}{2p-5}\times\frac{p-3}{2p-10}=-1$

M1 Use of $m_1m_2=-1$ with their m_{AC} and m_{BC}.

$p^2-p-6=-(4p^2-30p+50) \rightarrow 5p^2-31p+44(=0)$

A1

$p=4$ (Ignore $p=\frac{11}{5}$)

A1 Factors $(p-4)(5p-11)$, or formula or completing square must be seen.

Total: 4

11(b)(ii)

Mid-point of $AB=(7\frac{1}{2},\frac{1}{2})$

B1 SOI

$r^2=2\frac{1}{2}^2+2\frac{1}{2}^2=[\frac{50}{4}]$ or $r=\sqrt{(2\frac{1}{2}^2+2\frac{1}{2}^2)}=[\frac{5\sqrt{2}}{2}]$

M1 Or $r^2=\frac{1}{4}(5^2+5^2)=[\frac{50}{4}]$ etc.

Equation of circle is $(x-\text{their}7\frac{1}{2})^2+(y-\text{their}\frac{1}{2})^2=\text{their}\frac{50}{4}$

DM1 Must use r^2 not r or d or d^2

$x^2+y^2-15x-y+44=0$

A1 CAO

Total: 4

Question 1

Oct/Nov 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "A few sketches did not consist of straight lines, but the most common error was to have the vertex on the vertical axis."

Source: 9709_w22_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

1

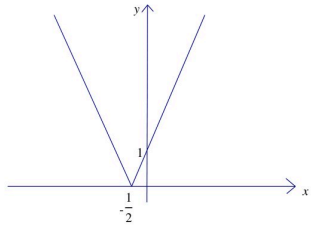
(a) Sketch the graph of $y = |2x + 1|$.
[1]

(b) Solve the inequality $3x + 5 < |2x + 1|$.
[3]

1(a)

Show a recognisable sketch graph of $y = |2x + 1|$

B1

1(a)	Show a recognisable sketch graph of $y = \|2x+1\|$	B1	 Ignore $y = 3x + 5$ if also drawn on the sketch.
		1	

Total: 1

1(b)

Find x-coordinate of intersection with $y = 3x + 5$

M1

Obtain $x = -\frac{6}{5}$

A1

State final answer $x < -\frac{6}{5}$ only

A1 Do not condone $\leq$ for $<$ in the final answer.

Alternative method 1 for question 1(b)

Solve the linear inequality $3x + 5 < -(2x + 1)$, or corresponding equation

M1 Must solve the relevant equation.

Obtain critical value $x = -\frac{6}{5}$

A1 Ignore -4 if seen.

State final answer $x < -\frac{6}{5}$ only

A1

Alternative method 2 for question 1(b)

Solve the quadratic inequality $(3x + 5)^2 < (2x + 1)^2$, or corresponding equation

M1 $5x^2 + 26x + 24 < 0$

Obtain critical value $x = -\frac{6}{5}$

A1 Ignore -4 if seen.

State final answer $x < -\frac{6}{5}$ only

A1

Total: 3

Question 3

Oct/Nov 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common error was to rewrite 5×3−x as 15−x."

Source: 9709_w22_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 3

Coaching With Shuja

3 Solve the equation $2^{3x-1} = 5(3^{-x})$. Give your answer in the form $\frac{\ln a}{\ln b}$, where a and b are integers.

[4]

3

Use law of logarithm of a product or power
M1 One correct application of a log law.

Obtain a correct linear equation in any form, e.g. $(3x-1)\ln 2 = \ln 5 - x\ln 3$
A1

Solve for x
M1 As far as $x = \ldots$ with only minor slips in processing.

Obtain answer $x = \frac{\ln 10}{\ln 24}$
A1

Alternative method for question 3

Use laws of indices to split at least one exponential term
M1 e.g. $\frac{2^{3x}}{2}$ or an arrangement with 8^x

Obtain $24^x = 10$
A1 OE

Solve for x
M1

Obtain answer $x = \frac{\ln 10}{\ln 24}$
A1

Total: 4

Coaching with Shuja

Question 5

Oct/Nov 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "There were several blank responses, some answers that gave a negative real value, and some that appeared to be guesses."

Source: 9709_w22_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

Coaching With Shuja

5 The complex numbers u and w are defined by $u = 2e^{\frac{i\pi}{6}}$ and $w = 3e^{\frac{5i\pi}{6}}$.

(a) Find $\frac{u^2}{w}$, giving your answer in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. Give the exact values of r and θ.

[3]

(b) State the least positive integer n such that both $\text{Im}\, w^n = 0$ and $\text{Re}\, w^n > 0$.

[1]

Coaching With Shuja

5(a)

State or imply $u^2 = 4e^{\frac{i\pi}{3}}$

B1

Obtain answer $y = \frac{4}{3}e^{\frac{i\pi}{6}}$

B1 + B1 For the modulus and the argument.

Total: 3

5(b)

State $n = 6$

B1

Total: 1

Coaching With Shuja

Question 6

Oct/Nov 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “Common errors came from slips in the algebra, in particular ( ) errors when squaring $2\cos 2\theta - 1$.”

Source: 9709_w22_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

6

(a) Prove the identity $\cos 4\theta + 4\cos 2\theta + 3 = 8\cos^4\theta$.
[4]

(b) Hence solve the equation $\cos 4\theta + 4\cos 2\theta = 4$ for $0^\circ \leq \theta \leq 180^\circ$.
[3]

Coaching With Shuja

6(a)

Express $\cos 4\theta$ in terms of $\cos 2\theta$ and/or $\sin 2\theta$

B1

Express $\cos 2\theta$ in terms of $\cos\theta$ and/or $\sin\theta$

B1 Anywhere

Expand to obtain a correct expression in terms of $\cos\theta$

B1 e.g. $2(2\cos^2\theta - 1)^2 - 1 + 4(2\cos^2\theta - 1) + 3$

Reduce correctly to $\cos 4\theta + 4\cos 2\theta + 3 = 8\cos^4\theta$

B1 AG

Total: 4

6(b)

Use the identity and carry out method to calculate a root

M1 $8\cos^4\theta - 3 = 4$

Obtain answer, e.g. 14.7°

A1

Obtain second answer, e.g. 165.3°, and no other in the given interval

A1 FT Ignore answers outside the given interval. Treat answers in radians as a misread.

Total: 3

Coaching With Shuja

Question 7

Oct/Nov 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(b) There were several blank responses to this part of the question."

> "(c) There were several blank responses to this part."

Source: 9709_w22_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

7 The equation of a curve is $y = \frac{x}{\cos^2 x}$, for $0 \leq x < \frac{1}{2}\pi$. At the point where $x = a$, the tangent to the curve has gradient equal to 12.

(a) Show that $a = \cos^{-1}\left(\sqrt{\frac{3}{4}\frac{\cos a + 2a\sin a}{12}}\right)$.

[3]

(b) Verify by calculation that a lies between 0.9 and 1.

[2]

(c) Use an iterative formula based on the equation in part (a) to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places.

[3]

Coaching With Shuja

7(a)

Use correct product or quotient rule

M1

Obtain correct derivative in any form

A1 e.g. $\frac{dy}{dx} = \frac{\cos^2 x + 2x \sin x \cos x}{\cos^4 x}$ or $\frac{dy}{dx} = \sec^2 x + 2x \sec^2 x \tan x$

Equate derivative at $x = a$ to 12 and obtain $a = \cos^{-1}\left(\sqrt{\frac{\cos a + 2a \sin a}{12}}\right)$

A1 AG

Total: 3

7(b)

Evaluate a relevant expression or pair of expressions at $a = 0.9$ and $a = 1$

M1 Must be calculated in radians.

Complete the argument correctly with correct calculated values

A1 e.g. $\frac{\cos 0.9 = 0.622 > 0.553}{0.9 < 0.985}$, $\frac{0.0846 > 0}{1 > 0.964}$ or $\frac{-0.0358 < 0}{\text{or could be looking at values of the gradient } 8.46\&14.1}$

Total: 2

7(c)

Use the process $a_{n+1} = \cos^{-1}\left(\sqrt{\frac{\cos a_n + 2a_n \sin a_n}{12}}\right)$ correctly at least once

M1 Must be working in radians.

Obtain final answer 0.97

A1

Show sufficient iterations to 4 d.p. to justify 0.97 to 2 d.p., or show there is a sign change in the interval $(0.965, 0.975)$

A1 e.g. $0.95, 0.9743, 0.9694, 0.9704$

Total: 3

Question 8

Oct/Nov 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(b) This part proved a challenge and there were many blank responses, often due to an incomplete response in part (a)."

Source: 9709_w22_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 8

Coaching With Shuja

8 In a certain chemical reaction the amount, x grams, of a substance is increasing. The differential equation satisfied by x and t, the time in seconds since the reaction began, is

$$\frac{dx}{dt} = kxe^{-0.1t},$$

where k is a positive constant. It is given that $x = 20$ at the start of the reaction.

(a) Solve the differential equation, obtaining a relation between x, t and k.
[5]

(b) Given that $x = 40$ when $t = 10$, find the value of k and find the value approached by x as t becomes large.
[3]

8(a)

Separate variables correctly
B1 $\int \frac{1}{x}dx = \int ke^{-0.1t}dt$

Obtain term $\ln x$
B1

Obtain term $-10ke^{-0.1t}$
B1 Not from $\int xe^{-0.1t}dt$

Use $x = 20, t = 0$ to evaluate a constant or as limits in a solution containing terms $a\ln x, be^{-0.1t}$ where $ab \neq 0$
M1

Obtain $\ln x = 10k(1 - e^{-0.1t}) + \ln 20$
A1 or equivalent ISW

Total: 5

8(b)

Use $x = 40, t = 10$ to find k or $10k$
M1 Available for their function of the correct structure even if they found no constant in (a).

Obtain $10k = 1.09654$
A1 or equivalent e.g. $10k = \frac{\ln 2}{1-e^{-1}}$

State that x tends to 59.9
A1 Need a number, not an expression for that value. 3 sf or better $59.87595\ldots$

Total: 3

Coaching With Shuja

Question 10

Oct/Nov 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(b) This part proved challenging with many candidates offering no attempt to use the binomial expansion."

> "A few candidates reached the correct conclusion but there were several errors in using the expansion and in simplifying the results."

Source: 9709_w22_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 10

Coaching With Shuja

Question 10

Let $f(x) = \frac{2x^2+7x+8}{(1+x)(2+x)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. **[5]**

10(a) State or imply the form $\frac{A}{1+x}+\frac{B}{2+x}+\frac{C}{(2+x)^2}$ **B1**

Use a correct method to find a constant **M1**

Obtain one of $A=3, B=-1$ and $C=-2$ **A1** (SR after B0 can score M1 A1 for one correct value)

Obtain a second value **A1**

Obtain the third value **A1** $\frac{A}{1+x}+\frac{Dx+E}{(2+x)^2}$, where $A=3, D=-1$ and $E=-4$, is awarded **B1 M1 A1 A1 A1** as above.

Total: 5

10(b) Use a correct method to find the first two terms of the expansion of $(1+x)^{-1}, (2+x)^{-1}, \left(1+\frac{1}{2}x\right)^{-1}, (2+x)^{-2}$ or $\left(1+\frac{1}{2}x\right)^{-2}$ **M1** (For the A, D, E form of fractions, award M1 A1 FT A1 FT for the expanded partial fractions, then if $D \neq 0$, M1 for multiplying out fully, and A1 for the final answer.)

Obtain correct unsimplified expansions up to the term in x^2 of each partial fraction **A3 FT**

$3\left(1-x+x^2\ldots..\right)-\frac{1}{2}\left(1-\frac{x}{2}+\frac{x^2}{4}\ldots..\right)-\frac{2}{4}\left(1-x+\frac{3}{4}x^2\right)$

Obtain final answer $2-\frac{9}{4}x+\frac{5}{2}x^2$ **A1**

Question 11

Oct/Nov 2022 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

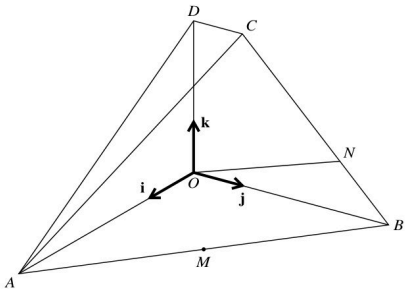
> "Question 11 This question was challenging for many candidates and there was a high proportion of blank responses to parts (b) and (c)."

Source: 9709_w22_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 11

Coaching With Shuja

Question 11

11



In the diagram, $OABCD$ is a solid figure in which $OA = OB = 4$ units and $OD = 3$ units. The edge OD is vertical, DC is parallel to OB and $DC = 1$ unit. The base, OAB, is horizontal and angle $AOB = 90°$. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OB and OD respectively. The midpoint of AB is M and the point N on BC is such that $CN = 2NB$.

(a) Express vectors $\overrightarrow{MD}$ and $\overrightarrow{ON}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. [4]

In the diagram, $OABCD$ is a solid figure in which $OA = OB = 4$ units and $OD = 3$ units. The edge OD is vertical, DC is parallel to OB and $DC = 1$ unit. The base, OAB, is horizontal and angle $AOB = 90°$. Unit vectors $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OB and OD respectively. The midpoint of AB is M and the point N on BC is such that $CN = 2NB$.

(a) Express vectors $\overrightarrow{MD}$ and $\overrightarrow{ON}$ in terms of $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$. **[4]**

(b) Calculate the angle in degrees between the directions of $\overrightarrow{MD}$ and $\overrightarrow{ON}$. **[3]**

(c) Show that the length of the perpendicular from M to ON is $\sqrt{\frac{22}{5}}$. **[4]**

11(a) State $\overrightarrow{OM} = 2\mathbf{i} + 2\mathbf{j}$ or equivalent **B1** (Can be implied by $\overrightarrow{MB} = -2\mathbf{i} + 2\mathbf{j}$ or $\overrightarrow{MA} = 2\mathbf{i} - 2\mathbf{j}$.)

Obtain $\overrightarrow{MD} = -2\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ **B1**

Use a correct method to find $\overrightarrow{ON}$ **M1** (e.g. $\overrightarrow{OC} + \frac{2}{3}\overrightarrow{CB}$)

Obtain answer $3\mathbf{j} + \mathbf{k}$ **A1**

Total: 4

11(b) Use the correct process for evaluating the scalar product of $\overrightarrow{MD}$ and $\overrightarrow{ON}$ **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and reach the inverse cosine of the result **M1** $\cos^{-1}\left(\frac{-6+3}{\sqrt{10}\sqrt{17}}\right)$

Obtain final answer 103.3° **A1**

Total: 3

11(c) Taking a general point P of ON to have position vector $\lambda(3\mathbf{j} + \mathbf{k})$, form an equation in λ by *either* equating the scalar product of $\overrightarrow{ON}$ and $\overrightarrow{MP}$ to zero, *or* applying Pythagoras to triangle OMP, or equating the derivative of $|\overrightarrow{MP}|$ to zero **M1** (e.g. $(-2)(0) - 2 + 3\lambda(\lambda)(1) = 0$)

Solve and obtain $\lambda = \frac{3}{5}$ **A1**

Substitute for λ and calculate $\overrightarrow{MP}$ **M1** $\overrightarrow{MP} = -2\mathbf{i} - \frac{1}{5}\mathbf{j} + \frac{3}{5}\mathbf{k}$

Obtain $\sqrt{\frac{22}{5}}$ **A1** (AG)

Alternative method for question 11(c)

Use a scalar product to find the projection OQ of OM on OM **M1**

Obtain $OQ = \frac{6}{\sqrt{10}}$ **A1**

Use Pythagoras in triangle OMQ to find MQ **M1**

Obtain $\sqrt{\frac{22}{5}}$ **A1** (AG)

Question 1

Oct/Nov 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common error was to rewrite 5×31−x as 151−x."

Source: 9709_w22_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

Question 1

Solve the equation $2^{3x-1} = 5(3^{1-x})$. Give your answer in the form $\frac{\ln a}{\ln b}$ where a and b are integers. **[4]**

Mark Scheme

1 Use law of the logarithm of a power or product **M1** (Ignoring the 3 or the 5 is not a misread.)

Obtain a correct linear equation in any form, e.g. $(3x-1)\ln 2 = \ln 5 + (1-x)\ln 3$ **A1** (Condone invisible brackets if they are used correctly later.)

Solve for x **M1** (Get as far as $x = \ldots$ Condone minor slips in the processing e.g. sign errors and losing a term that had been there, but award M0 for a fundamental error e.g. $3x\ln 2 + x\ln 3 = 3x\ln 6$ or ignoring the 3 or the 5 completely. Condone working in decimals.)

Obtain **final** answer $x = \frac{\ln 30}{\ln 24}$ **A1** (Do not ISW)

Alternative method for question 1

Use laws of indices to split at least one exponential term **M1** (e.g. $\frac{2^{3x}}{2}$ or an arrangement with 8^x and/or 3^x.)

Obtain $24^x = 30$ **A1** (Or equivalent e.g. Double exponent: use braces to clarify not for Double exponent: use braces to clarify (need two factors with the same index).)

Solve for x **M1** (Get as far as $x = \ldots$)

Obtain **final** answer $x = \frac{\ln 30}{\ln 24}$ **A1** (Do not ISW)

Question 2

Oct/Nov 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Very few candidates with the correct factors demonstrated an understanding that for a negative result they required one factor to be positive and the other negative."

Source: 9709_w22_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Coaching With Shuja

Question 2

The polynomial $2x^3 - x^2 + a$, where a is a constant, is denoted by p(x). It is given that $(2x + 3)$ is a factor of p(x).

(a) Find the value of a. **[2]**

(b) When a has this value, solve the inequality p(x) < 0. **[4]**

2(a) Substitute $x = -\frac{3}{2}$ and equate result to zero **M1** (Or divide by $2x + 3$ and set constant remainder equal to zero. Or state $(2x^3 - x^2 + a) = (2x + 3)(x^2 + px + q)$, compare coefficients and solve for p or q .)

Obtain $a = 9$ **A1**

Total: 2

2(b) Commence division by $(2x + 3)$ reaching a partial quotient $x^2 + kx$ ***M1** (The M1 is earned if inspection reaches an unknown factor: $x^2 + Bx + C$ and an equation in B and/or C, or an unknown factor $Ax^2 + Bx + 3$ and an equation in A and/or B.)

Obtain factorisation $(2x + 3)(x^2 - 2x + 3)$ **A1** (Allow if the correct quotient seen. Correct factors seen in (a) and quoted or used here scores M1 A1.)

Show that $x^2 - 2x + 3$ is always positive, or $2x^3 - x^2 + 9$ only intersects the x-axis once **DM1** (Must use their quadratic factor. SC If M0, allow B1 if state $x < -\frac{3}{2}$ and no error seen)

State **final** answer $x < -\frac{3}{2}$ from correct work **A1**

Total: 4

Question 4

Oct/Nov 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The question 4 specifies that the angle should be given correct to 2 decimal places, but writing an answer to 1 decimal place or more than 2 decimal places were common errors."

> "The most common errors were in adjusting from an expression in x in part (a) to an equation in 2x in part (b)."

Source: 9709_w22_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

Question 4

(a) Express $4\cos x - \sin x$ in the form $R\cos(x + \alpha)$, where $R > 0$ and $0° < \alpha < 90°$. State the exact value of R and give α correct to 2 decimal places. **[3]**

(b) Hence solve the equation $4\cos 2x - \sin 2x = 3$ for $0° < x < 180°$. **[5]**

Coaching With Shuja

4(a) State $R = \sqrt{17}$ **B1** (Allow if working from an incorrect expansion but not from decimals.)

Use correct trig formulae to find α (Correct expansion and correct expression for trig ratio for α) **M1** (NB: $\cos\alpha = 4$ and $\sin\alpha = 1$ scores M0A0. M0 for incorrect expansion of $\cos(x - \alpha)$. M1 for correct expression for trig ratio for α and no errors seen.)

Obtain $\alpha = 14.04^\circ$ **A1** (2 d.p. required. Allow M1A1 for correct answer with no working shown. Correct answer from incorrect working (e.g. $\tan^{-1}(-\frac{1}{4})$) is awarded M0A0. $180^\circ - \tan^{-1}(-\frac{1}{4})$ is awarded M1)

Total: 3

4(b) Evaluate $\cos^{-1}\left(\frac{3}{\sqrt{17}}\right)$ to at least 1 d.p. (43.31 38...°) **B1 FT** (FT *their* R. Accept awrt 43.3° or awrt 316.7°. Can be implied by subsequent working.)

Use correct method to find a value of x in the interval **M1** (Must be working with $2x$ and *their* α.)

Obtain answer, e.g. 14.6° **A1** (Accept overspecified answers but they need to be correct. (14.6388.... and 151.3249....).)

Use a correct method to find a second answer in the interval **M1** (Must be working with $2x$, *their* α and $360^\circ -$ *their* 43.3.)

Obtain second answer in the interval, e.g. 151.3°, and no other in the interval **A1** (Ignore answers outside the given interval. Treat answers in radians (0.255... and 2.64...) as a misread.)

Total: 5

Coaching With Shuja

Question 5

Oct/Nov 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common error was to use 6 in place of 6i when substituting values into the quadratic formula."

Source: 9709_w22_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

Coaching With Shuja

Question 5

(a) Solve the equation $z^2 - 6iz - 12 = 0$, giving the answers in the form $x + iy$, where x and y are real and exact. **[3]**

(b) On a sketch of an Argand diagram with origin O, show points A and B representing the roots of the equation in part (a). **[1]**

(c) Find the exact modulus and argument of each root. **[3]**

(d) Hence show that the triangle OAB is equilateral. **[1]**

Coaching With Shuja

5(a) Use quadratic formula, or completing the square $\left((z-3i)^2-3=0\right)$ **M1** and use $i^2=-1$ to find a root

Obtain a root, e.g. $\sqrt{3}+3i$ **A1** (Or exact 2 term equivalent e.g. $\frac{6i+\sqrt{12}}{2}$ ISW.)

Obtain the other root, e.g. $-\sqrt{3}+3i$ **A1** (Or exact 2 term equivalent ISW.)

Total: 3

5(b) Show points representing the roots correctly **B1 FT** (2 roots consistent with *their* (a) and with no errors seen on the diagram. B0 if they only have one root or more than 2 roots. Must match *their* scale and $1<\sqrt{3}<2$ (Linear scales seen or implied. Need some indication of scale (numbers or dashes). Scales along an axis must be approximately consistent but scales may be different on the 2 axes.)

Total: 1

5(c) State modulus of either root is $2\sqrt{3}$, or simplified exact equivalent **B1 FT** (ISW if converted to decimal. Ignore modulus of second root if seen. Follow *their* root(s) not on either axis (from (a) or (b)).)

Find the argument of one of *their* roots – get as far as $\tan^{-1}(\ldots)$ **M1** (SOI but must be correct for *their* root.)

Obtain correct arguments $\frac{1}{3}\pi$ and $\frac{2}{3}\pi$, or simplified exact equivalents **A1** (Must obtain values. Allow degrees.)

Total: 3

5(d) Give a **complete justification** that the **correct** triangle is equilateral **B1** (Check *their* diagram in (b). Possible justifications: 3 equal sides, or all angles equal to $\frac{\pi}{3}$, or isosceles and an angle of $\frac{\pi}{3}$.)

Total: 1

Question 6

Oct/Nov 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The other common error υυυρ was to use an incorrect pair of vectors."

> "(b) Although the question asks for an exact answer, many candidates could not see how to obtain this and gave a decimal answer instead."

Source: 9709_w22_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

Question 6

Relative to the origin O, the points A, B and C have position vectors given by

$$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 5 \\ 3 \\ -2 \end{pmatrix}.$$

(a) Using a scalar product, find the cosine of angle BAC. **[4]**

(b) Hence find the area of triangle ABC. Give your answer in a simplified exact form. **[4]**

Coaching With Shuja

Mark Scheme

Oct/Nov 2022 • Paper 3 (variant 32) • Q6

6(a) State or imply $\overrightarrow{AB}$ or $\overrightarrow{AC}$ correctly in component form **B1** $(\overrightarrow{AB} = 2\mathbf{i} - 2\mathbf{j} + \mathbf{k}, \overrightarrow{AC} = 4\mathbf{i} - 3\mathbf{k})$.

Using the correct process with relevant vectors to evaluate the scalar product $\overrightarrow{AB}.\overrightarrow{AC}$, **M1** (or $\overrightarrow{BA}.\overrightarrow{CA}$ (8 - 3 - 5).)

Using the correct process for the moduli, divide *their* scalar product by the product of *their* moduli to obtain $\cos\theta$ or θ **M1** $\left(\frac{5}{\sqrt{9}\sqrt{25}}\right)$ (Independent of the first M1.)

Obtain answer $\frac{1}{3}$ **A1** (ISW. Need to see a value for $\cos\theta$. Accept $\frac{5}{15}$ or 0.333 ($\cos^{-1}\frac{1}{3}$ alone is not sufficient))

Total: 4

6(b) Use correct method to find an **exact value** for the sine of angle BAC from *their* (a) **M1** $\left(\sqrt{1-\frac{1}{9}}\right)$ (or $\frac{1}{2}\sqrt{9}\sqrt{25} \times$ *their* $\sin\theta$) or $\frac{1}{2}\sqrt{9}\sqrt{25}(\cos^{-1}\frac{1}{3})$)

Obtain answer $\frac{2}{3}\sqrt{2}$, or equivalent **A1** (Only ISW)

Use correct area formula to find the area of triangle ABC with *their* versions of relevant vectors **M1** $\left(\frac{1}{2}\sqrt{9}\sqrt{25} \times$ *their* $\sin\theta\right)$ or $\frac{1}{2} \times$ *their* $\sqrt{24} \times$ *their* $\frac{1}{\sqrt{75}}$) (The length they use for *their* base must be found correctly.)

Obtain answer $5\sqrt{2}$ or $\sqrt{50}$ **A1** (Only ISW)

Alternative method 1 for question 6(b)

Use correct method to find the perpendicular distance from A to BC (or B to AC or C to AB) **M1** $\left(\begin{pmatrix}2+2\lambda\\-2+2\lambda\\1-4\lambda\end{pmatrix}\begin{pmatrix}2\\2\\-4\end{pmatrix}\right.$

= 0 \Rightarrow \lambda = \frac{1}{6}\right)$

Obtain $\frac{1}{6}\sqrt{75}$ **A1** $\left(\left|\frac{1}{6}\mathbf{i} - \frac{5}{6}\mathbf{j} + \frac{1}{6}\mathbf{k}\right|\right)$

Use correct area formula to find the area of triangle ABC **M1** $\left(\frac{1}{2} \times$ *their* $\sqrt{24} \times$ *their* $\frac{1}{\sqrt{75}}\right)$ (The length they use for *their* base must be found correctly.)

Obtain answer $5\sqrt{2}$ or $\sqrt{50}$ **A1**

Question 7

Oct/Nov 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The integration of tan2θ proved to be more challenging, with relatively few sin2θ © 2022 Cambridge International Advanced Subsidiary and Advanced Level 9709 Mathematics November 2022 Principal Examiner Report for Teachers 1 candidates recognising this as sec2θ."

Source: 9709_w22_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

Question 7

The variables x and θ satisfy the differential equation

$$x \sin^2 \theta \frac{dx}{d\theta} = \tan^2 \theta - 2 \cot \theta,$$

for $0 < \theta < \frac{1}{2}\pi$ and $x > 0$. It is given that $x = 2$ when $\theta = \frac{1}{4}\pi$.

(a) Show that $\frac{d}{d\theta}(\cot^2 \theta) = -\frac{2 \cot \theta}{\sin^2 \theta}$.

(You may assume without proof that the derivative of $\cot \theta$ with respect to θ is $-\operatorname{cosec}^2 \theta$.) **[1]**

(b) Solve the differential equation and find the value of x when $\theta = \frac{1}{6}\pi$. **[7]**

Coaching With Shuja

7(a) Show sufficient working to justify the given statement **B1** (e.g. see $2\cot\theta \times -\text{cosec}^2\theta$ in the working or express in terms of $\sin\theta$ and $\cos\theta$ and use quotient rule to obtain the given result. Solution must have θ present throughout and must reach the given answer.)

Total: 1

7(b) Separate variables correctly **B1** (Check for relevant working in (a))

(Integral: $\int x\,dx = \int \frac{\tan^2\theta - 2\cot\theta}{\sin^2\theta}\,d\theta$ Condone incorrect notation e.g. missing dx. Need either the integral sign or the dx, dθ.)

Obtain term $\frac{1}{2}x^2$ **B1**

Obtain terms $\tan\theta + \cot^2\theta$ **B1** + **B1** (Alternative: $\int \frac{2\cot\theta - \cos\cot\theta}{\sin^2\theta}\,d\theta = -\frac{1}{\sin^2\theta}(-C)$) Condone incorrect notation e.g. missing dx. Need either the integral sign or the dx, dθ.

Form an equation for the constant of integration, or use limits $x = 2, \theta = \frac{1}{4}\pi$, in a solution with **at least two correctly obtained terms** of the form $ax^2, b\tan\theta$ and $c\cot^2\theta$, where $abc \neq 0$ **M1** (Need to have 3 terms. Constant of correct form.)

State correct solution in any form, e.g. $\frac{1}{2}x^2 = \tan\theta + \cot^2\theta - 1$ **A1** (or $\frac{1}{2}x^2 = \tan\theta + \text{cosec}^2\theta - 1$) If everything else is correct, allow a correct final answer to imply this A1. If the user has an error elsewhere, allow this A1 to imply that this A1 is assumed.

Substitute $\theta = \frac{1}{6}\pi$ and obtain answer $x = 2.67$ **A1** $\left(2.6748\ldots \sqrt{\frac{18+2\sqrt{3}}{3}}\right)$ If see a correctly rounded value ISW.

Question 8

Oct/Nov 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors occurred in the course of substituting for dx, with several candidates having a factor of u or 1 in their integral."

Source: 9709_w22_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 32), Question 8

Coaching With Shuja

Question Paper Oct/Nov 2022 • Paper 3 (variant 32) • Q8

The diagram shows part of the curve $y = \sin\sqrt{x}$. This part of the curve intersects the x-axis at the point where $x = a$.

(a) State the exact value of a. **[1]**

(b) Using the substitution $u = \sqrt{x}$, find the exact area of the shaded region in the first quadrant bounded by this part of the curve and the x-axis. **[7]**

Coaching With Shuja

Mark Scheme

(a) State $(a) = \pi^2$ **B1** Allow 32400, 180^2. Accept $(x =)\pi^2$.

(b) State or imply $dx = 2u\,du$ or equivalent **B1** e.g. $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$. Incorrect statements e.g. $du = \frac{1}{2\sqrt{x}}$ is **B0**.

Substitute for x and dx throughout the integral **M1**

Obtain $\int 2u \sin u\,du$ **A1** Allow with missing du.

Commence integration of $\int ku \sin u\,du$ by parts and reach $\mp ku\cos u \pm \int k\cos u\,du$ **M1**

Obtain integral $-ku\cos u + k\sin u$ **A1**

Substitute limits $u = 0$ and $u = \sqrt{\text{their } a}$, $a \neq 0$, a in radians or $x = 0$ and *their* a in the complete integral **DM1** $-2\pi\cos\pi + 2\sin\pi(+0 - 2\sin 0)$. Need limits stated but condone if zeros not shown in substitution.

Obtain answer 2π **A1**

Total: 7

Coaching With Shuja

Question 9

Oct/Nov 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Some candidates found it challenging when using this to obtain the given answer; some answers contained $\sin(\pi-2\theta)$, and some claimed that $\sin(\pi-2\theta)=\sin\pi-\sin2\theta$."

Source: 9709_w22_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Question Paper Oct/Nov 2022 • Paper 3 (variant 32) • Q9

The diagram shows a semicircle with diameter AB, centre O and radius r. The shaded region is the minor segment on the chord AC and its area is one third of the area of the semicircle. The angle CAB is θ radians.

(a) Show that $\theta = \frac{1}{3}(\pi - 1.5 \sin 2\theta)$. **[4]**

(b) Verify by calculation that $0.5 < \theta < 0.7$. **[2]**

(c) Use an iterative formula based on the equation in part **(a)** to determine θ correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **[3]**

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Mark Scheme Oct/Nov 2022 • Paper 3 (variant 32) • Q9

(a) State or imply angle $AOC = \pi - 2\theta$ **B1** Might be seen on the printed diagram.

Use correct formulae for the area of a sector and triangle, or of a segment, and find the area of the shaded region **M1** $\frac{1}{2}r^2(\pi - 2\theta) - \frac{1}{2}r^2\sin(\pi - 2\theta)$ or $\frac{1}{2}\pi r^2 - [\frac{1}{2}r^2(2\theta) + \frac{1}{2}r^2\sin(\pi - 2\theta)]$. **M0** if subtraction the wrong way round.

Equate to $\frac{1}{6}\pi r^2$ and obtain a correct equation in any form **A1** e.g.
$\frac{1}{6}\pi r^2 = \frac{1}{2}r^2(\pi - 2\theta) - \frac{1}{2}r^2\sin(\pi - 2\theta)$.

Obtain $\theta = \frac{1}{3}(\pi - 1.5\sin 2\theta)$ correctly **A1 AG** Condone if state / imply $\sin(\pi - 2\theta) = \sin 2\theta$.

Total: 4

(b) Evaluate a relevant expression or pair of expressions at $\theta = 0.5$ and $\theta = 0.7$ **M1** Allow work on a smaller interval. Need to evaluate for both limits, with at least one correct. When using $x = f(x)$ embedded values are not sufficient e.g. $f(0.5)\ldots$ is accepted but $\frac{1}{3}(\pi - 1.5\sin 2 \times 0.5) = \ldots$ is not.

Complete the argument correctly with correct calculated values **A1** e.g. $0.5 < 0.626$, $0.7 > 0.554$ or $0.126 > 0$, $-0.146 < 0$. If using pairs then the pairing must be clear. Need to see the inequalities or an appropriate comment. Need to see values calculated to at least 2 sf.

Total: 2

(c) Use the iterative process $\theta_{n+1} = \frac{1}{3}(\pi - 1.5\sin 2\theta_n)$ correctly at least once **M1** i.e obtain one value and use that value to obtain a second value. Must be working in radians.

Obtain final answer 0.586 **A1**

Show sufficient iterations to 5 d.p. to justify 0.586 to 3 d.p., or show there is a sign change in the interval (0.5855, 0.5865). e.g. 0.5, 0.62646, 0.57225, 0.59195, 0.58416, 0.58715, 0.58599, 0.58644 or 0.6, 0.58118, 0.58833, 0.58553, 0.58661, 0.58619, 0.58636 or 0.7, 0.55447, 0.59958, 0.58133, 0.58827, 0.58556, 0.58661, 0.58620, 0.58636. Allow working to more than 5 dp, but not less. **A1**

Total: 3

Question 10

Oct/Nov 2022 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The A second integral proved to be more challenging; many candidates ignored the numerator and assumed that the answer would be of the form ptan−1q, or they recognised that it should be a 1 logarithm but they retained x or as a coefficient."

Source: 9709_w22_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

Let $f(x) = \frac{4-x+x^2}{(1+x)(2+x^2)}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Find the exact value of $\int_0^4 f(x)\,dx$. Give your answer as a single logarithm. **[5]**

(a) State or imply the form $\frac{A}{1+x} + \frac{Bx+C}{2+x^2}$ **B1**

Use a correct method for finding a constant **M1**

Obtain one of $A = 2$, $B = -1$ and $C = 0$ **A1 SC**: A maximum of **M1A1** is available for obtaining $A = 2$ after scoring **B0**.

Obtain a second value **A1**

Obtain the third value **A1**

Total: 5

(b) Integrate and obtain term $2\ln(1+x)$ **B1 FT** $A\ln(1+x)$

Integrate and obtain term of the form $k\ln(2+x^2)$ from an integral of the correct form **M1** Ignore any separate working relating to $C \neq 0$.

Obtain term $-\frac{1}{2}\ln(2+x^2)$ **A1 FT** $\frac{B}{2}\ln(2+x^2)$

Substitute limits in an integral containing terms of the form $a\ln(1+x) + b\ln(2+x^2)$, where $ab \neq 0$ **DM1** Ignore working relating to $C \neq 0$. $(2\ln 5 - 2\ln 1 - \frac{1}{4}\ln 18 + \frac{1}{4}\ln 2)$. Dependent on the first **M1**. Must be subtracting the correct way round. Must have an exact substitution.

Obtain answer $\ln\left(\frac{25}{3}\right)$ **A1 ISW**

Any exact equivalent e.g. $\ln 25\sqrt{2}/\sqrt{8}$ with no number in front of the logarithm.

Total: 5

Question 3

Oct/Nov 2022 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> “Otherwise, the most common error was a sign error in the ln cos x term.”

Source: 9709_w22_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 3

Coaching With Shuja

Find the exact value of $\int_0^{\pi/4} x \sec^2 x \, dx$. **[5]**

Coaching With Shuja

Commence integration by parts and reach $x \tan x \pm \int \tan x . 1 dx$ **M1**

Use a correct method to integrate $\tan x$ **M1**

Obtain integral $x \tan x - \ln \sec x$, or equivalent **A1**

Use limits correctly, having integrated twice **DM1**

Obtain answer $\frac{1}{4}\pi - \frac{1}{2}\ln 2$, or exact equivalent **A1**

Total: 5

Coaching With Shuja

Question 5

Oct/Nov 2022 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "(b) This proved challenging even for those with the required correct diagram, and many candidates made no attempt, seemingly being unable to locate the point P that would give the greatest argument."

Source: 9709_w22_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 5

Coaching With Shuja

(a) On a sketch of an Argand diagram, shade the region whose points represent complex numbers z satisfying the inequalities $|z + 2| < 2$ and $\text{Im}\, z > 1$. **[4]**

(b) Find the greatest value of $\arg z$ for points in the shaded region. **[2]**

Coaching With Shuja

(a) Show a circle with centre -2 **B1**

Show a circle with radius 2 and centre not the origin **B1**

Show the line $y = 1$ **B1**

Shade the correct region **B1**

Total: 4

(b) Identify the correct point and carry out a correct method to find the argument **M1**

Obtain answer $\frac{11}{12}\pi$ **A1** 2.88 radians or 165°.

Total: 2

Coaching With Shuja

Question 9

Oct/Nov 2022 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> “Question 9 (a) Many candidates did not seem well prepared to tackle this vector question.”

> “A large number of marks were lost throughout this question through carelessness and inaccuracies.”

> “Common errors υυυρ υυυρ υυυρ arose through using OA–OB for vector AB, confusing the position vector of the midpoint M of υυυυρ υυυυρ υυυρ υυυρ AC, OM , with the vector AM , or misunderstanding what the statement BN =2NC meant about the position of N.”

Source: 9709_w22_er.pdf, Page 30, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

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Question Paper Oct/Nov 2022 • Paper 3 (variant 33) • Q9

With respect to the origin O, the position vectors of the points A, B and C are given by

$$\overrightarrow{OA} = \begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 4 \\ -3 \\ -2 \end{pmatrix}$$

The midpoint of AC is M and the point N lies on BC, between B and C, and is such that $BN = 2NC$.

(a) Find the position vectors of M and N. **[3]**

(b) Find a vector equation for the line through M and N. **[2]**

(c) Find the position vector of the point Q where the line through M and N intersects the line through A and B. **[4]**

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(a) State $\overrightarrow{OM} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ **B1**

Use a correct method to find $\overrightarrow{ON}$ **M1**

Obtain answer $\begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$ **A1**

Total: 3

(b) Carry out a correct method to form a vector equation for MN **M1**

Obtain a correct equation in any form, e.g. $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}$ **A1 OE**

Total: 2

(c) [Detailed marking scheme provided in MS, involves finding intersection of two lines through parameter equations]

Coaching with Shuja

Question 10

Oct/Nov 2022 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "(b) This question proved very challenging for many candidates who were unable to separate the variables correctly."

Source: 9709_w22_er.pdf, Page 31, Section: Comments on specific questions – Paper 3 (variant 33), Question 10

Coaching With Shuja

Question Paper Oct/Nov 2022 • Paper 3 (variant 33) • Q10

A gardener is filling an ornamental pool with water, using a hose that delivers 30 litres of water per minute. Initially the pool is empty. At time t minutes after filling begins the volume of water in the pool is V litres. The pool has a small leak and loses water at a rate of 0.01V litres per minute.

The differential equation satisfied by V and t is of the form $\frac{dV}{dt} = a - bV$.

(a) Write down the values of the constants a and b. **[1]**

(b) Solve the differential equation and find the value of t when $V = 1000$. **[6]**

(c) Obtain an expression for V in terms of t and hence state what happens to V as t becomes large. **[2]**

Coaching With Shuja

(a) $a = 30$ and $b = 0.01$ **B1**

Total: 1

(b) Separate variables and integrate one side **M1**

Obtain terms $-100\ln(30 - 0.01V)$ and t, or equivalent **A1 FT + A1 FT** FT *their* a and b.

Evaluate a constant, or use $t = 0$, $V = 0$ as limits, in a solution containing terms $c\ln(30 - 0.01V)$ and dt where $cd \neq 0$ **M1**

Obtain solution $100\ln 30 - 100\ln(30 - 0.01V) = t$, or equivalent **A1**

Substitute $V = 1000$ and obtain answer $t = 40.5$ **A1**

Total: 6

(c) Obtain $V = 3000(1 - e^{-0.01t})$ **B1 OE**

State that V approaches 3000 **B1**

Total: 2

Coaching With Shuja

Question 1

Feb/Mar 2021 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "(b) The use of (−2x) in the expansion made this question challenging for some candidates, however, many correct answers were seen to this part."

Source: 9709_m21_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 12), Question 1

Question Paper

Feb/Mar 2021 • Paper 1 (variant 12) • Q1

(a) Find the first three terms in the expansion, in ascending powers of x, of $(1+x)^5$. [1]

(b) Find the first three terms in the expansion, in ascending powers of x, of $(1-2x)^6$. [2]

(c) Hence find the coefficient of x^2 in the expansion of $(1+x)^5(1-2x)^6$. [2]

Coaching With Shuja

(a)

$1 + 5x + 10x^2$ **B1**

Total: 1

(b)

$1 - 12x + 60x^2$ **B2,1,0** B2 all correct, B1 for two correct components.

Total: 2

(c)

$(1 + 5x + 10x^2)(1 - 12x + 60x^2)$ leading to $60 - 60 + 10$ **M1** 3 products required

10 **A1** Allow $10x^2$

Total: 2

Question 8

Feb/Mar 2021 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "© 2021 Cambridge International Advanced Subsidiary and Advanced Level 9709 Mathematics March 2021 Principal Examiner Report for Teachers (b) This was the most frequently omitted part of a question."

Source: 9709_m21_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

Coaching With Shuja

Question Paper Feb/Mar 2021 • Paper 1 (variant 12) • Q8

The points $A(7, 1)$, $B(7, 9)$ and $C(1, 9)$ are on the circumference of a circle.

(a) Find an equation of the circle. [5]

(b) Find an equation of the tangent to the circle at B. [2]

Coaching With Shuja

(a)

Centre of circle is $(4, 5)$ **B1 B1**

$r^2 = (7-4)^2 + (1-5)^2$ **M1** OE. Either using their centre and A or C or using A and C and dividing by 2.

$r = 5$ **A1 FT** FT on their $(4, 5)$ if used.

Equation is $(x-4)^2 + (y-5)^2 = 25$ **A1** OE. Allow 5^2 for 25.

Total: 5

(b)

Gradient of radius $= \frac{9-5}{7-4} = \frac{4}{3}$ **B1 FT** FT for use of their centre.

Equation of tangent is $y - 9 = -\frac{3}{4}(x-7)$ **B1** or $y = -\frac{3x}{4} + \frac{57}{4}$

Total: 2

Coaching with Shuja

Question 10

Feb/Mar 2021 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "The use of radians proved difficult for a minority of candidates."

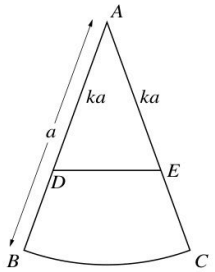
> "(b) This part was the most frequently omitted and the generalisation (ka)2 sinθ= 1 a2θ was rarely 2 seen."

> "Candidates who were able to see this from their working in part (a) usually went on to find a suitable inequality but few completely correct solutions were seen because negative values of k were often then included."

Source: 9709_m21_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

Coaching With Shuja

10



The diagram shows a sector ABC which is part of a circle of radius a. The points D and E lie on AB and AC respectively and are such that $AD = AE = ka$, where $k < 1$. The line DE divides the sector into two regions which are equal in area.

(a) For the case where angle $BAC = \frac{1}{6}\pi$ radians, find k correct to 4 significant figures. [5]

The diagram shows a sector $\boldsymbol{ABC}$ which is part of a circle of radius $\boldsymbol{a}$. The points $\boldsymbol{D}$ and $\boldsymbol{E}$ lie on $\boldsymbol{AB}$ and $\boldsymbol{AC}$ respectively and are such that $\boldsymbol{AD = AE = ka}$, where $\boldsymbol{k < 1}$. The line $\boldsymbol{DE}$ divides the sector into two regions which are equal in area.

(a) For the case where angle $\boldsymbol{BAC = \frac{1}{6}\pi}$ radians, find $\boldsymbol{k}$ correct to 4 significant figures. [5]

(b) For the general case in which angle $\boldsymbol{BAC = \theta}$ radians, where $\boldsymbol{0 < \theta < \frac{1}{2}\pi}$, it is given that $\boldsymbol{\frac{\theta}{\sin\theta} > 1}$.

Find the set of possible values of $\boldsymbol{k}$. [3]

(a)

$\triangle ADE = \frac{1}{2}(ka)^2 \sin\frac{\pi}{6}$ **M1** Attempt to find the area of $\triangle ADE$.

$\frac{1}{4}k^2a^2$ **A1** OE

Sector $ABC = \frac{1}{2}a^2\frac{\pi}{6}$ **B1**

$2 \times \frac{1}{4}k^2a^2 = \frac{1}{2}a^2\frac{\pi}{6}$ **M1** OE. For $2 \times \triangle ADE =$ sector ABC with at least one correct area.

$k = \sqrt{\frac{\pi}{6}} = 0.7236$ **A1**

Total: 5

(b)

$2 \times \frac{1}{2}(ka)^2 \sin\theta = \frac{1}{2}a^2\theta$ **M1**

$k^2 = \frac{\theta}{2\sin\theta}$ **A1**

$k^2 > \frac{1}{2}$ leading to $\frac{1}{\sqrt{2}} < k < 1$ **A1** OE. Accept $k > \frac{1}{\sqrt{2}}$ or $k > 0.707$ AWRT, with $k < 1$.

Total: 3

Coaching With Shuja

Question 1

Feb/Mar 2021 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "However, a few candidates mistakenly gave their final answer correct to 3 decimal places instead of 3 significant figures, as demanded by the question."

Source: 9709_m21_er.pdf, Page 5, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

Solve the equation $\ln(x^3 - 3) = 3\ln x - \ln 3$. Give your answer correct to 3 significant figures. [3]

Coaching With Shuja

Mark Scheme

Use law of the logarithm of a product or power **M1**

Obtain a correct equation free of logarithms, e.g. $x^3 - 3 = \frac{x^3}{3}$ **A1**

Obtain $x = 1.65$ **A1**

Total: 3

Question 4

Feb/Mar 2021 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "(b) The majority of candidates found this question challenging."

Source: 9709_m21_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

The variables x and y satisfy the differential equation

$$(1 - \cos x)\frac{dy}{dx} = y \sin x.$$

It is given that $y = 4$ when $x = \pi$.

(a) Solve the differential equation, obtaining an expression for y in terms of x. [6]

(b) Sketch the graph of y against x for $0 < x < 2\pi$. [1]

Coaching With Shuja

(a)

Separate variables correctly and attempt integration of at least one side **M1**

Obtain term $\ln y$ **A1**

Obtain term of the form $\pm \ln(1 - \cos x)$ **M1**

Obtain term $\ln(1 - \cos x)$ **A1**

Use $x = \pi$, $y = 4$ to evaluate a constant, or as limits, in a solution containing terms of the form $a \ln y$ and $b \ln(1 - \cos x)$ **M1**

Obtain final answer $y = 2(1 - \cos x)$ **A1** OE

Total: 6

(b)

Show a correct graph for $0 < x < 2\pi$ with the maximum at $x = \pi$ **B1 FT**

Total: 1

Coaching with Shuja

Question 5

Feb/Mar 2021 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Some candidates did not introduce 2θ, ☐ 11☐ continuing with x, and then struggled with the rest of the question."

Source: 9709_m21_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

Coaching With Shuja

Question Paper Feb/Mar 2021 • Paper 3 (variant 32) • Q5

(a) Express $\sqrt{7}\sin x + 2\cos x$ in the form $R\sin(x+\alpha)$, where $R > 0$ and $0° < \alpha < 90°$. State the exact value of R and give α correct to 2 decimal places. [3]

(b) Hence solve the equation $\sqrt{7}\sin 2\theta + 2\cos 2\theta = 1$, for $0° < \theta < 180°$. [5]

Coaching With Shuja

(a)

State $R = \sqrt{11}$ **B1**

Use trig formulae to find α **M1**

Obtain $\alpha = 37.09°$ **A1**

Total: 3

(b)

Evaluate $\sin^{-1}\left(\frac{1}{\sqrt{11}}\right)$ to at least 2 d.p. $(17.5484°)$ **B1 FT**

Use correct method to find a value of θ in the interval **M1**

Obtain answer, e.g. $62.7°$ **A1**

Use a correct method to obtain a second answer **M1**

Obtain second answer, e.g. $170.2°$, and no other in the interval **A1**

Total: 5

Coaching With Shuja

Question 6

Feb/Mar 2021 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 6 (a) Candidates found this partial fraction question to be more challenging, likely due to the presence of a the constant a."

> "Those who multiplied out and equated coefficients, or set x = 3a and , were 2 usually successful, but those who set x to just a numerical value usually struggled to then correctly find the constants in their partial fractions."

Source: 9709_m21_er.pdf, Page 6, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

Question Paper Feb/Mar 2021 • Paper 3 (variant 32) • Q6

Let $f(x) = \frac{5a}{(2x-a)(3a-x)}$, where a is a positive constant.

(a) Express $f(x)$ in partial fractions. [3]

(b) Hence show that $\int_a^{2a} f(x)\,dx = \ln 6$. [4]

Coaching With Shuja

(a)

Carry out a relevant method to determine constants A and B such that $\frac{5a}{(2x-a)(3a-x)} = \frac{A}{2x-a} + \frac{B}{3a-x}$ **M1**

Obtain $A = 2$ **A1**

Obtain $B = 1$ **A1**

Total: 3

(b)

Integrate and obtain terms $\ln(2x-a)$ and $-\ln(3a-x)$ **B1 FT B1 FT**

Substitute limits correctly in a solution containing terms of the form $b\ln(2x-a)$ and $c\ln(3a-x)$, where $bc \neq 0$ **M1**

Obtain the given answer showing full and correct working **A1**

Total: 4

Coaching With Shuja

Question 7

Feb/Mar 2021 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 7 (a) Most candidates knew how to show that the two lines did not intersect, although occasionally they had an arithmetical error present."

> "Few candidates were able to undertake this successfully, since they either used the position vectors or line vectors as opposed to the direction vectors, and related that they needed to not be equal to each other rather than not be a multiple of each other."

Source: 9709_m21_er.pdf, Page 7, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

Two lines have equations

$$\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} + s \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

and

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}.$$

(a) Show that the lines are skew. [5]

(b) Find the acute angle between the directions of the two lines. [3]

Coaching With Shuja

(a)

Express general point of a line in component form, e.g. $(1+2s, 3-s, 2+3s)$ or $(2+t, 1-t, 4+4t)$ **B1**

Equate at least two pairs of components and solve for s or for t **M1**

Obtain correct answer for s or for t **A1**

Verify that all three component equations are not satisfied **A1**

Show that the lines are not parallel and are thus skew **A1**

Total: 5

(b)

Carry out correct process for evaluating the scalar product of the direction vectors **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and evaluate the inverse cosine of the result **M1**

Obtain answer $19.1°$ or 0.333 radians **A1**

Total: 3

Question 9

Feb/Mar 2021 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "However, those candidates who found the four values of x and f(x) at x = 1 and x = 1.5 struggled with this question, as they often omitted any comparison, or, if they did undertake one, they did it between the x values and between the f(x) values instead of between x =1 and f(1) and between 1.5 and f(1.5)."

Source: 9709_m21_er.pdf, Page 7, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Let $f(x) = \frac{e^{2x}+1}{e^{2x}-1}$, for $x > 0$.

(a) The equation $x = f(x)$ has one root, denoted by α. Verify by calculation that α lies between 1 and 1.5. [2]

(b) Use an iterative formula based on the equation in part (a) to determine α correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

(c) Find $f'(x)$. Hence find the exact value of x for which $f'(x) = -8$. [6]

Coaching With Shuja

(a)

Calculate the values of a relevant expression or pair of expressions at $x = 1$ and $x = 1.5$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(b)

Use the iterative formula $x_{n+1} = \frac{e^{2x_n}+1}{e^{2x_n}-1}$, or equivalent, correctly at least once **M1**

Obtain final answer **1.20** **A1**

Show sufficient iterations to 4 d.p. to justify **1.20** to 2 d.p., or show there is a sign change in $(1.195, 1.205)$ **A1**

Total: 3

(c)

Use quotient rule **M1**

Obtain correct derivative in any form **A1**

Equate derivative to -8 and obtain a quadratic in e^{2x} **M1**

Obtain $2(e^{2x})^2 - 5e^{2x} + 2 = 0$ **A1** OE

Solve a 3-term quadratic in e^{2x} for x **M1**

Obtain answer $x = \frac{1}{2}\ln 2$, or exact equivalent, only **A1**

Total: 6

Coaching With Shuja

Question 10

Feb/Mar 2021 | Paper 3 (variant 32)

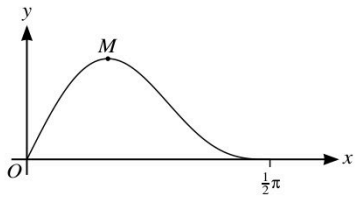
Examiner Report — Why students found this question difficult

> "Question 10 (a) This question was well answered by many candidates, although several did not establish a correct integral in terms of u."

Source: 9709_m21_er.pdf, Page 8, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

10



The diagram shows the curve $y = \sin 2x \cos^2 x$ for $0 \leqslant x \leqslant \frac{1}{2}\pi$, and its maximum point M.

(a) Using the substitution $u = \sin x$, find the exact area of the region bounded by the curve and the x-axis. [5]

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(a) Using the substitution $u = \sin x$, find the exact area of the region bounded by the curve and the x-axis. [5]

(b) Find the exact x-coordinate of M. [6]

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(a)

State or imply $du = \cos x\, dx$ **B1**

Using double angle formula for $\sin 2x$ and Pythagoras, express integral in terms of u and du **M1**

Obtain integral $\int (2u - 2u^3)\, du$ **A1** OE

Use limits $u = 0$ and $u = 1$ in an integral of the form $au^2 + bu^4$, where $ab \neq 0$ **M1**

Obtain answer $\frac{1}{2}$ **A1**

Total: 5

(b)

Use product rule **M1**

Obtain correct derivative in any form **A1**

Equate derivative to zero and use a double angle formula **M1**

Obtain an equation in one trig variable **DM1**

Obtain $4\sin^2 x = 1$, $4\cos^2 x = 3$ or $3\tan^2 x = 1$ **A1**

Obtain answer $x = \frac{1}{6}\pi$ **A1**

Total: 6

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Question 2

May/June 2021 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "A common error was to find the sum of 60 terms instead of the 60th term."

Source: 9709_s21_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 11), Question 2

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Question Paper May/June 2021 • Paper 1 (variant 11) • Q2

The sum of the first 20 terms of an arithmetic progression is 405 and the sum of the first 40 terms is 1410.

Find the 60th term of the progression. **[5]**

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$10(2a + 19d) = 405$ **B1**

$20(2a + 39d) = 1410$ **B1**

Solving simultaneously two equations obtained from using the correct sum formulae $[a = 6, d = 1.5]$ **M1**

Reach $a =$ or $d =$

Using the correct formula for 60th term with their a and d **M1**

60th term $= 94.5$ **A1 OE**, e.g. $\frac{189}{2}$

Total: 5

Question 4

May/June 2021 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

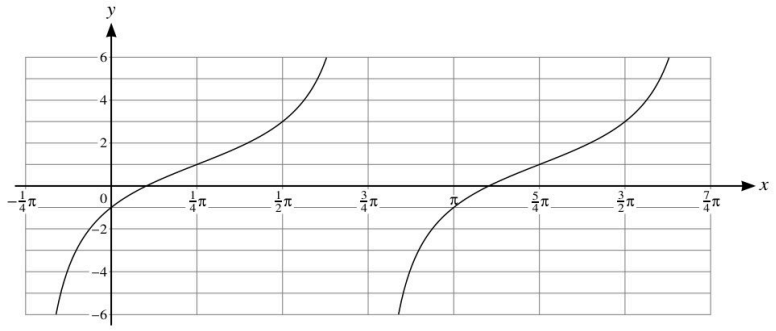
> "Most candidates attempted this question but very few candidates obtained fully correct answers."

Source: 9709_s21_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 4

Coaching With Shuja

Question 4

4



The diagram shows part of the graph of $y = a\tan(x - b) + c$.

Given that $0 < b < \pi$, state the values of the constants a, b and c. [3]

The diagram shows part of the graph of $y = a\tan(x - b) + c$.

Given that $0 < b < \pi$, state the values of the constants a, b and c. [3]

Question 4

M1 $a = 2$ **B1**

M1 $b = \frac{\pi}{4}$ **B1** or $\frac{2\pi}{8}$

A1 $c = 1$ **B1**

Total: 3

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Question 7

May/June 2021 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "A common error was to write 1– 2sin2θ 2sin2θ = 1 – then omit the 2 in order to reach the given answer."

Source: 9709_s21_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 7

Coaching With Shuja

Question Paper May/June 2021 • Paper 1 (variant 11) • Q7

(a) Prove the identity $\frac{1-2\sin^2\theta}{1-\sin^2\theta} \equiv 1 - \tan^2\theta$. [2]

(b) Hence solve the equation $\frac{1-2\sin^2\theta}{1-\sin^2\theta} = 2\tan^4\theta$ for $0° \leq \theta \leq 180°$. [3]

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(a) Reach $\frac{\cos^2\theta-\sin^2\theta}{\cos^2\theta}$ **or** $\frac{1-\sin^2\theta}{\cos^2\theta}$ **or** $\frac{\sin^2\theta+\cos^2\theta}{\cos^2\theta} - 2\tan^2\theta$ **or** $\sec^2\theta - \frac{2\sin^2\theta}{\cos^2\theta}$ **or** $2 - \sec^2\theta$ **or** $\frac{\cos 2\theta}{\cos^2\theta}$ **M1**
May start with $1 - \tan^2\theta$

$1 - \tan^2\theta$ **A1** AG, must show sufficient stages

Total: 2

(b) $1 - \tan^2\theta = 2\tan^4\theta \Rightarrow 2\tan^4\theta + \tan^2\theta - 1 = [0]$ **M1** Forming a 3-term quadratic in $\tan^2\theta$ or e.g. u

$\tan^2\theta = 0.5$ or -1 leading to $\tan\theta = [\pm]\sqrt{0.5}$ **M1**

$\theta = 35.3°$ and $144.7°$ (AWRT) **A1** Both correct. Radians 0.615, 2.53 scores A0.

Total: 3

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Question 8

May/June 2021 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "A common mistake 3 was to omit π in the circumference of the semicircles."

> "(c) This question proved to be challenging for most candidates."

Source: 9709_s21_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 8

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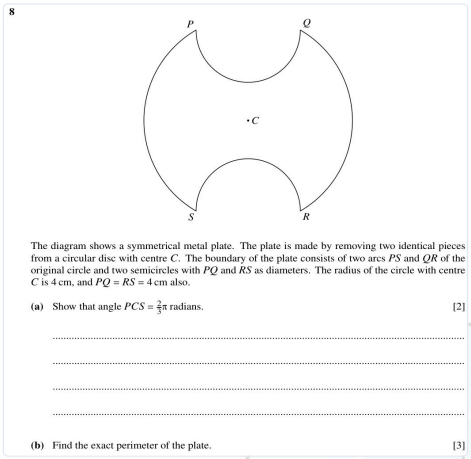
Question 8



The diagram shows a symmetrical metal plate. The plate is made by removing two identical pieces from a circular disc with centre C. The boundary of the plate consists of two arcs PS and QR of the original circle and two semicircles with PQ and RS as diameters. The radius of the circle with centre C is 4 cm, and $PQ = RS = 4$ cm also.

(a) Show that angle $PCS = \frac{2\pi}{3}$ radians. [2]

(b) Find the exact perimeter of the plate. [3]

(c) Show that the area of the plate is $\left(\frac{20\pi}{3} + 8\sqrt{3}\right)$ cm^2. [5]

(a) Either Let midpoint of PQ be H: $\sin HCP = \frac{2}{4} \Rightarrow$ Angle $HCP = \frac{\pi}{6}$ **M1**

Or $\sin PSQ = \frac{4}{8} \Rightarrow$ Angle $PSQ = \frac{\pi}{6}$ **M1**

Or using cosine rule: angle $PCQ = \frac{\pi}{3}$ **M1**

Or by inspection: triangle PCQ or PCT is equilateral so angle $PCQ = \frac{\pi}{3}$ **M1**

Angle $PCS = \pi - \frac{\pi}{6} - \frac{\pi}{6} = \frac{2\pi}{3}$ **A1** AG

Total: 2

(b) Perimeter $= 2 \times 4 \times \frac{2\pi}{3}$ or $8\pi - \frac{8\pi}{3}$ **M1** Length of two arcs PS and QR

$+2\pi \times 2$ **M1** Adding circumference of two semicircles

$\frac{28\pi}{3}$ **A1** Must be a single term

Total: 3

(c) Area sector $CPQ = \frac{1}{2} \times 4^2 \times \frac{\pi}{3} = \frac{8\pi}{3}$ **M1** Uses correct formula for sector

Area of segment of large circle beyond $CPQ = \frac{8\pi}{3} - \frac{1}{2} \times 4^2 \times \sin\left(\frac{\pi}{3}\right) = \frac{8\pi}{3} - 4\sqrt{3}$ **M1** Attempts to find area of segment

Area of small semicircle $= \pi \times 2$ or area of small circle $= \pi \times 2^2$ **M1**

Area of plate = Large circle $-[2\times]$ small semicircle $-[2\times]$ segment area **M1**

$\pi \times 4^2 - \pi \times 2^2 - 2 \times \left(\frac{8\pi}{3} - 4\sqrt{3}\right) = \frac{20\pi}{3} + 8\sqrt{3}$ **A1** AG

Alternative method for Question 8(c)

Area of sector $PCS = \frac{1}{2} \times 4^2 \times \frac{2\pi}{3} = \frac{16\pi}{3}$ **M1** Uses correct formula for sector

Area of triangle $PCQ = \frac{1}{2} \times 4^2 \times \sin\left(\frac{\pi}{3}\right) = 4\sqrt{3}$ **M1** Uses correct formula for triangle

Area of small semicircle $= \pi \times 2$ or area of circle $= \pi \times 2^2$ **M1**

Area of plate $= [2\times]$ large sector $+[2\times]$ triangle $-[2\times]$ small semicircle **M1**

$2\left(\frac{16\pi}{3}\right) + 2\left(4\sqrt{3}\right) - \pi \times 2^2 = \frac{20\pi}{3} + 8\sqrt{3}$ **A1** AG

Total: 5

Question 9

May/June 2021 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Common errors in incorrect responses included omitting to use a strict inequality and using x instead of y or f(x) to denote the range."

> "However, many candidates did not realise that only the answer x = 3 was valid because of the restriction on the domain of f(x), hence they could not be awarded the final mark."

Source: 9709_s21_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 9

Coaching With Shuja

Question 9

Functions f and g are defined as follows:

$$f(x) = (x-2)^2 - 4 \text{ for } x \geq 2,$$

$$g(x) = ax + 2 \text{ for } x \in \mathbb{R},$$

where a is a constant.

(a) State the range of f. [1]

(b) Find $f^{-1}(x)$. [2]

(c) Given that $a = -\frac{5}{3}$, solve the equation $f(x) = g(x)$. [3]

(d) Given instead that $gff^{-1}(12) = 62$, find the possible values of a. [5]

(a) Range of f is $f(x) \geq -4$ **B1** Allow y, f 'range' or $[-4, \infty)$

Total: 1

(b) $y = (x-2)^2 - 4 \Rightarrow (x-2)^2 = y+4 \Rightarrow x - 2 = \pm\sqrt{(y+4)}$ or $\pm\sqrt{(y+4)}$ **M1** May swap variables here

$[f^{-1}(x)] = \sqrt{(x+4)} + 2$ **A1**

Total: 2

(c) $(x - 2)^2 - 4 = -\frac{5}{3}x + 2 \Rightarrow x^2 - 4x + 4 - 4 = -\frac{5}{3}x + 2 [\Rightarrow x^2 - \frac{7}{3}x - 2 = 0]$ **M1** Equating and simplifying to a 3-term quadratic

$(3x+2)(x-3) = [0]$ or $\frac{7 \pm \sqrt{7^2 - 4(3)(-6)}}{6}$ **M1** Solving quadratic

$x = 3$ only **A1**

Total: 3

(d) $(3(3x+4))^{-0.5} - 1 = 0$ or $3(3x+4)^{-0.5} - 1 = 0$ **M1** Setting their $\frac{dy}{dx} = 0$

Solving as far as $x =$ **M1** Where $\frac{dy}{dx}$ contains $a(bx+c)^{-0.5}$, a, b, c any values

$x = \frac{5}{3}, y = 2(3x+4)^{0.5} - \frac{5}{3}x - \frac{13}{3}$ **A1**

Total: 3

Coaching With Shuja

Question 10

May/June 2021 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Question 10 (a) Many candidates did not realise the need to substitute y = 0 into the circle equation in order to find where the tangents cross the x-axis."

Source: 9709_s21_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 10

Question 10

The equation of a circle is $x^2 + y^2 - 4x + 6y - 77 = 0$.

(a) Find the x-coordinates of the points A and B where the circle intersects the x-axis. [2]

(b) Find the point of intersection of the tangents to the circle at A and B. [6]

(a) When $y = 0$: $x^2 - 4x - 77 = 0$ [$\Rightarrow (x+7)(x-11) = 0$ or $(x-2)^2 = 81$] **M1** Substituting $y = 0$

So x-coordinates are -7 and 11 **A1**

Total: 2

(b) Centre of circle C is $(2, -3)$ **B1**

Gradient of AC is $-\frac{1}{3}$ or Gradient of BC is $\frac{1}{3}$ **M1** For either gradient (M1 sign error, M0 if x-coordinate(s) in numerator)

Gradient of tangent at A is 3 or Gradient of tangent at B is -3 **M1** For either perpendicular gradient

Equations of tangents are $y = 3x + 21, y = -3x + 33$ **A1** For either equation

Meet when $3x + 21 = -3x + 33$ **M1** OR: (centre of circle has x coordinate 2) so x coordinate of point of intersection is 2

Coordinates of point of intersection $(2, 27)$ **A1**

Alternative method for Question 10(b)

Implicit differentiation: $2y\frac{dy}{dx}$ seen **B1**

$2x - 4 + 2y\frac{dy}{dx} + 6\frac{dy}{dx} = 0$ **M1** Fully differentiated $= 0$ with at least one term involving y differentiated correctly

Gradient of tangent at A is 3 or Gradient of tangent at B is -3 **M1** For either gradient

Equations of tangents are $y = 3x + 21, y = -3x + 33$ **A1** For either equation

Meet when $3x + 21 = -3x + 33$ **M1** OR: (centre of circle has x coordinate 2) so x coordinate of point of intersection is 2

Coordinates of point of intersection $(2, 27)$ **A1**

Total: 6

Question 11

May/June 2021 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "A common error seen was to omit the coefficient of 3 (not applying the chain rule)."

> "Most candidates showed their substitution of the limits clearly, though a common error was to assume the limit x = 0 gave a value of 0 for the expression."

Source: 9709_s21_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 11), Question 11

Coaching With Shuja

Question 11

The equation of a curve is $y = 2\sqrt{3x+4} - x$.

(a) Find the equation of the normal to the curve at the point $(4, 4)$, giving your answer in the form $y = mx + c$. [5]

(b) Find the coordinates of the stationary point. [3]

(c) Determine the nature of the stationary point. [2]

(d) Find the exact area of the region bounded by the curve, the x-axis and the lines $x = 0$ and $x = 4$. [4]

Mark Scheme May/June 2021 • Paper 1 (variant 11) • Q11

(a) $\frac{dy}{dx} = 3(3x+4)^{-0.5} - 1$ **B1 B1** All correct with 1 error, B2 if all correct

Gradient of tangent $= -\frac{1}{4}$ and Gradient of normal $= 4$ *M1 Substituting $x = 4$ into a differentiated expression and using $m_1 m_2 = -1$

Equation of line is $(y-4) = 4(x-4)$ or evaluate c **DM1** With $(4,4)$ and their gradient of normal

So $y = 4x - 12$ **A1**

Total: 5

(b) $3(3x+4)^{-0.5} - 1 = 0$ **M1** Setting their $\frac{dy}{dx} = 0$

Solving as far as $x =$ **M1** Where $\frac{dy}{dx}$ contains $a(bx+c)^{-0.5}$, a, b, c any values

$x = \frac{5}{3}, y = 2(3x+4)^{0.5} - \frac{5}{3}x - \frac{13}{3}$ **A1**

Total: 3

(c) $\frac{d^2y}{dx^2} = -\frac{9}{2}(3x+4)^{-1.5}$ **M1** Differentiating their $\frac{dy}{dx}$ OR checking $\frac{dy}{dx}$ to find $+ve$ and $-ve$ either side of their $x = \frac{5}{3}$

At $x = \frac{5}{3}$, $\frac{d^2y}{dx^2}$ is negative so the point is a maximum **A1**

Total: 2

(d) Area $= \left[2(3x+4)^{0.5} - x\right] - \frac{4}{9}(3x+4)^{1.5} - \frac{1}{2}x^2$ **B1 B1** For each correct term (unsimplified)

$\left(\frac{4}{9}(16)^{1.5} - \frac{1}{2}(4)^2\right) - \frac{4}{9}(4)^{1.5} = \frac{256}{9} - 8 - \frac{32}{9}$ **M1** Substituting limits 0 and 4 into an expression obtained by integrating y

$\frac{16}{8}$ or $\frac{152}{9}$ **A1** Or

Total: 4

Question 3

May/June 2021 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Candidates found part (c) to be more challenging with many candidates either omitting it or talking about a stationary point or an increasing function."

Source: 9709_s21_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

Question 3

The equation of a curve is $y = (x - 3)\sqrt{x + 1} + 3$. The following points lie on the curve. Non-exact values are rounded to 4 decimal places.

$$A(2, k) \quad B(2.9, 2.8025) \quad C(2.99, 2.9800) \quad D(2.999, 2.9980) \quad E(3, 3)$$

(a) Find k, giving your answer correct to 4 decimal places. [1]

(b) Find the gradient of AE, giving your answer correct to 4 decimal places. [1]

The gradients of BE, CE and DE, rounded to 4 decimal places, are 1.9748, 1.9975 and 1.9997 respectively.

(c) State, giving a reason for your answer, what the values of the four gradients suggest about the gradient of the curve at the point E. [2]

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(a) 1.2679 B1 AWRT. ISW if correct answer seen. $3 - \sqrt{3}$ scores B0

Total: 1

(b) 1.7321 B1 AWRT. ISW if correct answer seen.

Total: 1

(c) Sight of 2 or 2.0000 or two in reference to the gradient ***B1**

This is because the gradient at E is the limit of the gradients of the chords as the x-value tends to 3 or Δx tends to 0. **DB1** Allow it gets nearer/approaches/tends/almost/approximately 2

Total: 2

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Question 7

May/June 2021 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Some candidates took the more difficult approach of solving simultaneously the equation of the circle with line l and a few attempted implicit differentiation, however these approaches were rarely successful."

> "Many candidates did not appear to understand what was required in part (b) and gave either the equation of the original circle, the equation of a circle centre A or the equation of it's perpendicular."

Source: 9709_s21_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 7

Coaching With Shuja

Question 7

The point A has coordinates $(1, 5)$ and the line l has gradient $-\frac{2}{3}$ and passes through A. A circle has centre $(5, 11)$ and radius $\sqrt{52}$.

(a) Show that l is the tangent to the circle at A. [2]

(b) Find the equation of the other circle of radius $\sqrt{52}$ for which l is also the tangent at A. [3]

Coaching With Shuja

(a) $(5-1)^2+(11-5)^2=52$ or $\frac{11-5}{5-1}$ **M1** For substituting $(1,5)$ into circle equation or showing gradient $=\frac{3}{2}$

For both circle equation and gradient, and proving line is perpendicular and stating that A lies on the circle **A1** Clear reasoning.

Alternative method for Question 7(a)

$(x-5)^2+(y-11)^2=52$ and $y=-\frac{2}{3}(x-1)$ **M1** Both equations seen and attempt to solve.

Solving simultaneously to obtain $(y-5)^2=0$ or $(x-1)^2=0 \Rightarrow 1$ root or tangent **A1** Clear reasoning.

Alternative method for Question 7(a)

$\frac{dy}{dx}=\frac{10-2x}{2y-22}=\frac{10-2}{10-22}$ **M1** Attempting implicit differentiation of circle equation and substitute $x=1$ and $y=5$,

Showing gradient of circle at A is $\frac{3}{2}$ **A1** Clear reasoning.

Total: 2

(b) Centre is $(-3,-1)$ **B1 B1** B1 for each correct co-ordinate.

Equation is $(x+3)^2+(y+1)^2=52$ **B1 FT** FT their centre, but not if either $(1,5)$ or $(5,11)$. Do not accept $\sqrt{52}^2$.

Total: 3

Question 12

May/June 2021 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 12 Many candidates found this question to be challenging and either omitted it or were unable to make much progress."

> "Those who had found part (c) to be particularly challenging often found part (d) to be difficult also."

Source: 9709_s21_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 12

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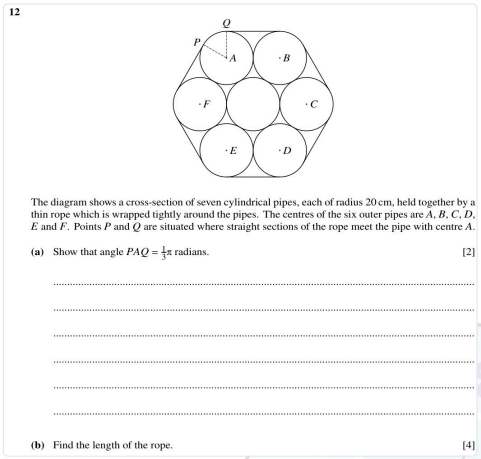
Question 12



The diagram shows a cross-section of seven cylindrical pipes, each of radius 20 cm, held together by a thin rope which is wrapped tightly around the pipes. The centres of the six outer pipes are A, B, C, D, E and F. Points P and Q are situated where straight sections of the rope meet the pipe with centre A.

(a) Show that angle $PAQ = \frac{\pi}{3}$ radians. **[2]**

(b) Find the length of the rope. **[4]**

(c) Find the area of the hexagon $ABCDEF$, giving your answer in terms of $\sqrt{3}$. **[2]**

(d) Find the area of the complete region enclosed by the rope. **[3]**

Mark Scheme May/June 2021 • Paper 1 (variant 12) • Q12

12(a)

M1 [By symmetry] $[6 \times P\hat{A}Q = 2\pi]$, $[P\hat{A}Q =]2\pi \div 6$

A1 AG Explaining that there are six sectors around the diagram that make up a complete circle.

Alternative method for Question 12(a)

M1 Using area or circumference of circle centre $A \div 6$: $\frac{400\pi}{6}$ or $\frac{40\pi}{6}$

A1 AG Justification for dividing by 6 followed by comparison with the sector area or arc length.

Alternative method for Question 12(a)

M1 Explain why $\triangle PAQ$ is an equilateral triangle

A1 AG Using $\triangle PAQ$ is an equilateral triangle $\therefore P\hat{A}Q = \frac{\pi}{3}$

Alternative method for Question 12(a)

M1 Using the internal angle of a regular hexagon $= \frac{2\pi}{3}$ Or $F\hat{A}O + O\hat{A}B = \frac{2\pi}{3}$, equilateral triangles

A1 AG $P\hat{A}Q = 2\pi - (\frac{\pi}{2} + \frac{2\pi}{3} + \frac{\pi}{2}) = \frac{\pi}{3}$

Total: 2

12(a) - Alternative method

M1 $\sin\theta = \frac{20}{40}$, with θ clearly identified

A1 AG $\theta = \frac{\pi}{6}, 2\theta = \frac{\pi}{3} = F\hat{A}O$ and by similar triangles $= P\hat{A}Q$

Total: 2

12(b)

B1 SOI Each straight section of rope has length 40 cm

M1 Use of $r\theta$ with $r = 20$ and θ in radians: Each curved section round each pipe has length $r\theta = 20 \times \frac{\pi}{3}$

DM1 $6\times$ (their straight section + their curved section): Total length $= 6\times$ (their $40) + k\pi$

A1 $240 + 40\pi$ or 366 (AWRT) (cm)

Or directly: $(6\times$ diameter$)$ + circumference

Total: 4

12(c)

B1 $[\text{Triangle area} =]\frac{1}{2} \times 40 \times 40 \times \sin(\frac{\pi}{3})$ or $\frac{1}{2} \times 40 \times 20\sqrt{3}$ or $400\sqrt{3}$ or 693 (AWRT)

B1 $[\text{Total area of hexagon} = 6 \times 400\sqrt{3} =]2400\sqrt{3}$

Condone $4800\frac{\sqrt{3}}{2}$

Alternative method for Question 12(c)

B1 $[\text{Trapezium area} =]\frac{1}{2} \times (40 + 80) \times 40\sin(\frac{\pi}{3})$ or $1200\sqrt{3}$ or 2080 (AWRT)

B1 $[\text{Total area of hexagon} = 2 \times 1200\sqrt{3} =]2400\sqrt{3}$

Alternative method for Question 12(c)

B1 Area of triangle $ABC = 400\sqrt{3}$ or 693 (AWRT) or $4\times$ Area of half of triangle $ABC = 4 \times 200\sqrt{3}$ or 1390 (AWRT) or Area of rectangle $ABDE = 1600\sqrt{3}$ or 2770 (AWRT)

B1 $[\text{Total area of hexagon} = 2 \times 400\sqrt{3} + 1600\sqrt{3} =]2400\sqrt{3}$ Or $[= 4 \times 200\sqrt{3} + 1600 =]2400\sqrt{3}$

Condone $4800\frac{\sqrt{3}}{2}$

If BoBo, SC **B1** can be scored for sight of 4160 (AWRT) as final answer.

Total: 2

12(d)

B1 SOI, e.g. by sight of 4800: Each rectangle area $= 40 \times 20 (= 800)$

B1 SOI Each sector area $= \frac{1}{2}r^2\theta = \frac{1}{2} \times 20^2 \times \frac{\pi}{3} [= \frac{200\pi}{3}]$

B1 $2400\sqrt{3} + 4800 + 400\pi$ or $10,200$ (cm^2) (AWRT)

Or directly: part **(c)** $+6800$ + area circle radius 20.

Total: 3

Question 4

May/June 2021 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Complete solutions were not always seen because many candidates did not correctly identify the second solution or gave additional incorrect solutions in the given range."

Source: 9709_s21_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 4

Coaching With Shuja

4 (a) Show that the equation

$$\frac{\tan x + \sin x}{\tan x - \sin x} = k,$$

where k is a constant, may be expressed as

$$\frac{1 + \cos x}{1 - \cos x} = k.$$

[2]

(b) Hence express $\cos x$ in terms of k. **[2]**

(c) Hence solve the equation

$$\frac{\tan x + \sin x}{\tan x - \sin x} = 4$$

for $-\pi < x < \pi$. **[2]**

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4(a)

M1 Multiply numerator and denominator by $\cos x$, or divide numerator and denominator by $\tan x$ or $\sin x$:

$$\frac{\sin x + \sin x \cos x}{\sin x - \sin x \cos x} [= k]$$

A1 AG, WWW

$$\frac{\sin x(1 + \cos x)}{\sin x(1 - \cos x)}$$

or

$$\frac{\cos x \cdot \cos x}{\frac{1}{\cos x} - 1} \cos x$$

or

$$\frac{\tan x(1 + \cos x)}{\tan x(1 - \cos x)}$$

leading to

$$\frac{1 + \cos x}{1 - \cos x} [= k]$$

Total: 2

4(b)

M1 Gather like terms on LHS and RHS: $k - k\cos x = 1 + \cos x$

A1 WWW, OE

$$k - 1 = (k + 1)\cos x$$

leading to

$$\cos x = \frac{k - 1}{k + 1}$$

Total: 2

4(c)

M1 Obtaining $\cos x$ from their (b) or (a): Expect $\cos x = \frac{3}{5}$

A1 AWRT. Accept $\pm 0.927 (only solutions in the given range)$

Total: 2

Question 6

May/June 2021 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "(b) Many candidates stated the transformation to be a translation but did not provide a precise enough description of their translation."

Source: 9709_s21_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 6

Coaching With Shuja

6 Functions f and g are both defined for $x \in \mathbb{R}$ and are given by

$$f(x) = x^2 - 2x + 5,$$

$$g(x) = x^2 + 4x + 13.$$

(a) By first expressing each of $f(x)$ and $g(x)$ in completed square form, express $g(x)$ in the form $f(x + p) + q$, where p and q are constants. **[4]**

(b) Describe fully the transformation which transforms the graph of $y = f(x)$ to the graph of $y = g(x)$. **[2]**

Coaching With Shuja

6(a)

$f(x) = (x-1)^2 + 4$ **B1**

$g(x) = (x+2)^2 + 9$ **B1**

$g(x) = f(x+3) + 5$ **B1 B1** B1 for each correct element. Accept $p = 3, q = 5$

Total: 4

6(b)

Translation or Shift **B1**

$\begin{pmatrix} -3 \\ 5 \end{pmatrix}$

or acceptable explanation **B1 FT** If given as 2 single translations both must be described correctly e.g. $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$ & $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$

FT from their $f(x+p)+q$ or their $f(x) \rightarrow g(x)$

Do not accept $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ or $\begin{pmatrix} -2 \\ 9 \end{pmatrix}$

Total: 2

Coaching with Shuja

Question 10

May/June 2021 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "This was the most frequently omitted question part."

Source: 9709_s21_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 10

Coaching With Shuja

10 Points $A(-2, 3)$, $B(3, 0)$ and $C(6, 5)$ lie on the circumference of a circle with centre D.

(a) Show that angle $ABC = 90°$. **[2]**

(b) Hence state the coordinates of D. **[1]**

(c) Find an equation of the circle. **[2]**

The point E lies on the circumference of the circle such that BE is a diameter.

(d) Find an equation of the tangent to the circle at E. **[5]**

Coaching With Shuja

10(a)

M1 Attempting to find required gradients, sides or vectors: Gradient of $AB = -\frac{3}{5}$, gradient of $BC = \frac{5}{3}$ or lengths of all 3 sides or vectors

A1 WWW: $m_{ab} \cdot m_{bc} = -1$ or Pythagoras or $\vec{AB} \cdot \vec{BC} = 0$ or $\cos ABC = 0$ from cosine rule

Total: 2

10(b)

B1 Centre = mid-point of $AC = (2, 4)$

Total: 1

10(c)

M1 Use of circle equation with their centre: $(x - \text{their } x_c)^2 + (y - \text{their } y_c)^2 = [r^2]$ or (their $x_c - x)^2 + (\text{their } y_c - y)^2 = [r^2]$

A1 $(x - 2)^2 + (y - 4)^2 = 17$

Accept $x^2 - 4x + y^2 - 8y + 3 = 0$ OE

Total: 2

10(d)

M1 Use of mid-point formula, vectors, steps on a diagram:

$$\begin{pmatrix} x+3 \\ y+0 \end{pmatrix} = (2, 4)$$

or $BE = 2BD = 2\,(-4)$ or Equation of BE is $y = -4(x - 3)$ or $y - 4 = -4(x - 2)$ leading to $y = -4x + 12$

May be seen to find x coordinate at E

A1 $E = (1, 8)$ or OE $= (3) + (-2) = (1)$

Accept without working for both marks SC **B2**

B1 Gradient of BD, $m_s = -4$ or gradient $AC = \frac{1}{4}$ = gradient of tangent

M1 A1 For M1, equation of tangent is $y - 8 = \frac{1}{4}(x - 1)$ OE

For M1, equation through their E or $(1, 8)$ (not A, B or C) and with gradient $\frac{-1}{\text{their} -4}$

Total: 5

Question 11

May/June 2021 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

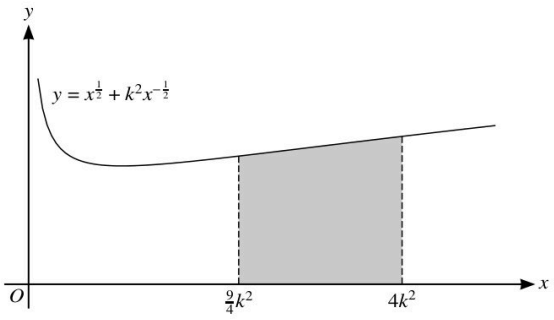
> "The power of was the source of error 2 for some; a common error was to only apply the power to k2 and not to the coefficient of k2."

Source: 9709_s21_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 11

Coaching With Shuja

11

11



The diagram shows part of the curve with equation $y = x^{\frac{1}{2}} + k^2x^{-\frac{1}{2}}$, where k is a positive constant.

(a) Find the coordinates of the minimum point of the curve, giving your answer in terms of k. [4]

The diagram shows part of the curve with equation $\boldsymbol{y = x^{\frac{1}{2}} + k^2x^{-\frac{1}{2}}}$, where $\boldsymbol{k}$ is a positive constant.

(a) Find the coordinates of the minimum point of the curve, giving your answer in terms of $\boldsymbol{k}$. **[4]**

The tangent at the point on the curve where $\boldsymbol{x = 4k^2}$ intersects the y-axis at $\boldsymbol{P}$.

(b) Find the y-coordinate of $\boldsymbol{P}$ in terms of $\boldsymbol{k}$. **[4]**

The shaded region is bounded by the curve, the x-axis and the lines $\boldsymbol{x = \frac{9}{4}k^2}$ and $\boldsymbol{x = 4k^2}$.

(c) Find the area of the shaded region in terms of $\boldsymbol{k}$. **[3]**

11(a)

$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} - \frac{1}{2}k^2x^{-3/2}$ **B1 B1** Allow any correct unsimplified form

Set to zero and one correct algebraic step towards the solutions: $\frac{1}{2}x^{-1/2} - \frac{1}{2}k^2x^{-3/2} = 0$ leading to $\frac{1}{2}x^{-1/2} = \frac{1}{2}k^2x^{-3/2}$ **M1** OE. Set to zero and one correct algebraic step towards the solutions. $\frac{dy}{dx}$ must only have 2 terms.

$(k^2, 2k)$ **A1**

Total: 4

11(b)

When $x = 4k^2$, $\frac{dy}{dx} = [\frac{1}{4k} - \frac{1}{16k}] = \frac{3}{16k}$ **B1** OE

$y = [2k + k^2 \times \frac{1}{2k}] = \frac{5k}{2}$ **B1** OE. Accept $2k + \frac{k}{2}$

Equation of tangent is $y - \frac{5k}{2} = \frac{3}{16k}(x - 4k^2)$ or $y = mx + c \rightarrow y = \frac{5k}{2} - \frac{3}{16k}(4k^2) + c$ **M1** Use of line equation with their gradient and $(4k^2, \text{their } y)$,

When $x = 0, y = [\frac{5k}{2} - \frac{3}{4}] = \frac{7k}{4}$ or from $y = mx + c, c = \frac{7k}{4}$ **A1** OE

Total: 4

11(c)

$$\int (x^{\frac{1}{2}} + k^2x^{-\frac{1}{2}})dx = \frac{2x^{\frac{3}{2}}}{3} + 2k^2x^{\frac{1}{2}}$$

B1 Any unsimplified form

$$[\frac{16k^3}{3} + 4k^3] - [\frac{9k^3}{4} + 3k^3]$$

M1 Apply limits $\frac{9}{4}k^2 \rightarrow 4k^2$ to an integration of y. M0 if volume attempted.

$$\frac{49k^3}{12}$$

A1 OE. Accept $4.08k^3$

Total: 3

Question 1

May/June 2021 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "For those candidates who attempted to remove the modulus signs by squaring both sides of the inequality, the most common error was not squaring the 2."

> "For candidates attempting to form two linear inequalities, the most common errors were sign errors and slips in the algebra."

Source: 9709_s21_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

1 Solve the inequality $2|3x - 1| < |x + 1|$. **[4]**

Coaching With Shuja

1

B1 State or imply non-modular inequality $2^2(3x-1)^2 < (x+1)^2$, or corresponding quadratic equation, or pair of linear equations

M1 Form and solve a 3-term quadratic, or solve two linear equations for x: e.g. $35x^2 - 26x + 3 = 0$

A1 Obtain critical values $x = \frac{3}{5}$ and $x = \frac{1}{7}$

Allow **0.143** or better

A1 State final answer $\frac{1}{7} < x < \frac{3}{5}$

Exact values required. Accept $x > \frac{1}{7}$ and $x < \frac{3}{5}$

Do not condone $<$ for $<$ in the final answer.
Fractions need not be in lowest terms.

Alternative method for Question 1

B1 Obtain critical value $x = \frac{3}{5}$ from a graphical method, or by solving a linear equation or linear inequality

B2 Obtain critical value $x = \frac{1}{7}$ similarly

Allow **0.143** or better

B1 State final answer $\frac{1}{7} < x < \frac{3}{5}$

OE. Exact values required. Accept $x > \frac{1}{7}$ and $x < \frac{3}{5}$

Do not condone $<$ for $<$ in the final answer.
Fractions need not be in lowest terms.

Total: 4

Question 3

May/June 2021 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "There were some sign errors in quoting the formulae, but the most common error was to use an incorrect formula such as cos(x−30°)=cosx−cos30°."

Source: 9709_s21_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 3

Coaching With Shuja

3 (a) Given that $\cos(x - 30^\circ) = 2\sin(x + 30^\circ)$, show that $\tan x = \frac{2-\sqrt{3}}{1-2\sqrt{3}}$. **[4]**

(b) Hence solve the equation

$$\cos(x - 30^\circ) = 2\sin(x + 30^\circ),$$

for $0^\circ < x < 360^\circ$. **[2]**

3(a)

M1 Use correct trig expansions and obtain an equation in $\sin x$ and $\cos x$

B1 FT Use correct exact trig ratios for 30° in their expansion: e.g. $\cos x(\frac{\sqrt{3}}{2} - 1) = \sin x(\sqrt{3} - \frac{1}{2})$

DM1 Obtain an equation in $\tan x$

Allow if their error in line 1 was a sign error

A1 AG: Obtain $\tan x = \frac{2-\sqrt{3}}{1-2\sqrt{3}}$ from correct working

Total: 4

3(b)

B1 Obtain answer in the given interval, e.g. 173.8°

Accept $174^\circ, 354^\circ$ or better

B1 Obtain a second answer and no other in the given interval, e.g. 353.8°

Ignore answers outside the given interval.
Treat answers in radians $(3.03 and 6.17)$ as a misread.

Total: 2

Coaching With Shuja

Question 4

May/June 2021 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(b) Many candidates found this definite integration question challenging."

Source: 9709_s21_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

Coaching With Shuja

4 (a) Prove that

$$\frac{1-\cos 2\theta}{1+\cos 2\theta} \equiv \tan^2\theta.$$

[2]

(b) Hence find the exact value of

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1-\cos 2\theta}{1+\cos 2\theta}\, d\theta.$$

[4]

Coaching With Shuja

4(a)

M1 Use correct double angle formula or t-substitution twice

A1 AG: Obtain $\frac{1-\cos 2\theta}{1+\cos 2\theta} = \tan^2\theta$ from correct working

Total: 2

4(b)

(No marking scheme provided for part b in the image)

Total:

Question 5

May/June 2021 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “Candidates who did this often made slips in simplifying (−2pi)2+4q, common errors being 4p2+4q and −4p+4q.”

Source: 9709_s21_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

Coaching With Shuja

Question Paper May/June 2021 • Paper 3 (variant 31) • Q5

(a) Solve the equation $z^2 - 2piz - q = 0$, where p and q are real constants. [2]

In an Argand diagram with origin O, the roots of this equation are represented by the distinct points A and B.

(b) Given that A and B lie on the imaginary axis, find a relation between p and q. [2]

(c) Given instead that triangle OAB is equilateral, express q in terms of p. [3]

(a) Use quadratic formula and $i^2 = -1$ **M1**

Obtain answers $pi + \sqrt{q - p^2}$ and $pi - \sqrt{q - p^2}$ **A1**

Accept $\frac{2pi \pm \sqrt{-4p^2 + 4q}}{2}$ and ISW

Total: 2

(b) State or imply that the discriminant must be negative **M1**

State condition $q < p^2$ **A1**

Total: 2

(c) Carry out a correct method for finding a relation, e.g. use the fact that the argument of one of the roots is $(\pm)60°$ **M1**

State a correct relation in any form, e.g. $\frac{p}{\sqrt{q - p^2}} = (\pm)\sqrt{3}$ **A1**

Simplify to $q = \frac{4}{3}p^2$ **A1**

Alternative method for Question 5(c)

Carry out a correct method for finding a relation, e.g. use the fact that the sides have equal length **M1**

State a correct relation in any form, e.g. $4(q - p^2) = p^2 + q - p^2$ **A1**

Simplify to $q = \frac{4}{3}p^2$ **A1**

Total: 3

Coaching with Shuja

Question 1

May/June 2021 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common error at this point was to forget to square the 3."

Source: 9709_s21_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

(1) Solve the inequality $|2x - 1| < 3|x + 1|$. [4]

Coaching With Shuja

(1) State or imply non-modular inequality $(2x-1)^2 < 3^2(x+1)^2$, or corresponding quadratic equation **B1**

e.g. $5x^2 + 22x + 8 = 0$ Allow recovery from 'invisible brackets' on RHS

Form and solve a 3-term quadratic in x **M1**

Obtain critical values $x = -4$ and $x = -\frac{2}{5}$ **A1**

State final answer $x < -4, x > -\frac{2}{5}$ **A1**

Do not condone $<$ for $<$, or $>$ for $>$ in the final answer. Allow 'or' but not 'and'.

$-\frac{2}{5} < x < -4$ scores A0.

Accept equivalent forms using brackets e.g. $x \in (-\infty, -4) \cup (-0.4, \infty)$

Alternative method for Question 1

Obtain critical value $x = -4$ from a graphical method, or by solving a linear equation or linear inequality **B1**

Obtain critical value $x = -\frac{2}{5}$ similarly **B2**

State final answer $x < -4, x > -\frac{2}{5}$ **B1**

Do not condone $<$ for $<$, or $>$ for $>$ in the final answer. Allow 'or' but not 'and'.

$-\frac{2}{5} < x < -4$ scores A0.

Accept equivalent forms using brackets e.g. $x \in (-\infty, -4) \cup (-0.4, \infty)$

Total: 4

Question 2

May/June 2021 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The half-line proved to be more challenging, with several candidates thinking that the line needed to pass through the centre of the circle."

Source: 9709_s21_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Coaching With Shuja

(2) On a sketch of an Argand diagram, shade the region whose points represent complex numbers z satisfying the inequalities $|z+1-i| \leq 1$ and $\arg(z-1) \leq \frac{3}{4}\pi$. [4]

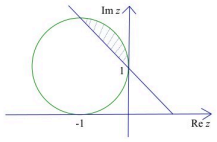
(2) Show a circle with centre $-1+i$. **B1**

Need some indication of scale or a correct label. Could just be mark(s) on the axes

Show a circle with radius 1 and centre not at the origin (or relevant part thereof). **B1**

Show correct half line from 1 (or relevant part thereof). **B1**

Shade the correct region on a correct diagram. **B1**

2	Show a circle with centre – 1 + i.	B1	Need some indication of scale or a correct label. Could just be mark(s) on the axes
	Show a circle with radius 1 and centre not at the origin (or relevant part thereof).	B1	
	Show correct half line from 1(or relevant part thereof) .	B1	
	Shade the correct region on a correct diagram.	B1	
		4	N.B. If they have very different scales on *their* 2 axes the diagram must match *their* scale - the 'circle' should be an ellipse. Allow freehand diagrams with clear correct intention.

Question	Answer	Marks	Guidance

N.B. If they have very different scales on their 2 axes the diagram must match their scale - the 'circle' should be an ellipse.

Allow freehand diagrams with clear correct intention.

Total: 4

Question 4

May/June 2021 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 4 This was a challenging question for many, with candidates not knowing how to use the given formulae or x x mistaking tan−1 for cot ."

Source: 9709_s21_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

(4) Using integration by parts, find the exact value of $\int_0^1 \tan^{-1}\left(\frac{x}{2}\right) dx$. [5]

Coaching With Shuja

(4) Commence integration and reach $ax\tan^{-1}\frac{x}{2}+b\int\frac{1}{c+x^2}dx$ **M1** (OE. Denominator might be $1+\frac{x^2}{4}$ or $2+\frac{x^2}{2}$.)

Obtain $x\tan^{-1}\left(\frac{1}{2}x\right)-\int x.\frac{1}{4+x^2}dx$ **A1** (OE)

Complete integration and obtain $x\tan^{-1}\left(\frac{1}{2}x\right)-\ln\left(4+x^2\right)$ **A1**

OE e.g. with $\ln\left|1+\frac{x^2}{4}\right|$

Substitute limits correctly in an expression of the form $px\tan^{-1}x+q\ln(c+x^2)$ **DM1**

$2\tan^{-1}1-\ln 8+\ln 4$ OE

Obtain final answer $\frac{1}{2}\pi-\ln 2$ **A1**

OE exact answer. Needs a value for $\tan^{-1}1$ and a single log term

Alternative method for Question 4

Use the substitution $\theta=\tan^{-1}\frac{x}{2}$ to obtain $\lambda\int 2\theta\sec^2\theta d\theta$ and reach **M1**

$p\theta\tan\theta+q\int\tan\theta d\theta$

Obtain $2\theta\tan\theta-2\int\tan\theta d\theta$ **A1**

Complete integration and obtain $2\theta\tan\theta+2\ln(\cos\theta)$ **A1**

Substitute correct limits correctly in an expression of the form $r\theta\tan\theta+s\ln(\cos\theta)$ **DM1**

Limits should be $\frac{\pi}{4}$ and 0. Limits must be in radians.

Obtain final answer $\frac{1}{2}\pi-\ln 2$ **A1**

OE exact answer. Need values for trig. functions and a single log term.

Total: 5

Question 3

May/June 2021 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 3 (a) Many candidates did not simplify their answer, for example leaving the exponential terms within a bracket, hence could not be awarded the final accuracy mark."

> "A common error involved differentiating the t term within the equation for x as 0, not 1."

Source: 9709_s21_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 33), Question 3

Coaching With Shuja

(3) The parametric equations of a curve are

$$x = t + \ln(t+2), \quad y = (t-1)e^{-2t},$$

where $t > -2$.

(a) Express $\frac{dy}{dx}$ in terms of t, simplifying your answer. [5]

(b) Find the exact y-coordinate of the stationary point of the curve. [2]

(3a) State $\frac{dx}{dt} = 1 + \frac{1}{t+2}$ **B1**

Use product rule **M1**

Obtain $\frac{dy}{dt} = e^{-2t} - 2(t-1)e^{-2t}$ **A1**

Use $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$ **M1**

Obtain correct answer in any simplified form, e.g. $\frac{(3-2t)(t+2)}{t+3}e^{-2t}$ **A1**

Total: 5

(3b) Equate derivative to zero and solve for t **M1**

Obtain $t = \frac{3}{2}$ and obtain answer $y = \frac{1}{2}e^{-3}$, or exact equivalent **A1**

Total: 2

Coaching With Shuja

Question 6

May/June 2021 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 6 (a) Many candidates' attempts to sketch both graphs showed they did not quite understand the properties of each type of graph and its transformations."

Source: 9709_s21_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 6

Coaching With Shuja

(6) (a) By sketching a suitable pair of graphs, show that the equation $\cot \frac{1}{2}x = 1 + e^{-x}$ has exactly one root in the interval $0 < x \leq \pi$. [2]

(b) Verify by calculation that this root lies between 1 and 1.5. [2]

(c) Use the iterative formula $x_{n+1} = 2\tan^{-1}\left(\frac{1}{1+e^{-x_n}}\right)$ to determine the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

Coaching With Shuja

(6a) Sketch a relevant graph, e.g. $y = \cot \frac{1}{2}x$ **B1**

Sketch a second relevant graph, e.g. $y = 1 + e^{-x}$, and justify the given statement **B1**

Total: 2

(6b) Calculate values of a relevant expression or pair of expressions at $x = 1$ and $x = 1.5$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(6c) Use the iterative formula correctly at least once **M1**

Obtain final answer 1.34 **A1**

Show sufficient iterations to 4 d.p. to justify 1.34 to 2 d.p. or show there is a sign change in the interval (1.335, 1.345) **A1**

Total: 3

Coaching With Shuja

Question 7

May/June 2021 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

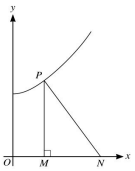
> "Question 7 (a) (i) Very few candidates were able to justify the statement, with most simply writing the given expression."

Source: 9709_s21_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 7

Coaching With Shuja

(7)

7



For the curve shown in the diagram, the normal to the curve at the point P with coordinates (x, y) meets the x-axis at N. The point M is the foot of the perpendicular from P to the x-axis.

The curve is such that for all values of x in the interval $0 \leqslant x < \frac{1}{2}\pi$, the area of triangle PMN is equal to $\tan x$.

(a) **(i)** Show that $\frac{MN}{y} = \frac{\mathrm{d}y}{\mathrm{d}x}$. [1]

(ii) Hence show that x and y satisfy the differential equation $\frac{1}{2}y^2\frac{\mathrm{d}y}{\mathrm{d}x} = \tan x$. [2]

For the curve shown in the diagram, the normal to the curve at the point $\boldsymbol{P}$ with coordinates $\boldsymbol{(x, y)}$ meets the x-axis at $\boldsymbol{N}$. The point $\boldsymbol{M}$ is the foot of the perpendicular from $\boldsymbol{P}$ to the x-axis.

The curve is such that for all values of $\boldsymbol{x}$ in the interval $\boldsymbol{0 \leq x < \frac{1}{2}\pi}$, the area of triangle $\boldsymbol{PMN}$ is equal to $\boldsymbol{\tan x}$.

(a) (i) Show that $\boldsymbol{\frac{MN}{y} = \frac{dy}{dx}}$. [1]

(ii) Hence show that $\boldsymbol{x}$ and $\boldsymbol{y}$ satisfy the differential equation $\boldsymbol{\frac{1}{2}y^2\frac{dy}{dx} = \tan x}$. [2]

(b) Given that $\boldsymbol{y = 1}$ when $\boldsymbol{x = 0}$, solve this differential equation to find the equation of the curve, expressing $\boldsymbol{y}$ in terms of $\boldsymbol{x}$. [6]

(7a)(i) Justify the given statement $\frac{MN}{y} = \frac{dy}{dx}$ **B1**

Total: 1

(7a)(ii) Express the area of PMN in terms of y and $\frac{dy}{dx}$ and equate to $\tan x$ **M1**

Obtain the given equation correctly **A1**

Total: 2

(7b) Separate variables and integrate at least one side **M1**

Obtain term $\frac{1}{6}y^3$ **A1**

Obtain term of the form $\pm \ln \cos x$ **M1**

Evaluate a constant or use $x = 0$ and $y = 1$ in a solution containing terms ay^3 and $\pm \ln \cos x$, or equivalent **M1**

Obtain correct answer in any form, e.g. $\frac{1}{6}y^3 = -\ln \cos x + \frac{1}{6}$ **A1**

Obtain final answer $y = \sqrt[3]{(1 - 6 \ln \cos x)}$ **A1**

OE

Total: 6

Coaching With Shuja

Question 8

May/June 2021 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "4 (b) Many candidates demonstrated excellent use of integration by parts although the conclusion proved challenging for most candidates."

Source: 9709_s21_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 8

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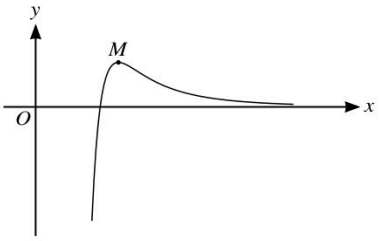
(8)

8



The diagram shows the curve $y = \dfrac{\ln x}{x^4}$ and its maximum point M.

(a) Find the exact coordinates of M. [4]

The diagram shows the curve $y = \frac{\ln x}{x^4}$ and its maximum point M.

(a) Find the exact coordinates of M. [4]

(b) By using integration by parts, show that for all $a > 1$, $\int_1^a \frac{\ln x}{x^4}\,dx < \frac{1}{6}$. [6]

(8a) Use quotient or product rule **M1**

Obtain correct derivative in any form **A1**

Equate derivative to zero and solve for x **M1**

Obtain $x = \sqrt[4]{e}$ and $y = \frac{1}{4e}$, or exact equivalents **A1**

Total: 4

(8b) Commence integration and reach $ax^{-3}\ln x + b\int x^{-3}.\frac{1}{x}dx$ **M1** (OE. Denominator might be $1 + \frac{x^2}{4}$ or $2 + \frac{x^2}{2}$.)

Obtain $-\frac{1}{3}x^{-3}\ln x + \frac{1}{3}\int x^{-3}.\frac{1}{x}dx$ **A1**

Complete integration and obtain $-\frac{1}{3}x^{-3}\ln x - \frac{1}{9}x^{-3}$ **A1**

Substitute limits correctly, having integrated twice **DM1**

Obtain answer $\frac{1}{9} - \frac{1}{3}a^{-3}\ln a - \frac{1}{9}a^{-3}$ **A1**

Justify the given statement **A1**

Total: 6

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Question 9

May/June 2021 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "(b) Many candidates did not give their answer in the correct form as a vector equation as required by the question."

Source: 9709_s21_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

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Question 9

The quadrilateral $ABCD$ is a trapezium in which AB and DC are parallel. With respect to the origin O, the position vectors of A, B and C are given by $\overrightarrow{OA} = -\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, $\overrightarrow{OB} = \mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\overrightarrow{OC} = 2\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$.

(a) Given that $\overrightarrow{DC} = 3\overrightarrow{AB}$, find the position vector of D. [3]

(b) State a vector equation for the line through A and B. [1]

(c) Find the distance between the parallel sides and hence find the area of the trapezium. [5]

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9(a)

B1 OE State or imply $\overrightarrow{AB} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$

M1 Carry out a correct method to find $\overrightarrow{OD}$

A1 OE Obtain answer $-4\mathbf{i} - \mathbf{j} + 3\mathbf{k}$

Total: 3

9(b)

B1FT State $\mathbf{r} = -\mathbf{i} + 2\mathbf{j} + 3\mathbf{k} + \lambda(2\mathbf{i} + \mathbf{j} - 2\mathbf{k})$

OE. The FT is on $\overrightarrow{AB}$.

Total: 1

9(c)

M1 For a general point P on AB, state $\overrightarrow{CP}$ or $\overrightarrow{DP}$ in component form, e.g. $\overrightarrow{CP} = (3 - 2\lambda, -\lambda, -6 + 2\lambda)$

DM1 Equate a relevant scalar product to zero or equate derivative of $|\overrightarrow{CP}|$ to zero or use Pythagoras in a relevant triangle and solve for λ

A1 Obtain $\lambda = 2$

A1 Show the perpendicular is of length 3

A1 Carry out a correct method to find the area of $ABCD$ and obtain the answer 18

Alternative method for Question 9(c)

M1 Use a scalar product to find the projection CN (or DN) of BC (or AD) on CD

A1 Obtain $CN = 3$ (or $DN = 3$)

DM1 Use Pythagoras to obtain BN (or AN)

Total: 5

Question 4

Oct/Nov 2020 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "A significant number incorrectly subtracted the terms to find d and hence thought that it equalled 7."

Source: 9709_w20_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

Coaching With Shuja

Question 4

The sum, S_n, of the first n terms of an arithmetic progression is given by

$$S_n = n^2 + 4n.$$

The kth term in the progression is greater than 200.

Find the smallest possible value of k. [5]

4

M1 S_k and S_{k+1}

A1 A1 $a = 5$, $d = 2$

M1 $a + (n-1)d > 200 \rightarrow 5 + 2(k-1) > 200$

Correct formula used with their a and d to form an equation or inequality with 200, condone use of n

A1 $(k =)99$

Condone > 99

Alternative method for question 4

M1 $\frac{n}{2}(2a + (n-1)d) = n^2 + 4n \rightarrow \left(\frac{d}{2} = 1, a - \frac{1}{2}d = 4\right)$

Equating two correct expressions of S_n and equating coefficients of n and n^2

A1 A1 $d = 2$, $a = 5$

M1 $a + (n-1)d > 200 \rightarrow 5 + 2(k-1) > 200$

Correct formula used with their a and d to form an equation or inequality with 200, condone use of n

A1 $(k =)99$

Condone > 99

Alternative method for question 4

M1 A1 $sum_k - sum_{k-1} \rightarrow k^2 + 4k - (k-1)^2 - 4(k-1)$

Using given formula with consecutive expressions subtracted. Allow $k+1$ and k.

M1 A1 $2k + 3 > 200$ or $= 200$

Simplifying to a linear equation or inequality

A1 $(k =)99$

Condone > 99

Total: 5

Question 6

Oct/Nov 2020 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "2 Some candidates found (b) difficult and often did not realise the need to cube root."

Source: 9709_w20_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 6

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(a) Prove the identity $\left(\frac{1}{\cos x} - \tan x\right)\left(\frac{1}{\sin x} + 1\right) \equiv \frac{1}{\tan x}$. [4]

(b) Hence solve the equation $\left(\frac{1}{\cos x} - \tan x\right)\left(\frac{1}{\sin x} + 1\right) = 2\tan^2 x$ for $0° \leq x \leq 180°$. [2]

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6(a)

B1 Uses "$\tan x = \sin x \div \cos x$" throughout

$$\left(\frac{1-\sin x}{\cos x}\right)\left(\frac{1+\sin x}{\sin x}\right) \text{ or } \frac{1-\sin^2 x}{\cos x \sin x}$$

M1 Correct algebra leading to two or four terms

$$\left(\frac{\cos^2 x}{\cos x \sin x}\right)$$

A1 OE. A correct expression which can be cancelled directly to $\frac{\cos x}{\sin x}$ e.g. $\frac{\cos x(1-\sin^2 x)}{\sin x(1-\sin^2 x)}$

$$\left(\frac{\cos^2 x}{\cos x \sin x}\right) = \left(\frac{\cos x}{\sin x}\right) = \frac{1}{\tan x}$$

A1 AG. Must show cancelling. If x is missing throughout their working withhold this mark.

Total: 4

6(b)

M1 Uses **(a)** $\rightarrow \frac{1}{\tan x} = 2\tan^2 x \tan^3 x = \frac{1}{2}$

Reducing to $\tan^2 x = k$.

A1 $(x =) 38.4^\circ$

AWRT.

Ignore extra answers outside the range 0 to 180° but A0 if within.

Total: 2

Question 8

Oct/Nov 2020 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "In part (b) candidates were generally confident in finding the arc length BD, however the length of DC was found to be more challenging."

Source: 9709_w20_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

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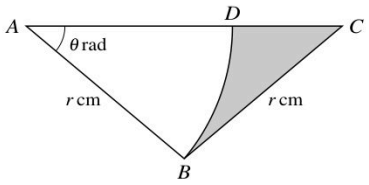
Question 8

8



In the diagram, ABC is an isosceles triangle with $AB = BC = r$ cm and angle $BAC = \theta$ radians. The point D lies on AC and ABD is a sector of a circle with centre A.

(a) Express the area of the shaded region in terms of r and θ. [3]

In the diagram, ABC is an isosceles triangle with $AB = BC = r$ cm and angle $BAC = \theta$ radians. The point D lies on AC and ABD is a sector of a circle with centre A.

(a) Express the area of the shaded region in terms of r and θ. [3]

(b) In the case where $r = 10$ and $\theta = 0.6$, find the perimeter of the shaded region. [4]

8(a)

M1 Use of correct formula for the area of triangle ABC

Use of $180 - 2\theta$ scores Mo. Condone $2\pi - 2\theta$

$$\frac{1}{2}r^2\sin(\pi - 2\theta) \text{ or } \frac{1}{2}r^2\sin 2\theta \text{ or } 2 \times \frac{1}{2}r \times r\cos\theta \times \sin\theta \text{ or } 2 \times \frac{1}{2}r\cos\theta \times r\sin\theta$$

A1 OE

B1 FT FT for *their* triangle area $-\frac{1}{2}r^2\theta$

(Condone use of 180 degrees for triangle area for B1)

Total: 3

8(b)

B1 SOI

Arc $BD = r\theta = 6$ cm

M1 Finding AC or $\frac{1}{2}AC$ (= 8.25)

$$AC = 2r\cos\theta = (2 \times 10\cos 0.6 = 20\cos 0.6 = 16.506)$$

or $\sqrt{(2r^2 - 2r^2\cos(\pi - 2\theta))}$ or $\frac{r \times \sin(\pi - 2\theta)}{\sin\theta}$

DM1 Subtracting r from *their* AC

or $r\cos\theta$ from *their* half AC (8.25-1.75)

$$DC = 2r\cos\theta - r \text{ or } \sqrt{(2r^2 - 2r^2\cos(\pi - 2\theta))} - r \text{ or } (= 6.506)$$

A1 AWRT

(Perimeter $= 10 + 6 + 6.506) = 22.5$

Total: 4

Question 9

Oct/Nov 2020 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "This question was challenging for many candidates with (b) and (c) being omitted by a significant number."

Source: 9709_w20_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 9

Coaching With Shuja

Question 9

A circle has centre at the point $B(5, 1)$. The point $A(-1, -2)$ lies on the circle.

(a) Find the equation of the circle. [3]

Point C is such that AC is a diameter of the circle. Point D has coordinates $(5, 16)$.

(b) Show that DC is a tangent to the circle. [4]

The other tangent from D to the circle touches the circle at E.

(c) Find the coordinates of E. [2]

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Mark Scheme Oct/Nov 2020 • Paper 1 (variant 12) • Q9

9(a)

B1 $r = \sqrt{(6^2 + 3^2)}$ or $r^2 = 45$

Sight of $r = 6.7$ implies B1

M1 Using centre given and ***their*** radius or r in correct formula

$(x-5)^2 + (y-1)^2 = r^2$ or $x^2 - 10x + y^2 - 2y = r^2 - 26$

A1 Do not allow $\sqrt{45}$ for r^2

$(x-5)^2 + (y-1)^2 = 45$ or $x^2 - 10x + y^2 - 2y = 19$

Total: 3

9(b)

B1 C has coordinates $(11, 4)$

B1 OE, Gradient of AB, BC or AC.

$$0.5$$

M1 Calculation of gradient needs to be shown for this M1.

Grad of $CD = -2$

A1 Clear reasoning needed.

$\left(\frac{1}{2} \times -2 = -1\right)$ then states + perpendicular $\rightarrow$ hence shown or tangent

Alternative method for question 9(b)

B1 C has coordinates $(11, 4)$

B1 OE, Gradient of AB, BC or AC.

Gradient of the perpendicular is -2

$\rightarrow$ Equation of the perpendicular is $y - 4 = -2(x - 11)$

M1 Use of $m_1 m_2 = -1$ with ***their*** gradient of AB, BC or AC and correct method for the equation of CD

.

Could use $D(5, 16)$ instead of $C(11, 4)$.

A1 Clear check and reasoning needed. Checks $D(5, 16)$ or checks gradient of CD and then states D lies on the line or CD has gradient -2 $\rightarrow$ hence shown or tangent

Alternative method for question 9(b)

B1 C has coordinates $(11, 4)$

B1 Finding $CD = \sqrt{180}$ and $BD = \sqrt{225}$

OE. Calculated from the co-ordinates of B, C & D without using r.

M1 Checking $(\textit{their}BD)^2 - (\textit{their}CD)^2$ is the same as $(\textit{their}r)^2$

A1 Triangle ACD could be used instead.

$\therefore$ Pythagoras valid $\therefore$ perpendicular $\rightarrow$ hence shown or tangent

Alternative method for question 9(b)

B1 C has coordinates $(11, 4)$

B1 Finding vectors $\overrightarrow{AC}$ and $\overrightarrow{CD}$ or $\overrightarrow{BC}$ and $\overrightarrow{CD}$

Must be correct pairing.

$$\left(= \begin{pmatrix} 6 \\ 3 \end{pmatrix} \text{ and } \begin{pmatrix} -6 \\ 12 \end{pmatrix}\right) \text{ or } \left(\begin{pmatrix} 12 \\ 6 \end{pmatrix} \text{ and } \begin{pmatrix} -6 \\ 12 \end{pmatrix}\right)$$

M1 Accept ***their*** $\overrightarrow{AC}$ and $\overrightarrow{CD}$ or ***their*** $\overrightarrow{BC}$ and $\overrightarrow{CD}$

Applying the scalar product to one of these pairs of vectors

Scalar product $= 0$ then states $\therefore$ perpendicular $\rightarrow$ hence shown or tangent

A1

Total: 4

9(c)

M1 From the first method, use alternate form from 9(b) to find the equation of the line through D and C perpendicular to AB, or use it to find the gradient of CD.

A1 Clear check and reasoning needed.

Question 10

Oct/Nov 2020 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Weaker candidates who did include the –x term were often unable to differentiate or integrate it correctly."

> "A significant number of candidates omitted (c), often after having been unable to find the correct coordinates of M in (b)."

Source: 9709_w20_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

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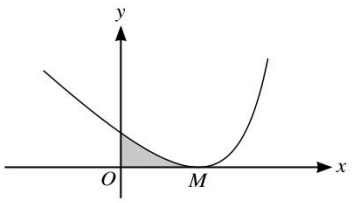
Question 10

10



The diagram shows part of the curve $y = \frac{2}{(3-2x)^2} - x$ and its minimum point M, which lies on the x-axis.

(a) Find expressions for $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and $\int y\,dx$. [6]

The diagram shows part of the curve $\boldsymbol{y = \frac{2}{(3-2x)^2} - x}$ and its minimum point $\boldsymbol{M}$, which lies on the x-axis.

(a) Find expressions for $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and $\int y\,dx$. [6]

(b) Find, by calculation, the x-coordinate of $\boldsymbol{M}$. [2]

(c) Find the area of the shaded region bounded by the curve and the coordinate axes. [2]

10(a)

B2, 1, 0 B2 for all three elements correct, B1 for two elements correct, B0 for only one or no elements correct.

$$\left(\frac{dy}{dx}\right) = [8] \times [(3-2x)^{-3}] + [-1] \quad \left(= \frac{8}{(3-2x)^3} - 1\right)$$

B1 FT FT providing ***their*** bracket is to a negative power

$$\frac{d^2y}{dx^2} = -3 \times 8 \times (3-2x)^{-4} \times (-2) \quad \left(= \frac{48}{(3-2x)^4}\right)$$

B1 B1 B1 Simplification not needed, B1 for each correct element

$$\int y dx = [(3-2x)^{-1}][2 \times (-1 \times -2)][-\frac{1}{2}x^2](+c) \quad \left(= -\frac{1}{3-2x} - \frac{1}{2}x^2 + c\right)$$

Total: 6

10(b)

M1 Setting ***their*** 2-term differential to 0 and attempts to solve as far as $x =$

$$\frac{dy}{dx} = 0 \rightarrow (3-2x)^3 = 8 \rightarrow 3 - 2x = k \rightarrow x =$$

A1 $\frac{1}{2}$

Alternative method for question 10(b)

M1 Setting y to 0 and attempts to solve a cubic as far as $x =$ (3 factors needed)

$$y = 0 \rightarrow \frac{2}{(3-2x)^2} - x = 0 \rightarrow (x-2)(2x-1)^2 = 0 \rightarrow x =$$

A1 $\frac{1}{2}$

Total: 2

10(c)

M1 Using ***their*** integral, ***their*** positive x limit from **part (b)** and 0 correctly.

$$\textbf{Area under curve} = \textit{their} \quad \left[\frac{1}{3-2x\left(\frac{1}{2}\right)} - \frac{1}{2}\right] - [\frac{1}{3-2\times 0} - 0]$$

A1 $\frac{1}{24}$

Total: 2

Question 11

Oct/Nov 2020 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (c) was found challenging by many, but those who attempted to draw the corresponding lines on their graphs were often successful."

Source: 9709_w20_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 11

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Question 11

A curve has equation $y = 3\cos 2x + 2$ for $0 \leq x \leq \pi$.

(a) State the greatest and least values of y. [2]

(b) Sketch the graph of $y = 3\cos 2x + 2$ for $0 \leq x \leq \pi$. [2]

(c) By considering the straight line $y = kx$, where k is a constant, state the number of solutions of the equation $3\cos 2x + 2 = kx$ for $0 \leq x \leq \pi$ in each of the following cases.

(i) $k = -3$ [1]

(ii) $k = 1$ [1]

(iii) $k = 3$ [1]

Functions f, g and h are defined for $x \in \mathbb{R}$ by

$$f(x) = 3\cos 2x + 2,$$

$$g(x) = f(2x) + 4,$$

$$h(x) = 2f\left(x + \frac{1}{2}\pi\right).$$

(d) Describe fully a sequence of transformations that maps the graph of $y = f(x)$ on to $y = g(x)$. [2]

(e) Describe fully a sequence of transformations that maps the graph of $y = f(x)$ on to $y = h(x)$. [2]

11(a)

B1 B1 $5, -1$

Sight of each value

Total: 2

11(b)

B1 Needs to be a curve, not straight lines.

One complete cycle starting and finishing at ***their*** largest value.

DB1 One complete cycle starting and finishing at $y = 5$ and going down to $y = -1$ and starting to level off at least one end.

Total: 2

11(c)(i)

0 solution

B1

Total: 1

11(c)(ii)

2 solutions

B1

Total: 1

11(c)(iii)

1 solution

B1

Total: 1

11(d)

B1 Stretch by (scale factor) $\frac{1}{2}$, parallel to x-axis or in x direction (or horizontally)

B1 Translation of $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$

Accept translation 4 units in positive y-direction.

Total: 2

11(e)

B1 Translation of $\begin{pmatrix} -\frac{\pi}{2} \\ 0 \end{pmatrix}$

Accept translation $-\frac{\pi}{2}$ units in x-direction.

B1 Stretch by (scale factor) 2 parallel to y-axis (or vertically).

Total: 2

Question 1

Oct/Nov 2020 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Many candidates used a wordy description but often did not give the required information in a mathematically acceptable form."

Source: 9709_w20_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 13), Question 1

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(a) Express $x^2 + 6x + 5$ in the form $(x + a)^2 + b$, where a and b are constants. [2]

(b) The curve with equation $y = x^2$ is transformed to the curve with equation $y = x^2 + 6x + 5$.

Describe fully the transformation(s) involved. [2]

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1(a)

B1 B1

$$[(x+3)^2][-4]$$

Total: 2

1(b)

B1 FT [Translation or shift] $\begin{pmatrix}-3\\-4\end{pmatrix}$

Accept [translation/shift] $\begin{pmatrix}-\textit{their } a\\ \textit{their } b\end{pmatrix}$

OR translation -3 units in x-direction and (translation) -4 units in y-direction.

Total: 2

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Question 2

Oct/Nov 2020 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Many candidates, however, did not realise that the limit of infinity would result in that part of the integral tending to zero."

Source: 9709_w20_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 13), Question 2

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Question Paper Oct/Nov 2020 • Paper 1 (variant 13) • Q2

The function f is defined by $f(x) = \frac{2}{(x+2)^2}$ for $x > -2$.

(a) Find $\int_1^\infty f(x)\,dx$. [3]

(b) The equation of a curve is such that $\frac{dy}{dx} = f(x)$. It is given that the point $(-1, -1)$ lies on the curve.

Find the equation of the curve. [2]

(a) $\frac{-2}{x+2}$

B1 Integrate $f(x)$. Accept $-2(x+2)^{-1}$. Can be unsimplified.

$0-\left(-\frac{2}{3}\right)=\frac{2}{3}$

M1 A1 Apply limit(s) to an integrated expansion. CAO for A1

Total: 3

(b) $-1=-2+c$

M1 Substitute $x=-1, y=-1$ into **their** integrated expression (c present)

$y=-2(x+2)^{-1}+1=-\frac{2}{x+2}+1$

A1 Accept $y=-2(x+2)^{-1}+1$. -2 must be resolved.

Total: 2

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Question 3

Oct/Nov 2020 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Common errors seen included not showing the factorisation (or equivalent) or 2 not giving the secondary solution."

Source: 9709_w20_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 13), Question 3

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Solve the equation $3\tan^2\theta + 1 = \frac{2}{\tan^2\theta}$ for $0^\circ < \theta < 180^\circ$. [5]

$3\tan^4\theta + \tan^2\theta - 2 (= 0)$

M1 SOI 3-term quartic, condone sign errors for this mark only

$(3\tan^2\theta - 2)(\tan^2\theta + 1) (= 0)$

M1 Attempt to factorise or solve 3-term quadratic in $\tan^2\theta$.

$\tan\theta = (\pm)\sqrt{\frac{2}{3}}$ or $(\pm)0.816$ or $(\pm)0.817$

A1 SOI Implied by final answer = 39.2° after 1st **M1** scored

$39.2^\circ, 140.8^\circ$

A1 A1 FT FT for 2nd solution = $180^\circ - 1st$ solution

Total: 5

Coaching With Shuja

Question 4

Oct/Nov 2020 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "The most common errors were that candidates omitted to show the factorisation (or equivalent), errors in collecting terms at the beginning, omitting a term or making sign errors and several left their answer as –22, 2 rather than applying correct inequalities."

Source: 9709_w20_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 4

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Question Paper Oct/Nov 2020 • Paper 1 (variant 13) • Q4

A curve has equation $y = 3x^2 - 4x + 4$ and a straight line has equation $y = mx + m - 1$, where m is a constant.

Find the set of values of m for which the curve and the line have two distinct points of intersection. [5]

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$3x^2 - 4x + 4 = mx + m - 1 \rightarrow 3x^2 - (4+m)x + (5-m)(= 0)$

M1 3-term quadratic

$b^2 - 4ac = (4+m)^2 - 4 \times 3 \times (5-m)$

M1 Find $b^2 - 4ac$ for **their** quadratic

$m^2 + 20m - 44$

A1

$(m+22)(m-2)$

A1 Or use of formula or completing square. This step must be seen

$m > 2, m < -22$

A1 Allow $x > 2$, $x < -22$

Total: 5

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Question 5

Oct/Nov 2020 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Weaker candidates sometimes struggled after this to find a suitable relationship that would lead them to a correct answer."

Source: 9709_w20_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 5

Coaching With Shuja

Question Paper

Oct/Nov 2020 • Paper 1 (variant 13) • Q5

In the expansion of $(a+bx)^7$, where a and b are non-zero constants, the coefficients of x, x^2 and x^4 are the first, second and third terms respectively of a geometric progression.

Find the value of $\frac{a}{b}$. [5]

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$[7Ca^6b(x)], [7C2a^5b^2\,(x^2)], [7C4a^3b^4\,(x^4)]$

B2, 1, 0 SOI, can be seen in an expansion.

$$\frac{7C2a^5b^2(x^2)}{7Ca^6b(x)} = \frac{7C4a^3b^4(x^4)}{7C2a^5b^2(x^2)} \rightarrow \frac{21a^2b^2}{35a^3b^4}$$

M1 A1 M1 for a correct relationship OE (FT from **their** 3 terms). For **A1** binomial coefficients must be correct & evaluated.

$$\frac{a}{b} = \frac{5}{9}$$

A1 OE

Total: 5

Question 6

Oct/Nov 2020 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Part (b) intended to aid candidates with (c) but most candidates did not notice the hint given in (b) and were unable to successfully answer (c)."

Source: 9709_w20_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 6

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Question Paper Oct/Nov 2020 • Paper 1 (variant 13) • Q6

The function f is defined by $f(x) = \frac{2x}{3x-1}$ for $x > \frac{1}{3}$.

(a) Find an expression for $f^{-1}(x)$. [3]

(b) Show that $\frac{2}{3} + \frac{2}{3(3x-1)}$ can be expressed as $\frac{2x}{3x-1}$. [2]

(c) State the range of f. [1]

Coaching With Shuja

(a) $y = \frac{2x}{3x-1} \rightarrow 3xy - y = 2x \rightarrow 3xy - 2x = y$ (or $-y = 2x - 3xy$)

***M1** For 1st two operations. Condone a sign error

$x(3y-2) = y \rightarrow x = \frac{y}{3y-2}$ (or $x = \frac{-y}{2-3y}$)

DM1 For 2nd two operations. Condone a sign error

$\left(f^{-1}(x)\right) = \frac{x}{3x-2}$

A1 Allow $\left(f^{-1}(x)\right) = \frac{-x}{2-3x}$

Total: 3

(b)

$$\left[\frac{2(3x-1)+2}{3(3x-1)}\right] = \left[\frac{6x}{3(3x-1)} = \frac{2x}{3x-1}\right]$$

B1 B1 AG, WWW. First **B1** is for a correct single unsimplified fraction. An intermediate step needs to be shown. Equivalent methods accepted.

Total: 2

Question 8

Oct/Nov 2020 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Careful reading of the question reveals there is only one stationary point, but many candidates did not reject one of the solutions."

> "Another common mistake was to forget to find the corresponding y value."

Source: 9709_w20_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 8

Coaching With Shuja

Question Paper Oct/Nov 2020 • Paper 1 (variant 13) • Q8

The equation of a curve is $y = 2x + 1 + \frac{1}{2x+1}$ for $x > -\frac{1}{2}$.

(a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. [3]

(b) Find the coordinates of the stationary point and determine the nature of the stationary point. [5]

Coaching With Shuja

(a) $\frac{dy}{dx} = [2]\left[-2(2x+1)^{-2}\right]$

B1 B1

$$\frac{d^2y}{dx^2} = 8(2x+1)^{-3}$$

B1

Total: 3

(b) Set **their** $\frac{dy}{dx} = 0$ and attempt solution

M1

$(2x+1)^2 = 1 \rightarrow 2x+1 = (\pm)1$ or $4x^2+4x = 0 \rightarrow (4)x(x+1) = 0$

M1 Solving as far as $x = \ldots$

$x = 0$

A1 WWW. Ignore other solution.

$(0, 2)$

A1 One solution only. Accept $x = 0, y = 2$ only.

$$\frac{d^2y}{dx^2} > 0 \text{ from a solution } x > -\frac{1}{2} \text{ hence minimum}$$

B1 Ignore other solution. Condone arithmetic slip in value of $\frac{d^2y}{dx^2}$.

Their $\frac{d^2y}{dx^2}$ must be of the form $k(2x+1)^{-3}$

Total: 5

Coaching With Shuja

Question 11

Oct/Nov 2020 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "In part (c), many candidates were able to find the gradient of CT and hence of AB but were unable to make further progress."

Source: 9709_w20_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 11

Coaching With Shuja

Question Paper Oct/Nov 2020 • Paper 1 (variant 13) • Q11

A circle with centre C has equation $(x-8)^2+(y-4)^2=100$.

(a) Show that the point $T(-6,6)$ is outside the circle. [3]

Two tangents from T to the circle are drawn.

(b) Show that the angle between one of the tangents and CT is exactly $45°$. [2]

The two tangents touch the circle at A and B.

(c) Find the equation of the line AB, giving your answer in the form $y=mx+c$. [4]

(d) Find the x-coordinates of A and B. [3]

Coaching With Shuja

(a) $(-6-8)^2+(6-4)^2$

M1 OE

$= 200$

A1

$$\sqrt{200} > 10, \text{ hence outside circle}$$

A1 AG ('Shown' not sufficient). Accept equivalents of $\sqrt{200} > 10$

Total: 3

(b) Gradient, m, of $CT = -\frac{1}{7}$

B1 OE

Attempt to find mid-point (M) of CT

Expect $(1, 5)$

*M1 Expect $1, 5$

M1 Solving as far as $x = \ldots$

Equation of AB is $y - 5 = 7(x-1)$

DM1 Through **their** $(1, 5)$ with gradient $-\frac{1}{m}$

$$y = 7x - 2$$

A1

Total: 4

Alternative method for question 11(b)

$(10\sqrt{2})^2 = 10^2 + TA^2$

M1

$TA = 10 \rightarrow 45^\circ$

A1 AG

Total: 2

(d) $(x-8)^2+(7x-2-4)^2 = 100$ or equivalent in terms of y

M1 Substitute **their** equation of AB into equation of circle.

$50x^2 - 100x (= 0)$

A1

$x = 0$ and 2

A1 WWW

Alternative method for question 11(d)

$$MC = \begin{pmatrix} 7 \\ -1 \end{pmatrix}$$

M1

$$\begin{pmatrix} 1 \\ 5 \end{pmatrix} + \begin{pmatrix} -1 \\ -7 \end{pmatrix} = \begin{pmatrix} -2 \\ -12 \end{pmatrix}, \begin{pmatrix} 1 \\ 5 \end{pmatrix} + \begin{pmatrix} 1 \\ 7 \end{pmatrix} = \begin{pmatrix} 12 \\ 12 \end{pmatrix}$$

A1

$x = 0$ and 2

A1

Total: 3

Question 1

Oct/Nov 2020 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "A few candidates chose to sketch a graph and reached a correct inequality as the sketch enabled them to visualise it."

Source: 9709_w20_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

Solve the inequality $2 - 5x > 2|x - 3|$. [4]

Coaching With Shuja

Mark Scheme Oct/Nov 2020 • Paper 3 (variant 31) • Q1

Make a recognisable sketch graph of $y = 2|x - 3|$ and the line $y = 2 - 5x$

B1 Need to see correct V at $x = 3$, roughly symmetrical, $x = 3$ stated, domain at least $(-2, 5)$.

Find x-coordinate of intersection with $y = 2 - 5x$

M1 Find point of intersection with $y = 2|x - 3|$ or solve $2 - 5x$ with $2(x - 3)$ or $-2(x - 3)$

Obtain $x = -\frac{4}{3}$

A1

State final answer $x < -\frac{4}{3}$

A1 Do not accept $x < -1.33$ [Do not condone $<$ for $<$ in the final answer.]

Alternative method for question 1

State or imply non-modular inequality/equality $(2 - 5x)^2 >, =, +, 2^2(x - 3)^2$, or corresponding quadratic equation, or pair of linear equations

$(2 - 5x) >, =, +2(x - 3)$

B1 Two correct linear equations only

Make reasonable attempt at solving a 3-term quadratic, or solve one linear equation, or linear inequality for x

M1 $21x^2 + 4x - 32 = (3x + 4)(7x - 8) = 0$ or $2 - 5x$ or $(-2 - 5x)$ with $2(x - 3)$ or $-2(x - 3)$

Obtain critical value $x = -\frac{4}{3}$

A1

State final answer $x < -\frac{4}{3}$

A1 Do not accept $x < -1.33$ [Do not condone $<$ for $<$ in the final answer.]

Total: 4

Coaching With Shuja

Question 5

Oct/Nov 2020 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “Many candidates were unable to sketch the graph ofy =cosec x.”

> “Common mistakes in sketching the exponential graph were drawing a straight line, not including y(0)2= and not indicating the asymptote at y =1 even for x > π.”

Source: 9709_w20_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

(a) By sketching a suitable pair of graphs, show that the equation $\csc x = 1 + e^{-\frac{x}{2}}$ has exactly two roots in the interval $0 < x < \pi$. **[2]**

(b) The sequence of values given by the iterative formula

$$x_{n+1} = \pi - \sin^{-1}\left(\frac{1}{e^{-\frac{1}{2}x_n} + 1}\right),$$

with initial value $x_1 = 2$, converges to one of these roots.

Use the formula to determine this root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

Coaching With Shuja

(5(a)) Sketch a relevant graph, e.g. $y = \csc x$ | **B1** | $\csc x$, U shaped, roughly symmetrical about $x = \frac{\pi}{2}$, $y(\frac{\pi}{2}) = 1$ and domain at least $(\frac{\pi}{6}, \frac{5\pi}{6})$ |

Sketch a second relevant graph, e.g. $y = 1 + e^{-\frac{x}{2}}$, and justify the given statement | **B1** | Exponential graph needs $y(0) = 2$, negative gradient, always increasing, and $y(\pi) > 1$. Needs to mark intersections with dots, crosses, or say roots at points of intersection, or equivalent |
| | | **2** |

(5(b)) | Use the iterative formula correctly at least twice | **M1** | $2, 2.3217, 2.2760, 2.2824\ldots$ Need to see 2 iterations and following value inserted correctly |
	Obtain final answer 2.28	**A1**	Must be supported by iterations
	Show sufficient iterations to at least 4 d.p. to justify 2.28 to 2 d.p., or show there is a sign change in the interval $(2.275, 2.285)$	**A1**	
		3

Coaching With Shuja

Question 7

Oct/Nov 2020 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "A fairly common mistake was to write 2x3 + x2+ 6x – 18 = (x – (–1 – √5i)) (x – (–1 + √5i)) (x – a) and to find a."

Source: 9709_w20_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

(a) Verify that $-1+\sqrt{5}i$ is a root of the equation $2x^3+x^2+6x-18=0$. **[3]**

(b) Find the other roots of this equation. **[4]**

(7(a)) | Substitute $-1+\sqrt{5}i$ in the equation and attempt expansions of x^2 and x^3 | **M1** | All working must be seen. Allow M1 if small errors in $1-2\sqrt{5}i-5$ or $1-\sqrt{5}i-\sqrt{5}i-5$ and $4-2\sqrt{5}i+10$ or $4-4\sqrt{5}i+2\sqrt{5}i+10$ |

| | Use $i^2=-1$ correctly at least once | **M1** | $1-5$ or $4+10$ seen |

| | Complete the verification correctly | **A1** | $2(14-2\sqrt{5}i)+(-4-2\sqrt{5}i)+6(-1+\sqrt{5}i)-18=0$ |

| | | **3** |

(7(b)) | State second root $-1-\sqrt{5}i$ | **B1** | |

| | Carry out a complete method for finding a quadratic factor with zeros $-1+\sqrt{5}i$ and $-1-\sqrt{5}i$ | **M1** | |

| | Obtain x^2+2x+6 | **A1** | |

| | Obtain root $x=\frac{3}{2}$ | **A1** | OE |

Alternative method for question 7(b)

| State second root $-1-\sqrt{5}i$ | **B1** | |

| | $(x+1-\sqrt{5}i)(x+1+\sqrt{5}i)(2x+a)=2x^3+x^2+6x-18$ | **M1** | |

| | $(1-\sqrt{5}i)(1+\sqrt{5}i)a=-18$ | **A1** | |

| | $6a=-18, a=-3$ leading to $x=\frac{3}{2}$ | **A1** | OE |

| | | **4** |

Coaching with Shuja

Question 9

Oct/Nov 2020 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common error was to multiply throughout by an expression involving (2 + 3x)3."

> "(b) This question proved more challenging for some candidates."

Source: 9709_w20_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 9

Let $f(x) = \frac{8+5x+12x^2}{(1-x)(2+3x)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. **[5]**

(9(a)) | State or imply the form $\frac{A}{1-x}+\frac{B}{2+3x}+\frac{C}{(2+3x)^2}$ | **B1** | |

| | Use a correct method for finding a coefficient | **M1** | |

| | Obtain one of $A=1, B=-1, C=6$ | **A1** | |

| | Obtain a second value | **A1** | |

| | Obtain the third value | **A1** | In the form $\frac{A}{1-x}+\frac{Dx+E}{(2+3x)^2}$, where $A=1, D=-3$ and $E=4$ can score B1 M1 A1 A1 A1 as above. |

| | | **5** |

(9(b)) | Use a correct method to find the first two terms of the expansion of $(1-x)^{-1}, (2+3x)^{-1}, (1+\frac{3}{2}x)^{-1}, (2+3x)^{-2}$ or $(1+\frac{3}{2}x)^{-2}$ | **M1** | Symbolic coefficients are not sufficient for the M1. $A\frac{1+(-1)(-x)+(-1)(-2)(-x)^2}{2...}$, $A=1$ |

| | | $\frac{B}{2}\left[\frac{1+(-1)(\frac{3x}{2})+(-1)(-2)(\frac{3x}{2})^2}{2...}\right]$, $B=1$ |

| | | $\frac{C}{4}\left[\frac{1+(-2)(\frac{3x}{2})+(-2)(-3)(\frac{3x}{2})^2}{2...}\right]$, $C=6$ |

| | Obtain correct un-simplified expansions up to the term in of each partial fraction | **A1 FT +** | $(1+x+x^2)+(-\frac{1}{2}+(\frac{3}{4})x-(\frac{9}{8})x^2)$ |

| | | **A1 FT +** | $(\frac{6}{4}-(\frac{18}{4})x+(\frac{81}{8})x^2)$ [The FT is on A, B, C] |

| | | **A1 FT** | $(1-\frac{1}{2}+\frac{4}{1})+(1+\frac{3}{4}-\frac{18}{4})x+(1-\frac{8}{8}+\frac{81}{8})x^2$ |

| | Obtain final answer $2-\frac{11}{4}x+10x^2$, or equivalent | **A1** | Allow unsimplified fractions. $(Dx+E)(1+(-2)(\frac{3x}{2})+(-2)(-3)(\frac{3x}{2})^2)$ where $D=-3, E=4$ can score B1 M1 A1 A1 A1 as above. The FT is on A, D, E. |

| | | **5** |

Coaching with Shuja

Question 10

Oct/Nov 2020 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

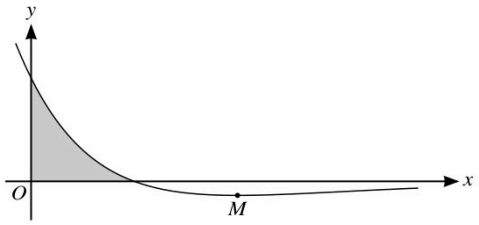
> "Question 10 (a) This should have been a straightforward application of the product rule for differentiation, but many –x candidates struggled because they were unable to differentiate e2 ."

> "Candidates often found the integration of e2 difficult, with similar errors to (a) seen."

Source: 9709_w20_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 10

Coaching With Shuja

10



The diagram shows the curve $y = (2 - x)e^{-\frac{1}{2}x}$, and its minimum point M.

(a) Find the exact coordinates of M. [5]

(a) Find the exact coordinates of ***M***. **[5]**

(b) Find the area of the shaded region bounded by the curve and the axes. Give your answer in terms of ***e***. **[5]**

Coaching with Shuja

(10(a)) | Use correct product or quotient rule | *M1 | $\frac{dy}{dx} = (-\frac{1}{2})(2-x)e^{-\frac{1}{2}x} - e^{-\frac{1}{2}x}$ M1 requires at least one of derivatives correct |

| | Obtain correct derivative in any form | A1 | |

| | Equate derivative to zero and solve for x | DM1 | |

| | Obtain $x = 4$ | A1 | ISW |

| | Obtain $y = -2e^{-2}$, or exact equivalent | A1 | |

| | | 5 |

(10(b)) | Commence integration and reach | *M1 | Condone omission of dx.
$a(2-x)e^{-\frac{1}{2}x} + b\int e^{-\frac{1}{2}x}dx$ |

| | Obtain $-2(2-x)e^{-\frac{1}{2}x} - 2\int e^{-\frac{1}{2}x}dx$ | A1 | OE |

| | Complete integration and obtain $2xe^{-\frac{1}{2}x}$ | A1 | OE |

| | Use correct limits, $x = 0$ and $x = 2$, correctly, having integrated twice | DM1 | Ignore omission of zeros and allow max of 1 error |

| | Obtain answer $4e^{-1}$, or exact equivalent | A1 | ISW |

Alternative method for question 10(b)

| | $\frac{d}{dx}(2xe^{-\frac{1}{2}x})$ | *M1 A1 | |

| | : $2xe^{-\frac{1}{2}x}$ | A1 | |

| | Use correct limits, $x = 0$ and $x = 2$, correctly, having integrated twice | DM1 | Ignore omission of zeros and allow max of 1 error |

| | Obtain answer $4e^{-1}$, or exact equivalent | A1 | ISW |

| | | 5 |

Coaching With Shuja

Question 1

Oct/Nov 2020 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A few candidates who used the correct method did not give the final answer to the required accuracy."

Source: 9709_w20_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

Solve the equation

$$\ln(1 + e^{-3x}) = 2.$$

Give the answer correct to 3 decimal places. **[3]**

Coaching With Shuja

1	State that $1 + e^{-3x} = e^2$	B1	With no errors seen to that point
	Use correct method to solve an equation of the form $e^{-3x} = a$, where $a > 0$, for x or equivalent	M1	$(e^{-3x} = 6.389\ldots)$ Evidence of method must be seen.
	Obtain answer $x = -0.618$ only	A1	Must be 3 decimal places
		3

Alternative method for question 1

State that $1 + e^{-3x} = e^2$	B1	
Rearrange to obtain an expression for e^x and solve an equation of the form $e^x = a$, where $a > 0$, or equivalent	M1	$e^x = 3\sqrt[3]{\frac{1}{e^2-1}}$
Obtain answer $x = -0.618$ only	A1	Must be 3 decimal places
	3

Coaching With Shuja

Question 2

Oct/Nov 2020 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "There were a number of slips in the arithmetic to simplify the coefficients, but the most common errors were using powers of x rather than 6x or using an incorrect index for the expansion."

Source: 9709_w20_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Coaching With Shuja

(a) Expand $\sqrt[4]{1+6x}$ in ascending powers of x, up to and including the term in x^3, simplifying the coefficients. **[4]**

(b) State the set of values of x for which the expansion is valid. **[1]**

Coaching With Shuja

(2(a)) | State a correct unsimplified version of the x or x^2 or x^3 term | **M1** | For the given expression |
	State correct first two terms $1 + 2x$	**A1**	
	Obtain the next two terms $-4x^2 + \frac{40}{3}x^3$	**A1** + **A1**	One mark for each correct term. ISW Accept $13\frac{1}{3}$. The question asks for simplified coefficients, so candidates should cancel fractions.
		4

(2(b)) | State answer $|x| < \frac{1}{6}$ | **B1** | OE. Strict inequality |
| | | **1** |

Question 4

Oct/Nov 2020 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A common error was to rewrite tan(θ+60°) as cos(θ+60°) 1 tanθ+tan60°."

> "The most common errors were due to premature approximation of decimal values or giving incorrect over-specified answers."

Source: 9709_w20_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

(a) Show that the equation $\tan(\theta + 60^\circ) = 2\cot\theta$ can be written in the form

$$\tan^2\theta + 3\sqrt{3}\tan\theta - 2 = 0. \quad [3]$$

(b) Hence solve the equation $\tan(\theta + 60^\circ) = 2\cot\theta$, for $0^\circ < \theta < 180^\circ$. **[3]**

Coaching With Shuja

Mark Scheme Oct/Nov 2020 • Paper 3 (variant 32) • Q4

(4(a)) | Use correct $\tan(A+B)$ formula and obtain an equation in $\tan\theta$ | **M1** | e.g. $\frac{\tan\theta+\tan 60^\circ}{1-\tan\theta\tan 60^\circ} = \frac{2}{\tan\theta}$ |

| | Use $\tan 60^\circ = \sqrt{3}$ and obtain a correct horizontal equation in any form | **A1** | e.g. $\tan\theta(\tan\theta+\sqrt{3}) = 2(1-\sqrt{3}\tan\theta)$ |

| | Reduce to $\tan^2\theta + 3\sqrt{3}\tan\theta - 2 = 0$ correctly | **A1** | AG |

| | | **3** |

(4(b)) | Solve the given quadratic to obtain a value for θ | **M1** | $\left(\tan\theta = \frac{-3\sqrt{3}\pm\sqrt{35}}{2} = 0.3599, -5.556\right)$ |

| | Obtain one correct answer e.g. $\theta = 19.8^\circ$ | **A1** | Accept 1d.p. or better. If over-specified must be correct. $19.797\ldots, 100.2029\ldots$ |

| | Obtain second correct answer $\theta = 100.2^\circ$ and no others in the given interval | **A1** | Ignore answers outside the given interval. |

| | | **3** |

Coaching With Shuja

Question 6

Oct/Nov 2020 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "There were several blank responses."

Source: 9709_w20_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

The complex number u is defined by

$$u = \frac{7+i}{1-i}.$$

(a) Express u in the form $x + iy$, where x and y are real. **[3]**

(b) Show on a sketch of an Argand diagram the points A, B and C representing u, $7+i$ and $1-i$ respectively. **[2]**

(c) By considering the arguments of $7+i$ and $1-i$, show that

$$\tan^{-1}\left(\frac{4}{3}\right) = \tan^{-1}\left(\frac{1}{7}\right) + \frac{1}{4}\pi. \quad [3]$$

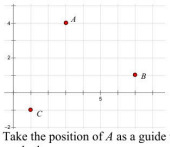
Coaching With Shuja

(6(a)) | Multiply numerator and denominator by $1+i$, or equivalent | **M1** | Must multiply out |
| | Obtain numerator $6+8i$ or denominator 2 | **A1** | |
| | Obtain final answer $u = 3+4i$ | **A1** | |

Alternative method for question 6(a)

Multiply out $(1-i)(x+iy) = 7+i$ and compare real and imaginary parts	**M1**	
	Obtain $x+y=7$ or $y-x=1$	**A1**	
	Obtain final answer $u = 3+4i$	**A1**	
		3

(6(b)) | Show the point A representing u in a relatively correct position | **B1 FT** | The FT is on $xy \neq 0$. |
| | Show the other two points B and C in relatively correct positions: approximately equal distance above / below real axis | **B1** |

6(a)	Multiply numerator and denominator by 1 + i, or equivalent	**M1**	Must multiply out
	Obtain numerator 6 + 8i or denominator 2	**A1**	
	Obtain final answer $u = 3 + 4i$	**A1**	
	Alternative method for question 6(a)		
	Multiply out $(1-i)(x+iy) = 7+i$ and compare real and imaginary parts	**M1**	
	Obtain $x+y=7$ or $y-x=1$	**A1**	
	Obtain final answer $u = 3 + 4i$	**A1**	
		3	
6(b)	Show the point A representing u in a relatively correct position	**B1 FT**	The FT is on $xy \neq 0$.
	Show the other two points B and C in relatively correct positions: approximately equal distance above / below real axis	**B1**	 Take the position of A as a guide to 'scale' if axes not marked
		2	

|
| | | **2** |

(6(c)) | State or imply $\arg(1-i) = -\frac{1}{4}\pi$ | **B1** | ArgC |
	Substitute exact arguments in $\arg(7+i) - \arg(1-i) = \arg u$	**M1**	Must see a statement about the relationship between the Args e.g. $\arg A = \arg B - \arg C$ or equivalent exact method
	Obtain $\tan^{-1}\left(\frac{4}{3}\right) = \tan^{-1}\left(\frac{1}{7}\right) + \frac{1}{4}\pi$ correctly	**A1**	Obtain given answer correctly from *their* $u = k(3+4i)$
		3

Question 7

Oct/Nov 2020 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “Common errors at this point were sign errors and putting the 3 in the wrong 3 place.”

> “The integration of cos−22x proved to be much more difficult.”

> “Common errors in attempts to complete the correct integration centred on the 2.”

Source: 9709_w20_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

(a) Solve the differential equation and obtain an expression for x in terms of t.

Total: 7

(b) State what happens to the value of x when t tends to infinity.

Total: 1

Question: The variables x and t satisfy the differential equation

$$e^{3t}\frac{dx}{dt} = \cos^2 2x$$

for $t \geq 0$. It is given that $x = 0$ when $t = 0$.

7(a) Correct separation of variables **B1**

$$\int \sec^2 2x \, dx = \int e^{-3t} \, dt$$

Needs correct structure

Obtain term $-\frac{1}{3}e^{-3t}$ **B1**

Obtain term of the form $k \tan 2x$ **M1** From correct working

Obtain term $\frac{1}{2}\tan 2x$ **A1**

Use $x = 0, t = 0$ to evaluate a constant, or as limits in a solution containing terms of the form $a \tan 2x$ and be^{-3t}, where $ab \neq 0$ **M1**

Obtain correct solution in any form **A1**

e.g. $\frac{1}{2}\tan 2x = -\frac{1}{3}e^{-3t} + \frac{1}{3}$

Obtain final answer $x = \frac{1}{2}\tan^{-1}\left(\frac{2}{3}(1 - e^{-3t})\right)$ **A1**

Total: 7

7(b) State that x approaches $\frac{1}{2}\tan^{-1}\left(\frac{2}{3}\right)$ **B1 FT**

Correct value. Accept $x \to 0.294$

The FT is dependent on letting $e^{-3t} \to 0$ in a solution containing e^{-3t}.

Total: 1

Coaching With Shuja

Question 9

Oct/Nov 2020 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “Apart from slips in the arithmetic, common errors were to use the © 2020 Cambridge International Advanced Level 9709 Mathematics November 2020 Principal Examiner Report for Teachers A B form + or to rewrite x2+4 as (x+2)2 or as (x−2)(x+2) .”

> “The integration of proved to be much more difficult, with only a minority of x2+4 candidates realising that this needed to be rewritten as two separate terms.”

Source: 9709_w20_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Question 9: Let $f(x) = \frac{4x+18}{(3x+2)(x^2+4)}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence find the exact value of $\int_0^2 f(x)\,dx$. **[6]**

Coaching With Shuja

9(a) State or imply the form

$$\frac{A}{3x+2}+\frac{Bx+C}{x^2+4}$$

B1

Use a correct method for finding a constant **M1**

Obtain one of $A=3, B=-1, C=3$ **A1**

Obtain a second value **A1**

Obtain the third value **A1**

Total: 5

9(b) Integrate and obtain $\ln(3x+2)\ldots$ **B1 FT** The FT is on A

State a term of the form $k\ln(x^2+4)$. **M1** From $\int \frac{\lambda x}{x^2+4}\,dx$

$\ldots-\frac{1}{2}\ln(x^2+4)\ldots$ **A1 FT** The FT is on B

$\ldots+\frac{3}{2}\tan^{-1}\frac{x}{2}$ **B1 FT** The FT is on C

Substitute limits correctly in an integral with at least two terms of the form $a\ln(3x+2), b\ln(x^2+4)$ and $c\tan^{-1}\left(\frac{x}{2}\right)$, and subtract in correct order **M1**

Using terms that have been obtained correctly from completed integrals

Obtain answer $\frac{3}{2}\ln 2+\frac{3}{8}\pi$, or exact 2-term equivalent **A1**

Total: 6

Question 10

Oct/Nov 2020 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

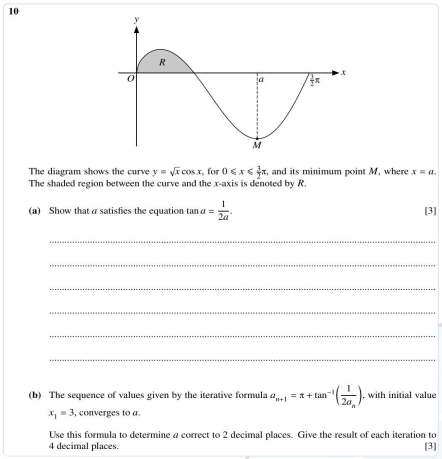
> “Some candidates made no attempt to rearrange their answer to the given form.”

> “The most common errors were to work in degrees, or to use an incorrect formula.”

Source: 9709_w20_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

Question 10:



The diagram shows the curve $y = \sqrt{x}\cos x$, for $0 < x \leq \frac{3\pi}{2}$, and its minimum point M, where $x = a$. The shaded region between the curve and the x-axis is denoted by R.

(a) Show that a satisfies the equation $\tan a = \frac{1}{2a}$. **[3]**

(b) The sequence of values given by the iterative formula $a_{n+1} = \pi + \tan^{-1}\left(\frac{1}{2a_n}\right)$, with initial value $x_1 = 3$, converges to a.

Use this formula to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

(c) Find the volume of the solid obtained when the region R is rotated completely about the x-axis. Give your answer in terms of π. **[6]**

10(a) Use correct product rule **M1**

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}\cos x - \sqrt{x}\sin x$$

Accept in a or in x

Obtain correct derivative in any form **A1**

Equate derivative to zero and obtain $\tan a = \frac{1}{2a}$ **A1**

Obtain given answer from correct working. The question says 'show that ...' so there should be an intermediate step e.g. $\cos x = 2x \sin x$.

Allow $\tan x = \frac{1}{2x}$

Total: 3

10(b) Use the iterative process correctly at least once (get one value and go on to use it in a second use of the formula) **M1**

Must be working in radians. Degrees gives 1, 12.6039, 5.4133, ... M0

Obtain final answer 3.29 **A1** Clear conclusion

Show sufficient iterations to at least 4 d.p. to justify 3.29, or show there is a sign change in the interval (3.285, 3.295) **A1**

3, 3.3067, 3.2917, 3.2923

Allow more than 4d.p. Condone truncation.

Total: 3

10(c) State or imply the indefinite integral for the volume is $\pi \int (\sqrt{x}\cos x)^2 \, dx$ **B1**

[If π omitted, or 2π or $\frac{1}{2}\pi$ used, give B0 and follow through. 4/6 available]

Use correct $\cos 2A$ formula, commence integration by parts and reach
$x(ax + b\sin 2x) + \int ax + b\sin 2x \, dx$ *M1

Alternative: $\frac{x^2}{4} + \frac{x}{4}\sin 2x - \int \frac{1}{4}\sin 2x \, dx$

Obtain $x\left(\frac{1}{2}x + \frac{1}{4}\sin 2x\right) - \int \frac{1}{2}x + \frac{1}{4}\sin 2x \, dx$ or equivalent **A1**

Complete integration and obtain $\frac{1}{4}x^2 + \frac{1}{4}x\sin 2x + \frac{1}{8}\cos 2x$ **A1** OE

Substitute limits $x = 0$ and $x = \frac{1}{2}\pi$, having integrated twice **DM1**

$$\pi\left[\frac{\pi^2}{8} + 0 - \frac{1}{4} - 0 - 0 - \frac{1}{4}\right]$$

Obtain answer $\frac{1}{16}\pi(\pi^2 - 4)$, or exact equivalent **A1** CAO

Total: 6

Question 1

Oct/Nov 2020 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "A few candidates chose to sketch a graph and reached a correct inequality as the sketch enabled them to visualise it."

Source: 9709_w20_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 33), Question 1

Coaching With Shuja

Question 1: Solve the inequality $2 - 5x > 2|x - 3|$. **[4]**

Coaching With Shuja

1 Make a recognisable sketch graph of $y = 2|x - 3|$ and the line $y = 2 - 5x$ **B1**

Need to see correct V at $x = 3$, roughly symmetrical, $x = 3$ stated, domain at least $(-2, 5)$.

Find x-coordinate of intersection with $y = 2 - 5x$ **M1**

Find point of intersection with $y = 2|x - 3|$ or solve $2 - 5x$ with $2(x - 3)$ or $-2(x - 3)$

Obtain $x = -\frac{4}{3}$ **A1**

State final answer $x < -\frac{4}{3}$ **A1**

Do not accept $x < -1.33$

[Do not condone $\leqslant$ for $<$ in the final answer.]

Alternative method for question 1

State or imply non-modular inequality/equality $(2 - 5x)^2 >, =, >, =, 2^2(x - 3)^2$ or corresponding quadratic equation, or pair of linear equations: $(2 - 5x) >, =, +2(x - 3)$ **B1**

Two correct linear equations only

Make reasonable attempt at solving a 3-term quadratic, or solve one linear equation, or linear inequality for x **M1**

$21x^2 + 4x - 32 = (3x + 4)(7x - 8) = 0$ or $2 - 5x$ or $-2(5x) - 2(x - 3)$

Obtain critical value $x = -\frac{4}{3}$ **A1**

State final answer $x < -\frac{4}{3}$ **A1**

Do not accept $x < -1.33$

[Do not condone $\leqslant$ for $<$ in the final answer.]

Total: 4

Question 5

Oct/Nov 2020 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> “Many candidates were unable to sketch the graph ofy =cosec x.”

> “Common mistakes in sketching the exponential graph were drawing a straight line, not including y(0)2= and not indicating the asymptote at y =1 even for x > π.”

Source: 9709_w20_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 33), Question 5

Question Paper Oct/Nov 2020 • Paper 3 (variant 33) • Q5

(a) By sketching a suitable pair of graphs, show that the equation $\operatorname{cosec} x = 1 + e^{-\frac{x}{2}}$ has exactly two roots in the interval $0 < x < \pi$. **[2]**

(b) The sequence of values given by the iterative formula

$$x_{n+1} = \pi - \sin^{-1}\left(\frac{1}{e^{-\frac{1}{2}x_n} + 1}\right)$$

with initial value $x_1 = 2$, converges to one of these roots.

Use the formula to determine this root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

Coaching With Shuja

Mark Scheme

5(a) Sketch a relevant graph, e.g. $y = \operatorname{cosec} x$ **B1**

$\operatorname{cosec} x$, U shaped, roughly symmetrical about $x = \frac{\pi}{2}$, $y\left(\frac{\pi}{2}\right) = 1$ and domain at least $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$.

Sketch a second relevant graph, e.g. $y = 1 + e^{-\frac{x}{2}}$, and justify the given statement **B1**

Exponential graph needs $y(0) = 2$, negative gradient, always increasing, and $y(\pi) > 1$

Needs to mark intersections with dots, crosses, or say roots at points of intersection, or equivalent

Total: 2

5(b) Use the iterative formula correctly at least twice **M1**

2, 2.3217, 2.2760, 2.2824...

Need to see 2 iterations and following value inserted correctly

Obtain final answer 2.28 **A1** Must be supported by iterations

Show sufficient iterations to at least 4 d.p. to justify 2.28 to 2 d.p., or show there is a sign change in the interval (2.275, 2.285) **A1**

Total: 3

Coaching with Shuja

Question 7

Oct/Nov 2020 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "A fairly common mistake was to write 2x3 + x2+ 6x – 18 = (x – (–1 – √5i)) (x – (–1 + √5i)) (x – a) and to find a."

Source: 9709_w20_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 33), Question 7

Coaching With Shuja

(a) Verify that $-1+\sqrt{5}i$ is a root of the equation $2x^3+x^2+6x-18=0$. **[3]**

(b) Find the other roots of this equation. **[4]**

7(a) Substitute $-1+\sqrt{5}i$ in the equation and attempt expansions of x^2 and x^3 **M1**

All working must be seen. Allow M1 if small errors in $1-2\sqrt{5}i-5$ or $1-\sqrt{5}i-\sqrt{5}i-5$ and $4-2\sqrt{5}i+10$ or $4-4\sqrt{5}i+2\sqrt{5}i+10$

Use $i^2=-1$ correctly at least once **M1**

$1-5$ or $4+10$ seen

Complete the verification correctly **A1**

$$2(14-2\sqrt{5}i)+(-4-2\sqrt{5}i)+6(-1+\sqrt{5}i)-18=0$$

Total: 3

7(b) State second root $-1-\sqrt{5}i$ **B1**

Carry out a complete method for finding a quadratic factor with zeros $-1+\sqrt{5}i$ and $-1-\sqrt{5}i$ **M1**

Obtain x^2+2x+6 **A1**

Obtain root $x=\frac{3}{2}$ **A1** OE

Alternative method for question 7(b)

State second root $-1-\sqrt{5}i$ **B1**

POR $(-1+\sqrt{5}i)(-1-\sqrt{5}i)a=9$ or **SOR** $(-1+\sqrt{5}i)+(-1-\sqrt{5}i)+a=-\frac{1}{2}$ **M1 A1**

Obtain root $x=\frac{3}{2}$ **A1** OE

Alternative method for question 7(b)

State second root $-1-\sqrt{5}i$ **B1**

$$(x+1-\sqrt{5}i)(x+1+\sqrt{5}i)(2x+a)=2x^3+x^2+6x-18$$

M1 A1

Obtain root $x=\frac{3}{2}$ **A1** OE

Total: 4

Question 9

Oct/Nov 2020 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The most common error was to multiply throughout by an expression involving (2 + 3x)3."

> "(b) This question proved more challenging for some candidates."

Source: 9709_w20_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

Coaching With Shuja

Question 9: Let $f(x) = \frac{8+5x+12x^2}{(1-x)(2+3x)^2}$.

(a) Express $f(x)$ in partial fractions. **[5]**

(b) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. **[5]**

9(a) State or imply the form

$$\frac{A}{1-x}+\frac{B}{2+3x}+\frac{C}{(2+3x)^2}$$

B1

Use a correct method for finding a coefficient **M1**

Obtain one of $A=1, B=-1, C=6$ **A1**

Obtain a second value **A1**

Obtain the third value **A1**

In the form $\frac{A}{1-x}+\frac{Dx+E}{(2+3x)^2}$, where $A=1, D=-3$ and $E=4$ can score **B1 M1 A1 A1 A1** as above.

Total: 5

9(b) Use a correct method to find the first two terms of the expansion of $(1-x)^{-1}, (2+3x)^{-1}$, or $\left(1+\frac{3}{2}x\right)^{-1}, (2+3x)^{-2}$ or $\left(1+\frac{3}{2}x\right)^{-2}$ **M1**

Symbolic coefficients are not sufficient for the M1:

$$\left[\frac{1+(-1)(-x)+(-1)(-2)(-x)^2}{2\ldots}\right] \quad A=1$$

$$\left[\frac{1+(-1)\left(\frac{3x}{2}\right)+(-1)(-2)\left(\frac{3x}{2}\right)^2}{2\ldots}\right] \quad B=1$$

$$\left[\frac{1+(-2)\left(\frac{3x}{2}\right)+(-2)(-3)\left(\frac{3x}{2}\right)^2}{4}\right] \quad C=6$$

Obtain correct un-simplified expansions up to the term in x^2 of each partial fraction **A1 FT** plus

$$\left(1+x+x^2\right)+\left(-\frac{1}{2}+\frac{3}{4}x-\frac{9}{8}x^2\right)$$

A1 FT plus

$$\left(6-\frac{18}{4}x+\frac{81}{8}x^2\right)$$

[The FT is on A, B, C]

Obtain final answer $2-\frac{11}{4}x+10x^2$, or equivalent **A1**

Allow unsimplified fractions: $(Dx+E)\frac{1+(-2)\left(\frac{3x}{2}\right)+(-2)(-3)\left(\frac{3x}{2}\right)^2}{4} \quad D=-3, E=4$

The FT is on A, D, E.

Total: 5

Question 10

Oct/Nov 2020 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 10 (a) This should have been a straightforward application of the product rule for differentiation, but many –x candidates struggled because they were unable to differentiate e2 ."

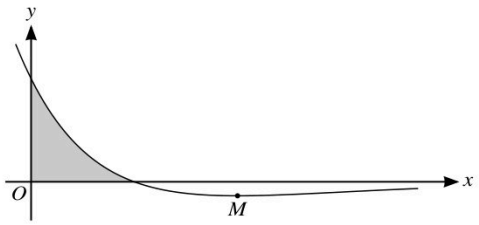
> "Candidates often found the integration of e2 difficult, with similar errors to (a) seen."

Source: 9709_w20_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 33), Question 10

Coaching With Shuja

Question 10:

10



The diagram shows the curve $y = (2 - x)e^{-\frac{1}{2}x}$, and its minimum point M.

(a) Find the exact coordinates of M. [5]

The diagram shows the curve $y = (2 - x)e^{-\frac{1}{2}x}$, and its minimum point M.

(a) Find the exact coordinates of M. **[5]**

(b) Find the area of the shaded region bounded by the curve and the axes. Give your answer in terms of e. **[5]**

10(a) Use correct product or quotient rule ***M1**

$$\frac{dy}{dx} = \left(-\frac{1}{2}\right)(2-x)e^{-\frac{1}{2}x} - e^{-\frac{1}{2}x}$$

M1 requires at least one of derivatives correct

Obtain correct derivative in any form **A1**

Equate derivative to zero and solve for x **DM1**

Obtain $x = 4$ **A1** ISW

Obtain $y = -2e^{-2}$, or exact equivalent **A1**

Total: 5

10(b) Commence integration and reach

$$a(2-x)e^{-\frac{1}{2}x} + b\int e^{-\frac{1}{2}x}\,dx$$

***M1**

Condone omission of dx

$$-2(2-x)e^{-\frac{1}{2}x} - 4e^{-\frac{1}{2}x} \quad \text{or} \quad 2xe^{-\frac{1}{2}x}$$

Obtain $-2(2-x)e^{-\frac{1}{2}x} - 2\int e^{-\frac{1}{2}x}\,dx$ **A1** OE

Complete integration and obtain $2xe^{-\frac{1}{2}x}$ **A1** OE

Use correct limits, $x = 0$ and $x = 2$, correctly, having integrated twice **DM1**

Ignore omission of zeros and allow max of 1 error

Obtain answer $4e^{-1}$, or exact equivalent **A1** ISW

Alternative method for question 10(b)

$$\frac{d}{dx}\left(2xe^{-\frac{1}{2}x}\right) = 2e^{-\frac{1}{2}x} - xe^{-\frac{1}{2}x}$$

***M1 A1**

$$\therefore 2xe^{-\frac{1}{2}x}$$

Use correct limits, $x = 0$ and $x = 2$, correctly, having integrated twice **DM1**

Ignore omission of zeros and allow max of 1 error

Obtain answer $4e^{-1}$, or exact equivalent **A1** ISW

Total: 5

Question 8

Feb/Mar 2019 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Many candidates did this but some only quoted x = 2 which gained no credit."

Source: 9709_m19_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

Coaching With Shuja

Question Paper Feb/Mar 2019 • Paper 1 (variant 12) • Q8

(i) Express $x^2 - 4x + 7$ in the form $(x + a)^2 + b$. [2]

The function f is defined by $f(x) = x^2 - 4x + 7$ for $x < k$, where k is a constant.

(ii) State the largest value of k for which f is a decreasing function. [1]

The value of k is now given to be 1.

(iii) Find an expression for $f^{-1}(x)$ and state the domain of f^{-1}. [3]

Coaching With Shuja

(i)

$(x-2)^2+3$ **B1 DB1** 2nd B1 dependent on ± 2 in first bracket

Total: 2

(ii)

Largest k is **2 B1** Accept $k \leq 2$. Must be in terms of k

Total: 1

(iii)

$y=(x-2)^2+3 \Rightarrow x-2=(\pm)\sqrt{y-3}$ **M1**

$f^{-1}(x)=2-\sqrt{x-3}$ for $x>4$ **A1B1**

Total: 3

Coaching With Shuja

Question 2

Feb/Mar 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Answer: 1.3195 (ii) Again most candidates gained both marks, although a few candidates did not use consistent symbols throughout their equations, whilst others gave a decimal answer rather than an exact answer."

Source: 9709_m19_er.pdf, Page 7, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Question Paper Feb/Mar 2019 • Paper 3 (variant 32) • Q2

The sequence of values given by the iterative formula

$$x_{n+1} = \frac{2x_n^6 + 12x_n}{3x_n^5 + 8},$$

with initial value $x_1 = 2$, converges to α.

(i) Use the formula to calculate α correct to 4 decimal places. Give the result of each iteration to 6 decimal places. [3]

(ii) State an equation satisfied by α and hence find the exact value of α. [2]

Coaching With Shuja

(i)

Use the iterative formula correctly at least once **M1**

Obtain answer $\mathbf{1.3195}$ **A1**

Show sufficient iterations to 6 d.p. to justify $\mathbf{1.3195}$ to 4 d.p., or show there is a sign change in $\mathbf{(1.31945, 1.31955)}$ **A1**

Total: 3

(ii)

State $x = \frac{2x^6+12x}{3x^6+8}$, or equivalent **B1**

State answer $\sqrt[5]{4}$, or exact equivalent **B1**

Total: 2

Coaching With Shuja

Question 3

Feb/Mar 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 3 (i) Most candidates were able to correctly expand sin(θ + 45°), but cos(θ + 60°) proved more difficult."

> "Answer: tan θ = (2√2 − 1)/(1 − √6) or equivalent (ii) Many candidates did not gain full marks due to incorrectly opting for (360° − θ*) instead of (180° + θ*) as their second answer."

Source: 9709_m19_er.pdf, Page 7, Section: Comments on specific questions – Paper 3 (variant 32), Question 3

Coaching With Shuja

Question Paper Feb/Mar 2019 • Paper 3 (variant 32) • Q3

(i) Given that $\sin(\theta + 45°) + 2\cos(\theta + 60°) = 3\cos\theta$, find the exact value of $\tan\theta$ in a form involving surds. You need not simplify your answer. [4]

(ii) Hence solve the equation $\sin(\theta + 45°) + 2\cos(\theta + 60°) = 3\cos\theta$ for $0° < \theta < 360°$. [2]

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(i)

Use trig formulae and obtain an equation in $\sin\theta$ and $\cos\theta$ **M1**

Obtain a correct equation in any form **A1**

Substitute exact trig ratios and obtain an expression for $\tan\theta$ **M1**

Obtain answer $\tan\theta = \frac{2\sqrt{2}-1}{1-\sqrt{6}}$, or equivalent **A1**

Total: 4

(ii)

State answer, e.g. $\theta = 128.4°$ **B1**

State second answer, e.g. $\theta = 308.4°$ **B1 ft**

Total: 2

Coaching With Shuja

Question 5

Feb/Mar 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 5 Nearly all candidates found this to be a difficult question."

> "Unless they realised that either implicit differentiation or the derivative of sin−1 (tan x) involving the chain rule was required, no progress was possible."

> "However whilst many candidates did reach this stage, the conversion to the given answer as a function of x proved too difficult for most."

Source: 9709_m19_er.pdf, Page 7, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

Coaching With Shuja

Question Paper Feb/Mar 2019 • Paper 3 (variant 32) • Q5

The variables x and y satisfy the relation $\sin y = \tan x$, where $-\frac{1}{2}\pi < y < \frac{1}{2}\pi$. Show that

$$\frac{dy}{dx} = \frac{1}{\cos x\sqrt{\cos 2x}}.$$

[5]

Mark Scheme Feb/Mar 2019 • Paper 3 (variant 32) • Q5

State $\cos y \frac{dy}{dx}$ as derivative of $\sin y$ **B1**

State correct derivative in terms of x and y, e.g. $\frac{\sec^2 x}{\cos y}$ **B1**

State correct derivative in terms of x, e.g. $\frac{\sec^2 x}{\sqrt{1-\tan^2 x}}$ **B1**

Use double angle formula **M1**

Obtain the given answer correctly **A1**

Total: 5

Coaching With Shuja

Question 2

May/June 2019 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "In both parts the candidates who chose to eliminate y in (i) often found the resulting algebra difficult to use to reach correct final answers."

Source: 9709_s19_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 11), Question 2

Coaching With Shuja

The line

$$4y = x + c,$$

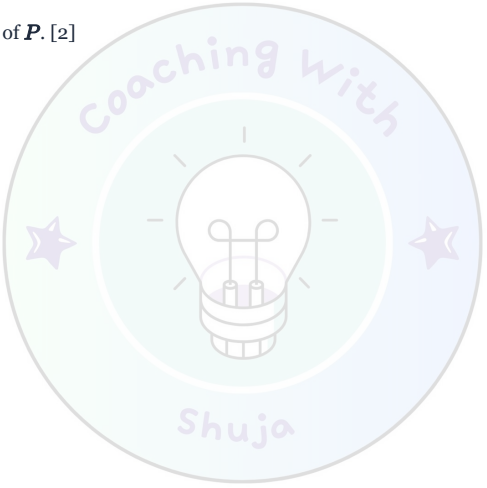
where c is a constant, is a tangent to the curve

$$y^2 = x + 3$$

at the point P on the curve.

(i) Find the value of c. [3]

(ii) Find the coordinates of P. [2]



(i)

Eliminate x or y to obtain a quadratic, e.g.

$$y^2 - 4y + c - 3 = 0$$

M1

Use discriminant $b^2 - 4ac = 0$, giving $4c - 28 = 0$ **M1**

Obtain

$$c = 7$$

A1

Total: 3

(ii)

Use $c = 7$ to obtain

$$y^2 - 4y + 4 = 0$$

M1

Obtain

$$P = (1, 2)$$

A1

Total: 2

Question 7

May/June 2019 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Question 7 (i) The components in the y and z directions were often found correctly for both vectors but the calculation of the component in the x direction proved to be very challenging for both the required vectors."

Source: 9709_s19_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 7

Coaching With Shuja

Question Paper May/June 2019 • Paper 1 (variant 11) • Q7

The diagram shows a three-dimensional shape in which the base $OABC$ and the upper surface $DEFG$ are identical horizontal squares. The parallelograms $OAED$ and $CBFG$ both lie in vertical planes. The point M is the mid-point of AF.

Unit vectors $\mathbf{i}$ and $\mathbf{j}$ are parallel to OA and OC respectively and the unit vector $\mathbf{k}$ is vertically upwards. The position vectors of A and D are given by

$$\overrightarrow{OA} = 8\mathbf{i}$$

and

$$\overrightarrow{OD} = 3\mathbf{i} + 10\mathbf{k}.$$

(i) Express each of the vectors $\overrightarrow{AM}$ and $\overrightarrow{GM}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. [3]

(ii) Hence find angle GMA. [4]

(i)

Obtain

$$\overrightarrow{AM} = 1.5\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$$

and

$$\overrightarrow{GM} = 6.5\mathbf{i} - 4\mathbf{j} - 5\mathbf{k}$$

B3

Total: 3

(ii)

Use the scalar product:

$$\overrightarrow{AM} \cdot \overrightarrow{GM} = 9.75 - 16 - 25 = -31.25$$

M1

Use the product of the moduli of $\overrightarrow{AM}$ and $\overrightarrow{GM}$ **M1**

Link correctly with

Math input error

M1

Obtain

Math input error

A1

Total: 4

Question 4

May/June 2019 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 4 (i) in particular proved to be challenging for candidates."

Source: 9709_s19_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

Coaching With Shuja

Angle x is such that

$$\sin x = a + b \quad \text{and} \quad \cos x = a - b,$$

where a and b are constants.

(i) Show that $a^2 + b^2$ has a constant value for all values of x. [3]

(ii) In the case where $\tan x = 2$, express a in terms of b. [2]

(i)

Expand

$$(a+b)^2 \quad \text{and} \quad (a-b)^2$$

correctly **B1**

Use $\sin^2 x + \cos^2 x = 1$ to form

$$(a+b)^2 + (a-b)^2 = 1$$

M1

Obtain

$$a^2 + b^2 = \frac{1}{2}$$

A1

Alternative method: use

$$a = \frac{1}{2}(\sin x + \cos x), \qquad b = \frac{1}{2}(\sin x - \cos x)$$

and hence obtain $a^2 + b^2 = \frac{1}{2}$. **B1 M1 A1**

Total: 3

(ii)

Use

$$\tan x = \frac{\sin x}{\cos x}$$

to form

$$\frac{a+b}{a-b} = 2$$

M1

Obtain

$$a = 3b$$

A1

Total: 2

Question 5

May/June 2019 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Common errors were: using 180°, or 2π rather than π; forgetting to 2 add the two radii for each sector; putting the 2 on the wrong side of the equation; and using the formula for area rather than perimeter."

Source: 9709_s19_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 5

Coaching With Shuja

Question Paper May/June 2019 • Paper 1 (variant 12) • Q5

The diagram shows a semicircle with diameter AB, centre O and radius r. The point C lies on the circumference and angle $AOC = \theta$ radians.

The perimeter of sector BOC is twice the perimeter of sector AOC. Find the value of θ correct to 2 significant figures. [5]

Coaching With Shuja

State the perimeter of sector AOC as

$$2r + r\theta$$

B1

State the angle COB as

$$\pi - \theta$$

B1

State the perimeter of sector BOC as

$$2r + r(\pi - \theta)$$

B1

Set up the equation

$$2r + r(\pi - \theta) = 2(2r + r\theta)$$

or equivalently

$$2 + \pi - \theta = 4 + 2\theta$$

M1

Obtain

$$\theta = \frac{\pi - 2}{3} = 0.38$$

to 2 significant figures. **A1**

Total: 5

Question 8

May/June 2019 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 8 (i) was very well done, with many candidates gaining full credit, although the other parts, particularly (iii), proved to be more challenging."

Source: 9709_s19_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

Coaching With Shuja

The position vectors of points A and B, relative to an origin O, are given by

$$\overrightarrow{OA} = \begin{pmatrix} 6 \\ -2 \\ -6 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OB} = \begin{pmatrix} 3 \\ k \\ -3 \end{pmatrix},$$

where k is a constant.

(i) Find the value of k for which angle AOB is $90°$. [2]

(ii) Find the values of k for which the lengths of OA and OB are equal. [2]

The point C is such that

$$\overrightarrow{AC} = 2\overrightarrow{CB}.$$

(iii) In the case where $k = 4$, find the unit vector in the direction of $\overrightarrow{OC}$. [4]

Coaching with Shuja

(i)

Use the scalar product condition for perpendicular vectors:

$$6 \cdot 3 + (-2)k + (-6)(-3) = 0$$

M1

Obtain

$$k = 18$$

A1

Total: 2

(ii)

Use equality of moduli:

$$36 + 4 + 36 = 9 + k^2 + 9$$

M1

Obtain

$$k = \pm\sqrt{58}$$

(or ± 7.62). **A1**

Total: 2

(iii)

For $k = 4$, find

$$\overrightarrow{AB} = \begin{pmatrix} -3 \\ 6 \\ 3 \end{pmatrix}$$

and use $\overrightarrow{AC} = \frac{2}{3}\overrightarrow{AB}$, then add to $\overrightarrow{OA}$ **M1**

Obtain

$$\overrightarrow{OC} = \begin{pmatrix} 4 \\ 2 \\ -4 \end{pmatrix}$$

A1

Divide by the modulus of $\overrightarrow{OC}$ **M1**

Obtain the unit vector

$$\frac{1}{6}\begin{pmatrix} 4 \\ 2 \\ -4 \end{pmatrix}$$

or

$$\frac{1}{6}(4\mathbf{i} + 2\mathbf{j} - 4\mathbf{k})$$

A1

Total: 4

Question 9

May/June 2019 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 9 This multi-stage question proved to be a real challenge for many candidates, although many fully correct solutions were also seen."

Source: 9709_s19_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 9

Coaching With Shuja

The curve C_1 has equation

$$y = x^2 - 4x + 7.$$

The curve C_2 has equation

$$y^2 = 4x + k,$$

where k is a constant. The tangent to C_1 at the point where $x = 3$ is also the tangent to C_2 at the point P.

Find the value of k and the coordinates of P. [8]

For C_1, find

$$\frac{dy}{dx} = 2x - 4$$

and hence the gradient at $x = 3$ is 2. **B1**

Use the point $(3, 4)$ and the gradient to find the tangent equation **M1**

Obtain

$$y - 4 = 2(x - 3), \quad \text{or} \quad y = 2x - 2$$

A1

Use the tangent with C_2, for example

$$(2x - 2)^2 = 4x + k,$$

leading to a quadratic in x. **M1**

Use the discriminant condition for tangency on the resulting three-term quadratic **M1**

Obtain

$$k = -5$$

A1

Substitute $k = -5$ to solve the quadratic, giving

$$4x^2 - 12x + 9 = 0$$

M1

Obtain

$$x = \frac{3}{2}, \qquad y = 1,$$

so

$$P = \left(\frac{3}{2}, 1\right).$$

A1

Total: 8

Question 10

May/June 2019 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Common errors in the formulation were: in (i) making the sum of the first 10 terms equal to the sum of the first 5 or first 15 terms; in (ii) forming an equation where the 10th term was 36 less, rather than 36 more, than the 4th term; and in (iii) multiplying the sum to infinity, rather than the sum of the first 4 terms, by 9."

Source: 9709_s19_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

Coaching With Shuja

Question Paper May/June 2019 • Paper 1 (variant 12) • Q10

(a) In an arithmetic progression, the sum of the first ten terms is equal to the sum of the next five terms. The first term is a.

(i) Show that the common difference of the progression is $\frac{1}{3}a$. [4]

(ii) Given that the tenth term is 36 more than the fourth term, find the value of a. [2]

(b) The sum to infinity of a geometric progression is 9 times the sum of the first four terms. Given that the first term is 12, find the value of the fifth term. [4]

Coaching With Shuja

Mark Scheme May/June 2019 • Paper 1 (variant 12) • Q10

(a)(i)

Use

$$S_{10} = S_{15} - S_{10}$$

or equivalently that the first 10 terms equal terms 11 to 15. **M1**

State

$$S_{10} = 5(2a + 9d)$$

B1

State the sum of the next five terms, for example

$$S_{15} - S_{10} = 7.5(2a + 14d) - 5(2a + 9d)$$

or

$$5\{2(a + 10d) + 4d\}/2$$

A1

Obtain the given result

$$d = \frac{1}{3}a$$

from convincing working. **A1**

Total: 4

(a)(ii)

Use the condition on the tenth and fourth terms:

$$a + 9d = a + 3d + 36$$

M1

With $d = \frac{1}{3}a$, obtain

$$a = 18$$

A1

Total: 2

(b)

Use

$$S_{\infty} = 9S_4$$

so

$$\frac{a}{1-r} = 9 \cdot \frac{a(1-r^4)}{1-r}$$

(or equivalent). **B1**

Eliminate a and $1 - r$ correctly to obtain

$$9(1 - r^4) = 1$$

M1

Hence

$$r^4 = \frac{8}{9}$$

A1

The fifth term is

$$12r^4 = 12 \cdot \frac{8}{9} = \frac{32}{3} = 10.7$$

A1

Total: 4

Question 3

May/June 2019 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "In part (ii), a few candidates, having found the correct two areas subtracted them, whilst others subtracted both areas from the area of triangle ABC, effectively finding the unshaded area."

Source: 9709_s19_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 3

Coaching With Shuja

Question Paper May/June 2019 • Paper 1 (variant 13) • Q3

The diagram shows triangle ABC which is right-angled at A. Angle $ABC = \frac{1}{5}\pi$ radians and $AC = 8$ cm. The points D and E lie on BC and BA respectively. The sector ADE is part of a circle with centre A and is such that BDC is the tangent to the arc DE at D.

(i) Find the length of AD. [3]

(ii) Find the area of the shaded region. [3]

Coaching With Shuja

(i)

State or imply

$$\angle EAD = \angle ACD = \frac{3\pi}{10}$$

or

$$\angle DAC = \frac{\pi}{5}$$

B1

Use a correct trigonometric equation, for example

$$AD = 8\sin\left(\frac{3\pi}{10}\right)$$

or

$$AD = 8\cos\left(\frac{\pi}{5}\right)$$

M1

Obtain

$$AD = 6.47$$

A1

Total: 3

(ii)

Find the sector area, for example

$$\frac{1}{2}(AD)^2\left(\pi - \frac{2\pi}{5}\right)$$

M1

Find the area of triangle ADC, for example

$$\frac{1}{2}\cdot AD\cdot 8\cdot\sin\left(\frac{\pi}{5}\right)$$

M1

Obtain the shaded area

35.0

(or **34.9**) square cm. **A1**

Total: 3

Question 4

May/June 2019 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Question 4 Many candidates found part (i) challenging."

Source: 9709_s19_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 4

Coaching With Shuja

The function f is defined by

$$f(x) = \frac{48}{x-1} \quad \text{for } 3 \leq x \leq 7.$$

The function g is defined by

$$g(x) = 2x - 4 \quad \text{for } a \leq x \leq b,$$

where a and b are constants.

(i) Find the greatest value of a and the least value of b which will permit the formation of the composite function gf. [2]

It is now given that the conditions for the formation of gf are satisfied.

(ii) Find an expression for $gf(x)$. [1]

(iii) Find an expression for $(gf)^{-1}(x)$. [2]

(i)

The range of f for $3 \leq x \leq 7$ is from 8 to 24.

Greatest value:

$$a = 8$$

B1

Least value:

$$b = 24$$

B1

Total: 2

(ii)

Obtain

$$gf(x) = \frac{96}{x-1} - 4$$

or equivalently

$$gf(x) = \frac{100-4x}{x-1}$$

B1

Total: 1

(iii)

Set

$$y = \frac{96}{x-1} - 4$$

and rearrange to make x the subject **M1**

Obtain

$$(gf)^{-1}(x) = \frac{96}{x+4} + 1$$

or equivalently

$$\frac{x+100}{x+4}$$

A1

Total: 2

Question 8

May/June 2019 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "This will eventually lead to four dx equations in four unknowns and relatively few candidates were successful when following this approach."

> "A common error made by some candidates was to forget the constant of integration and also to start by dy dy substituting =2 when x = –1 and =k when x = 3."

Source: 9709_s19_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 8

Question Paper May/June 2019 • Paper 1 (variant 13) • Q8

A curve is such that

$$\frac{dy}{dx} = 3x^2 + ax + b.$$

The curve has stationary points at $(-1, 2)$ and $(3, k)$.

Find the values of the constants a, b and k. [8]

Use the stationary points to form simultaneous equations:

$$f'(-1) = 0 \quad \Rightarrow \quad 3 - a + b = 0,$$

$$f'(3) = 0 \quad \Rightarrow \quad 27 + 3a + b = 0.$$

M1

Solve to obtain

$$a = -6$$

A1

Obtain

$$b = -9$$

A1

Integrate to get

$$f(x) = x^3 - 3x^2 - 9x + c$$

B1

Use $(-1, 2)$ to find c **M1**

Obtain

$$c = -3$$

A1

Use $x = 3$ in the curve equation to find k **M1**

Obtain

$$k = -30$$

A1

Total: 8

Question 9

May/June 2019 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> “Question 9 This question was quite challenging, although strong candidates were able to score well.”

> “A significant proportion of candidates made no attempt at all or some of the parts of this question.”

Source: 9709_s19_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 9

Coaching With Shuja

The function

$$f : x \mapsto p \sin^2 2x + q$$

is defined for

$$0 \leq x \leq \pi,$$

where p and q are positive constants. The diagram shows the graph of $y = f(x)$.

(i) In terms of p and q, state the range of f. [2]

(ii) State the number of solutions of the following equations.

(a) $f(x) = p + q$ [1]

(b) $f(x) = q$ [1]

(c) $f(x) = \frac{1}{2}p + q$ [1]

(iii) For the case where $p = 3$ and $q = 2$, solve the equation $f(x) = 4$, showing all necessary working. [5]

Coaching With Shuja

(i)

State the range as

$$q \leq f(x) \leq p + q$$

B1 B1

Total: 2

(ii)

For $f(x) = p + q$, the number of solutions is

2

B1

For $f(x) = q$, the number of solutions is

3

B1

For $f(x) = \frac{1}{2}p + q$, the number of solutions is

4

B1

Total: 3

(iii)

With $p = 3$ and $q = 2$, form

$$3\sin^2 2x + 2 = 4,$$

so

$$\sin^2 2x = \frac{2}{3}$$

M1

Obtain

$$\sin 2x = \pm\sqrt{\frac{2}{3}}$$

A1

Find at least two correct values for $2x$ in the required interval, for example

$$0.955,\ 2.19,\ 4.10,\ 5.33$$

A1

Obtain

$$x = 0.478,\ 1.09,\ 2.05,\ 2.66$$

A1 A1

Total: 5

Question 10

May/June 2019 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "There was a significant proportion of dx 2 candidates who made no attempt at part (iii)."

Source: 9709_s19_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 10

Coaching With Shuja

Question Paper May/June 2019 • Paper 1 (variant 13) • Q10

The diagram shows part of the curve with equation

$$y = (3x + 4)^{\frac{1}{2}}$$

and the tangent to the curve at the point A. The x-coordinate of A is 4.

(i) Find the equation of the tangent to the curve at A. [5]

(ii) Find, showing all necessary working, the area of the shaded region. [5]

(iii) A point is moving along the curve. At the point P the y-coordinate is increasing at half the rate at which the x-coordinate is increasing. Find the x-coordinate of P. [3]

(i)

At $x = 4$,

$$y = (3 \cdot 4 + 4)^{1/2} = 4$$

B1

Differentiate:

$$\frac{dy}{dx} = \frac{1}{2}(3x + 4)^{-1/2} \cdot 3$$

B1

At $x = 4$,

$$\frac{dy}{dx} = \frac{3}{8}$$

B1

Use a line through $(4, 4)$ with gradient $\frac{3}{8}$ **M1**

Obtain

$$y - 4 = \frac{3}{8}(x - 4)$$

or

$$y = \frac{3}{8}x + \frac{5}{2}$$

A1

Total: 5

(ii)

Find the area under the line from **0** to **4**, equal to

$$13$$

B1

Integrate the curve:

$$\int (3x + 4)^{1/2}\, dx = \frac{2}{9}(3x + 4)^{3/2}$$

B1 B1

Apply the limits **0** to **4** to obtain the area under the curve:

$$\frac{112}{9} = 12\frac{4}{9}$$

M1

Hence the shaded area is

$$13 - \frac{112}{9} = \frac{5}{9}$$

(or **0.556**). **A1**

Total: 5

(iii)

Use

$$\frac{dy}{dx} = \frac{1}{2}$$

B1

Set

$$\frac{3}{2}(3x+4)^{-1/2} = \frac{1}{2}$$

M1

Obtain

$$(3x+4)^{1/2} = 3,$$

so

$$x = \frac{5}{3}$$

A1

Total: 3

Question 1

May/June 2019 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common error was to ignore the modulus signs and use ordinates −3, −2, 0 and 4."

Source: 9709_s19_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

Use the trapezium rule with 3 intervals to estimate the value of

$$\int_0^3 |2^x - 4|\, dx.$$

[3]

State or imply the ordinates

$$3,\ 2,\ 0,\ 4$$

B1

Use the trapezium rule correctly with $h = 1$ and the four ordinates **M1**

Obtain

$$5.5$$

A1

Total: 3

Question 2

May/June 2019 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common errors in the working occurred when candidates started with −(−1)= before dealing with the 2."

Source: 9709_s19_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

Coaching With Shuja

Question Paper May/June 2019 • Paper 3 (variant 31) • Q2

Showing all necessary working, solve the equation

$$\ln(2x-3) = 2\ln x - \ln(x-1).$$

Give your answer correct to 2 decimal places. [4]

Coaching With Shuja

Use a logarithm law for a product, quotient or power **M1**

Obtain a correct equation free of logarithms, for example

$$(2x-3)(x-1)=x^2$$

or

$$x^2-5x+3=0$$

A1

Solve the three-term quadratic, obtaining at least one root **M1**

Obtain the valid answer only:

$$x=4.30$$

A1

Total: 4

Coaching With Shuja

Question 3

May/June 2019 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common errors were to introduce an additional term $+\cdots$ at the start of their work or to not differentiate the 1 on the right hand side of the equation."

Source: 9709_s19_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 3

Coaching With Shuja

Question Paper May/June 2019 • Paper 3 (variant 31) • Q3

Find the gradient of the curve

$$x^3 + 3xy^2 - y^3 = 1$$

at the point with coordinates $(1, 3)$. [4]

Coaching With Shuja

State or imply

$$3y^2 + 6xy\frac{dy}{dx}$$

as the derivative of $3xy^2$ **B1**

State or imply

$$3y^2\frac{dy}{dx}$$

as the derivative of y^3 **B1**

Equate the derivative of the left hand side to zero, substitute $(1, 3)$ and find the gradient **M1**

Obtain

$$\frac{10}{3}$$

A1

Total: 4

Coaching With Shuja

Question 4

May/June 2019 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 4 Most candidates were aware that θ = , but some found dealing with (θ + 45°) challenging."

Source: 9709_s19_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

Coaching With Shuja

By first expressing the equation

$$\cot\theta - \cot(\theta + 45°) = 3$$

as a quadratic equation in $\tan\theta$, solve the equation for

$$0° < \theta < 180°.$$

[6]

Use a correct trigonometric formula to obtain an equation in $\tan\theta$ **M1**

Obtain a correct equation in any form, for example

$$\frac{1}{\tan\theta} - \frac{1-\tan\theta}{1+\tan\theta} = 3$$

A1

Reduce to

$$2\tan^2\theta + 3\tan\theta - 1 = 0$$

A1

Solve the three-term quadratic and find a value of θ **M1**

Obtain

$$\theta = 15.7°$$

A1

Obtain

$$\theta = 119.3°$$

with no other values in the range. **A1**

Total: 6

Coaching with Shuja

Question 6

May/June 2019 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(ii) Most candidates did not use the result from part (i) to tackle this integral."

Source: 9709_s19_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

(i) By first expanding $\sin(2x + x)$, show that

$$\sin 3x \equiv 3\sin x - 4\sin^3 x.$$

[4]

(ii) Hence, showing all necessary working, find the exact value of

$$\int_0^{\pi/3} \sin^3 x \, dx.$$

[4]

(i)

State the correct expansion of

$$\sin(2x + x)$$

B1

Use trigonometric formulae and Pythagoras to express $\sin 3x$ in terms of $\sin x$ **M1**

Obtain a correct expression in any form **A1**

Obtain

$$\sin 3x \equiv 3\sin x - 4\sin^3 x$$

A1

Total: 4

(ii)

Use the identity, integrate, and obtain the terms

$$-\frac{1}{4}\cos x$$

and

$$\frac{1}{12}\cos 3x$$

B1 B1

Use the limits correctly in an integral of the form

$$a\cos x + b\cos 3x,$$

where $ab \neq 0$ **M1**

Obtain the exact value

$$\frac{5}{24}$$

A1

Total: 4

Question 1

May/June 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common error in expanding (1+3) was to use in place of 3."

Source: 9709_s19_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

Question Paper May/June 2019 • Paper 3 (variant 32) • Q1

Find the coefficient of x^3 in the expansion of

$$(3-x)(1+3x)^{\frac{1}{3}}$$

in ascending powers of x. [4]

Coaching With Shuja

State the unsimplified term in x^2, or its coefficient, in the expansion of $(1+3x)^{1/3}$ **B1**

State the unsimplified term in x^3, or its coefficient, in the expansion of $(1+3x)^{1/3}$ **B1**

Multiply by $(3-x)$ to give the two terms contributing to x^3 **M1**

Obtain the coefficient

$$6$$

A1

Total: 4

Question 2

May/June 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “Another common error was to use 9 =3×3.”

Source: 9709_s19_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Coaching With Shuja

Question Paper May/June 2019 • Paper 3 (variant 32) • Q2

Showing all necessary working, solve the equation

$$9^x = 3^x + 12.$$

Give your answer correct to 2 decimal places. [4]

Coaching With Shuja

State or imply, with $u = 3^x$,

$$u^2 - u - 12 = 0$$

B1

Solve for u or 3^x and obtain the positive root

$$u = 4$$

B1

Use a correct method to solve

$$3^x = 4$$

M1

Obtain

$$x = 1.26$$

only. **A1**

Total: 4

Coaching With Shuja

Question 5

May/June 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "However, the common error was to combine the quadratic factor with a linear factor + , in which case no further progress was possible."

Source: 9709_s19_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

Coaching With Shuja

Question Paper May/June 2019 • Paper 3 (variant 32) • Q5

Throughout this question the use of a calculator is not permitted.

It is given that the complex number $-1+\sqrt{3}i$ is a root of the equation

$$kx^3+5x^2+10x+4=0,$$

where k is a real constant.

(i) Write down another root of the equation. [1]

(ii) Find the value of k and the third root of the equation. [6]

(i)

Since the coefficients are real, another root is

$$-1-\sqrt{3}i$$

B1

Total: 1

(ii)

Substitute $x=-1+\sqrt{3}i$ in the equation and attempt expansions of x^2 and x^3 **M1**

Use $i^2=-1$ correctly at least once **M1**

Obtain

$$k=2$$

A1

Carry out a complete method for finding the quadratic factor with roots $-1+\sqrt{3}i$ and $-1-\sqrt{3}i$ **M1**

Obtain the quadratic factor

$$x^2+2x+4$$

A1

Obtain the third root

$$x=-\frac{1}{2}$$

A1

Total: 6

Question 6

May/June 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors occurred in the area of the sector, with many candidates using an angle of rather than2."

Source: 9709_s19_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

In the diagram, A is the mid-point of the semicircle with centre O and radius r. A circular arc with centre A meets the semicircle at B and C. The angle OAB is equal to x radians. The area of the shaded region bounded by AB, AC and the arc with centre A is equal to half the area of the semicircle.

(i) Use triangle OAB to show that

$$AB = 2r\cos x.$$

[1]

(ii) Hence show that

$$x = \cos^{-1}\sqrt{\frac{\pi}{16x}}.$$

[2]

(iii) Verify by calculation that x lies between 1 and 1.5. [2]

(iv) Use an iterative formula based on the equation in part (ii) to determine x correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

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(i)

Use correct trigonometry in triangle OAB to obtain

$$AB = 2r\cos x$$

B1

Total: 1

(ii)

Use the correct method for finding the area of the sector and the semicircle and form an equation in x **M1**

Obtain

$$x = \cos^{-1}\sqrt{\frac{\pi}{16x}}$$

correctly. **A1**

Total: 2

(iii)

Calculate values of a relevant expression at $x = 1$ and $x = 1.5$, working in radians **M1**

Use correct values and complete the sign-change or bounding argument correctly **A1**

Total: 2

(iv)

Use the iteration

$$x_{n+1} = \cos^{-1}\sqrt{\frac{\pi}{16x_n}}$$

correctly at least twice, working in radians **M1**

Obtain

$$x = 1.144$$

A1

Show sufficient iterations to at least 5 decimal places to justify the answer to 3 decimal places **A1**

Total: 3

Question 9

May/June 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "There was one very common error: candidates who reached a correct equation 3=2 often concluded that = ."

Source: 9709_s19_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

The points A and B have position vectors

$$\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$

and

$$3\mathbf{i} + \mathbf{j} + \mathbf{k}$$

respectively. The line l has equation

$$\mathbf{r} = 2\mathbf{i} + \mathbf{j} + \mathbf{k} + \mu(\mathbf{i} + \mathbf{j} + 2\mathbf{k}).$$

(i) Show that l does not intersect the line passing through A and B. [5]

(ii) The plane m is perpendicular to AB and passes through the mid-point of AB. The plane m intersects the line l at the point P. Find the equation of m and the position vector of P. [5]

(i)

Find a vector equation for the line AB **M1**

Obtain, for example,

$$\mathbf{r} = (\mathbf{i} + 2\mathbf{j} - \mathbf{k}) + \lambda(2\mathbf{i} - \mathbf{j} + 2\mathbf{k})$$

A1

Equate two pairs of components of general points on this line and on l, and solve for λ or μ **M1**

Obtain a correct value, for example $\lambda = 0$ and $\mu = -1$ from one pair of component equations **A1**

Verify that all three component equations are not satisfied, so the lines do not intersect **A1**

Total: 5

(ii)

State or imply that the midpoint of AB has position vector

$$2\mathbf{i} + \frac{3}{2}\mathbf{j}$$

B1

Use the normal direction parallel to AB and substitute the midpoint to find the constant in the plane equation **M1**

Obtain the plane equation

$$4x - 2y + 4z = 5$$

(or equivalent). **A1**

Substitute the components of l in the plane equation and solve for μ **M1**

Obtain

$$\mu = -\frac{1}{2}$$

and hence the position vector of P:

$$\frac{3}{2}\mathbf{i} + \frac{1}{2}\mathbf{j}$$

A1

Total: 5

Question 1

May/June 2019 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 1 Many fully correct answers were seen, and only a few candidates were unable to present their final answer to the accuracy required."

Source: 9709_s19_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 33), Question 1

Coaching With Shuja

Question Paper May/June 2019 • Paper 3 (variant 33) • Q1

Use logarithms to solve the equation

$$5^{3-2x} = 4(7^x),$$

giving your answer correct to 3 decimal places. [4]

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Use the logarithm law for a product or quotient **M1**

Use the logarithm law for powers twice **M1**

Obtain a correct linear equation in x, for example

$$(3-2x)\ln 5 = \ln 4 + x\ln 7$$

A1

Obtain

$$x = 0.666$$

A1

Total: 4

Coaching With Shuja

Question 4

May/June 2019 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Many candidates with a correct expression were unable to make any progress, for example inverting terms individually to change the equation from one in e–a to one in ea."

> "Although few candidates obtained the equation correctly, most realised it was a quadratic in ea and were able to solve it."

> "Most candidates did not reject one of the roots, and many did not give the final answer in exact form."

Source: 9709_s19_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 4

Coaching With Shuja

The equation of a curve is

$$y = \frac{1 + e^{-x}}{1 - e^{-x}}, \qquad x > 0.$$

(i) Show that $\frac{dy}{dx}$ is always negative. [3]

(ii) The gradient of the curve is equal to -1 when $x = a$.

Show that a satisfies the equation

$$e^{2a} - 4e^{a} + 1 = 0.$$

Hence find the exact value of a. [4]

(i)

Use the quotient rule or product rule **M1**

Obtain the correct derivative in any form **A1**

Reduce to

$$-\frac{2e^{-x}}{(1-e^{-x})^2}$$

or an equivalent expression, and explain why this is always negative for $x > 0$. **A1**

Total: 3

(ii)

Equate the derivative to -1 and obtain the given equation

$$e^{2a} - 4e^a + 1 = 0$$

B1

State or imply the quadratic

$$u^2 - 4u + 1 = 0,$$

where $u = e^a$. **B1**

Solve for a **M1**

Obtain

$$a = \ln(2 + \sqrt{3})$$

with no other value. **A1**

Total: 4

Question 4

Oct/Nov 2019 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "The use of rn−1 rather than rn was a common error made by candidates."

Source: 9709_w19_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 11), Question 4

Coaching With Shuja

4 A runner who is training for a long-distance race plans to run increasing distances each day for 21 days. She will run x km on day 1, and on each subsequent day she will increase the distance by 10% of the previous day's distance. On day 21 she will run 20 km.

(i) Find the distance she must run on day 1 in order to achieve this. Give your answer in km correct to 1 decimal place. **[3]**

(ii) Find the total distance she runs over the 21 days. **[2]**

Coaching With Shuja

(i) Identifies common ratio as 1.1 **B1**

Use of $x(1.1)^{20} = 20$ **M1** SOI

$$x = \frac{20}{(1.1)^{20}} = 3.0$$

A1 Accept 2.97

Total: 3

(ii) $their 3.0 \times \frac{[(1.1)^{21}-1]}{1.1-1} \rightarrow 192$ **M1** Correct formula used for M mark. **A1** Allow 2.97 used from (i); Accept 190 from $x = 2.97\ldots$

Total: 2

Coaching With Shuja

Question 6

Oct/Nov 2019 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Candidates who equated the gradient of the line to the gradient of the curve were often less successful and often found difficulty obtaining the quadratic in x only."

Source: 9709_w19_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 6

Coaching With Shuja

6 A straight line has gradient m and passes through the point $(0, -2)$. Find the two values of m for which the line is a tangent to the curve $y = x^2 - 2x + 7$ and, for each value of m, find the coordinates of the point where the line touches the curve. **[7]**

Equation of line is $y = mx - 2$ **B1** OR

$x^2 - 2x + 7 = mx - 2 \rightarrow x^2 - x(2 + m) + 9 = 0$ **M1**

Apply $b^2 - 4ac(-0) \rightarrow (2 + m)^2 - 4 \times 9(-0)$ ***M1**

$m = 4$ or -8 **A1**

$m = 4 \rightarrow x^2 - 6x + 9 = 0 \rightarrow x = 3$ **DM1**
$m = -8 \rightarrow x^2 + 6x + 9 = 0 \rightarrow x = -3$

$(3, 10), (-3, 22)$ **A1A1**

Alternative method for question 6

$$\frac{dy}{dx} = 2x - 2$$

B1

$2x - 2 = m$ **M1**

$x^2 - 2x + 7 = (2x - 2)x - 2 = 2x^2 - 2x - 2$ **M1**

$x^2 - 9 = 0 \rightarrow x = \pm 3$ **A1**

$(3, 10), (-3, 22)$ **A1A1**

When $x = 3, m = 4$; when $x = -3, m = -8$ **A1**

Total: 7

Question 10

Oct/Nov 2019 | Paper 1 (variant 11)

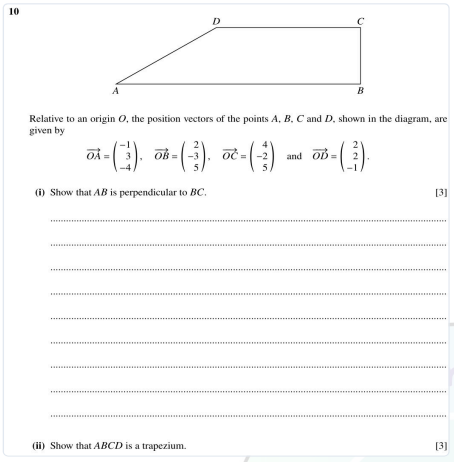
Examiner Report — Why students found this question difficult

> "This was the most frequently omitted question part."

Source: 9709_w19_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 10

Coaching With Shuja

10



Relative to an origin O, the position vectors of the points A, B, C and D, shown in the diagram, are given by

$$\overrightarrow{OA} = \begin{pmatrix} -1 \\ -3 \\ -4 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}, \quad \overrightarrow{OC} = \begin{pmatrix} 4 \\ -2 \\ 5 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OD} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}.$$

(i) Show that AB is perpendicular to BC. **[3]**

(ii) Show that $ABCD$ is a trapezium. **[3]**

(iii) Find the area of $ABCD$, giving your answer correct to 2 decimal places. **[3]**

(i)

$$\mathbf{AB} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} - \begin{pmatrix} -1 \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 3 \\ -6 \\ 9 \end{pmatrix}, \quad \mathbf{BC} = \begin{pmatrix} 4 \\ -2 \\ 5 \end{pmatrix} - \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

B1B1 Condone reversal of labels

$\mathbf{AB}.\mathbf{BC} = 6 - 6 \rightarrow 0$ (hence perpendicular) **B1** AG

(ii)

$$\mathbf{DC} = \begin{pmatrix} 4 \\ -2 \\ 5 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix}$$

B1 Or: $\mathbf{CD} = \begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$

$\mathbf{AB} = k\mathbf{DC}$ **M1** OE Expect $k = \frac{3}{2}$ Or: $\mathbf{DC}.\mathbf{BC} = 4 - 4 = 0$ hence BC is also perpendicular to DC Or: $\mathbf{AB}.\mathbf{DC} = 1$ or $\mathbf{AB}.\mathbf{DC} = -1$, angle between lines is 0 or 180

AB is parallel to DC, hence $ABCD$ is a trapezium **A1**

(iii) $|\mathbf{AB}| = \sqrt{9 + 36 + 81} = \sqrt{126} = 11.22$ **M1** Method for finding at least 2 magnitudes

$|\mathbf{DC}| = \sqrt{4 + 16 + 36} = \sqrt{56} = 7.483$

$|\mathbf{BC}| = \sqrt{4 + 1 + 0} = \sqrt{5} = 2.236$

Area $= \frac{1}{2}(their\,AB + their\,DC) \times their\,BC = 20.92$ **M1A1** OE

Question 3

Oct/Nov 2019 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 3 This question appeared to be more challenging for candidates, with many candidates not identifying the need to integrate or wrongly combining integration with the equation of a straight line."

Source: 9709_w19_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

3 A curve is such that $\frac{dy}{dx} = \frac{k}{\sqrt{x}}$, where k is a constant. The points $P(1, -1)$ and $Q(4, 4)$ lie on the curve. Find the equation of the curve. **[4]**

$$(y =)\frac{kx^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = \frac{k\sqrt{x}}{\frac{1}{2}}(+c)$$

B1 OE

Substitutes both points into an integrated expression with a '$+c$' and solve as far as a value for one variable. **M1** Expect to see $-1 = 2k + c$ and $4 = 4k + c$

$k = 2\frac{1}{2}$ **and** $c = -6$ **A1** WWW

$y = 5\sqrt{x} - 6$ **A1** OE From correct values of both k & c and correct integral.

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Question 4

Oct/Nov 2019 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Many candidates, however, did not find the correct length of AT or BT, often not realising that angles OAT and OBT were 90°."

Source: 9709_w19_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

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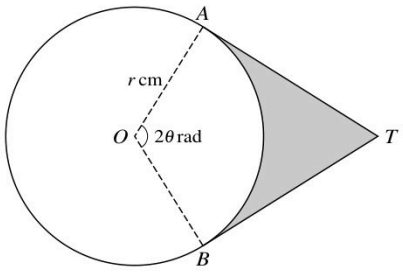
4

4



The diagram shows a circle with centre O and radius r cm. Points A and B lie on the circle and angle $AOB = 2\theta$ radians. The tangents to the circle at A and B meet at T.

(i) Express the perimeter of the shaded region in terms of r and θ. [3]

The diagram shows a circle with centre $\boldsymbol{O}$ and radius $\boldsymbol{r}$ cm. Points $\boldsymbol{A}$ and $\boldsymbol{B}$ lie on the circle and angle $\boldsymbol{AOB = 2\theta}$ radians. The tangents to the circle at $\boldsymbol{A}$ and $\boldsymbol{B}$ meet at $\boldsymbol{T}$.

(i) Express the perimeter of the shaded region in terms of $\boldsymbol{r}$ and $\boldsymbol{\theta}$. **[3]**

(ii) In the case where $\boldsymbol{r = 5}$ and $\boldsymbol{\theta = 1.2}$, find the area of the shaded region. **[4]**

Mark Scheme Oct/Nov 2019 • Paper 1 (variant 12) • Q4

(i) Arc length $AB = 2r\theta$ **B1**

$\tan\theta = \frac{AT}{r}$ or $\frac{BT}{r} \rightarrow AT$ or $BT = r\tan\theta$ **B1** Accept or $\sqrt{\left(\frac{r}{\cos\theta}\right)^2 - r^2}$ or $\frac{r\sin\theta}{\sin\left(\frac{\pi}{2}-\theta\right)}$ NOT $(90-\theta)$

$P = 2r\theta + 2r\tan\theta$ **B1FT** OE, FT for their arc length + $2\times$ their AT

Total: 3

(ii) [Markings indicate calculations for $r = 5, \theta = 1.2$] Area = (circle sector) - (triangle area) + 20.92

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Question 6

Oct/Nov 2019 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 6 This question, particularly parts (a) and (b)(ii) were challenging for candidates."

> "In part (a) common errors ( ) included attempting to expand tan 2x+1 , working in degrees, and using π− rather than π+ to find the required values."

Source: 9709_w19_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 6

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Question Paper Oct/Nov 2019 • Paper 1 (variant 12) • Q6

(a) Given that $x > 0$, find the two smallest values of x, in radians, for which $3\tan(2x + 1) = 1$. Show all necessary working. **[4]**

(b) The function f: $x \mapsto 3\cos^2 x - 2\sin^2 x$ is defined for $0 \leq x \leq \pi$.

(i) Express $f(x)$ in the form $a\cos^2 x + b$, where a and b are constants. **[1]**

(ii) Find the range of f. **[2]**

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Mark Scheme Oct/Nov 2019 • Paper 1 (variant 12) • Q6

(a) $(2x+1) = \tan^{-1}(\frac{1}{3}) = (0.322$ or 18.4 OR -0.339 rad or $8.7°)$ ***M1** Correct order of operations.
Allow degrees.

Either their $0.322 + \pi$ or 2π **DM1** Must be in radians
Or their $-0.339 + \frac{\pi}{2}$ or π

$x = 1.23$ **or** $x = 2.80$ **A1** AWRT for either correct answer, accept 0.39π or 0.89π

Total: 4 (For the second answer with no other answers between 0 and 2.8 **SC1** For both 1.2 and 2.8)

(b)(i) $5\cos^2 x - 2$ **B1** Allow $a = 5, b = -2$

Total: 1

(b)(ii) -2 **B1FT** FT for sight of their b

3 B1FT FT for sight of their $a + b$

Total: 2

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Question 10

Oct/Nov 2019 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (iii) was frequently omitted by candidates or the incorrect limits were used; using the coordinates of B or the y intercept of the curve when integrating the curve with respect to x was quite common."

> "Very few candidates were able to obtain the whole area including the triangle."

Source: 9709_w19_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

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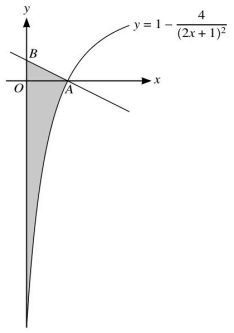

The diagram shows part of the curve $y = 1 - \frac{4}{(2x+1)^2}$. The curve intersects the x-axis at A. The normal to the curve at A intersects the y-axis at B.

(i) Obtain expressions for $\frac{dy}{dx}$ and $\int y\,dx$. [4]

The diagram shows part of the curve $\boldsymbol{y = 1 - \frac{4}{(2x+1)^2}}$. The curve intersects the $\boldsymbol{x}$-axis at $\boldsymbol{A}$. The normal to the curve at $\boldsymbol{A}$ intersects the $\boldsymbol{y}$-axis at $\boldsymbol{B}$.

(i) Obtain expressions for $\boldsymbol{\frac{dy}{dx}}$ and $\boldsymbol{\int y\,dx}$. **[4]**

(ii) Find the coordinates of $\boldsymbol{B}$. **[4]**

(iii) Find, showing all necessary working, the area of the shaded region. **[4]**

(i)

$$\frac{dy}{dx} = [0] + [(2x+1)^{-3}] \times [+16]$$

B2,1,0 OE. Full marks for 3 correct components. Withhold one mark for each error or omission.

$$\int y dx = [x] + [(2x+1)^{-1}] \times [+2](+c)$$

B2,1,0 OE. Full marks for 3 correct components. Withhold one mark for each error or omission.

Total: 4

(ii) At $A, x = \frac{1}{2}$. **B1** Ignore extra answer $x = -1.5$

$$\frac{dy}{dx} = 2 \rightarrow \text{Gradient of normal}(- = -\frac{1}{2})$$

***M1** With their positive value of x at A and their $\frac{dy}{dx}$, uses $m_1 m_2 = -1$

Equation of normal:

$$y - 0 = -\frac{1}{2}(x - \frac{1}{2}) \text{ or } y - 0 = -\frac{1}{2}(0 - \frac{1}{2}) \text{ or } 0 = -\frac{1}{2}x + \frac{1}{2} + c$$

DM1 Use of their x at A and their normal gradient.

$$B(0, \frac{1}{4})$$

A1

Total: 4

(iii)

$$\int_0^{1/2} 1 - \frac{4}{(2x+1)^2}(dx)$$

***M1** $\int y dx$ SOI with 0 and their positive x coordinate of A.

$$[\frac{1}{2} + 1] - [0 + 2] = (-\frac{1}{2})$$

DM1 Substitutes both 0 and their $\frac{1}{2}$ into their $\int y dx$ and subtracts.

Area of triangle above x-axis $= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{4} = (\frac{1}{16})$ **B1**

Total area of shaded region $= \frac{9}{16}$ **A1** OE (including AWRT 0.563)

Question 2

Oct/Nov 2019 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "A large number of candidates gave the incorrect answer of $x > -2$ for the domain of the inverse function, not recognising the significance of the function g being defined for $x > 4$."

Source: 9709_w19_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 13), Question 2

Coaching With Shuja

2 The function g is defined by $g(x) = x^2 - 6x + 7$ for $x > 4$. By first completing the square, find an expression for $g^{-1}(x)$ and state the domain of g^{-1}. **[5]**

$$(y =)[(x-3)^2][-2]$$

***B1** DB1 dependent on 3 in 1st bracket

$$x-3=(\pm)\sqrt{y+2} \quad \text{or} \quad y-3=(\pm)\sqrt{x+2}$$

M1 Correct order of operations

$$(g^{-1}(x))=3+\sqrt{x+2}$$

A1 Must be in terms of x

Domain (of g^{-1}) is $(x)>-1$ B1 Allow $(-1,\infty)$. Do not allow $y>-1$ or $g(x)>-1$ or $g^{-1}(x)>-1$

Total: 5

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Question 9

Oct/Nov 2019 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> “Another approach was to find two expressions for ar2 which leads to 5k −62 3k = 6k −4, but many candidates using this approach could not effectively manipulate the 3k algebra.”

Source: 9709_w19_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 9

Coaching With Shuja

9 The first, second and third terms of a geometric progression are $3k$, $5k - 6$ and $6k - 4$, respectively.

(i) Show that k satisfies the equation $7k^2 - 48k + 36 = 0$. **[2]**

(ii) Find, showing all necessary working, the exact values of the common ratio corresponding to each of the possible values of k. **[4]**

(iii) One of these ratios gives a progression which is convergent. Find the sum to infinity. **[2]**

(i) $\frac{5k-6}{3k} = \frac{6k-4}{5k-6} \rightarrow (5k-6)^2 = 3k(6k-4)$ **M1** OR any valid relationship

$25k^2 - 60k + 36 = 18k^2 - 12k \rightarrow 7k^2 - 48k + 36$ **A1** AG

Total: 2

(ii) $k = \frac{6}{7}, 6$ **B1B1** Allow $0.857(1)$ for $\frac{6}{7}$

When $k = \frac{6}{7}$, $r = -\frac{2}{3}$ **B1** Must be exact

When $k = 6$, $r = \frac{4}{3}$ **B1**

Total: 4

(iii) Use of $S_\infty = \frac{a}{1-r}$ with $r =$ **their** $-\frac{2}{3}$ and $a = 3 \times$ **their** $\frac{6}{7}$ **M1** Provided $0 < |$**their** $r - 2/3| < 1$

$\frac{18}{7} \div \left(1 + \frac{2}{3}\right) = \frac{54}{35}$ or **1.54 A1** FT if $0.857(1)$ has been used in part (ii).

Total: 2

Coaching With Shuja

Question 10

Oct/Nov 2019 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Few candidates successfully answered part (iii), several omitted this part."

Source: 9709_w19_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 10

Coaching With Shuja

10 Relative to an origin O, the position vectors of the points A, B and X are given by

$$\overrightarrow{OA} = \begin{pmatrix} -8 \\ -4 \\ 2 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 10 \\ 2 \\ 11 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OX} = \begin{pmatrix} -2 \\ -2 \\ 5 \end{pmatrix}.$$

(i) Find $\overrightarrow{AX}$ and show that AXB is a straight line. **[3]**

The position vector of a point C is given by $\overrightarrow{OC} = \begin{pmatrix} 1 \\ -8 \\ 3 \end{pmatrix}$.

(ii) Show that CX is perpendicular to AX. **[3]**

(iii) Find the area of triangle ABC. **[3]**

Coaching With Shuja

Mark Scheme Oct/Nov 2019 • Paper 1 (variant 13) • Q10

(i) $\overrightarrow{AX} = \begin{pmatrix}6\\2\\3\end{pmatrix}$, and one of $\overrightarrow{AB} = \begin{pmatrix}18\\6\\9\end{pmatrix}$, $\overrightarrow{XB} = \begin{pmatrix}12\\4\\6\end{pmatrix}$, $\overrightarrow{BX} = \begin{pmatrix}-12\\-4\\-6\end{pmatrix}$ **B1B1**

State $\overrightarrow{AB} = 3\overrightarrow{AX}$ (or $\overrightarrow{XB} = 2\overrightarrow{AX}$ or $\overrightarrow{AB} = \frac{3}{2}\overrightarrow{XB}$ etc) hence straight line **B1** WWW A conclusion (i.e. a straight line) is required.

OR $\frac{\overrightarrow{AX} \cdot \overrightarrow{AB}}{|\overrightarrow{AX}||\overrightarrow{AB}|} = 1 (\rightarrow \theta = 0)$ ***or*** $\frac{\overrightarrow{AX} \cdot \overrightarrow{BX}}{|\overrightarrow{AX}||\overrightarrow{BX}|} = -1 (\rightarrow \theta = 180)$

hence straight line

Total: 3

(ii) $\overrightarrow{CX} = \begin{pmatrix}-3\\6\\2\end{pmatrix}$ **B1**

$\overrightarrow{CX} \cdot \overrightarrow{AX} = -18 + 12 + 6$ **M1**

$= 0$ (hence CX is perpendicular to AX) **A1**

Total: 3

(iii) $|\overrightarrow{CX}| = \sqrt{3^2 + 6^2 + 2^2}$, $|\overrightarrow{AB}| = \sqrt{18^2 + 6^2 + 9^2}$ **M1**

Both attempted

Area $\triangle ABC = \frac{1}{2} \times$ **their 21** $\times$ **their 7** $= 73\frac{1}{2}$ **M1A1** Accept answers which round to **73.5**

Total: 3

Question 11

Oct/Nov 2019 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

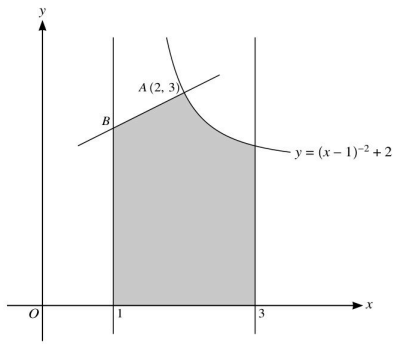
> "There were many candidates that provided good responses to this question by avoiding these common errors."

Source: 9709_w19_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 11

Coaching With Shuja

11

11



The diagram shows part of the curve $y = (x - 1)^{-2} + 2$, and the lines $x = 1$ and $x = 3$. The point A on the curve has coordinates $(2, 3)$. The normal to the curve at A crosses the line $x = 1$ at B.

(i) Show that the normal AB has equation $y = \frac{1}{2}x + 2$. [3]

The diagram shows part of the curve $\boldsymbol{y = (x-1)^{-2} + 2}$, and the lines $\boldsymbol{x = 1}$ and $\boldsymbol{x = 3}$. The point $\boldsymbol{A}$ on the curve has coordinates $\boldsymbol{(2, 3)}$. The normal to the curve at $\boldsymbol{A}$ crosses the line $\boldsymbol{x = 1}$ at $\boldsymbol{B}$.

(i) Show that the normal $\boldsymbol{AB}$ has equation $\boldsymbol{y = \frac{1}{2}x + 2}$. **[3]**

(ii) Find, showing all necessary working, the volume of revolution obtained when the shaded region is rotated through $\boldsymbol{360^\circ}$ about the $\boldsymbol{x}$-axis. **[8]**

(i) $\frac{dy}{dx} = -2(x-1)^{-3}$ **B1**

When $x = 2$, $m = -2 \rightarrow$ gradient of normal $= -\frac{1}{m}$ **M1** m must come from differentiation

Equation of normal is $y - 3 = \frac{1}{2}(x-2) \rightarrow y = \frac{1}{2}x + 2$ **A1** AG Through $(2, 3)$ with gradient $-\frac{1}{m}$. Simplify to AG

Total: 3

(ii) $(\pi)\int y_1^2(\mathrm{d}x), (\pi)\int y_2^2(\mathrm{d}x)$ *M1 Attempt to integrate y^2 for at least one of the functions

$(\pi)\left[(\frac{1}{2}x+2)^2\right.$ or $\left.(\frac{1}{4}x^2+2x+4)\right.$ **A1A1**

$(\pi)\left[(x-1)^{-4}+4(x-1)^{-2}+4\right]$ **A1A1** Must have at least 2 terms correct for each integral

$(\pi)\left[2(\frac{1}{2}x+2)^3\right.$ or $\left.\frac{1}{12}x^3+x^2+4x\right.$ **DM1** Apply limits to at least 1 integrated expansion

Attempt to add 2 volume integrals (or 1 volume integral + frustum) **DM1**

$\pi\left[7\frac{7}{12}+6\frac{7}{24}\right]$ **A1** 2 volume integrals (or 1 volume integral + frustum) added correctly
$= \frac{2}{3}+4+8-(\frac{1}{12}+1+4), \{\frac{-1}{24}-2+12-(\frac{-1}{3}-4+8)\}$

$13\frac{7}{8}\pi$ or $\frac{111}{8}\pi$ or 13.9π or 43.6 **A1** $\frac{2}{3}+4+8-(\frac{1}{12}+1+4), \{\frac{-1}{24}-2+12-(\frac{-1}{3}-4+8)\}$

Total: 8

Question 1

Oct/Nov 2019 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The next step of rearranging to make y the subject of the equation proved more challenging for candidates."

Source: 9709_w19_er.pdf, Page 19, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

1 Given that $\ln(1+e^{2y}) = x$, express y in terms of x. **[3]**

Coaching With Shuja

Mark Scheme

State $1 + e^{2y} = e^{x}$ **B1**

Make y the subject **M1** Rearrange to $e^{2y} = \ldots$ and use logs

Obtain answer $y = \frac{1}{2}\ln(e^{x} - 1)$ **A1** OE

Total: 3

Question 4

Oct/Nov 2019 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(iii) Very few correct answers were seen."

Source: 9709_w19_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

Coaching With Shuja

4 The number of insects in a population t weeks after the start of observations is denoted by N. The population is decreasing at a rate proportional to $Ne^{-0.02t}$. The variables N and t are treated as continuous, and it is given that when $t = 0$, $N = 1000$ and $\frac{dN}{dt} = -10$.

(i) Show that N and t satisfy the differential equation

$$\frac{dN}{dt} = -0.01e^{-0.02t}N.$$

[1]

(ii) Solve the differential equation and find the value of t when $N = 800$. **[6]**

(iii) State what happens to the value of N as t becomes large. **[1]**

(i) State $\frac{dN}{dt} = ke^{-0.02t}N$ and show $k = -0.01$ **B1** OE $(-10 = k \times 1 \times 1000)$

Total: 1

(ii) Separate variables correctly and integrate at least one side **B1** $\int \frac{1}{N} dN = \int -0.01e^{-0.02t} dt$

Obtain term $\ln N$ **B1** OE

Obtain term $0.5e^{-0.02t}$ **B1** OE

Use $N = 1000, t = 0$ to evaluate a constant, or as limits, in a solution with terms $a \ln N$ and $be^{-0.02t}$, where $ab \neq 0$ **M1**

Obtain correct solution in any form
e.g. $\ln N - \ln 1000 = 0.5(e^{-0.02t} - 1)$ **A1** $\ln 1000 - \frac{1}{2} = 6.41$

Substitute $N = 800$ and obtain $t = 29.6$ **A1**

Total: 6

(iii) State that N approaches $\frac{1000}{\sqrt{e}}$ **B1** Accept **606** or **607** or **606.5**

Total: 1

Coaching With Shuja

Question 7

Oct/Nov 2019 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Common errors were often due to slips with signs when forming equations or in the arithmetic."

> "The most common errors were sign errors and arithmetic errors in completing the vector product."

Source: 9709_w19_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

7 Two lines l and m have equations $\mathbf{r} = a\mathbf{i} + 2\mathbf{j} + 3\mathbf{k} + \lambda(\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})$ and $\mathbf{r} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k} + \mu(2\mathbf{i} - \mathbf{j} + \mathbf{k})$ respectively, where a is a constant. It is given that the lines intersect.

(i) Find the value of a. **[4]**

(ii) When a has this value, find the equation of the plane containing l and m. **[5]**

Coaching With Shuja

(i) Express general point of l or m in component form
e.g. $(a+\lambda, 2-2\lambda, 3+3\lambda)$ or $(2+2\mu, 1-\mu, 2+\mu)$ **B1**

Equate at least two pairs of corresponding components and solve for λ or for μ **M1**

Obtain **either** $\lambda = -2$ or $\mu = -5$ **A1**

or $\lambda = \frac{1}{3}a$ or $\mu = \frac{2}{3}a - 1$ **A1**

or $\lambda = \frac{1}{5}(a-4)$ or $\mu = \frac{1}{5}(3a-7)$ **A1**

Obtain $a = -6$ **A1**

Total: 4

(ii) Use scalar product to obtain a relevant equation in a, b and c, e.g. $a - 2b + 3c = 0$ **B1**

Obtain a second equation, e.g. $2a - b + c = 0$ and solve for one ratio **M1**

Obtain $a : b : c = 1 : 5 : 3$ **A1** OE

Substitute a relevant point and values of a, b, c in general equation and find d **M1**

Obtain correct answer $x + 5y + 3z = 13$ **A1FT** OE. The FT is on a from part (i), if used

Alternative method for question 7(ii)

Attempt to calculate vector product of relevant vectors, **M1** e.g. $(\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}).(2\mathbf{i} - \mathbf{j} + \mathbf{k})$

Obtain two correct components **A1**

Obtain correct answer, e.g. $\mathbf{i} + 5\mathbf{j} + 3\mathbf{k}$ **A1**

Substitute a relevant point and find d **M1**

Obtain correct answer $x + 5y + 3z = 13$ **A1FT** OE. The FT is on a from part (i), if used

Total: 5

Question 8

Oct/Nov 2019 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common error, apart from in arithmetic, was to omit either the term with denominator x, or the term with denominator x2."

Source: 9709_w19_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 8

Coaching With Shuja

8 Let $f(x) = \frac{x^2+x+6}{x^2(x+2)}$.

(i) Express $f(x)$ in partial fractions. **[5]**

(ii) Hence, showing full working, show that the exact value of $\int_1^4 f(x)\,dx$ is $\frac{9}{4}$. **[5]**

(i) State or imply the form $\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2}$ **B1**

Use a correct method for finding a constant **M1**

Obtain one of $A = -1$, $B = 3$, $C = 2$ **A1**

Obtain a second value **A1**

Obtain the third value **A1** Allow in the form $\frac{4x+B}{x^2} + \frac{C}{x+2}$

Total: 5

(ii) Integrate and obtain terms $\ln x - \frac{3}{x} + 2\ln(x+2)$ **B1FT** + The FT is on A, B, C; or on A, D, E.

Substitute limits correctly in an integral with terms $a\ln x$, $\frac{b}{x}$ and $c\ln(x+2)$, where $abc \neq 0$ **M1**
$-\ln 4 - \frac{3}{4} + 2\ln 6(+\ln 1) + 3 - 2\ln 3$

Obtain $\frac{9}{4}$ following full and exact working **A1** AG - work to combine or simplify logs is required

Total: 5

Coaching with Shuja

Question 2

Oct/Nov 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A few candidates used the product rule, but the derivative of (1 – x2)–1 proved challenging."

Source: 9709_w19_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Coaching With Shuja

2 The curve with equation $y = \frac{e^{-x}}{1-x^2}$ has a stationary point in the interval $-1 < x < 1$. Find $\frac{dy}{dx}$ and hence find the x-coordinate of this stationary point, giving the answer correct to 3 decimal places. **[5]**

3 The polynomial $x^4 + 3x^3 + ax + b$, where a and b are constants, is denoted by $p(x)$. When $p(x)$ is divided by $x^2 + x - 1$ the remainder is $2x + 3$. Find the values of a and b. **[5]**

Coaching With Shuja

2

Use correct quotient rule or correct product rule **M1**

Obtain correct derivative in any form **A1** $\frac{dy}{dx} = \frac{-2e^{-x}(1-x^2)+2xe^{-x}}{(1-x^2)^2}$

Equate derivative to zero and obtain a 3 term quadratic in x **M1** From correct work only
e.g. $2x^2 + 2x - 2 = 0$ or $x^2 + x - 1$ **A1**

Solve and obtain $x = 0.618$ only **A1** From correct work only

Total: 5

3

Divide $x^4 + 3x^3 + ax + b$ by $x^2 + x - 1$ to obtain remainder in the form $cx + d$ or divide and use remainder theorem directly **M1** [3 marks]

Quotient of form $x^2 + 2x + k$ and remainder $cx + d$ where c and d depend on a and k **A1**

Form two equations, e.g. $c = a + 2k + 2 = 2$ and $d = -k + b - 2 = 3$ [needs 3 marks] **M1**

Obtain both a and b **A1A1**

Total: 5

Coaching With Shuja

Question 3

Oct/Nov 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “Question 3 This question appeared to be particularly challenging for candidates.”

> “Having reached a linear remainder, many candidates did not go on to set their remainder identically equal to 2x+3, and to solve for a and b accordingly.”

> “A common error was for candidates to treat2x+3 as a factor.”

Source: 9709_w19_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 32), Question 3

1 Solve the equation $5\ln(4-3^x)=6$. Show all necessary working and give the answer correct to 3 decimal places. **[3]**

Question 3

Commence division and reach partial quotient $x^2 + kx$ | M1

Obtain correct quotient $x^2 + 2x - 1$ | A1

Set their linear remainder equal to $2x + 3$ and solve for a or for b | M1 | Remainder $= (a + 3)x + (b - 1)$

Obtain answer $a = -1$ | A1

Obtain answer $b = 4$ | A1

Alternative method for question 3

State $x^4 + 3x^3 + ax + b = (x^2 + x - 1)(x^2 + Ax + B) + 2x + 3$ and form and solve two equations in A and B | M1 | e.g. $3 = 1 + A$ and $0 = -1 + A + B$

Obtain $A = 2, B = -1$ | A1

Form and solve equations for a or b | M1 | e.g. $a = B - A + 2, b = -B + 3$

Obtain answer $a = -1$ | A1

Obtain answer $b = 4$ | A1

Total: 5

Coaching with Shuja

Question 4

Oct/Nov 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "(ii) Most candidates found this question challenging."

> "Many candidates did not work with enough accuracy in answering this question, given that part (i) required an answer to 3 decimal places, candidates should have identified that at least 3 decimal places was needed for this part."

Source: 9709_w19_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

(i) Express $(\sqrt{6})\sin x + \cos x$ in the form $R\sin(x + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. State the exact value of R and give α correct to 3 decimal places. **[3]**

(ii) Hence solve the equation $(\sqrt{6})\sin 2\theta + \cos 2\theta = 2$, for $0^\circ < \theta < 180^\circ$. **[4]**

Coaching With Shuja

4(i) | State $R = \sqrt{7}$ | B1

Use correct trig formulae to find α | M1 | e.g. $\tan\alpha = \frac{1}{\sqrt{6}}$, $\sin\alpha = \frac{1}{\sqrt{7}}$, or $\cos\alpha = \frac{\sqrt{6}}{\sqrt{7}}$

Obtain $\alpha = 22.208^\circ$ | A1 | ISW

4(ii) | Evaluate $\sin^{-1}\left(\frac{2}{\sqrt{7}}\right)$ to at least 1 d.p. | B1FT | 49.107° to 3 d.p. **B1 can be implied by correct answer(s) later. The FT is on their R**

SC: allow B1 for a correct alternative equation e.g. $3\tan^2\theta - 2\sqrt{6}\tan\theta + 1 = 0$

Use correct method to find a value of θ in the interval | M1 | Must get to θ

Obtain answer, e.g. 13.4° | A1 | Accept correct over-specified answers. $13.449\ldots, 54.3425\ldots$

Obtain second answer, e.g. 54.3° and no extras in the given interval | A1 | Ignore answers outside the given interval.

Total: 4

Coaching With Shuja

Question 6

Oct/Nov 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Most candidates went on to obtain kln sin1 θ but 2 only a minority had k = 2.Many candidates found difficulty in the correct simplification and removal of the 1 1 logarithms."

> "Very few fully correct solutions were seen as the majority of candidates did not attempt the final step to express the answer in terms of cosθ."

Source: 9709_w19_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

6 The variables x and θ satisfy the differential equation

$$\sin\frac{1}{2}\theta\frac{dx}{d\theta} = (x+2)\cos\frac{1}{2}\theta$$

for $0 < \theta < \pi$. It is given that $x = 1$ when $\theta = \frac{1}{3}\pi$. Solve the differential equation and obtain an expression for x in terms of $\cos\theta$. **[8]**

6 | **Separate variables correctly to obtain** $\int \frac{1}{x+2}dx = \int \cot \frac{1}{2}\theta d\theta$ | **B1** | Or equivalent integrands. Integral signs SOI

Obtain term $\ln(x+2)$ | **B1** | Modulus signs not needed.

Obtain term of the form $k \ln \sin \frac{1}{2}\theta$ | **M1**

Obtain term $2 \ln \sin \frac{1}{2}\theta$ | **A1**

Use $x = 1, \theta = \frac{1}{3}\pi$ **to evaluate a constant, or as limits, in an expression containing** $p \ln(x+2)$ **and** $q \ln\left(\sin \frac{1}{2}\theta\right)$ | **M1** | Reach $C =$ an expression or a decimal value

Obtain correct solution in any form e.g. $\ln(x+2) = 2 \ln \sin \frac{1}{2}\theta + \ln 12$ | **A1** | $\ln 12 = 2.4849\ldots$ Accept constant to at least 3 s.f. Accept with $\ln 3 - 2 \ln \frac{1}{2}$

Remove logarithms and use correct double angle formula | **M1** | Need correct algebraic process. $\left(\frac{x+2}{12} = \frac{1-\cos\theta}{2}\right)$

Obtain answer $x = 4 - 6\cos\theta$ | **A1**

Total: 8

Coaching With Shuja

Question 7

Oct/Nov 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 7 (a) Many candidates found this question challenging, often candidates would confuse their working involving z or z* rather than x and y."

> "Many candidates found drawing the line Imz=3 challenging, with several candidates drawing a second circle."

Source: 9709_w19_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

(a) Find the complex number z satisfying the equation

$$z + \frac{iz}{z^*} - 2 = 0,$$

where z^* denotes the complex conjugate of z. Give your answer in the form $x + iy$, where x and y are real. **[5]**

(b) (i) On a single Argand diagram sketch the loci given by the equations $|z - 2i| = 2$ and $\text{Im } z = 3$, where Im z denotes the imaginary part of z. **[2]**

(ii) In the first quadrant the two loci intersect at the point P. Find the exact argument of the complex number represented by P. **[2]**

7(a) | Substitute and obtain a correct horizontal equation in x and y in any form | B1 |
$zz^* + iz - 2z^* = 0 \Rightarrow x^2 + y^2 + ix - y - 2x + 2iy = 0$ Allow if still includes brackets and/or i^2

Use $i^2 = -1$ and equate real and imaginary parts to zero OE | M1 | For their horizontal equation

Obtain two correct equations e.g. $x^2 + y^2 - y - 2x = 0$ and $x + 2y = 0$ **| A1 |** Allow $ix + 2iy = 0$

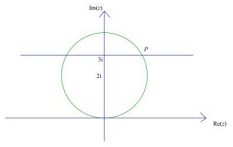
Solve for x or for y | DM1

Obtain answer $\frac{6}{5} - \frac{3}{5}i$ and no other | A1 | OE, condone $\frac{1}{5}(6 - 3i)$

Total: 5

7(b)(i) | Show a circle with centre $2i$ and radius 2 | B1

Show horizontal line $y = 3$ in first and second quadrant | B1 |

Question	Answer	Marks	Guidance
7(b)(i)	Show a circle with centre 2i and radius 2	B1	
	Show horizontal line $y = 3$ – in first and second quadrant	B1	
			SC: For clearly labelled axes not in the conventional directions, allow B1 for a fully 'correct' diagram.
		2	
7(b)(ii)	Carry out a complete method for finding the argument. (Not by measuring the sketch)	M1	$\left(z = \sqrt{3} + 3i\right)$ Must show working if using 1.7 in place of $\sqrt{3}$.
	Obtain answer $\frac{1}{3}\pi$ (or 60°)	A1	**SC:** Allow B2 for 60° with no working
		2	

SC: For clearly labelled axes not in the conventional directions, allow B1 for a fully 'correct' diagram.

Total: 2

7(b)(ii) | Carry out a complete method for finding the argument. (Not by measuring the sketch) | M1 | $(z = \sqrt{3} + 3i)$ Must show working if using 1.7 in place of $\sqrt{3}$.

Obtain answer $\frac{1}{3}\pi$ (or 60°) | A1 | SC: Allow B2 for 60° with no working

Total: 2

Question 8

Oct/Nov 2019 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "In x2 +2 x2 +2 this question many candidates did not include brackets, which was a common error seen."

Source: 9709_w19_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 8

Coaching With Shuja

8 Let

$$f(x) = \frac{2x^2 + x + 8}{(2x - 1)(x^2 + 2)}.$$

(i) Express $f(x)$ in partial fractions. **[5]**

(ii) Hence, showing full working, find

$$\int_1^5 f(x)\,dx,$$

giving the answer in the form $\ln c$, where c is an integer. **[5]**

8(i) | **State or imply the form** $\frac{4}{2x-1} + \frac{Bx+C}{x^2+2}$ | **B1**

Use a correct method for finding a constant | **M1**

Obtain one of $A = 4, B = -1, C = 0$ | **A1**

Obtain a second value | **A1**

Obtain the third value | **A1**

Total: 5

8(ii) | **Integrate and obtain term** $2\ln(2x-1)$ | **B1FT** | The FT is on A. $\frac{1}{2}4\ln(2x-1)$

Integrate and obtain term of the form $k\ln(x^2+2)$ | **M1** | From $\frac{nx}{x^2+2}$

Obtain term $-\frac{1}{2}\ln(x^2+2)$ | **A1FT** | The FT is on B

Substitute limits correctly in an integral of the form $a\ln(2x-1) + b\ln(x^2+2)$, **where** $ab \neq 0$ | **DM1** | $2\ln 9(-2\ln 1) - \frac{1}{2}\ln 27 + \frac{1}{2}\ln 3$

Obtain answer $\ln 27$ **after full and correct exact working** | **A1** | ISW

Total: 5

Coaching with Shuja

Question 2

Oct/Nov 2019 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Candidates doing so usually gained full marks, although a few candidates incorrectly used the remainder + 24 instead of – 24."

Source: 9709_w19_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 33), Question 2

Coaching With Shuja

2 The polynomial $6x^3 + ax^2 + bx - 2$, where a and b are constants, is denoted by $\mathbf{p}(x)$. It is given that $(2x + 1)$ is a factor of $\mathbf{p}(x)$ and that when $\mathbf{p}(x)$ is divided by $(x + 2)$ the remainder is -24. Find the values of a and b. **[5]**

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2 | Substitute $x = -\frac{1}{2}$, equate result to zero and obtain a correct equation, e.g. $-\frac{6}{8} + \frac{1}{4}a - \frac{1}{2}b - 2 = 0$ | B1

Substitute $x = -2$ and equate result to -24 | M1

Obtain a correct equation, e.g. $-48 + 4a - 2b - 2 = -24$ | A1

Solve for a or for b | DM1

Obtain $a = 5$ and $b = -3$ | A1

Total: 5

Coaching With Shuja

Question 4

Oct/Nov 2019 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "A common error was to find inverse tan directly from t2 instead of first obtaining t."

Source: 9709_w19_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 33), Question 4

Coaching With Shuja

(i) By first expanding $\tan(2x + x)$, show that the equation $\tan 3x = 3\cot x$ can be written in the form $\tan^4 x - 12\tan^2 x + 3 = 0$. **[4]**

(ii) Hence solve the equation $\tan 3x = 3\cot x$ for $0° < x < 90°$. **[3]**

4(i) | Use $\tan(A+B)$ formula to express the LHS in terms of $\tan 2x$ and $\tan x$ | M1

Using the $\tan 2A$ formula, express the entire equation in terms of $\tan x$ | M1

Obtain a correct equation in $\tan x$ in any form | A1

Obtain the given form correctly | A1 | AG

Total: 4

4(ii) | Use correct method to solve the given equation for x | M1

Obtain answer, e.g. $x = 26.8^\circ$ | A1

Obtain second answer, e.g. $x = 73.7^\circ$ and no other | A1 | Ignore answers outside the given interval

Total: 3

Coaching With Shuja

Question 5

Oct/Nov 2019 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 5 (i) Candidates appeared to find this question quite challenging."

Source: 9709_w19_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 33), Question 5

Coaching With Shuja

Question Paper Oct/Nov 2019 • Paper 3 (variant 33) • Q5

(i) By sketching a suitable pair of graphs, show that the equation $\ln(x+2) = 4e^{-x}$ has exactly one real root. **[2]**

(ii) Show by calculation that this root lies between $x = 1$ and $x = 1.5$. **[2]**

(iii) Use the iterative formula $x_{n+1} = \ln\left(\frac{4}{f(x_n)+2}\right)$ to determine the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

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5(i) | Sketch a relevant graph, e.g. $y = \ln(x + 2)$ | B1

Sketch a second relevant graph, e.g. $y = 4e^{-x}$, and justify the given statement | B1 |
Consideration of behaviour for $x < 0$ is needed for the second B1

Total: 2

5(ii) | Calculate the values of a relevant expression or pair of expressions at $x = 1$ and $x = 1.5$ | M1

Complete the argument correctly with correct calculated values | A1

Total: 2

5(iii) | Set up and use the iterative formula or similar. Must interpret the notation correctly. | M1

Give results of at least 2 iterations to 4 d.p. | M1

Obtain correct root to 2 d.p. | A1

Total: 3

Coaching With Shuja

Question 8

Oct/Nov 2019 | Paper 3 (variant 33)

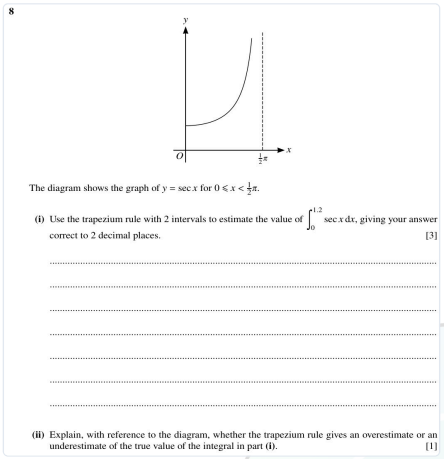
Examiner Report — Why students found this question difficult

> "Question 8 (i) Many candidates found this trapezium rule question challenging."

Source: 9709_w19_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 33), Question 8

Coaching With Shuja

8 The diagram shows the graph of $y = \sec x$ for $0 \leq x < \frac{1}{2}\pi$.


8

The diagram shows the graph of $y = \sec x$ for $0 \leqslant x < \frac{1}{2}\pi$.

(i) Use the trapezium rule with 2 intervals to estimate the value of $\int_0^{1.2} \sec x \, dx$, giving your answer correct to 2 decimal places. [3]

(ii) Explain, with reference to the diagram, whether the trapezium rule gives an overestimate or an underestimate of the true value of the integral in part (i). [1]

(i) Use the trapezium rule with 2 intervals to estimate the value of $\int_0^{1.2} \sec x \, dx$, giving your answer correct to 2 decimal places. **[3]**

(ii) Explain, with reference to the diagram, whether the trapezium rule gives an overestimate or an underestimate of the true value of the integral in part **(i)**. **[1]**

(iii) P is the point on the part of the curve $y = \sec x$ for $0 \leq x < \frac{1}{2}\pi$ at which the gradient is 2. By first differentiating $\frac{1}{\cos x}$, find the x-coordinate of P, giving your answer correct to 3 decimal places. **[6]**

(i) B1 State or imply ordinates $1, 1.2116\ldots, 2.7597\ldots$

M1 Use correct formula, or equivalent, with $h = 0.6$

A1 Obtain answer 1.85

Total: 3

(ii) B1 Explain why the rule gives an overestimate

Total: 1

(iii) M1 Differentiate using quotient or chain rule

A1 Obtain correct derivative in terms of $\sin x$ and $\cos x$

M1 Equate derivative to 2, use Pythagoras and obtain an equation in $\sin x$

A1 Obtain $2\sin^2 x + \sin x - 2 = 0$ OE

M1 Solve a 3-term quadratic for x

A1 Obtain answer $x = 0.896$ only

Total: 6

Coaching with Shuja

Question 9

Oct/Nov 2019 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The most common error occurred after finding the value (40−x) of the arbitrary constant when candidates attempted to exponentiate."

Source: 9709_w19_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

Coaching With Shuja

9 The variables x and t satisfy the differential equation $5\frac{dx}{dt} = (20 - x)(40 - x)$. It is given that $x = 10$ when $t = 0$.

(i) Using partial fractions, solve the differential equation, obtaining an expression for x in terms of t. **[9]**

(ii) State what happens to the value of x when t becomes large. **[1]**

(i) B1 Separate variables correctly and integrate one side

B1 Obtain term $0.2t$, or equivalent

M1 Carry out a relevant method to obtain A and B such that $\frac{1}{(20-x)(40-x)} \equiv \frac{A}{20-x} + \frac{B}{40-x}$ OE

A1 Obtain $A = \frac{1}{20}$ and $B = -\frac{1}{20}$

A1FT+A1FT Integrate and obtain terms $-\frac{1}{20}\ln(20-x) + \frac{1}{20}\ln(40-x)$ OE. The FT is on A and B + **A1FT**

DM1 Use $x = 10, t = 0$ to evaluate a constant, or as limits

A1 Obtain correct answer in any form

A1 Obtain final answer $x = \frac{60e^{4t}-40}{3e^{4t}-1}$ OE

Total: 9

(ii) B1 State that x approaches 20

Total: 1

Coaching With Shuja

Question 10

Oct/Nov 2019 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "(ii) Many candidates did not do the requested substitution and attempted integration by parts instead; candidates who used this method often made little progress."

Source: 9709_w19_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 10

Coaching With Shuja

10

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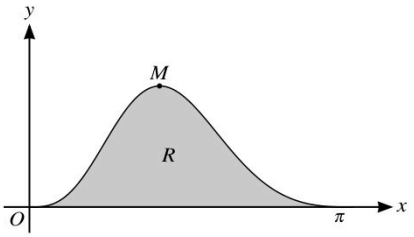


The diagram shows the graph of $y = e^{\cos x} \sin^3 x$ for $0 \leqslant x \leqslant \pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M. Show all necessary working and give your answer correct to 2 decimal places. [5]

The diagram shows the graph of $y = e^{\cos x} \sin^3 x$ for $0 \leq x \leq \pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M. Show all necessary working and give your answer correct to 2 decimal places. **[5]**

(ii) By first using the substitution $u = \cos x$, find the exact value of the area of R. **[7]**

(i) M1 Use product rule and chain rule at least once

A1 Obtain correct derivative in any form

M1 Equate derivative to zero, use Pythagoras and obtain an equation in $\cos x$

A1 Obtain $\cos^2 x + 3\cos x - 1 = 0$, or 3-term equivalent

A1 Obtain answer $x = 1.26$

Total: 5

(ii) M1 Using $du = \pm \sin x \, dx$ express integrand in terms of u and du

A1 Obtain integrand $e^u(u^2 - 1)$ OE

M1 Commence integration by parts and reach $ae^u(u^2 - 1) + b\int ue^u \, du$

A1 Obtain $e^u(u^2 - 1) - 2\int ue^u \, du$ OE

A1 Complete integration, obtaining $e^u(u^2 - 2u + 1)$ OE

DM1 Substitute limits $u = 1$ and $u = -1$ (or $x = 0$ and $x = \pi$), having integrated completely

A1 Obtain answer $\frac{4}{e}$, or exact equivalent

Total: 7

Coaching With Shuja

Question 3

Feb/Mar 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Although limits were usually correctly substituted, many candidates did not show sufficient working to convincingly obtain the given answer."

Source: 9709_m18_er.pdf, Page 9, Section: Comments on specific questions – Paper 3 (variant 32), Question 3

Coaching With Shuja

(i) Using the expansions of $\cos(3x + x)$ and $\cos(3x - x)$, show that

$$\frac{1}{2}(\cos 4x + \cos 2x) \equiv \cos 3x \cos x.$$

[3]

(ii) Hence show that

$$\int_{-\frac{1}{6}\pi}^{\frac{1}{6}\pi} \cos 3x \cos x \, dx = \frac{3}{8}\sqrt{3}.$$

[3]

(i)

State correct expansion of $\cos(3x + x)$ or $\cos(3x - x)$ **B1**

Substitute in $\frac{1}{2}(\cos 4x + \cos 2x)$ **M1**

Obtain the given identity correctly AG **A1**

Total: 3

(ii)

Obtain integral $\frac{1}{8}\sin 4x + \frac{1}{4}\sin 2x$ **B1**

Substitute limits correctly **M1**

Obtain the given answer following full, correct and exact working AG **A1**

Total: 3

Coaching With Shuja

Question 4

Feb/Mar 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Numerous other such basic ln errors were very prevalent with the result that few candidates had even a correct method for n and for A, with even fewer obtaining the correct answers."

Source: 9709_m18_er.pdf, Page 9, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

Question Paper

Feb/Mar 2018 • Paper 3 (variant 32) • Q4

The variables x and y satisfy the equation $y^n = Ax^3$, where n and A are constants. It is given that $y = 2.58$ when $x = 1.20$, and $y = 9.49$ when $x = 2.51$.

(i) Explain why the graph of $\ln y$ against $\ln x$ is a straight line. [2]

(ii) Find the values of n and A, giving your answers correct to 2 decimal places. [4]

Coaching With Shuja

(i)

State or imply $n \ln y = \ln A + 3 \ln x$ **B1**

State that the graph of $\ln y$ against $\ln x$ has an equation which is linear in $\ln y$ and $\ln x$, or has equation of the form $nY = \ln A + 3X$, where $Y = \ln y$ and $X = \ln x$, and is thus a straight line **B1**

Total: 2

(ii)

Substitute x- and y-values in $n \ln y = \ln A + 3 \ln x$ or in the given equation and solve for one of the constants **M1**

Obtain a correct constant, e.g. $n = 1.70$ **A1**

Solve for a second constant **M1**

Obtain the other constant, e.g. $A = 2.90$ **A1**

Total: 4

Coaching With Shuja

Question 5

Feb/Mar 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "☐ 1☐ (ii) Few candidates scored both marks, and many candidates who reached tant =☐– ☐ were unable ☐ 4☐ ☐ 1☐ to move to t =tant–1 ☐– ☐, the usual error being the omission of the minus sign."

Source: 9709_m18_er.pdf, Page 10, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

The parametric equations of a curve are

$$x = 2t + \sin 2t, \qquad y = 1 - 2\cos 2t,$$

for $-\frac{1}{2}\pi < t < \frac{1}{2}\pi$.

(i) Show that $\frac{dy}{dx} = 2\tan t$. [5]

(ii) Hence find the x-coordinate of the point on the curve at which the gradient of the normal is 2. Give your answer correct to 3 significant figures. [2]

(i)

State correct derivative of x or y with respect to t **B1**

Use $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$ **M1**

Obtain $\frac{dy}{dx} = \frac{4\sin 2t}{2+2\cos 2t}$, or equivalent **A1**

Use double angle formulae throughout **M1**

Obtain the given answer correctly AG **A1**

Total: 5

(ii)

State or imply $t = \tan^{-1}\left(-\frac{1}{4}\right)$ **B1**

Obtain answer $x = -0.961$ **B1**

Total: 2

Coaching With Shuja

Question 6

Feb/Mar 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 6 (i) Some candidates seemed to believe that this relationship was to be found within the differential equation, hence made no progress."

Source: 9709_m18_er.pdf, Page 10, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

Question Paper Feb/Mar 2018 • Paper 3 (variant 32) • Q6

The variables x and θ satisfy the differential equation

$$x\cos^2\theta\frac{dx}{d\theta} = 2\tan\theta + 1,$$

for $0 \leq \theta < \frac{1}{2}\pi$ and $x > 0$. It is given that $x = 1$ when $\theta = \frac{1}{4}\pi$.

(i) Show that $\frac{d}{d\theta}(\tan^2\theta) = \frac{2\tan\theta}{\cos^2\theta}$. [1]

(ii) Solve the differential equation and calculate the value of x when $\theta = \frac{1}{3}\pi$, giving your answer correct to 3 significant figures. [7]

Coaching With Shuja

(i)

Show sufficient working to justify the given statement AG **B1**

Total: 1

(ii)

Separate variables correctly and attempt integration of at least one side **B1**

Obtain term $\frac{1}{2}x^2$ **B1**

Obtain terms $\tan^2\theta$ and $\tan\theta$ **B1+B1**

Evaluate a constant, or use limits $x = 1, \theta = \frac{1}{4}\pi$, in a solution with two terms of the form ax^2 and $b\tan\theta$, where $ab \neq 0$ **M1**

State correct answer in any form, e.g. $\frac{1}{2}x^2 = \tan^2\theta + \tan\theta - \frac{3}{2}$ **A1**

Substitute $\theta = \frac{1}{3}\pi$ and obtain $x = 2.54$ **A1**

Total: 7

Coaching with Shuja

Question 7

Feb/Mar 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "© 2018 Cambridge International Advanced Level 9709 Mathematics March 2018 Principal Examiner Report for Teachers (iii) Another part that candidates found very difficult."

Source: 9709_m18_er.pdf, Page 10, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

Question Paper Feb/Mar 2018 • Paper 3 (variant 32) • Q7

(i) By sketching suitable graphs, show that the equation $e^{2x} = 6 + e^{-x}$ has exactly one real root. [2]

(ii) Verify by calculation that this root lies between 0.5 and 1. [2]

(iii) Show that if a sequence of values given by the iterative formula

$$x_{n+1} = \frac{1}{3}\ln(1 + 6e^{x_n})$$

converges, then it converges to the root of the equation in part (i). [2]

(iv) Use this iterative formula to calculate the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

Coaching With Shuja

(i)

Sketch a relevant graph, e.g. $y = e^{2x}$ **B1**

Sketch a second relevant graph, e.g. $y = 6 + e^{-x}$, and justify the given statement **B1**

Total: 2

(ii)

Calculate the value of a relevant expression or values of a pair of relevant expressions at $x = 0.5$ and $x = 1$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(iii)

State a suitable equation, e.g. $x = \frac{1}{3}\ln(1 + 6e^{x})$ **B1**

Rearrange this as $e^{2x} = 6 + e^{-x}$, or commence working vice versa **B1**

Total: 2

(iv)

Use the iterative formula correctly at least once **M1**

Obtain final answer 0.928 **A1**

Show sufficient iterations to 5 d.p. to justify 0.928 to 3 d.p., or show there is a sign change in the interval $(0.9275, 0.9285)$ **A1**

Total: 3

Coaching with Shuja

Question 9

Feb/Mar 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "(b) The majority of candidates realised that one of the other roots was 1 – 2i, but were then unclear how to establish the quadratic term, or the correct quadratic term but struggled with the long division."

Source: 9709_m18_er.pdf, Page 11, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

The complex number $1 + 2i$ is denoted by u.

(i) It is given that u is a root of the equation $2x^3 - x^2 + 4x + k = 0$, where k is a constant.

(a) Showing all working and without using a calculator, find the value of k. [3]

(b) Showing all working and without using a calculator, find the other two roots of this equation. [4]

(ii) On an Argand diagram sketch the locus of points representing complex numbers z satisfying the equation $|z - u| = 1$. Determine the least value of $\arg z$ for points on this locus. Give your answer in radians correct to 2 decimal places. [4]

Coaching With Shuja

(i)(a)

Substitute $x = 1 + 2i$ in the equation and attempt expansions of x^2 and x^3 **M1**

Use $i^2 = -1$ correctly at least once and solve for k **M1**

Obtain answer $k = 15$ **A1**

Total: 3

(i)(b)

State answer $1 - 2i$ **B1**

Carry out a complete method for finding a quadratic factor with zeros $1 + 2i$ and $1 - 2i$ **M1**

Obtain $x^2 - 2x + 5$ **A1**

Obtain root $-\frac{3}{2}$, or equivalent, via division or inspection **A1**

Total: 4

(ii)

Show a circle with centre $1 + 2i$ **B1**

Show a circle with radius 1 **B1**

Carry out a complete method for calculating the least value of $\arg z$ **M1**

Obtain answer 0.64 **A1**

Total: 4

Coaching with Shuja

Question 8

May/June 2018 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "The correct identification n of the first term proved difficult for all but the higher scoring candidates."

Source: 9709_s18_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 8

Coaching With Shuja

Question Paper May/June 2018 • Paper 1 (variant 11) • Q8

(a) A geometric progression has a second term of 12 and a sum to infinity of 54. Find the possible values of the first term of the progression. [4]

(b) The nth term of a progression is $p + qn$, where p and q are constants, and S_n is the sum of the first n terms.

(i) Find an expression, in terms of p, q and n, for S_n. [3]

(ii) Given that $S_4 = 40$ and $S_6 = 72$, find the values of p and q. [2]

Coaching With Shuja

(a)

Use

$$ar = 12$$

B1

Use

$$\frac{a}{1-r} = 54$$

B1

Eliminate a or r to obtain a three-term quadratic, e.g.

$$9r^2 - 9r + 2 = 0$$

M1

Obtain $r = \frac{2}{3}$ or $r = \frac{1}{3}$, and hence

$$a = 18 \quad \text{or} \quad a = 36$$

A1

Total: 4

(b)(i)

Use first term $p + q$ and common difference q, or use last term $p + qn$ **B1**

Use the sum formula **M1**

Obtain

$$S_n = \frac{n}{2}(2p + (n+1)q),$$

or equivalent **A1**

Total: 3

(b)(ii)

Use the expression from part (i) to form equations from $S_4 = 40$ and $S_6 = 72$ **M1**

Obtain

$$p = 5, \qquad q = 2$$

A1

Total: 2

Question 9

May/June 2018 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "(ii) For many candidates this proved to be the most difficult part of a question."

Source: 9709_s18_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 9

Coaching With Shuja

Functions f and g are defined for $x \in \mathbb{R}$ by

$$f : x \mapsto \frac{1}{2}x - 2,$$

$$g : x \mapsto 4 + x - \frac{1}{2}x^2.$$

(i) Find the points of intersection of the graphs of $y = f(x)$ and $y = g(x)$. [3]

(ii) Find the set of values of x for which $f(x) > g(x)$. [2]

(iii) Find an expression for $fg(x)$ and deduce the range of fg. [4]

The function h is defined by

$$h : x \mapsto 4 + x - \frac{1}{2}x^2 \quad \text{for } x \geq k.$$

(iv) Find the smallest value of k for which h has an inverse. [2]

Coaching with Shuja

(i)

Equate

$$4 + x - \frac{1}{2}x^2 = \frac{1}{2}x - 2$$

and form a three-term quadratic **M1**

Obtain the two x-values $x = 4$ and $x = -3$ **A1**

Obtain points of intersection

$$(4, 0) \quad \text{and} \quad (-3, -3.5)$$

A1

Total: 3

(ii)

State

$$x > 4$$

B1

State

$$x < -3$$

B1

Total: 2

(iii)

Obtain

$$fg(x) = \frac{x}{2} - \frac{x^2}{4}$$

B1

Complete the square or use calculus to find the maximum **M1**

Identify $x = 1$ or an equivalent completed-square form **A1**

Obtain maximum value $\frac{1}{4}$ and range

$$fg(x) \leq \frac{1}{4}$$

A1

Total: 4

(iv)

Use calculus, completing the square, or a sketch on h to find the turning point at $x = 1$ **M1**

Obtain

$$k = 1$$

A1

Total: 2

Question 4

May/June 2018 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (ii) proved much more challenging with a good number taking the phrase 'no solution' to mean that the discriminant had to be used rather than using the values obtained in part (i) to consider when the trigonometric equation would have no solution."

Source: 9709_s18_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

Coaching With Shuja

The function f is such that

$$f(x) = a + b\cos x$$

for $0 \leq x \leq 2\pi$. It is given that

$$f\left(\frac{\pi}{3}\right) = 5$$

and

$$f(\pi) = 11.$$

(i) Find the values of the constants a and b. [3]

(ii) Find the set of values of k for which the equation $f(x) = k$ has no solution. [3]

Coaching With Shuja

(i)

Use

$$a + \frac{1}{2}b = 5$$

B1

Use

$$a - b = 11$$

B1

Obtain

$$a = 7, \quad b = -4$$

B1

Total: 3

(ii)

Select the minimum value

$$a + b = 3$$

B1

Select the maximum value

$$a - b = 11$$

B1

Obtain

$$k < 3 \quad \text{or} \quad k > 11$$

B1

Total: 3

Coaching With Shuja

Question 10

May/June 2018 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 10 Part (i) of this question was very well done by the vast majority of candidates but a significant number struggled to draw the graphs requested in part (ii), and part (iii) proved too difficult except for the most able."

> "This was true even more so in part (iii) where all but the most able candidates struggled."

Source: 9709_s18_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

Coaching With Shuja

(i) Solve the equation

$$2\cos x + 3\sin x = 0,$$

for $0° \leq x \leq 360°$. [3]

(ii) Sketch, on the same diagram, the graphs of $y = 2\cos x$ and $y = -3\sin x$ for $0° \leq x \leq 360°$. [3]

(iii) Use your answers to parts (i) and (ii) to find the set of values of x for $0° \leq x \leq 360°$ for which

$$2\cos x + 3\sin x > 0.$$

[2]

Coaching With Shuja

(i)

Use

$$2\cos x = -3\sin x$$

to obtain

$$\tan x = -\frac{2}{3}$$

M1

Obtain

$$x = 146.3°$$

A1

Obtain

$$x = 326.3°$$

with no extra values in the range **A1**

Total: 3

(ii)

Sketch one complete cycle of $y = 2\cos x$, starting and finishing at the same positive y value and going below the x-axis by roughly the same distance **B1**

Sketch one complete cycle of $y = -3\sin x$, starting and finishing on the x-axis, inverted, and going below then above the x-axis by roughly the same distance **B1**

Give a fully correct sketch with the sine curve clearly having larger amplitude than the cosine curve **B1**

Total: 3

(iii)

State

$$0° \leq x < 146.3°$$

B1

State

$$326.3° < x \leq 360°$$

B1

Total: 2

Question 2

May/June 2018 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "The most common error was to not include the minus sign, or the 2, in the term to be cubed."

Source: 9709_s18_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 2

Coaching With Shuja

Find the coefficient of $\frac{1}{x}$ in the expansion of

$$\left(x - \frac{2}{x}\right)^5.$$

[3]

Coaching With Shuja

Select the relevant term

$$^5C_3x^2\left(-\frac{2}{x}\right)^3$$

(or equivalent) **B2**

Obtain coefficient

$$-80$$

B1

Total: 3

Question 4

May/June 2018 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "There was, 2 however, a very common error made, (−1)3 being evaluated as −1 instead of +1, which lost the last mark."

Source: 9709_s18_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 4

Coaching With Shuja

Question Paper May/June 2018 • Paper 1 (variant 13) • Q4

A curve with equation $y = f(x)$ passes through the point $A(3, 1)$ and crosses the y-axis at B. It is given that

$$f'(x) = (3x - 1)^{-1/3}.$$

Find the y-coordinate of B. [6]

Coaching With Shuja

Integrate to obtain

$$f(x) = \frac{1}{2}(3x-1)^{2/3} + c$$

B2

Use $x = 3$, $y = 1$ to find the constant **M1**

Obtain

$$c = -1$$

so that

$$y = \frac{1}{2}(3x-1)^{2/3} - 1$$

A1

Substitute $x = 0$ **M1**

Obtain the y-coordinate of B as

$$-\frac{1}{2}$$

A1

Total: 6

Coaching With Shuja

Question 6

May/June 2018 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "But it is still disappointing to see a number of candidates not using brackets in these situations and inevitably mistakes were made when simplifying the numerator and denominator which, of course, meant that the gradient did 1 not reduce to (independent of k)."

Source: 9709_s18_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 6

Coaching With Shuja

Question Paper May/June 2018 • Paper 1 (variant 13) • Q6

The coordinates of points A and B are $(-3k-1, k+3)$ and $(k+3, 3k+5)$ respectively, where k is a constant $(ke-1)$.

(i) Find and simplify the gradient of AB, showing that it is independent of k. [2]

(ii) Find and simplify the equation of the perpendicular bisector of AB. [5]

Coaching With Shuja

(i)

Use the gradient formula:

$$m = \frac{(3k+5)-(k+3)}{(k+3)-(-3k-1)}$$

M1

Obtain

$$m = \frac{1}{2}$$

A1

Total: 2

(ii)

Obtain the midpoint coordinates

$$(1-k, 2k+4)$$

B2

Use the gradient of the perpendicular bisector as -2 **M1**

Form an equation through the midpoint with gradient -2 **M1**

Obtain

$$2x + y = 6$$

A1

Total: 5

Question 8

May/June 2018 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "The first approach required candidates to choose one or other of the critical values and a significant number incorrectly chose 2."

Source: 9709_s18_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 8

Coaching With Shuja

(i) The tangent to the curve

$$y = x^3 - 9x^2 + 24x - 12$$

at a point A is parallel to the line $y = 2 - 3x$. Find the equation of the tangent at A. [6]

(ii) The function f is defined by

$$f(x) = x^3 - 9x^2 + 24x - 12 \quad \text{for } x > k,$$

where k is a constant. Find the smallest value of k for f to be an increasing function. [2]

Coaching With Shuja

(i)

Differentiate to obtain

$$\frac{dy}{dx} = 3x^2 - 18x + 24$$

M1 A1

Equate the derivative to -3 **M1**

Obtain

$$x = 3$$

A1

Obtain

$$y = 6$$

A1

Form the tangent equation

$$y - 6 = -3(x - 3),$$

or

$$y = -3x + 15$$

A1

Total: 6

(ii)

Factorise or solve

$$3x^2 - 18x + 24 = 3(x - 2)(x - 4)$$

M1

State the smallest value

$$k = 4$$

A1

Total: 2

Question 1

May/June 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Many candidates made no progress with the question at all because they started with the incorrect statement ln(x4 −4) =lnx4 −ln4."

Source: 9709_s18_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

Question Paper May/June 2018 • Paper 3 (variant 31) • Q1

Showing all necessary working, solve the equation

$$\ln(x^4 - 4) = 4\ln x - \ln 4,$$

giving your answer correct to 2 decimal places. [4]

Coaching With Shuja

Use a law for the logarithm of a product, quotient or power **M1**

Obtain a correct equation free of logarithms, e.g.

$$4(x^4 - 4) = x^4$$

A1

Solve for x **M1**

Obtain answer

$$x = 1.52$$

only **A1**

Total: 4

Coaching With Shuja

Question 2

May/June 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common errors were to use tanx = , or a completely false start sinx with expansions such as sin(x−60°)=sinx−sin60°."

Source: 9709_s18_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

Coaching With Shuja

(i) Given that

$$\sin(x - 60°) = 3\cos(x - 45°),$$

find the exact value of $\tan x$. [4]

(ii) Hence solve the equation

$$\sin(x - 60°) = 3\cos(x - 45°),$$

for $0° < x < 360°$. [2]

(i)

Use trigonometric formulae and obtain an equation in $\sin x$ and $\cos x$ **M1**

Obtain a correct equation in any form **A1**

Substitute exact trigonometric ratios and obtain an expression for $\tan x$ **M1**

Obtain

$$\tan x = \frac{\sqrt{3}+3\sqrt{2}}{1-3\sqrt{2}},$$

or equivalent **A1**

Total: 4

(ii)

State answer $x = 118.5°$

B1

State second answer $x = 298.5°$

B1

Total: 2

Coaching With Shuja

Question 5

May/June 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Many candidates did not seem to have a confident working knowledge of complex numbers or of vectors."

Source: 9709_s18_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

Coaching With Shuja

Let

$$I = \int_{1/4}^{3/4} \sqrt{\frac{x}{1-x}}\, dx.$$

(i) Using the substitution $x = \cos^2 \theta$, show that

$$I = \int_{\pi/6}^{\pi/3} 2\cos^2 \theta \, d\theta.$$

[4]

(ii) Hence find the exact value of I. [4]

(i)

State or imply

$$dx = -2\cos\theta\sin\theta\, d\theta,$$

or equivalent **B1**

Substitute for x and dx, and use Pythagoras **M1**

Obtain integrand $\pm 2\cos^2\theta$ **A1**

Justify the change of limits and obtain the given answer correctly **A1**

Total: 4

(ii)

Obtain an indefinite integral of the form

$$a\theta + b\sin 2\theta$$

M1

Obtain

$$\theta + \frac{1}{2}\sin 2\theta$$

A1

Use the correct limits correctly **M1**

Obtain

$$\frac{\pi}{6}$$

with no errors seen **A1**

Total: 4

Question 6

May/June 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "After a correct start, the most common error was to confuse k with a constant of integration."

Source: 9709_s18_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

Question Paper May/June 2018 • Paper 3 (variant 31) • Q6

In a certain chemical reaction the amount, x grams, of a substance is decreasing. The differential equation relating x and t, the time in seconds since the reaction started, is

$$\frac{dx}{dt} = -kx\sqrt{t},$$

where k is a positive constant. It is given that $x = 100$ at the start of the reaction.

(i) Solve the differential equation, obtaining a relation between x, t and k. [5]

(ii) Given that $t = 25$ when $x = 80$, find the value of t when $x = 40$. [3]

(i)

Separate variables correctly and integrate at least one side **B1**

Obtain term $\ln x$ **B1**

Obtain term

$$-\frac{2}{3}kt^{3/2},$$

or equivalent **B1**

Evaluate a constant, or use limits $x = 100$ and $t = 0$, in a solution containing terms $a \ln x$ and $bt^{3/2}$ **M1**

Obtain correct solution in any form, e.g.

$$\ln x = -\frac{2}{3}kt^{3/2} + \ln 100$$

A1

Total: 5

(ii)

Substitute $x = 80$ and $t = 25$ to form an equation in k **M1**

Substitute $x = 40$ and eliminate k **M1**

Obtain

$$t = 64.1$$

A1

Total: 3

Coaching with Shuja

Question 7

May/June 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "8i was a common error, and several candidates did not split 8 as 4×2."

> "(ii) There were many blank responses to this question implying candidates were not familiar with drawing an argand diagram."

Source: 9709_s18_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

(i) Showing all working and without using a calculator, solve the equation

$$z^2 + 2\sqrt{6}z + 8 = 0,$$

giving your answers in the form $x + iy$, where x and y are real and exact. [3]

(ii) Sketch an Argand diagram showing the points representing the roots. [1]

(iii) The points representing the roots are A and B, and O is the origin. Find angle AOB. [3]

(iv) Prove that triangle AOB is equilateral. [1]

(i)

Use the quadratic formula, completing the square, or the substitution $z = x + iy$ to find a root, using $i^2 = -1$ **M1**

Obtain one root, e.g.

$$-\sqrt{6} - \sqrt{2}i$$

A1

Obtain the other root, e.g.

$$-\sqrt{6} + \sqrt{2}i$$

A1

Total: 3

(ii)

Represent both roots in relatively correct positions on an Argand diagram **B1**

Total: 1

(iii)

State or imply a correct relevant length or angle, e.g. OA, OB, AB, or the angle between OA or OB and the real axis **B1**

Carry out a complete method for finding angle AOB **M1**

Obtain

Math input error

A1

Total: 3

(iv)

Give a complete justification that triangle AOB is equilateral **B1**

Total: 1

Question 8

May/June 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common error 1 was in division by ."

> "Most candidates used the limits correctly, although a minority did not appear 2 to consider the lower limit."

> "A large number of incorrect algebraic statements were seen."

Source: 9709_s18_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 8

Coaching With Shuja

The positive constant a is such that

$$\int_0^a xe^{-x/2}\,dx = 2.$$

(i) Show that a satisfies the equation

$$a = 2\ln(a+2).$$

[5]

(ii) Verify by calculation that a lies between 3 and 3.5. [2]

(iii) Use an iteration based on the equation in part (i) to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

Coaching With Shuja

(i)

Integrate by parts and reach an expression of the form

$$lxe^{-x/2} + m\int e^{-x/2}\,dx$$

M1

Obtain

$$-2xe^{-x/2} + 2\int e^{-x/2}\,dx$$

A1

Complete the integration and obtain

$$-2xe^{-x/2} - 4e^{-x/2},$$

or equivalent **A1**

Having integrated twice, use limits and equate the result to **2** **M1**

Obtain the given equation

$$a = 2\ln(a+2)$$

correctly **A1**

Total: 5

(ii)

Calculate values of a relevant expression at $a = 3$ and $a = 3.5$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(iii)

Use the iterative formula

$$a_{n+1} = 2\ln(a_n + 2)$$

correctly at least once **M1**

Obtain final answer **3.36** **A1**

Show sufficient iterations to 4 d.p. to justify **3.36** to 2 d.p., or show there is a sign change in the interval **(3.355, 3.365)** **A1**

Total: 3

Question 10

May/June 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "For both of these approaches, the most common error was to use the direction of the line, rather than the position vector of a point, to evaluate d."

Source: 9709_s18_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 10

Coaching With Shuja

The point P has position vector

$$3\mathbf{i} - 2\mathbf{j} + \mathbf{k}.$$

The line l has equation

$$\mathbf{r} = 4\mathbf{i} + 2\mathbf{j} + 5\mathbf{k} + \lambda(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}).$$

(i) Find the length of the perpendicular from P to l, giving your answer correct to 3 significant figures. [5]

(ii) Find the equation of the plane containing l and P, giving your answer in the form $ax + by + cz = d$. [5]

(i)

Find $\overrightarrow{PQ}$ for a general point Q on l, e.g.

$$(1+\lambda)\mathbf{i}+(4+2\lambda)\mathbf{j}+(4+3\lambda)\mathbf{k}$$

B1

Calculate the scalar product of $\overrightarrow{PQ}$ and a direction vector for l and equate to zero **M1**

Solve and obtain, for example, $\lambda=-\frac{3}{2}$ **A1**

Carry out a method to calculate PQ **M1**

Obtain answer **1.22** **A1**

Total: 5

(ii)

Use scalar product to obtain a relevant equation in a, b and c, e.g.

$$a+2b+3c=0$$

B1

Obtain a second relevant equation, e.g. using $\overrightarrow{PA}$,

$$a+4b+4c=0,$$

and solve for one ratio **M1**

Obtain

$$a:b:c=4:1:-2$$

A1

Substitute a relevant point and values of a, b, c in the general equation and find d **M1**

Obtain

$$4x+y-2z=8$$

A1

Total: 5

Question 1

May/June 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors centred in attempts to square the original equation."

Source: 9709_s18_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

Question Paper May/June 2018 • Paper 3 (variant 32) • Q1

Showing all necessary working, solve the equation

$$3|2^x - 1| = 2^x,$$

giving your answers correct to 3 significant figures. [4]

Coaching With Shuja

Either state or imply a non-modular equation, or pair of equations,

$$3(2^x - 1) = \pm 2^x$$

M1

Obtain

$$2^x = \frac{3}{2} \quad \text{and} \quad 2^x = \frac{3}{4},$$

or equivalent **A1**

Use a correct method for solving an equation of the form $2^x = a$, where $a > 0$ **M1**

Obtain final answers

$$x = 0.585 \quad \text{and} \quad x = -0.415$$

only **A1**

Total: 4

Coaching With Shuja

Question 2

May/June 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “Another common error was to 2 cos(θ+45°) claim that cot(θ+45°)= .”

Source: 9709_s18_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Coaching With Shuja

Question Paper May/June 2018 • Paper 3 (variant 32) • Q2

Showing all necessary working, solve the equation

$$\cot\theta + \cot(\theta + 45°) = 2,$$

for $0° < \theta < 180°$. [5]

Coaching With Shuja

Use the correct $\tan(A \pm B)$ formula and obtain an equation in $\tan\theta$ **M1**

Obtain a correct equation in any form **A1**

Reduce to

$$3\tan^2\theta = 1,$$

or equivalent **A1**

Obtain answer

$\theta = 30°$

A1

Obtain answer

$\theta = 150°$

and no others in the range **A1**

Total: 5

Coaching With Shuja

Question 4

May/June 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “The most common error was to omit the "–"in the integration, but the limits were usually applied correctly.”

Source: 9709_s18_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

(i) Show that

$$\frac{2\sin x - \sin 2x}{1 - \cos 2x} \equiv \frac{\sin x}{1 + \cos x}.$$

[4]

(ii) Hence, showing all necessary working, find

$$\int_{\pi/3}^{\pi/2} \frac{2\sin x - \sin 2x}{1 - \cos 2x}\, dx,$$

giving your answer in the form $\ln k$. [4]

(i)

Use correct double-angle formulae and express the left hand side in terms of $\cos x$ and $\sin x$ **M1**

Obtain a correct expression **A1**

Complete a method to get the correct denominator, e.g. by factorising to remove a factor of $1 - \cos x$ **M1**

Obtain the given right hand side correctly **A1**

Total: 4

(ii)

State an integral of the form

$$a \ln(1 + \cos x)$$

M1

Obtain integral

$$-\ln(1 + \cos x)$$

A1

Substitute correct limits in the correct order **M1**

Obtain

$$\ln\left(\frac{3}{2}\right)$$

A1

Total: 4

Question 5

May/June 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A disappointing feature of the work this time was the number of candidates who reached x3 =1 and concluded that x = ±1."

Source: 9709_s18_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

Coaching With Shuja

The equation of a curve is

$$x^2(x+3y)-y^3=3.$$

(i) Show that

$$\frac{dy}{dx}=\frac{x^2+2xy}{y^2-x^2}.$$

[4]

(ii) Hence find the exact coordinates of the two points on the curve at which the gradient of the normal is **1**. [4]

(i)

State or imply $3y^2\frac{dy}{dx}$ as the derivative of y^3 **B1**

State or imply $6xy + x^2\frac{dy}{dx}$ as part of the derivative of $3x^2y$ **B1**

Equate the derivative of the left hand side to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain the given answer **A1**

Total: 4

(ii)

Equate the tangent gradient to -1 and solve for y **M1**

Use $y = -2x$ or $y = 0$ to obtain an equation in x or y **M1**

Obtain point

$(1, -2)$

A1

Obtain point

$(\sqrt[3]{3}, 0)$

B1

Total: 4

Coaching With Shuja

Question 6

May/June 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common error was use of the incorrect radius for the semi-circles."

Source: 9709_s18_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

Question Paper May/June 2018 • Paper 3 (variant 32) • Q6

The diagram shows a triangle ABC in which $AB = AC = a$ and angle $BAC = \theta$ radians. Semicircles are drawn outside the triangle with AB and AC as diameters. A circular arc with centre A joins B and C. The area of the shaded segment is equal to the sum of the areas of the semicircles.

(i) Show that

$$\theta = \frac{1}{2}\pi + \sin\theta.$$

[3]

(ii) Verify by calculation that θ lies between 2.2 and 2.4. [2]

(iii) Use an iterative formula based on the equation in part (i) to determine θ correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

Coaching With Shuja

(i)

Use a correct method for finding the area of a segment and the area of a semicircle, and form an equation in θ **M1**

State a correct equation in any form **A1**

Obtain

$$\theta = \frac{1}{2}\pi + \sin\theta$$

A1

Total: 3

(ii)

Calculate values of a relevant expression at $\theta = 2.2$ and $\theta = 2.4$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(iii)

Use

$$\theta_{n+1} = \frac{1}{2}\pi + \sin\theta_n$$

correctly at least once **M1**

Obtain final answer 2.31 **A1**

Show sufficient iterations to 4 d.p. to justify 2.31 to 2 d.p., or show there is a sign change in the interval $(2.305, 2.315)$ **A1**

Total: 3

Question 7

May/June 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Very few candidates provided a correct demonstration that ∠AOB = π."

Source: 9709_s18_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

Question Paper May/June 2018 • Paper 3 (variant 32) • Q7

Throughout this question the use of a calculator is not permitted.

The complex numbers $-3\sqrt{3}+i$ and $\sqrt{3}+2i$ are denoted by u and v respectively.

(i) Find, in the form $x+iy$, where x and y are real and exact, the complex numbers uv and $\frac{u}{v}$. [5]

(ii) On a sketch of an Argand diagram with origin O, show the points A and B representing the complex numbers u and v respectively. Prove that angle $AOB = \frac{2}{3}\pi$. [3]

(i)

Substitute in uv, expand the product and use $i^2 = -1$ **M1**

Obtain

$$uv = -11 - 5\sqrt{3}\,i$$

A1

Substitute in $\frac{u}{v}$ and multiply numerator and denominator by the conjugate of v, or equivalent **M1**

Obtain numerator $-7 + 7\sqrt{3}\,i$ or denominator 7 **A1**

Obtain final answer

$$\frac{u}{v} = -1 + \sqrt{3}\,i$$

A1

Total: 5

(ii)

Show the points A and B representing u and v in relatively correct positions **B1**

Carry out a complete method for finding angle AOB, e.g. calculate Math input error **M1**

Prove

Math input error

A1

Total: 3

Coaching with Shuja

Question 1

May/June 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "However, there were a considerable number of candidates for whom it proved to be difficult or impossible."

Source: 9709_s18_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 1

Coaching With Shuja

Expand

$$\frac{4}{\sqrt{4-3x}}$$

in ascending powers of x, up to and including the term in x^2, simplifying the coefficients. [4]

Obtain a correct unsimplified version of the x or x^2 term of the expansion of

$$(4-3x)^{-1/2}$$

or

$$2\left(1-\frac{3x}{4}\right)^{-1/2}$$

M1

State correct first term **2 B1**

Obtain the next two terms

$$\frac{3}{4}x+\frac{27}{64}x^2$$

A2

Total: 4

Coaching With Shuja

Question 5

May/June 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 5 (i) Candidates found this question extremely difficult."

Source: 9709_s18_er.pdf, Page 31, Section: Comments on specific questions – Paper 3 (variant 33), Question 5

Coaching With Shuja

(i) By first expanding $(\cos^2 x + \sin^2 x)^3$, or otherwise, show that

$$\cos^6 x + \sin^6 x = 1 - \frac{3}{4}\sin^2 2x.$$

[4]

(ii) Hence solve the equation

$$\cos^6 x + \sin^6 x = \frac{2}{3},$$

for $0° < x < 180°$. [4]

(i)

Attempt the cubic expansion and equate to **1** **M1**

Obtain a correct equation **A1**

Use Pythagoras and a double-angle formula in the expansion **M1**

Obtain the given result correctly **A1**

Total: 4

(ii)

Use the identity and carry out a method for finding a root **M1**

Obtain answer **20.9°** **A1**

Obtain a second answer, e.g. **69.1°** **A1**

Obtain the remaining answers **110.9°** and **159.1°**, and no others in the given interval **A1**

Total: 4

Coaching With Shuja

Question 6

May/June 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Many candidates failed to realise how to extract an expression for y in terms of x from their correct ☐(2+y)☐ (2+y) ln☐ ☐ or expressions."

Source: 9709_s18_er.pdf, Page 31, Section: Comments on specific questions – Paper 3 (variant 33), Question 6

Coaching With Shuja

(i) Express

$$\frac{1}{4-y^2}$$

in partial fractions. [2]

(ii) The variables x and y satisfy the differential equation

$$x\frac{dy}{dx} = 4 - y^2,$$

and $y = 1$ when $x = 1$. Solve the differential equation, obtaining an expression for y in terms of x. [6]

Coaching With Shuja

(i)

Carry out a relevant method to find A and B such that

$$\frac{1}{4-y^2} \equiv \frac{A}{2+y} + \frac{B}{2-y}$$

M1

Obtain

$$A = B = \frac{1}{4}$$

A1

Total: 2

(ii)

Separate variables correctly and integrate at least one side to obtain one of the terms $a\ln x$, $b\ln(2+y)$ or $c\ln(2-y)$ **M1**

Obtain term $\ln x$ **B1**

Integrate and obtain terms

$$\frac{1}{4}\ln(2+y) - \frac{1}{4}\ln(2-y)$$

A1

Use $x = 1$, $y = 1$ to evaluate a constant, or use limits **M1**

Obtain a correct solution in any form, e.g.

$$\ln x = \frac{1}{4}\ln(2+y) - \frac{1}{4}\ln(2-y) - \frac{1}{4}\ln 3$$

A1

Rearrange to obtain

$$y = \frac{2(3x^4-1)}{3x^4+1},$$

or equivalent **A1**

Total: 6

Question 8

May/June 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "(ii) This question, along with Question 5(i) and Question 10(ii), proved very challenging for the majority of the candidates."

Source: 9709_s18_er.pdf, Page 32, Section: Comments on specific questions – Paper 3 (variant 33), Question 8

Coaching With Shuja

The equation of a curve is

$$2x^3 - y^3 - 3xy^2 = 2a^3,$$

where a is a non-zero constant.

(i) Show that

$$\frac{dy}{dx} = \frac{2x^2 - y^2}{y^2 + 2xy}.$$

[4]

(ii) Find the coordinates of the two points on the curve at which the tangent is parallel to the y-axis. [5]

Coaching With Shuja

(i)

State or imply $3y^2\frac{dy}{dx}$ as the derivative of y^3 **B1**

State or imply $3y^2 + 6xy\frac{dy}{dx}$ as the derivative of $3xy^2$ **B1**

Equate the derivative of the left hand side to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain the given answer **A1**

Total: 4

(ii)

Equate the denominator to zero and solve for y **M1**

Obtain $y = 0$ and $x = a$ **A1**

Obtain $y = -2x$ and substitute in the curve equation to find x or y **M1**

Obtain $x = -a$ **A1**

Obtain $y = 2a$ **A1**

Total: 5

Coaching with Shuja

Question 9

May/June 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "For some reason the trigonometry to calculate the greatest value of arg proved difficult."

Source: 9709_s18_er.pdf, Page 32, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

Coaching With Shuja

(a) Find the complex number z satisfying the equation

$$3z - iz^* = 1 + 5i,$$

where z^* denotes the complex conjugate of z. [4]

(b) On a sketch of an Argand diagram, shade the region whose points represent complex numbers z which satisfy both the inequalities

$$|z| \leq 3$$

and

$$\mathrm{Im}\, z \geq 2,$$

where $\mathrm{Im}\, z$ denotes the imaginary part of z. Calculate the greatest value of Math input error for points in this region. Give your answer in radians correct to 2 decimal places. [5]

Coaching With Shuja

(a)

Substitute $z = x + iy$ and obtain a correct equation in x and y **B1**

Use $i^2 = -1$ and equate real and imaginary parts **M1**

Obtain two correct equations, e.g.

$$3x - y = 1, \qquad 3y - x = 5$$

A1

Solve and obtain

$$z = 1 + 2i$$

A1

Total: 4

(b)

Show a circle with radius **3** **B1**

Show the line $y = 2$ extending in both quadrants **B1**

Shade the correct region **B1**

Carry out a complete method for finding the greatest value of Math input error **M1**

Obtain answer **2.41** **A1**

Total: 5

Coaching With Shuja

Question 10

May/June 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "(ii) All candidates found this question difficult."

Source: 9709_s18_er.pdf, Page 32, Section: Comments on specific questions – Paper 3 (variant 33), Question 10

Coaching With Shuja

The points A and B have position vectors $2\mathbf{i}+\mathbf{j}+3\mathbf{k}$ and $4\mathbf{i}+\mathbf{j}+\mathbf{k}$ respectively. The line l has equation

$$\mathbf{r}=4\mathbf{i}+6\mathbf{j}+t(\mathbf{i}+2\mathbf{j}-2\mathbf{k}).$$

(i) Show that l does not intersect the line passing through A and B. [5]

The point P, with parameter t, lies on l and is such that angle PAB is equal to $120°$.

(ii) Show that

$$3t^2+8t+4=0.$$

Hence find the position vector of P. [6]

(i)

Carry out a correct method for finding a vector equation for AB **M1**

Obtain

$$\mathbf{r} = 2\mathbf{i} + \mathbf{j} + 3\mathbf{k} + \lambda(2\mathbf{i} - 2\mathbf{k}),$$

or equivalent **A1**

Equate pairs of components for AB and l and solve for λ or t **M1**

Obtain a correct value for λ or t **A1**

Verify that all three component equations are not satisfied **A1**

Total: 5

(ii)

State or imply a direction vector for AP has components

$$(2 + t,\ 5 + 2t,\ -3 - 2t)$$

B1

State or imply that $\cos 120°$ equals the scalar product of $\overrightarrow{AP}$ and $\overrightarrow{AB}$ divided by the product of their moduli **M1**

Carry out the correct processes for finding the scalar product and product of moduli in terms of t **M1**

Obtain the given equation

$$3t^2 + 8t + 4 = 0$$

A1

Solve the quadratic and use a root to find a position vector for P **M1**

Obtain position vector

$$2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$$

from $t = -2$, having rejected $t = -\frac{2}{3}$ **A1**

Total: 6

Question 3

Oct/Nov 2018 | Paper 1 (variant 12)

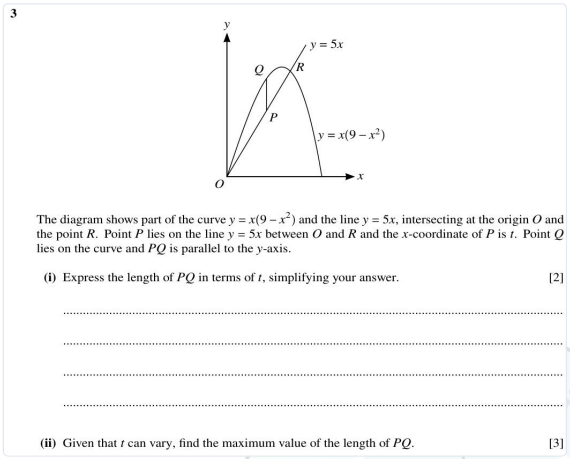
Examiner Report — Why students found this question difficult

> "Question 3 This question was slightly different to normal and many candidates struggled with it."

Source: 9709_w18_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

3



The diagram shows part of the curve $y = x(9 - x^2)$ and the line $y = 5x$, intersecting at the origin O and the point R. Point P lies on the line $y = 5x$ between O and R and the x-coordinate of P is t. Point Q lies on the curve and PQ is parallel to the y-axis.

(i) Express the length of PQ in terms of t, simplifying your answer. **[2]**

(ii) Given that t can vary, find the maximum value of the length of PQ. **[3]**

Mark Scheme

(i) B1 B1 P is $(t, 5t)$ Q is $(t, t(9-t^2)) \rightarrow 4t - t^3$ **B1** for both y coordinates which can be implied by subsequent working. **B1** for PQ allow $|4t - t^3|$ or $|t^3 - 4t|$.

Note: $4x - x^3$ from equating line and curve 0/2 even if x then replaced by t.

[2]

(ii) Finding the maximum: differentiate, set equal to zero, solve to find t, substitute back.

Total: 3

Question 4

Oct/Nov 2018 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (ii) proved to be more difficult, with many candidates failing to draw a curve which clearly turned at both ends."

Source: 9709_w18_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

Coaching With Shuja

4 Functions f and g are defined by

$f : x \mapsto 2 - 3\cos x$ for $0 \leq x \leq 2\pi$,

$g : x \mapsto \frac{1}{2}x$ for $0 \leq x \leq 2\pi$.

(i) Solve the equation $\mathrm{fg}(x) = 1$. **[3]**

(ii) Sketch the graph of $y = f(x)$. **[3]**

(i) B1 Correct $\mathbf{fg}(x) = 2 - 3\cos\left(\frac{1}{2}x\right)$

M1 M1 for correct order of operations to solve their $\mathbf{fg}(x) = 1$ as far as using inverse $\cos$ expect $1.23,$ (or 70.5°) condone $x =$.

A1 One solution only in the given range, ignore answers outside the range.

Answer in degrees A0.

Alternative:
Solve $f(y) = 1 \rightarrow y = 1.23 \rightarrow \frac{1}{2}x = 1.23$ B1M1
$\rightarrow x = 2.46$ A1

Total: 3

(ii) B1 One cycle of $\pm\cos$ curve, evidence of turning at the ends not required at this stage. Can be a poor curve but not an inverted "V". If horizontal axis is not labelled mark everything to the right of the vertical axis. If axis is clearly labelled mark $0 \rightarrow 2\pi$.

B1 Start and finish at roughly the same negative y value. Significantly more above the x axis than below or correct range implied by labels.

B1 Fully correct. Curves not lines.

Must be a reasonable curve clearly turning at both ends. Labels not required but must be appropriate if present.

Total: 3

Question 6

Oct/Nov 2018 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

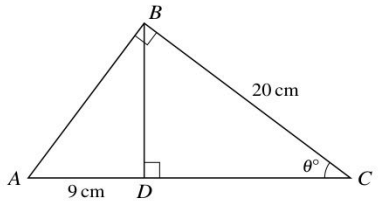
> "Weaker candidates were often unable to make the second connection correctly."

Source: 9709_w18_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 6

Coaching With Shuja

6

6



The diagram shows a triangle ABC in which $BC = 20$ cm and angle $ABC = 90°$. The perpendicular from B to AC meets AC at D and $AD = 9$ cm. Angle $BCA = \theta°$.

(i) By expressing the length of BD in terms of θ in each of the triangles ABD and DBC, show that $20\sin^2\theta = 9\cos\theta$. [4]

The diagram shows a triangle ABC in which $BC = 20$ cm and angle $ABC = 90°$. The perpendicular from B to AC meets AC at D and $AD = 9$ cm. Angle $BCA = \theta°$.

(i) By expressing the length of BD in terms of θ in each of the triangles ABD and DBC, show that $20\sin^2\theta = 9\cos\theta$. [4]

(ii) Hence, showing all necessary working, calculate θ. [3]

Mark Scheme

(i) In $\triangle ABD$, $\tan\theta = \frac{9}{BD} \rightarrow BD = \frac{9}{\tan\theta}$ or $9\tan(90-\theta)$ or $9\cot\theta$

or $\sqrt{(20\tan\theta)^2 - 9^2}$ (Pythag) or $\frac{9\sin(90-\theta)}{\sin\theta}$ (Sine rule) **B1**

In $\triangle DBC$, $\sin\theta = \frac{BD}{20} \rightarrow BD = 20\sin\theta$ **B1**

$20\sin\theta = \frac{9}{\tan\theta}$ **M1** Equates their expressions for BD and uses $\sin\theta\cos\theta = \tan\theta$ or $\cos\theta\sin\theta = \cot\theta$ if necessary.

$\rightarrow 20\sin^2\theta = 9\cos\theta$ **AG A1** Correct manipulation of their expression to arrive at given answer.

SC:
In $\triangle DBC$, $\sin\theta = \frac{BD}{20} \rightarrow BD = 20\sin\theta$ **B1**

In $\triangle ABD$, $BA = \frac{9}{\sin\theta}$ and $\cos\theta = \frac{BD}{BA}$

$\cos\theta = \frac{20\sin\theta}{9/\sin\theta} \rightarrow \cos\theta = \frac{20\sin^2\theta}{9}$ **M1**

$\rightarrow 20\sin^2\theta = 9\cos\theta$ **A1** Scores 3/4

Total: 4

(ii) Uses $s^2 + c^2 = 1 \rightarrow 20\cos^2\theta + 9\cos\theta - 20 (= 0)$ **M1** Uses $s^2 + c^2 = 1$ to form a three term quadratic in $\cos\theta$

$\rightarrow \cos\theta = 0.8$ **A1** www

$\rightarrow \theta = 36.9^\circ$ awrt **A1** www. Allow 0.644° awrt. Ignore 323.1° or 2.50°.

Note: correct answer without working scores 0/3.

Total: 3

Question 7

Oct/Nov 2018 | Paper 1 (variant 12)

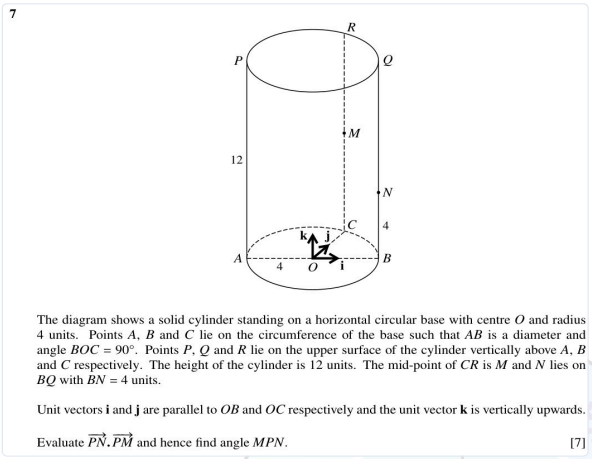
Examiner Report — Why students found this question difficult

> "A significant number, though, incorrectly rounded their final answer, with 31 and 40 quite common."

Source: 9709_w18_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 7

Coaching With Shuja

7



The diagram shows a solid cylinder standing on a horizontal circular base with centre O and radius 4 units. Points A, B and C lie on the circumference of the base such that AB is a diameter and angle $BOC = 90°$. Points P, Q and R lie on the upper surface of the cylinder vertically above A, B and C respectively. The height of the cylinder is 12 units. The mid-point of CR is M and N lies on BQ with $BN = 4$ units.

Unit vectors $\mathbf{i}$ and $\mathbf{j}$ are parallel to OB and OC respectively and the unit vector $\mathbf{k}$ is vertically upwards.

Evaluate $\overrightarrow{PN} \cdot \overrightarrow{PM}$ and hence find angle MPN. **[7]**

$\overrightarrow{PN} = 8\mathbf{i} - 8\mathbf{k}$ **B1**

$\overrightarrow{PM} = 4\mathbf{i} + 4\mathbf{j} - 6\mathbf{k}$ **B2,1,0** Loses 1 mark for each component incorrect

SC: $\overrightarrow{PN} = -8\mathbf{i} + 8\mathbf{k}$ and $\overrightarrow{PM} = -4\mathbf{i} - 4\mathbf{j} + 6\mathbf{k}$ scores 2/3.

$\overrightarrow{PN} \cdot \overrightarrow{PM} = 32 + 0 + 48 = 80$ **M1** Evaluates $x_1x_2 + y_1y_2 + z_1z_2$ for correct vectors or one or both reversed.

$|\overrightarrow{PN}| = \sqrt{128} \times \sqrt{68} (= 16\sqrt{34})$ **M1** Product of their moduli - may be seen in cosine rule

$\sqrt{128} \times \sqrt{68} \cos M\hat{P}N = 80$ **M1** All linked correctly.

Angle $M\hat{P}N = 31.0^\circ$ awrt **A1** Answer must come directly from +ve cosine ratio.

Cosine rule not accepted as a complete method. Allow 0.540° awrt.

Note: Correct answer from incorrect vectors scores A0 (XP)

Total: 7

Coaching With Shuja

Question 10

Oct/Nov 2018 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (i) proved to be more difficult for weaker candidates, who often knew what was required, equating and using the discriminant, but were unable to correctly carry out the algebraic manipulation needed."

Source: 9709_w18_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

Coaching With Shuja

10 The equation of a curve is $y = 2x + \frac{12}{x}$ and the equation of a line is $y + x = k$, where k is a constant.

(i) Find the set of values of k for which the line does not meet the curve. **[3]**

In the case where $k = 15$, the curve intersects the line at points A and B.

(ii) Find the coordinates of A and B. **[3]**

(iii) Find the equation of the perpendicular bisector of the line joining A and B. **[3]**

Coaching With Shuja

(i) $2x + \frac{12}{x} = k - x$ or $y = 2(k - y) + \frac{12}{k-y} \rightarrow$ **3** term quadratic. **M1** Attempt to eliminate y (or x) to form a 3 term quadratic.

Expect $3x^2 - kx + 12$ or $3y^2 - 5ky + (2k^2 + 12)(= 0)$

Use of $b^2 - 4ac \rightarrow k^2 - 144 < 0$ **DM1** Using the discriminant, allow $<, =; 0$: expect 12 and -12

$-12 < k < 12$ **A1** Do NOT accept $<$. Separate statements OK.

Total: 3

(ii) Using $k = 15$ in their 3 term quadratic **M1** From **(i)** or restart. Expect $3x^2 - 15x + 12$ or $3y^2 - 75y + 462(= 0)$

$x = 1, 4$ or $y = 11, 14$ **A1** Either pair of x or y values correct..

$(1, 14)$ and $(4, 11)$ **A1** Both pairs of coordinates

Total: 3

(iii) Gradient of $AB = -1 \rightarrow$ Perpendicular gradient $= +1$ **B1FT** Use of $m_1 m_2 = -1$ to give $+1$ or it from their A and B.

Finding their midpoint using their $(1, 14)$ and $(4, 11)$ **M1** Expect $(2\frac{1}{2}, 12\frac{1}{2})$

Equation: $y - 12\frac{1}{2} = (x - 2\frac{1}{2})$ $[y = x + 10]$ **A1** Accept correct unsimplified and isw

Total: 3

Coaching With Shuja

Question 11

Oct/Nov 2018 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

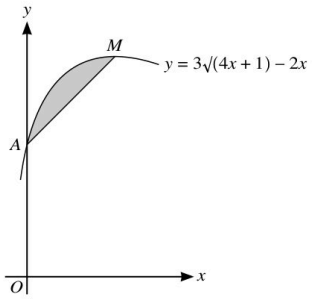
> "Question 11 Parts (i) and (ii) of this final question were very well done, but many candidates struggled to find the required area in part (iii)."

Source: 9709_w18_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 11

Coaching With Shuja

11

11



The diagram shows part of the curve $y = 3\sqrt{(4x + 1)} - 2x$. The curve crosses the y-axis at A and the stationary point on the curve is M.

(i) Obtain expressions for $\frac{dy}{dx}$ and $\int y\,dx$. [5]

The diagram shows part of the curve $\boldsymbol{y = 3\sqrt{4x + 1} - 2x}$. The curve crosses the $\boldsymbol{y}$-axis at $\boldsymbol{A}$ and the stationary point on the curve is $\boldsymbol{M}$.

(i) Obtain expressions for $\frac{dy}{dx}$ and $\int y\,dx$. **[5]**

(ii) Find the coordinates of $\boldsymbol{M}$. **[3]**

(iii) Find, showing all necessary working, the area of the shaded region. **[4]**

Total: 12

(i) $\frac{dy}{dx} = \frac{3}{2} \times (4x+1)^{-1/2}[\times 4][-2]\left(\frac{6}{\sqrt{4x+1}} - 2\right)$ **[B2,1,0 Looking for 3 components]**

$\int y\,dx = \left[3(4x+1)^{3/2} \div \frac{3}{2}\right][\div 4][-\frac{2x^2}{2}](+C)$

$= \left(\frac{(4x+1)^{3/2}}{2} - x^2\right)$ **[B1 B1 B1]**

B1 for $3(4x+1)^{3/2} \div \frac{3}{2}$ B1 for '$\div 4$' B1 for '$-\frac{2x^2}{2}$'. Ignore omission of $+C$. If included isw any attempt at evaluating. **[5]**

(ii) At M, $\frac{dy}{dx} = 0 \rightarrow \frac{6}{\sqrt{4x+1}} = 2$ **[M1]**

Sets their 2 term $\frac{dy}{dx}$ to 0 and attempts to solve (as far as $x = k$)

$x = 2, y = 5$ **[A1 A1]**

[3]

(iii) Area under the curve = $\left[\frac{1}{2}(4x+1)^{3/2} - x^2\right]_0^2$ **[M1]**

Uses their integral and their '2' and 0 correctly

$(13.5 - 4) - 0.5$ or $9.5 - 0.5 = 9$ **[A1]**

No working implies use of integration function on calculator M0A0.

Area under the chord = trapezium = $\frac{1}{2} \times 2 \times (3+5) = 8$ **[M1]**

Either using the area of a trapezium with their 2, 3 and 5 or $\int (their x + 3)\,dx$ using their '2' and 0 correctly.

Or $\left[\frac{x^2}{2} + 3x\right] = 8$ **[A1]**

Dependent on both method marks.

(Shaded area = $9 - 8) = 1$ **[A1]**

OR Area between the chord and the curve is:

$\int_0^2 \sqrt{3\sqrt{4x+1} - 2x - (x+3)}\,dx$ **[M1]**

Subtracts their line from given curve and uses their '2' and 0 correctly.

$= \int_0^2 \sqrt{3\sqrt{4x+1} - 3x - 3}\,dx$ **[A1]**

All integration correct and limits 2 and 0.

$= 3\left[\frac{1}{6}(4x+1)^{3/2} - \frac{x^2}{2} - x\right]_0^2$ **[M1]**

Evidence of substituting their '2' and 0 into their integral.

$= 3\left\{\left(\frac{27}{6} - 2 - 2\right) - \left(\frac{1}{6}\right)\right\}$ **[A1]**

No working implies use of a calculator M0A0.

$= 3\left(\frac{1}{2} - \frac{1}{6}\right) = 3\left(\frac{1}{3}\right) = 1$ **[A1]**

[4]

Question 1

Oct/Nov 2018 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> “A common mistake was to lose the minus sign.”

Source: 9709_w18_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 1

Coaching With Shuja

1 Find the coefficient of $\frac{1}{x^3}$ in the expansion of $\left(x - \frac{2}{x}\right)^5$. **[3]**

Mark Scheme

$7C5x^2\left(-\frac{2}{x}\right)^5$ **[B1 Can appear in an expansion. Allow 7C2]**

$21x - 32$ **[B1 Identified. Allow** $(21x^2) \times (-32x^{-5})$**. Implied by correct answer]**

-672 **[B1 Allow** $-\frac{672}{x^3}$**. If 0/3 scored, 672 scores SCB1]**

Total: 3

Question 3

Oct/Nov 2018 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "In addition to this, as reported in General comments above, many candidates obtained an incorrect final answer because they did not work with more than 3 significant figures in their calculations."

Source: 9709_w18_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 3

Coaching With Shuja

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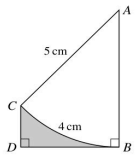

The diagram shows an arc BC of a circle with centre A and radius 5 cm. The length of the arc BC is 4 cm. The point D is such that the line BD is perpendicular to BA and DC is parallel to BA.

(i) Find angle BAC in radians. [1]

.....

.....

.....

.....

.....

(ii) Find the area of the shaded region BDC. [5]

The diagram shows an arc BC of a circle with centre A and radius 5 cm. The length of the arc BC is 4 cm. The point D is such that the line BD is perpendicular to BA and DC is parallel to BA.

(i) Find angle BAC in radians. **[1]**

(ii) Find the area of the shaded region BDC. **[5]**

(i) 0.8 [oe B1]

[1]

(ii) $BD = 5\sin$ **their 0.8 [M1 Expect 3.58(7). Methods using degrees are acceptable]**

$DC = 5 - 5\cos$ **their 0.8 [M1 Expect 1.51(6)]**

Sector $= \frac{1}{2} \times 5^2 \times$ **their0.8 [M1 Expect 10 for sector.]**

OR **Seg** $= \frac{1}{2} \times 5^2 \times$ **[their 0.8** $-\sin$ **their 0.8] [M1 Expect 1.03(3) for segment]**

Trap $= \frac{1}{2}(5 +$ **their**$DC) \times$ **their**BD **[M1 OR (for last 2 marks) if** X **is on** AB **and** XC **is parallel to** BD**:]**

OR $\triangle BDC = \frac{1}{2}$**their**$BD \times$ **their**CD

Shaded area $= 11.69 - 10$ **[A1** $BDCX =$ **(sector** $- \triangle AXC) = 5.43(8) - [10 - 6.24(9)] = 1.69$ cao M1A1]**

OR $2.71(9) - 1.03(3) = 1.69$ **[cao]**

Total: 5

Question 4

Oct/Nov 2018 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "This question should have provided a good source of marks to most candidates but, in practice, marks were lost through minor mistakes."

Source: 9709_w18_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 4

Coaching With Shuja

4 Two points A and B have coordinates $(-1, 1)$ and $(3, 4)$ respectively. The line BC is perpendicular to AB and intersects the x-axis at C.

(i) Find the equation of BC and the x-coordinate of C. **[4]**

(ii) Find the distance AC, giving your answer correct to 3 decimal places. **[2]**

Coaching With Shuja

(i) Gradient, m, of $AB = \frac{3}{4}$ **[B1]**

Equation of BC is $y - 4 = \frac{-4}{3}(x - 3)$ **[M1A1 Line through $(3, 4)$ with gradient $\frac{-1}{m}$ (M1). (Expect $y = \frac{-4}{3}x + 8$)]**

$x = 6$ [A1 Ignore any y coordinate given.]

Total: 4

Question 6

Oct/Nov 2018 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> “A significant number of candidates found BF incorrectly.”

Source: 9709_w18_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 6

Coaching With Shuja

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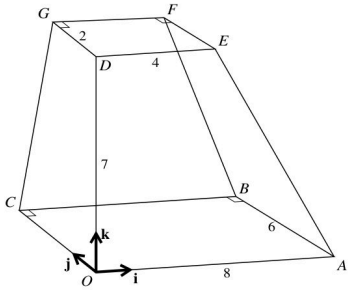

The diagram shows a solid figure $OABCDEFG$ with a horizontal rectangular base $OABC$ in which $OA = 8$ units and $AB = 6$ units. The rectangle $DEFG$ lies in a horizontal plane and is such that D is 7 units vertically above O and DE is parallel to OA. The sides DE and DG have lengths 4 units and 2 units respectively. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OC and OD respectively. Use a scalar product to find angle OBF, giving your answer in the form $\cos^{-1}\left(\frac{a}{b}\right)$, where a and b are integers.

[6]

The diagram shows a solid figure $OABCDEFG$ with a horizontal rectangular base $OABC$ in which $OA = 8$ units and $AB = 6$ units. The rectangle $DEFG$ lies in a horizontal plane and is such that D is 7 units vertically above O and DE is parallel to OA. The sides DE and DG have lengths 4 units and 2 units respectively. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OC and OD respectively. Use a scalar product to find angle OBF, giving your answer in the form $\cos^{-1}\left(\frac{a}{b}\right)$, where a and b are integers.

[6]

$(\mathbf{BO}) = -8\mathbf{i} - 6\mathbf{j}$ **[B1 OR** $(\mathbf{OB}) = 8\mathbf{i} + 6\mathbf{j}$**]**

$(\mathbf{BF}) = -6\mathbf{i} - 8\mathbf{i} + 7\mathbf{k} + 4\mathbf{i} + 2\mathbf{j} = -4\mathbf{i} - 4\mathbf{j} + 7\mathbf{k}$** [*B1OR*$(\mathbf{FB}) = 4\mathbf{i} + 4\mathbf{j} - 7\mathbf{k}$]**

$(\mathbf{BF}.\mathbf{BO}) = (-4)(-8) + (-4)(-6)$ **[M1 OR** $(\mathbf{FB}.\mathbf{OB})$ **Expect 56. Accept one reversed but award final A0]**

$|\mathbf{BF}| \times |\mathbf{BO}| = \sqrt{4^2 + 4^2 + 7^2} \times \sqrt{8^2 + 6^2}$ **[M1 Expect 90. At least one magnitude methodically correct]**

Angle $OBF = \cos^{-1}\left(\frac{\text{their}56}{\text{their}90}\right) = \cos^{-1}\left(\frac{56}{90}\right)$ or $\cos^{-1}\left(\frac{28}{45}\right)$ **[DM1A1 Or equivalent 'integer' fractions. All M marks dependent on use of** $(\pm)\mathbf{BO}$ **and** $(\pm)\mathbf{BF}$**. 3rd M mark dep on both preceding M marks]**

Total: 6

Coaching With Shuja

Question 8

Oct/Nov 2018 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> “Part (ii) proved to be dy more difficult.”

Source: 9709_w18_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 8

Coaching With Shuja

8 A curve passes through $(0, 11)$ and has an equation for which $\frac{dy}{dx} = ax^2 + bx - 4$, where a and b are constants.

(i) Find the equation of the curve in terms of a and b. **[3]**

(ii) It is now given that the curve has a stationary point at $(2, 3)$. Find the values of a and b. **[5]**

Coaching With Shuja

(i) $y = \frac{1}{3}ax^3 + \frac{1}{2}bx^2 - 4x(+c)$ **[B1]**

$11 = 0 + 0 + 0 + c$ **[M1 Sub** $x = 0, y = 11$ **into an integrated expression.** c **must be present]**

$y = \frac{1}{3}ax^3 + \frac{1}{2}bx^2 - 4x + 11$ **[A1]**

Total: 3

(ii) $4a + 2b - 4 = 0$ **[M1 Sub** $x = 2$, $\frac{dy}{dx} = 0$**]**

$\frac{1}{3}(8a) + 2b - 8 + 11 = 3$ **[M1 Sub** $x = 2, y = 3$ **into an integrated expression. Allow if 11 missing]**

Solve simultaneous equations **[DM1 Dep. on both M marks]**

$a = 3, b = -4$ **[A1A1 Allow if no working seen for simultaneous equations]**

Total: 5

Coaching With Shuja

Question 10

Oct/Nov 2018 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Some final answers were given without showing the working for substituting the limits (for which marks were lost) and 1 was seen occasionally as the lower limit."

Source: 9709_w18_er.pdf, Page 11, Section: Comments on specific questions – Paper 1 (variant 13), Question 10

Coaching With Shuja

10

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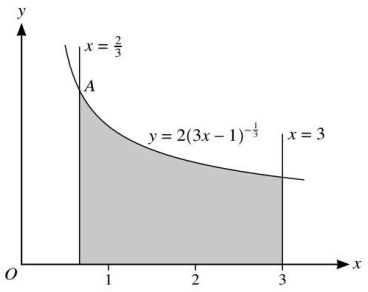


The diagram shows part of the curve $y = 2(3x - 1)^{-\frac{1}{3}}$ and the lines $x = \frac{2}{3}$ and $x = 3$. The curve and the line $x = \frac{2}{3}$ intersect at the point A.

(i) Find, showing all necessary working, the volume obtained when the shaded region is rotated through 360° about the x-axis. [5]

The diagram shows part of the curve $\boldsymbol{y = 2(3x - 1)^{-1/3}}$ and the lines $\boldsymbol{x = \frac{2}{3}}$ and $\boldsymbol{x = 3}$. The curve and the line $\boldsymbol{x = \frac{2}{3}}$ intersect at the point $\boldsymbol{A}$.

(i) Find, showing all necessary working, the volume obtained when the shaded region is rotated through $\boldsymbol{360^\circ}$ about the $\boldsymbol{x}$-axis. **[5]**

(ii) Find the equation of the normal to the curve at $\boldsymbol{A}$, giving your answer in the form $\boldsymbol{y = mx + c}$. **[5]**

Shuja

(i) $V = 4(\pi)\int(3x-1)^{-2/3}\,dx = 4(\pi)\left[\frac{(3x-1)^{1/3}}{1/3}\right]_{2/3}^{3}$ **[M1A1A1 Recognisable integration of** y^2 **(M1) Independent A1, A1 for []]**

$4(\pi)[2-1]$ **[DM1 Expect** $4(\pi)(3x-1)^{1/3}$**]**

4π or **12.6 [A1 Apply limits** $2/3 \rightarrow 3$**. Some working must be shown.]**

Total: 5

Coaching With Shuja

Question 1

Oct/Nov 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 1 This question proved to be challenging for candidates."

> "Many candidates failed to square the 2 outside the modulus sign or poor algebra prevented them from arriving at the correct equation."

Source: 9709_w18_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

1 Find the set of values of x satisfying the inequality $2|2x - a| < |x + 3a|$, where a is a positive constant.

[4]

EITHER: State or imply non-modular inequality **[B1]**

$2^2(2x-a)^2 < (x+3a)^2$, or corresponding quadratic equation, or pair of linear equations

$2(2x-a) = \pm(x+3a)$

Make reasonable attempt at solving a 3-term quadratic, or solve two linear equations for x **[M1]**

Obtain critical values $x = \frac{5}{3}a$ and $x = -\frac{1}{5}a$ **[A1]**

State final answer $-\frac{1}{5}a < x < \frac{5}{3}a$ **[A1]**

OR: Obtain critical value $x = \frac{5}{3}a$ from a graphical method, or by inspection, or **[B1]**

by solving a linear equation or an inequality

Obtain critical value $x = -\frac{1}{5}a$ similarly **[B2]**

State final answer $-\frac{1}{5}a < x < \frac{5}{3}a$ **[B1]**

[Do not condone $\leq$ for $<$ in the final answer.]

Total: 4

Coaching With Shuja

Question 2

Oct/Nov 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 2 This question was usually well done, however there were still many candidates who did not reach an b equation of the form ae2x = b, aex = be–x or aex = with the correct values of both a and b."

> "Most candidates knew they then needed to take lns but some did not show this step despite the additional instruction to show all necessary working."

Source: 9709_w18_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

2 Showing all necessary working, solve the equation $\frac{2e^{x}+e^{-x}}{e^{x}-e^{-x}} = 4$, giving your answer correct to 2 decimal places.

Total: 4

2 Rearrange the equation in the form $ae^{2x} = b$ or $ae^{x} = be^{-x}$ — **M1**

Obtain correct equation in either form with $a = 2$ and $b = 5$ — **A1**

Use correct method to solve for x — **M1**

Obtain answer $x = 0.46$ — **A1**

Total: 4

Question 3

Oct/Nov 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 3 (i) This part proved challenging more many candidate, for various reasons."

Source: 9709_w18_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 3

Coaching With Shuja

(i) By sketching a suitable pair of graphs, show that the equation $x^3 = 3 - x$ has exactly one real root.

Total: 2

(ii) Show that if a sequence of real values given by the iterative formula

$$x_{n+1} = \frac{2x_n^3 + 3}{3x_n^2 + 1}$$

converges, then it converges to the root of the equation in part (i).

Total: 2

(iii) Use this iterative formula to determine the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places.

Total: 3

Coaching With Shuja

3(i) Sketch a relevant graph, e.g. $y = x^3$ — **B1**

Sketch a second relevant graph, e.g. $y = 3 - x$, and justify the given statement — **B1** (Consideration of behaviour for $x < 0$ is needed for the second **B1**)

Total: 2

3(ii) State or imply the equation $x = \frac{2x^3+3}{3x^2+1}$ — **B1**

Rearrange this in the form $x^3 = 3 - x$, or commence work *vice versa* — **B1**

Total: 2

Question 5

Oct/Nov 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The integral in x proved more difficult, since instead of considering and x, some candidates x resorted to integrating by parts and increased the possibility of sign errors."

Source: 9709_w18_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

Coaching With Shuja

5 The coordinates (x, y) of a general point on a curve satisfy the differential equation

$$x\frac{dy}{dx} = (2 - x^2)y.$$

The curve passes through the point $(1, 1)$. Find the equation of the curve, obtaining an expression for y in terms of x.

Total: 7

5 Separate variables correctly and integrate at least one side — **B1**

Obtain term $\ln y$ — **B1**

Obtain terms $2\ln x - \frac{1}{2}x^2$ — **B1+B1**

Use $x = 1, y = 1$ to evaluate a constant, or as limits — **M1**

Obtain correct solution in any form, e.g. $\ln y = 2\ln x - \frac{1}{2}x^2 + \frac{1}{2}$ — **A1**

Rearrange as $y = x^2 \exp\left(\frac{1}{2} - \frac{1}{2}x^2\right)$, or equivalent — **A1**

Total: 7

Question 6

Oct/Nov 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 6 (i) Many candidates found it difficult to obtain the given expression, taking far more steps than were required."

Source: 9709_w18_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

(i) Show that the equation $(\sqrt{2})\csc x + \cot x = \sqrt{3}$ can be expressed in the form $R\sin(x - \alpha) = \sqrt{2}$, where $R > 0$ and $0° < \alpha < 90°$.

Total: 4

(ii) Hence solve the equation $(\sqrt{2})\csc x + \cot x = \sqrt{3}$, for $0° < x < 180°$.

Total: 4

6(i) Rearrange in the form $\sqrt{3}\sin x - \cos x = \sqrt{2}$ — **B1**

State $R = 2$ — **B1**

Use trig formulae to obtain a — **M1**

Obtain $a = 30°$ with no errors seen — **A1**

Total: 4

Coaching With Shuja

Question 7

Oct/Nov 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(ii) Nearly all candidates struggled with this question despite being given the substitution required."

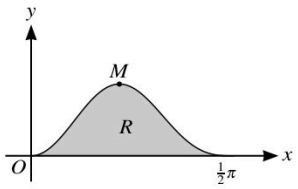
> "Common errors were poor substitution of du = cos x dx, inability to express cos2 x correctly in terms of u, substituting π limits 0 and for u or substituting limits 0 and 1 for x."

Source: 9709_w18_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

7

7



The diagram shows the curve $y = 5\sin^2 x \cos^3 x$ for $0 \leqslant x \leqslant \frac{1}{2}\pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M, giving your answer correct to 3 decimal places. [5]

The diagram shows the curve $y = 5\sin^2 x \cos^3 x$ for $0 \leq x \leq \frac{1}{2}\pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M, giving your answer correct to 3 decimal places.

Total: 5

(ii) Using the substitution $u = \sin x$ and showing all necessary working, find the exact area of R.

Total: 4

7(i) Use product rule — **M1***

Obtain correct derivative in any form — **A1**

Equate derivative to zero and obtain an equation in a single trig function — **depM1***

Obtain a correct equation, e.g. $3\tan^2 x = 2$ — **A1**

Obtain answer $x = 0.685$ — **A1**

Total: 5

7(ii) Use the given substitution and reach $a\int(u^2 - u^4)du$ — **M1**

Obtain correct integral with $a = 5$ and limits **0** and **1** — **A1**

Use correct limits in an integral of the form $a\left(\frac{1}{3}u^3 - \frac{1}{5}u^5\right)$ — **M1**

Obtain answer $\frac{2}{3}$ — **A1**

Total: 4

Coaching With Shuja

Question 8

Oct/Nov 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Few candidates managed to obtain the correct argument, the usual answer being $\theta = -1.05$, something that candidates arrived at from their calculators without any reference to the position of the complex number $-4\ 7 + i$ in the Argand diagram."

Source: 9709_w18_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 31), Question 8

Coaching With Shuja

(a) Showing all necessary working, express the complex number $\frac{2+3i}{1-2i}$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. Give the values of r and θ correct to 3 significant figures.

Total: 5

(b) On an Argand diagram sketch the locus of points representing complex numbers z satisfying the equation $|z - 3 + 2i| = 1$. Find the least value of $|z|$ for points on this locus, giving your answer in an exact form.

Total: 4

8(i) EITHER: Multiply numerator and denominator by $1 + 2i$, or equivalent, or equate to $x + iy$, obtain two equations in x and y and solve for x or y — **M1**

Obtain quotient $-\frac{4}{5} + \frac{7}{5}i$, or equivalent — **A1**

Use correct method to find either r or θ — **M1**

Obtain $r = 1.61$ — **A1**

Obtain $\theta = 2.09$ — **A1**

OR: Find modulus or argument of $2 + 3i$ or of $1 - 2i$ — **B1**

Use correct method to find r — **M1**

Obtain $r = 1.61$ — **A1**

Use correct method to find θ — **M1**

Obtain $\theta = 2.09$ — **A1**

Total: 5

8(ii) Show a circle with centre $3 - 2i$ — **B1**

Show a circle with radius 1 — **B1ft** (Centre not at the origin)

Carry out a correct method for finding the least value of $|z|$ — **M1**

Obtain answer $\sqrt{13} - 1$ — **A1**

Total: 4

Coaching With Shuja

Question 10

Oct/Nov 2018 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(iii) Those candidates who knew the formula for the distance of a point from a plane usually quickly scored full marks, whilst most others struggled."

Source: 9709_w18_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 31), Question 10

Coaching With Shuja

10 The planes m and n have equations $3x + y - 2z = 10$ and $x - 2y + 2z = 5$ respectively. The line l has equation $\mathbf{r} = 4\mathbf{i} + 2\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} + \mathbf{j} + 2\mathbf{k})$.

(i) Show that l is parallel to m.

Total: 3

(ii) Calculate the acute angle between the planes m and n.

Total: 3

(iii) A point P lies on the line l. The perpendicular distance of P from the plane n is equal to 2. Find the position vectors of the two possible positions of P.

Total: 4

Coaching With Shuja

10(i) EITHER: Expand scalar product of a normal to m and a direction vector of l — **M1**

Verify scalar product is zero — **A1**

Verify that one point of l does not lie in the plane — **A1**

OR: Substitute coordinates of a general point of l in the equation of the plane m — **M1**

Obtain correct equation in λ in any form — **A1**

Verify that the equation is not satisfied for any value of λ — **A1**

Total: 3

10(ii) Use correct method to evaluate a scalar product of normal vectors to m and n — **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and evaluate the inverse cosine of the result — **M1**

Obtain answer 74.5° or 1.30 radians — **A1**

Total: 3

10(iii) EITHER: Using the components of a general point P of l form an equation in λ by equating the perpendicular distance from n to 2 — **M1**

Obtain a correct modular or non-modular equation in any form — **A1**

Solve for λ and obtain a position vector for P, e.g. $7\mathbf{i} + 5\mathbf{j} + 7\mathbf{j}$ from $\lambda = 3$ — **A1**

Obtain position vector of the second point, e.g. $3\mathbf{i} + \mathbf{j} - \mathbf{k}$ from $\lambda = -1$ — **A1**

OR: Take a point Q on l, e.g. $(5, 3, 3)$ and form an equation in λ by equating the length of the projection of QP onto a normal to plane n to 2 — **M1**

Total: 4

Question 1

Oct/Nov 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors were in squaring the brackets and in simplifying to obtain a quadratic equation/inequality."

> "1 7 The incorrect answer – > x > was a common error."

Source: 9709_w18_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

1 Solve the inequality $3|2x - 1| > |x + 4|$.

Total: 4

Mark Scheme Oct/Nov 2018 • Paper 3 (variant 32) • Q1

1 State or imply non-modular inequality $3^2(2x-1)^2 > (x+4)^2$, or corresponding quadratic equation, or pair of linear equations/inequalities $3(2x-1) = \pm(x+4)$ — **B1**

Make reasonable attempt at solving a 3-term quadratic, or solve two linear equations for x — **M1**

Obtain critical values $x = \frac{7}{5}$ and $x = -\frac{1}{7}$ — **A1** (Accept **1.4** and **−0.143** or better for penultimate **A1** mark)

State final answer $x > \frac{7}{5}, x < -\frac{1}{7}$ — **A1** ('*and*' is **A0**, $\frac{7}{5} < x < -\frac{1}{7}$ is **A0**. Must be exact values. Must be strict inequalities in final answer)

Alternative

Obtain critical value $x = \frac{7}{5}$ from a graphical method or by inspection, or by solving a linear equation or an inequality — **B1** (or by inspection, or by solving a linear equation or an inequality)

Obtain critical value $x = -\frac{1}{7}$ similarly — **B2**

State final answer $x > \frac{7}{5}$ or $x < -\frac{1}{7}$ or equivalent — **B1** ([Do not condone $>$ for $<$, or $<$ for $>$.])

Total: 4

Coaching With Shuja

Question 4

Oct/Nov 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "© 2018 Cambridge International Advanced Level 9709 Mathematics November 2018 Principal Examiner Report for Teachers A significant minority of candidates made no progress at all because they attempted to apply incorrect rules of logarithms to the original fraction, or to ex +e−x = 4ex +4, in order to remove the exponentials."

Source: 9709_w18_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

Question 4

Showing all necessary working, solve the equation

$$\frac{e^x + e^{-x}}{e^x + 1} = 4,$$

giving your answer correct to 3 decimal places.

Marks: [5]

Question 4

B1 Substitute and obtain 3-term quadratic $3u^2 + 4u - 1 = 0$, or equivalent. e.g. $3(e^x)^2 + 4e^x - 1 = 0$

M1 Solve a 3-term quadratic for u

A1 Obtain root $\left(\sqrt{7} - 2\right)/3$, or decimal in $[0.21, 0.22]$

M1 Use correct method for finding x from a positive value of e^x. Must see some indication of method: use of $x = \ln u$

A1 Obtain answer $x = -1.536$ only. **CAO**. Must be 3 dp

Total: 5

Coaching With Shuja

Question 6

Oct/Nov 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "1 1 The majority of candidates completed ∫ dx correctly, but ∫ dy proved to be more difficult."

Source: 9709_w18_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

Question 6

A certain curve is such that its gradient at a general point with coordinates (x, y) is proportional to $\frac{y^2}{x}$. The curve passes through the points with coordinates $(1, 1)$ and $(e, 2)$. By setting up and solving a differential equation, find the equation of the curve, expressing y in terms of x.

Marks: [8]

Coaching With Shuja

Question 6

B1 State equation $\frac{dy}{dx} = k\frac{y^2}{x}$, or equivalent

B1 Separate variables correctly and integrate at least one side: $\int \frac{k}{x}dx = \int \frac{1}{y^2}dy$. Allow with incorrect value substituted for k

B1 + B1 Obtain terms $-\frac{1}{y}$ and $k\ln x$

M1 Use given coordinates correctly to find k and/or a constant of integration C in an equation containing terms $\frac{c}{x}$, $b\ln x$ and C

A1 + A1 Obtain $k = \frac{1}{2}$ and $c = -1$, or equivalent. $\frac{1}{2}\ln x = 1 - \frac{1}{y}$. A0 for fortuitous answers.

A1 Obtain answer $y = \frac{-1}{-1+\ln x}$ or equivalent, and **ISW**. $y = \frac{-1}{-1+\ln x}$

Alternative schemes:

B1 SC: MR of the fraction. $\frac{dy}{dx} = k\frac{y^2}{x^2}$. Separate variables and integrate: $-\frac{1}{y} = -k\left(\frac{1}{x} + C\right)$

B1 Separate variables and integrate

B1 + B1 Substitute to find k and/or c: $k = \frac{e}{2(e-1)}$, $c = \frac{2-e}{2(e-1)}$

M1 Substitute to find k and/or c

A1 + A1 Answer

A0 A0 for fortuitous answers.

Total: 8

Question 7

Oct/Nov 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The common errors were to attempt to use integration by parts, or to integrate the numerator and denominator separately."

> "A common error in the final stages of the solution was the incorrect statement ln(2+sinx)=ln2+ln(sinx)."

Source: 9709_w18_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

Question 7

A curve has equation $y = \frac{3\cos x}{2+\sin x}$, for $-\frac{1}{2}\pi < x < \frac{1}{2}\pi$.

(i) Find the exact coordinates of the stationary point of the curve.

Marks: [6]

(ii) Verify by calculation that a lies between 2.9 and 3.1.

Marks: [2]

(iii) Use an iterative formula based on the equation in part (i) to determine a correct to 2 decimal places. Give the result of each iteration to 4 decimal places.

Marks: [3]

Coaching With Shuja

Question 7(i)

M1 Use correct quotient or product rule

A1 Obtain correct derivative in any form: $\frac{dy}{dx} = \frac{-3\sin x(2+\sin x)-3\cos x\cos x}{(2+\sin x)^2}$

Condone invisible brackets if recovery implied later.

M1 Equate numerator to zero

M1 Use $\cos^2 x + \sin^2 x = 1$ and solve for $\sin x$: $-6\sin x - 3 = 0 \Rightarrow \sin x = \ldots$

A1 + A1 Obtain coordinates $x = -\frac{\pi}{6}$ and $y = \sqrt{3}$ **ISW**. From correct working. No others in range

SR: A candidate who only states the numerator of the derivative, but justifies this, can have full marks. Otherwise they score **M0 A0 M1 A0 A0**

Total: 6

Question 7(ii)

M1* State indefinite integral of the form $k\ln(2+\sin x)$

A1 Substitute limits correctly, equate result to 1 and obtain: $3\ln(2+\sin a) - 3\ln 2 = 1$

M1(dep*) Use correct method to solve for a. Allow for a correct method to solve an incorrect equation, so long as that equation has a solution.

$1 + \frac{1}{3}\sin a = e^{a/2} \Rightarrow a = \sin^{-1}\left[2\left(e^{a/2} - 1\right)\right]$

Can be implied by 52.3 degrees

A1 Obtain answer $a = 0.913$ or better. Ignore additional solutions. Must be in radians.

Total: 4

Question 9

Oct/Nov 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Many candidates did not attempt to find the greatest value of argz."

Source: 9709_w18_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Question 9(a)(i)

[Question part not shown in image - mark scheme only provided]

Question 9(a)(ii)

[Question part not shown in image - mark scheme only provided]

Question 9(b)

[Question part not shown in image - mark scheme only provided]

Coaching With Shuja

Question 9(a)(i)

M1 Multiply numerator and denominator by $1 + 2i$, or equivalent

Requires at least one of $2 + 10i + 12i^2$ and $1 - 4i^2$ together with use of $i^2 = -1$. Can be implied by $\frac{-10+10i}{5}$

A1 Obtain quotient $-2 + 2i$

Alternative:

M1 Equate to $x + iy$, obtain two equations in x and y and solve for x or for y: $x + 2y = 2$, $y - 2x = 6$

A1 Obtain quotient $-2 + 2i$

Total: 2

Question 9(a)(ii)

M1 Use correct method to find either r or θ

A1ft Obtain $r = 2\sqrt{2}$, or exact equivalent

A1ft Obtain $\theta = \frac{3}{4}\pi$ from exact work. ft on $k(-1 + i)$ for $k > 0$. Do not **ISW**

Total: 3

Question 9(b)

B1 Show a circle with centre $3i$

B1ft Show a circle with radius 1. Follow through their centre provided not at the origin. For clearly unequal scales, should be an ellipse

B1 All correct with even scales and shade the correct region

9(a)(i)	Multiply numerator and denominator by 1 + 2i, or equivalent	**M1**	Requires at least one of $2+10i+12i^2$ and $1-4i^2$ together with use of $i^2=-1$. Can be implied by $\frac{-10+10i}{5}$
	Obtain quotient – 2 + 2i	**A1**	
	Alternative		
	Equate to $x + iy$, obtain two equations in x and y and solve for x or for y	**M1**	$x+2y=2, \quad y-2x=6$
	Obtain quotient – 2 + 2i	**A1**	
		2	
9(a)(ii)	Use correct method to find either r or θ	**M1**	If only finding θ, need to be looking for θ in the correct quadrant
	Obtain $r = 2\sqrt{2}$, or exact equivalent	**A1ft**	ft their $x + iy$
	Obtain $\theta = \frac{3}{4}\pi$ from exact work	**A1ft**	ft on $k(-1 + i)$ for $k > 0$ Do not ISW
		3	

M1 Carry out a correct method for calculating greatest value of $\arg z$

e.g. $\arg z = \frac{\pi}{2} + \sin^{-1}\frac{1}{3}$

A1 Obtain answer 1.91

Total: 5

Question 10

Oct/Nov 2018 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 10 (i) Although there have been questions in the past using the form r.n= a.n, many candidates did not recognise this use of (r−a).n=0."

Source: 9709_w18_er.pdf, Page 30, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

Question 10(i)

[Question part not shown in provided images - mark scheme indicates vector work]

Question 10(ii)

Calculate the acute angle between l and p.

Marks: [4]

Question 10(iii)

Find the equation of the line which lies in p and intersects l at right angles.

Marks: [4]

Coaching With Shuja

Question 10(i)

M1* Substitute for $\mathbf{r}$ and expand the scalar product to obtain an equation in λ: e.g. $3(5+\lambda)+(-3-2\lambda)+(-1+\lambda)=5$ $(2\lambda=5-11)$. Must attempt to deal with $\mathbf{i}+2\mathbf{j}$

M1(dep*) Solve a linear equation for λ

A1 Obtain $\lambda=-3$ and position vector $\mathbf{r}_A=2\mathbf{i}+3\mathbf{j}-4\mathbf{k}$ for A. Accept coordinates

Total: 3

Question 10(ii)

B1 State or imply a normal vector of p is $3\mathbf{i}+\mathbf{j}+\mathbf{k}$, or equivalent

M1 Use correct method to evaluate a scalar product of relevant vectors: e.g. $(\mathbf{i}-2\mathbf{j}+\mathbf{k}).(3\mathbf{i}+\mathbf{j}+\mathbf{k})$

M1 Using the correct process for calculating the moduli, divide the scalar product by the product of the moduli and evaluate the inverse sine or cosine of the result: $\cos\theta=\frac{2}{\sqrt{6}\sqrt{11}}$. Second **M1** available if working with the wrong vectors

A1 Obtain answer 14.3° or 0.249 radians. Or better

Total: 4

Question 10(iii)

B1 Taking the direction vector of the line to be $a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$, state a relevant equation in a, b, c, e.g. $3a+b+c=0$

M1 State a second relevant equation, e.g. $a-2b+c=0$, and solve for one ratio, e.g. $a:b$

A1 Obtain $a:b:c=3:-2:-7$, or equivalent

A1ft State answer $\mathbf{r}=2\mathbf{i}+3\mathbf{j}-4\mathbf{k}+\mu(3\mathbf{i}-2\mathbf{j}-7\mathbf{k})$. Or equivalent. The f.t. is on $\mathbf{r}_A$. Requires "$\mathbf{r}=\ldots$ "

Alternative:

M1 Attempt to calculate the vector product of relevant vectors, e.g. $(3\mathbf{i}+\mathbf{j}+\mathbf{k})\times(\mathbf{i}-2\mathbf{j}+\mathbf{k})$

A1 Obtain two correct components of the product

A1 Obtain correct product, e.g. $3\mathbf{i}-2\mathbf{j}-7\mathbf{k}$

A1ft State answer $\mathbf{r}=2\mathbf{i}+3\mathbf{j}-4\mathbf{k}+\mu(3\mathbf{i}-2\mathbf{j}-7\mathbf{k})$. Or equivalent. The f.t. is on $\mathbf{r}_A$. Requires "$\mathbf{r}=\ldots$ "

Total: 4

Question 1

Oct/Nov 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 1 This question proved to be challenging for candidates."

> "Many candidates failed to square the 2 outside the modulus sign or poor algebra prevented them from arriving at the correct equation."

Source: 9709_w18_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 33), Question 1

Coaching With Shuja

Question 1

Find the set of values of x satisfying the inequality $2|2x - a| < |x + 3a|$, where a is a positive constant.

Marks: [4]

Coaching With Shuja

Question 1

B1 EITHER: State or imply non-modular inequality:

$$2^2(2x-a)^2 < (x+3a)^2,$$

or corresponding quadratic equation, or pair of linear equations:

$$2(2x-a) = \pm(x+3a)$$

M1 Make reasonable attempt at solving a 3-term quadratic, or solve two linear equations for x

A1 Obtain critical values $x = \frac{5}{3}a$ and $x = -\frac{1}{5}a$

A1 State final answer $-\frac{1}{5}a < x < \frac{5}{3}a$

B1 OR: Obtain critical value $x = \frac{5}{3}a$ from a graphical method, or by inspection, or by solving a linear equation or an inequality

B2 Obtain critical value $x = -\frac{1}{5}a$ similarly

B1 State final answer $-\frac{1}{5}a < x < \frac{5}{3}a$

[Do not condone $\leq$ for $<$ in the final answer.]

Total: 4

Coaching With Shuja

Question 2

Oct/Nov 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 2 This question was usually well done, however there were still many candidates who did not reach an b equation of the form ae2x = b, aex = be–x or aex = with the correct values of both a and b."

> "Most candidates knew they then needed to take lns but some did not show this step despite the additional instruction to show all necessary working."

Source: 9709_w18_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 33), Question 2

Question 2

Showing all necessary working, solve the equation

$$\frac{2e^{x}+e^{-x}}{e^{x}-e^{-x}}=4,$$

giving your answer correct to 2 decimal places.

Marks: [4]

Question 2

M1 Rearrange the equation in the form $ae^{2x} = b$ or $ae^{x} = be^{-x}$

A1 Obtain correct equation in either form with $a = 2$ and $b = 5$

M1 Use correct method to solve for x

A1 Obtain answer $x = 0.46$

Total: 4

Question 3

Oct/Nov 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 3 (i) This part proved challenging more many candidate, for various reasons."

Source: 9709_w18_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 33), Question 3

Coaching With Shuja

Question 3(i)

By sketching a suitable pair of graphs, show that the equation $x^3 = 3 - x$ has exactly one real root.

Marks: [2]

Question 3(ii)

Show that if a sequence of real values given by the iterative formula

$$x_{n+1} = \frac{2x_n^3 + 3}{3x_n^2 + 1}$$

converges, then it converges to the root of the equation in part (i).

Marks: [2]

Question 3(iii)

Use this iterative formula to determine the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places.

Marks: [3]

Coaching With Shuja

Question 3(i)

B1 Sketch a relevant graph, e.g. $y = x^3$

B1 Sketch a second relevant graph, e.g. $y = 3 - x$, and justify the given statement. Consideration of behaviour for $x < 0$ is needed for the second B1

Total: 2

Question 3(ii)

B1 State or imply the equation $x = (2x^3 + 3)/(3x^2 + 1)$

B1 Rearrange this in the form $x^3 = 3 - x$, or commence work *vice versa*

Total: 2

Coaching With Shuja

Question 5

Oct/Nov 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The integral in x proved more difficult, since instead of considering and x, some candidates x resorted to integrating by parts and increased the possibility of sign errors."

Source: 9709_w18_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 33), Question 5

Coaching With Shuja

5 The coordinates (x, y) of a general point on a curve satisfy the differential equation

$$x\frac{dy}{dx} = (2 - x^2)y$$

The curve passes through the point $(1, 1)$. Find the equation of the curve, obtaining an expression for y in terms of x. **[7]**

Mark Scheme Oct/Nov 2018 • Paper 3 (variant 33) • Q5

5 Separate variables correctly and integrate at least one side | **B1**

Obtain term ln y | **B1**

Obtain terms $2\ln x - \frac{1}{2}x^2$ | **B1+B1**

Use $x = 1, y = 1$ to evaluate a constant, or as limits | **M1**

Obtain correct solution in any form, e.g. | **A1**
$\ln y = 2\ln x - \frac{1}{2}x^2 + \frac{1}{2}$

Rearrange as $y = x^2 \exp\left(\frac{1}{2} - \frac{1}{2}x^2\right)$, or equivalent | **A1**

Total: 7

Coaching With Shuja

Question 6

Oct/Nov 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 6 (i) Many candidates found it difficult to obtain the given expression, taking far more steps than were required."

Source: 9709_w18_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 33), Question 6

Coaching With Shuja

(i) Show that the equation $(\sqrt{2})\csc x + \cot x = \sqrt{3}$ can be expressed in the form $R\sin(x - \alpha) = \sqrt{2}$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. **[4]**

(ii) Hence solve the equation $(\sqrt{2})\csc x + \cot x = \sqrt{3}$, for $0^\circ < x < 180^\circ$. **[4]**

Coaching With Shuja

6(i) Rearrange in the form $\sqrt{3}\sin x - \cos x = \sqrt{2}$ | **B1**

State $R = 2$ | **B1**

Use trig formulae to obtain α | **M1**

Obtain $\alpha = 30^\circ$ with no errors seen | **A1**

Total: 4

6(ii) [Part (ii) marking scheme not clearly visible in MS image]

Question 7

Oct/Nov 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "(ii) Nearly all candidates struggled with this question despite being given the substitution required."

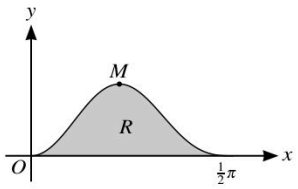
> "Common errors were poor substitution of du = cos x dx, inability to express cos2 x correctly in terms of u, substituting π limits 0 and for u or substituting limits 0 and 1 for x."

Source: 9709_w18_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 33), Question 7

Coaching With Shuja

7

7



The diagram shows the curve $y = 5\sin^2 x \cos^3 x$ for $0 \leqslant x \leqslant \frac{1}{2}\pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M, giving your answer correct to 3 decimal places. [5]

The diagram shows the curve $y = 5\sin^2 x \cos^3 x$ for $0 \leq x \leq \frac{1}{2}\pi$, and its maximum point M. The shaded region R is bounded by the curve and the x-axis.

(i) Find the x-coordinate of M, giving your answer correct to 3 decimal places. **[5]**

(ii) Using the substitution $u = \sin x$ and showing all necessary working, find the exact area of R. **[4]**

Coaching with Shuja

7(i) Use product rule | **M1***

Obtain correct derivative in any form | **A1**

Equate derivative to zero and obtain an equation in a single trig function | **depM1***

Obtain a correct equation, e.g. $3\tan^2 x = 2$ | **A1**

Obtain answer $x = 0.685$ | **A1**

Total: 5

7(ii) Use the given substitution and reach $a\int(u^2 - u^4)du$ | **M1**

Obtain correct integral with $a = 5$ and limits **0** and **1** | **A1**

Use correct limits in an integral of the form $a\left(\frac{1}{3}u^3 - \frac{1}{5}u^5\right)$ | **M1**

Obtain answer $\frac{2}{3}$ | **A1**

Total: 4

Coaching With Shuja

Question 8

Oct/Nov 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Few candidates managed to obtain the correct argument, the usual answer being $\theta = -1.05$, something that candidates arrived at from their calculators without any reference to the position of the complex number $-4\ 7 + i$ in the Argand diagram."

Source: 9709_w18_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 33), Question 8

Coaching With Shuja

8(a) Showing all necessary working, express the complex number $\frac{2+3i}{1-2i}$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. Give the values of r and θ correct to 3 significant figures. **[5]**

8(b) On an Argand diagram sketch the locus of points representing complex numbers z satisfying the equation $|z - 3 + 2i| = 1$. Find the least value of $|z|$ for points on this locus, giving your answer in an exact form. **[4]**

Coaching With Shuja

8(i) EITHER: Multiply numerator and denominator by $1 + 2i$, or equivalent, or equate to $x + iy$, obtain two equations in x and y and solve for x or for y | **M1**

Obtain quotient $-\frac{4}{5} + \frac{7}{5}i$, or equivalent | **A1**

Use correct method to find either r or θ | **M1**

Obtain $r = 1.61$ | **A1**

Obtain $\theta = 2.09$ | **A1**

OR: Find modulus or argument of $2 + 3i$ or of $1 - 2i$ | **B1**

Use correct method to find r | **M1**

Obtain $r = 1.61$ | **A1**

Use correct method to find θ | **M1**

Obtain $\theta = 2.09$ | **A1**

Total: 5

8(ii) Show a circle with centre $3 - 2i$ | **B1**

Show a circle with radius 1 | **B1ft** (Centre not at the origin)

Carry out a correct method for finding the least value of $|z|$ | **M1**

Obtain answer $\sqrt{13} - 1$ | **A1**

Total: 4

Coaching With Shuja

Question 10

Oct/Nov 2018 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "(iii) Those candidates who knew the formula for the distance of a point from a plane usually quickly scored full marks, whilst most others struggled."

Source: 9709_w18_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 33), Question 10

10 The planes m and n have equations $3x + y - 2z = 10$ and $x - 2y + 2z = 5$ respectively. The line l has equation $\mathbf{r} = 4\mathbf{i} + 2\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} + \mathbf{j} + 2\mathbf{k})$.

(i) Show that l is parallel to m. **[3]**

(ii) Calculate the acute angle between the planes m and n. **[3]**

(iii) A point P lies on the line l. The perpendicular distance of P from the plane n is equal to 2. Find the position vectors of the two possible positions of P. **[4]**

10(i) EITHER: Expand scalar product of a normal to m and a direction vector of l | **M1**

Verify scalar product is zero | **A1**

Verify that one point of l does not lie in the plane | **A1**

OR: Substitute coordinates of a general point of l in the equation of the plane m | **M1**

Obtain correct equation in λ in any form | **A1**

Verify that the equation is not satisfied for any value of λ | **A1**

Total: 3

10(ii) Use correct method to evaluate a scalar product of normal vectors to m and n | **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and evaluate the inverse cosine of the result | **M1**

Obtain answer 74.5° or 1.30 radians | **A1**

Total: 3

10(iii) EITHER: Using the components of a general point P of l form an equation in λ by equating the perpendicular distance from n to 2 | **M1**

Obtain a correct modular or non-modular equation in any form | **A1**

Solve for λ and obtain a position vector for P, e.g. $7\mathbf{i}+5\mathbf{j}+7\mathbf{j}$ from $\lambda = 3$ | **A1**

Obtain position vector of the second point, e.g. $3\mathbf{i}+\mathbf{j}-\mathbf{k}$ from $\lambda = -1$ | **A1**

OR: Take a point Q on l, e.g. $(5, 3, 3)$ and form an equation in λ by equating the length of the projection of QP onto a normal to plane n to 2 | **M1**

Obtain a correct modular or non-modular equation in any form | **A1**

Obtain a correct modular or non-modular equation in any form | **A1**

Obtain position vector of the second point, e.g. $3\mathbf{i}+\mathbf{j}-\mathbf{k}$ from $\lambda = -1$ | **A1**

Total: 4

Question 1

Feb/Mar 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 1 This question was intended as a straightforward start to the paper but the marks scored were surprisingly disappointing with the average being about two marks out of four."

Source: 9709_m17_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 12), Question 1

Coaching With Shuja

Question Paper Feb/Mar 2017 • Paper 1 (variant 12) • Q1

Find the set of values of k for which the equation $2x^2 + 3kx + k = 0$ has distinct real roots. [4]

Coaching With Shuja

Mark Scheme

Feb/Mar 2017 • Paper 1 (variant 12) • Q1

$(3k)^2 - 4 \times 2 \times k$ **M1** Attempt $b^2 - 4ac$

$9k^2 - 8k > 0$ soi **A1** Must involve correct inequality. Allow $9k^2 - 8k \geq 0$

$0, \frac{8}{9}$ soi **A1**

$k < 0, k > \frac{8}{9}$ or 0.889 **A1** Allow $(-\infty, 0), (\frac{8}{9}, \infty)$

Total: 4

Coaching With Shuja

Question 3

Feb/Mar 2017 | Paper 1 (variant 12)

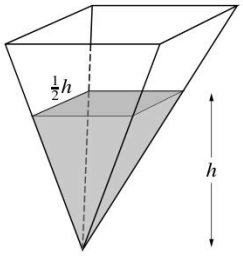
Examiner Report — Why students found this question difficult

> "A significant proportion of candidates applied the Chain Rule incorrectly but those who followed a systematic approach were usually successful."

Source: 9709_m17_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

3



The diagram shows a water container in the form of an inverted pyramid, which is such that when the height of the water level is h cm the surface of the water is a square of side $\frac{1}{2}h$ cm.

(i) Express the volume of water in the container in terms of h. [1]

The diagram shows a water container in the form of an inverted pyramid, which is such that when the height of the water level is h cm the surface of the water is a square of side $\frac{1}{2}h$ cm.

(i) Express the volume of water in the container in terms of h. [1]

[The volume of a pyramid having a base area A and vertical height h is $\frac{1}{3}Ah$.]

Water is steadily dripping into the container at a constant rate of $20\ \text{cm}^3$ per minute.

(ii) Find the rate, in cm per minute, at which the water level is rising when the height of the water level is 10 cm. [4]

(i)

$V = \frac{1}{12}h^3$ OE **B1**

Total: 1

(ii)

$\frac{dV}{dh} = \frac{1}{4}h^2$ or $\frac{dh}{dV} = 4(12v)^{-2/3}$ **M1A1** Attempt differentiation

$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{4}{h^2} \times 20$ soi **DM1** Use chain rule correctly with $\frac{dV}{dt} = 20$

$\left(\frac{dh}{dt}\right) = \frac{4}{10^2} \times 20 = 0.8$ or equivalent fraction **A1**

Total: 4

Question 5

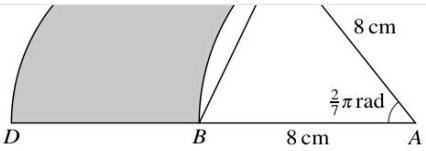
Feb/Mar 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (ii) was even less successful and candidates struggled both with accuracy and radians, as well as in identifying a strategy for finding the coordinates of B."

Source: 9709_m17_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 12), Question 5

Coaching With Shuja



In the diagram, $AB = AC = 8$ cm and angle $CAB = \frac{2}{7}\pi$ radians. The circular arc BC has centre A, the circular arc CD has centre B and ABD is a straight line.

(i) Show that angle $CBD = \frac{9}{14}\pi$ radians. [1]

The diagram shows the graphs of $y = \tan x$ and $y = \cos x$ for $0 \leq x \leq \pi$. The graphs intersect at points A and B.

(i) Find by calculation the x-coordinate of A. [4]

(ii) Find by calculation the coordinates of B. [3]

Coaching with Shuja

(i)

$\tan x = \cos x \rightarrow \sin x = \cos^2 x$ **M1** Use $\tan = \sin / \cos$ and multiply by $\cos$

$\sin x = 1 - \sin^2 x$ **M1** Use $\cos^2 x = 1 - \sin^2 x$

$\sin x = 0.618$; allow $\frac{-1+\sqrt{5}}{2}$ **M1** Attempt solution of quadratic in $\sin x$. Ignore solution -1.618

x-coord of $A = \sin^{-1} 0.618 = 0.666$ cao **A1** Must be radians. Accept 0.212π

Total: 4

(ii)

EITHER: x-coord of B is $\pi -$ their 0.666 **M1** Expect $2.475(3)$; must be radians throughout

y-coord of B is $\tan(\text{their } 2.475)$ or $\cos(\text{their } 2.475)$ **M1**

$x = 2.48,\ y = -0.786$ or -0.787 cao **A1** Accept $x = 0.788\pi$

OR: y-coord of B is $-(\cos$ or $\tan)(\text{their } 0.666)$ **M1**

x-coord of B is $\cos^{-1}(\text{their } y)$ or $\pi + \tan^{-1}(\text{their } y)$ **M1**

$x = 2.48,\ y = -0.786$ or -0.787 **A1** Accept $x = 0.788\pi$

Total: 3

Question 1

Feb/Mar 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 1 Many candidates produced a completely correct solution, however, some either failed to give their answer to 3 decimal places or rounded it incorrectly."

Source: 9709_m17_er.pdf, Page 7, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

Question Paper Feb/Mar 2017 • Paper 3 (variant 32) • Q1

Solve the equation $\ln(1+2^x)=2$, giving your answer correct to 3 decimal places. [3]

Coaching With Shuja

Mark Scheme

Feb/Mar 2017 • Paper 3 (variant 32) • Q1

Remove logarithm and obtain $1 + 2^x = e^2$ **B1**

Use correct method to solve an equation of the form $2^x = a$, where $a > 0$ **M1**

Obtain answer $x = 2.676$ **A1**

Total: 3

Question 4

Feb/Mar 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The common errors seen were the negative sign within the expression for α, so α = tan−1(−15/8), or a muddled tan−1, so α = tan−1(8/15) or the failure to give the value of α to 2 decimal places."

> "Answer: R = 17, α = 61.93° (ii) Many candidates successfully found the solution x = 7.2°, however, few candidates were able to find the second solution."

Source: 9709_m17_er.pdf, Page 8, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

(i) Express $8\cos\theta - 15\sin\theta$ in the form $R\cos(\theta + \alpha)$, where $R > 0$ and $0° < \alpha < 90°$, stating the exact value of R and giving the value of α correct to 2 decimal places. [3]

(ii) Hence solve the equation

$$8\cos 2x - 15\sin 2x = 4,$$

for $0° < x < 180°$. [4]

(i)

State $R = 17$ **B1**

Use trig formula to find α **M1**

Obtain $\alpha = 61.93°$ with no errors seen **A1**

Total: 3

(ii)

Evaluate $\cos^{-1}(4/17)$ to at least 1 d.p. ($76.39°$ to 2 d.p.) **B1**

Use a correct method to find a value of x in the interval $0° < x < 180°$ **M1**

Obtain answer, e.g. $x = 7.2°$ **A1**

Obtain second answer, e.g. $x = 110.8°$ and no others **A1**

Total: 4

Coaching with Shuja

Question 5

Feb/Mar 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 5 Most candidates were unable to progress far with this question, although usually the correct product rule was applied apart from some weak attempts at the differentiation of e–ax."

Source: 9709_m17_er.pdf, Page 8, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

Coaching With Shuja

Question Paper

Feb/Mar 2017 • Paper 3 (variant 32) • Q5

The curve with equation $y = e^{-ax} \tan x$, where a is a positive constant, has only one point in the interval $0 < x < \frac{1}{2}\pi$ at which the tangent is parallel to the x-axis. Find the value of a and state the exact value of the x-coordinate of this point. [7]

Coaching With Shuja

Mark Scheme

Use product rule **M1**

Obtain correct derivative in any form **A1**

Equate derivative to zero, use Pythagoras and obtain a quadratic equation in $\tan x$ **M1**

Obtain $\tan^2 x - a\tan x + 1 = 0$, or equivalent **A1**

Use the condition for a quadratic to have only one root **M1**

Obtain answer $a = 2$ **A1**

Obtain answer $x = \frac{1}{4}\pi$ **A1**

Total: 7

Coaching With Shuja

Question 7

Feb/Mar 2017 | Paper 3 (variant 32)

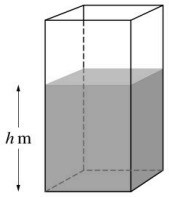
Examiner Report — Why students found this question difficult

> "Unfortunately a common error was the omission of the constant 10, despite the integration producing a lnu and a u term."

Source: 9709_m17_er.pdf, Page 9, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

7



A water tank has vertical sides and a horizontal rectangular base, as shown in the diagram. The area of the base is $2\,\text{m}^2$. At time $t = 0$ the tank is empty and water begins to flow into it at a rate of $1\,\text{m}^3$ per hour. At the same time water begins to flow out from the base at a rate of $0.2\sqrt{h}\,\text{m}^3$ per hour, where h m is the depth of water in the tank at time t hours.

(i) Form a differential equation satisfied by h and t, and show that the time T hours taken for the depth of water to reach 4 m is given by

$$T = \int_0^4 \frac{10}{5 - \sqrt{h}}\,\mathrm{d}h.$$ [3]

A water tank has vertical sides and a horizontal rectangular base, as shown in the diagram. The area of the base is $\mathbf{2\,m^2}$. At time $\boldsymbol{t = 0}$ the tank is empty and water begins to flow into it at a rate of $\mathbf{1\,m^3}$ per hour. At the same time water begins to flow out from the base at a rate of $\mathbf{0.2\sqrt{h}\,m^3}$ per hour, where $\boldsymbol{h}$ m is the depth of water in the tank at time $\boldsymbol{t}$ hours.

(i) Form a differential equation satisfied by $\boldsymbol{h}$ and $\boldsymbol{t}$, and show that the time $\boldsymbol{T}$ hours taken for the depth of water to reach $\mathbf{4}$ m is given by

$$\boldsymbol{T = \int_0^4 \frac{10}{5 - \sqrt{h}}\,dh.}$$

[3]

(ii) Using the substitution $\boldsymbol{u = 5 - \sqrt{h}}$, find the value of $\boldsymbol{T}$. [6]

(i)

State or imply $\frac{dV}{dt} = 2\frac{dh}{dt}$ **B1**

State or imply $\frac{dV}{dt} = 1 - 0.2\sqrt{h}$ **B1**

Obtain the given answer correctly **B1**

Total: 3

(ii)

State or imply $\frac{du}{dh} = -\frac{1}{2\sqrt{h}}$, or equivalent **B1**

Substitute for h and dh throughout **M1**

Obtain $T = \int_3^5 \frac{20(5-u)}{u}\, du$, or equivalent **A1**

Integrate and obtain terms $100\ln u - 20u$, or equivalent **A1**

Substitute limits $u = 3$ and $u = 5$ correctly **M1**

Obtain answer **11.1**, with no errors seen **A1**

Total: 6

Coaching with Shuja

Question 8

Feb/Mar 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "(ii) Candidates found this section difficult, since many failed to use the fact that another root of p(z) was the conjugate of u."

Source: 9709_m17_er.pdf, Page 9, Section: Comments on specific questions – Paper 3 (variant 32), Question 8

Coaching With Shuja

Question Paper

Feb/Mar 2017 • Paper 3 (variant 32) • Q8

Throughout this question the use of a calculator is not permitted.

The polynomial $z^4 + 3z^2 + 6z + 10$ is denoted by $p(z)$. The complex number $-1 + i$ is denoted by u.

(i) Showing all your working, verify that u is a root of the equation $p(z) = 0$. [3]

(ii) Find the other three roots of the equation $p(z) = 0$. [7]

Coaching With Shuja

(i)

Substitute $z = -1 + i$ and attempt expansions of the z^2 and z^4 terms **M1**

Use $i^2 = -1$ at least once **M1**

Complete the verification correctly **A1**

Total: 3

(ii)

State second root $z = -1 - i$ **B1**

Carry out a complete method for finding a quadratic factor with zeros $-1 + i$ and $-1 - i$ **M1**

Obtain $z^2 + 2z + 2$, or equivalent **A1**

Attempt division of $p(z)$ by $z^2 + 2z + 2$ and reach a partial quotient $z^2 + kz$ **M1**

Obtain quadratic factor $z^2 - 2z + 5$ **A1**

Solve 3-term quadratic and use $i^2 = -1$ **M1**

Obtain roots $1 + 2i$ and $1 - 2i$ **A1**

Total: 7

Coaching with Shuja

Question 6

May/June 2017 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Question 6 (i) This proved to be an unexpectedly difficult question."

Source: 9709_s17_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 6

Coaching With Shuja

The horizontal base of a solid prism is an equilateral triangle of side x cm. The sides of the prism are vertical. The height of the prism is h cm and the volume of the prism is $2000\ \text{cm}^3$.

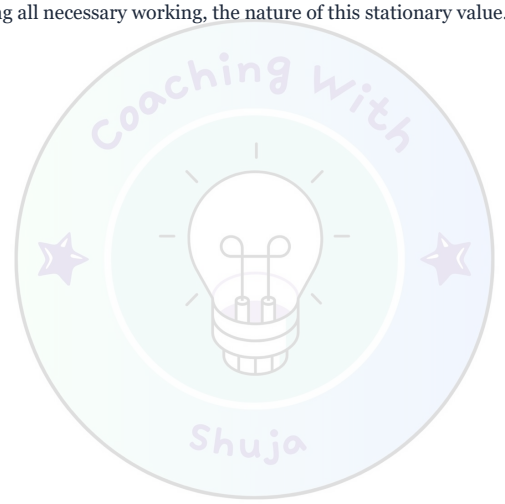
(i) Express h in terms of x and show that the total surface area of the prism, $A\ \text{cm}^2$, is given by

$$A = \frac{\sqrt{3}}{2}x^2 + \frac{24000}{\sqrt{3}}x^{-1}.$$

[3]

(ii) Given that x can vary, find the value of x for which A has a stationary value. [3]

(iii) Determine, showing all necessary working, the nature of this stationary value. [2]



(i)

Use

$$\left(\frac{1}{2}x \cdot \frac{\sqrt{3}}{2}x\right)h = 2000$$

to obtain

$$h = \frac{8000}{\sqrt{3}x^2}$$

M1

Use

$$A = 3xh + 2\left(\frac{1}{2}x \cdot \frac{\sqrt{3}}{2}x\right)$$

M1

Substitute for h to obtain

$$A = \frac{\sqrt{3}}{2}x^2 + \frac{24000}{\sqrt{3}}x^{-1}$$

A1

Total: 3

(ii)

Obtain

$$\frac{dA}{dx} = \sqrt{3}x - \frac{24000}{\sqrt{3}}x^{-2}$$

B1

Set $\frac{dA}{dx} = 0$ and attempt to solve for x **M1**

Obtain $x = 20$ **A1**

Total: 3

(iii)

Obtain or use

$$\frac{d^2A}{dx^2} = \sqrt{3} + \frac{48000}{\sqrt{3}}x^{-3} > 0$$

M1

Conclude that the stationary value is a minimum **A1**

Total: 2

Question 9

May/June 2017 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Although the correct result could be reached by considering a flowchart and the associated inverse operations, few candidates who selected this method could reach a correct solution."

Source: 9709_s17_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 9

Coaching With Shuja

The function f is defined by

$$f : x \mapsto \frac{2}{3-2x} \quad \text{for } x \in \mathbb{R},\ x \neq \frac{3}{2}.$$

(i) Find an expression for $f^{-1}(x)$. [3]

The function g is defined by

$$g : x \mapsto 4x + a \quad \text{for } x \in \mathbb{R},$$

where a is a constant.

(ii) Find the value of a for which $gf(-1) = 3$. [3]

(iii) Find the possible values of a given that the equation $f^{-1}(x) = g^{-1}(x)$ has two equal roots. [4]

Coaching with Shuja

(i)

Start with

$$y = \frac{2}{3-2x}$$

and rearrange, e.g.

$$y(3-2x) = 2, \qquad 2x = 3 - \frac{2}{y}$$

M1

Use the correct order of operations **M1**

Obtain

$$f^{-1}(x) = \frac{3}{2} - \frac{1}{x}$$

A1

Total: 3

(ii)

Use $gf(-1) = 3$ and obtain $f(-1) = \frac{2}{5}$ **M1**

Form an equation in a, e.g.

$$\frac{8}{5} + a = 3$$

M1

Obtain $a = \frac{7}{5}$ **A1**

Total: 3

(iii)

Find

$$g^{-1}(x) = \frac{x-a}{4}$$

and equate this to $f^{-1}(x)$ **M1**

Obtain a quadratic with a in a coefficient, e.g.

$$x^2 - (6+a)x + 4 = 0$$

M1

Use the discriminant condition for equal roots and solve a three-term quadratic, e.g.

$$(a+6)^2 - 16 = 0$$

M1

Obtain

$$a = -2 \quad \text{or} \quad a = -10$$

A1

Total: 4

Question 1

May/June 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (ii) 1 proved to be more of a challenge for weaker candidates with some failing to realise that the coefficient of x in the original expansion was now needed so that it could be multiplied by 3x2 and a combined coefficient of x found."

Source: 9709_s17_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 1

Coaching With Shuja

(i) Find the coefficient of x in the expansion of

$$\left(2x - \frac{1}{x}\right)^5.$$

[2]

(ii) Hence find the coefficient of x in the expansion of

$$(1 + 3x^2)\left(2x - \frac{1}{x}\right)^5.$$

[4]

(i)

Obtain coefficient of x as **80 B2**

Total: 2

(ii)

Obtain coefficient of $\frac{1}{x}$ in

$$\left(2x - \frac{1}{x}\right)^5$$

as **-40 B2**

Link the appropriate two terms only:

$$(1 \times 80) + (3 \times (-40))$$

M1

Obtain coefficient of x as **-40 A1**

Total: 4

Coaching with Shuja

Question 3

May/June 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 3 Part (i) proved to be a real challenge for many candidates and some very poor algebraic manipulation was seen, such as simply squaring each term rather than fully expanding both brackets."

> "Many candidates though, did make some progress, but the final step of factorising the numerator to $(1-\sin\theta)(1-\sin\theta)$ rather than the more usual $(\sin\theta-1)(\sin\theta-1)$ proved too difficult for many."

> "A very small minority missed this part of the question out if they had failed to prove the first part or failed to use the given result and made no progress."

Source: 9709_s17_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

(i) Prove the identity

$$\left(\frac{1}{\cos\theta} - \tan\theta\right)^2 \equiv \frac{1-\sin\theta}{1+\sin\theta}.$$

[3]

(ii) Hence solve the equation

$$\left(\frac{1}{\cos\theta} - \tan\theta\right)^2 = \frac{1}{2},$$

for $0° \leq \theta \leq 360°$. [3]

Coaching With Shuja

(i)

Eliminate $\tan\theta$ by replacing it with $\frac{\sin\theta}{\cos\theta}$, leading to a function of $\sin\theta$ and/or $\cos\theta$ only **M1**

Use $\sin^2\theta + \cos^2\theta = 1$, leading to a function of $\sin\theta$ only **M1**

Show the factorisation step and obtain

$$\frac{(1-\sin\theta)^2}{1-\sin^2\theta} = \frac{1-\sin\theta}{1+\sin\theta}$$

A1

Total: 3

(ii)

Use part (i) to obtain

$$2 - 2\sin\theta = 1 + \sin\theta,$$

so that $\sin\theta = \frac{1}{3}$ **M1**

Obtain $\theta = 19.5°$ **A1**

Obtain $\theta = 160.5°$ and no extra answers in the range **A1**

Total: 3

Coaching With Shuja

Question 5

May/June 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Some weaker candidates failed to use the equation of a straight line for the tangent and instead integrated."

Source: 9709_s17_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 5

Coaching With Shuja

A curve has equation

$$y = 3 + \frac{12}{2 - x}.$$

(i) Find the equation of the tangent to the curve at the point where the curve crosses the x-axis. [5]

(ii) A point moves along the curve in such a way that the x-coordinate is increasing at a constant rate of 0.04 units per second. Find the rate of change of the y-coordinate when $x = 4$. [2]

Coaching With Shuja

(i)

Find that the curve crosses the x-axis at $(6, 0)$ **B1**

Differentiate to obtain

$$\frac{dy}{dx} = 12(2 - x)^{-2}$$

B2

Use $\frac{dy}{dx}$ at $x = 6$ and the correct form of a straight-line equation **M1**

Obtain

$$y = \frac{3}{4}x - \frac{9}{2},$$

or equivalent **A1**

Total: 5

(ii)

At $x = 4$, use $\frac{dy}{dx} = 3$ and multiply by $\frac{dx}{dt} = 0.04$ **M1**

Obtain

$$\frac{dy}{dt} = 0.12$$

A1

Total: 2

Coaching with Shuja

Question 8

May/June 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> “Part (ii) proved to be much more difficult.”

Source: 9709_s17_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

Coaching With Shuja

Relative to an origin O, the position vectors of three points A, B and C are given by

$$\overrightarrow{OA} = 3\mathbf{i} + p\mathbf{j} - 2p\mathbf{k},$$

$$\overrightarrow{OB} = 6\mathbf{i} + (p+4)\mathbf{j} + 3\mathbf{k}$$

and

$$\overrightarrow{OC} = (p-1)\mathbf{i} + 2\mathbf{j} + q\mathbf{k},$$

where p and q are constants.

(i) In the case where $p = 2$, use a scalar product to find angle AOB. [4]

(ii) In the case where $\overrightarrow{AB}$ is parallel to $\overrightarrow{OC}$, find the values of p and q. [4]

Coaching With Shuja

(i)

Use scalar product correctly, e.g.

$$3 \times 6 + 2 \times 6 + (-4) \times 3 = 18$$

M1

Use a correct method for a modulus, obtaining $|\overrightarrow{OA}| = \sqrt{29}$ and $|\overrightarrow{OB}| = 9$ **M1**

Link correctly:

Math input error

M1

Obtain

Math input error

or **1.19** radians **A1**

Total: 4

(ii)

Use

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 3\mathbf{i} + 4\mathbf{j} + (3 + 2p)\mathbf{k}$$

M1

Compare components, with $\overrightarrow{OC}$ containing p and q **M1**

Obtain $p = \frac{5}{2}$ **A1**

Obtain $q = 4$ **A1**

Total: 4

Question 9

May/June 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> “Part (iv) proved to be difficult for all but the best candidates.”

Source: 9709_s17_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 9

Coaching With Shuja

Question Paper May/June 2017 • Paper 1 (variant 12) • Q9

The equation of a curve is

$$y = 8\sqrt{x} - 2x.$$

(i) Find the coordinates of the stationary point of the curve. [3]

(ii) Find an expression for $\frac{d^2y}{dx^2}$ and hence, or otherwise, determine the nature of the stationary point. [2]

(iii) Find the values of x at which the line $y = 6$ meets the curve. [3]

(iv) State the set of values of k for which the line $y = k$ does not meet the curve. [1]

(i)

Obtain

$$\frac{dy}{dx} = 4x^{-1/2} - 2$$

B1

Set this to zero to obtain $\sqrt{x} = 2$ **B1**

Obtain stationary point $(4, 8)$ **B1**

Total: 3

(ii)

Obtain

$$\frac{d^2y}{dx^2} = -2x^{-3/2}$$

B1

Evaluate at $x = 4$ to get $-\frac{1}{4}$, and conclude the stationary point is a maximum **B1**

Total: 2

(iii)

Recognise a quadratic in $\sqrt{x}$, for example by setting $u = \sqrt{x}$ to obtain

$$2u^2 - 8u + 6 = 0$$

M1

Obtain $u = 1$ and $u = 3$ **A1**

Obtain

$$x = 1 \quad \text{and} \quad x = 9$$

A1

Total: 3

(iv)

State

$$k > 8$$

B1

Total: 1

Question 10

May/June 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 10 Part (i) of this question proved more challenging than expected."

> "Most were aware that the inverse function would be a reflection of the original function in the line y = x, it was drawing the original function which proved to be the challenge."

Source: 9709_s17_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

Coaching With Shuja

The function f is defined by

$$f(x) = 3\tan\left(\frac{1}{2}x\right) - 2, \quad -\frac{1}{2}\pi \leq x \leq \frac{1}{2}\pi.$$

(i) Solve the equation $f(x) + 4 = 0$, giving your answer correct to 1 decimal place. [3]

(ii) Find an expression for $f^{-1}(x)$ and find the domain of f^{-1}. [5]

(iii) Sketch, on the same diagram, the graphs of $y = f(x)$ and $y = f^{-1}(x)$. [3]

(i)

Attempt to obtain

$$\tan\left(\frac{1}{2}x\right) = k$$

from

$$3\tan\left(\frac{1}{2}x\right) + 2 = 0$$

M1

Use inverse tangent, giving $\frac{1}{2}x = -0.588\ldots$ **M1**

Obtain

$$x = -1.2$$

A1

Total: 3

(ii)

Start from

$$\frac{y+2}{3} = \tan\left(\frac{1}{2}x\right)$$

M1

Apply inverse tangent and multiply by **2** **M1**

Obtain

$$f^{-1}(x) = 2\tan^{-1}\left(\frac{x+2}{3}\right)$$

A1

State the domain of f^{-1} as

$$-5 \leq x \leq 1$$

B2

Total: 5

(iii)

Sketch a tangent graph for $y = f(x)$ through the first, third and fourth quadrants **B1**

Sketch an inverse tangent graph for $y = f^{-1}(x)$ through the first, second and third quadrants **B1**

Show the two curves clearly symmetrical about $y = x$, approximately in the correct domain and range and not intersecting **B1**

Total: 3

Question 8

May/June 2017 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "In part (ii) a significant proportion of candidates first tried approaches involving equations of lines, before many, after making no progress, resorted to a method based on equating the distance between (a, b) and (–1, 1) with the distance between (–1, 1) and (10, –1) and then substituting for b using the relationship found in part (i)."

Source: 9709_s17_er.pdf, Page 11, Section: Comments on specific questions – Paper 1 (variant 13), Question 8

Coaching With Shuja

Question Paper May/June 2017 • Paper 1 (variant 13) • Q8

$A(-1, 1)$ and $P(a, b)$ are two points, where a and b are constants. The gradient of AP is 2.

(i) Find an expression for b in terms of a. [2]

$B(10, -1)$ is a third point such that $AP = AB$.

(ii) Calculate the coordinates of the possible positions of P. [6]

(i)

Use

$$\frac{b-1}{a+1} = 2$$

M1

Obtain

$$b = 2a + 3$$

A1

Total: 2

(ii)

Obtain

$$AB^2 = 11^2 + (-2)^2 = 125$$

B1

Use

$$(a+1)^2 + (b-1)^2 = 125$$

B1

Substitute $b = 2a + 3$ to form a quadratic equation in a **M1**

Simplify and attempt to solve, e.g.

$$5a^2 + 10a - 120 = 0$$

M1

Obtain $a = 4$ or $a = -6$ **A1**

Obtain $b = 11$ or $b = -9$, giving possible positions

$$(4, 11) \quad \text{and} \quad (-6, -9)$$

A1

Total: 6

Question 9

May/June 2017 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Question 9 In part (i) many candidates did not read the question carefully enough, as reported in the Key Messages section above, and did not present their answers in the required form."

> "In part (ii) most candidates realised that 1/3 was the value of p required but did not always express the answer perfectly."

Source: 9709_s17_er.pdf, Page 11, Section: Comments on specific questions – Paper 1 (variant 13), Question 9

Coaching With Shuja

(i) Express $9x^2 - 6x + 6$ in the form $(ax + b)^2 + c$, where a, b and c are constants. [3]

The function f is defined by

$$f(x) = 9x^2 - 6x + 6 \quad \text{for } x \geq p,$$

where p is a constant.

(ii) State the smallest value of p for which f is a one-one function. [1]

(iii) For this value of p, obtain an expression for $f^{-1}(x)$, and state the domain of f^{-1}. [4]

(iv) State the set of values of q for which the equation $f(x) = q$ has no solution. [1]

Coaching With Shuja

(i)

Obtain

$$9x^2 - 6x + 6 = (3x - 1)^2 + 5$$

B3

Total: 3

(ii)

State the smallest value

$$p = \frac{1}{3}$$

B1

Total: 1

(iii)

Start from

$$y = (3x - 1)^2 + 5$$

and obtain

$$3x - 1 = \pm\sqrt{y - 5}$$

B1

Rearrange to

$$x = \frac{1 \pm \sqrt{y - 5}}{3}$$

B1

Use the restricted domain to remove the negative branch and obtain

$$f^{-1}(x) = \frac{1 + \sqrt{x - 5}}{3}$$

B1

State the domain of f^{-1} as

$$x \geq 5$$

B1

Total: 4

(iv)

State

$$q < 5$$

B1

Total: 1

Question 10

May/June 2017 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "A greater proportion of candidates, however, opted for the more difficult route which had a number of pitfalls on the way."

Source: 9709_s17_er.pdf, Page 12, Section: Comments on specific questions – Paper 1 (variant 13), Question 10

Coaching With Shuja

(a) Fig. 1 shows part of the curve $y = x^2 - 1$ and the line $y = h$, where h is a constant.

(i) The shaded region is rotated through $360°$ about the y-axis. Show that the volume of revolution, V, is given by

$$V = \pi\left(\frac{1}{2}h^2 + h\right).$$

[3]

(ii) Find, showing all necessary working, the area of the shaded region when $h = 3$. [4]

(b) Fig. 2 shows a cross-section of a bowl containing water. When the height of the water level is h cm, the volume, V cm^3, of water is given by

$$V = \pi\left(\frac{1}{2}h^2 + h\right).$$

Water is poured into the bowl at a constant rate of 2 cm^3 s^{-1}. Find the rate, in cm s^{-1}, at which the height of the water level is increasing when the height of the water level is 3 cm. [4]

Coaching with Shuja

(a)(i)

Attempt to integrate with respect to y using

$$V = \pi \int (y+1)\,dy$$

M1

Obtain

$$V = \pi \left[\frac{y^2}{2} + y\right]$$

A1

Use limits **0** to h to obtain

$$V = \pi \left(\frac{1}{2}h^2 + h\right)$$

A1

Total: 3

(a)(ii)

Attempt to integrate the area with respect to y:

$$\int (y+1)^{1/2}\,dy$$

M1

Obtain

$$\frac{2}{3}(y+1)^{3/2}$$

A1

Use limits **0** to **3** **M1**

Obtain area

$$\frac{14}{3}$$

A1

Total: 4

(b)

Differentiate with respect to h **M1**

Obtain

$$\frac{dV}{dh} = \pi(h+1)$$

so that, when $h = 3$, $\frac{dV}{dh} = 4\pi$ **A1**

Use

$$\frac{dh}{dt} = \frac{dV/dt}{dV/dh}$$

with $\frac{dV}{dt} = 2$ **M1**

Obtain

$$\frac{dh}{dt} = \frac{1}{2\pi}$$

(or **0.159**) **A1**

Total: 4

Question 2

May/June 2017 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common error was to omit the 6 in (6x)2 and (6x)3 ."

Source: 9709_s17_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

Coaching With Shuja

Expand

$$(1+6x)^{-1/3}$$

in ascending powers of x, up to and including the term in x^3, simplifying the coefficients. [4]

State a correct unsimplified version of the x, x^2 or x^3 term in the expansion of $(1+6x)^{-1/3}$ **M1**

State correct first two terms

$$1-2x$$

A1

Obtain term

$$8x^2$$

A1

Obtain term

$$-\frac{112}{3}x^3$$

in the final answer **A1**

Total: 4

Coaching With Shuja

Question 3

May/June 2017 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "After correct substitution of limits, some candidates struggled to combine the two terms correctly."

Source: 9709_s17_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 31), Question 3

Coaching With Shuja

It is given that

$$x = \ln(1 - y) - \ln y,$$

where $0 < y < 1$.

(i) Show that

$$y = \frac{e^{-x}}{1 + e^{-x}}.$$

[2]

(ii) Hence show that

$$\int_0^1 y \, dx = \ln\left(\frac{2e}{e+1}\right).$$

[4]

Coaching With Shuja

(i)

Remove logarithms correctly and obtain

$$e^x = \frac{1-y}{y}$$

B1

Obtain the given answer

$$y = \frac{e^{-x}}{1+e^{-x}}$$

following full working **B1**

Total: 2

(ii)

State an integral of the form

$$k\ln(1+e^{-x})$$

where $k = \pm 1$ **M1**

State the correct integral

$$-\ln(1+e^{-x})$$

A1

Use limits correctly **M1**

Obtain the given answer

$$\ln\left(\frac{2e}{e+1}\right)$$

following full working **A1**

Total: 4

Question 4

May/June 2017 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “dy (ii) Many candidates did not connect the gradient of the normal being 1 with =−1 and therefore dx made no progress here.”

Source: 9709_s17_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

Coaching With Shuja

The parametric equations of a curve are

$$x = \ln\cos\theta,$$

$$y = 3\theta - \tan\theta,$$

where $0 \leq \theta < \frac{1}{2}\pi$.

(i) Express $\frac{dy}{dx}$ in terms of $\tan\theta$. [5]

(ii) Find the exact y-coordinate of the point on the curve at which the gradient of the normal is equal to 1. [3]

Coaching With Shuja

(i)

Use the chain rule to differentiate x, obtaining

$$\frac{dx}{d\theta} = -\tan\theta$$

M1

State

$$\frac{dy}{d\theta} = 3 - \sec^2\theta$$

B1

Use

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

M1

Obtain a correct expression in any form, e.g.

$$\frac{3 - \sec^2\theta}{-\tan\theta}$$

A1

Obtain

$$\frac{dy}{dx} = \frac{\tan^2\theta - 2}{\tan\theta}$$

A1

Total: 5

(ii)

Equate the tangent gradient to -1 and obtain an equation in $\tan\theta$ **M1**

Solve the quadratic

$$\tan^2\theta + \tan\theta - 2 = 0$$

M1

Obtain $\theta = \frac{\pi}{4}$ and hence

$$y = \frac{3\pi}{4} - 1$$

A1

Total: 3

Question 5

May/June 2017 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "© 2017 Cambridge International Advanced Level 9709 Mathematics June 2017 Principal Examiner Report for Teachers (iii) Again, many candidates did not attempt this question, possibly because they simply did not continue after not completing the first part."

Source: 9709_s17_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

Coaching With Shuja

The diagram shows a semicircle with centre O, radius r and diameter AB. The point P on its circumference is such that the area of the minor segment on AP is equal to half the area of the minor segment on BP. The angle AOP is x radians.

(i) Show that x satisfies the equation

$$x = \frac{1}{3}(\pi + \sin x).$$

[3]

(ii) Verify by calculation that x lies between 1 and 1.5. [2]

(iii) Use an iterative formula based on the equation in part (i) to determine x correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

Coaching With Shuja

(i)

Use the correct sector formula at least once and form an equation in r and x **M1**

Obtain a correct equation in any form, e.g.

$$x - \sin x = \frac{1}{2}(\pi - x - \sin x)$$

A1

Rearrange to obtain

$$x = \frac{1}{3}(\pi + \sin x)$$

A1

Total: 3

(ii)

Calculate values of a relevant expression at $x = 1$ and $x = 1.5$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(iii)

Use the iterative formula, for example

$$x_{n+1} = \frac{1}{3}(\pi + \sin x_n),$$

correctly at least once **M1**

Obtain final answer **1.374** **A1**

Show sufficient iterations to 5 d.p. to justify **1.374** to 3 d.p., or show there is a sign change in the interval $(1.3745, 1.3755)$ **A1**

Total: 3

Question 6

May/June 2017 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(ii) This proved to be one of the most difficult questions on the paper."

> "Many candidates did not attempt to use the fact that the equation r.n = a.n gives you the distance of the plane from the origin if n is the unit normal vector."

> "A few candidates reached the correct answer, sometimes via quite difficult routes."

> "Many candidates did not appear to recognise that the equation of a parallel plane would be of the form 2x+2y −z =d Answer: (ii) 2x+2y −z =2"

Source: 9709_s17_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching with Shuja

The plane with equation

$$2x + 2y - z = 5$$

is denoted by m. Relative to the origin O, the points A and B have coordinates $(3, 4, 0)$ and $(-1, 0, 2)$ respectively.

(i) Show that the plane m bisects AB at right angles. [5]

A second plane p is parallel to m and nearer to O. The perpendicular distance between the planes is 1.

(ii) Find the equation of p, giving your answer in the form $ax + by + cz = d$. [3]

(i)

State or obtain coordinates $(1, 2, 1)$ for the midpoint of AB **B1**

Verify that this midpoint lies on m **B1**

State or imply a correct normal vector to the plane, e.g.

$$2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$

B1

State or imply a direction vector for AB, e.g.

$$-4\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$$

B1

Confirm that m is perpendicular to AB **B1**

Total: 5

(ii)

State or imply that the perpendicular distance of m from the origin is $\frac{5}{3}$, or an unsimplified equivalent **B1**

State or imply that p has an equation of the form

$$2x + 2y - z = k$$

B1

Obtain

$$2x + 2y - z = 2$$

B1

Total: 3

Question 7

May/June 2017 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Some marks were lost needlessly through arithmetic errors and poor use of brackets."

> "Some candidates lost marks because they used the trigonometry of a right angled triangle without demonstrating that they had a right angle triangle."

Source: 9709_s17_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

Question Paper May/June 2017 • Paper 3 (variant 31) • Q7

Throughout this question the use of a calculator is not permitted.

The complex numbers u and w are defined by

$$u = -1 + 7i \quad \text{and} \quad w = 3 + 4i.$$

(i) Showing all your working, find in the form $x + iy$, where x and y are real, the complex numbers $u - 2w$ and $\frac{u}{w}$. [4]

In an Argand diagram with origin O, the points A, B and C represent the complex numbers u, w and $u - 2w$ respectively.

(ii) Prove that angle $AOB = \frac{1}{4}\pi$. [2]

(iii) State fully the geometrical relation between the line segments OB and CA. [2]

Coaching With Shuja

(i)

State

$$u - 2w = -7 - i$$

B1

Multiply numerator and denominator of $\frac{u}{w}$ by $3 - 4i$, or equivalent **M1**

Simplify the numerator to $25 + 25i$ or the denominator to 25 **A1**

Obtain

$$\frac{u}{w} = 1 + i$$

A1

Total: 4

(ii)

Find the argument of $\frac{u}{w}$ **M1**

Since Math input error, obtain the given answer **A1**

Total: 2

(iii)

State that OB and CA are parallel **B1**

State that $CA = 2OB$, or equivalent **B1**

Total: 2

Coaching With Shuja

Question 1

May/June 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The false statement ln(x2 + 1)=lnx2 +ln1 was a common error."

Source: 9709_s17_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

Question Paper May/June 2017 • Paper 3 (variant 32) • Q1

Solve the equation

$$\ln(x^2+1) = 1 + 2\ln x,$$

giving your answer correct to 3 significant figures. [3]

Coaching With Shuja

Use a law of logarithms for a power or a quotient **M1**

Remove logarithms and obtain a correct equation in x, e.g.

$$x^2 + 1 = ex^2$$

A1

Obtain answer $x = 0.763$ and no other **A1**

Total: 3

Question 2

May/June 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Common errors included slips in the arithmetic, errors in squaring 2 brackets, or squaring only the modular side of the inequality."

Source: 9709_s17_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Coaching With Shuja

Solve the inequality

$$|x - 3| < 3x - 4.$$

[4]

Coaching With Shuja

Either state or imply the non-modular inequality

$$(x-3)^2 < (3x-4)^2,$$

or the corresponding equation **B1**

Make a reasonable attempt at solving a three-term quadratic **M1**

Obtain critical value

$$x = \frac{7}{4}$$

A1

State final answer

$$x > \frac{7}{4}$$

only **A1**

Total: 4

Question 3

May/June 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors were slips in the arithmetic and algebra, which prevented several candidates from reaching a correct final answer."

> "Many candidates considered only the positive square root and consequently did not find any solution in the given range."

Source: 9709_s17_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 32), Question 3

(i) Express the equation

$$\cot\theta - 2\tan\theta = \sin 2\theta$$

in the form

$$a\cos^4\theta + b\cos^2\theta + c = 0,$$

where a, b and c are constants to be determined. [3]

(ii) Hence solve the equation

$$\cot\theta - 2\tan\theta = \sin 2\theta$$

for $90° < \theta < 180°$. [2]

Coaching With Shuja

(i)

Use correct formulae to express the equation in terms of $\cos\theta$ and $\sin\theta$ **M1**

Use Pythagoras and express the equation in terms of $\cos\theta$ only **M1**

Obtain a correct three-term equation, e.g.

$$2\cos^4\theta + \cos^2\theta - 2 = 0$$

A1

Total: 3

(ii)

Solve a three-term quadratic in $\cos^2\theta$ for $\cos\theta$ **M1**

Obtain answer

$$\theta = 152.1°$$

only **A1**

Total: 2

Coaching With Shuja

Question 4

May/June 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common error dx d 1 was to obtain ln(2t−1)= ."

Source: 9709_s17_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

The parametric equations of a curve are

$$x = t^2 + 1,$$

$$y = 4t + \ln(2t - 1).$$

(i) Express $\frac{dy}{dx}$ in terms of t. [3]

(ii) Find the equation of the normal to the curve at the point where $t = 1$. Give your answer in the form $ax + by + c = 0$. [3]

(i)

State

$$\frac{dy}{dt} = 4 + \frac{2}{2t-1}$$

B1

Use

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

M1

Obtain

$$\frac{dy}{dx} = \frac{8t-2}{2t(2t-1)},$$

or equivalent **A1**

Total: 3

(ii)

Use a correct method to find the gradient of the normal at $t = 1$ **M1**

Use a correct method to form an equation for the normal at $t = 1$ **M1**

Obtain

$$x + 3y - 14 = 0$$

A1

Total: 3

Question 5

May/June 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Several candidates never formed a differential equation in y and t and made no progress at all."

Source: 9709_s17_er.pdf, Page 30, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

Coaching With Shuja

In a certain chemical process a substance A reacts with and reduces a substance B. The masses of A and B at time t after the start of the process are x and y respectively. It is given that

$$\frac{dy}{dt} = -0.2xy$$

and

$$x = \frac{10}{(1+t)^2}.$$

At the beginning of the process $y = 100$.

(i) Form a differential equation in y and t, and solve this differential equation. [6]

(ii) Find the exact value approached by the mass of B as t becomes large. State what happens to the mass of A as t becomes large. [2]

Coaching With Shuja

(i)

State

$$\frac{dy}{dt} = -\frac{2y}{(1+t)^2}$$

B1

Separate variables correctly and attempt integration of one side **M1**

Obtain term $\ln y$, or equivalent **A1**

Obtain term

$$\frac{2}{1+t}$$

A1

Use $y = 100$ and $t = 0$ to evaluate a constant, or use limits, in an expression containing terms of the form $a \ln y$ and $\frac{b}{1+t}$ **M1**

Obtain a correct solution in any form, e.g.

$$\ln y = \frac{2}{1+t} - 2 + \ln 100$$

A1

Total: 6

(ii)

State that the mass of B approaches

$$\frac{100}{e^2}$$

B1

State or imply that the mass of A tends to zero **B1**

Total: 2

Question 7

May/June 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common error here was that candidates could not integrate $\sec\theta\tan\theta$ – they often offered a complicated expression involving logarithms."

Source: 9709_s17_er.pdf, Page 30, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

(i) Prove that if

$$y = \frac{1}{\cos\theta}$$

then

$$\frac{dy}{d\theta} = \sec\theta\tan\theta.$$

[2]

(ii) Prove the identity

$$\frac{1+\sin\theta}{1-\sin\theta} \equiv 2\sec^2\theta + 2\sec\theta\tan\theta - 1.$$

[3]

(iii) Hence find the exact value of

$$\int_0^{\frac{1}{4}\pi} \frac{1+\sin\theta}{1-\sin\theta}\, d\theta.$$

[4]

Coaching with Shuja

(i)

Use the quotient rule or chain rule **M1**

Obtain the given answer correctly **A1**

Total: 2

(ii)

Multiply numerator and denominator of the left hand side by $1 + \sin\theta$ **M1**

Use Pythagoras and express the result in terms of $\sec\theta$ and $\tan\theta$ **M1**

Complete the proof **A1**

Total: 3

(iii)

Use the identity and obtain the integral

$$\int (2\tan\theta + 2\sec\theta - 1)\, d\theta$$

B2

Use the correct limits in an integral containing terms $a\tan\theta$ and $b\sec\theta$ **M1**

Obtain

$$2\sqrt{2} - \frac{\pi}{4}$$

A1

Total: 4

Question 8

May/June 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common error was to mishandle the extracted factors of 2 and 5."

Source: 9709_s17_er.pdf, Page 31, Section: Comments on specific questions – Paper 3 (variant 32), Question 8

Coaching With Shuja

Let

$$f(x) = \frac{5x^2 - 7x + 4}{(3x + 2)(x^2 + 5)}.$$

(i) Express $f(x)$ in partial fractions. [5]

(ii) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. [5]

(i)

State or imply the form

$$\frac{A}{3x+2}+\frac{Bx+C}{x^2+5}$$

B1

Use a relevant method to determine a constant **M1**

Obtain one of $A=2$, $B=1$, $C=-3$ **A1**

Obtain a second value **A1**

Obtain the third value **A1**

Total: 5

(ii)

Use a correct method to find the first two terms of the expansion of $(3x+2)^{-1}$ or $(x^2+5)^{-1}$ **M1**

Obtain correct unsimplified expansions up to the term in x^2 of each partial fraction **A2**

Multiply out up to the term in x^2 by $Bx+C$, where $BC\neq 0$ **M1**

Obtain final answer

$$\frac{2}{5}-\frac{13}{10}x+\frac{237}{100}x^2$$

A1

Total: 5

Question 9

May/June 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 9 (i) Although very straight forward, this question was not similar to anything that the candidates had seen on recent papers, and many of them struggled to find a correct approach."

> "Many candidates made no attempt to find the position vector of the reflection of A."

> "The most common error in applying the formula for distance of a point from a plane was to use the modulus of i+2j+4k in the denominator rather than the modulus of their normal vector."

Source: 9709_s17_er.pdf, Page 31, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Relative to the origin O, the point A has position vector

$$\overrightarrow{OA} = \mathbf{i} + 2\mathbf{j} + 4\mathbf{k}.$$

The line l has equation

$$\mathbf{r} = 9\mathbf{i} - \mathbf{j} + 8\mathbf{k} + \lambda(3\mathbf{i} - \mathbf{j} + 2\mathbf{k}).$$

(i) Find the position vector of the foot of the perpendicular from A to l. Hence find the position vector of the reflection of A in l. [5]

(ii) Find the equation of the plane through the origin which contains l. Give your answer in the form $ax + by + cz = d$. [3]

(iii) Find the exact value of the perpendicular distance of A from this plane. [3]

Coaching with Shuja

(i)

Find $\overrightarrow{AP}$ for a general point P on l with parameter λ, e.g.

$$(8+3\lambda)\mathbf{i}+(-3-\lambda)\mathbf{j}+(4+2\lambda)\mathbf{k}$$

B1

Equate the scalar product of $\overrightarrow{AP}$ and the direction vector of l to zero and solve for λ **M1**

Obtain $\lambda=-\frac{5}{2}$ and foot of perpendicular

$$\frac{3}{2}\mathbf{i}+\frac{3}{2}\mathbf{j}+3\mathbf{k}$$

A1

Carry out a complete method for finding the position vector of the reflection of A in l **M1**

Obtain

$$2\mathbf{i}+\mathbf{j}+2\mathbf{k}$$

A1

Total: 5

(ii)

Use a scalar product to obtain an equation in a, b and c, e.g.

$$3a-b+2c=0$$

B1

Form a second relevant equation, e.g.

$$9a-b+8c=0,$$

and solve for one ratio **M1**

Obtain $a:b:c=1:1:-1$ and state plane equation

$$x+y-z=0$$

A1

Total: 3

(iii)

Apply the perpendicular distance formula using $A(1,2,4)$ and plane $x+y-z=0$ **M1**

Obtain a correct expression, e.g.

$$\frac{|1+2-4|}{\sqrt{1^2+1^2+(-1)^2}}$$

A1

Obtain

$$\frac{1}{\sqrt{3}}$$

A1

Total: 3

Question 10

May/June 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors were to 1 see 2 in place of , or sign errors."

Source: 9709_s17_er.pdf, Page 32, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

The diagram shows the curve

$$y = x^2 \cos 2x$$

for $0 \leq x \leq \frac{1}{4}\pi$. The curve has a maximum point at M where $x = p$.

(i) Show that p satisfies the equation

$$p = \frac{1}{2}\tan^{-1}\left(\frac{1}{p}\right).$$

[3]

(ii) Use the iterative formula

$$p_{n+1} = \frac{1}{2}\tan^{-1}\left(\frac{1}{p_n}\right)$$

to determine the value of p correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

(iii) Find, showing all necessary working, the exact area of the region bounded by the curve and the x-axis. [5]

Coaching With Shuja

(i)

Use the product rule correctly **M1**

Obtain a correct derivative in any form, e.g.

$$y' = 2x\cos 2x - 2x^2 \sin 2x$$

A1

Equate to zero and derive the given equation **A1**

Total: 3

(ii)

Use the iterative formula correctly at least once, e.g.

$$0.5 \to 0.55357 \to 0.53261 \to 0.54070 \to 0.53755$$

M1

Obtain final answer **0.54 A1**

Show sufficient iterations to 4 d.p. to justify **0.54** to 2 d.p., or show there is a sign change in the interval **(0.535, 0.545) A1**

Total: 3

(iii)

Integrate by parts and reach an expression of the form

$$ax^2 \sin 2x + b\int x \sin 2x\, dx$$

M1

Obtain

$$\frac{1}{2}x^2 \sin 2x - \frac{1}{2}\int x \sin 2x\, dx$$

A1

Complete the integration and obtain

$$\frac{1}{2}x^2 \sin 2x + \frac{1}{2}x\cos 2x - \frac{1}{4}\sin 2x,$$

or equivalent **A1**

Substitute limits $x = 0$ and $x = \frac{1}{4}\pi$, having integrated twice **M1**

Obtain

$$\frac{\pi^2 - 8}{32}$$

A1

Total: 5

Question 3

May/June 2017 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "It was disappointing to see some candidates going straight from their quadratic equation to the solution, hence risking losing the method mark should any error occur."

Source: 9709_s17_er.pdf, Page 34, Section: Comments on specific questions – Paper 3 (variant 33), Question 3

Coaching With Shuja

Question Paper May/June 2017 • Paper 3 (variant 33) • Q3

Using the substitution $u = e^x$, solve the equation

$$4e^{-x} = 3e^x + 4.$$

Give your answer correct to 3 significant figures. [4]

Coaching With Shuja

Rearrange as

$$3u^2 + 4u - 4 = 0,$$

or

$$3e^{2x} + 4e^x - 4 = 0,$$

or equivalent **B1**

Solve a three-term quadratic for e^x or for u **M1**

Obtain

$$e^x = \frac{2}{3}$$

A1

Obtain answer $x = -0.405$ and no other **A1**

Total: 4

Coaching With Shuja

Question 5

May/June 2017 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Question 5 (i) Although many candidates could differentiate the ln function correctly using the chain rule, either by finding the derivative of 1 + 3cos2x or by converting the expression into one involving cos 2x, many found it challenging to progress to an expression in tan x as the question demanded."

> "© 2017 Cambridge International Advanced Level 9709 Syllabus June 2017 Principal Examiner Report for Teachers (ii) Few candidates had a suitable expression from (i) to enable them to proceed to solving a 3-term quadratic in this part, although it was quite possible to solve from the initial correct derivative in terms of sin x and cos x."

Source: 9709_s17_er.pdf, Page 34, Section: Comments on specific questions – Paper 3 (variant 33), Question 5

Coaching With Shuja

A curve has equation

$$y = \frac{2}{3}\ln(1 + 3\cos^2 x)$$

for $0 \leq x \leq \frac{1}{2}\pi$.

(i) Express $\frac{dy}{dx}$ in terms of $\tan x$. [4]

(ii) Hence find the x-coordinate of the point on the curve where the gradient is -1. Give your answer correct to 3 significant figures. [2]

Coaching With Shuja

(i)

Use the chain rule **M1**

Obtain a correct derivative in any form **A1**

Use correct trigonometry to express the derivative in terms of $\tan x$ **M1**

Obtain

$$\frac{dy}{dx} = \frac{-4\tan x}{4+\tan^2 x},$$

or equivalent **A1**

Total: 4

(ii)

Equate derivative to -1 and solve a three-term quadratic for $\tan x$ **M1**

Obtain answer $x = 1.11$ and no other in the given interval **A1**

Total: 2

Coaching With Shuja

Question 6

May/June 2017 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "In fact a considerable number of candidates made no attempt at this section or at (ii)."

Source: 9709_s17_er.pdf, Page 35, Section: Comments on specific questions – Paper 3 (variant 33), Question 6

Coaching With Shuja

The equation

$$\cot x = 1 - x$$

has one root in the interval $0 < x < \pi$, denoted by ω.

(i) Show by calculation that ω is greater than 2.5. [2]

(ii) Show that, if a sequence of values in the interval $0 < x < \pi$ given by the iterative formula

$$x_{n+1} = \pi + \tan^{-1}\left(\frac{1}{1 - x_n}\right)$$

converges, then it converges to ω. [2]

(iii) Use this iterative formula to determine ω correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

Coaching With Shuja

(i)

Calculate the value of a relevant expression at $x = 2.5$ and at another relevant value, e.g. $x = 3$ **M1**

Complete the argument correctly with correct calculated values **A1**

Total: 2

(ii)

State a suitable equation, e.g.

$$\tan^{-1}\left(\frac{1}{1-x}\right) = x - \pi$$

B1

Rearrange this as

$$\cot x = 1 - x,$$

or commence working vice versa **B1**

Total: 2

(iii)

Use the iterative formula correctly at least once **M1**

Obtain final answer **2.576** only **A1**

Show sufficient iterations to 5 d.p. to justify **2.576** to 3 d.p., or show there is a sign change in the interval $(2.5755, 2.5765)$ **A1**

Total: 3

Coaching With Shuja

Question 7

May/June 2017 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "(iii) Few candidates were able to explain whether the estimate with two intervals would be greater than or less than the estimate with four intervals."

Source: 9709_s17_er.pdf, Page 35, Section: Comments on specific questions – Paper 3 (variant 33), Question 7

Coaching With Shuja

The diagram shows a sketch of the curve

$$y = \frac{e^{x/2}}{x}$$

for $x > 0$, and its minimum point M.

(i) Find the x-coordinate of M. [4]

(ii) Use the trapezium rule with two intervals to estimate the value of

$$\int_1^3 \frac{e^{x/2}}{x}\, dx,$$

giving your answer correct to 2 decimal places. [3]

(iii) The estimate found in part (ii) is denoted by E. Explain, without further calculation, whether another estimate found using the trapezium rule with four intervals would be greater than E or less than E. [1]

(i)

Use the correct quotient rule or product rule **M1**

Obtain a correct derivative in any form **A1**

Equate derivative to zero and solve for x **M1**

Obtain $x = 2$ **A1**

Total: 4

(ii)

State or imply ordinates $1.6487\ldots$, $1.3591\ldots$, $1.4938\ldots$ **B1**

Use the correct trapezium rule formula, or equivalent, with $h = 1$ and three ordinates **M1**

Obtain answer 2.93 only **A1**

Total: 3

(iii)

Explain why the estimate with four intervals would be less than E **B1**

Total: 1

Coaching With Shuja

Question 8

May/June 2017 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Only a few candidates were unable to separate the variables correctly."

Source: 9709_s17_er.pdf, Page 35, Section: Comments on specific questions – Paper 3 (variant 33), Question 8

Coaching With Shuja

Question Paper May/June 2017 • Paper 3 (variant 33) • Q8

In a certain chemical reaction, a compound A is formed from a compound B. The masses of A and B at time t after the start of the reaction are x and y respectively and the sum of the masses is equal to 50 throughout the reaction. At any time the rate of increase of the mass of A is proportional to the mass of B at that time.

(i) Explain why

$$\frac{dx}{dt} = k(50 - x),$$

where k is a constant. [1]

It is given that $x = 0$ when $t = 0$, and $x = 25$ when $t = 10$.

(ii) Solve the differential equation in part (i) and express x in terms of t. [8]

Coaching With Shuja

(i)

Justify the given differential equation **B1**

Total: 1

(ii)

Separate variables correctly and attempt to integrate one side **B1**

Obtain term kt, or equivalent **B1**

Obtain term $-\ln(50-x)$, or equivalent **B1**

Evaluate a constant, or use $x=0$, $t=0$ as limits in a solution containing terms $a\ln(50-x)$ and bt **M1**

Obtain solution

$$-\ln(50-x)=kt-\ln 50,$$

or equivalent **A1**

Use $x=25$, $t=10$ to determine k **M1**

Obtain correct solution in any form, e.g.

$$\ln 50-\ln(50-x)=\frac{t}{10}\ln 2$$

A1

Obtain

$$x=50\left(1-e^{-0.0693t}\right),$$

or equivalently

$$x=50\left(1-2^{-t/10}\right)$$

A1

Total: 8

Question 9

May/June 2017 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "A few candidates incorrectly introduced a constant term within their partial fraction, but fortunately most were able to recover by finding that the value of this constant was zero."

Source: 9709_s17_er.pdf, Page 36, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

Coaching With Shuja

Let

$$f(x) = \frac{3x^2 - 4}{x^2(3x + 2)}.$$

(i) Express $f(x)$ in partial fractions. [5]

(ii) Hence show that

$$\int_1^2 f(x)\,dx = \ln\left(\frac{25}{8}\right) - 1.$$

[5]

(i)

State or imply the form

$$\frac{A}{x}+\frac{B}{x^2}+\frac{C}{3x+2}$$

B1

Use a relevant method to determine a constant **M1**

Obtain one of the values $A=3$, $B=-2$, $C=-6$ **A1**

Obtain a second value **A1**

Obtain the third value **A1**

Total: 5

(ii)

Integrate and obtain terms

$$3\ln x+\frac{2}{x}-2\ln(3x+2)$$

B3

Use limits correctly, having integrated all the partial fractions, in a solution containing terms $a\ln x+b\ln(3x+2)+\frac{c}{x}$ **M1**

Obtain the given answer following full and exact working **A1**

Total: 5

Coaching With Shuja

Question 6

Oct/Nov 2017 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "The use of rounded fractions made it very difficult to reach the required form."

Source: 9709_w17_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 6

Coaching With Shuja

6 The points $A(1, 1)$ and $B(5, 9)$ lie on the curve $6y = 5x^2 - 18x + 19$.

(i) Show that the equation of the perpendicular bisector of AB is $2y = 13 - x$. **[4]**

The perpendicular bisector of AB meets the curve at C and D.

(ii) Find, by calculation, the distance CD, giving your answer in the form $\sqrt{\left(\frac{p}{q}\right)}$, where p and q are integers. **[5]**

6(i) Mid-point of $AB = (3, 5)$ | **B1**

Gradient of $AB = 2$ | **B1**

Eqn of perp. bisector is $y - 5 = -\frac{1}{2}(x - 3) \rightarrow 2y = 13 - x$ | **M1A1**

AG For **M1** FT from mid-point and gradient of AB

Total: 4

6(ii) $-3x + 39 = 5x^2 - 18x + 19 \rightarrow (5)(x^2 - 3x - 4) = (0)$ | **M1**

$x = 4$ or -1 | **A1**

$y = 4\frac{1}{2}$ or 7 | **A1**

$CD^2 = 5^2 + 2\frac{1}{2}^2 \rightarrow CD = \sqrt{\frac{125}{4}}$ | **M1A1**

Or equivalent integer fractions ISW

Total: 5

Coaching With Shuja

Question 2

Oct/Nov 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> “Drawing the required lines in part (ii) proved much more of a challenge for many.”

Source: 9709_w17_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 12), Question 2

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2 A function f is defined by $f : x \mapsto 4 - 5x$ for $x \in \mathbb{R}$.

(i) Find an expression for $f^{-1}(x)$ and find the point of intersection of the graphs of $y = f(x)$ and $y = f^{-1}(x)$. **[3]**

(ii) Sketch, on the same diagram, the graphs of $y = f(x)$ and $y = f^{-1}(x)$, making clear the relationship between the graphs. **[3]**

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Mark Scheme

Oct/Nov 2017 • Paper 1 (variant 12) • Q2

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Question 4

Oct/Nov 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Assuming that the radius was simply 6 was a common mistake among π weaker candidates as was the use of 45° rather than radians in the formula rθ."

Source: 9709_w17_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

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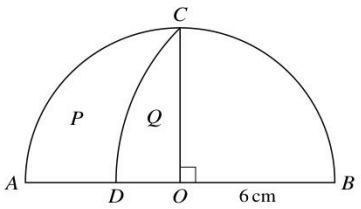
4 The diagram shows a semicircle with centre O and radius 6 cm. The radius OC is perpendicular to the diameter AB. The point D lies on AB, and DC is an arc of a circle with centre B.

4



The diagram shows a semicircle with centre O and radius 6 cm. The radius OC is perpendicular to the diameter AB. The point D lies on AB, and DC is an arc of a circle with centre B.

(i) Calculate the length of the arc DC. [3]

(i) Calculate the length of the arc DC. [3]

(ii) Find the value of $\frac{\text{area of region } P}{\text{area of region } Q}$, giving your answer correct to 3 significant figures. [4]

4(i) Pythagoras $\rightarrow r = \sqrt{72}$ OE

or $\cos 45 = \frac{6}{r} \rightarrow r = \frac{6}{\cos 45} = 6\sqrt{2}$ **M1** Correct method leading to $r =$

Arc $DC = \sqrt{72} \times \frac{1}{4}\pi = \frac{3\sqrt{2}}{2}\pi, 2.12\pi, 6.66$ **M1 A1** Use of $s = r\theta$ with their r (NOT 6) and $\frac{1}{4}\pi$

Total: 3

4(ii) Area of sector BDC is $\frac{1}{2} \times 72 \times \frac{1}{4}\pi (= 9\pi$ or $28.274\ldots)$ *M1 Use of $\frac{1}{2}r^2\theta$ with their r (NOT 6) and $\frac{1}{4}\pi$

Area $Q = 9\pi - 18(10.274\ldots)$ **DM1** Subtracts their $\frac{1}{2} \times 6 \times 6$ from their $\frac{1}{2}r^2\theta$

Area P is $(\frac{1}{4}\pi 6^2 -$ **area** $Q) = 18$ **M1** Uses $\{\frac{1}{4}\pi 6^2 -$ **(their area Q using** $\sqrt{72})\}$

Ratio is $\frac{18}{9\pi - 18}\left(\frac{18}{10.274}\right) \rightarrow 1.75$ **A1**

Total: 4

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Question 5

Oct/Nov 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 5 The vast majority of candidates were able to make some progress in attempting to show the given result in part (i) but the use of 2x in the question proved a significant challenge, especially for weaker candidates."

Source: 9709_w17_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 5

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5 (i) Show that the equation $\cos 2x(\tan^2 2x + 3) + 3 = 0$ can be expressed as $2\cos^2 2x + 3\cos 2x + 1 = 0$. [3]

(ii) Hence solve the equation $\cos 2x(\tan^2 2x + 3) + 3 = 0$ for $0° \leq x \leq 180°$. [4]

5(i) EITHER: Uses $\tan^2 2x = \frac{\sin^2 2x}{\cos^2 2x}$ **M1** Replaces $\tan^2 2x$ by $\frac{\sin^2 2x}{\cos^2 2x}$ not $\frac{\sin^2}{\cos^2} 2x$

Uses $\sin^2 2x = (1 - \cos^2 2x)$ **M1** Replaces $\sin^2 2x$ by $(1 - \cos^2 2x)$

$\rightarrow 2\cos^2 2x + 3\cos 2x + 1 = 0$ **A1)** AG. All correct

OR: **(M1** Replaces $\tan^2 2x$ by $\sec^2 2x - 1$

$\tan^2 2x = \sec^2 2x - 1$

$\sec^2 2x = \frac{1}{\cos^2 2x}$ **M1** Replaces $\sec^2 2x$ by $\frac{1}{\cos^2 2x}$

Multiply through by $\cos^2 2x$ and rearrange

$\rightarrow 2\cos^2 2x + 3\cos 2x + 1 = 0$ **A1)** AG. All correct

Total: 3

5(ii) $\cos 2x = -\frac{1}{2}, -1$ **M1** Uses (i) to get values for $\cos 2x$. Allow incorrect sign(s).

$2x = 120^\circ, 240^\circ$ or $2x = 180^\circ \circ 1$ **A1 A1 FT** A1 for 60° or 120° FT for $180^\circ - 1$st answer

$x = 60^\circ$ or 120°

or $x = 90^\circ$ **A1** Any extra answer(s) in given range only penalise fourth mark so max 3/4.

Total: 4

Question 6

Oct/Nov 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (b) in contrast proved to be perhaps the most challenging part of a question on the paper."

> "A significant number though failed to understand the situation described and perhaps the use of a sketch □2+4□ would have helped."

Source: 9709_w17_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 6

6 (a) The function f, defined by $f : x \mapsto a + b\sin x$ for $x \in \mathbb{R}$, is such that $f(\frac{\pi}{6}) = 4$ and $f(\frac{\pi}{2}) = 3$.

(i) Find the values of the constants a and b. [3]

(ii) Evaluate $ff(0)$. [2]

(b) The function g is defined by $g : x \mapsto c + d\sin x$ for $x \in \mathbb{R}$. The range of g is given by $-4 \leq g(x) \leq 10$. Find the values of the constants c and d. [3]

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6(a)(i) $4 = a + \frac{1}{2}b$ **M1** Forming simultaneous equations and eliminating one of the variables – probably a. May still include $\sin\frac{\pi}{2}$ and / or $\sin\frac{\pi}{6}$

$3 = a + b$

$\rightarrow a = 5, b = -2$ **A1 A1**

Total: 3

6(a)(ii) $ff(x) = a + b\sin(a + b\sin x)$ **M1** Valid method for ff. Could be $f(0) = N$ followed by $f(N) = M$.

$ff(0) = 5 - 2\sin 5 = 6.92$ **A1**

Total: 2

6(b) EITHER:

$10 = c + d$ and $-4 = c - d$ **(M1** Either pair of equations stated.

$10 = c - d$ and $-4 = c + d$

$c = 3, d = 7, -7$ or ± 7 **A1 A1)** Either pair solved ISW

Alternately $c = 3$ B1, range $= 14$ M1 $\rightarrow d = 7, -7$ or ± 7 A1

OR: **(M1 A1 A1)** Either of these diagrams can be awarded M1. Correct values of c and/or d can be awarded the A1, A1

Question 9

Oct/Nov 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Weaker candidates often used the two given vectors and were therefore unable to make progress."

> "A very common error was to simply assert that the scalar product was 0 rather than showing this clearly as the result of the 3 component products."

> "Another very common mistake was to stop when the scalar product was shown to be 0 rather than going on to explain that this proved that the given angle was 90°."

> "Many candidates did not appreciate the fact that because OA was parallel to it, CB could be obtained by multiplying each component by 3 and instead attempted to work with the magnitudes of the vectors."

Source: 9709_w17_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 9

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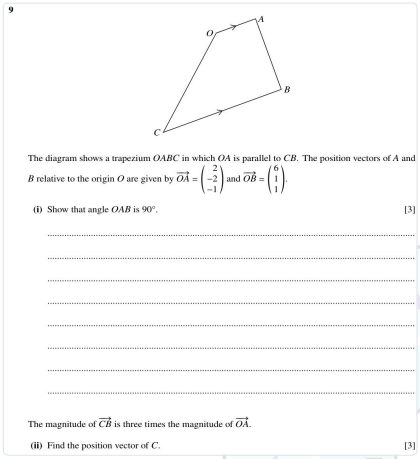
9 The diagram shows a trapezium $OABC$ in which OA is parallel to CB. The position vectors of A and B relative to the origin O are given by $\overrightarrow{OA} = \begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 6 \\ 1 \\ 1 \end{pmatrix}$.



(i) Show that angle OAB is $90°$. [3]

(ii) Find the position vector of C. [3]

(iii) Find the exact area of the trapezium $OABC$, giving your answer in the form $a\sqrt{b}$, where a and b are integers. [3]

9(i) $\overrightarrow{AB} = \begin{pmatrix}4\\3\\2\end{pmatrix}$ or $\overrightarrow{BA} = \begin{pmatrix}-4\\-3\\-2\end{pmatrix}$ **M1** Use of $\mathbf{b}-\mathbf{a}$ or $\mathbf{a}-\mathbf{b}$

e.g. $\overrightarrow{AO}\cdot\overrightarrow{AB} = -8+6+2 = 0 \rightarrow \angle OAB = 90^\circ$ AG **M1 A1** Use of dot product with either $\overrightarrow{AO}$ or $\overrightarrow{OA}$& either $\overrightarrow{AB}$ or $\overrightarrow{BA}$. Must see 3 component products

OR Correct use of Pythagoras. In both methods must state angle or $\Theta = 90^\circ$ or similar for A1

$|\overrightarrow{OA}| = 3, |\overrightarrow{OB}| = \sqrt{38}, |\overrightarrow{AB}| = \sqrt{29}$

$\overrightarrow{OA}^2 + \overrightarrow{AB}^2 = \overrightarrow{OB}^2 \rightarrow \angle OAB = 90^\circ$ AG

Total: 3

9(ii) $\overrightarrow{CB} = \begin{pmatrix}6\\-6\\-3\end{pmatrix}$ or $\overrightarrow{BC} = \begin{pmatrix}-6\\6\\3\end{pmatrix}$ **B1** Must correctly identify the vector.

$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC}$(or $-\overrightarrow{CB}$) $= \begin{pmatrix}0\\7\\4\end{pmatrix}$ **M1 A1** Correct link leading to $\overrightarrow{OC}$

Total: 3

9(iii) Method 1 Areas

Area under line $= \int(3-2x)dx = 3x - x^2$ **B1**

$= [(3-1) - (\frac{3}{4} - \frac{1}{16})]$ **M1** Apply their limits (e.g. $\frac{1}{4} \rightarrow 1$) after integn.

Area under curve $= \int(4-3x^{1/2})dx = 4x - 2x^{3/2}$ **B1**

$[(4-2) - (1-\frac{1}{4})]$ **M1** Apply their limits (e.g. $\frac{1}{4} \rightarrow 1$) after integration.

Required area $= \frac{21}{16} - \frac{5}{4} = \frac{1}{16}$ (or 0.0625) **A1)**

OR: $+/-\int(3-2x) - (4-3x^{1/2}) = +/-\int(-1-2x+3x^{1/2})$ *(M1 Substract functions and then attempt integration

$+/-[-x-x^2+\frac{3x^{3/2}}{3/2}]$ **A2, 1, 0 FT** FT on their subtraction. Deduct 1 mark for each term incorrect

$+/-[-1-1+2-(\frac{1}{4}+\frac{1}{16}+\frac{8}{)}] = \frac{1}{16}$ (or 0.0625) **DM1 A1)** Apply their limits $\frac{1}{4} \rightarrow 1$

Question 2

Oct/Nov 2017 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "A common error at this stage was to lose the solution zero."

Source: 9709_w17_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 2

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2 Find the set of values of a for which the curve $y = -\frac{2}{x}$ and the straight line $y = ax + 3a$ meet at two distinct points. [4]

Mark Scheme Oct/Nov 2017 • Paper 1 (variant 13) • Q2

2 $ax + 3a = -\frac{2}{x} \rightarrow ax^2 + 3ax + 2 (= 0)$ *M1 Rearrange into a 3-term quadratic.

Apply $b^2 - 4ac > 0$ SOI **DM1** Allow $>$. If no inequalities seen, **M1** is implied by 2 correct final answers in a or x.

$a < 0, a > \frac{8}{9}$ (or 0.889) OE **A1 A1** For final answers accept $0 > a > \frac{8}{9}$ but not $<, >$.

Question 3

Oct/Nov 2017 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Part (ii) was found more challenging with a minority of candidates not able to make headway and to distinguish which combinations of terms were required.."

> "A common mistake following this was to equate the two terms rather than to add the terms and equate to zero."

Source: 9709_w17_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 3

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3 (i) Find the term independent of x in the expansion of $\left(\frac{2}{x} - 3x\right)^6$. [2]

(ii) Find the value of a for which there is no term independent of x in the expansion of $(1 + ax^2)\left(\frac{2}{x} - 3x\right)^6$. [3]

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3(i) $\mathbf{6C3}\left(\frac{2}{x}\right)^{3}(-3x)^{3}$ SOI also allowed if seen in an expansion **M1** Both x's can be missing.

$\mathbf{-4320}$ Identified as answer **A1** Cannot be earned retrospectively in (ii).

Total: 2

3(ii) $\mathbf{6C2}\left(\frac{2}{x}\right)^{4}[(-3x)]^{2}$ SOI clearly identified as critical term **M1** Both x's and minus sign can be missing.

$\mathbf{15a \times 16 \times 9 - their4320(= 0)}$ **A1 FT** FT on their 4320.

$\boldsymbol{a = 2}$ **A1**

Total: 3

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Question 7

Oct/Nov 2017 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "It was surprising, therefore, that a significant number of candidates failed to show that angle ABP = 0.6435 radians."

Source: 9709_w17_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 7

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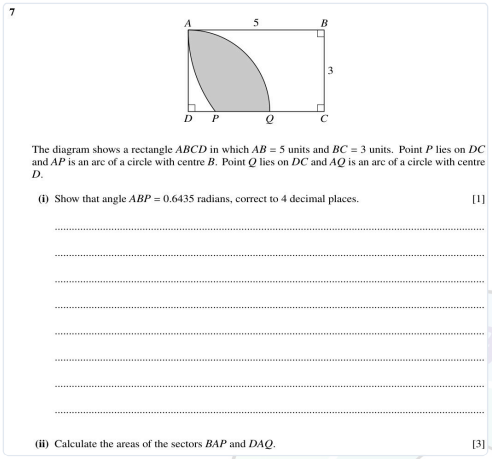
Question Paper Oct/Nov 2017 • Paper 1 (variant 13) • Q7

7 The diagram shows a rectangle $ABCD$ in which $AB = 5$ units and $BC = 3$ units. Point P lies on DC and AP is an arc of a circle with centre B. Point Q lies on DC and AQ is an arc of a circle with centre D.



(i) Show that angle $ABP = 0.6435$ radians, correct to 4 decimal places. [1]

(ii) Calculate the areas of the sectors BAP and DAQ. [3]

(iii) Calculate the area of the shaded region. [3]

Mark Scheme Oct/Nov 2017 • Paper 1 (variant 13) • Q7

7(i) $\sin^{-1}\left(\frac{3}{5}\right) = 0.6435$ AG **M1** OR $(PBC =) \cos^{-1}\left(\frac{3}{5}\right) = 0.9273 \Rightarrow (ABP) = \frac{\pi}{2} - 0.9273 = 0.6435$

or other valid method. Check working and diagram for evidence of incorrect method

Total: 1

7(ii) Use (once) of sector area $= \frac{1}{2}r^2\theta$ **M1**

Area sector $BAP = \frac{1}{2} \times 5^2 \times 0.6435 = 8.04$ **A1**

Area sector $DAQ = \frac{1}{2} \times \frac{1}{2}\pi \times 3^2 = 7.07$, Allow $\frac{9\pi}{4}$ **A1**

Total: 3

7(iii) EITHER: **(M1** Use of correct strategy

Region = sect + sect − (rect − A) or sect − [rect − (sect + A)]

(Area $\triangle BPC =$) $\frac{1}{2} \times 3 \times 4 = 6$ Seen **A1**

$8.04 + 7.07 - (15 - 6) = 6.11$ **A1)**

OR1: **(M1** Use of correct strategy

Region = sector ADQ - (trap $ABPD$ - sector ABP).

(Area trap $ABPD =$) $\frac{1}{2}(5 + 1) \times 3 = 9$ Seen **A1**

$7.07 - (9 - 8.04) = 7.07 - 0.96 = 6.11$ **A1)**

OR2: **(M1** Use of correct strategy

Area segment $AP = 2.5686$ Area segment $AQ = 0.5438$ **M1** Use of correct strategy

Region = segment AP+ segment $AQ + \triangle APQ$.

(Area $\triangle APQ =$) $\frac{1}{2} \times 2 \times 3 = 3$ Seen **A1**

$2.57 + 0.54 + 3 = 6.11$ **A1)**

Total: 3

Question 8

Oct/Nov 2017 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

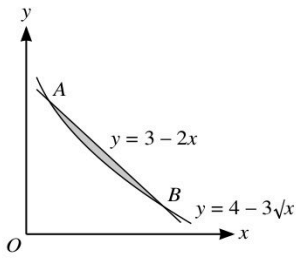
> “A common error was to square individual terms.”

Source: 9709_w17_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 8

Coaching With Shuja

8 The diagram shows parts of the graphs of $y = 3 - 2x$ and $y = 4 - 3\sqrt{x}$ intersecting at points A and B.

8



The diagram shows parts of the graphs of $y = 3 - 2x$ and $y = 4 - 3\sqrt{x}$ intersecting at points A and B.

(i) Find by calculation the x-coordinates of A and B. [3]

(i) Find by calculation the x-coordinates of A and B. [3]

(ii) Find, showing all necessary working, the area of the shaded region. [5]

Mark Scheme Oct/Nov 2017 • Paper 1 (variant 13) • Q8

8(i) EITHER: **(M1** Form 3-term quad \& attempt to solve for $\sqrt{x}$.

$4 - 3\sqrt{x} = 3 - 2x \rightarrow 2x - 3\sqrt{x} + 1 (= 0)$ or e.g. $2k^2 - 3k + 1 (= 0)$

$\sqrt{x} = \frac{1}{2}, 1$ **A1** Or $k = \frac{1}{2}$ or 1 (where $k = \sqrt{x}$).

$x = \frac{1}{4}, 1$ **A1)**

OR1: **(M1**

$(3\sqrt{x})^2 = (1 + 2x)^2$

$4x^2 - 5x + 1 (= 0)$ **A1**

$x = \frac{1}{4}, 1$ **A1)**

OR2: **(M1** Eliminate x

$\frac{3-y}{2} = \left(\frac{4-y}{3}\right)^2 (\rightarrow 2y^2 - 7y + 5 (= 0))$ **M1**

$y = \frac{5}{2}, 1$ **A1**

$x = \frac{1}{4}, 1$ **A1)**

Total: 3

8(ii) EITHER: **(B1**

Area under line $= \int (3 - 2x) dx = 3x - x^2$

$= [(3 - 1) - (\frac{3}{4} - \frac{1}{16})]$ **M1** Apply their limits (e.g. $\frac{1}{4} \rightarrow 1$) after integn.

Area under curve $= \int (4 - 3x^{1/2}) dx = 4x - 2x^{3/2}$ **B1**

$[(4 - 2) - (1 - \frac{1}{4})]$ **M1** Apply their limits (e.g. $\frac{1}{4} \rightarrow 1$) after integration.

Required area $= \frac{21}{16} - \frac{5}{4} = \frac{1}{16}$ (or 0.0625) **A1)**

OR: *(*\M1 Substract functions and then attempt integration

$+/- \int (3 - 2x) - (4 - 3x^{1/2}) = +/- \int (-1 - 2x + 3x^{1/2})$

$+/- [-x - x^2 + \frac{3x^{3/2}}{3/2}]$ **A2, 1, 0 FT** FT on their subtraction. Deduct 1 mark for each term incorrect

$+/- [-1 - 1 + 2 - (-\frac{1}{4} + \frac{1}{16} + \frac{8}{)}] = \frac{1}{16}$ (or 0.0625) **DM1 A1)** Apply their limits $\frac{1}{4} \rightarrow 1$

Total: 5

Question 9

Oct/Nov 2017 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "In contrast to part (i), part (ii) was not done well, what 2 most candidates did not appreciate was that the ratio of |CD| : |BC|, which most candidates found in part 3 (i), applied in part (ii) also, and was in fact intended as a helpful lead-in to part (ii)."

Source: 9709_w17_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 9

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9 Relative to an origin O, the position vectors of the points A, B and C are given by

$$\overrightarrow{OA} = \begin{pmatrix} 8 \\ -6 \\ 5 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} -10 \\ 3 \\ -13 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix}.$$

A fourth point, D, is such that the magnitudes $|\overrightarrow{AB}|$, $|\overrightarrow{BC}|$ and $|\overrightarrow{CD}|$ are the first, second and third terms respectively of a geometric progression.

(i) Find the magnitudes $|\overrightarrow{AB}|$, $|\overrightarrow{BC}|$ and $|\overrightarrow{CD}|$. **[5]**

(ii) Given that D is a point lying on the line through B and C, find the two possible position vectors of the point D. **[4]**

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9(i) $\overrightarrow{AB} = \pm/-\begin{pmatrix}-18\\9\\-18\end{pmatrix}$, $\overrightarrow{BC} = \pm/-\begin{pmatrix}12\\-6\\12\end{pmatrix}$ **B1 B1** Allow i, j, k form throughout.

$|AB| = 27$, $|BC| = 18$ **B1 FT** FT on their AB, their OD. **B1 FT**

$|\overrightarrow{CD}| = \left(\frac{18}{27}\right) \times 18$ OR $\left(\frac{18}{27}\right) \times 27 = 12$ **B1**

Total: 5

9(ii) $\overrightarrow{CD} = (\pm)$**their** $\frac{18}{27} \times$ **their** $\overrightarrow{BC}$ SOI **M1**

$\overrightarrow{OD} =$

(\pm) their \frac{18}{27}

$\begin{pmatrix}2\\-3\\-1\end{pmatrix}$

$\begin{pmatrix}12\\-6\\12\end{pmatrix}$

=

$\begin{pmatrix}10\\-7\\7\end{pmatrix}$

,

$\begin{pmatrix}-6\\-9\end{pmatrix}$

$ **M1 A1 A1**

Other methods possible for $\overrightarrow{OD}$, e.g. $\overrightarrow{OB} + \frac{5}{2}\overrightarrow{CD}$, $\overrightarrow{OB} + \frac{1}{2}\overrightarrow{CD}$ (One soln **M2A1**, 2nd soln A1) OR $\overrightarrow{OB} + \frac{5}{3}\overrightarrow{BC}$, $\overrightarrow{OB} + \frac{1}{3}\overrightarrow{BC}$ (One soln **M2A1**, 2nd soln A1)

Total: 4

Question 2

Oct/Nov 2017 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 2 This was another question that many candidates found difficult."

Source: 9709_w17_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

Coaching With Shuja

2 Two variable quantities x and y are believed to satisfy an equation of the form $y = C(a^x)$, where C and a are constants. An experiment produced four pairs of values of x and y. The table below gives the corresponding values of x and $\ln y$.

x	0.9	1.6	2.4	3.2
$\ln y$	1.7	1.9	2.3	2.6

By plotting $\ln y$ against x for these four pairs of values and drawing a suitable straight line, estimate the values of C and a. Give your answers correct to 2 significant figures. **[5]**

Question	Answer	Marks

Coaching With Shuja

Question 4

Oct/Nov 2017 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "A large proportion of candidates found it difficult to progress beyond the first three marks since they did not recognise that sec2x could be expressed as 1/cos2x."

Source: 9709_w17_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

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(i) Prove the identity $\tan(45^\circ + x) + \tan(45^\circ - x) \equiv 2\sec 2x$. **[4]**

(ii) Hence sketch the graph of $y = \tan(45^\circ + x) + \tan(45^\circ - x)$ for $0^\circ \leq x \leq 90^\circ$. **[3]**

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4(i) Use correct $\tan(A \pm B)$ formula and express the LHS in terms of $\tan x$ **M1**

Using $\tan 45^\circ = 1$ express LHS as a single fraction **A1**

Use Pythagoras or correct double angle formula **M1**

Obtain given answer **A1**

Total: 4

4(ii) Show correct sketch for one branch **B1**

Both branches correct and nothing else seen in the interval **B1**

Show asymptote at $x = 45^\circ$ **B1**

Total: 3

Coaching With Shuja

Question 6

Oct/Nov 2017 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "A common error was failing to write 1/cos2y as sec2y, or not realising it integrated to tany."

Source: 9709_w17_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

6 The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = 4\cos^2 y \tan x,$$

for $0 < x < \frac{1}{2}\pi$, and $x = 0$ when $y = \frac{1}{4}\pi$. Solve this differential equation and find the value of x when $y = \frac{3}{8}\pi$. **[8]**

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Question Answer Marks

[Mark scheme not fully visible in provided image]

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Question 1

Oct/Nov 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A few candidates lost marks through not working to sufficient accuracy, or incorrect rounding of their final answer."

Source: 9709_w17_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

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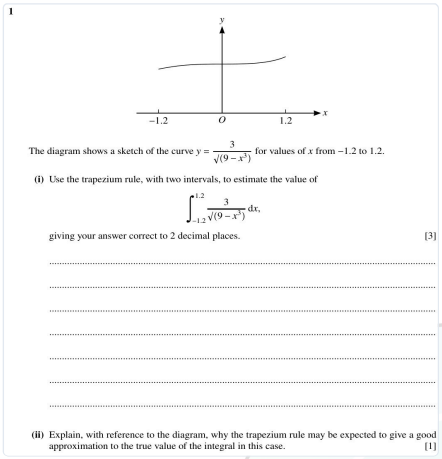
1



The diagram shows a sketch of the curve $y = \frac{3}{\sqrt{9-x^3}}$ for values of x from -1.2 to 1.2.

(i) Use the trapezium rule, with two intervals, to estimate the value of

$$\int_{-1.2}^{1.2} \frac{3}{\sqrt{9-x^3}}\,dx,$$

giving your answer correct to 2 decimal places. **[3]**

(ii) Explain, with reference to the diagram, why the trapezium rule may be expected to give a good approximation to the true value of the integral in this case. **[1]**

Mark Scheme

Oct/Nov 2017 • Paper 3 (variant 32) • Q1

1(i) State or imply ordinates $0.915929\ldots, 1, 1.112485\ldots$ **B1**

Use correct formula, or equivalent, with $h = 1.2$ and three ordinates **M1**

Obtain answer **2.42** only **A1**

Total: 3

1(ii) Justify the given statement **B1**

Total: 1

Coaching With Shuja

Question 2

Oct/Nov 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A common error at the very end of the solution was the acceptance of a negative value of x as a possible solution."

Source: 9709_w17_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Coaching With Shuja

2 Showing all necessary working, solve the equation $2\log_3 x = 3 + \log_2(x+1)$, giving your answer correct to 3 significant figures. **[5]**

2 Use law for the logarithm of a power or a quotient on the given equation **M1**

Use $\log_3 8 = 3$ or $2^3 = 8$ **M1**

Obtain $x^2 - 8x - 8 = 0$, or horizontal equivalent **A1**

Solve a 3-term quadratic equation **M1**

Obtain final answer $x = 8.90$ only **A1**

Total: 5

Question 6

Oct/Nov 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors were sign errors and transcription errors in rearranging the terms."

Source: 9709_w17_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

6 The equation of a curve is $x^3y - 3xy^3 = 2a^4$, where a is a non-zero constant.

(i) Show that $\frac{dy}{dx} = \frac{3x^2y - 3y^3}{9xy^2 - x^3}$. **[4]**

(ii) Hence show that there are only two points on the curve at which the tangent is parallel to the x-axis and find the coordinates of these points. **[4]**

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Mark Scheme Oct/Nov 2017 • Paper 3 (variant 32) • Q6

EITHER: Use scalar product to obtain an equation in a, b and c, e.g. $a + b + 3c = 0$ **B1**

Form a second relevant equation, e.g. $2a - 2b + c = 0$, and solve for one ratio, e.g. $a : b$ **M1**

Obtain final answer $a : b : c = 7 : 5 : -4$ **A1**

Use coordinates of A and values of a, b and c in general equation and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR1: Calculate the vector product of relevant vectors, e.g. $(\mathbf{i} + 3\mathbf{k}) \times (2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ **(*M1**

Obtain two correct components **A1**

Obtain correct answer, e.g. $7\mathbf{i} + 5\mathbf{j} - 4\mathbf{k}$ **A1**

Substitute coordinates of A in plane equation with their normal and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR2: Using relevant vectors, form a two-parameter equation for the plane **(*M1**

State a correct equation, e.g. $\mathbf{r} = 3\mathbf{i} + 2\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} + \mathbf{j} + 3\mathbf{k}) + \mu(2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ **A1 FT**

State 3 correct equations in x, y, z, λ and μ **A1 FT**

Eliminate λ and μ **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR3: Use the direction vector of the line of intersection of the two planes as normal vector to the plane **(*M1**

Two correct components **A1**

Three correct components **A1**

Substitute coordinates of A in plane equation with their normal and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

Total: 7

Question 8

Oct/Nov 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Some candidates lost marks through inaccurate algebraic manipulation or numerical errors, even though they were using a correct method."

> "The expansions of –1 –2 □ 2 □ □ 2 □ □1+ x□ and □1+ x□ were often correct, but the candidates often did not deal with the factors □ 3 □ □ 3 □ 1 1 1 of and correctly: the most common errors were to multiply by 3 and by 9, or to use in place 3 9 3 1 of ."

Source: 9709_w17_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 32), Question 8

Coaching With Shuja

8 Let $f(x) = \frac{8x+9x+8}{(1-x)(2x+3)^2}$.

(i) Express $f(x)$ in partial fractions. **[5]**

(ii) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. **[5]**

Mark Scheme Oct/Nov 2017 • Paper 3 (variant 32) • Q8

EITHER: Use scalar product to obtain an equation in a, b and c, e.g. $a + b + 3c = 0$ **B1**

Form a second relevant equation, e.g. $2a - 2b + c = 0$, and solve for one ratio, e.g. $a : b$ **M1**

Obtain final answer $a : b : c = 7 : 5 : -4$ **A1**

Use coordinates of A and values of a, b and c in general equation and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR1: Calculate the vector product of relevant vectors, e.g. $(\mathbf{i} + 3\mathbf{k}) \times (2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ **(*M1**

Obtain two correct components **A1**

Obtain correct answer, e.g. $7\mathbf{i} + 5\mathbf{j} - 4\mathbf{k}$ **A1**

Substitute coordinates of A in plane equation with their normal and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR2: Using relevant vectors, form a two-parameter equation for the plane **(*M1**

State a correct equation, e.g. $\mathbf{r} = 3\mathbf{i} + 2\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} + \mathbf{j} + 3\mathbf{k}) + \mu(2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ **A1 FT**

State 3 correct equations in x, y, z, λ and μ **A1 FT**

Eliminate λ and μ **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

OR3: Use the direction vector of the line of intersection of the two planes as normal vector to the plane **(*M1**

Two correct components **A1**

Three correct components **A1**

Substitute coordinates of A in plane equation with their normal and find d **DM1**

Obtain answer $7x + 5y - 4z = 27$, or equivalent **A1 FT**

Total: 7

Question 9

Oct/Nov 2017 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "There were many incorrect answers, partly because some candidates struggled to key this function correctly on their calculators (often not using brackets when they needed to), and partly because candidates were not considering an appropriate function."

Source: 9709_w17_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Question 9

It is given that $\int_1^a x^2 \ln x \, dx = 2$, where $a > 1$.

(i) Show that $a^{3/2} = \frac{7+2a^3}{3\ln a}$. **[5]**

(ii) Show by calculation that a lies between 2 and 4. **[2]**

(iii) Use the iterative formula

$$a_{n+1} = \left(\frac{7+2a_n^3}{3\ln a_n}\right)^{3/4}$$

to determine a correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **[3]**

Coaching With Shuja

9(iii)

M1 Use the iterative formula correctly at least once

A1 Obtain final answer 3.031

A1 Show sufficient iterations to 5 d.p. to justify 3.031 to 3 d.p., or show there is a sign change in the interval (3.0305, 3.0315)

Total: 3

Coaching With Shuja

Question 4

Oct/Nov 2017 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "A large proportion of candidates found it difficult to progress beyond the first 3 marks since they did not recognise that sec2x could be expressed as 1/cos2x."

Source: 9709_w17_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 33), Question 4

Coaching With Shuja

(i) Prove the identity $\tan(45^\circ + x) + \tan(45^\circ - x) \equiv 2\sec 2x$. **[4]**

(ii) Hence sketch the graph of $y = \tan(45^\circ + x) + \tan(45^\circ - x)$ for $0^\circ \leq x \leq 90^\circ$. **[3]**

Coaching With Shuja

4(i)

M1 Use correct $\tan(A \pm B)$ formula and express the LHS in terms of $\tan x$

A1 Using $\tan 45^\circ = 1$ express LHS as a single fraction

M1 Use Pythagoras or correct double angle formula

A1 Obtain given answer

Total: 4

4(ii)

B1 Show correct sketch for one branch

B1 Both branches correct and nothing else seen in the interval

B1 Show asymptote at $x = 45^\circ$

Total: 3

Coaching With Shuja

Question 6

Oct/Nov 2017 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "A common error was failing to write 1/cos2y as sec2y, or not realising it integrated to tany."

Source: 9709_w17_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 33), Question 6

Coaching With Shuja

Question 6

The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = 4\cos^2 y \tan x,$$

for $0 \leq x < \frac{1}{2}\pi$, and $x = 0$ when $y = \frac{1}{4}\pi$. Solve this differential equation and find the value of x when $y = \frac{1}{3}\pi$. **[8]**

Coaching With Shuja

Question	Answer	Marks
	Obtain answer $2x-2y+z=9$, or equivalent	**A1)**
	OR3: Using the relevant point and relevant vectors, form a determinant equation for the plane	**(M1**
	State a correct equation, e.g. $\begin{vmatrix} x-3 & y+2 & z+1 \\ -1 & 1 & 4 \\ 2 & 1 & -2 \end{vmatrix}=0$	**A1**
	Attempt to expand the determinant	**M1**
	Obtain two correct cofactors	**A1**
	Obtain answer $-2x+2y-z=-9$, or equivalent	**A1)**
		5

Coaching with Shuja

Question 1

Feb/Mar 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Some candidates struggled with minus signs and either made both terms positive or both terms negative."

Source: 9709_m16_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 12), Question 1

Coaching With Shuja

(i) Find the coefficients of x^4 and x^5 in the expansion of $(1-2x)^5$. [2]

(ii) It is given that, when $(1+px)(1-2x)^5$ is expanded, there is no term in x^5. Find the value of the constant p. [2]

(i)

$80x^4$, $-32x^5$ **B1B1** Fully simplified

Total: 2

(ii)

$(-32+80p)x^5=0$ **M1** Attempt to multiply relevant terms and put $=0$

$p=2/5$ or $32/80$ OE **A1**

Total: 2

Question 2

Feb/Mar 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "The number of minus signs involved in substituting –1 also proved challenging for some but many fully correct answers were seen."

Source: 9709_m16_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 12), Question 2

Coaching With Shuja

Question Paper Feb/Mar 2016 • Paper 1 (variant 12) • Q2

A curve for which $\frac{dy}{dx} = 3x^2 - \frac{2}{x^3}$ passes through $(-1, 3)$. Find the equation of the curve. [4]

Coaching With Shuja

Mark Scheme Feb/Mar 2016 • Paper 1 (variant 12) • Q2

$y = \frac{3x^3}{3} - \frac{2x^{-2}}{-2} + c$ **B1B1**

$3 = -1 + 1 + c$ **M1** Sub $x = -1, y = 3$. c must be present

$y = x^3 + x^{-2} + 3$ **A1** Accept $c = 3$ WWW

Total: 4

Question 3

Feb/Mar 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 3 The majority of candidates gained full marks on this question however some weaker candidates struggled to interpret the given information and were unable to form the required equations."

Source: 9709_m16_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

Question Paper Feb/Mar 2016 • Paper 1 (variant 12) • Q3

The 12th term of an arithmetic progression is 17 and the sum of the first 31 terms is 1023. Find the 31st term. [5]

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$a + 11d = 17$ **B1**

$\frac{31}{2}(2a + 30d) = 1023$ **B1**

Solve simultaneous equations **M1**

$d = 4,\ a = -27$ **A1** At least one correct

31st term = 93 A1

Total: 5

Question 4

Feb/Mar 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (b) proved more challenging than expected with many candidates unable to factorise the equation."

> "Attempting to square each term was a common mistake of weaker candidates."

> "Stronger candidates often lost marks due to cancelling and losing one of the solutions or giving answers in degrees rather than radians."

Source: 9709_m16_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

Coaching With Shuja

Question Paper Feb/Mar 2016 • Paper 1 (variant 12) • Q4

(a) Solve the equation $\sin^{-1}(3x) = -\frac{1}{3}\pi$, giving the solution in an exact form. [2]

(b) Solve, by factorising, the equation $2\cos\theta\sin\theta - 2\cos\theta - \sin\theta + 1 = 0$ for $0 \leq \theta \leq \pi$. [4]

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(a)

$3x = -\sqrt{3}/2$ **M1** Accept -0.866 at this stage

$x = -\frac{\sqrt{3}}{6}$ OE **A1** Or $\frac{-3}{6\sqrt{3}}$ or $\frac{-1}{2\sqrt{3}}$

Total: 2

(b)

$(2\cos\theta - 1)(\sin\theta - 1) = 0$ **M1** Reasonable attempt to factorise and solve

$\cos\theta = 1/2$ or $\sin\theta = 1$ **A1** Award B1B1 WWW

$\theta = \pi/3$ or $\pi/2$ **A1A1** Allow $1.05, 1.57$. SCA1 for both $60°, 90°$

Total: 4

Coaching With Shuja

Question 6

Feb/Mar 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 6 The majority of candidates scored very well on this question although weaker ones struggled in part (i) as they were unaware of the formulae for the volume and surface area of a cylinder."

Source: 9709_m16_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 12), Question 6

Coaching With Shuja

Question Paper Feb/Mar 2016 • Paper 1 (variant 12) • Q6

A vacuum flask (for keeping drinks hot) is modelled as a closed cylinder in which the internal radius is r cm and the internal height is h cm. The volume of the flask is $1000\ \text{cm}^3$. A flask is most efficient when the total internal surface area, $A\ \text{cm}^2$, is a minimum.

(i) Show that $A = 2\pi r^2 + \frac{2000}{r}$. [3]

(ii) Given that r can vary, find the value of r, correct to 1 decimal place, for which A has a stationary value and verify that the flask is most efficient when r takes this value. [5]

Coaching With Shuja

(i)

$A = 2\pi r^2 + 2\pi rh$ **B1**

$\pi r^2 h = 1000 \rightarrow h = \frac{1000}{\pi r^2}$ **M1**

Sub for h into $A \rightarrow A = 2\pi r^2 + \frac{2000}{r}$ AG **A1**

Total: 3

(ii)

$\frac{dA}{dr} = 0 \Rightarrow 4\pi r - \frac{2000}{r^2} = 0$ **M1A1** Attempt differentiation and set $= 0$

$r = 5.4$ **DM1 A1** Reasonable attempt to solve to $r^3 = \frac{500}{\pi}$

$\frac{d^2A}{dr^2} = 4\pi + \frac{4000}{r^3}$

> 0 hence MIN hence MOST EFFICIENT AG **B1** Or convincing alternative method

Total: 5

Coaching with Shuja

Question 7

Feb/Mar 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

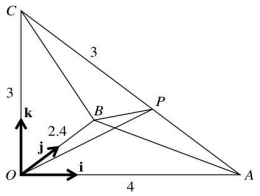
> "Question 7 This question proved to be one of the more difficult ones on the paper."

> "Many candidates struggled with this and 5 some weaker ones used a circular argument with the answers from part (ii)."

Source: 9709_m16_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 12), Question 7

Coaching With Shuja

7



The diagram shows a pyramid $OABC$ with a horizontal triangular base OAB and vertical height OC. Angles AOB, BOC and AOC are each right angles. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OB and OC respectively, with $OA = 4$ units, $OB = 2.4$ units and $OC = 3$ units. The point P on CA is such that $CP = 3$ units.

(i) Show that $\overrightarrow{CP} = 2.4\mathbf{i} - 1.8\mathbf{k}$. [2]

(ii) Express $\overrightarrow{OP}$ and $\overrightarrow{BP}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. [2]

(iii) Use a scalar product to find angle BPC. [4]

(i)

$CP = \frac{3}{5}CA$ soi **M1**

$\overrightarrow{CP} = \frac{3}{5}(4\mathbf{i} - 3\mathbf{k}) = 2.4\mathbf{i} - 1.8\mathbf{k}$ AG **A1**

Total: 2

(ii)

$\overrightarrow{OP} = 2.4\mathbf{i} + 1.2\mathbf{k}$ **B1**

$\overrightarrow{BP} = 2.4\mathbf{i} - 2.4\mathbf{j} + 1.2\mathbf{k}$ **B1**

Total: 2

(iii)

$\overrightarrow{BP} \cdot \overrightarrow{CP} = 5.76 - 2.16 = 3.6$ **M1** Use of $x_1x_2 + y_1y_2 + z_1z_2$

$|BP||CP| = \sqrt{2.4^2 + 2.4^2 + 1.2^2}\sqrt{2.4^2 + 1.8^2}$ **M1** Product of moduli

$\cos BPC = \frac{3.6}{\sqrt{12.96}\sqrt{9}} = \frac{1}{3}$ **M1** All linked correctly

Angle $BPC = 70.5°$ or 1.23 rads cao **A1**

Total: 4

Coaching with Shuja

Question 8

Feb/Mar 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "In part (ii) a significant number of candidates did not realise the need to complete the square first in order to find the inverse function and so made no progress."

Source: 9709_m16_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

Coaching With Shuja

Question Paper

The function f is such that $f(x) = a^2x^2 - ax + 3b$ for $x \leq \frac{1}{2a}$, where a and b are constants.

(i) For the case where $f(-2) = 4a^2 - b + 8$ and $f(-3) = 7a^2 - b + 14$, find the possible values of a and b. [5]

(ii) For the case where $a = 1$ and $b = -1$, find an expression for $f^{-1}(x)$ and give the domain of f^{-1}. [5]

(i)

$2a + 4b = 8$ **M1** Substitute in -2 and -3

$2a^2 + 3a + 4b = 14$ **A1**

$2a^2 + 3a + (8 - 2a) = 14 \rightarrow (a + 2)(2a - 3) = 0$ **M1** Substitute linear into quadratic and attempt solution

$a = -2$ or $3/2$ **A1**

$b = 3$ or $5/4$ **A1** If A0A0 scored allow SCA1 for either $(-2, 3)$ or $(3/2, 5/4)$

Total: 5

(ii)

$y = \left(x - \frac{1}{2}\right)^2 - \frac{13}{4}$ **M1A1** Attempt completing the square

$x - \frac{1}{2} = (\pm)\sqrt{y + \frac{13}{4}}$ OE **DM1**

$f^{-1}(x) = \frac{1}{2} - \sqrt{x + \frac{13}{4}}$ OE **A1** Must be a function of x

Domain of f^{-1} is $x \geq -\frac{13}{4}$ **B1**

Total: 5

Coaching With Shuja

Question 9

Feb/Mar 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 9 Part (a) was generally well done but part (b) proved to be a significant challenge even for many good candidates."

> "In part (ii) a surprisingly large number of candidates failed to use the 1 formula absinCfor the area of a triangle."

Source: 9709_m16_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 12), Question 9

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(a)

9 **(a)**

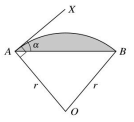


Fig. 1

In Fig. 1, OAB is a sector of a circle with centre O and radius r. AX is the tangent at A to the arc AB and angle $BAX = \alpha$.

(i) Show that angle $AOB = 2\alpha$. [2]

(ii) Find the area of the shaded segment in terms of r and α. [2]

(b)

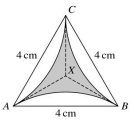
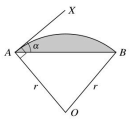


Fig. 2

In Fig. 2, ABC is an equilateral triangle of side 4 cm. The lines AX, BX and CX are tangents to the equal circular arcs AB, BC and CA. Use the results in part **(a)** to find the area of the shaded region, giving your answer in terms of π and $\sqrt{3}$. [6]

9 **(a)**



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(b)

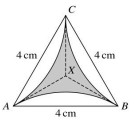

Fig. 2

In Fig. 2, ABC is an equilateral triangle of side 4 cm. The lines AX, BX and CX are tangents to the equal circular arcs AB, BC and CA. Use the results in part **(a)** to find the area of the shaded region, giving your answer in terms of π and $\sqrt{3}$. [6]

(a)(i)

$BAO = OBA = \frac{\pi}{2} - \alpha$ **M1** Allow use of $90°$ or $180°$

$AOB = \pi - \left(\frac{\pi}{2} - \alpha\right) - \left(\frac{\pi}{2} - \alpha\right) = 2\alpha$ AG **A1** Or other valid reasoning

Total: 2

(a)(ii)

$\frac{1}{2}r^2(2\alpha) - \frac{1}{2}r^2 \sin 2\alpha$ OE **B2,1,0** SCB1 for reversed subtraction

Total: 2

(b)

Use of $\alpha = \frac{\pi}{6}$, $r = 4$ **B1B1**

1 segment $S=\left(\frac{1}{2}\right)4^2\left(\frac{\pi}{3}\right)-\left(\frac{1}{2}\right)4^2\sin\frac{\pi}{3}=\frac{8\pi}{3}-4\sqrt{3}**M1**FTtheir(ii),\alpha,r$

Area $ABC, T = \left(\frac{1}{2}\right)4^2 \sin\frac{\pi}{3} = 4\sqrt{3}$ **B1**

$T-3S=\left(\frac{1}{2}\right)4^2\sin\frac{\pi}{3}-3\left[\left(\frac{1}{2}\right)4^2\left(\frac{\pi}{3}\right)-\left(\frac{1}{2}\right)4^2\sin\frac{\pi}{3}\right]$ **M1**

$16\sqrt{3} - 8\pi$ cao **A1**

Total: 6

Question 4

Feb/Mar 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "(b) Most candidates obtained the correct answer, but often failed to justify this and so lost marks."

Source: 9709_m16_er.pdf, Page 8, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

Question Paper Feb/Mar 2016 • Paper 3 (variant 32) • Q4

The polynomial $4x^3 + ax + 2$, where a is a constant, is denoted by $p(x)$. It is given that $(2x + 1)$ is a factor of $p(x)$.

(i) Find the value of a. [2]

(ii) When a has this value,

(a) factorise $p(x)$, [2]

(b) solve the inequality $p(x) > 0$, justifying your answer. [3]

Coaching With Shuja

(i)

Substitute $x = -\frac{1}{2}$ and equate to zero, or divide by $(2x + 1)$ and equate constant remainder to zero **M1**

$a = 3$ **A1**

Total: 2

(ii)(a)

Commence division by $(2x + 1)$ reaching a partial quotient of $2x^2 + kx$ **M1**

$(2x + 1)(2x^2 - x + 2)$ **A1**

Total: 2

(ii)(b)

State or imply critical value $x = -\frac{1}{2}$ **B1**

Show that $2x^2 - x + 2$ is always positive, or that the gradient of $4x^3 + 3x + 2$ is always positive **B1***

Justify final answer $x > -\frac{1}{2}$ **B1(dep*)**

Total: 3

Coaching with Shuja

Question 8

Feb/Mar 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "(ii) Few candidates made much progress with this part, incorrectly using either i + 2j –k or 5i + 3j + k in trying to establish the direction of the line m."

Source: 9709_m16_er.pdf, Page 9, Section: Comments on specific questions – Paper 3 (variant 32), Question 8

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Question Paper Feb/Mar 2016 • Paper 3 (variant 32) • Q8

The line l has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$. The plane p has equation $\mathbf{r} \cdot \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} = 6$.

(i) Show that l is parallel to p. [3]

(ii) A line m lies in the plane p and is perpendicular to l. The line m passes through the point with coordinates $(5, 3, 1)$. Find a vector equation for m. [6]

Coaching With Shuja

(i)

EITHER: Substitute for $\mathbf{r}$ in the given equation of p and expand scalar product **M1**

Obtain equation in λ in any correct form **A1**

Verify this is not satisfied for any value of λ **A1**

OR: Expand scalar product of the normal to p and the direction vector of l **M1**

Verify scalar product is zero **A1**

Verify that one point of l does not lie in the plane **A1**

Total: 3

(ii)

Calling the unknown direction vector $a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$, state equation $2a+b+3c=0$ **B1**

State equation $2a-b-c=0$ **B1**

Solve for one ratio, e.g. $a:b:c=1:4:-2$, or equivalent **M1 A1**

Form line equation with relevant vectors **M1**

$\mathbf{r}=5\mathbf{i}+3\mathbf{j}+\mathbf{k}+\mu(\mathbf{i}+4\mathbf{j}-2\mathbf{k})$, or equivalent **A1**

Total: 6

Coaching with Shuja

Question 9

Feb/Mar 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "C+Dx (ii) Unless candidates had found the constant D in (i) to be zero their integration of the term x2 +2 proved difficult."

Source: 9709_m16_er.pdf, Page 9, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

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Let $f(x) = \frac{3x^3+6x-8}{x(x^2+2)}$.

(i) Express $f(x)$ in the form $A + \frac{B}{x} + \frac{Cx+D}{x^2+2}$. [5]

(ii) Show that $\int_1^2 f(x)\,dx = 3 - \ln 4$. [5]

(i)

State or obtain $A = 3$ **B1**

Use a relevant method to find a constant **M1**

Obtain one of $B = -4$, $C = 4$ and $D = 0$ **A1**

Obtain a second value **A1**

Obtain the third value **A1**

Total: 5

(ii)

Integrate and obtain $3x - 4\ln x$ **B1**

Integrate and obtain term of the form $k\ln(x^2 + 2)$ **M1**

Obtain term $2\ln(x^2 + 2)$ **A1**

Substitute limits in an integral of the form $ax + b\ln x + c\ln(x^2 + 2)$, where $abc \neq 0$ **M1**

Obtain given answer $3 - \ln 4$ after full and correct working **A1**

Total: 5

Coaching with Shuja

Question 5

May/June 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 5 This question was very well done by stronger candidates but weaker ones struggled and either missed out the second part completely or failed to justify the given answer."

Source: 9709_s16_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 5

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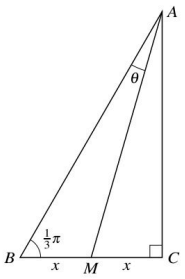
5



In the diagram, triangle ABC is right-angled at C and M is the mid-point of BC. It is given that angle $ABC = \frac{1}{3}\pi$ radians and angle $BAM = \theta$ radians. Denoting the lengths of BM and MC by x,

(i) find AM in terms of x, [3]

(ii) show that $\theta = \frac{1}{6}\pi - \tan^{-1}\left(\frac{1}{2\sqrt{3}}\right)$. [2]

In the diagram, triangle $\boldsymbol{ABC}$ is right-angled at $\boldsymbol{C}$ and $\boldsymbol{M}$ is the mid-point of $\boldsymbol{BC}$. It is given that angle $\boldsymbol{ABC = \frac{\pi}{3}}$ radians and angle $\boldsymbol{BAM = \theta}$ radians. Denoting the lengths of $\boldsymbol{BM}$ and $\boldsymbol{MC}$ by $\boldsymbol{x}$,

(i) find $\boldsymbol{AM}$ in terms of $\boldsymbol{x}$, [3]

(ii) show that $\boldsymbol{\theta = \frac{\pi}{6} - \tan^{-1}\left(\frac{1}{2\sqrt{3}}\right)}$. [2]

(i)

$\tan\left(\frac{\pi}{3}\right) = \frac{AC}{2x}$ or equivalent trig ratio **B1**

$AC = 2\sqrt{3}x$ or $AB = 4x$ **M1**

$AM = \sqrt{13}x$ or $\frac{\sqrt{52}}{2}x$ or $3.61x$ **A1**

Total: 3

(ii)

$\tan(\widehat{MAC}) = \frac{x}{AC}$ **M1** Their AC must be a function of x.

$\theta = \frac{\pi}{6} - \tan^{-1}\left(\frac{1}{2\sqrt{3}}\right)$ **A1** AG. Justifies $\frac{\pi}{6}$ and links MAC and θ.

Total: 2

Coaching With Shuja

Question 6

May/June 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Candidates could usually find PQ and PT but QT proved much more of a challenge."

> "Part (ii) was more successfully answered with full marks often seen but a significant number did fail to give the final answer to 2 significant figures as instructed in the question."

Source: 9709_s16_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 6

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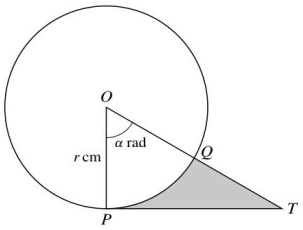
6



The diagram shows a circle with radius r cm and centre O. The line PT is the tangent to the circle at P and angle $POT = \alpha$ radians. The line OT meets the circle at Q.

(i) Express the perimeter of the shaded region PQT in terms of r and α. [3]

(ii) In the case where $\alpha = \frac{1}{3}\pi$ and $r = 10$, find the area of the shaded region correct to 2 significant figures. [3]

$$\frac{1+\cos\theta}{1-\cos\theta} \equiv 4$$

The diagram shows a circle with radius r cm and centre O. The line PT is the tangent to the circle at P and angle $POT = \theta$ radians. The line OT meets the circle at Q.

(i) Express the perimeter of the shaded region PQT in terms of r and θ. [3]

(ii) In the case where $\theta = \frac{\pi}{3}$ and $r = 10$, find the area of the shaded region correct to 2 significant figures. [3]

(i)

$PT = r\tan\theta$ **B1**

$QT = OT - OQ = \frac{r}{\cos\theta} - r$ or equivalent **B1**

Perimeter is sum of the three parts including $r\theta$ **B1**

Total: 3

(ii)

Area of triangle $= \frac{1}{2} \times 10 \times 10\tan\frac{\pi}{3}$ **M1** Correct formula used.

Area of sector $= \frac{1}{2} \times 10^2 \times \frac{\pi}{3}$ **M1** Correct formula used.

Shaded region has area **34 A1** 2 s.f.

Total: 3

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Question 7

May/June 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Some weaker candidates failed to use the identity or misread the question and ignored the sinθ and so the relatively straightforward task of isolating tanθ 3 became much more difficult."

Source: 9709_s16_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 7

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(i) Prove the identity

$$\frac{1+\cos\theta}{1-\cos\theta}-\frac{1-\cos\theta}{1+\cos\theta}\equiv\frac{4}{\sin\theta\tan\theta}.$$

[4]

(ii) Hence solve, for $0°<\theta<360°$, the equation

$$\sin\theta\left(\frac{1+\cos\theta}{1-\cos\theta}-\frac{1-\cos\theta}{1+\cos\theta}\right)=3.$$

[3]

(i)

Attempt at combining fractions **M1**

LHS $= \frac{(1+c)^2-(1-c)^2}{(1-c)(1+c)}$ **A1 A1** A1 for numerator, A1 for denominator.

$= \frac{4c}{1-c^2} = \frac{4c}{s^2}$ **A1** Essential step for award.

$= \frac{4}{\sin\theta\tan\theta}$ AG

Total: 4

(ii)

Use part (i) to eliminate sine correctly **M1**

$\sin\theta \times \frac{4}{\sin\theta\tan\theta} = 3$, hence $\tan\theta = \frac{4}{3}$ **A1**

$\theta = 53.1°$ and $233.1°$ **A1**

Total: 3

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Question 10

May/June 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Some weaker 8 candidates struggled with the term and were unable to successfully differentiate, or more commonly, x 64 square and integrate it."

> "In part (iii) many candidates failed to clearly show both of the limits substituted and because the integral was often incorrect it was not possible to award any method marks."

Source: 9709_s16_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

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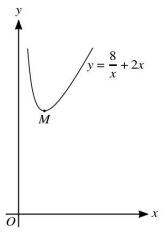
10



The diagram shows the part of the curve $y = \frac{8}{x} + 2x$ for $x > 0$, and the minimum point M.

(i) Find expressions for $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and $\int y^2\,dx$. [5]

(ii) Find the coordinates of M and determine the coordinates and nature of the stationary point on the part of the curve for which $x < 0$. [5]

(iii) Find the volume obtained when the region bounded by the curve, the x-axis and the lines $x = 1$ and $x = 2$ is rotated through $360°$ about the x-axis. [2]

The diagram shows the part of the curve $y = \frac{8}{x} + 2x$ for $x > 0$, and the minimum point M.

(i) Find expressions for $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and $\int y^2\,dx$. [5]

(ii) Find the coordinates of M and determine the coordinates and nature of the stationary point on the part of the curve for which $x < 0$. [5]

(iii) Find the volume obtained when the region bounded by the curve, the x-axis and the lines $x = 1$ and $x = 2$ is rotated through $360°$ about the x-axis. [2]

(i)

$\frac{dy}{dx} = -8x^{-2} + 2$ **B1** Unsimplified OK.

$\frac{d^2y}{dx^2} = 16x^{-3}$ **B1** Unsimplified OK.

$\int y^2\,dx = -64x^{-1} + 32x + \frac{4}{3}x^3$ **B3** B1 for each term, unsimplified OK.

Total: 5

(ii)

Set $\frac{dy}{dx} = 0$ leading to $x = \pm 2$ **M1**

$M = (2, 8)$ **A1**

Other turning point is $(-2, -8)$ **A1**

If $x = -2$, $\frac{d^2y}{dx^2} < 0$ **M1** Using their second derivative if kx^{-3} and $x < 0$.

Hence maximum **A1**

Total: 5

(iii)

Volume $= \pi \times$ [**their integral of** y^2] from 1 to 2 **M1** Evidence of using limits 1 and 2.

$\frac{220\pi}{3}$, 73.3π or 230 **A1**

Total: 2

Question 11

May/June 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Part (ii) was found to be the most challenging part of the question for candidates but there were many who were able to equate, re-arrange and then use the discriminant to obtain the required equation."

Source: 9709_s16_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 12), Question 11

Coaching With Shuja

Question Paper May/June 2016 • Paper 1 (variant 12) • Q11

The function f is defined by $f : x \mapsto 6x - x^2 - 5$ for $x \in \mathbb{R}$.

(i) Find the set of values of x for which $f(x) \leq 3$. [3]

(ii) Given that the line $y = mx + c$ is a tangent to the curve $y = f(x)$, show that $4c = m^2 - 12m + 16$. [3]

The function g is defined by $g : x \mapsto 6x - x^2 - 5$ for $x \geq k$, where k is a constant.

(iii) Express $6x - x^2 - 5$ in the form $a - (x - b)^2$, where a and b are constants. [2]

(iv) State the smallest value of k for which g has an inverse. [1]

(v) For this value of k, find an expression for $g^{-1}(x)$. [2]

Coaching With Shuja

(i)

$6x - x^2 - 5 \leq 3$ **M1**

$x^2 - 6x + 8 \geq 0$, with critical values $x = 2, x = 4$ **A1**

$x \leq 2, x \geq 4$ **A1** Condone strict inequalities; accept recognisable notation.

Total: 3

(ii)

Equate $mx + c$ and $6x - x^2 - 5$ and set to zero **M1**

Use discriminant with values independent of x **DM1**

$4c = m^2 - 12m + 16$ **A1** AG, with $= 0$ appearing before last line.

Total: 3

(iii)

$6x - x^2 - 5 = 4 - (x - 3)^2$ **B1 B1 4** for B1 and $-(x - 3)^2$ for B1.

Total: 2

(iv)

$k = 3$ **B1**

Total: 1

(v)

$g^{-1}(x) = 3 + \sqrt{4 - x}$ **M1 A1** Correct order of operations.

Total: 2

Coaching With Shuja

Question 5

May/June 2016 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "The most common error was the failure to apply the chain rule when finding the first and/or second derivative and thereby losing a factor of 2."

Source: 9709_s16_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 5

Coaching With Shuja

Question Paper May/June 2016 • Paper 1 (variant 13) • Q5

A curve has equation $y = 8x + (2x - 1)^{-1}$. Find the values of x at which the curve has a stationary point and determine the nature of each stationary point, justifying your answers. [7]

$\frac{dy}{dx} = 8 - 2(2x-1)^{-2}$ **B2,1,0**

Set to zero, simplify and attempt to solve **M1** e.g. $(2x-1)^2 = \frac{1}{4}$ or $16x^2 - 16x + 3 = 0$.

$x = \frac{1}{4}$ and $x = \frac{3}{4}$ **A1** Needs both x values; ignore y values.

$\frac{d^2y}{dx^2} = 8(2x-1)^{-3}$ **B1** FT to $k(2x-1)^{-3}$ where $k > 0$.

When $x = \frac{1}{4}$, $\frac{d^2y}{dx^2} = -64$ and/or < 0, maximum **DB1**

When $x = \frac{3}{4}$, $\frac{d^2y}{dx^2} = 64$ and/or > 0, minimum **DB1**

Total: 7

Coaching With Shuja

Question 7

May/June 2016 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Question 7 This question provided a challenge even for some candidates of high ability."

Source: 9709_s16_er.pdf, Page 11, Section: Comments on specific questions – Paper 1 (variant 13), Question 7

Coaching With Shuja

Question Paper May/June 2016 • Paper 1 (variant 13) • Q7

The point $P(x, y)$ is moving along the curve $y = x^2 - \frac{10}{3}x^{\frac{3}{2}} + 5x$ in such a way that the rate of change of y is constant. Find the values of x at the points at which the rate of change of x is equal to half the rate of change of y. [7]

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$\frac{dy}{dx} = 2x - 5x^{1/2} + 5$ **B1**

$\frac{dy}{dx} = 2$ **B1**

Equate their $\frac{dy}{dx}$ to 2 **M1**

$2x - 5x^{1/2} + 3 = 0$ or equivalent 3-term quadratic **A1**

Attempt to solve for $x^{1/2}$, e.g. $(2x^{1/2} - 3)(x^{1/2} - 1) = 0$ **DM1** Dependent on 3-term quadratic.

$x^{1/2} = \frac{3}{2}$ and 1 **A1**

$x = \frac{9}{4}$ and 1 **A1**

Total: 7

Coaching With Shuja

Question 10

May/June 2016 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Part (ii) proved to be quite challenging for most candidates."

Source: 9709_s16_er.pdf, Page 11, Section: Comments on specific questions – Paper 1 (variant 13), Question 10

Coaching With Shuja

Question Paper May/June 2016 • Paper 1 (variant 13) • Q10

The function f is such that $f(x) = 2x + 3$ for $x \geq 0$. The function g is such that $g(x) = ax^2 + b$ for $x \leq q$, where a, b and q are constants. The function fg is such that $fg(x) = 6x^2 - 21$ for $x \leq q$.

(i) Find the values of a and b. [3]

(ii) Find the greatest possible value of q. [2]

It is now given that $q = -3$.

(iii) Find the range of fg. [1]

(iv) Find an expression for $(fg)^{-1}(x)$ and state the domain of $(fg)^{-1}$. [3]

(i)

$2(ax^2 + b) + 3 = 6x^2 - 21$ **M1**

$a = 3$ **A1**

$b = -12$ **A1**

Total: 3

(ii)

$3x^2 - 12 \geq 0$ or $6x^2 - 21 \geq 3$ **M1** FT from their a, b.

$x \leq -2$, so greatest $q = -2$ **A1** Must be in terms of q.

Total: 2

(iii)

Range is $y \geq 33$ **B1** Accept $[33, \infty)$.

Total: 1

(iv)

$y = 6x^2 - 21$ implies $x = \pm\sqrt{\frac{y+21}{6}}$ **M1**

$(fg)^{-1}(x) = -\sqrt{\frac{x+21}{6}}$ **A1** Must be a function of x.

Domain is $x \geq 33$ **B1** FT from their part (iii), but x essential.

Total: 3

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Question 11

May/June 2016 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "In part (i) a significant number of candidates did not take note of the need to find the exact area and so lost marks by, for example, using a calculator to find angles and so introducing rounding errors."

Source: 9709_s16_er.pdf, Page 12, Section: Comments on specific questions – Paper 1 (variant 13), Question 11

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Question Paper May/June 2016 • Paper 1 (variant 13) • Q11

Triangle ABC has vertices at $A(-2, -1)$, $B(4, 6)$ and $C(6, -3)$.

(i) Show that triangle ABC is isosceles and find the exact area of this triangle. [6]

(ii) The point D is the point on AB such that CD is perpendicular to AB. Calculate the x-coordinate of D. [6]

Coaching With Shuja

(i)

$AB^2 = 6^2 + 7^2 = 85$ **B1**

$BC^2 = 2^2 + 9^2 = 85$, hence triangle is isosceles **B1**

$AC^2 = 8^2 + 2^2 = 68$ **B1**

$M = (2, -2)$ or $BM^2 = 85 - \left(\frac{1}{2}\sqrt{68}\right)^2$ **B1** Where M is midpoint of AC.

$BM^2 = 8^2 + \sqrt{68}^2$ or $85 - 17 = 68$ **B1**

Area of triangle $ABC = \frac{1}{2}\sqrt{68}\sqrt{68} = 34$ **B1**

Total: 6

(ii)

Gradient of $AB = \frac{7}{6}$ **B1**

Equation of AB is $y + 1 = \frac{7}{6}(x + 2)$ **M1**

Gradient of $CD = -\frac{6}{7}$ **M1**

Equation of CD is $y + 3 = -\frac{6}{7}(x - 6)$ **M1**

Solve simultaneous equations **M1**

$x = \frac{34}{5}$ **A1**

Total: 6

Coaching With Shuja

Question 6

May/June 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "2 ☐ x ☐ Many candidates did not appear to understand what they needed to do, or did not give sufficient explanation to support the given answer."

Source: 9709_s16_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

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(i) By sketching a suitable pair of graphs, show that the equation

$$5e^{-x} = \sqrt{x}$$

has one root. [2]

(ii) Show that, if a sequence of values given by the iterative formula

$$x_{n+1} = \frac{1}{2}\ln\left(\frac{25}{x_n}\right)$$

converges, then it converges to the root of the equation in part (i). [2]

(iii) Use this iterative formula, with initial value $x_1 = 1$, to calculate the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

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(i)

Make recognisable sketch of a relevant graph **B1**

Sketch the other relevant graph and justify the given statement **B1**

Total: 2

(ii)

State $x = \frac{1}{2}\ln(25/x)$ **B1**

Rearrange this in the form $5e^{-x} = \sqrt{x}$ **B1**

Total: 2

(iii)

Use the iterative formula correctly at least once **M1**

Obtain final answer **1.43 A1**

Show sufficient iterations to 4 d.p. to justify **1.43** to 2 d.p., or show there is a sign change in the interval $(1.425, 1.435)$ **A1**

Total: 3

Coaching with Shuja

Question 7

May/June 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(ii) This part of the question proved to be challenging for many candidates It required an dy understanding that for the tangent to be parallel to the x-axis, they needed =0and therefore dx x2 −2xy =0."

Source: 9709_s16_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

Question Paper May/June 2016 • Paper 3 (variant 31) • Q7

The equation of a curve is $x^3 - 3x^2y + y^3 = 3$.

(i) Show that $\frac{dy}{dx} = \frac{x^2-2xy}{x^2-y^2}$. [4]

(ii) Find the coordinates of the points on the curve where the tangent is parallel to the x-axis. [5]

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(i)

State or imply $6xy + 3x^2\frac{dy}{dx}$ as derivative of $3x^2y$ **B1**

State $3y^2\frac{dy}{dx}$ as derivative of y^3 **B1**

Equate attempted derivative of the LHS to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain the given answer **A1**

Total: 4

(ii)

Equate numerator to zero **M1***

Obtain $x = 2y$, or equivalent **A1**

Obtain an equation in x or y **DM1***

Obtain the point $(-2, -1)$ **A1**

State the point $(0, 1.44)$ **B1**

Total: 5

Coaching With Shuja

Question 9

May/June 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common error was to include the position vector of D in the working, when it is not given that D lies on the plane."

Source: 9709_s16_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 31), Question 9

Coaching With Shuja

With respect to the origin O, the points A, B, C, D have position vectors given by

$$\overrightarrow{OA} = \mathbf{i} + 3\mathbf{j} + 2\mathbf{k},$$

$$\overrightarrow{OB} = 2\mathbf{i} + \mathbf{j} - \mathbf{k},$$

$$\overrightarrow{OC} = 2\mathbf{i} + 4\mathbf{j} + \mathbf{k},$$

$$\overrightarrow{OD} = -3\mathbf{i} + \mathbf{j} + 2\mathbf{k}.$$

(i) Find the equation of the plane containing A, B and C, giving your answer in the form $ax + by + cz = d$. [6]

(ii) The line through D parallel to OA meets the plane with equation $x + 2y - z = 7$ at the point P. Find the position vector of P and show that the length of DP is $2\sqrt{14}$. [5]

(i)

Obtain a vector parallel to the plane, e.g. $\overrightarrow{AB} = \mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$ **B1**

Use scalar product to obtain an equation in a, b, c, e.g. $a - 2b - 3c = 0$, $a + b - c = 0$, or $3b + 2c = 0$ **M1**

State two correct equations **A1**

Solve to obtain ratio $a : b : c$ **M1**

Obtain $a : b : c = 5 : -2 : 3$ **A1**

Obtain equation $5x - 2y + 3z = 5$, or equivalent **A1**

Total: 6

(ii)

Correctly form an equation for the line through D parallel to OA **M1**

Obtain a correct equation e.g. $\mathbf{r} = -3\mathbf{i} + \mathbf{j} + 2\mathbf{k} + \lambda(\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$ **A1**

Substitute components in the equation of the plane and solve for λ **M1**

Obtain $\lambda = 2$ and position vector $-\mathbf{i} + 7\mathbf{j} + 6\mathbf{k}$ for P **A1**

Obtain the given answer correctly **A1**

Total: 5

Coaching with Shuja

Question 10

May/June 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The half line for arg$z-1$ = proved to be more challenging."

Source: 9709_s16_er.pdf, Page 25, Section: Comments on specific questions – Paper 3 (variant 31), Question 10

Coaching With Shuja

Question Paper May/June 2016 • Paper 3 (variant 31) • Q10

(a) Showing all your working and without the use of a calculator, find the square roots of the complex number $7-(6\sqrt{2})i$. Give your answers in the form $x+iy$, where x and y are real and exact. [5]

(b)(i) On an Argand diagram, sketch the loci of points representing complex numbers w and z such that $|w-1-2i|=1$ and $\arg(z-1)=\frac{3}{4}\pi$. [4]

(b)(ii) Calculate the least value of $|w-z|$ for points on these loci. [2]

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(a)

Square $x + iy$ and equate real and imaginary parts to 7 and $-6\sqrt{2}$ respectively **M1**

Obtain equations $x^2 - y^2 = 7$ and $2xy = -6\sqrt{2}$ **A1**

Eliminate one variable and find an equation in the other **M1**

Obtain $x^4 - 7x^2 - 18 = 0$ or $y^4 + 7y^2 - 18 = 0$, or 3-term equivalent **A1**

Obtain answers $\pm(3 - \sqrt{2}i)$ **A1**

Total: 5

(b)(i)

Show point representing $1 + 2i$ **B1**

Show circle with radius 1 and centre $1 + 2i$ **B1**

Show a half-line from the point representing 1 **B1**

Show line making the correct angle with the real axis **B1**

Total: 4

(b)(ii)

State or imply the relevance of the perpendicular from $1 + 2i$ to the line **M1**

Obtain answer $\sqrt{2} - 1$ or 0.414 **A1**

Total: 2

Coaching With Shuja

Question 1

May/June 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “On the left the omission of brackets in (3x−1)ln4 was a common error.”

Source: 9709_s16_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

Coaching With Shuja

Question Paper May/June 2016 • Paper 3 (variant 32) • Q1

Use logarithms to solve the equation

$$4^{3x-1} = 3(5^x),$$

giving your answer correct to 3 decimal places. [4]

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Use a law of logarithms for a product, power or quotient **M1**

Obtain a correct linear equation, e.g.

$$(3x - 1)\ln 4 = \ln 3 + x \ln 5$$

A1

Solve a linear equation for x **DM1**

Obtain answer $x = 0.975$ **A1**

Total: 4

Question 9

May/June 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A common error was to equate CD to AB rather than BA."

> "A common error was to think that the equality of a pair of parallel sides was sufficient."

Source: 9709_s16_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

The points A, B and C have position vectors, relative to the origin O, given by

$$\overrightarrow{OA} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k},$$

$$\overrightarrow{OB} = 4\mathbf{j} + \mathbf{k}$$

and

$$\overrightarrow{OC} = 2\mathbf{i} + 5\mathbf{j} - \mathbf{k}.$$

A fourth point D is such that the quadrilateral $ABCD$ is a parallelogram.

(i) Find the position vector of D and verify that the parallelogram is a rhombus. [5]

(ii) The plane p is parallel to OA and the line BC lies in p. Find the equation of p, giving your answer in the form $ax + by + cz = d$. [5]

(i)

Either state or imply $\overrightarrow{AB}$ or $\overrightarrow{BC}$ in component form, or state the position vector of the midpoint of AC **B1**

Use a correct method for finding the position vector of D **M1**

Obtain answer $\mathbf{3i} + \mathbf{3j} + \mathbf{k}$, or equivalent **A1**

Either use the correct process for the moduli and compare lengths of a pair of adjacent sides, e.g. AB and BC **M1**

Show that $ABCD$ has a pair of adjacent sides that are equal **A1**

Alternatively, calculate $\overrightarrow{AC} \cdot \overrightarrow{BD}$, or equivalent, and show that the diagonals are perpendicular.

Total: 5

(ii)

Either state

$$a + 2b + 3c = 0$$

or

$$2a + b - 2c = 0$$

B1

Obtain two relevant equations and solve for one ratio, e.g. $a : b$ **M1**

Obtain $a : b : c = -7 : 8 : -3$, or equivalent **A1**

Substitute coordinates of a relevant point in $-7x + 8y - 3z = d$ and evaluate **M1**

Obtain answer

$$-7x + 8y - 3z = 29,$$

or equivalent **A1**

Total: 5

Coaching With Shuja

Question 10

May/June 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Those with unequal scales on the axes found this difficult."

> "A common error was to take the locus to be a circle with centre at the above point (ii) This was well answered by those with a correct diagram."

Source: 9709_s16_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

(a) Showing all necessary working, solve the equation

$$iz^2 + 2z - 3i = 0,$$

giving your answers in the form $x + iy$, where x and y are real and exact. [5]

(b)

(i) On a sketch of an Argand diagram, show the locus representing complex numbers satisfying the equation

$$|z| = |z - 4 - 3i|.$$

[2]

(ii) Find the complex number represented by the point on the locus where $|z|$ is least. Find the modulus and argument of this complex number, giving the argument correct to 2 decimal places. [3]

Coaching With Shuja

(a)

Either use the quadratic formula to solve for z **M1**

Use $i^2 = -1$ **M1**

Obtain a correct answer in any form, simplified as far as

$$\frac{-2 \pm i\sqrt{8}}{2i}$$

A1

Multiply numerator and denominator by i, or equivalent **M1**

Obtain final answers $\sqrt{2} + i$ and $-\sqrt{2} + i$ **A1**

Alternatively, substitute $x + iy$ and equate real and imaginary parts to zero, obtaining

$$2xy + 2x = 0$$

and

$$x^2 - y^2 + 2y - 3 = 0,$$

or equivalent, then solve for x and y.

Total: 5

(b)(i)

Either show the point representing $4 + 3i$ in relatively correct position **B1**

Show the perpendicular bisector of the line segment joining this point to the origin **B1**

Alternatively, obtain the correct Cartesian equation of the locus in any form, e.g.

$$8x + 6y = 25,$$

and show this line.

Total: 2

(b)(ii)

State or imply the relevant point is represented by $2 + 1.5i$, or is at $(2, 1.5)$ **B1**

Obtain modulus 2.5 **B1**

Obtain argument 0.64, or $36.9°$ **B1**

Total: 3

Question 5

May/June 2016 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> “Another common error in the x term was a muddle of signs.”

Source: 9709_s16_er.pdf, Page 31, Section: Comments on specific questions – Paper 3 (variant 33), Question 5

Coaching With Shuja

Question Paper May/June 2016 • Paper 3 (variant 33) • Q5

The variables x and y satisfy the differential equation

$$\frac{dy}{dx} = e^{-2y}\tan^2 x,$$

for $0 \leq x < \frac{1}{2}\pi$, and it is given that $y = 0$ when $x = 0$. Solve the differential equation and calculate the value of y when $x = \frac{1}{4}\pi$. [8]

Coaching With Shuja

Mark Scheme May/June 2016 • Paper 3 (variant 33) • Q5

Separate variables and make a reasonable attempt at integration of either integral **M1**

Obtain term $\frac{1}{2}e^{2y}$ **B1**

Use Pythagoras **M1**

Obtain terms $\tan x - x$ **A1**

Evaluate a constant or use $x = 0$, $y = 0$ as limits in a solution containing terms $ae^{\pm 2y}$ and $b\tan x$, where $abe0$ **M1**

Obtain correct solution in any form, e.g.

$$\frac{1}{2}e^{2y} = \tan x - x + \frac{1}{2}$$

A1

Set $x = \frac{1}{4}\pi$ and use a correct method to solve an equation of the form $e^{\pm 2y} = a$, or $e^{\pm y} = a$, where $a > 0$ **M1**

Obtain answer $y = 0.179$ **A1**

Total: 8

Coaching with Shuja

Question 7

Oct/Nov 2016 | Paper 1 (variant 11)

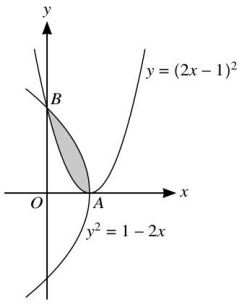
Examiner Report — Why students found this question difficult

> "The integration of (1 – 2x)1/2 proved to be quite challenging for many candidates especially the division by 3/2."

Source: 9709_w16_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 7

Coaching With Shuja

7



The diagram shows parts of the curves $y = (2x - 1)^2$ and $y^2 = 1 - 2x$, intersecting at points A and B.

(i) State the coordinates of A. [1]

(ii) Find, showing all necessary working, the area of the shaded region. [6]

The diagram shows parts of the curves $\boldsymbol{y = (2x - 1)^2}$ and $\boldsymbol{y^2 = 1 - 2x}$, intersecting at points $\boldsymbol{A}$ and $\boldsymbol{B}$.

(i) State the coordinates of $\boldsymbol{A}$. **[1]**

(ii) Find, showing all necessary working, the area of the shaded region. **[6]**

7(i)

B1 $A = (1/2, 0)$. Accept $x = 0$ at $y = 0$

Total: 1

7(ii)

$$\int (1-2x)^{1/2} dx = \left[\frac{(1-2x)^{3/2}}{3/2}\right] [\div(-2)]$$

B1B1 May be seen in a single expression

$$\int (2x-1)^2 dx = \left[\frac{(2x-1)^3}{3}\right] [\div 2]$$

B1B1 May use $\int y\, dy$, may expand

$$[0-(-1/3)] - [0-(-1/6)]$$

$$1/6$$

M1 Correct use of their limits

A1 [6]

Total: 6

Coaching with Shuja

Question 1

Oct/Nov 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Some struggled with the power of – but the majority 2 of candidates were able to obtain full marks."

Source: 9709_w16_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 12), Question 1

Coaching With Shuja

Question 1

A curve is such that $\frac{dy}{dx} = \frac{8}{\sqrt{4x+1}}$. The point $(2, 5)$ lies on the curve. Find the equation of the curve. **[4]**

Coaching With Shuja

1

$$(y') = 8(4x+1)^{-1/2} \div 1/2 \div 4 (+c)$$

B1 Correct integrand (unsimplified) without $\div 4 \div 4$. Ignore c.

B1

Uses $x = 2$ and $y = 5$

M1 Substitution of correct values into an integrand to find c.

$$c = -7$$

A1 $y = 4\sqrt{4x+1} - 7$

[4]

Total: 4

Coaching With Shuja

Question 2

Oct/Nov 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 2 Obtaining full marks for this question proved to be more difficult for candidates than expected."

> "A great many candidates were then confident in part (ii) 'in finding the inverse tan before dividing by 2 and obtaining the solution in the first quadrant but very few were able to correctly identify the solution in the fourth quadrant."

Source: 9709_w16_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 12), Question 2

Question 2

(i) Express the equation $\sin 2x + 3\cos 2x = 3(\sin 2x - \cos 2x)$ in the form $\tan 2x = k$, where k is a constant. **[2]**

(ii) Hence solve the equation for $-90^\circ \leq x < 90^\circ$. **[3]**

Coaching With Shuja

2(i)

$$2\sin 2x = 6\cos 2x$$

$$\tan 2x = k$$

$$\rightarrow \tan 2x = 3 \text{ or } k = 3$$

M1 Expand and collect as far as $\tan 2x =$ a constant from $\sin \div \cos$ soi cwo

A1 [2]

2(ii)

$$x = (\tan^{-1}(\text{their } k)) \div 2$$

$$(71.6^\circ \text{ or } -108.4^\circ) \div 2$$

$$x = 35.8^\circ, -54.2^\circ$$

$$x = 0.624^\circ, -0.946^\circ$$

$$x = 0.198\pi^\circ, -0.301\pi^\circ$$

M1 Inverse then $\div 2$. soi.

A1 A1 [3] on 1st answer $+/-90^\circ$ if in given range but no extra solutions in the given range.

Both SR A1A0

Total: 3

Question 3

Oct/Nov 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "This was usually done successful but some © 2016 Cambridge International Advanced Subsidiary and Advanced Level 9709 Mathematics November 2016 Principal Examiner Report for Teachers weaker candidates struggled to correctly evaluate it with 64 – 8(5 – k) sometimes becoming 64 – 40 – 8k."

Source: 9709_w16_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

Question 3

A curve has equation $y = 2x^2 - 6x + 5$.

(i) Find the set of values of x for which $y > 13$. **[3]**

(ii) Find the value of the constant k for which the line $y = 2x + k$ is a tangent to the curve. **[3]**

3(i)

$$2x^2 - 6x + 5 > 13$$

$$2x^2 - 6x - 8(> 0)$$

$$(x =) -1 \text{ and } 4$$

$$x > 4, x < -1$$

M1 Sets to 0 + attempts to solve. Both values required

A1 Allow all recognisable notation.

A1 [3]

3(ii)

$$2x^2 - 6x + 5 = 2x + k$$

$$\rightarrow 2x^2 - 8x + 5 - k(= 0)$$

M1 Equates and sets to 0.

Use of $b^2 - 4ac$

DM1 Use of discriminant

$$\rightarrow -3$$

A1 [3]

OR

$$\frac{dy}{dx} = 4x - 6$$

$$4x - 6 = 2$$

$$x = 2$$

$$x = 2 \rightarrow y = 1$$

$$\text{Using their } (2, 1) \text{ in } y = 2x + k$$

M1 Sets (their $\frac{dy}{dx}$) = 2

M1 Uses their $x = 2$ and their $y = 1$

DM1 Uses their $(2, 1)$ in $y = 2x^2 - 6x + 5$ or $y = 2x - 6x + 5$

$$\text{or } y = 2x^2 - 6x + 5$$

$$\rightarrow k = -3$$

A1 [3]

Total: 3

Question 5

Oct/Nov 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 5 This question was found to be difficult by many candidates although a good number of fully correct solutions were seen."

Source: 9709_w16_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 5

Coaching With Shuja

The line $\frac{x}{a} + \frac{y}{b} = 1$, where a and b are positive constants, intersects the x- and y-axes at the points A and B respectively. The mid-point of AB lies on the line $2x + y = 10$ and the distance $AB = 10$. Find the values of a and b.

Total: 6

Coaching With Shuja

(a) $A(a, 0)$ and $B(0, b)$ **B1** soi

$a^2 + b^2 = 100$ **M1*** Uses Pythagoras with their A \& B.

M has coordinates $\left(\frac{a}{2}, \frac{b}{2}\right)$ **B1*** on their A and B.

M lies on $2x + y = 10$

$\rightarrow a + \frac{b}{2} = 10$ **M1*** Subs into given line, using their M, to link a and b.

Sub $\rightarrow a^2 + (20 - 2a)^2 = 100$ **DM1** Forms quadratic in a or in b.

or $\left(10 - \frac{b}{2}\right)^2 + b^2 = 100$

$\rightarrow a = 6, b = 8$. **A1** cao **Total: 6**

Coaching With Shuja

Question 8

Oct/Nov 2016 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 8 In part (a) many candidates failed to fully appreciate the situation described."

> "The vast majority realised that an arithmetic series would occur but a significant number incorrectly used d = +5 rather than d = –5 because the distance covered each day was reducing."

> "Putting the 8 on the wrong side of the equation was a common error leading to the common ratio being 2."

Source: 9709_w16_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

Coaching With Shuja

(a) A cyclist completes a long-distance charity event across Africa. The total distance is 3050 km. He starts the event on May 1st and cycles 200 km on that day. On each subsequent day he reduces the distance cycled by 5 km.

(i) How far will he travel on May 15th? **[2]**

(ii) On what date will he finish the event? **[3]**

(b) A geometric progression is such that the third term is 8 times the sixth term, and the sum of the first six terms is $31\frac{1}{2}$. Find

(i) the first term of the progression, **[4]**

(ii) the sum to infinity of the progression. **[1]**

Coaching With Shuja

(a)(i) $200 + (15-1)(\pm/-5)$ **M1** Use of nth term with $a = 200, n = 14$ or 15 and $d = +/-5$.

$= 130$ **A1 [2]**

(ii) $\frac{n}{2}[400 + (n-1)(\pm/-5)] = (3050)$ **M1** Use of S_n, $a = 200$ and $d = +/-5$.

$\rightarrow 5n^2 - 405n + 6100 (= 0)$ **A1**

$\rightarrow 20$ **A1 [3]**

(b)(i) $ar^2, ar^5 \rightarrow r = 1/2$ **M1 A1** Both terms correct.

$\frac{63}{2} = \frac{a(1-1/2^6)}{1/2} \rightarrow a = 16$ **M1 A1** Use of $S_n = 31.5$ with a numeric r. **[4]**

(ii) Sum to infinity $= \frac{16}{1/2} = 32$ **B1*** for their a and r with $|r| < 1$. **[1]**

Coaching With Shuja

Question 1

Oct/Nov 2016 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Very many candidates failed to score the final mark, often giving the answer as $k > \pm 2$ or assuming that the correct region for k was between the critical values, giving the answer as $-2 < k < 2$."

Source: 9709_w16_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 1

Coaching With Shuja

Find the set of values of k for which the curve $y = kx^2 - 3x$ and the line $y = x - k$ do not meet. **[3]**

Coaching With Shuja

Mark Scheme

$kx^2 - 3x = x - k \Rightarrow kx^2 - 4x + k = 0$ **M1** Eliminate y and rearrange into 3-term quad

$(-4)^2 - 4(k)(k)$ soi **M1** $b^2 - 4ac$.

$k > 2$, $k < -2$ cao **A1** Allow $(2, \infty)$ etc. Allow $2 < k < -k$

Total: 3

Coaching With Shuja

Question 3

Oct/Nov 2016 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Some candidates lost marks by giving answers correct to 3 significant figures when angles in degrees need to be given correct to 1 decimal place."

Source: 9709_w16_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 3

Coaching With Shuja

Question Paper Oct/Nov 2016 • Paper 1 (variant 13) • Q3

Showing all necessary working, solve the equation $6\sin^2 x - 5\cos^2 x = 2\sin^2 x + \cos^2 x$ for $0^\circ \leq x \leq 360^\circ$. **[4]**

Coaching With Shuja

Mark Scheme

$4\sin^2 x = 6\cos^2 x \Rightarrow \tan^2 x = \frac{3}{2}$ or $4\sin^2 x = 6(1 - \sin^2 x)$ **M1** Or $4(1 - \cos^2 x) = 6\cos^2 x$

$[\tan x = (\pm)1.225$ or $\sin x = (\pm)0.7746$ or $\cos x = (\pm)0.6325]$

$x = 50.8$ (Allow 0.886 (rad)) **A1*** Another angle correct

$x = 50.8^\circ, 129.2^\circ, 230.8^\circ, 309.2^\circ$ **A1**

$[0.886, 2.25/6, 4.03, 5.40$ (rad)] **A1** Ft from 1st angle (Allow radians) All 4 angles correct in degrees

Total: 4

Coaching With Shuja

Question 5

Oct/Nov 2016 | Paper 1 (variant 13)

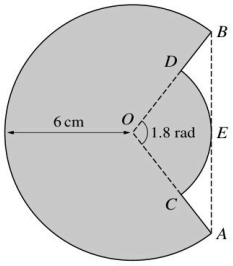
Examiner Report — Why students found this question difficult

> "Question 5 Part (i) of the question was intended as a straightforward lead into part (ii) but many candidates failed to see that all that was required was to find 6cos(0.9)."

Source: 9709_w16_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 5

Coaching With Shuja

5



The diagram shows a major arc AB of a circle with centre O and radius 6 cm. Points C and D on OA and OB respectively are such that the line AB is a tangent at E to the arc CED of a smaller circle also with centre O. Angle $COD = 1.8$ radians.

(i) Show that the radius of the arc CED is 3.73 cm, correct to 3 significant figures. [2]

(ii) Find the area of the shaded region. [4]

The diagram shows a major arc AB of a circle with centre O and radius 6 cm. Points C and D on OA and OB respectively are such that the line AB is a tangent at E to the arc CED of a smaller circle also with centre O. Angle $COD = 1.8$ radians.

(i) Show that the radius of the arc CED is 3.73 cm, correct to 3 significant figures. **[2]**

(ii) Find the area of the shaded region. **[4]**

(i) $\cos 0.9 = OE/6$ or $\sin\left(\frac{\pi}{2} - 0.9\right)$ oe **M1**

$OE = 6\cos 0.9 = 3.73$ oe **A1** AG **[2]**

(ii) Use of $(2\pi - 1.8)$ or equivalent method **M1** Expect 4.48

Area of large sector $= 1/2 \times 6^2 \times (2\pi - 1.8)$ oe **M1** Or $\pi 6^2 - 1/6 \times 1.8$. Expect 80.70 Expect 12.52

Area of small sector $1/2 \times 3.73^2 \times 1.8$

Total area $= 80.7(0) + 12.5(2) = 93.2$ **A1** Other methods possible

[4]

Coaching With Shuja

Question 6

Oct/Nov 2016 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "This proved to be quite a challenging question."

Source: 9709_w16_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 6

Coaching With Shuja

Question Paper Oct/Nov 2016 • Paper 1 (variant 13) • Q6

Three points, A, B and C, are such that B is the mid-point of AC. The coordinates of A are $(2, m)$ and the coordinates of B are $(n, -6)$, where m and n are constants.

(i) Find the coordinates of C in terms of m and n. **[2]**

The line $y = x + 1$ passes through C and is perpendicular to AB.

(ii) Find the values of m and n. **[5]**

Coaching With Shuja

(i) $\frac{2+x}{2} = n \Rightarrow x = 2n - 2$ **B1** No MR for $(1/2(2+n), 1/2(m-6))$

$\frac{m+y}{2} = -6 \Rightarrow y = -12 - m$ **B1** Expect $(2n-2, -12-m)$ **[2]**

(ii) Slope of $AB = \frac{2+x}{2} = n$; Perpendicular slope $= -1$; Point C on line:
$-12 - m = (2n-2) + 1 = 2n - 1$; So $m + 2n = -11$; Also $\frac{m-(-6)}{2-n} = -1$ so
$m + 6 = -(2-n) = n - 2$ so $m = n - 8$; Solving: $n - 8 + 2n = -11$ so $3n = -3$ so $n = -1$ and
$m = -9$. **[5]**

Coaching With Shuja

Question 8

Oct/Nov 2016 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Question 8 In part (i) many candidates lost marks by choosing to give their answer in the form a(x + b)2 + c rather than (ax + b)2 + c."

> "Many candidates did not show the ± sign for the square root – but on this occasion the + sign was the correct sign which meant that some candidates gained the mark somewhat fortunately by default."

Source: 9709_w16_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 8

Coaching With Shuja

Question Paper

Oct/Nov 2016 • Paper 1 (variant 13) • Q8

(i) Express $4x^2 + 12x + 10$ in the form $(ax + b)^2 + c$, where a, b and c are constants. **[3]**

(ii) Functions f and g are both defined for $x > 0$. It is given that $f(x) = x^2 + 1$ and $fg(x) = 4x^2 + 12x + 10$. Find $g(x)$. **[1]**

(iii) Find $(\mathrm{fg})^{-1}(x)$ and give the domain of $(\mathrm{fg})^{-1}$. **[4]**

Coaching With Shuja

(i) $(2x+3)^2+1$ Cannot score retrospectively in (iii) **B1B1B1** For $a=2, b=3, c=1$ **[3]**

(ii) $g(x)=2x+3$ cao **B1** In (ii),(iii) Allow if from $4\left(x+\frac{3}{2}\right)^2+1$ **[1]**

(iii) $y=(2x+3)^2+1 \Rightarrow 2x+3=(\pm)\sqrt{y-1}$ or ft from (i) **M1**

$x=(\pm)\frac{1}{2}\sqrt{y-1}-\frac{3}{2}$ or ft from (i) **M1** Or with x/y transposed.

$(fg)^{-1}(x)=\frac{1}{2}\sqrt{x-1}-\frac{3}{2}$ cao Note alt. method $g^{-1}[f^{-1}(x)]$ **A1** Must be a function of x. Allow $y=\ldots$

Domain is $(x)>10$ **B1** Allow $(10,\infty), 10<x<\infty$ etc. but not with y or f or g involved. Not >10 **[4]**

Coaching With Shuja

Question 9

Oct/Nov 2016 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Question 9 It was anticipated that part (a) would be a good source of marks for most candidates - but this did not altogether prove to be the case with a surprising number trying to solve $6/(1 - r) = 12/(1 - r)$."

> "However some candidates lost marks by giving θ correct to 2 significant figures whilst others gave answers in degrees rather than radians."

Source: 9709_w16_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 9

Coaching With Shuja

Question Paper Oct/Nov 2016 • Paper 1 (variant 13) • Q9

(a) Two convergent geometric progressions, P and Q, have the same sum to infinity. The first and second terms of P are 6 and $6r$ respectively. The first and second terms of Q are 12 and $-12r$ respectively. Find the value of the common sum to infinity. **[3]**

(b) The first term of an arithmetic progression is $\cos\theta$ and the second term is $\cos\theta + \sin^2\theta$, where $0 \leq \theta \leq \pi$. The sum of the first 13 terms is 52. Find the possible values of θ. **[5]**

Coaching With Shuja

(a) $\frac{6}{1-r} = \frac{12}{1+r}$ **M1**

$r = \frac{1}{3}$ **A1**

$S = 9$ **A1** **[3]**

(b) $\frac{13}{2}\left[2\cos\theta + 12\sin^2\theta\right] = 52$ **M1*** Use of correct formula for sum of AP

$2\cos\theta + 12(1-\cos^2\theta) = 8 \rightarrow 6\cos^2\theta - \cos\theta - 2(=0)$ **DM1** Use $s^2 = 1 - c^2$ \& simplify to 3-term quad

$\cos\theta = 2/3$ or $-1/2$ soi **A1**

$\theta = 0.841$ **,** 2.09 Dep on previous A1 **A1A1** Accept 0.268π, $2\pi/3$. SRA1 for 48.2°, 120° Extra solutions in range -1

[5]

Coaching With Shuja

Question 11

Oct/Nov 2016 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Those with the correct derivative were usually able to reach (kx – 3)2 = 1 but a disappointing number subsequently forgot the ± and so obtained only one stationary value."

> "expressing the function over a single denominator x – 3) little or no progress was usually made with integrating the resulting function."

Source: 9709_w16_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 11

Question Paper Oct/Nov 2016 • Paper 1 (variant 13) • Q11

(i) Find the x-coordinates of the stationary points in terms of k, and determine the nature of each stationary point, justifying your answers.

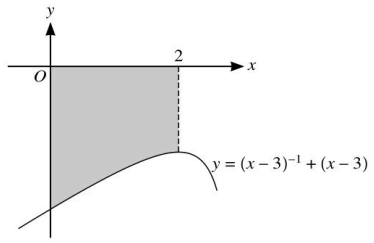
(ii) The diagram shows part of the curve for the case when $k = 1$. Showing all necessary working, find the volume obtained when the region between the curve, the x-axis, the y-axis and the line $x = 2$, shown shaded in the diagram, is rotated through 360° about the x-axis.

$$y = (kx - 3)^{-1} + (kx - 3)$$

11 A curve has equation $y = (kx - 3)^{-1} + (kx - 3)$, where k is a non-zero constant.

(i) Find the x-coordinates of the stationary points in terms of k, and determine the nature of each stationary point, justifying your answers. [7]

(ii)



The diagram shows part of the curve for the case when $k = 1$. Showing all necessary working, find the volume obtained when the region between the curve, the x-axis, the y-axis and the line $x = 2$, shown shaded in the diagram, is rotated through 360° about the x-axis. [5]

Total: 12 marks

Shuja

(i)

M1 Attempt $\frac{dy}{dx} = -k(kx-3)^{-2} + k = 0$

$(kx-3)^2 = 1$ or $k^3x^2 - 6k^2x + 8k = 0$

$x = \frac{2}{k}$ or $\frac{4}{k}$

$\frac{d^2y}{dx^2} = 2k^2(kx-3)^{-3}$

DM1 When $x = \frac{2}{k}$: $\frac{d^2y}{dx^2} = (-2k^2) < 0$ --- MAX (all previous)

DM1 When $x = \frac{4}{k}$: $\frac{d^2y}{dx^2} = (2k^2) > 0$ --- MIN (working correct)

A1A1 Legitimately obtained

B1 Must contain $(kx-3)^{-2}$ + other term(s)

Total: 7 marks

(ii)

M1 Attempt to expand y^2 and then integrate

$V = (\pi)\left[(x-3)^{-1} + (x-3)^2 + 2\right]dx$

$= (\pi)\left[-(x-3)^{-1} + \frac{(x-3)^3}{3} + (2x)\right]$ (Condone missing $2x$)

A1 Or $\left[-(x-3)^{-1} + \frac{x^3}{3} - 3x^2 + 9x + 2x\right]$

$= (\pi)\left[1 - \frac{1}{3} + 4 - \left(-\frac{1}{9} + 0\right)\right]$

$= 40\pi/3$ or 41.9

DM1 Apply limits $0 \rightarrow 2$

A1 2 missing $\rightarrow 28\pi/3$ scores M1A0A1M1A0

Total: 5 marks

Question 1

Oct/Nov 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common errors involved incorrect statements such as 8×3x =24x and 24x −3x =21x."

Source: 9709_w16_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

(1) Solve the equation $\frac{3^x+2}{3^x-2} = 8$, giving your answer correct to 3 decimal places.

Total: 3 marks

Mark Scheme Oct/Nov 2016 • Paper 3 (variant 31) • Q1

Solve for 3^x and obtain $3^x = \frac{18}{7}$

B1 Use correct method for solving an equation of the form $3^x = a$, where $a > 0$

M1 Obtain answer $x = 0.860$ 3 d.p. only

A1

Total: 3 marks

Question 2

Oct/Nov 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Many candidates lost marks in this question due to errors in their algebra and arithmetic when multiplying their expansion by (2−x)."

Source: 9709_w16_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

Coaching With Shuja

(2) Expand $(2-x)(1+2x)^{-2}$ in ascending powers of x, up to and including the term in x^2, simplifying the coefficients.

Total: 4 marks

(2)

B1 State correct unsimplified first two terms of the expansion of $(1+2x)^{-2}$, e.g. $1+(-\frac{2}{1})(2x)$

B1 State correct unsimplified term in x^2, e.g. $(-\frac{3}{2})(-\frac{1}{2}-1)(2x)^2/2!$

M1 Obtain sufficient terms of the product of $(2-x)$ and the expansion up to the term in x^2

A1 Obtain **final** answer $2-7x+18x^2$ (Do not ISW)

Total: 4 marks

Question 4

Oct/Nov 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “Question 4 Many candidates found this question challenging.”

Source: 9709_w16_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

Coaching With Shuja

(4) The equation of a curve is $xy(x - 6y) = 9a^3$, where a is a non-zero constant. Show that there is only one point on the curve at which the tangent is parallel to the x-axis, and find the coordinates of this point.

Total: 7 marks

EITHER:

B1 State $2xy + x^2\frac{dy}{dx}$, or equivalent, as derivative of x^2y

B1 State $6y^2 + 12xy\frac{dy}{dx}$, or equivalent, as derivative of $6xy^2$

B1 Differentiating LHS using correct product rule, state term $xy(1 - 6\frac{dy}{dx})$, or equivalent

B1 State term $(y + x\frac{dy}{dx})(x - 6y)$, or equivalent

M1* Equate attempted derivative of LHS to zero and set $\frac{dy}{dx}$ equal to zero

A1 Obtain a horizontal equation, e.g. $6y^2 - 2xy = 0$ (from correct work only)

A1 Explicitly reject $y = 0$ as a possibility $py^2 - qxy = 0$

DM1 Obtain an equation in x or y

A1 Obtain answer $(-3a, -a)$

OR:

B1 Rearrange to $y = \frac{9a^3}{x(x-6y)}$ and use correct quotient rule to obtain $-\frac{9a^3}{x^2(x-6y)^2}$....

B1 State term $(x - 6y) + x(1 - 6y')$, or equivalent

B1 Justify division by $x(x - 6y)$

M1* Set $\frac{dy}{dx}$ equal to zero

A1 Obtain a horizontal equation, e.g. $6y^2 - 2xy = 0$ (from correct work only)

A1 Obtain an equation in x or y

DM1 Obtain answer $(-3a, -a)$

Total: 7 marks

Question 5

Oct/Nov 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common errors involved losing the denominator when combining the two terms on the left hand side, and sign errors in removing brackets (which were often omitted)."

> "(ii) Many candidates did not see the link between the two parts of this question."

> "The lower limit of the integral often appeared to have been ignored, and several candidates struggled to deal with the logarithms correctly."

Source: 9709_w16_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

Coaching With Shuja

(i) Prove the identity $\tan 2\theta - \tan\theta \equiv \tan\theta \sec 2\theta$.

(ii) Hence show that $\int_0^{\frac{\pi}{6}} \tan\theta \sec 2\theta \, d\theta = \frac{1}{2}\ln\frac{2}{5}$.

Total: 8 marks

Coaching With Shuja

(i)

EITHER: Use $\tan 2A$ formula to express LHS in terms of $\tan\theta$

M1 Express as a single fraction in any correct form

A1 Use Pythagoras or $\cos 2A$ formula

M1 Obtain the given result correctly

A1

OR: Express LHS in terms of $\sin 2\theta, \cos 2\theta, \sin\theta$ and $\cos\theta$

M1 Express as a single fraction in any correct form

A1 Use Pythagoras or $\cos 2A$ formula or $\sin(A-B)$ formula

M1 Obtain the given result correctly

A1

Total: 4 marks

(ii)

M1* Integrate and obtain a term of the form $a\ln(\cos 2\theta)$ or $b\ln(\cos\theta)$ (or secant equivalents)

A1 Obtain integral $-\frac{1}{2}\ln(\cos 2\theta) + \ln(\cos\theta)$, or equivalent

DM1 Substitute limits correctly (expect to see use of **both** limits)

A1 Obtain the given answer following full and correct working

Total: 4 marks

Question 6

Oct/Nov 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "There were several candidates who clearly did not understand the notation sin−1(θ), the most common error being to 1 1 interpret it as ."

> "sinθ sin There were several candidates who struggled with the algebra, particularly among those who 3 3 3 tackled the problem in reverse – the error = + was common."

Source: 9709_w16_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

(i) By sketching a suitable pair of graphs, show that the equation

$$\operatorname{cosec}\frac{1}{2}x = \frac{1}{3}x + 1$$

has one root in the interval $0 < x \leq \pi$.

(ii) Show by calculation that this root lies between 1.4 and 1.6.

(iii) Show that, if a sequence of values in the interval $0 < x < \pi$ given by the iterative formula

$$x_{n+1} = 2\sin^{-1}\left(\frac{3}{x_n + 3}\right)$$

converges, then it converges to the root of the equation in part (i).

(iv) Use this iterative formula to calculate the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places.

Total: 9 marks

Coaching With Shuja

(i)

B1 Make recognizable sketch of a relevant graph

B1 Sketch the other relevant graph and justify the given statement

Total: 2 marks

(ii)

M1 Use calculations to consider the value of a relevant expression at $x = 1.4$ and $x = 1.6$, or the values of relevant expressions at $x = 1.4$ and $x = 1.6$

A1 Complete the argument correctly with correct calculated values

Total: 2 marks

(iii)

B1 State $x = 2\sin^{-1}\left(\frac{3}{x+3}\right)$

B1 Rearrange this in the form $\operatorname{cosec}\frac{1}{2}x = \frac{1}{3}x + 1$

B1 If working in reverse, need $\sin\frac{x}{2} = \left(\frac{3}{x+3}\right)$ for first **B1**

Total: 2 marks

(iv)

M1 Use the iterative formula correctly at least once

A1 Obtain final answer 1.471

A1 Show sufficient iterations to 5 d.p. to justify 1.471 to 3 d.p., or show there is a sign change in the interval (1.4705, 1.4715)

Total: 3 marks

Question 7

Oct/Nov 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "There were some sign errors at this stage, but the most common error was incorrect 1 1 1 1 x x x 1 – x differentiation of e2 , obtaining either e2 , 2e2 or occasionally e 2 ."

Source: 9709_w16_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

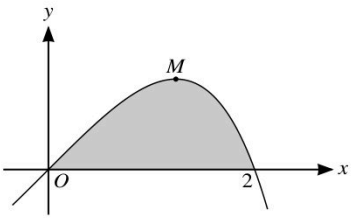
Coaching With Shuja

(i) Find the exact x-coordinate of M.

(ii) Find the exact value of the area of the shaded region bounded by the curve and the positive x-axis.

The diagram shows part of the curve $y = (2x - x^2)e^{\frac{1}{2}x}$ and its maximum point M.

7



The diagram shows part of the curve $y = (2x - x^2)e^{\frac{1}{2}x}$ and its maximum point M.

(i) Find the exact x-coordinate of M. [4]

(ii) Find the exact value of the area of the shaded region bounded by the curve and the positive x-axis. [5]

Total: 9 marks

Shuja

(i)

M1 Use the correct product rule

A1 Obtain correct derivative in any form, e.g. $(2-2x)e^{\frac{1}{2}x}+\frac{1}{2}(2x-x^2)e^{\frac{1}{2}x}$

M1 Equate derivative to zero and solve for x

A1 Obtain $x=\sqrt{5}-1$ only

Total: 4 marks

(ii)

M1* Integrate by parts and reach $a(2x-x^2)e^{\frac{1}{2}x}+b\int(2-2x)e^{\frac{1}{2}x}\,dx$

A1 Obtain $2e^{\frac{1}{2}x}(2x-x^2)-2\int(2-2x)e^{\frac{1}{2}x}\,dx$, or equivalent

A1 Complete the integration correctly, obtaining $(12x-2x^2-24)e^{\frac{1}{2}x}$, or equivalent

DM1 Use limits $x=0, x=2$ correctly having integrated by parts twice

A1 Obtain answer $24-8e$, or exact simplified equivalent

Total: 5 marks

Coaching with Shuja

Question 8

Oct/Nov 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "A common error was to have the normal vector of the first plane as $3\mathbf{i}+\mathbf{j}-2\mathbf{k}$."

Source: 9709_w16_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 8

Coaching With Shuja

(i) Show that the planes are perpendicular.

(ii) Find a vector equation for the line of intersection of the two planes.

Two planes have equations $3x + y - z = 2$ and $x - y + 2z = 3$.

Total: 9 marks

(i)

B1 State or imply a correct normal vector to either plane, e.g. $3\mathbf{i}+\mathbf{j}-\mathbf{k}$ or $\mathbf{i}-\mathbf{j}+2\mathbf{k}$

M1 Use correct method to calculate their scalar product

A1 Show value is zero and planes are perpendicular

Total: 3 marks

(ii)

EITHER: Carry out a complete strategy for finding a point on l the line of intersection

M1 Obtain such a point, e.g. $(0,7,5),(1,0,1),(5/4,-7/4,0)$

A1 State two equations for a direction vector $a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$ for l, e.g. $3a+b-c=0$ and $a-b+2c=0$

B1 Solve for one ratio, e.g. $a:b$

M1 Obtain $a:b:c=1:-7:-4$, or equivalent

A1 State a correct answer, e.g. $\mathbf{r}=7\mathbf{j}+5\mathbf{k}+\lambda(\mathbf{i}-7\mathbf{j}-4\mathbf{k})$

EITHER2: Obtain a second point on l, e.g. $(1,0,1)$

B1 Subtract vectors and obtain a direction vector for l

M1 Obtain $\mathbf{i}+\mathbf{k}+\lambda(\mathbf{i}-7\mathbf{j}-4\mathbf{k})$

A1 State a correct answer, e.g. $\mathbf{r}=\mathbf{i}+\mathbf{k}+\lambda(\mathbf{i}-7\mathbf{j}-4\mathbf{k})$

OR2: Attempt to find the vector product of the two normal vectors

M1 Obtain two correct components of the product

A1 Obtain $\mathbf{i}-7\mathbf{j}-4\mathbf{k}$, or equivalent

A1 State a correct answer, e.g. $\mathbf{r}=7\mathbf{j}+5\mathbf{k}+\lambda(\mathbf{i}-7\mathbf{j}-4\mathbf{k})$

OR1: Express one variable in terms of a second variable

M1 Obtain a correct simplified expression, e.g. $y=7-7x$

M1 Express the same variable in terms of the second

A1 Obtain a correct simplified expression e.g. $z=5-4x$

M1 Form a vector equation for the line

A1 Obtain a correct equation, e.g. $\mathbf{r}=7\mathbf{j}+5\mathbf{k}+\lambda(\mathbf{i}-7\mathbf{j}-4\mathbf{k})$

OR2: Express one variable in terms of a second variable

M1 Obtain a correct simplified expression, e.g. $z=5-4x$

M1 Express the same variable in terms of the third

A1 Obtain a correct simplified expression e.g. $z=(7+4y)/7$

M1 Form a vector equation for the line

A1 Obtain a correct equation, e.g. $\mathbf{r} = \frac{7}{4}\mathbf{i} + \frac{7}{4}\mathbf{k} + \lambda(- \frac{7}{4}\mathbf{i} + \frac{7}{4}\mathbf{j} + \mathbf{k})$

Total: 6 marks

Question 9

Oct/Nov 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The second inequality describes two half lines which should start at the origin; it was a common error to see them starting at the centre of the circle."

Source: 9709_w16_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 9

Coaching With Shuja

Question Paper Oct/Nov 2016 • Paper 3 (variant 31) • Q9

9 Throughout this question the use of a calculator is not permitted.

(a) Solve the equation $(1+2i)w^2 + 4w - (1-2i) = 0$, giving your answers in the form $x + iy$, where x and y are real. **Total: 5**

(b) On a sketch of an Argand diagram, shade the region whose points represent complex numbers satisfying the inequalities $|z - 1 - i| \leq 2$ and $-\frac{1}{4}\pi \leq \arg z \leq \frac{1}{4}\pi$. **Total: 5**

(a) M1 EITHER: Use quadratic formula to solve for w

M1 Use $i^2 = -1$

A1 Obtain one of the answers $w = \frac{1}{2i+1}$ and $w = -\frac{5}{2i+1}$

M1 Multiply numerator and denominator of an answer by $-2i + 1$, or equivalent

A1 Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$

OR1: **M1** Multiply the equation by $1 - 2i$

M1 Use $i^2 = -1$

A1 Obtain $5w^2 + 4w(1 - 2i) - (1 - 2i)^2 = 0$, or equivalent

M1 Use quadratic formula or factorise to solve for w

A1 Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$

OR2: **M1** Substitute $w = x + iy$ and form equations for real and imaginary parts

M1 Use $i^2 = -1$

A1 Obtain $(x^2 - y^2) - 4xy + 4x - 1 = 0$ and $2(x^2 - y^2) + 2xy + 4y + 2 = 0$ o.e.

M1 Form equation in x only or y only and solve

A1 Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$ **[5]**

(b) B1 Show a circle with centre $1 + i$

B1 Show a circle with radius 2

B1 Show half-line $\arg z = \frac{1}{4}\pi$

B1 Show half-line $\arg z = -\frac{1}{4}\pi$

B1 Shade the correct region **[5]**

Coaching with Shuja

Question 10

Oct/Nov 2016 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “Many candidates did not realise that ∫ dx requires x(4−x) the use of partial fractions.”

Source: 9709_w16_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 31), Question 10

10 A large field of area $4\ \text{km}^2$ is becoming infected with a soil disease. At time t years the area infected is $x\ \text{km}^2$ and the rate of growth of the infected area is given by the differential equation

$$\frac{dx}{dt} = kx(4 - x),$$

where k is a positive constant. It is given that when $t = 0$, $x = 0.4$ and that when $t = 2$, $x = 2$.

(i) Solve the differential equation and show that $k = \frac{1}{4}\ln 3$. **Total: 9**

(ii) Find the value of t when 90% of the area of the field is infected. **Total: 2**

(i) M1 Separate variables correctly and integrate at least one side

A1 Integrate and obtain term kt, or equivalent

M1** Carry out a relevant method to obtain A and B such that

$$\frac{1}{x(4-x)} = \frac{A}{x} + \frac{B}{4-x},$$

or equivalent

A1 Obtain $A = B = \frac{1}{4}$, or equivalent

A1** Integrate and obtain terms $\frac{1}{4}\ln x - \frac{1}{4}\ln(4-x)$, or equivalent

DM1 EITHER: Use a pair of limits in an expression containing $p\ln x$, $q\ln(4-x)$ and rt and evaluate a constant. Obtain correct answer in any form, e.g. $\ln x - \ln(4-x) = 4kt - \ln 9$, or

$$\ln\left(\frac{x}{4-x}\right) = 4kt - 8k$$

DM1 Use a second pair of limits and determine k. Obtain the given exact answer correctly

A1

OR: **M1**** Use both pairs of limits in a definite integral. Obtain the given exact answer correctly

A1 Substitute k and either pair of limits in an expression containing $p\ln x$, $q\ln(4-x)$ and rt and evaluate a constant. Obtain

$$\ln\frac{x}{4-x} = t\ln 3 - \ln 9,$$

or equivalent **[9]**

(ii) M1 Substitute $x = 3.6$ and solve for t

A1 Obtain answer $t = 4$ **[2]**

Question 1

Oct/Nov 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors involved incorrect statements such as 8×3x =24x and 24x −3x =21x."

Source: 9709_w16_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 32), Question 1

1 Solve the equation

$$\frac{3^x + 2}{3^x - 2} = 8,$$

giving your answer correct to 3 decimal places. **Total: 3**

Mark Scheme

B1 Solve for 3^x and obtain $3^x = \frac{18}{7}$

M1 Use correct method for solving an equation of the form $3^x = a$, where $a > 0$

A1 Obtain answer $x = 0.860$ 3 d.p. only **[3]**

Question 2

Oct/Nov 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Many candidates lost marks in this question due to errors in their algebra and arithmetic when multiplying their expansion by (2−x)."

Source: 9709_w16_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

2 Expand $(2-x)(1+2x)^{-2}$ in ascending powers of x, up to and including the term in x^2, simplifying the coefficients. **Total: 4**

B1 State correct unsimplified first two terms of the expansion of $(1+2x)^{-2}$, e.g. $1+(-2)(2x)$

B1 State correct unsimplified term in x^2, e.g. $\left(-\frac{3}{2}\right)\left(-\frac{3}{2}-1\right)(2x)^2/2!$

M1 Obtain sufficient terms of the product of $(2-x)$ and the expansion up to the term in x^2

A1 Obtain **final** answer $2-7x+18x^2$. Do not ISW **[4]**

Coaching With Shuja

Question 4

Oct/Nov 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 4 Many candidates found this question challenging."

Source: 9709_w16_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

4 The equation of a curve is $xy(x - 6y) = 9a^3$, where a is a non-zero constant. Show that there is only one point on the curve at which the tangent is parallel to the x-axis, and find the coordinates of this point.
Total: 7

Mark Scheme

B1 EITHER: EITHER: State $2xy + x^2\frac{dy}{dx}$, or equivalent, as derivative of x^2y

B1 State $6y^2 + 12xy\frac{dy}{dx}$, or equivalent, as derivative of $6xy^2$

OR: Differentiating LHS using correct product rule, state term $xy\left(1 - 6\frac{dy}{dx}\right)$, or equivalent

B1 State term $\left(y + x\frac{dy}{dx}\right)(x - 6y)$, or equivalent

B1

M1** Equate attempted derivative of LHS to zero and set $\frac{dy}{dx}$ equal to zero

A1 Obtain a horizontal equation, e.g. $6y^2 - 2xy = 0$ (from correct work only)

A1 Explicitly reject $y = 0$ as a possibility $py^2 - qxy = 0$

DM1 Obtain an equation in x or y

A1 Obtain answer $(-3a, -a)$

OR: Rearrange to $y = \frac{9a^3}{x(x-6y)}$ and use correct quotient rule to obtain $-\frac{9a^3}{x^2(x-6y)^2}\cdots$

B1 State term $(x - 6y) + x(1 - 6y')$, or equivalent

B1 Justify division by $x(x - 6y)$

M1** Set $\frac{dy}{dx}$ equal to zero

A1 Obtain a horizontal equation, e.g. $6y^2 - 2xy = 0$ (from correct work only)

DM1 Obtain an equation in x or y

A1 Obtain answer $(-3a, -a)$ **[7]**

Coaching With Shuja

Question 5

Oct/Nov 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors involved losing the denominator when combining the two terms on the left hand side, and sign errors in removing brackets (which were often omitted)."

> "(ii) Many candidates did not see the link between the two parts of this question."

> "The lower limit of the integral often appeared to have been ignored, and several candidates struggled to deal with the logarithms correctly."

Source: 9709_w16_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

Coaching With Shuja

(i) Prove the identity $\tan 2\theta - \tan\theta \equiv \tan\theta \sec 2\theta$. **Total: 4**

(ii) Hence show that

$$\int_0^{\frac{1}{6}\pi} \tan\theta \sec 2\theta \, d\theta = \frac{1}{2}\ln\frac{3}{2}.$$

Total: 4

(i) M1 EITHER: Use $\tan 2A$ formula to express LHS in terms of $\tan\theta$

A1 Express as a single fraction in any correct form

M1 Use Pythagoras or $\cos 2A$ formula

A1 Obtain the given result correctly

OR: Express LHS in terms of $\sin 2\theta$, $\cos 2\theta$, $\sin\theta$ and $\cos\theta$

A1 Express as a single fraction in any correct form

M1 Use Pythagoras or $\cos 2A$ formula or $\sin(4-B)$ formula

A1 Obtain the given result correctly **[4]**

(ii) M1** Integrate and obtain a term of the form $a\ln(\cos 2\theta)$ or $b\ln(\cos\theta)$ (or secant equivalents)

A1 Obtain integral $-\frac{1}{2}\ln(\cos 2\theta) + \ln(\cos\theta)$, or equivalent

DM1 Substitute limits correctly (expect to see use of **both** limits)

A1 Obtain the given answer following full and correct working **[4]**

Coaching with Shuja

Question 6

Oct/Nov 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "There were several candidates who clearly did not understand the notation sin−1(θ), the most common error being to 1 1 interpret it as ."

> "sinθ sin There were several candidates who struggled with the algebra, particularly among those who 3 3 3 tackled the problem in reverse – the error = + was common."

Source: 9709_w16_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

(i) By sketching a suitable pair of graphs, show that the equation

$$\operatorname{cosec}\frac{1}{2}x = \frac{1}{3}x + 1$$

has one root in the interval $0 < x \leq \pi$. **Total: 2**

(ii) Show by calculation that this root lies between 1.4 and 1.6. **Total: 2**

(iii) Show that, if a sequence of values in the interval $0 < x \leq \pi$ given by the iterative formula

$$x_{n+1} = 2\sin^{-1}\left(\frac{3}{x_n + 3}\right)$$

converges, then it converges to the root of the equation in part (i). **Total: 2**

(iv) Use this iterative formula to calculate the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **Total: 3**

(i) B1 Make recognizable sketch of a relevant graph

B1 Sketch the other relevant graph and justify the given statement **[2]**

(ii) M1 Use calculations to consider the value of a relevant expression at $x = 1.4$ and $x = 1.6$, or the values of relevant expressions at $x = 1.4$ and $x = 1.6$

A1 Complete the argument correctly with correct calculated values **[2]**

(iii) B1 State $x = 2\sin^{-1}\left(\frac{3}{x+3}\right)$

B1 Rearrange this in the form $\cosec \frac{1}{2}x = \frac{1}{3}x + 1$ **[2]**

If working in reverse, need $\sin \frac{x}{2} = \left(\frac{3}{x+3}\right)$ for first **B1**

(iv) M1 Use the iterative formula correctly at least once

A1 Obtain final answer 1.471

A1 Show sufficient iterations to 5 d.p. to justify 1.471 to 3 d.p., or show there is a sign change in the interval (1.4705, 1.4715) **[3]**

Coaching With Shuja

Question 7

Oct/Nov 2016 | Paper 3 (variant 32)

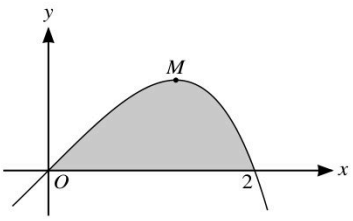
Examiner Report — Why students found this question difficult

> "There were some sign errors at this stage, but the most common error was incorrect 1 1 1 1 x x x 1 – x differentiation of e2 , obtaining either e2 , 2e2 or occasionally e 2 ."

Source: 9709_w16_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 32), Question 7

Coaching With Shuja

7



The diagram shows part of the curve $y = (2x - x^2)e^{\frac{1}{2}x}$ and its maximum point M.

(i) Find the exact x-coordinate of M. [4]

(ii) Find the exact value of the area of the shaded region bounded by the curve and the positive x-axis. [5]

The diagram shows part of the curve $\boldsymbol{y = (2x - x^2)e^{\frac{x}{2}}}$ and its maximum point $\boldsymbol{M}$.

(i) Find the exact $\boldsymbol{x}$-coordinate of $\boldsymbol{M}$. **Total: 4**

(ii) Find the exact value of the area of the shaded region bounded by the curve and the positive $\boldsymbol{x}$-axis.
Total: 5

(i) M1 Use the correct product rule

A1 Obtain correct derivative in any form, e.g. $(2-2x)e^{\frac{x}{2}} + \frac{1}{2}(2x-x^2)e^{\frac{x}{2}}$

M1 Equate derivative to zero and solve for x

A1 Obtain $x = \sqrt{5} - 1$ only **[4]**

(ii) M1** Integrate by parts and reach $a(2x-x^2)e^{\frac{x}{2}} + b\int(2-2x)e^{\frac{x}{2}}dx$

A1 Obtain $2e^{\frac{x}{2}}(2x-x^2) - 2\int(2-2x)e^{\frac{x}{2}}dx$, or equivalent

A1 Complete the integration correctly, obtaining $(12x-2x^2-24)e^{\frac{x}{2}}$, or equivalent

DM1 Use limits $x = 0, x = 2$ correctly having integrated by parts twice

A1 Obtain answer $24 - 8e$, or **exact** simplified equivalent **[5]**

Coaching With Shuja

Question 8

Oct/Nov 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A common error was to have the normal vector of the first plane as $3\mathbf{i}+\mathbf{j}-2\mathbf{k}$."

Source: 9709_w16_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 32), Question 8

8 Two planes have equations $3x + y - z = 2$ and $x - y + 2z = 3$.

(i) Show that the planes are perpendicular. **[3]**

(ii) Find a vector equation for the line of intersection of the two planes. **[6]**

Coaching With Shuja

Mark Scheme Oct/Nov 2016 • Paper 3 (variant 32) • Q8

8 (i) State or imply a correct normal vector to either plane, e.g. $3\mathbf{i}+\mathbf{j}-\mathbf{k}$ or $\mathbf{i}-\mathbf{j}+2\mathbf{k}$ **M1**

Use correct method to calculate their scalar product **M1**

Show value is zero and planes are perpendicular **A1** [3]

(ii) EITHER: Carry out a complete strategy for finding a point on l the line of intersection **M1**

Obtain such a point, e.g. $(0,7,5)$, $(1,0,1)$, $(5/4,-7/4,0)$ **A1**

EITHER: State two equations for a direction vector $a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$ for l, e.g. $3a+b-c=0$ and $a-b+2c=0$ **B1**

Solve for one ratio, e.g. $a:b$ **M1**

Obtain $a:b:c=1:-7:-4$, or equivalent **A1**

State a correct answer, e.g. $\mathbf{r}=7\mathbf{j}+5\mathbf{k}+\lambda(\mathbf{i}-7\mathbf{j}-4\mathbf{k})$ *A1^$

OR1: Obtain a second point on l, e.g. $(1,0,1)$ **B1**

Subtract vectors and obtain a direction vector for l **M1**

Obtain $\mathbf{i}-7\mathbf{j}+4\mathbf{k}$, or equivalent **A1**

State a correct answer, e.g. $\mathbf{r}=\mathbf{i}+\mathbf{k}+\lambda(\mathbf{i}-7\mathbf{j}+4\mathbf{k})$ *A1^$

OR2: Attempt to find the vector product of the two normal vectors **M1**

Obtain two correct components of the product **A1**

Obtain $\mathbf{i}-7\mathbf{j}-4\mathbf{k}$, or equivalent **A1**

State a correct answer, e.g. $\mathbf{r}=7\mathbf{j}+5\mathbf{k}+\lambda(\mathbf{i}-7\mathbf{j}-4\mathbf{k})$ *A1^$

OR1: Express one variable in terms of a second variable **M1**

Obtain a correct simplified expression, e.g. $y=7-7x$ **A1**

Express the third variable in terms of the second **M1**

Obtain a correct simplified expression, e.g. $z=5-4x$ **A1**

Form a vector equation for the line **M1**

Obtain a correct equation, e.g. $\mathbf{r}=7\mathbf{j}+5\mathbf{k}+\lambda(\mathbf{i}-7\mathbf{j}-4\mathbf{k})$ *A1^$

OR2: Express one variable in terms of a second variable **M1**

Obtain a correct simplified expression, e.g. $z=5-4x$ **A1**

Express the same variable in terms of the third **M1**

Obtain a correct simplified expression e.g. $z=(7+4y)/7$ **A1**

Form a vector equation for the line **M1**

Obtain a correct equation, e.g. $\mathbf{r} = \frac{1}{4}\mathbf{i} - \frac{7}{4}\mathbf{j} + \lambda(\frac{11}{4}\mathbf{i} + \frac{7}{4}\mathbf{j} + \mathbf{k})$ ** A1* [6]

Question 9

Oct/Nov 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The second inequality describes two half lines which should start at the origin; it was a common error to see them starting at the centre of the circle."

Source: 9709_w16_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Question Paper Oct/Nov 2016 • Paper 3 (variant 32) • Q9

9 Throughout this question the use of a calculator is not permitted.

(a) Solve the equation $(1+2i)w^2 + 4w - (1-2i) = 0$, giving your answers in the form $x + iy$, where x and y are real. [5]

(b) On a sketch of an Argand diagram, shade the region whose points represent complex numbers satisfying the inequalities $|z - 1 - i| \leq 2$ and $-\frac{1}{4}\pi \leq \arg z \leq \frac{1}{4}\pi$. [5]

Coaching With Shuja

9 (a) EITHER: Use quadratic formula to solve for w **M1**

Use $i^2 = -1$ **M1**

Obtain one of the answers $w = \frac{1}{2i+1}$ and $w = -\frac{5}{2i+1}$ **A1**

Multiply numerator and denominator of an answer by $-2i + 1$, or equivalent **M1**

Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$ **A1**

OR1: Multiply the equation by $1 - 2i$ **M1**

Use $i^2 = -1$ **M1**

Obtain $5w^2 + 4w(1 - 2i) - (1 - 2i)^2 = 0$, or equivalent **A1**

Use quadratic formula or factorise to solve for w **M1**

Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$ **A1**

OR2: Substitute $w = x + iy$ and form equations for real and imaginary parts **M1**

Use $i^2 = -1$ **M1**

Obtain $(x^2 - y^2) - 4xy + 4x - 1 = 0$ and $2(x^2 - y^2) + 2xy + 4y + 2 = 0$ o.e. **A1**

Form equation in x only or y only and solve **M1**

Obtain final answers $\frac{1}{3} - \frac{2}{3}i$ and $-1 + 2i$ **A1** [5]

(b) Show a circle with centre $1 + i$ **B1**

Show a circle with radius 2 **B1**

Show half-line arg $z = \frac{1}{4}\pi$ **B1**

Show half-line arg $z = -\frac{1}{4}\pi$ **B1**

Shade the correct region **B1** [5]

Question 10

Oct/Nov 2016 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Many candidates did not realise that ∫ dx requires x(4−x) the use of partial fractions."

Source: 9709_w16_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

10 A large field of area $4\ \text{km}^2$ is becoming infected with a soil disease. At time t years the area infected is $x\ \text{km}^2$ and the rate of growth of the infected area is given by the differential equation

$$\frac{dx}{dt} = kx(4 - x),$$

where k is a positive constant. It is given that when $t = 0$, $x = 0.4$ and that when $t = 2$, $x = 2$.

(i) Solve the differential equation and show that $k = \frac{1}{4}\ln 3$. [9]

(ii) Find the value of t when 90% of the area of the field is infected. [2]

10 (i) Separate variables correctly and integrate at least one side **M1**

Integrate and obtain term kt, or equivalent **A1**

Carry out a relevant method to obtain A and B such that

$$\frac{1}{x(4-x)} = \frac{A}{x} + \frac{B}{4-x},$$

or equivalent *M1$

Obtain $A = B = \frac{1}{4}$, or equivalent **A1**

Integrate and obtain terms $\frac{1}{4}\ln x - \frac{1}{4}\ln(4-x)$, or equivalent *A1^$

EITHER: Use a pair of limits in an expression containing $p\ln x$, $q\ln(4-x)$ and rt and evaluate a constant **DM1**

Obtain correct answer in any form, e.g. $\ln x - \ln(4-x) = 4kt - \ln 9$, or

$$\ln\left(\frac{x}{4-x}\right) = 4kt - 8k$$

A1

Use a second pair of limits and determine k **DM1**

Obtain the given exact answer correctly **A1**

OR: Use both pairs of limits in a definite integral *M1$

Obtain the given exact answer correctly **A1**

Substitute k and either pair of limits in an expression containing $p\ln x$, $q\ln(4-x)$ and rt and evaluate a constant **DM1**

Obtain $\ln\frac{x}{4-x} = t\ln 3 - \ln 9$, or equivalent **A1** [9]

(ii) Substitute $x = 3.6$ and solve for t **M1**

Obtain answer $t = 4$ **A1** [2]

Question 1

Oct/Nov 2016 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Most candidates quickly produced the correct solution, although a number struggled with the algebra at the end."

Source: 9709_w16_er.pdf, Page 31, Section: Comments on specific questions – Paper 3 (variant 33), Question 1

Coaching With Shuja

1 It is given that $z = \ln(y+2) - \ln(y+1)$. Express y in terms of z. **[3]**

Coaching With Shuja

Mark Scheme

1 Use law of the logarithm of a quotient **M1**

Remove logarithms and obtain a correct equation, e.g. $e^z = \frac{y+2}{y+1}$ **A1**

Obtain answer $y = \frac{2-e^z}{e^z-1}$, or equivalent **A1** [3]

Question 4

Oct/Nov 2016 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Unfortunately many candidates who opted for the long division approach became confused as to whether they were subtracting or adding rows, so few candidates finished with the correct remainder."

Source: 9709_w16_er.pdf, Page 32, Section: Comments on specific questions – Paper 3 (variant 33), Question 4

Coaching With Shuja

4 The polynomial $4x^4 + ax^2 + 11x + b$, where a and b are constants, is denoted by $p(x)$. It is given that $p(x)$ is divisible by $x^2 - x + 2$.

(i) Find the values of a and b. [5]

(ii) When a and b have these values, find the real roots of the equation $p(x) = 0$. [2]

Coaching With Shuja

4 (i) Commence division by $x^2 - x + 2$ and reach a partial quotient $4x^2 + kx$ **M1**

Obtain quotient $4x^2 + 4x + a - 4$ or $4x^2 + 4x + b/2$ **A1**

Equate x or constant term to zero and solve for a or b **M1**

Obtain $a = 1$ **A1**

Obtain $b = -6$ **A1** [5]

(ii) Show that $x^2 - x + 2 = 0$ has no real roots **B1**

Obtain roots $\frac{1}{2}$ and $-\frac{3}{2}$ from $4x^2 + 4x - 3 = 0$ **B1** [2]

Question 8

Oct/Nov 2016 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> “A few candidates made arithmetical errors, while others started from an incorrect form, opting for C/(x2 + 4) instead of (Bx + C) /(x2 + 4) as one of their partial fractions.”

Source: 9709_w16_er.pdf, Page 34, Section: Comments on specific questions – Paper 3 (variant 33), Question 8

Coaching With Shuja

8 Let $f(x) = \frac{3x^2+x+6}{(x+2)(x^2+4)}$.

(i) Express $f(x)$ in partial fractions. [5]

(ii) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. [5]

8 (i) State or imply the form

$$\frac{A}{x+2}+\frac{Bx+C}{x^2+4}$$

B1

Use a correct method to determine a constant **M1**

Obtain one of $A=2, B=1, C=-1$ **A1**

Obtain a second value **A1**

Obtain a third value **A1** [5]

Question 10

Oct/Nov 2016 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "After achieving the correct answer it was disappointing to see some candidates for some unknown reason decide to reject their negative solution."

Source: 9709_w16_er.pdf, Page 35, Section: Comments on specific questions – Paper 3 (variant 33), Question 10

Coaching With Shuja

10 The line l has vector equation $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + \mathbf{k} + \lambda(2\mathbf{i} - \mathbf{j} + \mathbf{k})$.

(i) Find the position vectors of the two points on the line whose distance from the origin is $\sqrt{(10)}$. [5]

(ii) The plane p has equation $ax + y + z = 5$, where a is a constant. The acute angle between the line l and the plane p is equal to $\sin^{-1}\left(\frac{2}{3}\right)$. Find the possible values of a. [5]

Coaching With Shuja

10 (i) Express general point of l in component form e.g. $(1+2\lambda, 2-\lambda, 1+\lambda)$ **B1**

Using the correct process for the modulus form an equation in λ *M1$

Reduce the equation to a quadratic, e.g. $6\lambda^2 + 2\lambda - 4 = 0$ **A1**

Solve for λ (usual requirements for solution of a quadratic) **DM1**

Obtain final answers $-\mathbf{i} + 3\mathbf{j}$ and $\frac{5}{3}\mathbf{i} + \frac{5}{3}\mathbf{j} + \frac{5}{3}\mathbf{k}$ **A1** [5]

(ii) Using the correct process, find the scalar product of a direction vector for l and a normal for p **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and equate the result to $\frac{2}{3}$ **M1**

State a correct equation in any form, e.g.

$$\frac{2a - 1 + 1}{\sqrt{(a^2+1+1)}.\sqrt{2^2+(-1)^2+1}} = \pm\frac{2}{3}$$

A1

Solve for a^2 **M1**

Obtain answer $a = \pm 2$ **A1** [5]

Question 2

May/June 2015 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 2 Many candidates did not understand the meaning of the function 'modulus' and it was very common to see ordinates with the values −9 and −7."

Source: 9709_s15_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

Coaching With Shuja

Use the trapezium rule with three intervals to find an approximation to

$$\int_0^3 |3^x - 10|\, dx.$$

[4]

Attempt calculation of at least 3 ordinates **M1**

Obtain **9**, **7**, **1**, **17** **A1**

Use trapezium rule with $h = 1$ **M1**

Obtain $\frac{1}{2}(9 + 14 + 2 + 17)$ or equivalent and hence **21** **A1**

Total: 4

Coaching With Shuja

Question 3

May/June 2015 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 3 1 Many candidates decided to express the first term in the expression as and to then expand this (1−2x2)2 denominator, hence making no progress."

Source: 9709_s15_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 3

Coaching With Shuja

Show that, for small values of x^2,

$$(1-2x^2)^{-2}-(1+6x^2)^{\frac{2}{3}} \approx kx^4,$$

where the value of the constant k is to be determined. [6]

Mark Scheme May/June 2015 • Paper 3 (variant 31) • Q3

Obtain correct unsimplified version of x^2 or x^4 term in $(1-2x^2)^{-2}$ **M1**

Obtain $1+4x^2$ **A1**

Obtain $+12x^4$ **A1**

Obtain correct unsimplified version of x^2 or x^4 term in $(1+6x^2)^{\frac{2}{3}}$ **M1**

Obtain $1+4x^2-4x^4$ **A1**

Combine expansions to obtain $k=16$ with no error seen **A1**

Total: 6

Question 9

May/June 2015 | Paper 3 (variant 31)

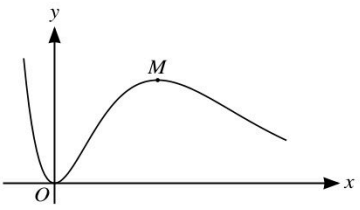
Examiner Report — Why students found this question difficult

> "In addition with a more difficult example, following exactly what the candidate is doing may not be so easy."

Source: 9709_s15_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 31), Question 9

Coaching With Shuja

9



The diagram shows the curve $y = x^2e^{2-x}$ and its maximum point M.

(i) Show that the x-coordinate of M is 2. [3]

(ii) Find the exact value of $\int_0^2 x^2e^{2-x}\,dx$. [6]

The diagram shows the curve $y = x^2e^{2-x}$ and its maximum point M.

(i) Show that the x-coordinate of M is 2. [3]

(ii) Find the exact value of $\int_0^2 x^2e^{2-x}\,dx$. [6]

Coaching with Shuja

(i)

Use product rule to find first derivative **M1**

Obtain $2xe^{2-x} - x^2e^{2-x}$ **A1**

Confirm $x = 2$ at M **A1**

Total: 3

(ii)

Attempt integration by parts and reach $\pm x^2e^{2-x} \pm \int 2xe^{2-x}\,dx$ **M1***

Obtain $-x^2e^{2-x} + \int 2xe^{2-x}\,dx$ **A1**

Attempt integration by parts and reach $-x^2e^{2-x} \pm 2xe^{2-x} \pm 2e^{2-x}$ **M1***

Obtain $-x^2e^{2-x} - 2xe^{2-x} - 2e^{2-x}$ **A1**

Use limits 0 and 2 having integrated twice **M1 dep***

Obtain $2e^2 - 10$ **A1**

Total: 6

Coaching With Shuja

Question 5

Oct/Nov 2015 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Some candidates lost marks by only failing to find the coordinates of the stationary points."

Source: 9709_w15_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 5

Coaching With Shuja

5 A curve has equation $y = \frac{8}{x} + 2x$.

(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. **[3]**

(ii) Find the coordinates of the stationary points and state, with a reason, the nature of each stationary point. **[5]**

Total: 8

5

(i) $\frac{dy}{dx} = -\frac{8}{x^2} + 2$ **cao B1B1**

$\frac{d^2y}{dx^2} = \frac{16}{x^3}$ **cao B1**

[3]

(ii) $-\frac{8}{x^2} + 2 = 0 \rightarrow 2x^2 - 8 = 0$ **M1** (Set = 0 and rearrange to quadratic form)

$x = \pm 2$ **A1**

$y = \pm 8$ **A1** (If A0A0 scored, SCA1 for just (2, 8))

$\frac{d^2y}{dx^2} > 0$ when $x = 2$ hence **MINIMUM B1** (FT for "correct" conclusion if $\frac{d^2y}{dx^2}$ incorrect or any valid method inc. a good sketch)

$\frac{d^2y}{dx^2} < 0$ when $x = -2$ hence **MAXIMUM B1**

[5]

Total: 8

Coaching With Shuja

Question 11

Oct/Nov 2015 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

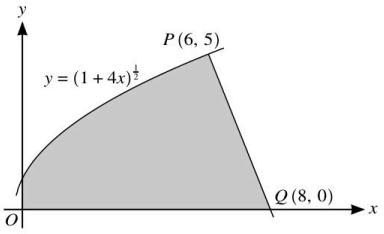
> "Those who chose to use the equation of PQ often found the integral and substitution very difficult to complete."

Source: 9709_w15_er.pdf, Page 3, Section: Comments on specific questions – Paper 1 (variant 11), Question 11

Coaching With Shuja

Question 11

11



The diagram shows part of the curve $y = (1 + 4x)^{\frac{1}{2}}$ and a point $P\,(6, 5)$ lying on the curve. The line PQ intersects the x-axis at $Q\,(8, 0)$.

(i) Show that PQ is a normal to the curve. [5]

(ii) Find, showing all necessary working, the exact volume of revolution obtained when the shaded region is rotated through 360° about the x-axis. [7]

The diagram shows part of the curve $y = (1 + 4x)^{\frac{1}{2}}$ and a point $P(6, 5)$ lying on the curve. The line PQ intersects the x-axis at $Q(8, 0)$.

(i) Show that PQ is a normal to the curve. **[5]**

(ii) Find, showing all necessary working, the exact volume of revolution obtained when the shaded region is rotated through 360° about the x-axis. **[7]**

Question 11 Solutions

(i)

$$\frac{dy}{dx} = \left[\frac{2}{2}(1+4x)^{-1/2}\right] \times [4]$$

At $x = 6$, $\frac{dy}{dx} = \frac{2}{5}$ **[B1]**

Gradient of normal at $P = -\frac{1}{2}$ **[B1]**

Gradient of $PQ = -\frac{5}{2}$ hence PQ is a normal, or $m_1 m_2 = -1$ **[B1]**

OR equation of norm: $y - 5 = their\ \left(-\frac{5}{2}\right)(x-6)$ When $y = 0$, $x = 8$ hence result **[B1]**

[5] [B1B1]

(ii) Vol for curve $= (\pi)\int(1+4x)$ and attempt to integrate y^2 **[M1]**

$$= (\pi)[x + 2x^2] \text{ ignore } `+c` **[A1]**$$

= (\pi)[6 + 72 - 0] **[DM1]**

$$= 78(\pi)$$

Vol for line $= \frac{1}{3} \times (\pi) \times 5^2 \times 2$ **[M1]**

$$= \frac{50}{3}(\pi) **[A1]**

Total Vol $= 78\pi + 50\pi/3 = 94\frac{2}{3}\pi$ (or $284\pi/3$) **[A1]**

[7]

Question 1

Oct/Nov 2015 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "A more common mistake was $35x = 10 \Rightarrow x = 3.5$."

Source: 9709_w15_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 12), Question 1

Coaching With Shuja

1 Functions f and g are defined by

$$f : x \mapsto 3x + 2, \quad x \in \mathbb{R},$$

$$g : x \mapsto 4x - 12, \quad x \in \mathbb{R}.$$

Solve the equation $f^{-1}(x) = gf(x)$. **[4]**

1 $f : x \mapsto 3x + 2$, $g : x \mapsto 4x - 12$ **B1**

$f^{-1}(x) = \frac{x-2}{3}$ **B1**

$gf(x) = 4(3x + 2) - 12$ **M1** (Equates, collects terms, +soln)

Equate $\rightarrow x = \frac{2}{7}$ **A1**

[4]

Total: 4

Coaching With Shuja

Question 3

Oct/Nov 2015 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 3 This question set in context proved to be a real challenge for many candidates with a number making no attempt to answer it."

Source: 9709_w15_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 3

Coaching With Shuja

3

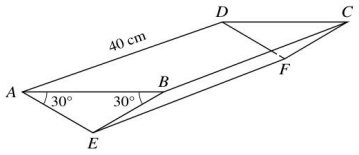


Fig. 1

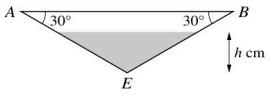


Fig. 2

Fig. 1 shows an open tank in the shape of a triangular prism. The vertical ends ABE and DCF are identical isosceles triangles. Angle ABE = angle BAE = 30°. The length of AD is 40 cm. The tank is fixed in position with the open top $ABCD$ horizontal. Water is poured into the tank at a constant rate of 200 $\text{cm}^3\ \text{s}^{-1}$. The depth of water, t seconds after filling starts, is h cm (see Fig. 2).

(i) Show that, when the depth of water in the tank is h cm, the volume, $V\ \text{cm}^3$, of water in the tank is given by $V = (40\sqrt{3})h^2$. [3]

(ii) Find the rate at which h is increasing when $h = 5$. [3]

$\left(\frac{1}{\quad}\right)^2 \quad 1 - \cos x$

Fig. 1 shows an open tank in the shape of a triangular prism. The vertical ends $\boldsymbol{ABE}$ and $\boldsymbol{DCF}$ are identical isosceles triangles. Angle $\boldsymbol{ABE}$ = angle $\boldsymbol{BAE = 30^\circ}$. The length of $\boldsymbol{AD}$ is **40** cm. The tank is fixed in position with the open top $\boldsymbol{ABCD}$ horizontal. Water is poured into the tank at a constant rate of $\boldsymbol{200}\ \text{cm}^3\ \text{s}^{-1}$. The depth of water, $\boldsymbol{t}$ seconds after filling starts, is $\boldsymbol{h}$ cm (see Fig. 2).

(i) Show that, when the depth of water in the tank is $\boldsymbol{h}$ cm, the volume, $\boldsymbol{V}\ \text{cm}^3$, of water in the tank is given by $\boldsymbol{V = (40\sqrt{3})h^2}$. **[3]**

(ii) Find the rate at which $\boldsymbol{h}$ is increasing when $\boldsymbol{h = 5}$. **[3]**

Total: 6

3

(i) $\tan 60^\circ = \frac{x}{h} \rightarrow x = h \tan 60^\circ$ **B1** (Any correct unsimplified length)

$A = h \times x$ **M1** (Correct method for area)

$V = 40\sqrt{(3h^2)}$ **A1** (ag)

[3]

(ii) $\frac{dV}{dh} = 80\sqrt{(3h)}$ **B1**

If $h = 5$, $\frac{dh}{dt} = \frac{1}{2\sqrt{3}}$ or **0.289** **M1A1** (M1: must be ÷, not ×)

[3]

Total: 6

Coaching With Shuja

Question 4

Oct/Nov 2015 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 4 In part (i) many candidates found proving the given identity very difficult."

> "Many candidates obtained cos2x – 2cosx + 1 = 0 for the numerator but factorised it as (cosx – 1)(cosx – 1) rather than (1 – cosx)(1 – cosx) and then struggled to obtain the given answer correctly."

Source: 9709_w15_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

Coaching With Shuja

4

(i) Prove the identity

$$\left(\frac{1}{\sin x} - \frac{1}{\tan x}\right) \equiv \frac{1-\cos x}{1+\cos x}$$

[4]

(ii) Hence solve the equation

$$\left(\frac{1}{\sin x} - \frac{1}{\tan x}\right)^2 = \frac{2}{5}$$

for $0 \leq x \leq 2\pi$. **[3]**

Total: 7

Coaching With Shuja

4

(i) $\left(\frac{1}{\sin x} - \frac{1}{\tan x}\right) = \left(\frac{1}{s} - \frac{c}{s}\right)$ **M1** (Use of $\tan = \sin / \cos$)

$\frac{(1-c)^2}{s^2} = \frac{(1-c)^2}{1-c^2}$ **M1** (Use of $s^2 = 1 - c^2$)

$= \frac{(1-c)(1-c)}{(1-c)(1+c)}$ or $\frac{(1-c)^2}{(1-c)(1+c)}$ **A1**

$\equiv \frac{1-\cos x}{1+\cos x}$ **A1** (ag)

[4]

(ii) $\left(\frac{1}{\sin x} - \frac{1}{\tan x}\right)^2 = \frac{2}{5}$

$\frac{1-\cos x}{1+\cos x} = \frac{2}{5} \rightarrow \cos x = \frac{3}{7}$ **M1** (Making $\cos x$ the subject)

$\rightarrow x = 1.13$ or 5.16 **A1A1** ($2\pi - 1^{st}$ answer)

[3]

Total: 7

Coaching With Shuja

Question 6

Oct/Nov 2015 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 6 Many candidates struggled with part (i) due to not appreciating that it was the lengths of AB and BC that were equal and not the gradients."

> "Many candidates were able to score full marks in part (ii) even if they had made no progress with part (i)."

Source: 9709_w15_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 6

Coaching With Shuja

6 Points A, B and C have coordinates $A(-3, 7)$, $B(5, 1)$ and $C(-1, k)$, where k is a constant.

(i) Given that $AB = BC$, calculate the possible values of k. **[3]**

The perpendicular bisector of AB intersects the x-axis at D.

(ii) Calculate the coordinates of D. **[5]**

Total: 8

Coaching With Shuja

6

(i) $AB = 10$ **B1**

$6^2 + (k-1)^2 = 10^2$ **M1** (Use of Pythagoras)

$k = -7$ and 9 **A1**

[3]

(ii) m of $AB = -\frac{3}{4}$, m perp $= \frac{4}{3}$ **B1M1** (M1: Use of $m_1 m_2 = -1$)

$M = (1, 4)$

Eqn $y - 4 = \frac{4}{3}(x - 1)$ **B1**

Set y to 0, $\rightarrow x = -2$ **M1A1** (Complete method leading to D)

[5]

Total: 8

Coaching With Shuja

Question 8

Oct/Nov 2015 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "In part (iii) many candidates seemed unable to deal with f(x + a) correctly and f(x + a) = f(x) + f(a) was sometimes seen."

> "This meant that no progress could be made with the question although the vast majority realised that b2 – 4ac < 0 was needed."

Source: 9709_w15_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

8 The function f is defined, for $x \in \mathbb{R}$, by $f : x \mapsto x^2 + ax + b$, where a and b are constants.

(i) In the case where $a = 6$ and $b = -8$, find the range of f. **[3]**

(ii) In the case where $a = 5$, the roots of the equation $f(x) = 0$ are k and $-2k$, where k is a constant. Find the values of b and k. **[3]**

(iii) Show that if the equation $f(x + a) = a$ has no real roots, then $a^2 < 4(b - a)$. **[3]**

Total: 9

Coaching With Shuja

8

(i) $x^2 + 6x - 8 = (x+3)^2 - 17$ or $2x + 6 = 0 \rightarrow x = -3 \rightarrow y = -17$ **B1B1** (B1 for $(x+3)^2$; B1 for -17 or B1 for $x = -3$, B1 $y = -17$)

$\rightarrow$ Range $f(x) \geq -17$ **B1** (Following through visible method)

[3]

(ii) $(x-k)(x+2k) = 0$ **M1** (Realises the link between roots and the equation)

$\equiv x^2 + 5x + b = 0$ **A1** (comparing coefficients of x)

$\rightarrow k = 5$ **A1**

$\rightarrow b = -2k^2 = -50$ **A1**

[3]

(iii) $(x+a)^2 + a(x+a) + b = a$ **M1** (Replaces "x" by "$x+a$" in 2 terms)

Uses $b^2 - 4ac \rightarrow 9a^2 - 4(2a^2 + b - a)$ **DM1** (Any use of discriminant)

$\rightarrow a^2 < 4(b-a)$ **A1**

[3]

Total: 9

Coaching With Shuja

Question 9

Oct/Nov 2015 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "This also led to lost marks in the second and third parts of the question."

Source: 9709_w15_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 9

Coaching With Shuja

9 The curve $y = f(x)$ has a stationary point at $(2, 10)$ and it is given that $f''(x) = \frac{12}{x^5}$.

(i) Find $f(x)$. **[6]**

(ii) Find the coordinates of the other stationary point. **[2]**

(iii) Find the nature of each of the stationary points. **[2]**

Total: 10

Coaching With Shuja

9

(i) $f'(x) = -\frac{6}{x^2}$ $(+c)$ **B1** (Correct integration)

$= 0$ when $x = 2 \rightarrow c = \frac{3}{2}$ **M1A1** (Uses $x = 2$, $f'(x) = 0$)

$f(x) = \frac{6}{x} + \frac{3x}{2}$ $(+A)$ **B1B1** (For each integral)

$= 10$ when $x = 2 \rightarrow A = 4$ **A1**

[6]

(ii) $-\frac{6}{x^2} + \frac{3}{2} = 0 \rightarrow x = \pm 2$ **M1** (Sets their 2 term $f'(x)$ to 0)

Other point is $(-2, -2)$ **A1**

[2]

(iii) At $x = 2$, $f''(x) = 1.5$ Min **B1**

At $x = -2$, $f''(x) = -1.5$ Max **B1**

[2]

Total: 10

Coaching With Shuja

Question 10

Oct/Nov 2015 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

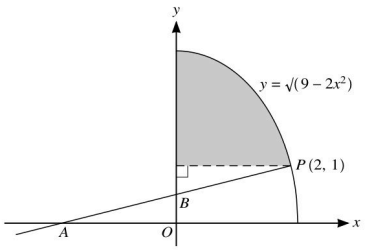
> "A common mistake was forgetting to multiply by (−4x) in the differentiation or multiplying by −4, x , 2x or −2x instead."

> "Those who realised what was required were usually successful although some struggled to obtain x on its own and then square it, whereas those finding x2 to begin with were much more successful."

Source: 9709_w15_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

10

10



The diagram shows part of the curve $y = \sqrt{(9 - 2x^2)}$. The point $P(2, 1)$ lies on the curve and the normal to the curve at P intersects the x-axis at A and the y-axis at B.

(i) Show that B is the mid-point of AP. [6]

The shaded region is bounded by the curve, the y-axis and the line $y = 1$.

(ii) Find, showing all necessary working, the exact volume obtained when the shaded region is rotated through 360° about the **y-axis**. [5]

The diagram shows part of the curve $\boldsymbol{y = \sqrt{9 - 2x^2}}$. The point $\boldsymbol{P(2, 1)}$ lies on the curve and the normal to the curve at $\boldsymbol{P}$ intersects the $\boldsymbol{x}$-axis at $\boldsymbol{A}$ and the $\boldsymbol{y}$-axis at $\boldsymbol{B}$.

(i) Show that $\boldsymbol{B}$ is the mid-point of $\boldsymbol{AP}$. **[6]**

The shaded region is bounded by the curve, the $\boldsymbol{y}$-axis and the line $\boldsymbol{y = 1}$.

(ii) Find, showing all necessary working, the exact volume obtained when the shaded region is rotated through $\boldsymbol{360^\circ}$ about the $\boldsymbol{y}$-axis. **[5]**

Total: 11

10

(i) $\frac{dy}{dx} = \frac{1}{2\sqrt{9-2x^2}} \times (-4x)$ **B1** (Without "$\times -4x$"; Allow even if B0 above)

B1

At P, $x = 2$, $m = -4$ Normal grad $= \frac{1}{4}$ **M1** (For $m_1 m_2 = -1$ calculus needed)

Eqn AP, $y - 1 = \frac{1}{4}(x - 2)$ **M1** (Normal, not tangent)

$\rightarrow A(-2, 0)$ or $B(0, \frac{1}{2})$ **A1A1** (Full justification)

Midpoint AP also $(0, \frac{1}{2})$ **A1**

[6]

(ii) $\int x^2 dy = \int \frac{9-y^2}{2} dy$ **M1** (Attempt to integrate x^2)

$= \frac{9y-y^3}{2} - \frac{y^3}{6}$ **A1** (Correct integration)

Upper limit $= 3$ **B1** (Evaluates upper limit)

Uses limits 1 to 3 **M1** (Uses both limits correctly)

$\rightarrow$ volume $= 4\frac{2}{3}\pi$ **A1**

[5]

Total: 11

Coaching With Shuja

Question 1

Oct/Nov 2015 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Other common errors included setting the discriminant to ' = 0 ' or even ' > 0 ' as well as errors made when removing the bracket from 36 – 4(c + 7)."

Source: 9709_w15_er.pdf, Page 8, Section: Comments on specific questions – Paper 1 (variant 13), Question 1

Question Paper Oct/Nov 2015 • Paper 1 (variant 13) • Q1

A line has equation $y = 2x - 7$ and a curve has equation $y = x^2 - 4x + c$, where c is a constant. Find the set of possible values of c for which the line does not intersect the curve. **[3]**

Coaching With Shuja

Mark Scheme

$x^2 - 4x + c = 2x - 7 \Rightarrow x^2 - 6x + c + 7 (= 0)$ **M1** All terms on one side

$36 - 4(c + 7) < 0$ **DM1** Apply $b^2 - 4ac < 0$. Allow $\leq$.

$c > 2$ **A1** **[3]**

Coaching With Shuja

Question 3

Oct/Nov 2015 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Others tried to work with the second derivative but could make no progress."

Source: 9709_w15_er.pdf, Page 9, Section: Comments on specific questions – Paper 1 (variant 13), Question 3

Coaching With Shuja

(i) Express $3x^2 - 6x + 2$ in the form $a(x + b)^2 + c$, where a, b and c are constants. **[3]**

(ii) The function f, where $f(x) = x^3 - 3x^2 + 7x - 8$, is defined for $x \in \mathbb{R}$. Find $f'(x)$ and state, with a reason, whether f is an increasing function, a decreasing function or neither. **[3]**

Coaching With Shuja

(i) $[3][(x-1)^2][-1]$ **B1B1B1 [3]**

(ii) $f'(x) = 3x^2 - 6x + 7$ **B1** FT their (i) + 5

$= 3(x-1)^2 + 4$ **B1√**

> 0 hence increasing **DB1** Dep B1√ unless other valid reason **[3]**

Coaching With Shuja

Question 6

Oct/Nov 2015 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Question 6 This question was set in context and a very large proportion of candidates made incorrect interpretations."

> "In addition, d = +0.04 was another common error."

Source: 9709_w15_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 6

A ball is such that when it is dropped from a height of 1 metre it bounces vertically from the ground to a height of 0.96 metres. It continues to bounce on the ground and each time the height the ball reaches is reduced. Two different models, $\boldsymbol{A}$ and $\boldsymbol{B}$, describe this.

Model $\boldsymbol{A}$: The height reached is reduced by 0.04 metres each time the ball bounces.

Model $\boldsymbol{B}$: The height reached is reduced by 4% each time the ball bounces.

(i) Find the total distance travelled vertically (up and down) by the ball from the 1st time it hits the ground until it hits the ground for the 21st time,

(a) using model $\boldsymbol{A}$, **[3]**

(b) using model $\boldsymbol{B}$. **[3]**

(ii) Show that, under model $\boldsymbol{B}$, even if there is no limit to the number of times the ball bounces, the total vertical distance travelled after the first time it hits the ground cannot exceed 48 metres. **[2]**

Coaching with Shuja

(i) (a) $1.92 + 1.84 + 1.76 + \ldots$ OE

$$\frac{20}{2}[2 \times 1.92 + 19 \times (-0.08)]$$

OE **B1** OR $a = 0.96, d = -0.04$ & ans

23.2 cao **M1** doubled/adjusted **A1** Corr formula used with corr d & their a, n; $a = 1, n = 21 \Rightarrow 12.6$ (25.2), $a = 0.96, n = 21 \Rightarrow 11.76$ (23.52) **[3]**

(b) $1.92 + 1.92(0.96) + 1.92(0.96)^2 + \ldots$ **B1**

$$\frac{1.92(1 - .96^{20})}{1 - .96}$$

M1 OR $a = .96, r = .96$ & ans /doubled/adjusted **A1** Corr formula used with $r = .96$ & their a, n; $a = .96, n = 21 \Rightarrow 13.82$ (27.63), $a = 1, n = 21 \Rightarrow 14.39$ (28.78) **[3]**

(ii)

$$\frac{1.92}{1 - .96} = 48$$

or

$$\frac{0.96}{1 - 0.96} = 24$$

& then Double AG **M1A1 [2]** $a = 1 \Rightarrow 25$ (50) but must be doubled for M1;
$\frac{1.92(1-0.96^n)}{1-0.96} < 48 \Rightarrow 0.96^n > 0$ (www) 'which is true' scores SCB1

Question 8

Oct/Nov 2015 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Question 8 Part (i) was found very difficult with most candidates failing to score any marks."

Source: 9709_w15_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 8

Coaching With Shuja

Question Paper Oct/Nov 2015 • Paper 1 (variant 13) • Q8

The function f is defined by $f(x) = 3x + 1$ for $x \leq a$, where a is a constant. The function g is defined by $g(x) = -1 - x^2$ for $x \leq -1$.

(i) Find the largest value of a for which the composite function gf can be formed. **[2]**

For the case where $a = -1$,

(ii) solve the equation $\mathrm{fg}(x) + 14 = 0$, **[3]**

(iii) find the set of values of x which satisfy the inequality $\mathrm{gf}(x) \leq -50$. **[4]**

Mark Scheme

(i) $3x + 1 \leq -1$ (Accept $3x + 1 = -1$, $3a + 1 = -1$) **M1** Do not allow gf in (i) to score in (iii)

$x \leq -2/3 \Rightarrow$ largest value of a is $-2/3$ (in terms of a) **A1** Accept $a \leq -2/3$ and $a = -2/3$ **[2]**

(ii) $\mathrm{fg}(x) = 3(-1 - x^2) + 1$ **B1** No marks in this part for gf used

$\mathrm{fg}(x) + 14 = 0 \Rightarrow 3x^2 = 12$ oe (2 terms) **B1**

$x = -2$ only **B1** **[3]**

(iii) $\mathrm{gf}(x) = -1 - (3x + 1)^2$ oe **B1** No marks in this part for gf used

$\mathrm{gf}(x) \leq -50 \Rightarrow (3x + 1)^2 \geq 49$ (Allow $<$ or $=$ 3x + 1 ≥ 7 or $3x + 1 \leq -7$ (one sufficient) www **M1** OR attempt soln of $9x^2 + 6x - 48$ + / / $\leq$ / $>$ 0 **A1**

$x \leq -8/3$ only www **A1** OR $x - 2 \geq$ or $3x + 8 \leq 0$(one suffic) **[4]**

Coaching With Shuja

Question 4

Oct/Nov 2015 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Common errors included truncation to 2.218 and giving the answer 2.22 to 3 s.f."

Source: 9709_w15_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

Coaching With Shuja

The equation $x^3 - x^2 - 6 = 0$ has one real root, denoted by α.

(i) Find by calculation the pair of consecutive integers between which α lies. **[2]**

(ii) Show that, if a sequence of values given by the iterative formula

$$x_{n+1} = \sqrt{x_n + \frac{6}{x_n}}$$

converges, then it converges to α. **[2]**

(iii) Use this iterative formula to determine α correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **[3]**

Coaching With Shuja

Mark Scheme

Oct/Nov 2015 • Paper 3 (variant 31) • Q4

(i) Evaluate, or consider the sign of, $x^3 - x^2 - 6$ for two integer values of x, or equivalent **M1** Obtain the pair $x = 2$ and $x = 3$, with no errors seen **A1 [2]**

(ii) State a suitable equation, e.g. $x = \sqrt{(x + (6/x))}$ **B1**

Rearrange this as $x^3 - x^2 - 6 = 0$, or work vice versa **B1 [2]**

(iii) Use the iterative formula correctly at least once **M1** Obtain final answer 2.219 **A1** Show sufficient iterates to 5 d.p. to justify 2.219 to 3 d.p., or show there is a sign change in the interval (2.2185, 2.2195) **A1 [3]**

Coaching With Shuja

Question 6

Oct/Nov 2015 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "A few candidates with correct working did not factorise the quotient $8x^2-2x-1$."

Source: 9709_w15_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

Question Paper Oct/Nov 2015 • Paper 3 (variant 31) • Q6

The polynomial $8x^3 + ax^2 + bx - 1$, where a and b are constants, is denoted by $\mathbf{p(x)}$. It is given that $(x + 1)$ is a factor of $\mathbf{p(x)}$ and that when $\mathbf{p(x)}$ is divided by $(2x + 1)$ the remainder is 1.

(i) Find the values of a and b. **[5]**

(ii) When a and b have these values, factorise $\mathbf{p(x)}$ completely. **[3]**

Mark Scheme

Oct/Nov 2015 • Paper 3 (variant 31) • Q6

(i) Substitute $x = -1$, equate to zero and simplify at least as far as $-8 + a - b - 1 = 0$ **B1**

Substitute $x = -\frac{1}{2}$ and equate the result to 1 **M1**

Obtain a correct equation in any form, e.g. $-1 + \frac{1}{4}a - \frac{1}{2}b - 1 = 1$ **A1**

Solve for a or for b **M1** Obtain $a = 6$ and $b = -3$ **A1 [5]**

(ii) Commence division by $(x + 1)$ reaching a partial quotient $8x^2 + kx$ **M1**

Obtain quadratic factor $8x^2 - 2x - 1$ **A1** Obtain factorisation $(x + 1)(4x + 1)(2x - 1)$ **A1 [3]**

[The M1 is earned if inspection reaches an unknown factor $8x^2 + Bx + C$ and an equation in B and/or C, or an unknown factor $Ax^2 + Bx - 1$ and an equation in A and/or B.]

[If linear factors are found by the factor theorem, give B1B1 for $(2x - 1)$ and $(4x + 1)$, and B1 for the complete factorisation.]

Coaching With Shuja

Question 4

Oct/Nov 2015 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Common errors included truncation to 2.218 and giving the answer 2.22 to 3 s.f."

Source: 9709_w15_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 32), Question 4

Coaching With Shuja

Question Paper Oct/Nov 2015 • Paper 3 (variant 32) • Q4

The equation $x^3 - x^2 - 6 = 0$ has one real root, denoted by α.

(i) Find by calculation the pair of consecutive integers between which α lies. **[2]**

(ii) Show that, if a sequence of values given by the iterative formula

$$x_{n+1} = \sqrt{x_n + \frac{6}{x_n}}$$

converges, then it converges to α. **[2]**

(iii) Use this iterative formula to determine α correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **[3]**

Mark Scheme

Oct/Nov 2015 • Paper 3 (variant 32) • Q4

(i) Evaluate, or consider the sign of, $x^3 - x^2 - 6$ for two integer values of x, or equivalent **M1** Obtain the pair $x = 2$ and $x = 3$, with no errors seen **A1 [2]**

(ii) State a suitable equation, e.g. $x = \sqrt{(x + (6/x))}$ **B1**

Rearrange this as $x^3 - x^2 - 6 = 0$, or work vice versa **B1 [2]**

(iii) Use the iterative formula correctly at least once **M1** Obtain final answer 2.219 **A1** Show sufficient iterates to 5 d.p. to justify 2.219 to 3 d.p., or show there is a sign change in the interval (2.2185, 2.2195) **A1 [3]**

Coaching With Shuja

Question 6

Oct/Nov 2015 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A few candidates with correct working did not factorise the quotient $8x^2-2x-1$."

Source: 9709_w15_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

Question Paper Oct/Nov 2015 • Paper 3 (variant 32) • Q6

The polynomial $8x^3 + ax^2 + bx - 1$, where a and b are constants, is denoted by $\mathbf{p(x)}$. It is given that $(x + 1)$ is a factor of $\mathbf{p(x)}$ and that when $\mathbf{p(x)}$ is divided by $(2x + 1)$ the remainder is 1.

(i) Find the values of a and b. **[5]**

(ii) When a and b have these values, factorise $\mathbf{p(x)}$ completely. **[3]**

Coaching With Shuja

Mark Scheme

Oct/Nov 2015 • Paper 3 (variant 32) • Q6

(i) Substitute $x = -1$, equate to zero and simplify at least as far as $-8 + a - b - 1 = 0$ **B1**

Substitute $x = -\frac{1}{2}$ and equate the result to 1 **M1**

Obtain a correct equation in any form, e.g. $-1 + \frac{1}{4}a - \frac{1}{2}b - 1 = 1$ **A1**

Solve for a or for b **M1** Obtain $a = 6$ and $b = -3$ **A1 [5]**

(ii) Commence division by $(x + 1)$ reaching a partial quotient $8x^2 + kx$ **M1**

Obtain quadratic factor $8x^2 - 2x - 1$ **A1** Obtain factorisation $(x + 1)(4x + 1)(2x - 1)$ **A1 [3]**

[The M1 is earned if inspection reaches an unknown factor $8x^2 + Bx + C$ and an equation in B and/or C, or an unknown factor $Ax^2 + Bx - 1$ and an equation in A and/or B.]

[If linear factors are found by the factor theorem, give B1B1 for $(2x - 1)$ and $(4x + 1)$, and B1 for the complete factorisation.]

Coaching With Shuja

Question 2

Oct/Nov 2015 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The most common error in the expansion was to use powers of x rather than of 9x."

Source: 9709_w15_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 2

Coaching With Shuja

2 Given that $\sqrt[3]{(1+9x)} \approx 1 + 3x + ax^2 + bx^3$ for small values of x, find the values of the coefficients a and b. **[3]**

Coaching With Shuja

2 Either State correct unsimplified x^2 or x^3 term **M1**

Obtain $a = -9$ **A1**

Obtain $b = 45$ **A1**

Or Use chain rule to differentiate twice to obtain form $k(1+9x)^{-\frac{5}{3}}$ **M1**

Obtain $f''(x) = -18(1+9x)^{-\frac{5}{3}}$ and hence $a = -9$ **A1**

Obtain $f'''(x) = 270(1+9x)^{-\frac{8}{3}}$ and hence $b = 45$ **A1** **[3]**

Question 6

Oct/Nov 2015 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> “Candidates who did not recognise sinA that they could use sec2B =1+tan2B often made no progress after forming an equation in cosB and ( ) sinB.”

Source: 9709_w15_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 6

6 The angles A and B are such that

$$\sin(A + 45^\circ) = (2\sqrt{2})\cos A \quad \text{and} \quad 4\sec^2 B + 5 = 12\tan B.$$

Without using a calculator, find the exact value of $\tan(A - B)$. **[8]**

Coaching With Shuja

6 State or imply $\sin A \times \cos 45 + \cos A \times \sin 45 = 2\sqrt{2}\cos A$ **B1**

Divide by $\cos A$ to find value of $\tan A$ **M1**

Obtain $\tan A = 3$ **A1**

Use identity $\sec^2 B = 1 + \tan^2 B$ **B1**

Solve three-term quadratic equation and find $\tan B$ **M1**

Obtain $\tan B = \frac{3}{2}$ only **A1**

Substitute **numerical values** in $\frac{\tan A - \tan B}{1 + \tan A \tan B}$ **M1**

Obtain $\frac{3}{11}$ **A1 [8]**

Coaching With Shuja

Question 7

Oct/Nov 2015 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Many candidates did not use the most efficient route to finding the values of the constants - several worked through the solution of three 3 equations in three unknowns rather than making the substitutions x =−1, x =2, and x =– ."

> "3 Many candidates who started with roots –1, 2 and – did not show where these values came from 4 ☐ 3☐ and they went on to use the incorrect factorised form (x+1)(x–2) ☐x+ ☐ so they gained very ☐ 4☐ little credit."

> "A small number of candidates made no progress at all with the integral, not recognising the potential use of partial fractions."

Source: 9709_w15_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 7

Coaching with Shuja

(i) Show that $(x+1)$ is a factor of $4x^3 - x^2 - 11x - 6$. **[2]**

(ii) Find $\int \frac{4x^2+9x-1}{4x^3-x^2-11x-6}\,dx$. **[8]**

(i) Either Substitute $x = -1$ and evaluate **M1**

Obtain **0** and conclude $x + 1$ is a factor **A1**

Or Divide by $x + 1$ and obtain a constant remainder **M1**

Obtain remainder $= 0$ and conclude $x + 1$ is a factor **A1 [2]**

(ii) Attempt division, or equivalent, at least as far as quotient $4x^2 + kx$ **M1**

Obtain complete quotient $4x^2 - 5x - 6$ **A1**

State form $\frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{4x+3}$ **A1**

Use relevant method for finding at least one constant **M1**

Obtain one of $A = -2, B = 1, C = 8$ **A1**

Obtain all three values **A1**

Integrate to obtain three terms each involving natural logarithm of linear form **M1**

Obtain $-2\ln(x+1) + \ln(x-2) + 2\ln(4x+3)$, condoning no use of modulus signs and absence of $+c$ **A1 [8]**

Coaching With Shuja

Question 9

Oct/Nov 2015 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The straight line proved to be more difficult - the most common errors were to draw it as a second circle, or to draw a horizontal straight line."

Source: 9709_w15_er.pdf, Page 30, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

Coaching With Shuja

Question Paper Oct/Nov 2015 • Paper 3 (variant 33) • Q9

(a) It is given that $(1+3i)w = 2+4i$. Showing all necessary working, prove that the exact value of $|w^2|$ is 2 and find $\arg(w^2)$ correct to 3 significant figures. **[6]**

(b) On a single Argand diagram sketch the loci $|z| = 5$ and $|z-5| = |z|$. Hence determine the complex numbers represented by points common to both loci, giving each answer in the form $re^{i\theta}$. **[4]**

Mark Scheme

Oct/Nov 2015 • Paper 3 (variant 33) • Q9

(a) Either Find w using conjugate of $1+3i$ **M1**

Obtain $\frac{7-i}{5}$ or equivalent **A1**

Square $x+iy$ form to find w^2 **M1**

Obtain $w^2=\frac{48-14i}{25}$ and confirm modulus is **2** **A1**

Use correct process for finding argument of w^2 **M1**

Obtain -0.284 radians or -16.3° **A1**

Or 1 Find w using conjugate of $1+3i$ **M1**

Obtain $\frac{7-i}{5}$ or equivalent **A1**

Find modulus of w and hence of w^2 **M1**

Confirm modulus is **2** **A1**

Find argument of w and hence of w^2 **M1**

Obtain -0.284 radians or -16.3° **A1**

Or 2 Square both sides to obtain $(-8+6i)w^2=-12+16i$ **B1**

Find w^2 using relevant conjugate **M1**

Use correct process for finding modulus of w^2 **M1**

Confirm modulus is **2** **A1**

Use correct process for finding argument of w^2 **M1**

Obtain -0.284 radians or -16.3° **A1**

Or 3 Find modulus of LHS and RHS **M1**

Find argument of LHS and RHS **M1**

Obtain $\sqrt{10}e^{i.2491}w=\sqrt{20}e^{i.1.1071}$ or equivalent **A1**

Obtain $w=\sqrt{2}e^{-0.1419i}$ or equivalent **A1**

Use correct process for finding w^2 **M1**

Obtain **2** and -0.284 radians or -16.3° **A1**

Or 4 Find moduli of $2+4i$ and $1+3i$ **M1**

Obtain $\sqrt{20}$ and $\sqrt{10}$ **A1**

Obtain $|w^2|=2$ correctly **A1**

Find $\arg(2+4i)$ and $\arg(1+3i)$ **M1**

Use correct process for $\arg(w^2)$ **A1**

Obtain -0.284 radians or -16.3° **A1**

Or 5 Let $w=a+ib$, form and solve simultaneous equations in a and b **M1**

$a=\frac{7}{5}$ and $b=-\frac{1}{5}$ **A1**

Find modulus of w and hence of w^2 **M1**

Confirm modulus is **2** **A1**

Find argument of w and hence of w^2 **M1**

Obtain -0.284 radians or -16.3° **A1**

Or 6 Find w using conjugate of $1+3i$ **M1**

Obtain $\frac{7-i}{5}$ or equivalent **A1**

Use $|w^2|=w\bar{w}$ **M1**

Confirm modulus is **2** **A1**

Find argument of w and hence of w^2 **M1**

Obtain -0.284 radians or -16.3° **A1** **[6]**

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(b) Draw circle with centre the origin and radius $\mathbf{5}$ **B1**

Draw straight line parallel to imaginary axis in correct position **B1**

Use relevant trigonometry on a correct diagram to find argument(s) **M1**

Obtain $\mathbf{5}e^{\pm\frac{\pi}{3}}$ or equivalents in required form **A1 [4]**

Question 10

Oct/Nov 2015 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The common errors were to omit the constant of proportionality, to use t as the constant, or to include an additional factor of t."

> "The most common error was in finding the value for k, usually because candidates overlooked the information that 60 was the rate of change of the number of plants, not a value for N or for t."

Source: 9709_w15_er.pdf, Page 31, Section: Comments on specific questions – Paper 3 (variant 33), Question 10

10 Naturalists are managing a wildlife reserve to increase the number of plants of a rare species. The number of plants at time t years is denoted by N, where N is treated as a continuous variable.

(i) It is given that the rate of increase of N with respect to t is proportional to $(N - 150)$. Write down a differential equation relating N, t and a constant of proportionality. **[1]**

(ii) Initially, when $t = 0$, the number of plants was 650. It was noted that, at a time when there were 900 plants, the number of plants was increasing at a rate of 60 per year. Express N in terms of t. **[7]**

(iii) The naturalists had a target of increasing the number of plants from 650 to 2000 within 15 years. Will this target be met? **[2]**

Mark Scheme

(i) State $\frac{dN}{dt} = k(N - 150)$ **B1 [1]**

(ii) Substitute $\frac{dN}{dt} = 60$ and $N = 900$ to find value of k **M1**

Obtain $k = 0.08$ **A1**

Separate variables and obtain general solution involving $\ln(N - 150)$ **M1***

Obtain $\ln(N - 150) = 0.08t + c$ (following their k) or $\ln(N - 150) = kt + c$ **A1***

Substitute $t = 0$ and $N = 650$ to find c **dep M1***

Obtain $\ln(N - 150) = 0.08t + \ln 500$ or equivalent **A1**

Obtain $N = 500e^{0.08t} + 150$ **A1 [7]**

(iii) Either Substitute $t = 15$ to find N or solve for t with $N = 2000$ **M1**

Obtain Either $N = 1810$ or $t = 16.4$ and conclude target not met **A1 [2]**

Coaching with Shuja

Question 1

May/June 2014 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 1 (i) Many candidates seemed unable to express both sin2α and secα in terms of sinα and cosα."

> "(ii) Here, many candidates did not use the double angle formula, and of those that did, many subsequently made errors."

Source: 9709_s14_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

(i) Simplify $\sin 2\alpha \sec \alpha$. [2]

(ii) Given that $3\cos 2\beta + 7\cos \beta = 0$, find the exact value of $\cos \beta$. [3]

(i)

State $\sin 2\alpha = 2\sin\alpha\cos\alpha$ and $\sec\alpha = \frac{1}{\cos\alpha}$ **B1**

Obtain $2\sin\alpha$ **B1**

Total: 2

(ii)

Use $\cos 2\beta = 2\cos^2\beta - 1$ or equivalent to produce correct equation in $\cos\beta$ **B1**

Solve three-term quadratic equation for $\cos\beta$ **M1**

Obtain $\cos\beta = \frac{1}{3}$ only **A1**

Total: 3

Coaching With Shuja

Question 4

May/June 2014 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 4 Few candidates were successful at this question since the separation of variables was nearly always not 6e3x undertaken, and even when it was the integration of proved too difficult."

Source: 9709_s14_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

Coaching With Shuja

Question Paper May/June 2014 • Paper 3 (variant 31) • Q4

The variables x and y are related by the differential equation

$$\frac{dy}{dx} = \frac{6ye^{3x}}{2 + e^{3x}}.$$

Given that $y = 36$ when $x = 0$, find an expression for y in terms of x. [6]

Coaching With Shuja

Mark Scheme May/June 2014 • Paper 3 (variant 31) • Q4

Separate variables correctly and recognisable attempt at integration of at least one side **M1**

Obtain $\ln y$, or equivalent **B1**

Obtain $\ln(2+e^{3x})^k$ **B1**

Use $y(0)=36$ to find constant in $\ln y = A\ln(2+e^{3x})+c$ or equivalent **M1***

Obtain equation correctly without logarithms from $\ln y = A\ln(2+e^{3x})+c$ *M1

Obtain $y=4(2+e^{3x})^2$ **A1**

Total: 6

Coaching With Shuja

Question 6

May/June 2014 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 6 (i) Few candidates were able to use correctly both the product and power laws of logarithms."

> "1 (iii) Most candidates did not realise that only x = 2 was a possible solution, with x = being rejected."

Source: 9709_s14_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

It is given that $2\ln(4x-5)+\ln(x+1)=3\ln 3$.

(i) Show that $16x^3-24x^2-15x-2=0$. [3]

(ii) By first using the factor theorem, factorise $16x^3-24x^2-15x-2$ completely. [4]

(iii) Hence solve the equation $2\ln(4x-5)+\ln(x+1)=3\ln 3$. [1]

Coaching With Shuja

(i)

Use law for the logarithm for a product or quotient or exponentiation and for a power **M1**

Obtain $(4x-5)^2(x+1)=27$ **B1**

Obtain given equation correctly $16x^3-24x^2-15x-2=0$ **A1**

Total: 3

(ii)

Obtain $x=2$ is root or $(x-2)$ is a factor, or likewise with $x=-\frac{1}{4}$ **B1**

Divide by $(x-2)$ to reach a quotient of the form $16x^2+kx$ **M1**

Obtain quotient $16x^2+8x+1$ **A1**

Obtain $(x-2)(4x+1)^2$ or $(x-2)$, $(4x+1)$, $(4x+1)$ **A1**

Total: 4

(iii)

State $x=2$ only **A1**

Total: 1

Coaching with Shuja

Question 8

May/June 2014 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 8 (i) The sketching of y =cosecx proved difficult, although better success was forthcoming for y = x(π−x)."

Source: 9709_s14_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 8

Coaching With Shuja

(i) By sketching each of the graphs $y = \operatorname{cosec} x$ and $y = x(\pi - x)$ for $0 < x < \pi$, show that the equation

$$\operatorname{cosec} x = x(\pi - x)$$

has exactly two real roots in the interval $0 < x < \pi$. [3]

(ii) Show that the equation $\operatorname{cosec} x = x(\pi - x)$ can be written in the form

$$x = \frac{1 + x^2 \sin x}{\pi \sin x}.$$

[2]

(iii) The two real roots of the equation $\operatorname{cosec} x = x(\pi - x)$ in the interval $0 < x < \pi$ are denoted by α and β, where $\alpha < \beta$.

(a) Use the iterative formula

$$x_{n+1} = \frac{1 + x_n^2 \sin x_n}{\pi \sin x_n}$$

to find α correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

(b) Deduce the value of β correct to 2 decimal places. [1]

Coaching With Shuja

(i)

Sketch $y = \operatorname{cosec} x$ for at least $0 < x < \pi$ **B1**

Sketch $y = x(\pi - x)$ for at least $0 < x < \pi$ **B1**

Justify statement concerning two roots, with evidence of 1 and $\frac{1}{4}\pi^2$ for y-values on graph via scales **B1**

Total: 3

(ii)

Use $\operatorname{cosec} x = \frac{1}{\sin x}$ and commence rearrangement **M1**

Obtain given equation correctly, showing sufficient detail **A1**

Total: 2

(iii)(a)

Use the iterative formula correctly at least once **M1**

Obtain final answer 0.66 **A1**

Show sufficient iterations to 4 decimal places to justify answer or show a sign change in the interval $(0.655, 0.665)$ **A1**

Total: 3

(iii)(b)

Obtain 2.48 **B1**

Total: 1

Coaching With Shuja

Question 4

Oct/Nov 2014 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "A common error in part (ii) was to write down the first three terms without actually finding the sum of these terms."

Source: 9709_w14_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 4

Coaching With Shuja

4 (a) The functions f and g are defined for $x \geq 0$ by

$$f : x \mapsto (ax + b)^{\frac{1}{3}}, \text{ where } a \text{ and } b \text{ are positive constants,}$$

$$g : x \mapsto x^2.$$

Given that $fg(1) = 2$ and $gf(9) = 16$,

(i) calculate the values of a and b. **[4]**

(ii) obtain an expression for $f^{-1}(x)$ and state the domain of f^{-1}. **[4]**

(b) A point P travels along the curve $y = (7x^2 + 1)^{\frac{1}{3}}$ in such a way that the x-coordinate of P at time t minutes is increasing at a constant rate of 8 units per minute. Find the rate of increase of the y-coordinate of P at the instant when P is at the point $(3, 4)$. **[5]**

Coaching With Shuja

4 (i) $S_P = \frac{-4}{1-\frac{1}{2}}$, $S_P = \frac{-9}{1-\frac{1}{3}}$ **M1** At least one correct

$S_P = 4, S_O = \frac{9}{2}$ **A1** At least one correct

$S_R = 5$ **cao A1 [3]**

(ii) $\frac{4}{1-r} =$ their S_R **M1**

$r = \frac{1}{5}$ **A1 [2]**

Total: 5

Question 10

Oct/Nov 2014 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "Inevitably, there were elements of confusion particularly among those who found the expression difficult to express correctly."

Source: 9709_w14_er.pdf, Page 11, Section: Comments on specific questions – Paper 1 (variant 13), Question 10

Coaching With Shuja

10 (a) The functions f and g are defined for $x \geq 0$ by

$$f : x \mapsto (ax + b)^{\frac{1}{3}}, \text{ where } a \text{ and } b \text{ are positive constants,}$$

$$g : x \mapsto x^2.$$

Given that $fg(1) = 2$ and $gf(9) = 16$,

(i) calculate the values of a and b. **[4]**

(ii) obtain an expression for $f^{-1}(x)$ and state the domain of f^{-1}. **[4]**

(b) A point P travels along the curve $y = (7x^2 + 1)^{\frac{1}{3}}$ in such a way that the x-coordinate of P at time t minutes is increasing at a constant rate of 8 units per minute. Find the rate of increase of the y-coordinate of P at the instant when P is at the point $(3, 4)$. **[5]**

Coaching With Shuja

(a) (i) $(a+b)^{\frac{1}{3}} = 2$, $(9a+b)^{\frac{1}{3}} = 16$ **B1B1**

Ignore $2^{\cdots}$ soln $(-9, 17)$ throughout

$a+b=8, 9a+b=64$ **M1** Cube etc. & attempt to solve

$a=7, b=1$ **A1** Correct answers without any working 0/4

[4]

(ii) $x = (7y+1)^{\frac{1}{3}}$ (x/y interchange as first or last step) **B1** ft on from **their** a, b or in terms of a, b

$x^3 = 7y+1$ or $y^3 = 7x+1$ **B1** ft on from **their** a, b or in terms of a, b

$$f^{-1}(x) = \frac{1}{7}(x^3 - 1)$$

cao B1 A function of x required

Domain of f^{-1} is $x \geq 1$ **cao B1** Accept $>$. Must be x

[4]

(b)

$$\frac{dy}{dx} = \left[\frac{1}{3}(7x^2+1)^{-\frac{2}{3}}\right] \times [14x]$$

B1B1

When $x=3$, $\frac{dy}{dx} = \frac{1}{3} \times (64)^{-\frac{2}{3}} \times 42 = \left(-\frac{7}{8}\right)$ **M1** Substitute $x=3$

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} = \frac{7}{8} \times 8$$

DM1 Use chain rule

$$= 7$$

A1 [5]

Total: 13

Question 2

Oct/Nov 2014 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 2 In general this was poorly answered."

> "Common errors seen in the attempts to apply the trapezium rule included the use of abscissae (x-coordinates) instead of ordinates (y-coordinates); incorrect weightings of the ordinates; 3 or 5 ordinates instead of 4; an incorrect value for the interval width; and ordinates calculated at unequal intervals."

Source: 9709_w14_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

Coaching With Shuja

(i) Use the trapezium rule with 3 intervals to estimate the value of

$$\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \operatorname{cosec} x \, dx,$$

giving your answer correct to 2 decimal places. **[3]**

(ii) Using a sketch of the graph of $y = \operatorname{cosec} x$, explain whether the trapezium rule gives an overestimate or an underestimate of the true value of the integral in part **(i)**. **[2]**

(i) State or imply ordinates $\mathbf{2, 1.1547\ldots, 1, 1.1547\ldots}$ **B1**

Use correct formula, or equivalent, with $h = \frac{1}{6}\pi$ and four ordinates **M1**

Obtain answer **1.95 A1 [3]**

(ii) Make recognisable sketch of $y = \operatorname{cosec} x$ for the given interval **B1**

Justify a statement that the estimate will be an overestimate **B1 [2]**

Question 4

Oct/Nov 2014 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "However, finding the derivatives of x and y with respect to dx t proved difficult for some candidates."

Source: 9709_w14_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

The parametric equations of a curve are

$$x = \frac{1}{\cos^3 t}, \quad y = \tan^3 t,$$

where $0 \leq t < \frac{1}{2}\pi$.

(i) Show that $\frac{dy}{dx} = \sin t$. **[4]**

(ii) Hence show that the equation of the tangent to the curve at the point with parameter t is $y = x \sin t - \tan t$. **[3]**

Mark Scheme

(i) Use chain rule correctly at least once **M1**

Obtain either $\frac{dx}{dt} = \frac{3\sin t}{\cos^4 t}$ or $\frac{dy}{dt} = 3\tan^2 \sec^2 t$, or equivalent **A1**

Use $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ **M1**

Obtain the given answer **A1 [4]**

(ii) State a correct equation for the tangent in any form **B1**

Use Pythagoras **M1**

Obtain the given answer **A1 [3]**

Question 5

Oct/Nov 2014 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "A common error here was to take -3 to be ± 3i rather than ±i 3."

Source: 9709_w14_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

Throughout this question the use of a calculator is not permitted.

The complex numbers w and z satisfy the relation

$$w = \frac{z+i}{iz+2}.$$

(i) Given that $z = 1 + i$, find w, giving your answer in the form $x + iy$, where x and y are real. **[4]**

(ii) Given instead that $w = z$ and the real part of z is negative, find z, giving your answer in the form $x + iy$, where x and y are real. **[4]**

(i) Substitute $z = 1 + i$ and obtain $w = \frac{2+i}{1+i}$ **B1**

EITHER: Multiply numerator and denominator by the conjugate of the denominator, or equivalent **M1**

Simplify numerator to $3 + i$ or denominator to **2** **A1**

Obtain final answer $\frac{3}{2} + \frac{1}{2}i$, or equivalent **A1**

OR: Obtain two equations in x and y, and solve for x or for y **M1**

Obtain $x = \frac{3}{2}$ or $y = \frac{1}{2}$, or equivalent **A1**

Obtain final answer $\frac{3}{2} + \frac{1}{2}i$, or equivalent **A1** **[4]**

Coaching With Shuja

Question 6

Oct/Nov 2014 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “Question 6 (i) Though some could make no progress at all, the majority did try to integrate by parts.”

Source: 9709_w14_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

It is given that $\int_1^a \ln(2x)\,dx = 1$, where $a > 1$.

(i) Show that $a = \frac{1}{2}\exp\left(1 + \frac{\ln 2}{a}\right)$, where $\exp(x)$ denotes e^x. **[6]**

(ii) Use the iterative formula

$$a_{n+1} = \frac{1}{2}\exp\left(1 + \frac{\ln 2}{a_n}\right)$$

to determine the value of a correct to 2 decimal places. Give the result of each iteration to 4 decimal places. **[3]**

(i) Integrate and reach $bx \ln 2x - c|x\frac{dx}{x}$, or equivalent **M1***

Obtain $x \ln 2x - \int x . \frac{dx}{x}$, or equivalent **A1**

Obtain integral $x \ln 2x - x$, or equivalent **A1**

Substitute limits correctly and equate to **1**, having integrated twice **M1(dep*)**

Obtain a correct equation in any form, e.g. $a \ln 2a - a + 1 - \ln 2 = 1$ **A1**

Obtain the given answer **A1 [6]**

(ii) Use the iterative formula correctly at least once **M1**

Obtain final answer **1.94 A1**

Show sufficient iterations to 4 d.p. to justify **1.94** to 2d.p. or show that there is a sign change in the interval **(1.935, 1.945)**. **A1 [3]**

Coaching With Shuja

Question 2

Oct/Nov 2014 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> “Some candidates focused on the cos2θ and appeared to forget the sinθ, and the incorrect result ( ) d dθcos2θ=sin2θ was a common error.”

Source: 9709_w14_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 33), Question 2

A curve is defined for $0 < \theta < \frac{1}{2}\pi$ by the parametric equations

$$x = \tan\theta, \quad y = 2\cos^2\theta \sin\theta.$$

Show that $\frac{dy}{dx} = 6\cos^3\theta - 4\cos^3\theta$. **[5]**

Coaching With Shuja

Mark Scheme

Use correct product rule or correct chain rule to differentiate y **M1**

Use $\frac{dy}{dx} = \frac{d\theta/dx}{d\theta}$ **M1***

Obtain $\frac{-4\cos\theta\sin^2\theta + 2\cos^3\theta}{\sec^2\theta}$ or equivalent **A1**

Express $\frac{dy}{dx}$ in terms of $\cos\theta$ **DM1***

Confirm given answer $6\cos^5\theta - 4\cos^3\theta$ legitimately **A1 [5]**

Coaching With Shuja

Question 5

Oct/Nov 2014 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Many candidates lost marks in part (ii) because they simply wrote down the given answer without explaining that they were using the general result arg (wz) = arg w + arg z and they did not explain what tan−13 and 5 tan−11 represented or where the 1 π comes from."

Source: 9709_w14_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 5

Coaching With Shuja

The complex numbers w and z are defined by $w = 5 + 3i$ and $z = 4 + i$.

(i) Express $\frac{iw}{z}$ in the form $x + iy$, showing all your working and giving the exact values of x and y. **[3]**

(ii) Find wz and hence, by considering arguments, show that

$$\tan^{-1}\left(\frac{3}{5}\right) + \tan^{-1}\left(\frac{1}{4}\right) = \frac{1}{4}\pi. \quad [4]$$

(i) State or imply $iw = -3 + 5i$ **B1**

Carry out multiplication by $\frac{4-i}{4-i}$ **M1**

Obtain final answer $-\frac{7}{17} + \frac{23}{17}i$ or equivalent **A1 [3]**

(ii) Multiply w by z to obtain $17 + 17i$ **B1**

State $\arg w = \tan^{-1}\frac{3}{5}$ or $\arg z = \tan^{-1}\frac{1}{4}$ **B1**

State $\arg wz = \arg w + \arg z$ **M1**

Confirm given result $\tan^{-1}\frac{3}{5} + \tan^{-1}\frac{1}{4} = \frac{1}{4}\pi$ legitimately **A1 [4]**

Coaching With Shuja

Question 6

Oct/Nov 2014 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "Amongst those candidates who demonstrated an understanding of the basic structure of the method, common errors involved working with the wrong number of intervals, not starting from x = 0 and using 0, 1, 2 and 3 as values for x rather than 0, 0.1, 0.2 and 0.3."

> "A 1 1 1 1 common error was to state ( ) = =1+ x2 + x4."

Source: 9709_w14_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 6

Coaching With Shuja

Question Paper Oct/Nov 2014 • Paper 3 (variant 33) • Q6

It is given that $I = \int_0^{0.3} (1 + 3x^2)^{-2}\, dx$.

(i) Use the trapezium rule with 3 intervals to find an approximation to I, giving the answer correct to 3 decimal places. **[3]**

(ii) For small values of x, $(1 + 3x^2)^{-2} \approx 1 + ax^2 + bx^4$. Find the values of the constants a and b.

Hence, by evaluating $\int_0^{0.3} (1 + ax^2 + bx^4)\, dx$, find a second approximation to I, giving the answer correct to 3 decimal places. **[5]**

Coaching With Shuja

(i) State or imply correct ordinates $1, 0.94259\ldots, 0.79719\ldots, 0.62000\ldots$ **B1**

Use correct formula or equivalent with $h = 0.1$ and four y values **M1**

Obtain **0.255** with no errors seen **A1 [3]**

(ii) Obtain or imply $a = -6$ **B1**

Obtain x^4 term including correct attempt at coefficient **M1**

Obtain or imply $b = 27$ **A1**

Either Integrate to obtain $x - 2x^3 + \frac{27}{5}x^5$, following their values of a and b **B1***

Obtain **0.259** **B1**

Or Use correct trapezium rule with at least 3 ordinates **M1**

Obtain **0.259** (from 4) **A1 [5]**

Coaching with Shuja

Question 8

Oct/Nov 2014 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> “Integrating xsin x proved to be more difficult.”

Source: 9709_w14_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 8

Coaching With Shuja

The variables x and y are related by the differential equation

$$\frac{dy}{dx} = \frac{1}{5}y^{\frac{1}{2}} \sin\left(\frac{1}{3}x\right).$$

(i) Find the general solution, giving y in terms of x. **[6]**

(ii) Given that $y = 100$ when $x = 0$, find the value of y when $x = 25$. **[3]**

Coaching With Shuja

(i) Sensibly separate variables and attempt integration of at least one side **M1**

Obtain $2y^{\frac{1}{2}} = \ldots$ or equivalent **A1**

Correct integration by parts of $x \sin \frac{1}{3}x$ as far as $ax \cos \frac{1}{3}x \pm \left| b \cos \frac{1}{3}x dx \right|$ **M1**

Obtain $-3x \cos \frac{1}{3}x + \int 3 \cos \frac{1}{3}x dx$ or equivalent **A1**

Obtain $-3x \cos \frac{1}{3}x + 9 \sin \frac{1}{3}x$ or equivalent **A1**

Obtain $y = \left(-\frac{3}{10}x \cos \frac{1}{3}x + \frac{9}{10} \sin \frac{1}{3}x + c\right)^2$ or equivalent **A1 [6]**

(ii) Use $x = 0$ and $y = 100$ to find constant **M1***

Substitute **25** and calculate value of y **DM1***

Obtain **203** **A1 [3]**

Coaching With Shuja

Question 9

Oct/Nov 2014 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The question specifies accuracy to a number of decimal places, but some candidates appeared to be working to that number of significant figures and lost marks as a result."

> "( ) (iv) A small number of candidates spotted the link between x3 +ln(x+1) =40and ey −13 +y = 40, but many either worked through a second set of iterations or made no progress with this part."

Source: 9709_w14_er.pdf, Page 30, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

(i) Sketch the curve $y = \ln(x + 1)$ and hence, by sketching a second curve, show that the equation

$$x^3 + \ln(x + 1) = 40$$

has exactly one real root. State the equation of the second curve. **[3]**

(ii) Verify by calculation that the root lies between 3 and 4. **[2]**

(iii) Use the iterative formula

$$x_{n+1} = \sqrt[3]{40 - \ln(x_n + 1)},$$

with a suitable starting value, to find the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places. **[3]**

(iv) Deduce the root of the equation

$$(e^y - 1)^3 + y = 40,$$

giving the answer correct to 2 decimal places. **[2]**

Coaching With Shuja

Mark Scheme Oct/Nov 2014 • Paper 3 (variant 33) • Q9

(i) Sketch increasing curve with correct curvature passing through origin, for $x \geq 0$ **B1**
Recognisable sketch of $y = 40 - x^3$, with equation stated, for $x > 0$ **B1**
Indicate in some way the one intersection, dependent on both curves being roughly correct and both existing for some $x < 0$ **B1 [3]**

(ii) Consider signs of $x^3 + \ln(x+1) - 40$ at 3 and 4 or equivalent or compare values of relevant expressions for $x = 3$ and $x = 4$
M1
Complete argument correctly with correct calculations (-11.6 and 25.6) **A1 [2]**

(iii) Use the iterative formula correctly at least once **M1**
Obtain final answer **3.377 A1**
Show sufficient iterations to justify accuracy to 3 d.p. or show sign change in interval $(3.3765, 3.3775)$ **A1 [3]**

(iv) Attempt value of $\ln(x+1)$ **M1**
Obtain **1.48 A1 [2]**

Coaching With Shuja

Question 1

May/June 2013 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “Another common error was to express the remainder as .”

Source: 9709_s13_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

Find the quotient and remainder when $2x^2$ is divided by $x + 2$. [3]

Coaching With Shuja

Carry out division or equivalent at least as far as two terms of quotient **M1**

Obtain quotient $\mathbf{2x - 4}$ **A1**

Obtain remainder **8** **A1**

Total: 3

Question 4

May/June 2013 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "This is something that most candidates fail to utilise, namely that where questions are displayed in sections (i), (ii), etc., then there is often a direct link between the individual sections, for example Question 4 (i) and (ii), Question 6 (i) and (ii), Question 9 (i), (b) (ii) and (iii)."

Source: 9709_s13_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

Coaching With Shuja

(i) Solve the equation $|4x - 1| = |x - 3|$. [3]

(ii) Hence solve the equation $|4^{y+1} - 1| = |4^y - 3|$ correct to 3 significant figures. [3]

(i)

EITHER: State or imply non-modular equation $(4x-1)^2=(x-3)^2$ or pair of linear equations $4x-1=\pm(x-3)$ **B1**

Solve a three-term quadratic equation or two linear equations **M1**

Obtain $-\frac{2}{3}$ and $\frac{4}{5}$ **A1**

OR: Obtain value $-\frac{2}{3}$ from inspection or solving linear equation **B1**

Obtain value $\frac{4}{5}$ similarly **B2**

Total: 3

(ii)

State or imply at least $4^y=\frac{4}{5}$, following a positive answer from part (i) **B1 FT**

Apply logarithms and use $\log_a b=\frac{\log b}{\log a}$ property **M1**

Obtain -0.161 and no other answer **A1**

Total: 3

Coaching With Shuja

Question 6

May/June 2013 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 6 (i) This question proved very difficult for nearly all the candidates, with OA believed usually to be 1 PQ, and the normal to the plane being either the actual line vector or the cross product of OP 2 and OQ."

Source: 9709_s13_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

The points P and Q have position vectors, relative to the origin O, given by

$$\overrightarrow{OP} = 7\mathbf{i} + 7\mathbf{j} - 5\mathbf{k}$$

and

$$\overrightarrow{OQ} = -5\mathbf{i} + \mathbf{j} + \mathbf{k}.$$

The mid-point of PQ is the point A. The plane Π is perpendicular to the line PQ and passes through A.

(i) Find the equation of Π, giving your answer in the form $ax + by + cz = d$. [4]

(ii) The straight line through P parallel to the x-axis meets Π at the point B. Find the distance AB, correct to 3 significant figures. [5]

Coaching With Shuja

(i)

State or imply A is $(1, 4, -2)$ **B1**

State or imply $\overrightarrow{QP} = 12\mathbf{i} + 6\mathbf{j} - 6\mathbf{k}$ or equivalent **B1**

Use QP as normal and A as mid-point to find equation of plane **M1**

Obtain $12x + 6y - 6z = 48$ or equivalent **A1**

Total: 4

(ii)

State equation of PB is $\mathbf{r} = 7\mathbf{i} + 7\mathbf{j} - 5\mathbf{k} + \lambda\mathbf{i}$ **B1**

Set up and solve a relevant equation for λ **M1**

Obtain $\lambda = -9$ and hence B is $(-2, 7, -5)$ **A1**

Use correct method to find distance between A and B **M1**

Obtain 5.20 **A1**

Total: 5

Coaching With Shuja

Question 7

May/June 2013 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 7 (a) This question proved difficult as few realised that they needed to expressw =u+v i, together with its conjugate w*, and take real and imaginary parts."

Source: 9709_s13_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

(a) Without using a calculator, solve the equation

$$3w + 2iw^* = 17 + 8i,$$

where w^* denotes the complex conjugate of w. Give your answer in the form $a + bi$. [4]

(b) In an Argand diagram, the loci

$$\arg(z - 2i) = \frac{1}{6}\pi \quad \text{and} \quad |z - 3| = |z - 3i|$$

intersect at the point P. Express the complex number represented by P in the form $re^{i\theta}$, giving the exact value of θ and the value of r correct to 3 significant figures. [5]

Coaching With Shuja

(a)

State or imply $3(a+bi)+2i(a-bi)=17+8i$ **B1**

Consider real and imaginary parts to obtain two linear equations in a and b **M1***

Solve two simultaneous linear equations for a or b **M1 dep***

Obtain $7-2i$ **A1**

Total: 4

(b)

EITHER: Show or imply a triangle with side 2 **B1**

State at least two of the angles $\frac{\pi}{2}$, $\frac{\pi}{3}$ and $\frac{\pi}{12}$ **B1**

State or imply argument is $\frac{\pi}{4}$ **B1**

Use sine rule or equivalent to find r **M1**

Obtain $6.69e^{\frac{1}{4}\pi i}$ **A1**

OR: State $y=x$ **B1**

State $x^2+(y-2)^2=\frac{1}{3}y^2$ or equivalent **B1**

State or imply argument is $\frac{\pi}{4}$ **B1**

Solve for x or y **M1**

Obtain $6.69e^{\frac{1}{4}\pi i}$ **A1**

Total: 5

Coaching With Shuja

Question 8

May/June 2013 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(ii) Many candidates either failed to substitute correctly for cos2(4x) in terms of u or muddled their substitution of dx, the latter usually resulting in another cos2(4x) appearing in the numerator instead of a cos(4x) cancelling from the numerator and the denominator."

Source: 9709_s13_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 8

Coaching With Shuja

(a) Show that

$$\int_2^4 4x \ln x \, dx = 56 \ln 2 - 12.$$

[5]

(b) Use the substitution $u = \sin 4x$ to find the exact value of

$$\int_0^{\frac{1}{24}\pi} \cos^3 4x \, dx.$$

[5]

(a)

Carry out integration by parts and reach $ax^2 \ln x + b\int x\,dx$ **M1***

Obtain $2x^2 \ln x - \int 2x\,dx$ **A1**

Obtain $2x^2 \ln x - x^2$ **A1**

Use limits, having integrated twice **M1 dep***

Confirm given result $56 \ln 2 - 12$ **A1**

Total: 5

(b)

State or imply $\frac{du}{dx} = 4\cos 4x$ **B1**

Carry out complete substitution except limits **M1**

Obtain $\frac{1}{4}\int(1 - u^2)\,du$ or equivalent **A1**

Integrate to obtain form $k_1 u + k_2 u^3$ with non-zero constants k_1, k_2 **M1**

Use appropriate limits to obtain $\frac{11}{96}$ **A1**

Total: 5

Coaching With Shuja

Question 9

May/June 2013 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Answer: 0.6435 (ii) Again most candidates obtained the answer 1.80, however the other answer proved more difficult, since instead of $2\pi - \cos^{-1}(2/5) + \alpha$ candidates assumed the answer was $2\pi - (\cos^{-1}(2/5) + \alpha)$."

Source: 9709_s13_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 9

Coaching With Shuja

(i) Express $4\cos\theta + 3\sin\theta$ in the form $R\cos(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{1}{2}\pi$. Give the value of α correct to 4 decimal places. [3]

(ii) Hence

(a) solve the equation $4\cos\theta + 3\sin\theta = 2$ for $0 < \theta < 2\pi$, [4]

(b) find

$$\int \frac{50}{(4\cos\theta + 3\sin\theta)^2}\, d\theta.$$

[3]

(i)

State or imply $R = 5$ **B1**

Use relevant trigonometry to find α **M1**

Obtain $\alpha = 0.6435$ **A1**

Total: 3

(ii)(a)

Carry out appropriate method to find one value in given range **M1**

Obtain **1.80** **A1**

Carry out appropriate method to find second value in given range **M1**

Obtain **5.77** and no other value **A1**

Total: 4

(ii)(b)

Express integrand as $k\sec^2(\theta - \text{their } \alpha)$ for any constant k **M1**

Integrate to obtain result $k\tan(\theta - \text{their } \alpha)$ **A1**

Obtain correct answer $2\tan(\theta - 0.6435)$ **A1**

Total: 3

Coaching With Shuja

Question 10

May/June 2013 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Those that they found difficult were Question 4 (ii) (a moduli equation requiring substitution), Question 6 (vector equation of a plane), Question 9 (ii) (b) (integration using substitution of an earlier derived relation) and Question 10 (i) (establishing a differential equation and solving it)."

Source: 9709_s13_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 10

Coaching With Shuja

Liquid is flowing into a small tank which has a leak. Initially the tank is empty and, t minutes later, the volume of liquid in the tank is V cm^3. The liquid is flowing into the tank at a constant rate of 80 cm^3 per minute. Because of the leak, liquid is being lost from the tank at a rate which, at any instant, is equal to kV cm^3 per minute where k is a positive constant.

(i) Write down a differential equation describing this situation and solve it to show that

$$V = \frac{1}{k}(80 - 80e^{-kt}).$$

[7]

(ii) It is observed that $V = 500$ when $t = 15$, so that k satisfies the equation

$$k = \frac{4 - 4e^{-15k}}{25}.$$

Use an iterative formula, based on this equation, to find the value of k correct to 2 significant figures. Use an initial value of $k = 0.1$ and show the result of each iteration to 4 significant figures. [3]

(iii) Determine how much liquid there is in the tank 20 minutes after the liquid started flowing, and state what happens to the volume of liquid in the tank after a long time. [2]

(i)

State $\frac{dV}{dt} = 80 - kV$ **B1**

Correctly separate variables and attempt integration of one side **M1**

Obtain $a\ln(80 - kV) = t$ or equivalent **M1***

Obtain $-\frac{1}{k}\ln(80 - kV) = t$ or equivalent **A1**

Use $t = 0$ and $V = 0$ to find constant of integration or as limits **M1 dep***

Obtain $-\frac{1}{k}\ln(80 - kV) = t - \frac{1}{k}\ln 80$ or equivalent **A1**

Obtain given answer $V = \frac{1}{k}(80 - 80e^{-kt})$ correctly **A1**

Total: 7

(ii)

Use iterative formula correctly at least once **M1**

Obtain final answer **0.14 A1**

Show sufficient iterations to 4 s.f. to justify answer to 2 s.f. or show a sign change in the interval $(0.135, 0.145)$ **A1**

Total: 3

(iii)

State a value between **530** and **540 cm^3** inclusive **B1**

State or imply that volume approaches **569 cm^3**, allowing any value between 567 and 571 inclusive **B1**

Total: 2

Question 6

Oct/Nov 2013 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 6 Part (i) was poorly answered and very few correctly argued proofs were seen."

Source: 9709_w13_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

Question Paper Oct/Nov 2013 • Paper 3 (variant 32) • Q6

A tank containing water is in the form of a cone with vertex C. The axis is vertical and the semi-vertical angle is 60°, as shown in the diagram. At time $t = 0$, the tank is full and the depth of water is H. At this instant, a tap at C is opened and water begins to flow out. The volume of water in the tank decreases at a rate proportional to $\sqrt{h}$, where h is the depth of water at time t. The tank becomes empty when $t = 60$.

(i) Show that h and t satisfy a differential equation of the form

$$\frac{dh}{dt} = -Ah^{-\frac{3}{2}},$$

where A is a positive constant.

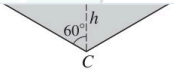
Total: 4

(ii) Solve the differential equation given in part **(i)** and obtain an expression for t in terms of h and H.

Total: 6

(iii) Find the time at which the depth reaches $\frac{1}{2}H$.

Total: 1



A tank containing water is in the form of a cone with vertex C. The axis is vertical and the semi-vertical angle is 60°, as shown in the diagram. At time $t = 0$, the tank is full and the depth of water is H. At this instant, a tap at C is opened and water begins to flow out. The volume of water in the tank decreases at a rate proportional to $\sqrt{h}$, where h is the depth of water at time t. The tank becomes empty when $t = 60$.

(i) Show that h and t satisfy a differential equation of the form

$$\frac{dh}{dt} = -Ah^{-\frac{3}{2}},$$

where A is a positive constant. [4]

(ii) Solve the differential equation given in part **(i)** and obtain an expression for t in terms of h and H. [6]

(iii) Find the time at which the depth reaches $\frac{1}{2}H$. [1]

(i) B1 State or imply $AB = 2r\cos\theta$ or $AB^2 = 2r^2 - 2r^2\cos(\pi - 2\theta)$

M1 Use correct formula to express the area of sector ABC in terms of r and θ

M1 Use correct area formulae to express the area of a segment in terms of r and θ

A1 State a correct equation in r and θ in any form

A1 Obtain the given answer

[SR: If the complete equation is approached by adding two sectors to the shaded area above BO and OC give the first **M1** as on the scheme, and the second **M1** for using correct area formulae for a triangle AOB or AOC, and a sector AOB or AOC.]

(ii) M1 Use the iterative formula correctly at least once

A1 Obtain final answer 0.95

A1 Show sufficient iterations to 4 d.p. to justify 0.95 to 2 d.p., or show there is a sign change in the interval $(0.945, 0.955)$

Total: 3

Coaching With Shuja

Question 10

Oct/Nov 2013 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 10 (i) Most candidates found this to be a difficult problem."

> "(iii) There were very few correct solutions."

Source: 9709_w13_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

7 Let $f(x) = \frac{2x^2 - 7x - 1}{(x-2)(x^2+3)}$.

(i) Express $f(x)$ in partial fractions. [5]

(ii) Hence obtain the expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. [5]

8 **Throughout this question the use of a calculator is not permitted.**

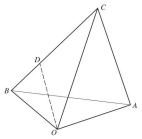
(a) The complex numbers u and v satisfy the equations

$$u + 2v = 2\mathrm{i} \quad \text{and} \quad \mathrm{i}u + v = 3.$$

Solve the equations for u and v, giving both answers in the form $x + \mathrm{i}y$, where x and y are real. [5]

(b) On an Argand diagram, sketch the locus representing complex numbers z satisfying $|z + \mathrm{i}| = 1$ and the locus representing complex numbers w satisfying $\arg(w - 2) = \frac{1}{4}\pi$. Find the least value of $|z - w|$ for points on these loci. [5]

9



The diagram shows three points A, B and C whose position vectors with respect to the origin O are given by $\overrightarrow{OA} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}$ and $\overrightarrow{OC} = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}$. The point D lies on BC, between B and C, and is such that $CD = 2DB$.

(i) Find the equation of the plane ABC, giving your answer in the form $ax + by + cz = d$. [6]

(ii) Find the position vector of D. [1]

(iii) Show that the length of the perpendicular from A to OD is $\frac{1}{3}\sqrt{(65)}$. [4]

A tank containing water is in the form of a cone with vertex C. The axis is vertical and the semi-vertical angle is 60°, as shown in the diagram. At time $t = 0$, the tank is full and the depth of water is H. At this instant, a tap at C is opened and water begins to flow out. The volume of water in the tank decreases at a rate proportional to $\sqrt{h}$, where h is the depth of water at time t. The tank becomes empty when $t = 60$.

(i) Show that h and t satisfy a differential equation of the form

$$\frac{dh}{dt} = -Ah^{-\frac{3}{2}},$$

where A is a positive constant. **[4]**

(ii) Solve the differential equation given in part (i) and obtain an expression for t in terms of h and H. **[6]**

(iii) Find the time at which the depth reaches $\frac{1}{4}H$. **[1]**

10 (i) State or imply $V = \pi h^3$ **B1**

State or imply $\frac{dV}{dt} = -k\sqrt{h}$ **B1**

Use $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$, or equivalent **M1**

Obtain the given equation **A1 [4]**

[The **M1** is only available if $\frac{dV}{dh}$ is in terms of h and has been obtained by a correct method.]

[Allow **B1** for $\frac{dV}{dt} = k\sqrt{h}$ but withhold the final **A1** until the polarity of the constant $\frac{k}{3\pi}$ has been justified.]

(ii) Separate variables and integrate at least one side **M1**

Obtain terms $\frac{2}{5}h^2$ and $-At$, or equivalent **A1**

Use $t = 0, h = H$ in a solution containing terms of the form ah^2 and $bt + c$ **M1**

Use $t = 60, h = 0$ in a solution containing terms of the form ah^2 and $bt + c$ **M1**

Obtain a correct solution in any form, e.g. $\frac{2}{5}h^2 = \frac{1}{150}H^2t + \frac{2}{5}H^2$ **A1**

(iii) Obtain final answer $t = 60\left[1 - \left(\frac{h}{H}\right)^2\right]$, or equivalent **A1 [6]**

(iii) Substitute $h = \frac{1}{2}H$ and obtain answer $t = 49.4$ **B1 [1]**

Coaching with Shuja

Question 2

May/June 2012 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Common errors were to incorrectly substitute x = 4 into the differential or to assume that the rate of change of the y-coordinate was equal to the gradient divided by 0.12 instead of being multiplied by 0.12 Answers: (i) y = 2 −x−1.5 (ii) 0.105 x"

Source: 9709_s12_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 12), Question 2

Coaching With Shuja

Question Paper May/June 2012 • Paper 1 (variant 12) • Q2

The equation of a curve is $y = 4\sqrt{x} + \frac{2}{\sqrt{x}}$.

(i) Obtain an expression for $\frac{dy}{dx}$. [3]

(ii) A point is moving along the curve in such a way that the x-coordinate is increasing at a constant rate of 0.12 units per second. Find the rate of change of the y-coordinate when $x = 4$. [2]

Coaching With Shuja

(i)

$y = 4x^{1/2} + 2x^{-1/2}$ **M1** Reducing their power by 1 once.

$\frac{dy}{dx} = 4 \cdot \frac{1}{2}x^{-1/2} + 2\left(-\frac{1}{2}\right)x^{-3/2}$ **A1**

$\frac{dy}{dx} = \frac{2}{\sqrt{x}} - \frac{1}{x^{3/2}}$ **A1** Allow unsimplified.

Total: 3

(ii)

$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ used **M1** Must be used correctly.

$= \frac{7}{8} \times 0.12 = 0.105$ **A1** Fraction or decimal.

Total: 2

Coaching With Shuja

Question 7

May/June 2012 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 7 (a) This part was poorly answered and it was rare to see a correct answer to this challenging question."

Source: 9709_s12_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 7

Coaching With Shuja

(a) In an arithmetic progression, the sum of the first n terms, denoted by S_n, is given by

$$S_n = n^2 + 8n.$$

Find the first term and the common difference. [3]

(b) In a geometric progression, the second term is 9 less than the first term. The sum of the second and third terms is 30. Given that all the terms of the progression are positive, find the first term. [5]

(a)

$S_1 = 9$, hence $a = 9$ **B1**

$S_2 = 20$, hence $a + (a + d) = 20$ or $a + d = 11$ **M1** Realises that S_2 is $a + (a + d)$.

$d = 2$ **A1**

Total: 3

(b)

$ar = a - 9$ **B1**

$ar + ar^2 = 30$ **B1**

Eliminates a leading to $3r^2 + 13r - 10 = 0$, or eliminates r leading to $2a^2 - 57a + 81 = 0$ **M1** Complete elimination of r or a.

$r = \frac{2}{3}$ **A1**

$a = 27$ **A1** Condone 27 or 1.5.

Total: 5

Coaching With Shuja

Question 8

May/June 2012 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "A $\begin{pmatrix} 3 \\ \\ \end{pmatrix}$ common error was to interpret 3i – 4k as the column vector $\begin{pmatrix} \\ -4 \\ \end{pmatrix}$."

> "(iii) Candidates were accurate in using AB = OB - OA, but again a large number were unable to proceed to the unit vector."

Source: 9709_s12_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

Coaching With Shuja

Question Paper May/June 2012 • Paper 1 (variant 12) • Q8

(i) Find the angle between the vectors $\mathbf{3i - 4k}$ and $\mathbf{2i + 3j - 6k}$. [4]

The vector $\overrightarrow{OA}$ has a magnitude of 15 units and is in the same direction as the vector $\mathbf{3i - 4k}$. The vector $\overrightarrow{OB}$ has a magnitude of 14 units and is in the same direction as the vector $\mathbf{2i + 3j - 6k}$.

(ii) Express $\overrightarrow{OA}$ and $\overrightarrow{OB}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. [3]

(iii) Find the unit vector in the direction of $\overrightarrow{AB}$. [3]

Coaching With Shuja

(i)

Dot product $= 6 + 24 = 30$ **M1** Uses $x_1x_2 + y_1y_2 + z_1z_2$.

$= \sqrt{25} \times \sqrt{49}\cos\theta$ **M1** Method for modulus.

Links everything correctly **M1**

Angle $= 31°$ or $0.54(1)$ radians **A1**

Total: 4

(ii)

$\overrightarrow{OA} = (3\mathbf{i} - 4\mathbf{k}) \times (15 \div 5)$ **M1** M mark for multiplying by $15 \div 5$ or $14 \div 7$.

$\overrightarrow{OA} = 9\mathbf{i} - 12\mathbf{k}$ **A1**

$\overrightarrow{OB} = 4\mathbf{i} + 6\mathbf{j} - 12\mathbf{k}$ **A1**

Total: 3

(iii)

$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = -5\mathbf{i} + 6\mathbf{j}$ **M1** Correct use for either AB or BA.

Magnitude $= \sqrt{61}$ or 7.81 **M1** Complete method for unit vector.

Unit vector $= \frac{-5\mathbf{i}+6\mathbf{j}}{\sqrt{61}}$ **A1**

Total: 3

Coaching With Shuja

Question 9

May/June 2012 | Paper 1 (variant 12)

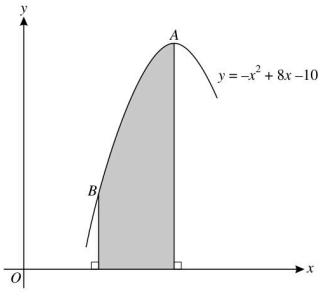
Examiner Report — Why students found this question difficult

> "Finding the coordinates of B proved challenging for the majority of candidates, most of whom took the gradient of the curve as 2 instead of the gradient of the line AB as 2."

Source: 9709_s12_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 9

Coaching With Shuja

9



The diagram shows part of the curve $y = -x^2 + 8x - 10$ which passes through the points A and B. The curve has a maximum point at A and the gradient of the line BA is 2.

(i) Find the coordinates of A and B. [7]

(ii) Find $\int y\,dx$ and hence evaluate the area of the shaded region. [4]

The diagram shows part of the curve $y = -x^2 + 8x - 10$ which passes through the points A and B. The curve has a maximum point at A and the gradient of the line BA is 2.

(i) Find the coordinates of A and B. [7]

(ii) Find $\int y\,dx$ and hence evaluate the area of the shaded region. [4]

(i)

$y = -x^2 + 8x - 10$ **B1**

$\frac{dy}{dx} = -2x + 8$ **M1**

$= 0$ when $x = 4$, A is $(4, 6)$ **A1**

Equation of AB is $y - 6 = 2(x - 4)$ **M1** Correct form of equation.

Solve simultaneous equations with equation of curve **M1** Eliminates x or y completely.

Obtain $x^2 - 6x + 8 = 0$ or $y^2 - 8y + 12 = 0$ **A1** Method for quadratic equation $= 0$.

$B = (2, 2)$ **A1** Must not be guessed from diagram.

Total: 7

(ii)

$\int(-x^2 + 8x - 10)\,dx = -\frac{x^3}{3} + 4x^2 - 10x$ **B2,1** Three terms, loses 1 for each error.

Uses their x limits 2 to 4 **M1** Uses x limits correctly.

Area $= 9\frac{1}{3}$ **A1** Allow $\pm$, but 2 must have been correctly found, not guessed.

Total: 4

Question 10

May/June 2012 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "The most common error was a simple algebraic slip in transferring data from one side of an equation to the other."

> "A few candidates incorrectly left the answer in terms of y instead of x."

> "Many candidates however did not realise that the lines y = f(x) and y = f -1(x) have the line y = x as a line of symmetry."

Source: 9709_s12_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 10

Coaching With Shuja

Functions f and g are defined by

$$f : x \mapsto 2x + 5 \quad \text{for } x \in \mathbb{R},$$

$$g : x \mapsto \frac{8}{x - 3} \quad \text{for } x \in \mathbb{R},\ x \neq 3.$$

(i) Obtain expressions, in terms of x, for $f^{-1}(x)$ and $g^{-1}(x)$, stating the value of x for which $g^{-1}(x)$ is not defined. [4]

(ii) Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the same diagram, making clear the relationship between the two graphs. [3]

(iii) Given that the equation $fg(x) = 5 - kx$, where k is a constant, has no solutions, find the set of possible values of k. [5]

Coaching With Shuja

(i)

$f^{-1}(x) = \frac{1}{2}(x-5)$ **B1**

$g^{-1}(x) = \frac{8}{x} + 3$ **M1 A1** Attempt at making x the subject. Correct, but allow if a linear denominator.

$x \neq 0$ **B1**

Total: 4

(ii)

Sketch $y = f(x)$ with positive gradient and positive y-intercept **B1**

Sketch $y = f^{-1}(x)$ with positive gradient and positive y-intercept **B1**

States or shows the line $y = x$ as a line of symmetry **B1**

Total: 3

(iii)

$fg(x) = \frac{16}{x-3} + 5$ **B1**

Forms equation $\frac{16}{x-3} + 5 = 5 - kx$ **M1** Must lead to a quadratic.

$kx^2 - 3kx + 16 = 0$ **M1**

Uses $b^2 = 4ac$ leading to $k = \frac{64}{9}$ or 0 **A1** Use of $b^2 - 4ac$ even if inequality sign is wrong.

Set of values $0 < k < \frac{64}{9}$ **A1**

Total: 5

Coaching With Shuja

Question 1

May/June 2012 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 1 This question proved difficult for many since, instead of removing the modulus sign and obtaining the equations 2x = 14 and 2x =−6, they tried to expand (4 - 2x)2."

Source: 9709_s12_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

Question Paper May/June 2012 • Paper 3 (variant 31) • Q1

Solve the equation $|4 - 2x| = 10$, giving your answer correct to 3 significant figures. [3]

Coaching With Shuja

State or imply $4 - 2x = -10$ and $4 - 2x = 10$ **B1**

Use correct method for solving equation of form $|2x + a| = b$ **M1**

Obtain **3.81** **A1**

Total: 3

Question 3

May/June 2012 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(ii) (a) Again most candidates were successful, although some did not progress beyond (x – 2)(x2 + 2x – 8)."

Source: 9709_s12_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 3

Coaching With Shuja

Question Paper May/June 2012 • Paper 3 (variant 31) • Q3

The polynomial $p(x)$ is defined by

$$p(x) = x^3 - 3ax + 4a,$$

where a is a constant.

(i) Given that $(x - 2)$ is a factor of $p(x)$, find the value of a. [2]

(ii) When a has this value,

(a) factorise $p(x)$ completely, [3]

(b) find all the roots of the equation $p(x^2) = 0$. [2]

Coaching With Shuja

(i)

Substitute $x = 2$ and equate to zero, or divide by $x - 2$ and equate constant remainder to zero, or equivalent **M1**

Obtain $a = 4$ **A1**

Total: 2

(ii)(a)

Find further quadratic or linear factor by division, inspection or factor theorem or equivalent **M1**

Obtain $x^2 + 2x - 8$ or $x + 4$ **A1**

State $(x - 2)^2(x + 4)$ or equivalent **A1**

Total: 3

(ii)(b)

State any two of the four or six roots **B1**

State all roots $\pm\sqrt{2}$, $\pm 2i$, provided two are purely imaginary **B1**

Total: 2

Coaching With Shuja

Question 5

May/June 2012 | Paper 3 (variant 31)

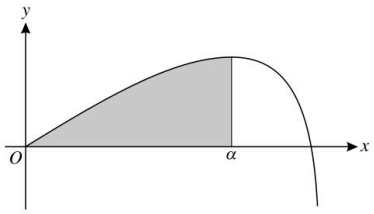
Examiner Report — Why students found this question difficult

> "Question 5 1 1 (i) Differentiating tan x proved difficult, although those that managed to reach the form bsec2 x 2 2 1 were able to continue to solve for cos x."

> "2 1 (ii) Whilst a few candidates managed to obtain −16cos xusually the sign or the coefficient was 2 1 incorrect, it was rare to see 2ln(cos x)."

Source: 9709_s12_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

5



The diagram shows the curve

$$y = 8\sin\tfrac{1}{2}x - \tan\tfrac{1}{2}x$$

for $0 \leqslant x < \pi$. The x-coordinate of the maximum point is α and the shaded region is enclosed by the curve and the lines $x = \alpha$ and $y = 0$.

(i) Show that $\alpha = \frac{2}{3}\pi$. [3]

(ii) Find the exact value of the area of the shaded region. [4]

The diagram shows the curve

$$y = 8\sin\frac{1}{2}x - \tan\frac{1}{2}x$$

for $0 \leq x < \pi$. The x-coordinate of the maximum point is α and the shaded region is enclosed by the curve and the lines $x = \alpha$ and $y = 0$.

(i) Show that $\alpha = \frac{2}{3}\pi$. [3]

(ii) Find the exact value of the area of the shaded region. [4]

(i)

Differentiate to obtain $4\cos\frac{1}{2}x - \frac{1}{2}\sec^2\frac{1}{2}x$ **B1**

Equate to zero and find value of $\cos\frac{1}{2}x$ **M1**

Obtain $\cos\frac{1}{2}x = \frac{1}{2}$ and confirm $\alpha = \frac{2\pi}{3}$ **A1**

Total: 3

(ii)

Integrate to obtain $-16\cos\frac{1}{2}x$ **B1**

Obtain $+2\ln\left(\cos\frac{1}{2}x\right)$ or equivalent **B1**

Use limits 0 and $\frac{2\pi}{3}$ in $a\cos\frac{1}{2}x + b\ln\left(\cos\frac{1}{2}x\right)$ **M1**

Obtain $8 + 2\ln\frac{1}{2}$ or exact equivalent **A1**

Total: 4

Coaching With Shuja

Question 6

May/June 2012 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Another common error was to still see the constant term of 45 present following differentiation."

Source: 9709_s12_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

The equation of a curve is $3x^2 - 4xy + y^2 = 45$.

(i) Find the gradient of the curve at the point $(2, -3)$. [4]

(ii) Show that there are no points on the curve at which the gradient is 1. [3]

Coaching With Shuja

(i)

Obtain $2y\frac{dy}{dx}$ as derivative of y^2 **B1**

Obtain $-4y - 4x\frac{dy}{dx}$ as derivative of $-4xy$ **B1**

Substitute $x = 2$ and $y = -3$ and find value of $\frac{dy}{dx}$, dependent on at least one B1 being earned and $\frac{d}{dx}(45) = 0$ **M1**

Obtain $\frac{12}{7}$ or equivalent **A1**

Total: 4

(ii)

Substitute $\frac{dy}{dx} = 1$ in an expression involving $\frac{dy}{dx}$, x and y and obtain $ay = bx$ **M1**

Obtain $y = x$ or equivalent **A1**

Use $y = x$ in original equation and demonstrate contradiction **A1**

Total: 3

Coaching With Shuja

Question 8

May/June 2012 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 8 The attempts at this question, apart from the occasional fully correct solution for both parts, were frequently incorrect."

Source: 9709_s12_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 8

Coaching With Shuja

The point P has coordinates $(-1, 4, 11)$ and the line l has equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}.$$

(i) Find the perpendicular distance from P to l. [4]

(ii) Find the equation of the plane which contains P and l, giving your answer in the form $ax + by + cz = d$, where a, b, c and d are integers. [5]

Coaching With Shuja

(i)

Obtain $\pm \begin{pmatrix} 2 \\ -1 \\ -15 \end{pmatrix}$ for vector PA, where A is point on line, or equivalent **B1**

Use scalar product to find cosine of angle between PA and line **M1**

Obtain $\frac{42}{\sqrt{230}\sqrt{14}}$ or equivalent **A1**

Use trigonometry to obtain $\sqrt{104}$ or 10.2 or equivalent **A1**

Alternative methods using foot of perpendicular, vector product, Pythagoras, or triangle area are accepted with equivalent marks.

Total: 4

(ii)

Use scalar product to obtain a relevant equation in a, b, c, e.g. $2a + b + 3c = 0$ or $2a - b - 15c = 0$ **M1**

State two correct equations in a, b and c **A1**

Solve simultaneous equations to obtain one ratio **M1**

Obtain $a : b : c = -3 : 9 : -1$ or equivalent **A1**

Obtain equation $-3x + 9y - z = 28$ or equivalent **A1**

Alternative methods using vector product or two-parameter plane equation are accepted.

Total: 5

Question 9

May/June 2012 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 9 B C Many candidates failed to state a correct general form such asA+ + ."

Source: 9709_s12_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 9

Coaching With Shuja

By first expressing $\frac{4x^2+5x+3}{2x^2+5x+2}$ in partial fractions, show that

$$\int_0^4 \frac{4x^2+5x+3}{2x^2+5x+2}\,dx = 8 - \ln 9.$$

[10]

Mark Scheme May/June 2012 • Paper 3 (variant 31) • Q9

State or imply form $A + \frac{B}{2x+1} + \frac{C}{x+2}$ **B1**

State or obtain $A = 2$ **B1**

Use correct method for finding B or C **M1**

Obtain $B = 1$ **A1**

Obtain $C = -3$ **A1**

Obtain $2x + \frac{1}{2}\ln(2x+1) - 3\ln(x+2)$ **B3** Deduct B1 for each error or omission.

Substitute limits in expression containing $a\ln(2x+1) + b\ln(x+2)$ **M1**

Show full and exact working to confirm that $8 + \frac{1}{2}\ln 9 - 3\ln 6 + 3\ln 2$, or an equivalent expression, simplifies to given result $8 - \ln 9$ **A1**

Total: 10

Coaching With Shuja

Question 2

Oct/Nov 2012 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “Common errors included taking ln(5x – 5) to be x In 5 – In 5 and, after letting y = 5X, taking 5x–1 to be y–1.”

Source: 9709_w12_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

Coaching With Shuja

Solve the equation

$$5^{x-1} = 5^x - 5,$$

giving your answer correct to 3 significant figures. [4]

Coaching With Shuja

EITHER Use laws of indices correctly and solve for 5^x or for 5^{-x} or for 5^{x-1} **M1**

$$\frac{5}{1-\frac{1}{\sqrt{5}}}$$

Obtain 5^x or for 5^{-x} or for 5^{x-1} in any correct form, e.g. 5^x **= A1**

Use correct method for solving $5^y = a$, or $5^{-x} = a$, or $5^{x-1} = a$, where $a \geq 0$ **M1**

Obtain answer $x = 1.14$ **A1**

OR Use an appropriate iterative formula, e.g. $x_{n+1} =$, correctly, at least once **M1**

Obtain answer **1.14 A1**

Show sufficient iterations to at least 3 d.p. to justify **1.14** to 2 d.p., or show there is a sign change in the interval $(1.135, 1.145)$ **A1**

Show there is no other root **A1** [4]

[For the solution $x = 1.14$ with no relevant working give **B1**, and a further **B1** if **1.14** is shown to be the only solution.]

Coaching With Shuja

Question 5

Oct/Nov 2012 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "However, the majority of candidates were challenged by this question."

> "Lack of an overall strategy was a 1–sinx problem in part (iii) and in part (iv) most candidates did not use the result of part (i) to write down the integral of sec X tan x."

Source: 9709_w12_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

Coaching With Shuja

(i) By differentiating $\frac{1}{\cos x}$, show that if $y = \sec x$ then $\frac{dy}{dx} = \sec x \tan x$. [2]

(ii) Show that

$$\frac{1}{\sec x - \tan x} \equiv \sec x + \tan x.$$

[1]

(iii) Deduce that

$$\frac{1}{(\sec x - \tan x)^2} = 2\sec^2 x - 1 + 2\sec x \tan x.$$

[2]

(iv) Hence show that

$$\int_0^{\frac{\pi}{4}} \frac{1}{(\sec x - \tan x)^2}\, dx = \frac{1}{4}(8\sqrt{2} - \pi).$$

[3]

Coaching with Shuja

(i) Use correct quotient or chain rule **A1**

Obtain the given answer correctly having shown sufficient working **A1**

(ii) Use a valid method, e.g. multiply numerator and denominator by $\sec x + \tan x$, and a version of Pythagoras to justify the given identity **B1** [1]

(iii) Substitute, expand $(\sec x + \tan x)^2$ and use Pythagoras once **M1**

Obtain given identity **A1** [2]

(iv) Obtain integral $2\tan x - x + 2\sec x$ **B1**

Use correct limits correctly. In an expression of the form $a\tan x + bx + c\sec x$, or equivalent, where $abc \neq 0$ **M1**

Obtain the given answer correctly **A1** [3]

Coaching With Shuja

Question 6

Oct/Nov 2012 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 6 This question was poorly answered by most candidates."

> "1–y2 These candidates usually obtained a correct solution in some form, but the removal of logarithms involved in making y the subject proved difficult for some."

Source: 9709_w12_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

The variables x and y are related by the differential equation

$$x\frac{dy}{dx} = 1 - y^2.$$

When $x = 2$, $y = 0$. Solve the differential equation, obtaining an expression for y in terms of x. [8]

Coaching With Shuja

Mark Scheme

Oct/Nov 2012 • Paper 3 (variant 31) • Q6

Separate variables correctly and attempt integration of one side **B1**

Obtain term $\ln x$ **B1**

State or imply and use a relevant method to find A or B **M1**

Obtain $A = \frac{1}{2}$, $B = \frac{1}{2}$ **A1**

Integrate and obtain $-\frac{1}{2}\ln(1-y) + \frac{1}{2}\ln(1+y)$, or equivalent **A1**

[If the integral is directly stated as $k_1 \ln$ or $k_2 \ln$ give **M1**, and then **A2** for $k_1 = \frac{1}{2}$ or $k_2 = -\frac{1}{2}$]

Evaluate a constant, or use limits $x = 2, y = 0$ in a solution containing terms $a\ln x$, $b\ln(1-y)$ and $c\ln(1+y)$, where $abc \neq 0$ **M1**

[This M mark is not available if the integral of $1/(1-y^2)$ is initially taken to be of the form $k\ln(1-y^2)$]

Obtain solution in any correct form, e.g. $\frac{1}{2}\ln\frac{x}{2} = \ln x - \ln 2$ **A1**

Rearrange and obtain $y =$, or equivalent, free of logarithms **A1** [8]

Coaching with Shuja

Question 7

Oct/Nov 2012 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “Part (ii) on the other hand was found to be difficult.”

Source: 9709_w12_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

The equation of a curve is $\ln(xy) - y^3 = 1$.

(i) Show that

$$\frac{dy}{dx} = \frac{y}{x(3y^3 - 1)}.$$

[4]

(ii) Find the coordinates of the point where the tangent to the curve is parallel to the y-axis, giving each coordinate correct to 3 significant figures. [4]

Mark Scheme

(i) EITHER: State or imply $\frac{x}{xy}\frac{dy}{dx}$ as derivative of $\ln xy$, or equivalent **B1**

State or imply $3y^2\frac{dy}{dx}$ as derivative of y^3, or equivalent **B1**

Equate derivative of LHS to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain the given answer **A1**

OR Obtain $xy = \exp(1+y^3)$ and state or imply $y' + x\frac{dy}{dx}$ as derivative of xy **B1**

State or imply $3y^2\frac{dy}{dx}\exp(1+y^3)$ as derivative of $(1+y^3)$ **B1**

Equate derivatives and solve for $\frac{dy}{dx}$ **M1**

Obtain the given answer **A1** [4]

(ii) Equate denominator to zero and solve for y **M1***

Obtain $y = 0.693$ only **A1**

Substitute found value in the equation and solve for x **M1**(dep*)

Obtain $x = 5.47$ only **A1** [4]

Coaching With Shuja

Question 10

Oct/Nov 2012 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 10 In general this question was poorly answered."

> "Yet many candidates did not grasp the geometry of the situation and worked with irrelevant pairs of direction vectors such as AB and AC, or OA and OB."

Source: 9709_w12_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 10

Coaching With Shuja

10 With respect to the origin O, the points A, B and C have position vectors given by

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 2 \\ -1 \\ 7 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 1 \\ -5 \\ -3 \end{pmatrix}$$

The plane m is parallel to $\overrightarrow{OC}$ and contains A and B.

(i) Find the equation of m, giving your answer in the form $ax + by + cz = d$. [6]

(ii) Find the length of the perpendicular from C to the line through A and B. [5]

Coaching With Shuja

10 (i) EITHER Use scalar product of relevant vectors, or subtract point equations to form two equations in a, b, c, e.g. $a - 5 - 4 = 0$ and $a = 3c - 0$ **M1**

State two correct equations in a, b, c **A1**

OR 1: Obtain $b \cdot (-3) : 0, 1$, or equivalent **A1**

OR 1: Attempt to calculate vector product of relevant vectors, e.g. $(1 - 5 - 4) \times (1 - 1 - 3k)$ **M2***

OR 1: Attempt to form 2-parameter equation of the plane with relevant vectors, e.g. $r = 3i - 2j + 4k + s(1 - 5 - 4) + t(1 - 1 - 3k)$ **M2***

OR 2: Attempt to calculate vector product of relevant vectors, e.g. $r = 3i - 2j + 4k$ State 3 equations in x, y, z, and i **A1**

OR 2: Eliminate 2 and u **A1**

OR 2: Obtain equation $3x + z = 13$, or equivalent **M1 (dep*) A1**

Obtain answer $3x + z = 13$, or equivalent **A1 [6]**

(ii) EITHER Find $\overrightarrow{CP}$ for a point P on AB with a parameter t, e.g. $2i + 3j + 7k + t(4i + j + 3k)$ **B1** *

Either: Equate scalar product $\overrightarrow{CP} \cdot \overrightarrow{AB}$ to zero and form an equation in t **M1**

Or 1: Use Pythagoras in triangle $CPA (or CPB)$ and form an equation in t **M1**

Or 2: Use Pythagoras in triangle ABC to find the length of CP **M1**

Solve and obtain correct value of t, e.g. $t = -2$ **A1**

Obtain answer $3\sqrt{2}$ (4.24), or equivalent **A1**

OR 1: State $\overrightarrow{AC}$ (or $\overrightarrow{BC}$) and $\overrightarrow{AB}$ in component form **B1** *

Using a relevant scalar product find the cosine of $\angle CAB$ (or $\angle CBA$) **M1**

Obtain correct value of $\cos A$ or $\cos B$ **A1 [6]**

OR 2: State $\overrightarrow{AC}$ (or $\overrightarrow{BC}$) and $\overrightarrow{AB}$ in component form **B1** *

Using a relevant scalar product find the length of the projection $\overrightarrow{AC}$ (or BC) on $\overrightarrow{AB}$ **M1**

Obtain answer $3\sqrt{2}$ (4.24), or equivalent **A1**

OR 3: State $\overrightarrow{AC}$ (or $\overrightarrow{BC}$) and $\overrightarrow{AB}$ in component form **B1** *

Calculate their vector product, e.g. $(-21 - 3j - 7k) \cdot (-4i - j - 3k)$ **M1**

Obtain correct product, e.g. $-2i + 13j - 5k$ **A1**

Divide modulus of the product by the modulus of $\overrightarrow{AB}$ **M1**

Obtain answer $3\sqrt{2}$ (4.24), or equivalent **A1**

OR 4: State two of $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{AC}$ in component form **B1** *

Use cosine formula in triangle ABC to find $\cos A$ or $\cos B$ **M1**

Obtain answer $3\sqrt{2}$ (4.24), or equivalent **A1 [5]**

Question 3

May/June 2011 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 3 (i) This proved a difficult question as many candidates did not realise that the vector AB was the normal to the plane and point B was in the plane."

> "(ii) Few candidates could give a correct vector representing the y-axis and many made errors in the evaluation of the scalar product of the vectors OA and OB."

Source: 9709_s11_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 31), Question 3

Coaching With Shuja

Question Paper May/June 2011 • Paper 3 (variant 31) • Q3

Points A and B have coordinates $(-1, 2, 5)$ and $(2, -2, 11)$ respectively. The plane p passes through B and is perpendicular to AB.

(i) Find an equation of p, giving your answer in the form $ax + by + cz = d$. [3]

(ii) Find the acute angle between p and the y-axis. [4]

Coaching With Shuja

(i)

Obtain $\begin{pmatrix} 3 \\ -4 \\ 6 \end{pmatrix}$ or its negative as normal to plane **B1**

Form equation of p as $3x - 4y + 6z = k$ or $-3x + 4y - 6z = k$ and use relevant point to find k **M1**

Obtain $3x - 4y + 6z = 80$ or $-3x + 4y - 6z = -80$ **A1**

Total: 3

(ii)

State the direction vector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ or equivalent **B1**

Carry out correct process for finding scalar product of two relevant vectors **M1**

Use correct complete process with moduli and scalar product and evaluate $\sin^{-1}$ or $\cos^{-1}$ of result **M1**

Obtain $30.8°$ or 0.538 radians **A1**

Total: 4

Question 4

May/June 2011 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(ii) Few candidates were able to spot the relationship between the roots of f(x) = 0 and the equation in 3 this section."

Source: 9709_s11_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

Coaching With Shuja

Question Paper May/June 2011 • Paper 3 (variant 31) • Q4

The polynomial $f(x)$ is defined by

$$f(x) = 12x^3 + 25x^2 - 4x - 12.$$

(i) Show that $f(-2) = 0$ and factorise $f(x)$ completely. [4]

(ii) Given that

$$12 \times 27^y + 25 \times 9^y - 4 \times 3^y - 12 = 0,$$

state the value of 3^y and hence find y correct to 3 significant figures. [3]

(i)

Verify that $-96 + 100 + 8 - 12 = 0$ **B1**

Attempt to find quadratic factor by division by $(x + 2)$, reaching a partial quotient $12x^2 + kx$, inspection or use of an identity **M1**

Obtain $12x^2 + x - 6$ **A1**

State $(x + 2)(4x + 3)(3x - 2)$ **A1**

Total: 4

(ii)

State $3^y = \frac{2}{3}$ and no other value **B1**

Use correct method for finding y from equation of form $3^y = k$, where $k > 0$ **M1**

Obtain -0.369 and no other value **A1**

Total: 3

Coaching with Shuja

Question 6

May/June 2011 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(ii) The iteration work was usually sound, but many candidates failed to give the final answer to the requested number of decimal places once convergence had been established."

Source: 9709_s11_er.pdf, Page 23, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

10 The number of birds of a certain species in a forested region is recorded over several ye... t years, the number of birds is N, where N is treated as a continuous variable. The varia... number of birds is modelled by

$$\frac{dN}{dt} = \frac{N(1800 - N)}{3600}.$$

It is given that $N = 300$ when $t = 0$.

(i) Find an expression for N in terms of t. [9]

(ii) According to the model, how many birds will there be after a long time? [1]

The diagram shows a circle with centre O and radius 10 cm. The chord AB divides the circle into two regions whose areas are in the ratio $1 : 4$ and it is required to find the length of AB. The angle AOB is θ radians.

(i) Show that $\theta = \frac{2}{5}\pi + \sin\theta$. [3]

(ii) Showing all your working, use an iterative formula, based on the equation in part (i), with an initial value of 2.1, to find θ correct to 2 decimal places. Hence find the length of AB in centimetres correct to 1 decimal place. [5]

Coaching with Shuja

(i)

State or imply area of segment is $\frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta$ or $50\theta - 50\sin\theta$ **B1**

Attempt to form equation from area of segment $= \frac{1}{5}$ of area of circle, or equivalent **M1**

Confirm given result $\theta = \frac{2\pi}{5} + \sin\theta$ **A1**

Total: 3

(ii)

Use iterative formula correctly at least once **M1**

Obtain value for θ of **2.11** **A1**

Show sufficient iterations to justify value of θ or show sign change in interval $(2.105, 2.115)$ **A1**

Use correct trigonometry to find an expression for the length of AB **M1** e.g. $20\sin 1.055$ or $\sqrt{200 - 200\cos 2.11}$

Hence **17.4** **A1**

Iterations: $2.1 \to 2.1198 \to 2.1097 \to 2.1149 \to 2.1122$

Total: 5

Coaching with Shuja

Question 9

May/June 2011 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "(ii) Whilst many candidates knew to use their result from part (i) in part (ii) they were often unable to 1 successfully convert $8\cos^4\theta - 3 = 1$ to $\cos 4\theta =$."

Source: 9709_s11_er.pdf, Page 24, Section: Comments on specific questions – Paper 3 (variant 31), Question 9

Coaching With Shuja

(i) Prove the identity

$$\cos 4\theta + 4\cos 2\theta \equiv 8\cos^4\theta - 3.$$

[4]

(ii) Hence

(a) solve the equation $\cos 4\theta + 4\cos 2\theta = 1$ for $-\frac{1}{2}\pi \leq \theta \leq \frac{1}{2}\pi$, [3]

(b) find the exact value of $\int_0^{\frac{1}{4}\pi} \cos^4\theta \, d\theta$. [3]

Coaching With Shuja

(i)

Express $\cos 4\theta$ as $2\cos^2 2\theta - 1$ or $\cos^2 2\theta - \sin^2 2\theta$ or $1 - 2\sin^2 2\theta$ **B1**

Express $\cos 4\theta$ in terms of $\cos\theta$ **M1**

Obtain $8\cos^4\theta - 8\cos^2\theta + 1$ **A1**

Use $\cos 2\theta = 2\cos^2\theta - 1$ to obtain given answer $8\cos^4\theta - 3$ **A1** AG

Total: 4

(ii)(a)

State or imply $\cos^4\theta = \frac{1}{2}$ **B1**

Obtain **0.572 B1**

Obtain **−0.572 B1**

Total: 3

(ii)(b)

Integrate and obtain form $k_1\theta + k_2\sin 4\theta + k_3\sin 2\theta$ **M1**

Obtain $\frac{3}{8}\theta + \frac{1}{32}\sin 4\theta + \frac{1}{4}\sin 2\theta$ **A1**

Obtain $\frac{3\pi}{32} + \frac{1}{4}$ following completely correct work **A1**

Total: 3

Coaching With Shuja

Question 1

May/June 2010 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “Only a a few candidates who reached x = and x = 7a were able to combine them into the correct final solution.”

Source: 9709_s10_er.pdf, Page 18, Section: Comments on specific questions – Paper 3 (variant 31), Question 1

Coaching With Shuja

Question Paper May/June 2010 • Paper 3 (variant 31) • Q1

Solve the inequality $|x + 3a| > 2|x - 2a|$, where a is a positive constant. [4]

Coaching With Shuja

Mark Scheme

May/June 2010 • Paper 3 (variant 31) • Q1

EITHER: State or imply non-modular inequality $(x+3a)^2 > 4(x-2a)^2$, or corresponding quadratic equation, or pair of linear equations $x+3a = \pm 2(x-2a)$ **B1**

Make reasonable solution attempt at a 3-term quadratic, or solve two linear equations **M1**

Obtain critical values $x = \frac{1}{3}a$ and $x = 7a$ **A1**

State answer $\frac{1}{3}a < x < 7a$ **A1**

OR: Obtain the critical value $x = 7a$ from a graphical method, or by inspection, or by solving a linear equation or inequality **B1**

Obtain the critical value $x = \frac{1}{3}a$ similarly **B2**

State answer $\frac{1}{3}a < x < 7a$ **B1**

Do not condone $\leq$ for $<$. Accept 0.33 for $\frac{1}{3}$.

Total: 4

Coaching With Shuja

Question 4

May/June 2010 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Question 4 Most candidates could state the required expansions, but too many then did not realise that this meant that they had the cos 2x and cos 4x required for the next part."

Source: 9709_s10_er.pdf, Page 19, Section: Comments on specific questions – Paper 3 (variant 31), Question 4

Coaching With Shuja

(i) Using the expansions of $\cos(3x - x)$ and $\cos(3x + x)$, prove that

$$\frac{1}{2}(\cos 2x - \cos 4x) \equiv \sin 3x \sin x.$$

[3]

(ii) Hence show that

$$\int_{\frac{1}{6}\pi}^{\frac{1}{3}\pi} \sin 3x \sin x \, dx = \frac{1}{8}\sqrt{3}.$$

[3]

(i)

State correct expansion of $\cos(3x - x)$ or $\cos(3x + x)$ **B1**

Substitute expansions in $\frac{1}{2}(\cos 2x - \cos 4x)$, or equivalent **M1**

Simplify and obtain the given identity correctly **A1**

Total: 3

(ii)

Obtain integral $\frac{1}{4}\sin 2x - \frac{1}{8}\sin 4x$ **B1**

Substitute limits correctly in an integral of the form $a\sin 2x + b\sin 4x$ **M1**

Obtain given answer following full, correct and exact working **A1**

Total: 3

Coaching With Shuja

Question 6

May/June 2010 | Paper 3 (variant 31)

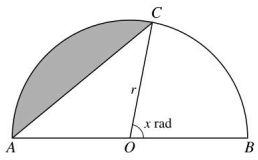
Examiner Report — Why students found this question difficult

> "Furthermore, relating a θ in their sector formula to something usually involving $(x - \pi)$, which was the approach that was usually adopted, proved difficult."

Source: 9709_s10_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 6

Coaching With Shuja

6



The diagram shows a semicircle ACB with centre O and radius r. The angle BOC is x radians. The area of the shaded segment is a quarter of the area of the semicircle.

(i) Show that x satisfies the equation

$$x = \tfrac{3}{4}\pi - \sin x.$$ [3]

(ii) This equation has one root. Verify by calculation that the root lies between 1.3 and 1.5. [2]

(iii) Use the iterative formula

$$x_{n+1} = \tfrac{3}{4}\pi - \sin x_n$$

to determine the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

The diagram shows a semicircle ACB with centre O and radius r. The angle BOC is x radians. The area of the shaded segment is a quarter of the area of the semicircle.

(i) Show that x satisfies the equation

$$x = \frac{3}{4}\pi - \sin x.$$

[3]

(ii) This equation has one root. Verify by calculation that the root lies between 1.3 and 1.5. [2]

(iii) Use the iterative formula

$$x_{n+1} = \frac{3}{4}\pi - \sin x_n$$

to determine the root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

(i)

Using the formulae $\frac{1}{2}r^2\theta$ and $\frac{1}{2}r^2\sin\theta$, or equivalent, form an equation **M1**

Obtain a correct equation in r and x and/or $x/2$ in any form **A1**

Obtain the given equation correctly **A1**

Total: 3

(ii)

Consider the sign of $x - \frac{3}{4}\pi + \sin x$ at $x = 1.3$ and $x = 1.5$, or equivalent **M1**

Complete the argument with correct calculations **A1**

Total: 2

(iii)

Use the iterative formula correctly at least once **M1**

Obtain final answer 1.38 **A1**

Show sufficient iterations to at least 4 d.p. to justify its accuracy to 2 d.p., or show there is a sign change in the interval $(1.375, 1.385)$ **A1**

Total: 3

Coaching With Shuja

Question 7

May/June 2010 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Part (iii) of this question was something that candidates found difficult."

Source: 9709_s10_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

The complex number $2 + 2i$ is denoted by u.

(i) Find the modulus and argument of u. [2]

(ii) Sketch an Argand diagram showing the points representing the complex numbers 1, i and u. Shade the region whose points represent the complex numbers z which satisfy both the inequalities $|z - 1| \leq |z - i|$ and $|z - u| \leq 1$. [4]

(iii) Using your diagram, calculate the value of $|z|$ for the point in this region for which $\arg z$ is least. [3]

Coaching With Shuja

(i)

Obtain modulus $\sqrt{8}$ **B1**

Obtain argument $\frac{1}{4}\pi$ or $45°$ **B1**

Total: 2

(ii)

Show 1, i and u in relatively correct positions on an Argand diagram **B1**

Show the perpendicular bisector of the line joining 1 and i **B1**

Show a circle with centre u and radius 1 **B1**

Shade the correct region **B1**

Total: 4

(iii)

State or imply relevance of the appropriate tangent from O to the circle **B1 FT**

Carry out complete strategy for finding $|z|$ for the critical point **M1**

Obtain answer $\sqrt{7}$ **A1**

Total: 3

Coaching With Shuja

Question 10

May/June 2010 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "Part (iii) again saw many candidates fail to choose the correct vectors from which to find relevant simultaneous equations or the vector product."

Source: 9709_s10_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 31), Question 10

Coaching With Shuja

The lines l and m have vector equations

$$\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k} + s(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$$

and

$$\mathbf{r} = 4\mathbf{i} + 6\mathbf{j} + \mathbf{k} + t(2\mathbf{i} + 2\mathbf{j} + \mathbf{k})$$

respectively.

(i) Show that l and m intersect. [4]

(ii) Calculate the acute angle between the lines. [3]

(iii) Find the equation of the plane containing l and m, giving your answer in the form $ax + by + cz = d$. [5]

Coaching with Shuja

(i)

Express general point of l or m in component form, e.g. $(1+s, 1-s, 1+2s)$ or $(4+2t, 6+2t, 1+t)$ **B1**

Equate at least two corresponding pairs of components and solve for s or t **M1**

Obtain $s = -1$ or $t = -2$ **A1**

Verify that all three component equations are satisfied **A1**

Total: 4

(ii)

Carry out correct process for evaluating the scalar product of the direction vectors of l and m **M1**

Using the correct process for the moduli, divide the scalar product by the product of the moduli and evaluate the inverse cosine of the result **M1**

Obtain answer $74.2°$ or 1.30 radians **A1**

Total: 3

(iii)

EITHER: Use scalar product to obtain $a - b + 2c = 0$ and $2a + 2b + c = 0$ **B1**

Solve and obtain one ratio, e.g. $a : b$ **M1**

Obtain $a : b : c = 5 : -3 : -4$, or equivalent **A1**

Substitute coordinates of a relevant point and values for a, b and c in general equation of plane and evaluate d **M1**

Obtain answer $5x - 3y - 4z = -2$, or equivalent **A1**

OR: Attempt to calculate vector product of direction vectors of l and m **M1**

Obtain two correct components of the product **A1**

Obtain correct product, e.g. $-5\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$ **A1**

Form a plane equation and use coordinates of a relevant point to calculate d **M1**

Obtain answer $-5x + 3y + 4z = 2$, or equivalent **A1**

Total: 5

Question 1

Oct/Nov 2010 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Question 1 Many candidates did not appreciate that the function could not be integrated as it stood but the bracket $3\left(1-\frac{1}{x}\right)$ needed to be expanded first."

Source: 9709_w10_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 11), Question 1

Find $\int \left(x + \frac{1}{x}\right)^2 dx$

Total: 3

Coaching With Shuja

Mark Scheme

M1 Integration of $\int \left(x + \frac{1}{x}\right)^2 dx$

$$= \frac{x^3}{3} - \frac{1}{x} + 2x + (c)$$

B1 × 3 (3 marks)

co Omission of middle term of expansion can still get 2/3.

Total: 3

Question 2

Oct/Nov 2010 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "There were a few incorrect evaluations of | | but the most | | | âŽ3⎠ âŽ3⎠ common mistake was to forget to raise a to the power of 3 which gave answers of 20 × −5 instead of 20 × (−5)3."

Source: 9709_w10_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 11), Question 2

Coaching With Shuja

Question Paper Oct/Nov 2010 • Paper 1 (variant 11) • Q2

In the expansion of $(1+ax)^6$, where a is a constant, the coefficient of x is -30. Find the coefficient of x^3.

Total: 4

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$(1+ax)^6$

Term in $x = 6ax$

B1 Equate with $-30 \rightarrow a = -5$

B1√

Term in $x^3 = \frac{6.5.4}{3!}a^3$

B1 $\rightarrow$ coefficient of -2500

B1√

co $\surd$ from his answer for $6ax$

co $\surd$ for $20 \times a^3$

Total: 4

Coaching With Shuja

Question 3

Oct/Nov 2010 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> “A fairly common mistake, however, was to see (2x + 3)2 − 2x instead of (2x + 3)2 − 2(2x + 3); algebraic mistakes were also made in the simplification of this expression.”

> “Expressing a quadratic expression (the correct expression is 4x2 + 8x + 3) in the form a(x + b)2 + c needs a lot of care and a significant number of candidates made algebraic errors.”

Source: 9709_w10_er.pdf, Page 1, Section: Comments on specific questions – Paper 1 (variant 11), Question 3

Functions f and g are defined for $x \in \mathbb{R}$ by

$$f : x \mapsto 2x + 3,$$

$$g : x \mapsto x^2 - 2x.$$

Express $gf(x)$ in the form $a(x + b)^2 + c$, where a, b and c are constants.

Total: 5

$f : x \mapsto 2x + 3$

$g : x \mapsto x^2 - 2x$

$gf(x) = (2x + 3)^2 - 2(2x + 3)$

$= 4x^2 + 8x + 3$

M1 Must be f into g, not g into f.

A1 co

3 × B1√ Allow all these as $\surd$ for either fg or gf.

$= 4(x + 1)^2 - 1$

Total: 5

Coaching With Shuja

Question 4

Oct/Nov 2010 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> "Question 4 Part (i) proved to be difficult for candidates and there were relatively few candidates who scored full marks for this part."

Source: 9709_w10_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 4

Coaching With Shuja

(i) Prove the identity $\frac{\sin x \tan x}{1-\cos x} \equiv 1 + \frac{1}{\cos x}$.

Total: 3

(ii) Hence solve the equation $\frac{\sin x \tan x}{1-\cos x} + 2 = 0$, for $0° \leq x < 360°$.

Total: 3

(i)

$$\frac{\sin x \tan x}{1-\cos x} = \frac{\sin x \cdot \frac{\sin x}{\cos x}}{1-\cos x}$$

M1 Use of $\tan x = \sin x \div \cos x$

$$= \frac{\sin^2 x}{\cos x(1-\cos x)}$$

M1 Use of $\sin^2 x = 1 - \cos^2 x$

$$= \frac{(1-\cos x)(1+\cos x)}{\cos x(1-\cos x)} = \frac{1}{\cos x} + 1$$

M1 Realising the need to use difference of 2 squares. Answer given.

Total: 3

(ii)

$$\frac{1}{\cos x} + 1 + 2 = 0$$

$$\rightarrow \cos x = -\frac{1}{3}$$

$$\rightarrow x = 109.5^\circ \text{ or } 250.5^\circ$$

M1 Uses part (i) with $\cos x$ as subject.

A1 A1√ co. $\surd$ for $360^\circ - 1^{st}$ answer.

Total: 3

Question 5

Oct/Nov 2010 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

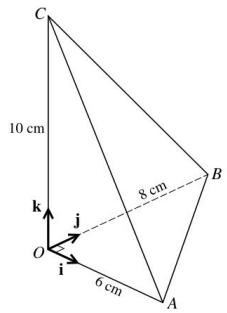
> "Question 5 This question was found more difficult than usual for questions of this type for two reasons."

Source: 9709_w10_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 5

Coaching With Shuja

Question 5

5



The diagram shows a pyramid $OABC$ with a horizontal base OAB where $OA = 6$ cm, $OB = 8$ cm and angle $AOB = 90°$. The point C is vertically above O and $OC = 10$ cm. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OB and OC as shown.

Use a scalar product to find angle ACB. [6]

Use a scalar product to find angle ACB. **Total: [6]**

5

$\overrightarrow{AC} = -6\mathbf{i} + 10\mathbf{k}$ **B1**

$\overrightarrow{BC} = -8\mathbf{j} + 10\mathbf{k}$ **B1**

$\overrightarrow{AC} \cdot \overrightarrow{BC} = 100$ **M1**

$\overrightarrow{AC} \cdot \overrightarrow{BC} = \sqrt{136}\sqrt{164}\cos ACB$ **M1**

M1 (All linked correctly – for ACB only)

Angle $ACB = 48.0^\circ$ **A1** co

Total: [6]

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Question 7

Oct/Nov 2010 | Paper 1 (variant 11)

Examiner Report — Why students found this question difficult

> “Question 7 1 1 Part (i) was found difficult.”

> “In part (ii) many candidates did not appreciate the âŽ2 / significance of the word ‘exact’ in the question and gave the answer as a decimal number.”

> “In this question, the first two parts are quite helpful in sketching the graph in part (iii) but many candidates did not seem to appreciate this.”

Source: 9709_w10_er.pdf, Page 2, Section: Comments on specific questions – Paper 1 (variant 11), Question 7

Coaching With Shuja

Question 7

A function f is defined by $f : x \mapsto 3 - 2\tan\left(\frac{1}{2}x\right)$ for $0 \leq x < \pi$.

(i) State the range of f. **Total: [1]**

(ii) State the exact value of $f\left(\frac{2\pi}{3}\right)$. **Total: [1]**

(iii) Sketch the graph of $y = f(x)$. **Total: [2]**

(iv) Obtain an expression, in terms of x, for $f^{-1}(x)$. **Total: [3]**

7 $x \mapsto 3 - 2\tan\left(\frac{1}{2}x\right)$

(i) Range of $f \leq 3$ **B1** co. Allow $<$ [1]

(ii) $f\left(\frac{2\pi}{3}\right) = 3 - 2\sqrt{3}$ **B1** co [1]

(iii) **B2**, 1, 0 Indep. Starting at $y = 3$. Shape correct – no turning points. Tending tangentially towards $x = \pi$. [2]

(iv) $y = 3 - 2\tan\left(\frac{x}{2}\right)$ **M1**, **M1** (Attempt at making x the subject. Order of operations all ok.)

$\rightarrow f^{-1}(x) = 2\tan^{-1}\left(\frac{3-x}{2}\right)$ **A1** co – but with x, not y. [3]

Coaching With Shuja

Question 1

Oct/Nov 2010 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "The most common errors occurred in finding the third term and (−2x2)2 was often replaced by −4x4 or by ±2x4."

Source: 9709_w10_er.pdf, Page 4, Section: Comments on specific questions – Paper 1 (variant 12), Question 1

Coaching With Shuja

Question 1

(i) Find the first 3 terms in the expansion, in ascending powers of x, of $(1 - 2x^2)^8$. **Total: [2]**

(ii) Find the coefficient of x^4 in the expansion of $(2 - x^2)(1 - 2x^2)^8$. **Total: [2]**

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1

(i) $1 + 8(-2x^2) + {}^8C_2(-2x^2)^2$ **B2**, 1

$1 - 16x^2 + 112x^4$ **A1** [2]

Loses 1 for each error

(ii) $(2 - x^2) \times \text{their}(1 - 16x^2 + 112x^4)$ **M1**

$(2 \times \text{their } 112) - \text{their}(-16)$ **A1**$\surd$ [2]

Must consider exactly 2 terms

240 A1

Coaching With Shuja

Question 4

Oct/Nov 2010 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 4 (i) Many candidates failed to realise that angle AOB equalled $2\pi - 2.3 - 2.3$, and many others disregarded the instruction to provide an answer correct to 4 significant figures."

> "(ii) A large number of candidates incorrectly assumed that the shaded region was itself a sector of a circle with centre C."

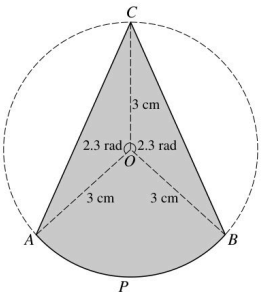
> "Another common error was to find the non-shaded region within the circle."

Source: 9709_w10_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 4

Coaching With Shuja

Question 4

4



The diagram shows points A, C, B, P on the circumference of a circle with centre O and radius 3 cm. Angle AOC = angle BOC = 2.3 radians.

(i) Find angle AOB in radians, correct to 4 significant figures. [1]

(ii) Find the area of the shaded region $ACBP$, correct to 3 significant figures. [4]

The diagram shows points $\boldsymbol{A}$, $\boldsymbol{C}$, $\boldsymbol{B}$, $\boldsymbol{P}$ on the circumference of a circle with centre $\boldsymbol{O}$ and radius 3 cm. Angle $\boldsymbol{AOC}$ = angle $\boldsymbol{BOC} = \mathbf{2.3}$ radians.

(i) Find angle $\boldsymbol{AOB}$ in radians, correct to 4 significant figures. **Total: [1]**

(ii) Find the area of the shaded region $\boldsymbol{ACBP}$, correct to 3 significant figures. **Total: [4]**

4

(i) 1.683(18 . . .) B1 [1]

(ii) (2) $\times \frac{1}{2} \times 3^2 \sin 2.3$ M1

$\frac{1}{2} \times 3^2 \times$ **their 1.683 M1** NB M0 if using angle of 2.3

Triangle $AOC + COB +$ sector **M1** Two correct triangles + sector

14.3 A1 co [4]

Question 5

Oct/Nov 2010 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "Question 5 (i) This was well answered with most candidates finding the common difference as −7 (though +7 was a common error) and then using the correct formula for the sum of m terms."

> "(ii) This presented more difficulty and the algebraic nature of the question led to many candidates omitting it altogether."

Source: 9709_w10_er.pdf, Page 5, Section: Comments on specific questions – Paper 1 (variant 12), Question 5

Question 5

(a) The first and second terms of an arithmetic progression are 161 and 154 respectively. The sum of the first m terms is zero. Find the value of m. **Total: [3]**

(b) A geometric progression, in which all the terms are positive, has common ratio r. The sum of the first n terms is less than 90% of the sum to infinity. Show that $r^n > 0.1$. **Total: [3]**

5

(a) $d = -7$ used **B1** co

$\frac{m}{2}[322 + (m-1)(-7)] = 0$ **M1**

47 A1 [3]

Condone omission of $(m/2)$. Statement co (condone $m = 0$)

(b) $\frac{a(1-r^n)}{1-r} < \frac{0.9a}{1-r}$ **M1** Allow for $=, <, >, \leq, \geq$

$1 - r^n < 0.9$ **M1** Needs inequality sign correct

$r^n > 0.1$ **A1** co [3]

Coaching With Shuja

Question 8

Oct/Nov 2010 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

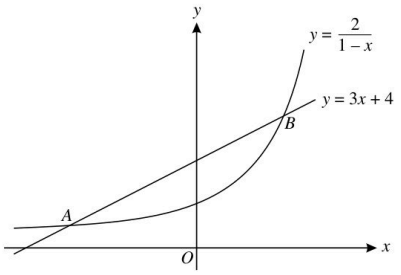
> “⎛ x –x y –y ⎞ (ii) A few candidates lost marks through incorrectly assuming that the mid-point is ⎜ 1 2, 1 2 ⎟ âŽ 2 2 ⎝ ( ) ( ) ⎠ or that the distance is x +x 2 + y +y 2 .”

Source: 9709_w10_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 8

Coaching With Shuja

Question 8

8



The diagram shows part of the curve $y = \frac{2}{1-x}$ and the line $y = 3x + 4$. The curve and the line meet at points A and B.

(i) Find the coordinates of A and B. [4]

(ii) Find the length of the line AB and the coordinates of the mid-point of AB. [3]

The diagram shows part of the curve $\boldsymbol{y = \frac{2}{1-x}}$ and the line $\boldsymbol{y = 3x + 4}$. The curve and the line meet at points $\boldsymbol{A}$ and $\boldsymbol{B}$.

(i) Find the coordinates of $\boldsymbol{A}$ and $\boldsymbol{B}$. **Total: [4]**

(ii) Find the length of the line $\boldsymbol{AB}$ and the coordinates of the mid-point of $\boldsymbol{AB}$. **Total: [3]**

Shuja

8

(i) $3x^2 + x - 2 = 0$ **M1A1**

$(x+1)(3x-2) \rightarrow x = -1$ or $\frac{2}{3}$ **M1**

Eliminates x or y. Sets quadratic to 0. Attempt to solve **their** equation co

$(-1, 1), (\frac{2}{3}, 6)$ **A1** [4]

(ii) $AB^2 = (5/3)^2 + 5^2$ **M1** $\sqrt{\text{their coordinates from (i)}}$

$AB = 5.27(0\ldots)$ **A1** Or $(5\sqrt{10})/3$ oe ft from **their (i)**

mid-point $= (-1/6, 7/2)$ **B1**$\sqrt{}$ [3]

Question 9

Oct/Nov 2010 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

> "(ii) Less than a half of all solutions correctly found vector BG, with the coefficients of j and k causing most difficulty."

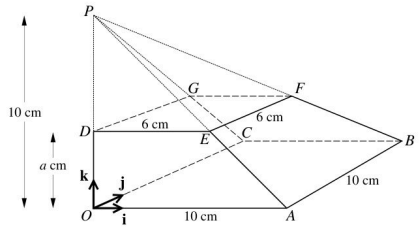
> "The use of BG.AB instead of BG.BA, thereby giving the obtuse angle, was another common error."

Source: 9709_w10_er.pdf, Page 6, Section: Comments on specific questions – Paper 1 (variant 12), Question 9

Coaching With Shuja

Question 9

9



The diagram shows a pyramid $OABCP$ in which the horizontal base $OABC$ is a square of side 10 cm and the vertex P is 10 cm vertically above O. The points D, E, F, G lie on OP, AP, BP, CP respectively and $DEFG$ is a horizontal square of side 6 cm. The height of $DEFG$ above the base is a cm. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OC and OD respectively.

(i) Show that $a = 4$. [2]

(ii) Express the vector $\overrightarrow{BG}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. [2]

(iii) Use a scalar product to find angle GBA. [4]

The diagram shows a pyramid $OABCP$ in which the horizontal base $OABC$ is a square of side 10 cm and the vertex P is 10 cm vertically above O. The points D, E, F, G lie on OP, AP, BP, CP respectively and $DEFG$ is a horizontal square of side 6 cm. The height of $DEFG$ above the base is a cm. Unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to OA, OC and OD respectively.

(i) Show that $a = 4$. **Total: [2]**

(ii) Express the vector $\overrightarrow{BG}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. **Total: [2]**

(iii) Use a scalar product to find angle GBA. **Total: [4]**

9

(i) $\frac{10-a}{10} = \frac{6}{10}$ oe **M1**

$a = 4$ **A1** AG [2]

or PDE is isos hence $PD = 6$ (M1)

(ii) $\overrightarrow{BG} = -10\mathbf{i} - 10\mathbf{j} + 4\mathbf{k} + 6\mathbf{j}$ **B2**, 1

$= -10\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$ [2]

Any acceptable notation. Loses 1 for each error.

(iii) $\overrightarrow{BG} \cdot \overrightarrow{BA} = 40$ **M1** Use of $x_1x_2 + y_1y_2 + z_1z_2$

$\cos GBA = \frac{40}{\sqrt{132}\sqrt{100}}$ **M1** Modulus worked correctly for either. All ok – must be using $\pm\overrightarrow{BG} \cdot \pm\overrightarrow{AB}$.

$GBA = 69.6^\circ$ **A1** Must be the acute angle [4]

DM1

Question 11

Oct/Nov 2010 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

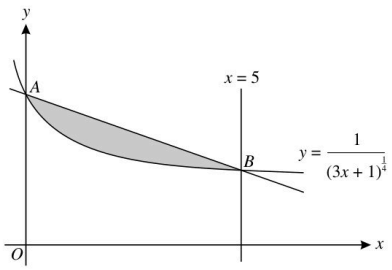
> "The use of limits was generally accurate, though âŽ automatic omission of a value at x = 0 remains a common error."

Source: 9709_w10_er.pdf, Page 7, Section: Comments on specific questions – Paper 1 (variant 12), Question 11

Coaching With Shuja

Question 11

11



The diagram shows part of the curve $y = \frac{1}{(3x+1)^{\frac{1}{3}}}$. The curve cuts the y-axis at A and the line $x = 5$ at B.

(i) Show that the equation of the line AB is $y = -\frac{1}{10}x + 1$. [4]

(ii) Find the volume obtained when the shaded region is rotated through 360° about the x-axis. [9]

The diagram shows part of the curve $\boldsymbol{y = \frac{1}{(3x+1)^{1/3}}}$. The curve cuts the y-axis at $\boldsymbol{A}$ and the line $\boldsymbol{x = 5}$ at $\boldsymbol{B}$.

(i) Show that the equation of the line $\boldsymbol{AB}$ is $\boldsymbol{y = -\frac{1}{10}x + 1}$. **Total: [4]**

(ii) Find the volume obtained when the shaded region is rotated through $\boldsymbol{360^\circ}$ about the x-axis. **Total: [9]**

Shuja

11

(i) $A = (0, 1)$ **B1**

$B = (5, \frac{1}{2})$ **B1**

$y - 1 = -\frac{1}{10}(x - 0)$ **M1** ft their A, B

$y = -\frac{1}{10}x + 1$ **A1** AG [4]

(ii) Curve: $(\pi) \int_0^5 (3x + 1)^{-1/2} dx$ **M1** Attempt $\int y^2 dx$ (π not vital)

$\frac{2\pi}{3}[(3x + 1)^{1/2}]_0^5$ **A1A1** (π not vital). 2nd A mark is for $\div 3$.

$\frac{2\pi}{3}[4 - 1]$ **DM1** Application of limits to **their** integral (in either integral). Limits 0 to 5 only.

$[2\pi]$ [9]

Line: $(\pi) \int_0^5 \left(\frac{1}{100}x^2 - \frac{1}{5}x + 1\right) dx$ **M1** Attempt $\int y^2 dx$ (π not vital)

$(\pi)\left[\frac{1}{300}x^3 - \frac{1}{10}x^2 + x\right]_0^5$ **A2**, 1

Also directly $-\frac{10}{3}\left(-\frac{1}{10}x + 1\right)^3$ or $-\frac{10}{3}\left[-\left(\frac{1}{2} + 1\right)^3 - (-1)^3\right]$ (π not vital)

$(\pi)\left[\frac{125}{300} - \frac{25}{10} + 5\right]$ – applying limits to **their** integral

$\frac{35\pi}{12}$ **DM1** Subtraction of **their** volumes co

Volume $= \frac{35\pi}{12} - 2\pi = \frac{11\pi}{12}$ **A1** [9]

Coaching With Shuja

Question 8

Oct/Nov 2010 | Paper 1 (variant 13)

Examiner Report — Why students found this question difficult

> "A common misconception was to state the area of the rhombus as 5 × 5."

Source: 9709_w10_er.pdf, Page 10, Section: Comments on specific questions – Paper 1 (variant 13), Question 8

Coaching With Shuja

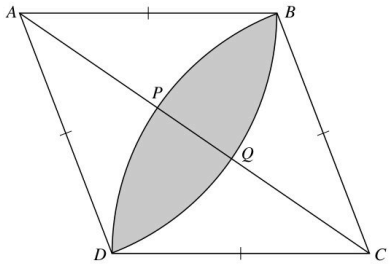
Question Paper Oct/Nov 2010 • Paper 1 (variant 13) • Q8

The diagram shows a rhombus $ABCD$. Points P and Q lie on the diagonal AC such that BPD is an arc of a circle with centre C and BQD is an arc of a circle with centre A. Each side of the rhombus has length 5 cm and angle $BAD = 1.2$ radians.

(i) Find the area of the shaded region $BPDQ$. **[4]**

(ii) Find the length of PQ. **[4]**

8



The diagram shows a rhombus $ABCD$. Points P and Q lie on the diagonal AC such that BPD is an arc of a circle with centre C and BQD is an arc of a circle with centre A. Each side of the rhombus has length 5 cm and angle $BAD = 1.2$ radians.

(i) Find the area of the shaded region $BPDQ$. [4]

(ii) Find the length of PQ. [4]

Shuja

(i)

B1 $\frac{1}{2} \times 5^2 \times 1.2$

B1 $\frac{1}{2} \times 5^2 \times \sin 1.2$

M1 $2[\frac{1}{2} \times 5^2 \times 1.2 - \frac{1}{2} \times 5^2 \times \sin 1.2]$

A1 6.70

Total: 4

Subtraction and multiplication by 2. Accept 6.7 or anything rounding to 6.70.

(ii)

M1 $5 \cos 0.6$

M1 $5 - \text{//}5 \cos 0.6\text{//}$

M1 $10(1 - \cos 0.6)$

A1 1.75

Total: 4

Subtraction from 5. Multiplication by 2.

Coaching With Shuja

Question 2

Oct/Nov 2010 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “A common error was to start by replacing ln(1 + x2) with ln 1 + ln x2, and sometimes a further error led to a quadratic in In x.”

Source: 9709_w10_er.pdf, Page 18, Section: Comments on specific questions – Paper 3 (variant 31), Question 2

Solve the equation

$$\ln(1+x^2) = 1 + 2\ln x,$$

giving your answer correct to 3 significant figures. **[4]**

Mark Scheme

M1 Use law for the logarithm of a power, a quotient, or a product correctly at least once

M1 Use $\ln e = 1$ or $e = \exp(1)$

A1 Obtain a correct equation free of logarithms, e.g. $1 + x^2 = ex^2$

A1 Solve and obtain answer $x = 0.763$ only

Total: 4

[For the solution $x = 0.763$ with no relevant working give B1, and a further B1 if 0.763 is shown to be the only root.]

[Treat the use of logarithms to base 10 with answer 0.333 only, as a misread.]

[SR: Allow iteration, giving B1 for an appropriate formula, e.g. $x_{n+1} = \exp((\ln(1 + x_n^2) - 1)/2)$, **M1** for using it correctly once, **A1** for 0.763, and **A1** for showing the equation has no other root but 0.763.]

Coaching With Shuja

Question 3

Oct/Nov 2010 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common errors arose in the algebraic work leading to the value of tan θ."

Source: 9709_w10_er.pdf, Page 18, Section: Comments on specific questions – Paper 3 (variant 31), Question 3

Coaching With Shuja

Solve the equation

$$\cos(\theta + 60^\circ) = 2\sin\theta,$$

giving all solutions in the interval $0^\circ \leq \theta \leq 360^\circ$. **[5]**

Mark Scheme

M1 Attempt use of $\cos(A+B)$ formula to obtain an equation in $\cos\theta$ and $\sin\theta$

M1 Use trig formula to obtain an equation in $\tan\theta$ (or $\cos\theta$, $\sin\theta$ or $\cot\theta$)

A1 Obtain $\tan\theta = 1/(4+\sqrt{3})$ or equivalent (or find $\cos\theta$, $\sin\theta$ or $\cot\theta$)

A1 Obtain answer $\theta = 9.9^\circ$

A1 Obtain $\theta = 189.9^\circ$, and no others in the given interval

Total: 5

[Ignore answers outside the given interval. Treat answers in radians as a misread (0.173, 3.31).]

[The other solution methods are *via* $\cos\theta = \pm(4+\sqrt{3})/\sqrt{1+(4+\sqrt{3})^2}$ or $\sin\theta = \pm 1/\sqrt{1+(4+\sqrt{3})^2}$.]

Coaching With Shuja

Question 5

Oct/Nov 2010 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> "The most common errors in faulty attempts were (a) replacing (4−4sin2θ) with 2 − 2 sinθ, (b) replacing dx by dθ, and (c) omitting a justification of the change in limits."

Source: 9709_w10_er.pdf, Page 19, Section: Comments on specific questions – Paper 3 (variant 31), Question 5

Let $I = \int_0^{\pi/6} \frac{x^2}{\sqrt{4 - x^2}}\, dx$.

(i) Using the substitution $x = 2\sin\theta$, show that

$$I = \int_0^{\pi/6} 4\sin^2\theta\, d\theta. \quad **[3]**$$

(ii) Hence find the exact value of I. **[4]**

(i)

B1 State or imply $dx = 2\cos\theta d\theta$, or $\frac{dx}{d\theta} = 2\cos\theta$, or equivalent

M1 Substitute for x and dx throughout the integral

A1 Obtain the given answer correctly, having changed limits and shown sufficient working

Total: 3

(ii)

B1 Replace integrand by $2 - 2\cos 2\theta$, or equivalent

B1√ Obtain integral $2\theta - \sin 2\theta$, or equivalent

M1 Substitute limits correctly in an integral of the form $a\theta + b\sin 2\theta$, where $ab \neq 0$

A1 Obtain answer $\frac{1}{3}\pi - \frac{\sqrt{3}}{2}$ or exact equivalent

Total: 4

[The f.t. is on integrands of the form $a + c\cos 2\theta$, where $ac \neq 0$.]

Coaching With Shuja

Question 7

Oct/Nov 2010 | Paper 3 (variant 31)

Examiner Report — Why students found this question difficult

> “In this passage of work common errors included ' −2(2 − 2λ) = −4 − 4λ ' and 1 ‘12λ + 2 = 0, ∴λ= '.”

Source: 9709_w10_er.pdf, Page 20, Section: Comments on specific questions – Paper 3 (variant 31), Question 7

Coaching With Shuja

Question Paper Oct/Nov 2010 • Paper 3 (variant 31) • Q7

With respect to the origin O, the points A and B have position vectors given by $\vec{OA} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and $\vec{OB} = 3\mathbf{i} + 4\mathbf{j}$. The point P lies on the line AB and OP is perpendicular to AB.

(i) Find a vector equation for the line AB. **[1]**

(ii) Find the position vector of P. **[4]**

(iii) Find the equation of the plane which contains AB and which is perpendicular to the plane OAB, giving your answer in the form $ax + by + cz = d$. **[4]**

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(i) State correct equation in any form, e.g. $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k} + \lambda(2\mathbf{i} + 2\mathbf{j} - 2\mathbf{k})$
B1 [1]

(ii) EITHER: Equate a relevant scalar product to zero and form an equation in λ
M1 OR 1: Equate derivative of OP^2 (or OP) to zero and form an equation in λ
M1 OR 2: Use Pythagoras in OAP or OBP and form an equation in λ
M1 State a correct equation in any form
A1 Solve and obtain $\lambda = -\frac{1}{6}$ or equivalent
A1 Obtain final answer $\vec{OP} = \frac{2}{3}\mathbf{i} + \frac{5}{3}\mathbf{j} + \frac{7}{3}\mathbf{k}$, or equivalent
A1 [4]

(iii) EITHER: State or imply $\vec{OP}$ is a normal to the required plane
M1 State normal vector $2\mathbf{i} + 5\mathbf{i} + 7\mathbf{k}$, or equivalent
A1√ Substitute coordinates of a relevant point in $2x + 5y + 7z = d$ and evaluate d
M1 Obtain answer $2x + 5y + 7z = 26$, or equivalent
A1 OR 1: Find a vector normal to plane AOB and calculate its vector product with a direction vector for the line AB
M1* A1 Substitute coordinates of a relevant point in $2x + 5y + 7z = d$ and evaluate d
M1(dep*) Obtain answer $2x + 5y + 7z = 26$, or equivalent
A1 OR 2: Set up and solve simultaneous equations in a, b, c derived from zero scalar products of $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ with (i) a direction vector for line AB, (ii) a normal to plane OAB
M1* A1 Substitute coordinates of a relevant point in $2x + 5y + 7z = d$ and evaluate d
M1(dep*) Obtain answer $2x + 5y + 7z = 26$, or equivalent
A1 OR 3: With Q on plane, use Pythagoras in OPQ to form an equation in x, y and z
M1* A1√ Form a correct equation
M1(dep*) Reduce to linear form
A1 Obtain answer $2x + 5y + 7z = 26$, or equivalent
A1 OR 4: Find a vector normal to plane AOB and form a 2-parameter equation with relevant vectors, e.g., $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k} + \lambda(2\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}) + \mu(8\mathbf{i} - 6\mathbf{j} + 2\mathbf{k})$
**** M1 ***** A1 ** *State three correct equations in* x, y, z, λ *and* μ
**** M1 (dep*) ** *Eliminate* λ *and* μ ** A1 ** *Obtain answer*** $2x + 5y + 7z = 26$, or equivalent
A1 [4]

Question 2

Oct/Nov 2010 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "A common error was to start by replacing ln(1 + x2) with ln 1 + ln x2, and sometimes a further error led to a quadratic in 1 In x."

Source: 9709_w10_er.pdf, Page 22, Section: Comments on specific questions – Paper 3 (variant 32), Question 2

Solve the equation

$$\ln(1+x^2) = 1 + 2\ln x,$$

giving your answer correct to 3 significant figures. **[4]**

Mark Scheme

Oct/Nov 2010 • Paper 3 (variant 32) • Q2

M1 Use law for the logarithm of a power, a quotient, or a product correctly at least once

M1 Use $\ln e = 1$ or $e = \exp(1)$

A1 Obtain a correct equation free of logarithms, e.g. $1 + x^2 = ex^2$

A1 Solve and obtain answer $x = 0.763$ only

Total: 4

[For the solution $x = 0.763$ with no relevant working give B1, and a further B1 if 0.763 is shown to be the only root.]

[Treat the use of logarithms to base 10 with answer 0.333 only, as a misread.]

[SR: Allow iteration, giving B1 for an appropriate formula, e.g. $x_{n+1} = \exp((\ln(1 + x_n^2) - 1)/2)$, **M1** for using it correctly once, **A1** for 0.763, and **A1** for showing the equation has no other root but 0.763.]

Coaching With Shuja

Question 3

Oct/Nov 2010 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors arose in the algebraic work leading to the value of tan θ."

Source: 9709_w10_er.pdf, Page 18, Section: Comments on specific questions – Paper 3 (variant 32), Question 3

Coaching With Shuja

Solve the equation

$$\cos(\theta + 60^\circ) = 2\sin\theta,$$

giving all solutions in the interval $0^\circ \leq \theta \leq 360^\circ$. **[5]**

Coaching With Shuja

Mark Scheme

M1 Attempt use of $\cos(A+B)$ formula to obtain an equation in $\cos\theta$ and $\sin\theta$

M1 Use trig formula to obtain an equation in $\tan\theta$ (or $\cos\theta$, $\sin\theta$ or $\cot\theta$)

A1 Obtain $\tan\theta = 1/(4+\sqrt{3})$ or equivalent (or find $\cos\theta$, $\sin\theta$ or $\cot\theta$)

A1 Obtain answer $\theta = 9.9^\circ$

A1 Obtain $\theta = 189.9^\circ$, and no others in the given interval

Total: 5

[Ignore answers outside the given interval. Treat answers in radians as a misread (0.173, 3.31).]

[The other solution methods are *via* $\cos\theta = \pm(4+\sqrt{3})/\sqrt{1+(4+\sqrt{3})^2}$ or $\sin\theta = \pm 1/\sqrt{1+(4+\sqrt{3})^2}$.]

Coaching With Shuja

Question 5

Oct/Nov 2010 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The most common errors in faulty attempts were (a) replacing (4−4sin2θ) with 2 − 2 sinθ, (b) replacing dx by dθ, and (c) omitting a justification of the change in limits."

Source: 9709_w10_er.pdf, Page 19, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

Let $I = \int_0^{\pi/6} \frac{x^2}{\sqrt{4 - x^2}}\, dx$.

(i) Using the substitution $x = 2\sin\theta$, show that

$$I = \int_0^{\pi/6} 4\sin^2\theta\, d\theta. \quad **[3]**$$

(ii) Hence find the exact value of I. **[4]**

(i)

B1 State or imply $dx = 2\cos\theta d\theta$, or $\frac{dx}{d\theta} = 2\cos\theta$, or equivalent

M1 Substitute for x and dx throughout the integral

A1 Obtain the given answer correctly, having changed limits and shown sufficient working

Total: 3

(ii)

B1 Replace integrand by $2 - 2\cos 2\theta$, or equivalent

B1√ Obtain integral $2\theta - \sin 2\theta$, or equivalent

M1 Substitute limits correctly in an integral of the form $a\theta + b\sin 2\theta$, where $ab \textcircled{b} 0$

A1 Obtain answer $\frac{1}{3}\pi - \frac{\sqrt{3}}{2}$ or exact equivalent

Total: 4

[The f.t. is on integrands of the form $a + c\cos 2\theta$, where $ac \textcircled{b} 0$.]

Coaching With Shuja

Question 10

Oct/Nov 2010 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “Only a few candidates answered part (iv) correctly.”

Source: 9709_w10_er.pdf, Page 21, Section: Comments on specific questions – Paper 3 (variant 32), Question 10

Coaching With Shuja

10 A certain substance is formed in a chemical reaction. The mass of substance formed t seconds after the start of the reaction is x grams. At any time the rate of formation of the substance is proportional to $(20 - x)$. When $t = 0$, $x = 0$ and $\frac{dx}{dt} = 1$.

(i) Show that x and t satisfy the differential equation

$$\frac{dx}{dt} = 0.05(20 - x).$$

[2]

(ii) Find, in any form, the solution of this differential equation. [5]

(iii) Find x when $t = 10$, giving your answer correct to 1 decimal place. [2]

(iv) State what happens to the value of x as t becomes very large. [1]

Coaching with Shuja

(i) State or imply $\frac{dx}{dt} = k(20 - x)$ **B1**

Show that $k = 0.05$ **B1** [2]

(ii) Separate variables correctly and integrate both sides **B1**

Obtain term $-\ln(20 - x)$, or equivalent **B1**

Obtain term $\frac{1}{20}t$, or equivalent **B1**

Evaluate a constant or use limits $t = 0, x = 0$ in a solution containing terms $a\ln(20 - x)$ and bt **M1***

Obtain correct answer in any form, e.g. $\ln 20 - \ln(20 - x) = \frac{1}{20}t$ **A1** [5]

(iii) Substitute $t = 10$ and calculate x **M1(dep*)$**

Obtain answer $x = 7.9$ **A1** [2]

(iv) State that x approaches 20 **B1** [1]

Coaching With Shuja

Question 1

Oct/Nov 2010 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "A few candidates were able to write down an expression involving $\binom{-3}{1}$ $\binom{-3}{2}$ unsimplified binomial coefficients and but they did not gain any credit until they expanded these coefficients."

Source: 9709_w10_er.pdf, Page 26, Section: Comments on specific questions – Paper 3 (variant 33), Question 1

Coaching With Shuja

1 Expand $(1+2x)^{-3}$ in ascending powers of x, up to and including the term in x^2, simplifying the coefficients. **[3]**

Coaching With Shuja

Mark Scheme

Oct/Nov 2010 • Paper 3 (variant 33) • Q1

Obtain $\mathbf{1 - 6x}$ **B1**

State correct unsimplified $\boldsymbol{x^2}$ term. Binomial coefficients must be expanded. **M1**

Obtain $\mathbf{\ldots + 24x^2}$ **A1** [3]

Coaching With Shuja

Question 2

Oct/Nov 2010 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The use of product or quotient rule to dt dx dt obtain or proved to be more difficult."

> "Although many candidates did not state the rule that they were dt dx using, it was often clear that the rule was being misused, with confusion between terms and sign errors."

> "Many candidates did not recognise the expression as a quotient at dx −2t all – they treated the numerator as a constant and obtained = for their derivative."

> "This was a particular problem dt for candidates who had used the product rule, for whom the incorrect statement ( ) 1 2t +3 2t +3 2 = – was a common error."

Source: 9709_w10_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 33), Question 2

Coaching with Shuja

2 The parametric equations of a curve are

$$x = \frac{t}{2t+3}, \quad y = e^{-2t}.$$

Find the gradient of the curve at the point for which $t = 0$. [5]

Mark Scheme

Use of correct quotient or product rule to differentiate x or t **M1**

Obtain correct $\frac{3}{(2t+3)^2}$ or unsimplified equivalent **A1**

Obtain $-2e^{-2t}$ for derivative of y **B1**

Use $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ or equivalent **M1**

Obtain -6 **cwo A1** [5]

Alternative:

Eliminate parameter and attempt differentiation $\left(y = e^{\frac{-6x}{1-2x}}\right)$ **B1**

Use correct quotient or product rule **M1**

Use chain rule **M1**

Obtain $\frac{dy}{dx} = \frac{-6}{(1-2x)^2}e^{\frac{-6x}{1-2x}}$ **A1**

Obtain -6 **cwo A1** [5]

Coaching With Shuja

Question 4

Oct/Nov 2010 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The most common errors by this method were to omit a factor of 3 in the differentiation or to lose it from the trigonometric term resulting in an expression of the form kcos3xsinx."

> "Several candidates struggled to make any progress with the integral, including some who had started part (i) by using the correct double angle form."

Source: 9709_w10_er.pdf, Page 27, Section: Comments on specific questions – Paper 3 (variant 33), Question 4

4 It is given that $f(x) = 4\cos^2 3x$.

(i) Find the exact value of $f'\left(\frac{1}{6}\pi\right)$. **[3]**

(ii) Find $\int f(x)\,dx$. **[3]**

(i) Obtain derivative of form $k\cos 3x \sin 3x$, any constant k **M1**

Obtain $-24\cos 3x \sin 3x$ or unsimplified equivalent **A1**

Obtain $-6\sqrt{3}$ or exact equivalent **A1** [3]

(ii) Express integrand in the form $a + b\cos 6x$, where $ab \neq 0$ **M1**

Obtain $2 + 2\cos 6x$ o.e. **A1**

Obtain $2x + \frac{1}{3}\sin 6x$ or equivalent, condoning absence of $+c$, ft on a, b **A1√** [3]

Coaching With Shuja

Question 7

Oct/Nov 2010 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> “Where the process was not properly 5( ) understood, the most common error was to see calculation of values for a = 1+lnn for integer n 3 values of n.”

Source: 9709_w10_er.pdf, Page 28, Section: Comments on specific questions – Paper 3 (variant 33), Question 7

Coaching With Shuja

7 **(i)** Given that $\int_1^a \frac{\ln x}{x^2}\,dx = \frac{2}{3}$, show that $a = \frac{3}{2}(1 + \ln a)$. **[5]**

(ii) Use an iteration formula based on the equation $a = \frac{3}{2}(1 + \ln a)$ to find the value of a correct to 2 decimal places. Use an initial value of **4** and give the result of each iteration to 4 decimal places. **[3]**

Mark Scheme Oct/Nov 2010 • Paper 3 (variant 33) • Q7

(i) Attempt integration by parts **M1**

Obtain $-x^{-1}\ln x + \int \frac{1}{x^2}\,dx$, $\frac{x\ln x - x}{x^2} + 2\int \frac{\ln x}{x^2}\,dx - 2\int \frac{1}{x^2}\,dx$ or equivalent **A1**

Obtain $-x^{-1}\ln x - x^{-1}$ or equivalent **A1**

Use limits correctly, equate to $\frac{2}{3}$ and attempt rearrangement to obtain a in terms of $\ln a$ **M1**

Obtain given answer $a = \frac{3}{2}(1 + \ln a)$ correctly **A1** [5]

(ii) Use valid iterative formula correctly at least once **M1**

Obtain final answer **3.96 A1**

Show sufficient iterations to > 4 dp to justify accuracy to 2 dp or show sign change in interval $(3.955, 3.965)$ **A1** [3]

$[4 \to 3.9772 \to 3.9676 \to 3.9636 \to 3.9619]$

SR: Use of $a_{n+1} = e^{(3a_n - 1)}$ to obtain **0.50** also earns **3/3**.

Coaching with Shuja

Question 8

Oct/Nov 2010 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The most common error was to see α=52.2° rather than the 2 decimal places asked for in the question."

Source: 9709_w10_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 8

Coaching With Shuja

8 (i) Express $(\sqrt{6})\cos\theta + (\sqrt{10})\sin\theta$ in the form $R\cos(\theta - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. Give the value of α correct to 2 decimal places. [3]

(ii) Hence, in each of the following cases, find the smallest positive angle θ which satisfies the equation

(a) $(\sqrt{6})\cos\theta + (\sqrt{10})\sin\theta = -4$. [2]

(b) $(\sqrt{6})\cos\frac{\theta}{2} + (\sqrt{10})\sin\frac{\theta}{2} = 3$. [4]

Coaching With Shuja

(i) Obtain or imply $R = 4$ **B1**

Use appropriate trigonometry to find α **M1**

Obtain $\alpha = 52.24$ or better from correct work **A1** [3]

(ii) (a) State or imply $\theta - \alpha = \cos^{-1}(-4 \div R)$ **M1**

Obtain **232.2** or better **A1** [2]

(b) Attempt at least one value using $\cos^{-1}(3 \div R)$ **M1**

Obtain one correct value e.g. $\pm 41.41^\circ$ **A1**

Use $\frac{1}{2}\theta - \alpha = \cos^{-1}\left(\frac{3}{R}\right)$ to find θ **M1**

Obtain **21.7** **A1** [4]

Coaching With Shuja

Question 9

Oct/Nov 2010 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

> "The most common error in the integration was the mistaken use of ( ) logarithms to obtain a term in ln 2A−5 ."

Source: 9709_w10_er.pdf, Page 29, Section: Comments on specific questions – Paper 3 (variant 33), Question 9

Coaching With Shuja

9 A biologist is investigating the spread of a weed in a particular region. At time t weeks after the start of the investigation, the area covered by the weed is A m^2. The biologist claims that the rate of increase of A is proportional to $\sqrt{2A-5}$.

(i) Write down a differential equation representing the biologist's claim. [1]

(ii) At the start of the investigation, the area covered by the weed was 7 m^2 and, 10 weeks later, the area covered was 27 m^2. Assuming that the biologist's claim is correct, find the area covered 20 weeks after the start of the investigation. [9]

Coaching With Shuja

(i) State $\frac{dA}{dt} = k\sqrt{2A-5}$ **B1** [1]

(ii) Separate variables correctly and attempt integration of each side **M1**

Obtain $(2A-5)^{\frac{1}{2}} = \ldots$ or equivalent **A1**

Obtain $= kt$ or equivalent **A1**

Use $t = 0$ and $A = 7$ to find value of arbitrary constant **M1**

Obtain $C = 3$ or equivalent **A1**

Use $t = 10$ and $A = 27$ to find k **M1**

Obtain $k = 0.4$ or equivalent **A1**

Substitute $t = 20$ and values for C and k to find value of A **M1**

Obtain **63 cwo A1** [9]

Coaching With Shuja

Question 3

Oct/Nov 2009 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 3 (i) The implicit differentiation of y3 was often correct, but applying the product rule to x2y caused more dy problems, and many candidates made a sign error at this stage, with −2xy +x2 being a very dx dy common error."

Source: 9709_w09_er.pdf, Page 17, Section: Comments on specific questions – Paper 3 (variant 32), Question 3

Coaching With Shuja

The equation of a curve is $x^3 - x^2y - y^3 = 3$.

(i) Find $\frac{dy}{dx}$ in terms of x and y. [4]

(ii) Find the equation of the tangent to the curve at the point $(2, 1)$, giving your answer in the form $ax + by + c = 0$. [2]

(i) State $2xy + x^2\frac{dy}{dx}$ as derivative of x^2y **B1**

State $3y^2\frac{dy}{dx}$ as derivative of y^3 **B1**

Equate derivative of LHS to zero and solve for $\frac{dy}{dx}$ **M1**

Obtain answer $\frac{3x^2-2xy}{x^2+3y^2}$, or equivalent **A1** [4]

(ii) Find gradient of tangent at $(2, 1)$ and form equation of tangent **M1**

Obtain answer $8x - 7y - 9 = 0$, or equivalent **A1** [2]

Total: 6

Coaching With Shuja

Question 5

Oct/Nov 2009 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "The standard of presentation in solutions to this problem was disappointing, with many candidates forming correct expressions by substituting x = −2 but not immediately equating their expressions to zero."

Source: 9709_w09_er.pdf, Page 18, Section: Comments on specific questions – Paper 3 (variant 32), Question 5

Coaching With Shuja

Question Paper Oct/Nov 2009 • Paper 3 (variant 32) • Q5

The polynomial $2x^3 + ax^2 + bx - 4$, where a and b are constants, is denoted by $p(x)$. The result of differentiating $p(x)$ with respect to x is denoted by $p'(x)$. It is given that $(x + 2)$ is a factor of $p(x)$ and of $p'(x)$.

(i) Find the values of a and b. [5]

(ii) When a and b have these values, factorise $p(x)$ completely. [3]

Mark Scheme

Oct/Nov 2009 • Paper 3 (variant 32) • Q5

(i) Substitute $x = -2$, equate to zero and state a correct equation, e.g. $-16 + 4a - 2b - 4 = 0$ **B1**

Differentiate $p(x)$, substitute $x = -2$ and equate to zero **M1**

Obtain a correct equation, e.g. $24 - 4a + b = 0$ **A1**

Solve for a or for b **M1**

Obtain $a = 7$ and $b = 4$ **A1** [5]

(ii) EITHER: State or imply $(x + 2)^2$ is a factor **B1**

Attempt division by $(x + 2)^2$ reaching a quotient $2x + k$ or use inspection with unknown factor $cx + d$ reaching $c = 2$ or $d = -1$ **M1**

Obtain factorisation $(x + 2)^2(2x - 1)$ **A1**

OR: Attempt division by $(x + 2)$ **M1**

Obtain quadratic factor $2x^2 + 3x - 2$ **A1**

Obtain factorisation $(x + 2)(x + 2)(2x - 1)$ **A1** [3]

[The M1 is earned if division reaches a partial quotient of $2x^2 + kx$, or if inspection has an unknown factor of $2x^2 + ex + f$ and an equation in e and/or f, or if two coefficients with the correct moduli are stated without working.]

Total: 8

Coaching With Shuja

Question 6

Oct/Nov 2009 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Question 6 (i) This part of the question proved to be particularly difficult for many candidates, either through algebraic errors or through errors in completing the substitution."

> "Common errors included using $x2 = 2\tan 2\theta$, and simply replacing dx by dθ."

> "(ii) There were many correct answers to this part, but many candidates did not appreciate that the first part of the question assisted them with the second and produced false answers, often involving a ∫ logarithm."

Source: 9709_w09_er.pdf, Page 18, Section: Comments on specific questions – Paper 3 (variant 32), Question 6

Coaching With Shuja

(i) Use the substitution $x = 2\tan\theta$ to show that

$$\int_0^2 \frac{8}{(4+x^2)^2}\,dx = \int_0^{\pi/4} \cos^2\theta\,d\theta.$$

[4]

(ii) Hence find the exact value of

$$\int_0^2 \frac{8}{(4+x^2)^2}\,dx.$$

[4]

(i) State or imply $\frac{dx}{d\theta} = 2\sec^2\theta$ or $dx = 2\sec^2\theta\, d\theta$ **B1**

Substitute for x and dx throughout **M1**

Obtain any correct form in terms of θ **A1**

Obtain the given form correctly (including the limits) **A1** [4]

(ii) Use $\cos 2\theta$ formula, replacing integrand by $a + b\cos 2\theta$, where $ab \neq 0$ **M1***

Integrate and obtain $\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta$ **A1**

Use limits $\theta = 0$ and $\theta = \frac{1}{4}\pi$ **M1(dep*)**

Obtain answer $\frac{1}{8}(\pi + 2)$, or exact equivalent **A1** [4]

Total: 8

Coaching With Shuja

Question 8

Oct/Nov 2009 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> “Common errors included 3 correct deduction that 3A = 2, but then concluding that A = or starting with an initial form 2 omitting one of the terms in the linear numerator.”

> “The term 2x−1 ( ) caused many more difficulties.”

> “Common errors included confusion over how to deal with 32+x2 2x−1 the constants in the numerator resulting in a starting point of ( ), use of the incorrect form 92+x2 −1 ☐ x2 ☐ (2 + x2)−1 = 2☐14+ ☐ , and errors in multiplying their expansion by (2x – 1).”

Source: 9709_w09_er.pdf, Page 19, Section: Comments on specific questions – Paper 3 (variant 32), Question 8

Coaching With Shuja

(i) Express $\frac{1+x}{(1-x)(2+x^2)}$ in partial fractions. [5]

(ii) Hence obtain the expansion of $\frac{1+x}{(1-x)(2+x^2)}$ in ascending powers of x, up to and including the term in x^2. [5]

Coaching With Shuja

(i) State or imply partial fractions are of the form $\frac{A}{1-x} + \frac{Bx+C}{2+x^2}$ **B1**

Use a relevant method to determine a constant **M1**

Obtain $A = \frac{1}{3}$, $B = \frac{1}{3}$ and $C = \frac{1}{3}$ **A1** + **A1** + **A1** [5]

Total: 5

Question 9

Oct/Nov 2009 | Paper 3 (variant 32)

Examiner Report — Why students found this question difficult

> "Common errors included incorrect or non-existent attempts to separate the variables in the given equation, or omission of a constant of ☐ 3A ☐ integration."

Source: 9709_w09_er.pdf, Page 19, Section: Comments on specific questions – Paper 3 (variant 32), Question 9

Coaching With Shuja

Question Paper Oct/Nov 2009 • Paper 3 (variant 32) • Q9

The temperature of a quantity of liquid at time t is θ. The liquid is cooling in an atmosphere whose temperature is constant and equal to A. The rate of decrease of θ is proportional to the temperature difference $(\theta - A)$. Thus θ and t satisfy the differential equation

$$\frac{d\theta}{dt} = -k(\theta - A),$$

where k is a positive constant.

(i) Find, in any form, the solution of this differential equation, given that $\theta = 4A$ when $t = 0$. [5]

(ii) Given also that $\theta = 3A$ when $t = 1$, show that $k = \ln \frac{3}{2}$. [1]

(iii) Find θ in terms of A when $t = 2$, expressing your answer in its simplest form. [3]

Coaching With Shuja

(i) Separate variables correctly **B1**

Integrate and obtain term $\ln(\theta - A)$, or equivalent **B1**

Integrate and obtain term $-kt$, or equivalent **B1**

Use $\theta = 4A$, $t = 0$ to determine a constant, or as limits **M1**

Obtain correct answer in any form, e.g. $\ln(\theta - A) = -kt + \ln 3A$, with no errors seen **A1** [5]

(ii) Substitute $\theta = 3A$, $t = 1$ and justify the given statement **B1** [1]

(iii) Substitute $t = 2$ and solve for θ in terms of A **M1**

Remove logarithms **M1**

Obtain answer $\theta = \frac{9}{4}A$, or equivalent, with no errors seen **A1** [3]

[The M marks are only available if the solution to part (i) contains terms $a\ln(\theta - A)$ and $br.t.]

Total: 9

Coaching with Shuja

Question 3

Oct/Nov 2017 | Paper 1 (variant 12)

Examiner Report — Why students found this question difficult

Coaching With Shuja

(a) Each year, the value of a certain rare stamp increases by 5% of its value at the beginning of the year. A collector bought the stamp for $10000 at the beginning of 2005. Find its value at the beginning of 2015 correct to the nearest 100. **[2]**

(b) The sum of the first n terms of an arithmetic progression is $\frac{1}{2}n(3n+7)$. Find the 1st term and the common difference of the progression. **[4]**

\section*{Question 3}

\subsection*{3(a)}

\begin{tabular}{|l|l|l|l|}

Answer & Marks & Guidance \

$r = (1.05 \text{ or } 105\%)^{10 \text{ or } 11}$ & **B1** & Used to multiply repeatedly or in any GP formula. \

New value = Misplaced & & **B1** & Uses $n = 1$ to find a \
$a = 5$ & & \

$n = 2 \rightarrow 13$ & **B1** & Correct S_n for any other value of n (e.g. $n = 2$) \

$a + (a + d) = 13 \rightarrow d = 3$ & **M1 A1** & Correct method leading to $d =$ \

OR: & & \
$\binom{n}{2}(2a + (n-1)d) = \binom{n}{2}(3n + 7)$ & & $\binom{n}{2}$ maybe be ignored \

$\therefore dn + 2a - d = 3n + 7 \rightarrow dn = 3n \rightarrow d = 3$ & (***M1A1** & Method mark awarded for equating terms in n from correct S_n formula. \

$2a - (\text{their } 3) = 7$, & **DM1 A1)** & \
$a = 5$ & & \

& 4 & \

\end{tabular}

Total: 6

Question 2

Oct/Nov 2017 | Paper 3 (variant 33)

Examiner Report — Why students found this question difficult

Coaching With Shuja

2 Two variable quantities x and y are believed to satisfy an equation of the form $y = C(a^x)$, where C and a are constants. An experiment produced four pairs of values of x and y. The table below gives the corresponding values of x and $\ln y$.

x	0.9	1.6	2.4	3.2
$\ln y$	1.7	1.9	2.3	2.6

By plotting $\ln y$ against x for these four pairs of values and drawing a suitable straight line, estimate the values of C and a. Give your answers correct to 2 significant figures. **[5]**

Coaching With Shuja

\section*{Question 2}

\begin{tabular}{|l|l|l|}

Answer & Marks & Guidance \

Plot the four points and draw straight line & **B1** & \

State or imply that $\ln y = \ln C + x \ln a$ & **B1** & \

Carry out a completely correct method for finding $\ln C$ or $\ln a$ & **M1** & \

Obtain answer $C = 3.7$ & **A1** & \

Obtain answer $a = 1.5$ & **A1** & \

& 5 & \

\end{tabular}

Total: 5

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